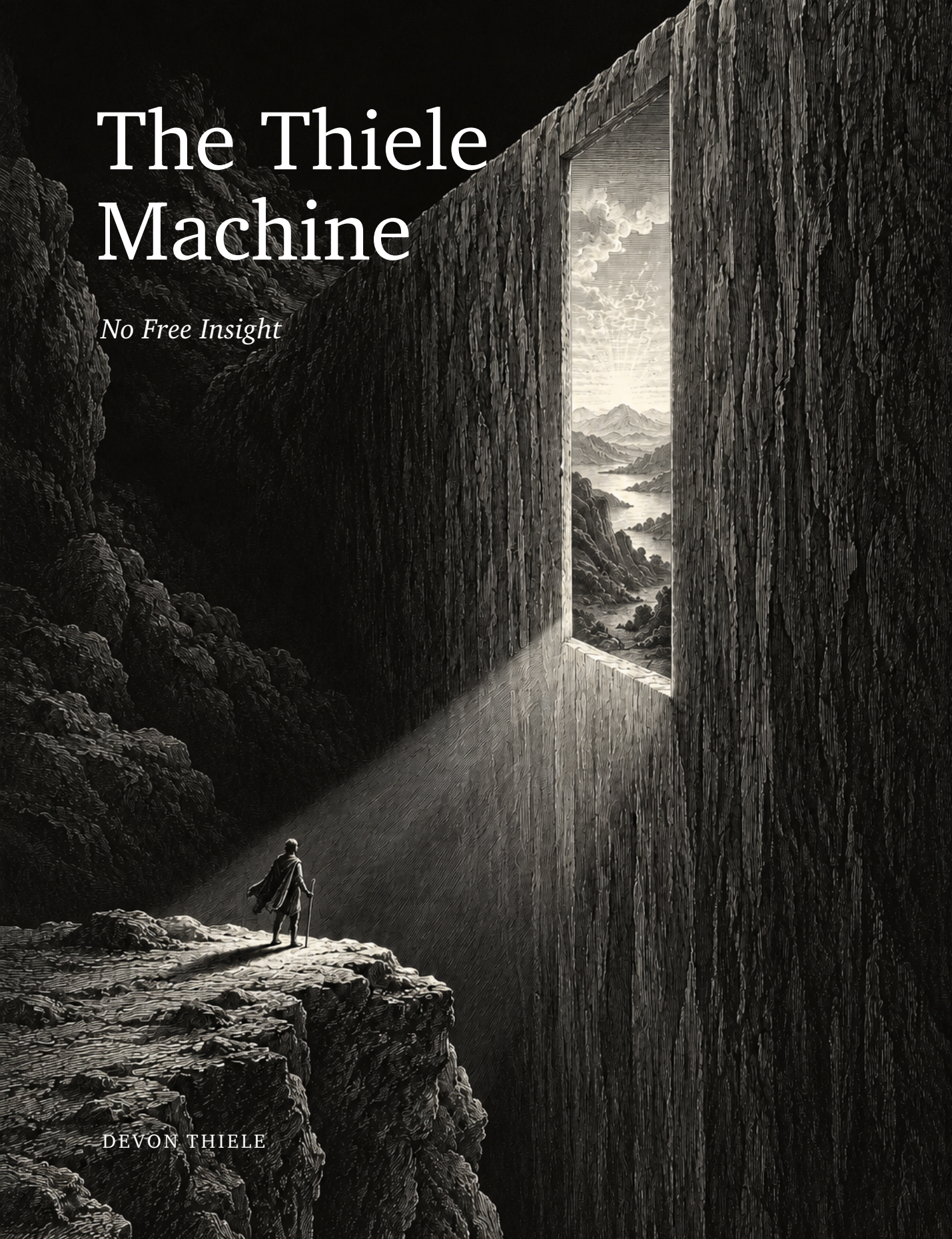
# The Thiele Machine

*No Free Insight*



DEVON THIELE

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# The Thiele Machine

*No Free Insight*

*The Thiele Machine: No Free Insight*

Devon Thiele

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The source of this book, its proofs and its build are public: github.com/sethirus/The-Thiele-Machine. Archived at doi.org/10.5281/zenodo.17316437.

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The cover and part illustrations were generated with AI in the engraving tradition of Gustave Doré. They are new illustrations, not reproductions of his work.

For Julie, and for my dad.

Julie,

Three years ago, a few months into the rendering engine, I came out of my office and told you I thought I was onto something. You said okay. You didn't ask what it was, or what it was for, or when it would be done, and you didn't ask me for anything I could have shown you, because there was nothing to show. There was a program that had done something it shouldn't have been able to do, and a man who didn't yet know what a Turing machine was.

You gave me the three years anyway. The nights after the lot closed. The weekends. The hours I was in the house and not in the house, because my head was in a wall I couldn't get past. Every dead end, and there were more of those than there are pages here. You never once asked me to stop, and you never once said it was fine when it wasn't. You just kept giving me the next evening.

This book is what I did with them. It's the one thing I've built from nothing that I'm sure of, because I threw out every piece of it I couldn't prove, and what's left holds. I know that isn't what you gave the years for. You gave them for me. But it's what I have to show, and I wanted it to be worth showing.

Dad,

The name on the cover was yours first. I asked you before I used it and you said go ahead. You and Mom taught me that nothing is true just because somebody says it is. This book is that lesson, six hundred pages long. I put our name on it because I wanted one thing in the world with that name on it that nobody handed me, that I couldn't have faked, and that I can stand behind in front of anyone. Here it is.

Julie carries the name now too. So the machine has it third. This is for the two of you who had it first.

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# The point

The whole book is four steps, and I'll set them out first so you know what you're carrying. Each step is motivated by the one before it, and the book goes up them in order.

*One: nothing becomes something for free.* Take a blank cell: a square of paper with nothing on it, a bead that could sit in either of two dips, a bit of memory nobody has set. Now make it a 0, and keep it a 0. Something is the case now that wasn't a moment ago. For a total deterministic rule, a reading newly raised and then held requires a merge somewhere in the run or endlessly fresh states. Physical dissipation requires a separate model; the thermal models discussed below impose further costs under their own hypotheses.

*Two: so what a computation makes the case is a resource.* Computer science charges a computation for two things, how long it runs and how much space it takes, and the bits it erases have been counted too. What no model I found charges is the moment a reading goes up and stays. The Thiele Machine is a model of computation that does: the step that raises a reading that stays is the step that's charged. Those steps are a few events in a run, not the billions of overwrites a computer makes every second; what they write doesn't stay, and the toll leaves them to the erasure count and to physics. Put the record inside the machine, then choose a window that shows only the working part. Through that window you can watch the computation while missing what the machine kept about it. That is the picture behind "computation is the slice." I will have to earn it by giving the window and the moves precise rules; calling something a window is not enough.

To show the definition can be met, Part III builds one example. It starts with two counters and a line number. The computing moves add, subtract and jump. The record moves check a claim, select a passed check and certify it. While the example is live, the counters and line follow their own computing rules; looking only there leaves out the record and its ledger. A failed record move can stop the example, so that boundary belongs in the statement too. Theorem D.4 says exactly which other computers admit the same construction and which do not. The Thiele Machine is the definition, not the example. Here the point is the view: working states can look alike while the full states differ in what was kept.

The counting underneath needs its own conditions too: states, a next-state rule, and a reading that stays. Once those are given, what follows is a theorem. Whether a physical process fits that description is the further question stated in the Conclusion.

*Three: the structure a computation makes is made of exactly this.* Structure in general is a relation kept under the changes you choose; Part II takes that word from the people who wrestled with it. The part a computation makes, and can't take back, is a distinction held in its state, and a 0 is the smallest one there is: the ways a cell could be, sorted into piles, and one pile held. A grouping, an order, a relation a later move leans on, a fact somebody found out and wrote down: each is a distinction that stays, paid for the same way as the 0. Insight, in the sense this book can price, is a kept distinction. The toll charges the moment it's raised; holding it up afterward adds nothing to the cost. That's what the book's other name, No Free Insight, means.

*Four: established is the sharpest case.* Two computers can end with the same answer, and only one of them checked it. The difference between them is a distinction one of them never kept. A record that a claim was checked and closed is a 0 that stays, with a check in front of it, and it's paid for the same way. I build a machine for that case, all the way down, because it's the one where you can point at the kept 0 and at what it cost. It's the sharpest case of what keeping takes, but it isn't the point. The point is the 0.

Step one is where the toll gets its reason. If it breaks, the toll goes back to being only my choice, though the latch, the window, the diagonal and the earned record are proved without it and would stand. So that's where I start: the blank cell, taken apart as far down as I can get before I hit something that holds or something I can't get under. I'll tell you which.

One warning about how this went, because it explains the shape of everything after it. Most of the time I had the hunch first and found the names later. Over and over, somebody had been here before me. Sometimes that made my day. Sometimes it pissed me off. You'll see both.

### Where a 0 comes from

Before any machine comes the cell. What does it take for there to be a 0 at all?

It takes three things. First, something that can be more than one way: a bead in a dish of water, a speck of magnetised metal, a charge on a tiny capacitor, a square of paper. Nature hands you that much, and it isn't a 0 yet. It's stuff that can sit in a lot of arrangements.

Second, a sorting: somebody says which arrangements count as 0, which count as 1, and which count as blank. The sorting is the first piece of structure Part II gets to: the ways a thing could be, cut into piles. Without it there's no 0, just a bead somewhere. A 0 and a 1 aren't things; they're the two sides of a line somebody drew across the ways a thing can be.

Third, something that keeps the chosen arrangement readable for the time you need it. In the bead experiment below, that something is a hill between two wells. A high hill makes escape rare; it does not make escape impossible. At a barrier of about eight times the thermal energy, the paper estimates a characteristic escape time of roughly fifty minutes. Lower the barrier to about twice that energy and the estimate is about ten seconds. Those are waiting-time scales, not appointments at which the bead must jump. The barrier height is not a bill for heat already given off.

A latch, a refreshed charge or a written mark needs its own account of how it holds. The common question is simple: which later changes are allowed, and do they leave the answer readable? In the mathematical examples I can require a yes to stay yes under every allowed move. The bead is a physical example of retention for a time, with a chance of failure.

Now for the moment itself: the cell is blank and you write a 0. Everything that step can take comes next, in Part I, and the rest of the book only ever adds detail to it.

### Why it has to cost

Now what it takes to keep that 0, made exact, and it's the part I'm proudest of, because once something stays, the choosing stops and the counting takes over. And the counting is the one behind musical chairs.

Do it with chairs when the room runs out. Draw a box on the floor, three chairs inside, one chair outside by the door. Four people, one per chair: the four states the machine could be in. The three inside already read yes. One step moves everybody at once, each to the chair the rule sends them to. Nobody inside may leave, because yes is permanent. The person by the door has to come in, because this is the step that flips them to yes. Now finish the move, one person per chair. You can't. Four people, three inside chairs: somebody shares. Maybe the newcomer ends up on a lap, maybe two insiders squeeze onto one chair; either way, two starting places share a destination. Somebody secretly picks a starting chair, the rule moves them, and you're told only where they ended up: at a shared chair, you can't say where they started. That's the pigeonhole principle, which is old and not mine, and the chapter "Why the toll isn't arbitrary" (Section 3) proves it with exactly three conditions on a rule that gives every state a next one: finitely many states that read yes, a yes that stays, and one flip. There's no physics in it anywhere, only counting.

So, the deepest thing I can say about keeping: not every forgetting keeps anything, and wiping a scratch pad keeps nothing. But every keeping pays. On a finite machine that keeps running, keeping pays by forgetting: the step that makes something stay for good lands two states on one, somewhere, maybe not where you happen to be standing, and afterward nothing can tell which one it was. With endless fresh states

it can dodge the merge only by spending fresh states it never gets back. A verdict keeps the decision and lets go of the arguing.

Part I proves the count on any machine, prices it on finite ones, and keeps the heat floor as an assumption with its name on it. I never let those two swap places.

An ordinary computational model need not carry a record of which checks ran or an account of establishing that record. Ordinary computers can implement both. What I want is a model that puts the record and its charge in the transition rules themselves. That's the Thiele Machine.

### What the usual models don't count

That's step one, and step two is what it means for computing.

Computer science has always counted what a computation costs. How many steps it takes is time. How many cells it needs is space. Whole fields rest on those two counts, and on the models built to make them exact: the random-access machine of the computer scientists Stephen Cook and Robert Reckhow (1973), which charges each instruction a unit; the cell-probe model (1981) of Andrew Yao, which counts probes of memory cells, leaving the computation between probes free; the input-output model of Alok Aggarwal and Jeffrey Vitter (1988), which counts only the blocks moved between a small fast memory and a large slow one. The random-access machine computes exactly what Turing's machine computes, and the other two leave the computing between the moves they count out of the count. They differ in what they count, and the counting is the reason to have them. Manuel Blum gave two axioms in 1967 for a general class of complexity measures. They let him prove results across different machines; a proposed cost still has to meet those axioms to belong to that class.

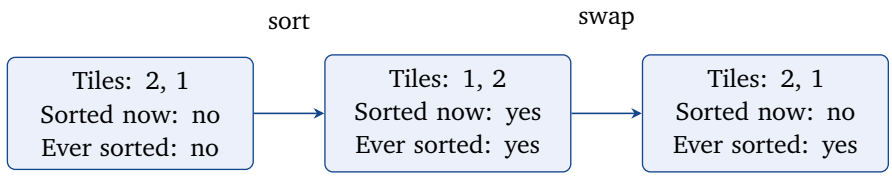
Now look again at what step one found keeping takes. Every time a computation makes something the case and keeps it, a bit set, a list left in order, an answer written down, it pays, in a merge or in fresh states. What it pays isn't the time count: a slow step and a fast one can keep the same thing. It isn't the space count either, though fresh states are one of the two ways to pay it: a machine with a fixed memory pays it by merging, over and over, in the same cells, and its space count never moves. The merge half of it has been counted before. Ming Li and Paul Vitányi (1996) took as axioms of a model of computation that a step that can be undone costs nothing and that every bit a computation irreversibly erases costs one unit, and Erik Demaine and three coauthors (2016) put that count on the random-access machine, beside time and space. What none of them charges, as far as I found, is the moment a chosen reading goes up and stays, whichever way keeping it is paid. That's a third thing a computation does, and it's what this model adds. It isn't a cost in Blum's sense once the computing itself costs nothing. Blum asks that "this run

costs exactly $m$" be decidable, and then it isn't: a program that runs another program at no cost and makes one move costing one unit if that one halts costs exactly one exactly when the other halts, so deciding its cost would decide halting. (Price every move at one and a run's cost is its step count, which is a cost in Blum's sense. And a run that never stops can still show a finite count; leave its cost undefined there, as Blum does, and his first axiom holds. It's the decidability that fails.)

So the Thiele Machine, all of it, is a model of computation, math you can reason about, the way Turing's machine is. It has five parts and one rule: a set of states, a set of moves, a step that takes a state and a move to the next state, a price on each move, a whole number of units of cost, and a record: a reading of the state into an ordered set, so that for any two records you can ask whether one sits at or below the other. The rule, which I call the toll, says that a step that carries the record somewhere not at or below where it stood costs at least one unit. The smallest record has two places, no and yes. There the toll says that turning no into yes costs at least one, and most of Part I is told with that record, because it is the one you can point at. The toll sits on the raise, not on the merge, so it charges for keeping something whichever way that's paid, in a merge or in fresh states. Two more pieces come in when a result needs them: a counter in the state that adds up the costs and only climbs, which is the ledger, and a window, what an outside observer gets to see. The toll's first name, the one I gave it before I understood it, is No Free Insight.

Put a dial with four positions beside a slider with three stops. ROTATE moves the dial one position around for free and leaves the slider alone. ADVANCE moves the slider toward its last stop, leaves the dial alone and costs one. At the last stop the reading is yes; at the other two it is no. ADVANCE pays when it first reaches yes and still costs one when pressed at the last stop. The price belongs to the button. Look only at the dial: ROTATE moves the picture and ADVANCE leaves it alone. Those are the horizontal and vertical moves of the sliced definition below. There are twelve full configurations, four dial positions for each slider stop. We have met the rules without introducing a checker or a claim.

It computes nothing new, any more than the random-access machine does. What it adds is a place, in the step itself, for what a computation made the case and what that cost. The reading can be any yes-or-no reading at all; the toll charges every step that turns it on. The readings this book cares about are the ones that, once yes, stay yes, because those are the kept distinctions and keeping is what the merge or the fresh states pay for. A cell's "it's a 0 now" stays only if the later moves preserve that value. Likewise, a sorted list can be scrambled by the next write. A flag saying "this list was sorted earlier" answers a different question.




The tiles can lose their order. A separate history flag can keep the fact that they were ordered earlier. Read one, and you have not read the other.

"This claim was checked and closed" can be a historical reading of that kind. The model doesn't care which, and for readings that stay, Part I derives the toll from what keeping takes, on stated premises. A reading that flips back and forth does not by itself retain whether an event ever happened: Part II calls it a toggle and proves it's the latch of no event (Section 10).

The one in the toll is one unit of cost, in the model's own count, and it's one because that's the least a whole-number count can charge and still not be free, and the toll only says *not free*. A unit of cost isn't a joule. What ties it to heat is Landauer's principle, named, through the merge a raise has to make when only finitely many states read yes, and nothing else.

And there's one thing to hold on to through the whole book, because it's where readers go wrong and where I went wrong for a long time. *The ledger is a count, not a lock.* It doesn't stop anybody from doing anything. It adds up the prices you chose for the moves that ran, writes the sum down where the machine can't lose it, and the toll makes each raise of the reading you chose cost at least one. It doesn't measure merges, fresh states or heat: a merge that keeps nothing can cost nothing, fresh states are taken at every step and the toll asks for a unit only at the raise, and where a merge raises nothing, a cost of one unit per raise leaves that merge out. What ties the count to what keeping takes is a theorem, not a reading of a dial: on a machine with finitely many states reading yes, any price list that charges merges has to charge each raise of a reading that stays at least one. What a kept reading *means*, whether a reading that's up stands for anything, is a separate question with a separate answer, and Part III gives it for one case: an order the step enforces, which is the lock. A lock would work at a price of zero. The price isn't there to make the lock work. It's there because, on a machine where every state has a next one, the raise of a reading that stays merges or takes fresh states, whether anybody writes it down or not.

### What I found, what I assumed and what I chose

I didn't invent a machine. I found one. That's how I see this work, and I'll be straight about the difference between that conviction and a theorem before I do anything else.

That sentence walks straight into one of the oldest fights in mathematics, so I'll take a short trip into it first. G. H. Hardy, a Cambridge number theorist who wrote *A Mathematician's Apology* in 1940 to defend doing mathematics for its own sake, came down on the found side about as hard as anyone ever has: mathematical reality, he wrote, lies outside the people who study it, and the theorems they like to call their creations are really just notes on what they saw there. The German mathematician Leopold Kronecker supposedly split the difference: God made the whole numbers, and everything else is the work of people. Supposedly, because push on that line and it gets thin fast. It comes down through Heinrich Weber, another German mathematician, in a memorial piece written after Kronecker died, pointing back to a lecture from 1886, and that's the only source I could find anybody citing. It's a great line with one witness. (The record chapter of Part I, Section 2, stops early on to ask how many witnesses a claim ought to need. The answer is more than one.) None of this settles whether I found the Thiele Machine or made it up. A proof doesn't settle it either: showing what follows from the premises doesn't decide whether you found the theorem or made it. That's why it's labelled a conviction. What it changes is how I hold the thing: like something that was already there, which means I can be wrong about its shape and the shape won't care. And to be exact about what "found" covers, it covers the parts the counting forces on you, the latch shape and the merge or the fresh states. The event to record, the price list, and putting the count inside the step are choices, mine, and the three lists below say which is which.

The three lists draw the line between what nothing let me choose, what I had to assume, and what I did choose, because everything after this depends on where it falls. Read the first list as the ground, the second as what the ground is taken to stand on, and the third as the model built on it.

*Found: what counting forces on any machine.*

- Run a machine that gives each state exactly one next state. It takes fresh states until it first comes back to one it has used; that return lands on its start or is a merge, two states with one next state; and after it the run only repeats steps it already took. There's no fourth thing a step can do, and a run merges at most once (Theorem 1.1).
- So any run that never comes back to its start merges somewhere or takes a fresh state at every step. A reading that goes from no to yes and never comes back rules the return out, so every run that raises it merges or takes fresh states (Proposition 3.5). The sharper fact is about the move: when only finitely many states read yes, the move that raises the reading merges two states (Theorem 3.1 for a finite machine, Proposition 3.4 when only the yes-states are finitely many). A machine that can stop, with a state that has no next state at all, slips past

this, and Part I says exactly how (Theorem 1.2). All of that is counting, with no physics in it.

- Over any machine at all, a record that the computation drives and that never comes down is a latch on one event: it switches on the first time something happens and nothing switches it off (Observation 10.1). That's counting too, given the two conditions in it.

*Assumed: what the found list is taken to stand on, in the world.*

- A merge of states the machine really could have been in, done in surroundings at some temperature, has a floor under its heat, on average, however slowly it's done, in a memory whose states are alike apart from what they record. That's Landauer's principle, the premise from thermodynamics, named and assumed and never proved here, with its critics named.
- A real device can tell apart only finitely many states in any bounded place, so its fresh states run out. Nobody has measured where that ceiling sits.
- The clean start: a raised record means the event happened only because the record was down at the start, and that start is put in by hand. It's a past hypothesis in miniature. The past hypothesis, the philosopher David Albert's name (2000) for an assumed low-entropy beginning of the universe, is an idea that goes back to Boltzmann; several accounts of why records point to the past lean on it, and every theorem here about records gets its arrow from an assumed clean start (Section 10).

*Chosen: what the model adds, each of which could have gone another way.*

- To put the account inside the step, as part of what a step is. Two things go there. That the record stays and the computing can't reach it has to: the record is one of the five parts, a reading of the state, and a rule about the state kept by the step holds of every program, where a log a program keeps holds of that program only.
- The toll itself: a whole unit, at least one, on the step that raises a reading, whichever way keeping is paid. On a machine with finitely many states reading yes, for a reading that stays, it follows from a price on merging, and that price is a choice too.
- Which readings matter: the ones that never come down. The toll charges any reading's rise; permanence is what makes the rise a case of keeping, and permanence is a choice. So is one next state for every state, the condition the stopping hole gets around.
- Which event gets a record, and what price list the record pays. The latch leaves both unforced. And on a machine that pays in fresh states, where the write merges nothing, the record stays up because I declare that it does (Theorem 10.3).

- On the machine of Part III, where its ordinary computer ends and its record begins, and that its step never reads the flag: the rest of the state moves by its own rule and the ledger adds the instruction's price, whichever way the flag reads.

The machine Part III builds is one worked demonstration of the third list, on one reading. Nothing in the first list needs it.

**What structure is made of**

Step three says that a 0 is the smallest structure that sorts, and that the rest of structure is the same thing at a bigger size.

A sorting of a pile into groups is a distinction: these go together, those don't. Keep it and it's a distinction that stays, paid for the same way. An ordering is every pair compared and the answers kept. A fact somebody found out and wrote down is a distinction between the ways things could have been and the way they were, kept. A machine can carry any of those in its state and in its rules, and each one it keeps is paid for the way the 0 is. Part II goes through the people who wrestled with the word *structure*, from Pythagoras to category theory, and comes out at that: structure is distinctions held in a state.

That gives *insight* an exact meaning, a narrower one than you'd like, and I'll be honest about how narrow. Finding something out doesn't have to pay the merge. Bennett showed that you can write down what you see on a blank page without merging anything, so Landauer's floor isn't owed; the page is what you spend, and it can be handed back while the original is still there. Making it stay is what pays. Learning borrows, committing pays. The insight the toll prices is the committed kind, a distinction made to stay. Whether that narrower thing deserves the word is a fair thing to argue about, and Part I argues it, where learning and committing get pulled apart.

Then Part II asks what follows, on any machine, once a distinction is kept and the keeping is charged. The ledger is itself a distinction that stays: a number that only climbs takes a fresh value for every unit it counts. A window that shows only part of a state can hide the very distinction that was paid for, and when it shows a step that paid and one that didn't the same, before and after, no price read off the window can be exact. A later step can be entitled to something because a distinction stayed, and the entitlement has an exact count. Over any machine at all, a reading the machine's own steps turn on and nothing turns off is a latch, the blank cell's 0 one level up. And there's a wall no level gets past: no state codes every question about the states.

### Established

Step four is the sharpest case, and it's the one I build a machine for.

Give two workers the same messy list. The first sorts it, then runs a separate test that compares each neighbouring pair and reports whether the order is right. The test passes. The second copies in that same sorted list and runs no test. Ask only for the final list. The bytes agree. Reading them can tell you whether they are sorted now; it cannot tell you whether the earlier test ran.

I have specified the output being compared, not every register and every line of program memory. If those retain the difference, the full machine can still tell the histories apart. If a cleanup makes the full endpoints agree too, then nothing in those endpoints can answer which history occurred. An ordinary computer can retain that evidence when its program and interface require it. The question is what this task asks it to keep.

*Established* is the name for the difference. An answer is established when a check that could have said no said yes, about that very answer, and the result was made to stay. That last part is a kept distinction, a 0 in some cell that says "checked," and it's paid for the same way as every other 0. Part III opens by taking that apart (Section 14): what "could have said no" means, what settling a question is, and why the people who thought hardest about knowledge kept landing near it.

How far the example reaches, to other models of computation and to universal machines that run it as a guest and earn its flag again, is proved too; Part III states it (Section 16) and the appendices build it. That's the reach of the case, not what the book is about.

### The four steps of the argument

The four steps above are the argument. The proofs establish the counts and constructions; the claims section at the back lists them with what each stands on.

Argued only: that the structure a computation makes is kept distinctions, priced by the toll, while a reading that undoable moves keep never rises and is never charged; that's a choice of what to price, and on a finite machine the second half is proved.

Argued only: that an account of computation *should* carry what a computation made the case, and what it cost, in how it steps. The proofs show that it hangs together, what it follows from, and what it costs; whether it ought to be adopted is a claim about what matters, which no theorem decides.

My proposed reason for choosing certification is that its records get copied and relied on by parties who didn't produce them. That is the pointer thesis in Part III.

It connects the machine to how people use records; the counting arguments work with whichever record is chosen.

The results also have a history. By my own sorting at the back (Appendix B), each piece has its neighbours standing right there. Start from the top. The small machine and its universal programs carry the reference monitor's every-program move (Anderson in 1972) to a record that's priced and earned; the two-player game check in the appendices (Appendix G) is careful engineering on known mathematics. The latch has the 1918 flip-flop and write-once memory for neighbours. The window fact is an old fact about functions, from Edward Moore's 1956 thought experiments and Rudolf Kalman's observability, with a price put on it. The toll derived from finiteness, permanence and a price on merging is a ledger form of the core underneath. And that core, what step one says keeping takes, is old mathematics: the pigeonhole principle, Dedekind's 1888 definition of infinite and the finite heart of Ernst Zermelo's 1896 recurrence argument, with Landauer and Bennett on merging and Thomas Ouldridge and David Wolpert's 2023 price in entropy on the merges a finite automaton can't avoid. For deterministic machines the arrangement is old too: Ming Li and Paul Vitányi (1996) wrote that a bounded memory forces erasure in the long run, and Demaine and his coauthors (2016) that extra space only postpones it. What I'd defend most is there, at the bottom, in the form I didn't find: one law with three columns and no fourth, stated alike for deterministic, noisy and quantum steps, with two small exact lemmas, the count of fresh states for a bounded memory and every scale paying the same way, and a toll on a reading that stays, which charges for keeping whichever column pays for it.

The route through the book is the route through the model: follow a step, see which distinctions survive it, and count what the next step has to work with. Then put the record and its price into the step itself. The example at the end shows those choices can all be made at once.

### How the book goes

Part I is what keeping costs. It covers a distinction that stays, and what every step of every machine does to the states it has used; the smallest thing that writes what keeping takes into its own step, as one rule, the toll; whether the rule is forced, and on which premises; and why finding out only borrows, where committing pays.

Part II is structure, as kept distinctions: the word itself, from the people who wrestled with it, and then what follows on any machine once a distinction is kept and the keeping charged: the ledger, the window, entitlement, the latch, and the wall no level gets past.

Part III is the sharpest case: what established means, the machine that keeps it, how far the machine reaches, five real systems that keep records and whether they

price the write, the thesis about which kept distinction is worth paying for, and what its ledger has to do with heat.

Part IV brings the results together: what each establishes, which conditions it uses, and how the constructions fit. After that comes how I got here, for anyone who wants the story, and then the appendices, which carry the full constructions the main text only states.

PART I

# What keeping costs

*The cut remains.*

Part I is step one, and the rule step two needs. It starts at the smallest structure that sorts: a cell that's blank, and then a 0. First comes what keeping takes, before any price list: what a distinction that stays is, what every step of every machine does to the states it has used, and what it costs for something to go from nothing to something and stay. Then the smallest machine that writes what keeping takes into its own step, as one rule, the toll. Then whether the rule is forced, and on which premises. Then the one thing that looks like it ought to pay a merge and doesn't have to: finding out, which borrows fresh states instead.

CHAPTER 1

# A distinction that stays

A cell can be blank, or 0, or 1, and somebody has drawn the lines between those. Write a 0 in it and it stays a 0, and that's the event this whole book is about, nothing becoming something; this chapter is what it takes, before any machine and before any price list.

What this chapter counts is a distinction that stays. A cell on a tape that reads 0, and not 1, and not blank, is one; a bead sitting in the left well and not the right is one; a flag that is up is one. Anything the world can tell apart from something else and keeps telling apart counts. It needs distinguishable arrangements, a reading that groups them, and rules or a physical mechanism that retain the chosen answer. The bead uses a hill for the last part. The point said so in pictures ("Where a 0 comes from"). Here I want two things exactly: what every step of every machine does to the states it has used, and what it costs for a state to be taken at all. Both answers are proofs, and every one of them is short enough to follow on the page. Nothing in this chapter needs a price list, and the toll, named up front, is stated exactly only in the next chapter, which writes what keeping takes into a machine's step; the one after asks whether it had to.

## 1.1 Every step does one of three things

Forget the cell for a moment, and forget finiteness. Take any machine at all, with any number of states and any rule that gives each state exactly one next state, and watch it run from any start. Keep a list of every state it has been in. Now look at the next step. There are exactly three things it can do.

It can land on a state the run has not used yet, which uses up a fresh state on this step. The only way to stop using up fresh states is to come back to a state the run has used: to the start, and a run that does that has kept nothing, or to another one, which is the next thing a step can do.

It can come back, for the first time in the run, to a state the run has used before, and not the state it started from. Then it came from a different state than the one that first got there, so two different states have led to the same place, and that is what I mean by a merge: a literal collision, two different states with the same next

state. Under Landauer's premise a merge of states that were really possible carries a floor of heat, and no amount of going slowly lowers it.

Or it can land back on the very state it started from, or take a step the run has already taken. Then the machine is going round a loop, and this is certain: a machine with one rule for the next state, once it stands where it stood before, does again what it did before, forever. Every later state is one it already had, and every later step is one it already took, and it is computing nothing new.

There is no fourth thing a step can do, and every run has the same shape: fresh states, then at most one return, and after it nothing but steps already taken. The proof is one line: the first time a run comes back to a state it has used, either it came back to its very first state, which is a pure loop from the start, or the step that brought it back lands on the same state as the step that first got there, which is a merge. So a run reveals at most one pair of distinct visited sources with a shared successor. Later laps can visit that same landing again; they reveal no additional pair.

□ **Theorem 1.1** (The step law)**.** Let a step give every state of a set exactly one next state, where the states can be told apart, and run it from any start $x_0$, writing $x_0, x_1, x_2, \ldots$ for the run. Either every step is fresh, landing on a state the run has not visited, or there is a first step that lands on a visited state, from $x_n$ to $x_{n+1} = x_j$ with $j \leq n$, and then: every step before it is fresh; it lands on the start ($j = 0$), or it is a merge, coming from a different state than the one that first reached $x_j$, so that two distinct states share a next state; and from $x_j$ on the run is periodic, so every later state is one the run already had and every later step, from $x_k$ to $x_{k+1}$ with $k > n$, is a step the run already took. In a run at most one pair of distinct states shares a next state, $x_{j-1}$ and $x_n$, when $j > 0$. Later laps may revisit the shared destination, but introduce no further pair of distinct visited sources with a shared successor.

*Proof.* If no state repeats, every step is fresh. Otherwise take the first step that lands on a visited state, from $x_n$ to $x_{n+1}$, and the position $j \leq n$ with $x_j = x_{n+1}$; it is unique, because $x_0, \ldots, x_n$ are all different. Every earlier step is fresh, by the choice of $n$. If $j > 0$, compare $x_{j-1}$, which first reached $x_j$, with $x_n$: both land on $x_j$. They differ, because $x_n = x_{j-1}$ with $j - 1 < n$ would be an earlier repeat. So two distinct states share a next state: a merge. Either way, since one rule gives the next state, $x_{n+1} = x_j$ gives $x_{n+1+k} = x_{j+k}$ for every $k$, so the run repeats with period $p = n + 1 - j$ from position $j$ on. For $k > n$ the step from $x_k$ to $x_{k+1}$ is then the step from $x_{k-p}$ to $x_{k-p+1}$, taken earlier, and every later state is one of $x_j, \ldots, x_n$. No later step introduces a new such pair: it repeats a transition already seen. If $j > 0$, the

only two distinct states with a shared next state in the run are $x_{j-1}$ and $x_n$; if $j = 0$, there is no such pair. ■

The step law is "nothing is kept for free" as a theorem with two hypotheses: you can tell states apart, and each state has one next state. The only step that neither uses up a fresh state nor merges is a step on a loop, and a loop makes nothing new. Anything new is a state never used before or one that two states have led to, and that's what keeping takes. Physical dissipation is a separate question, answered here only under the specified thermal dynamics and distributional premises. That one is heat against time: I prove it below for Bennett's walker, and the cited physicists prove related bounds for specified thermal stochastic dynamics. The reading that stays up is the way to force "new": a reading that was no and is now yes can never come back to the state it started on, so the run can't be a loop through its start. Its first return to any state it has used, if it ever has one, is a merge; if it never has one, every state it takes is fresh. Two chapters on, that's the first part of Proposition 3.5, where the fresh states get counted exactly. (After a merge the run can settle onto a loop among states that read yes and idle there, making nothing new. The keeping was paid for at the merge.)

There is one hole, and it's worth seeing exactly. Take a machine that stops, with two states. The first reads no, the step takes it to the second, which reads yes, and the second has no next step at all. The reading is raised, there is no merge, no fresh state after that, nothing. The step law assumed every state has a next state, and a machine that stops its clock dodges it. Close the hole either way and what keeping takes comes back. Say the halted state's next state is itself, which is the usual way to give a stopped machine a step, and the halt is a merge: the moving state and the halted state both land on the halted state. A physical device can also hold a reading through its storage arrangement. What that arrangement costs is a question about the device; the partial transition rule has not answered it. Here I can settle the state-counting question once holding is included in a total next-state rule.

**Theorem 1.2** (Halting)**.** A partial step, one that may have no next state, can raise a permanent reading on two states with no two states sharing a next state. Totalising it, with the halted state stepping to itself, makes the raise a merge: the halted state and the state that steps onto it share a next state. In general, once every halted state steps to itself, any other state that steps onto a halted state shares that landing with it. □

*Proof.* States $a$ and $b$, reading no on $a$ and yes on $b$; $a$ steps to $b$ and $b$ has no next state. No two states step to the same state, because only $a$ steps at all. Totalise:

$b$ steps to $b$, and now $a$ and $b$ both step to $b$, a merge of two distinct states. The general case is the same two lines: if $c \neq h$ steps to the halted state $h$ and $h$ steps to $h$, then $c$ and $h$ share a next state. ■

Stepping the halted state to itself is one way to totalise. Step it to a fresh sink state instead and the merge only moves: the sink steps to itself, and the halted state lands there too. On finitely many states reading yes, every way to totalise that keeps the reading up merges somewhere (Proposition 3.4, two chapters on). An endless counter offers another arrangement: the reading stays put while the full state keeps changing. I want to count that arrangement now, because the screen and the box are doing different things.

*The camera stops; the box keeps ticking*

At three o'clock the screen says 42. The program is finished. Stop filming and you have the computation from start to finish. Leave the camera running and the box is still on the desk. At 3:01 it says 42. At 3:02 it says 42.

Let the box's next-state rule keep ticking. It can stay exactly where it is, or change something behind that screen. Imagine a wheel inside it, turning from "first holding state" to "second holding state" while the display never moves. The screen gives you 42 each time. Open the box and the wheel may tell you which tick you are looking at.

Start the count one tick before completion. That unfinished state is $u$. Call the total next-state rule $F$, so $F(u)$ is the first completed state and $F^2(u)$ is the next one. Gather the full states that keep the answer into a set $H$, and require the rule to stay in that set once it enters. Give $H$ room for $K$ configurations. Two bits can have four configurations; these are the arrangements of the box I am counting.

Now ask it to hold for $L$ samples while every distinct source in that stretch, including $u$, has a distinct successor. Suppose two holding samples repeated the same full state. Follow their equal successors backwards. Each cancellation takes one step off both sides, until the unfinished state equals a state already holding the answer. That is impossible. Every holding sample needs its own full configuration.

□ **Theorem 1.3** (Room to hold an answer)**.** Let $F : X \to X$ be total, with distinguishable states, and let $H \subseteq X$ have $K$ elements. Suppose $u \notin H$, $F(u) \in H$, and $F(H) \subseteq H$. For $L \geq 0$, put

$$D_L = \{u, F(u), \ldots, F^{L-1}(u)\}, \qquad D_0 = \varnothing.$$

If $F$ is one-to-one on $D_L$, then the $L$ held states $F(u), \ldots, F^L(u)$ are distinct, and $L \leq K$. In particular, among $u, F(u), \ldots, F^K(u)$ there are two distinct states with the

same successor. If instead $F$ is one-to-one everywhere and the held region is allowed to be infinite, the orbit from $u$ never repeats.

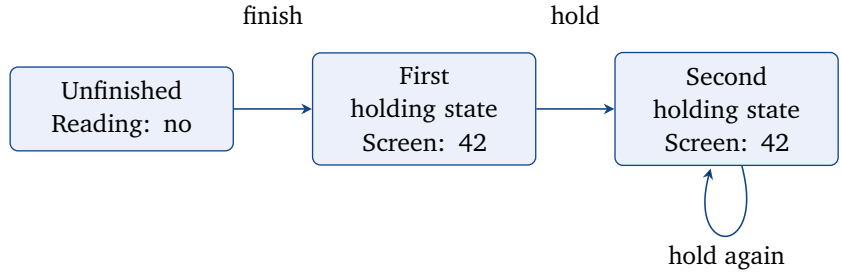
*Proof.* Suppose the same full state occurs twice in the displayed stretch, first at position $i$ and later at position $j$. If the earlier position is the start, I have already equated the unfinished state $u$ with a held state, which is impossible.

Otherwise look one tick behind both occurrences. Those two predecessors belong to $D_L$, and they have the same successor. The one-to-one rule on $D_L$ says that the predecessors must therefore be equal. Move back another tick on both sides and repeat. You eventually reach position zero on the earlier side and position $j - i > 0$ on the later side. In symbols, the repeated backward comparison has given $u = F^{j-i}(u)$. The left side is unfinished; the right side has entered $H$ and stayed there. That is the contradiction.

So all $L$ held samples are distinct. There are only $K$ states available in $H$, giving $L \leq K$. If the sources through $F^K(u)$ had no shared successor, use $L = K + 1$: you would need $K + 1$ held states where only $K$ exist. Finally, when the step is one-to-one everywhere, the backward comparison works for any two finite positions in the run. No repetition is possible, no matter how long you keep watching. ∎

Draw the smallest delayed example:

$$u \longrightarrow h_1 \longrightarrow h_2 \longrightarrow h_2.$$




Only the unfinished state enters the first holding state. The two holding states share a successor. The collision is later than the raise.

The screen says 42 at both $h_1$ and $h_2$. Only $u$ leads to $h_1$; the entry is clear. The two arrows meet a tick later, when $h_1$ and $h_2$ both lead to $h_2$. With $K$ holding configurations, the run exposes a collision within its first $K + 1$ transitions from $u$.

You can reach the bound exactly. Hand me a guest run that halts after $T \geq 1$ steps, and let me use $K \geq 1$ holding states. Line them up:

$$\text{prep} \longrightarrow a_0 \longrightarrow \cdots \longrightarrow a_{T-1} \longrightarrow h_1 \longrightarrow \cdots \longrightarrow h_K \longrightarrow h_K.$$

The preparation state shows nothing yet. Each $a_j$ shows that step of the guest run. Every $h_j$ shows its final answer. There are $1 + T + K$ full states. Starting at the last

running state, the first $K$ sources have different successors. Add the next source and the arrows meet. The box uses every holding state before exposing the collision.

The camera can stop at 42. The box that keeps showing it still has a next step. Include those steps and you can count exactly how long it can keep its full states apart. [1]

## 1.2 What a kept distinction is

Now I turn to the cell. Blank is one of its logical states, alongside zero and one. Give it a write-once rule: write zero when blank, and leave either written value alone. That is what "stays" means in this example. The same rule sends blank and zero to zero, so it has a collision. Whether that collision removes uncertainty depends on which inputs were possible in the preparation. Its physical heat needs the device and protocol as well.

And the cell isn't the lowest level. Describe whatever holds it, a bead in two dips, the molecules that jostle the bead, whatever holds those, and you've described states that can be told apart and a way they change: a machine of the same kind, charged the same way, because a merge one level up forces, one level down, either a merge or a state never used before, as long as the level above is exact about the level below and the level below stays among the states the one above describes (Theorem 1.10). Every level you can describe is one of these. Under all of them there's nothing told apart at all, and that's the nothing a blank cell stands for.

Preparation is a resource assumption too. A supplied known blank is a specified initial condition, not evidence that a positive heat cost was paid earlier. If an apparatus must reset an unknown input distribution for reuse, its reset can incur an entropy-loss bound under Landauer's premises. Reeb and Wolf's precise finite-dimensional formulation, for example, assumes an initially thermal reservoir uncorrelated with the system and joint unitary evolution; its heat bound is conditional on that setup. Inferring a past or future payment from a known state alone would need more assumptions about preparation, correlations, reuse and the thermal protocol.

The cell has three states: blank, zero, one. The reading is "the cell is definite," no on blank and yes on the other two, and a written cell stays written, so the reading is permanent. The move is "write a 0," which takes blank to zero and leaves a written cell alone. Now look at where each state lands. Blank lands on zero, and zero lands on zero, so two different states have led to the same place, and writing a 0 on nothing is, by itself, a merge. For an input distribution with positive uncertainty between blank and 0, this map removes entropy. Landauer's thermal premises then

[1] This is a state-capacity count. A ledger price requires a pricing rule; a heat bound requires physical premises.

give a heat bound for that loss. For a known blank, the input entropy is zero and this argument gives no positive bound; it also implies no earlier payment for knowing. The history in which the cell was blank and the history in which it already held a 0 are now the same history. This is not an analogy to the step law; it's the step law with three states filled in.

The obvious way round it is a fourth state that remembers: "0, and it was blank before." Now blank lands on a state nothing else has landed on. But the remembering state has to go somewhere on the next write too, and it's written, so it has to stay written. Every written state already has something landing on it: zero and one land on themselves, and the remembering state is where blank landed. So the remembering state has to share. In the smallest version it lands on itself, and then blank and the remembering state land on the same state. No finite extension of the cell gets out of the merge, and the reason is counting (Proposition 3.4, two chapters on), not bad luck. The only way to write with no merge, once every state has a next one, is a supply of fresh states that never runs out: a state never used before for every write, forever, such as a new cell of an endless tape for each one (it comes back two chapters on, as the first of three ways out of a theorem). That isn't getting out of what keeping takes. It dodges the floor by taking a state it has never used, at every write, without limit, and each write can dissipate under the stated thermal-model assumptions. Or, in a form I need below, a tape that stretches both ways with the machine stepping one cell along it. Every cell has one before it and one after it, nothing ever shares a landing, the reading "past the start" goes up on the first step and never comes down, and the machine never returns. This is the bilateral shift, and it pays at every step by taking a cell it has never used.

**Theorem 1.4** (The write of a 0 merges, and only fresh states avoid it)**.** (1) On the three states blank, zero, one, with reading "definite," writing a 0 is permanent, raises the reading from blank, and merges blank with zero. (2) Adding a fourth state that remembers the cell was blank, with written cells staying written, does not avoid the merge: on the four states the write still merges two distinct states. The same holds on any finite set of states. (3) On an endless tape, writing adds one cell and is one-to-one; on the integers, the step "one cell along" is a bijection, raises "past the start" on the first step, keeps it up, and after any positive number of steps is away from where it started. □

*Proof.* (1) Blank and zero both step to zero, and blank is not zero; zero and one step to themselves, so a written cell stays written. (2) Blank steps to the remembering state, and permanence sends each of the three written states, zero, one and the remembering state, to a written state. So four states land among the three written

ones, and two of them share a landing. Whenever only finitely many states are written the same count works: permanence sends the finitely many written states among themselves, blank lands among them too, so one more state lands among them than there are, and two share a landing (Proposition 3.4, two chapters on, states it in general). (3) "Add one" on the counting numbers is one-to-one and never returns to zero; "add one" on the integers is one-to-one and onto, sends 0 to 1, sends every number that is at least 1 to one that is at least 1, and after $k > 0$ steps has moved by $k$. ■

The theorem leans on one thing, and I want it in the open. The write is a move defined on every state of the cell, written ones included, and a written cell stays written, so the state blank merges with is a cell that was already written. A write defined only on blank, with the machine simply stopping when it meets a written cell, merges nothing. That's the stopping hole from above (Theorem 1.2): it buys the 0 by having no next step at all.

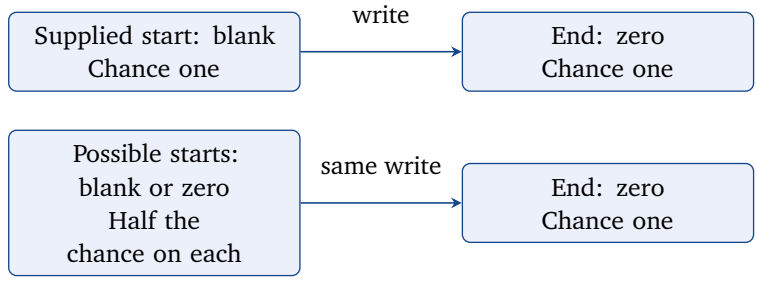
The theorem buys this: a fact is a distinction that stays, and a kept distinction is a state the run has used and can't give back. The machine is the thing whose own state includes the used states it can't give back.

It doesn't stop at the cell. The tape is a set of distinctions that stayed. So is the rule for the next state, and so is the question you ask of a state. Each of the five parts of the model is itself a definite thing where there could have been anything; the model doesn't charge for them, but they're the same kind of thing it charges for. And the model can run itself: a host that obeys the toll and runs a guest move for move, carrying the guest's prices on its own ledger, pays at least one unit there for every step where the guest's record went up, and Part III states one (Section 16; Appendix C builds it). The tower is the same kind of machine above and below. What no level does is hold all the questions you can ask of it: for any way of coding readings as states, there's a reading no state codes, the one that answers each state with the opposite of what that state's own coded reading says of it. That's Cantor's diagonal, the shape of Russell's 1902 letter to Frege (Section 5), and the subject of Section 13. Put the two together and you have the regress exactly: every level runs the one below that way and pays again, in its own ledger, for every rise of the lower level's record, and no level ever holds every question about itself.

□ **Theorem 1.5** (No level holds its own questions)**.** For any set of states and any assignment of a yes/no reading to each state, some yes/no reading on the states is assigned to no state.

*Proof.* Take the reading that says, of each state, the opposite of what that state's assigned reading says of the state itself. If some state were assigned it, that state would answer itself both ways. ∎

Write a 0 on a cell known to be blank and the entropy-loss premise sets no positive floor: the input and output entropies are both zero (Theorem 3.17). The thermal implementation can still have a finite-time cost under the specified dynamical assumptions. A writer that must handle both blank and 0 with one protocol faces a different question: under the uniform two-input distribution its logical map removes one bit, so Landauer's premises give an average heat bound. This does not charge that average to every particular input.




The first preparation loses no uncertainty. The second loses one bit. The transition rule is the same; the input preparation is different.

Keeping an exact answer among $N$ independently possible alternatives needs at least $N$ representational states, or $\lceil \log_2 N \rceil$ bits in a fixed-length binary encoding. That is a capacity bound. Identifying an arbitrary one of three possibilities can require two binary questions in the worst case; testing just "is this cell blank?" is a different, binary task. Neither count proves that a known blank was prepared by an erasure or that a heat bill has already been paid.

## 1.3 A reversible machine keeps nothing

Bennett's reversible machine never lets two of its states lead to the same next state; that's what reversible means. Put one on finitely many states and watch it. By the pigeonhole it has to revisit a state within as many steps as there are states. The step law says the first revisit is a merge or a return to the start, and a merge is impossible for it, so it returns to its start, and once it's back at its start it does everything again. Any reading it raised has come back down. A reversible finite machine with a next state for every state keeps nothing, and that's a theorem, not a slogan. One that stops slips past, as the stopping hole above says, and keeps what it halted on. It is the recurrence Zermelo threw at Boltzmann (Section 3 tells that story), in its finite form, proved from the step law and the pigeonhole and nothing else. What Bennett's machine keeps is the fresh states it's handed: on an endless tape it can

keep its whole history and never come back, taking a state it has never used at every step, and the tape was paid for before the first step by whoever built it. A reversible machine dodges the merge and its floor. For a specified thermal implementation, a finite-time change of distribution can also require entropy production. Fresh states and physical dissipation are different quantities; contact with surroundings alone supplies no numerical bound.

□ **Theorem 1.6** (Reversible keeps nothing)**.** Let a one-to-one step act on finitely many states. From any start the run returns to the start within as many steps as there are states, and any reading that is no at the start and never goes from yes to no is no at every step.

*Proof.* Among the first (number of states) plus one states of the run, two are equal, by the pigeonhole. Take two equal positions and cancel the one-to-one step back from both the same number of times, until the earlier one reaches the start; the later one then equals the start. So the run returns to the start after at most that many steps, and from then on repeats. If the reading were yes at some step, it would stay yes from there, in particular at a later multiple of the period, where the state is the start, which reads no. ■

### 1.4 Noisy steps

Real states are noisy. The bead hops on its own. So take the step to be a table of chances: from each state, the chance of landing on each state next, every row adding to one. A permanent reading is a set of states you never leave once you're in it. And "no merge" has an exact noisy meaning: the step preserves the even spread. Start with everybody spread evenly over the states, take one step, and they're still spread evenly; which is to say every column of the table adds to one as well as every row. Such a step squeezes nothing on average.

Then the cell's count goes through with chances in place of people. A step that preserves the even spread cannot raise a permanent reading from any state outside it, and not merely with small chance but with none. Count the chance flowing into the yes-states from everywhere. Column by column that's one per yes-state, so it equals the number of yes-states. The yes-states alone already send exactly that much, because they never leave. What's left over for the outside states is zero, and it's a sum of chances, so each is zero. The count never used that the spread was even, only that every state had some chance and the step gave the spread back. A lopsided memory that keeps its own lopsided spread can't raise a reading that stays either: weight each state by its chance and the same sum comes out.

And it runs the other way too. A step that can raise a permanent reading does not preserve the even spread, so after one step from the even spread the crowd is more bunched up than it was. Here is a way to measure that bunching. Draw a state twice, independently, using the same spread of chances. Ask whether the two draws match. For two equally likely states, the matching pairs have chances one quarter and one quarter, so a match occurs half the time. Put three quarters of the chance on one state and one quarter on the other: the matching chances are now nine sixteenths and one sixteenth, adding to ten sixteenths. Bunching has made a match more likely.

In general, a state with chance $q$ contributes $q^2$ to this matching chance: $q$ for the first draw, times $q$ for the independent second draw. Add those squares over all states. I call that total the *collision weight*. The step just described raises it above the value for an even spread.

There is another measure of the same change, Shannon entropy, which measures uncertainty in the spread. It is largest when all states are equally likely, so leaving that even spread lowers it. This is the quantity that enters the heat bound under Landauer's premise. Section 3.7 works through its meaning in bits; the proof here establishes the drop. Finally, when a step preserves the even spread, it cannot increase the collision weight of *any* starting spread. That last claim needs a calculation too.

**Theorem 1.7** (Noisy steps)**.** Let a step on finitely many states be a table of chances with rows adding to one, and let a set of states be closed: from inside, the chance of leaving is zero. (1) If every column also adds to one, the chance of entering the closed set from any state outside it is zero. (2) If some state outside can enter the closed set with positive chance, then some column does not add to one, and one step from the even spread strictly raises the collision weight, the sum of the squared chances, which is least for the even spread, and it leaves a Shannon entropy strictly below the logarithm of the number of states. (3) If every column adds to one, one step never raises the collision weight of any spread. □

Part (1) needs only the closed set to be finite, the rest as large as you like, and part (3) needs no finiteness at all; part (2) needs the even spread, so it needs finitely many states.

*Proof.* (1) Count the total chance sent into the closed set. Each of its columns sums to one, so the total equals the number of states in that set. The rows belonging to states already inside send all their chance inside, and they already supply that entire total. Nothing remains for an outside row to contribute. Because every contribution is nonnegative, each outside contribution is zero.

(2) Suppose instead that an outside state contributes a positive chance of entry. Then the conclusion of (1) fails, so its column-sum condition cannot hold: at least one column sum differs from one. Call the number of states $N$ and the sum of column $j$ $c_j$. Starting evenly puts chance $1/N$ at each source, so the chance at destination $j$ after the step is $c_j/N$. The column sums themselves add to $N$, because the $N$ rows each add to one. Expand the following squares and use that total:

$$\sum_j (c_j - 1)^2 = \sum_j c_j^2 - 2\sum_j c_j + N = \sum_j c_j^2 - N.$$

At least one square is positive. Thus $\sum_j c_j^2 > N$. Dividing by $N^2$ says that the new collision weight is greater than $1/N$, the old even spread's collision weight. Now compare the uncertainty before and after. Write $q_j = c_j/N$ for the new chances. At least one differs from $1/N$. Shannon's entropy weights each possible state's logarithmic uncertainty by its chance. Using the natural logarithm, written ln, its value is

$$H(q) = -\sum_{j:q_j>0} q_j \ln q_j.$$

The sum leaves out states with zero chance; they contribute nothing. An even spread has $N$ equal terms, giving $H = \ln N$. Subtract the new entropy from that value. Since the chances add to one, you can put $\ln N$ inside the sum:

$$\ln N - H(q) = \sum_{j:q_j>0} q_j \ln(Nq_j).$$

I need to show that this difference is positive, even though some individual terms may be negative.

Use the elementary logarithm inequality $\ln y \le y - 1$, with equality only at $y = 1$. One way to see it is to look at $y - 1 - \ln y$: its slope is $1 - 1/y$, negative below one and positive above one, so its least value is zero at one. Apply the inequality with $y = 1/(Nq_j)$ and multiply by $q_j$. Rearranging gives

$$q_j \ln(Nq_j) \ge q_j - 1/N,$$

with equality only when $q_j = 1/N$. A zero chance contributes zero to the entropy difference, which is strictly greater than its comparison term $-1/N$. Add the comparisons over all $N$ states. Their right sides sum to $1 - N/N = 0$. At least one comparison is strict because the new spread is not even, so the entropy difference is positive.

(3) Fix a destination column. Because its entries add to one, its new chance is an average of the old chances, with those entries as weights. An average of squares is never less than the square of the average. To see why, call the weights $w_i$, the old

chances $p_i$, and their weighted average $m$. Expanding gives

$$\sum_i w_i (p_i - m)^2 = \sum_i w_i p_i^2 - m^2 \geq 0, \qquad \sum_i w_i = 1, \quad m = \sum_i w_i p_i.$$

So the new chance squared is at most the weighted sum of the old chances squared. Add this comparison over all destination columns. Each old chance squared gets weights from its whole row, whose sum is one. The result is exactly the old collision weight, bounding the new one from above. ∎

## 1.5 Unitary steps

The quantum version needs a different picture. Draw a sheet through the centre of a room. Call every direction along the sheet YES, and the direction straight through it NO. Turn the whole arrangement without stretching it. Can the sheet stay within its old plane while the direction through it turns into that plane? No. The turn keeps right angles. If the sheet still fills that plane, the direction that was at right angles to the whole sheet is still at right angles to it.

Here is what carries from that picture into the quantum statement. A quantum state is described by a vector. A reversible quantum step preserves lengths and overlaps between vectors; such a step is called *unitary*. The YES part is a *subspace*: a collection of directions that also contains their linear combinations. The directions with zero overlap with every YES direction make up its *orthogonal complement*. These are the NO states in this argument. Permanence means that a vector in the YES subspace stays in it after the step.

There is one detail to settle before the right-angle argument works. The old YES directions must fill the whole YES subspace after the step, leaving no missing direction. Choose a finite list of directions from which every YES vector can be assembled, with none of the listed directions assembled from the others. Such a list is called an independent spanning list, or a basis. A unitary step keeps the list independent, since any relation between its images could be reversed to give the same relation before the step. The images are still inside the YES subspace, and there are just as many independent directions as before. They therefore still span it. That is why the step fills the YES subspace, rather than merely landing somewhere inside it.

Now take a NO vector. Before the step it has zero overlap with every YES vector. After the step those overlaps are still zero, and the stepped YES vectors still fill the whole YES subspace. The stepped NO vector is therefore still NO. Repeat the step and the same argument applies again.

Finite dimension was used exactly once: a full finite list of independent YES directions still fills the YES subspace after the step. Drop it and the bilateral shift,

which is unitary on an endless ladder of states, raises "past the start" on the first step and keeps it forever, taking a state it has never used every time. A reading that stays, which no unitary step can raise in finite dimension, is paid for either by fresh states, a ladder of states that never comes back around, or by a step that squeezes the even spread, which, when the squeezed states were really possible, owes Landauer's floor. Not every step that fails to be unitary squeezes: one that keeps the even spread, as pure dephasing and an unread projective measurement do, can't raise a reading that stays either. The balance argument is the same one used for noisy steps. Choose a complete set of mutually perpendicular directions, some filling the YES subspace and the rest filling its orthogonal complement. Spread the chances equally over those directions. This is the fully mixed state: choosing a different complete perpendicular set gives the same mixture.

Suppose there are $N$ directions altogether and $m$ of them are YES. The mixture has chance $m/N$ of YES. A step that preserves this mixture must leave that chance at $m/N$. But the YES part already supplies all of it: it started with weight $m/N$, and permanence keeps that whole weight inside YES. Every contribution from the other directions is nonnegative, so none can contribute any chance of entering YES. You can include any chosen NO vector among the perpendicular directions of the complement. The same balance then rules out entry from that vector. That is why the outside cannot get in. In finite dimension nothing is kept without one of the two. Which way a quantum measurement pays when its outcome is kept isn't settled here. The collapse models people test keep the even spread too, so the heat the experiments look for is the heat of the noise those models add, not a Landauer price on a kept outcome; I come back to them below.

□ **Theorem 1.8** (Unitary steps)**.** Let a unitary map a finite-dimensional subspace into itself. Then it maps the orthogonal complement into itself: a state orthogonal to the subspace remains orthogonal after every step.

*Proof.* Call the step $U$ and the YES subspace $Y$. Choose a basis of $Y$, meaning an independent list from which every vector in $Y$ can be assembled. The step sends that list to an equally long independent list inside $Y$, so its images are another basis of $Y$.

Take any vector $x$ orthogonal to $Y$. To test whether $Ux$ is still orthogonal, choose any vector $y$ in $Y$. Because the images fill $Y$, there is a vector $z$ in $Y$ with $Uz = y$. Unitarity preserves the overlap, written with angle brackets, so

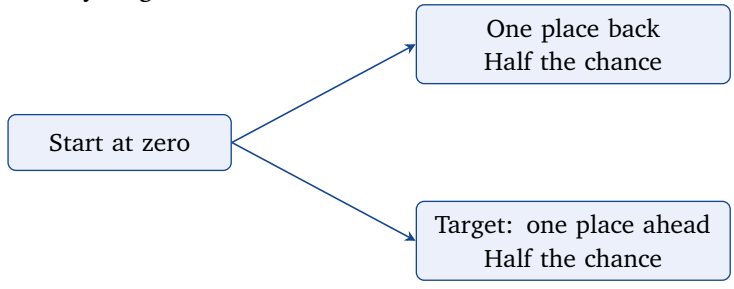
$$\langle Ux, y\rangle = \langle Ux, Uz\rangle = \langle x, z\rangle = 0.$$

The last equality holds because $x$ is orthogonal to every vector in $Y$, including $z$. The choice of $y$ was arbitrary, so $Ux$ is orthogonal to all of $Y$. This is the same condition I started with, and it survives every further step. ∎

## 1.6 Going faster gives off more heat

Everything so far was what keeping takes. What comes next is a dissipation result for a particular stochastic model. Bennett's Brownian computer is a walker on a line that is pushed forward a little: at each tick it steps forward with some chance and back with the rest. Its speed is the difference between the two. A forward translation is one-to-one, and so is a backward translation. The random choice between them is a stochastic rule, not one deterministic translation; the deterministic merge count does not apply to it as though it were. The entropy it gives off per tick is Schnakenberg's formula for a biased walk, the speed times the logarithm of the forward-to-backward ratio, and that number is strictly positive at any speed other than zero, because both factors have the same sign. The only walker that gives off nothing is the one with no drift, as likely to step back as forward. This is the point that going slower doesn't get you out of paying, now as a theorem: at any speed you actually run it, every tick gives off heat on average.

The exchange rate has a floor as well. Heat per net step forward, counted in units of $kT$, is the logarithm of the forward-to-backward ratio, and that logarithm is at least the speed. So walking a distance costs at least distance times speed in heat and takes distance over speed in time, and heat times time is at least the distance squared. At low speed, halve the speed and the heat roughly halves while the time doubles; the product never drops below the square of how far you went. Going slowly makes the mean heat for a specified amount of directed progress smaller and its drift time longer. These are averages. With no bias, the walker still moves, and it can reach a nearby target on the first tick.




The mean displacement is zero. Half the paths reach this target on the first tick. Zero drift does not mean no movement or no completed paths.

Below, distance divided by speed is the time scale for the mean displacement to advance that distance. It is not the duration of every path that happens to finish.

□ **Theorem 1.9** (Heat against time)**.** For a walker stepping forward with chance $p$ and back with chance $1-p$, with $0<p<1$, the entropy it hands to its surroundings per tick, $(2p-1)\ln\frac{p}{1-p}$, is zero at $p=\frac{1}{2}$ and strictly positive otherwise. With speed $v=2p-1$ in $(0,1)$ and distance $N\geq 0$, the mean heat in units of $kT$ for drift progress $N$, $N\ln\frac{1+v}{1-v}$, times the drift time $N/v$ is at least $N^2$.

*Proof.* For $p>\frac{1}{2}$ both factors are positive and for $p<\frac{1}{2}$ both are negative, by the monotonicity of the logarithm; at $p=\frac{1}{2}$ the first factor is zero. For the floor, $\ln(1+v)\geq 0$ and $\ln(1-v)\leq -v$ since $\ln y\leq y-1$, so $\ln\frac{1+v}{1-v}\geq v$, and the product is $N^2$ times that logarithm divided by $v$. ■

That is the surroundings' share of the entropy production. The whole of it adds the change in the walker's own spread, which on a line keeps growing but ever more slowly, so per tick that share goes to zero and the long-run rate is the heat's.

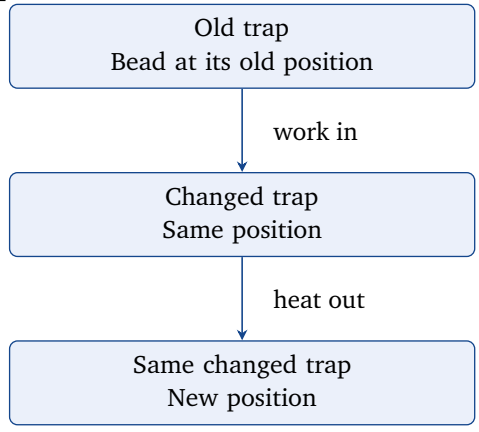
The sharp floor is twice that. The difference $\ln\frac{1+v}{1-v}-2v$ is zero at $v=0$ and its derivative, $\frac{2}{1-v^2}-2$, is never negative on $(0,1)$, so the logarithm is at least $2v$ and the product is at least $2N^2$. I state $N^2$ because the one-line bound gives it, and what matters is the square, not the factor in front of it.

The walker is the case proved here. Other finite-time results use other, explicitly specified dynamics. Drag a bead held in a spring-like trap through water from one place to another in a set time and the least work you can get away with is the drag times the distance squared, divided by the time plus twice the drag over the stiffness, which at long times is the drag times the distance squared over the time (Schmiedl and Seifert, 2007, who worked out the best protocols for doing it in finite time). Within their overdamped diffusion or Markov jump models, change a probability distribution into a different one in a set time and the entropy produced is at least a fixed fraction of the square of how far apart the two spreads are, divided by the time and by how mobile the states are: the diffusion, for a bead that drifts (Aurell, Mejía-Monasterio and Muratore-Ginanneschi, 2011), and how busy the jumping is, for states that jump (Shiraishi, Funo and Saito, 2018). Those are their results, and I cite them as physics. These bounds require their stated dynamics, a nonzero change of distribution, and finite activity or mobility. For the jump-process bound, local detailed balance identifies the usual thermal entropy production; without it the cited result uses a different entropy-production quantity. A thermal environment alone does not supply these assumptions. The bounds concern entropy produced, not positive heat on every step: a step that leaves the spread exactly as it was owes nothing on these bounds, and the heat a single step gives off can be negative when the device's own entropy goes up. An idealised machine

cut off from everything, frictionless billiard balls bouncing their way through a computation (Edward Fredkin and Tommaso Toffoli's, 1982), could in principle run without dissipating at all; Bennett (2003) wrote that such ballistic machines compute at constant velocity without dissipating any energy, and that they must be kept isolated from heat baths and noise. Keeping one on course against noise means correcting its errors, and whether that brings the dissipation back I don't settle here; in 2003 Bennett called the thermodynamics of error correction a subject that appears to be in its infancy. The bead of Jun, Gavrilov and Bechhoefer (2014) shows the other half. Put through the same moves as an erasure but with no tilt, so nothing is squashed, its work extrapolated to the slow limit came out at $0.05 \pm 0.03\,kT$, within two error bars of zero: no floor under it, which is what the premise says. Keeping something costs, on top of that, either a merge, which under Landauer's premise owes a floor of $kT \ln 2$ per bit that no slowness removes, or a state the device has never used before. Taking a fresh state escapes the floor and does not settle the implementation's finite-time dissipation. And a finite set of states, where every state has a next one, forces the merge.

I also built a numerical bead to see the steps, using a quartic double well. It follows the experiment's sequence: lower the barrier for one second, ramp the tilt, remove the tilt, then raise the barrier for one second. This is a chosen potential and a numerical approximation, not the measured trap. The starting positions are sampled from a numerical approximation to the thermal distribution in that potential, with equal chances of starting in either well.

Keep three accounts as it moves. Changing the trap while holding the position fixed does work on the bead. Moving the bead in the fixed trap transfers heat to the bath. Whatever is left changes the bead's potential energy. These are the steps the script actually adds up:




Work in minus heat out equals the change in stored energy. Either transfer can be negative on an individual step.

The work account includes removing the tilt and restoring the starting trap. The heat account uses the potential difference during each displacement. Those two accounts balance on every simulated path to rounding precision. That checks the accounting; it does not show that the time steps are small enough to reproduce continuous motion exactly.

Here is what came out, with 3,000 paths at each setting and a time step of half a millisecond. Each setting uses a specified tilt, chosen earlier to make roughly nine erasures in ten work. Success means ending in the right well. The last two columns average only those successes; energies are in units of $kT$. The time column is the tilt ramp, with the two one-second barrier ramps in addition.

| Tilt time (s) | Success rate | Heat out on success | Work in on success |
|---|---|---|---|
| 2 | 0.909 | 2.517 | 2.501 |
| 5 | 0.916 | 1.466 | 1.490 |
| 10 | 0.913 | 1.073 | 1.095 |
| 20 | 0.899 | 0.830 | 0.907 |
| 40 | 0.893 | 0.773 | 0.782 |
| 80 | 0.869 | 0.693 | 0.719 |

Heat and work are close in some rows and less close in others. The difference is measured by the simulation, not supplied as a constant correction. At the last setting the sampling standard errors are about $0.023\,kT$ for heat and $0.011\,kT$ for work. They do not include error from the time step or the model.

Now try to make a floor out of those six points. Fit the successful heat to a constant plus a number divided by the tilt time: the constant is $0.674\,kT$, with sampling error about $0.015\,kT$. Halve the time step and repeat the six settings: it becomes $0.732\,kT$, with sampling error about $0.016\,kT$. The second run also samples different paths, so this comparison mixes sampling and numerical error. It is enough to stop me treating a fitted constant near $\ln 2$ as a confirmation. The barrier ramps are still one second long, the tilt changes between settings, and this fit does not take the whole protocol to a slow limit.

There is a useful theoretical comparison, but it is a comparison with work. For the continuous thermal model, beginning in equilibrium and ending with the same symmetric trap, the detailed Jarzynski relation gives the successful paths a mean-work bound $kT\ln(2p)$, where $p$ is the success probability. Averaging the successful and failed paths gives the bound $kT[\ln 2 + p\ln p + (1-p)\ln(1-p)]$. At nine successes in ten these are about $0.588\,kT$ and $0.368\,kT$. Bérut, Petrosyan and Ciliberto's 2013 paper, Detailed Jarzynski Equality applied to a Logically Irreversible Procedure, works through the success-conditioned relation. Neither formula is a heat bound for this simulation without also accounting for its change in stored energy. Nor does an exact bookkeeping identity make the discrete simulation obey every continuum fluctuation relation exactly.

The script is `scripts/bead_erasure_sim.py`. It reports heat, work, stored-energy change, sampling errors and the descriptive fits, and accepts a smaller time step for checking them. The slower settings here use less mean work and release less mean heat. That is what this calculation demonstrates. It does not determine a universal floor or reproduce the laboratory's fitted constant.

| | |
|---|---|
| A merge, the floor | Jun, Gavrilov and Bechhoefer (2014): one bead in a feedback trap. Squash two wells to one: $0.71 \pm 0.03\,kT$ of work, in the slow limit. Run the same moves without the tilt, so nothing is squashed: $0.05 \pm 0.03\,kT$, also in the slow limit, within two error bars of zero. $\ln 2$ is 0.693. Same bead, same trap. |
| The floor, three more ways | Bérut and others (2012), a colloidal bead; Hong, Lambson, Dhuey and Bokor (2016), nanomagnetic memory bits, within two standard deviations of Landauer; Gaudenzi, Burzurí, Maegawa, van der Zant and Luis (2018), molecular spins at one kelvin. |
| Lopsided wells | Gavrilov and Bechhoefer (2016), the feedback trap of Jun and others, with an asymmetric double well: the average work to erase came out below $kT \ln 2$. The floor as stated is for a symmetric memory. |
| Time against heat | Bérut and others (2012; Bérut, Petrosyan and Ciliberto 2015): heat $= kT \ln 2 + B/\tau$ fits with $B = 8.15\,kT \cdot$ s; left free, the floor of the fit comes out at $0.72\,kT$. The floor is reached only as the time goes to infinity. |
| The distinction itself | The same bead: the paper estimates a characteristic Kramers escape time of about 3,000 seconds at $8\,kT$, and about ten seconds at $2.2\,kT$. Those are statistical time scales. The barrier height controls retention; it is not itself a measured heat payment. |
| The loop | Rauer, Erne, Schweigler, Cataldini, Tajik and Schmiedmayer (2018): recurrences of coherence in an isolated gas of thousands of atoms. A collective quantity of a closed system comes back around; the full state of the gas wasn't shown to return, and couldn't be. |
| Heat of collapse | Vinante and others (2017): a cantilever at millikelvin shows a non-thermal force noise of unknown origin, compatible with the collapse heating one model predicts. Donadi, Piscicchia, Curceanu, Diósi, Laubenstein and Bassi (2021): the radiation a gravity-driven collapse would emit was not seen at Gran Sasso, ruling out that model's parameter-free form. These test the noise a collapse model adds, not a Landauer price on a kept outcome. |

The ceiling on fresh states Bekenstein's proposed bound uses energy and size; it is not a measured state count.

The cited erasure experiments test specified memories and protocols, including symmetric-memory Landauer bounds and heat versus duration. They are not an exhaustive survey of physical computation, nor an experimental proof of the abstract state-space dichotomy. Bekenstein's bound relates entropy to energy and size under its physical assumptions; it does not supply a finite enumeration of arbitrary device states. The finite-state theorems here take their finite set as a mathematical hypothesis.

## 1.7 Every scale pays the same way

One more question, and it's the one that decides whether any of this is about the world or only about my pictures. A state is a description. The dial on a lab bench is a state, and so is each atom in it, and so is each atom together with the light it gave off. Describe the same thing coarser or finer and you get different states. Does what keeping takes change with the zoom?

It doesn't, and the reason fits in one picture. Draw the fine states, and draw a fence around each group of them that counts as one coarse state. Say the coarse picture merges: two coarse states step to one coarse state. Then, when the fine states are finitely many, the fine step can't keep them all apart either. Count them. If the fine step sent different fine states to different fine states and none of them stepped out of the fenced set, it would be a reshuffle of the fine states, with every fine state landed on exactly once afterwards. But some fenced group is now landed on by nothing, because the coarse step missed a coarse state (two coarse states went to one, so on a finite set some coarse state is stepped to by nothing), and so nothing lands on any fine state in that group. A reshuffle leaves no state out. So something gave: the fine step merges too, or some fine state stepped out of the fenced set, or the fenced set has no end. The three ways come back one level down, though stepping out of the fenced set isn't taking a fresh state; it's where the coarse picture stops tracking the fine one.

□ **Theorem 1.10** (Every scale pays the same way)**.** Let a fine machine have one rule $F$ on a set $X$ of states, let $A$ be a set of its states, and let a coarse description be a map $\pi$ from $X$ to a set $S$ of coarse states that takes $A$ onto $S$, with a coarse rule $f$ on $S$ such that $\pi(F(x)) = f(\pi(x))$ for every $x$ in $A$. If the coarse rule merges two coarse states, then at least one of these holds: $F$ merges two states of $A$; $F$ takes

some state of $A$ outside $A$; or some run of $F$ from a state of $A$ stays in $A$ and never comes back to a state it has visited.

*Proof.* Suppose the fine machine avoids all three ways out. It stays inside $A$, keeps different states apart, and every run eventually visits a state it has visited before. Follow one run to its first repeat. The two visits have the same state, so their predecessors must agree too: different predecessors are never allowed to land together. Walk backwards like that until the earlier visit reaches the start. The later visit has reached the same start. The run is a loop from the start, with no loose approach leading into it.

Now watch that loop through the coarse window. Every fine step gives the coarse step specified by $f$. When the fine state returns to its start, its picture returns too. Every coarse state has a fine state above it, because $\pi$ takes $A$ onto $S$. So every coarse state also lies on a loop.

That rules out a coarse merge. Suppose two different coarse starts had the same next state. Call the lengths of their loops $L_1$ and $L_2$. Wait $L_1 L_2$ steps from the starts: that is a whole number of circuits for both loops, so each must be back at its own start. But after their first step they were together. The same rule cannot separate them afterwards, so at the end of that wait they are still together. Their starts must have been the same after all.

The coarse rule does merge, by hypothesis. The fine machine therefore cannot avoid all three alternatives. In the last alternative, every tick visits another state of $A$, so no finite collection can hold the run. ∎

When I say a coarse description is exact about a set of fine states, I mean what the theorem asks: it follows the fine step on the set, and every coarse state describes some state in it. Put it next to the permanent flip. On a finite coarse picture, a reading that goes up and stays up is raised by a merge (the cell above, on three states; Theorem 3.1, two chapters on, in general). By this theorem, at every finer description that the coarse one is exact about, keeping that reading is again a merge, though not always where the reading rose, or a step off the described states, or a run that takes a fresh described state at every step. The dial and its light are the example. In the dial's own picture, two blends of readings decay to one either-or, a merge (Section 18). In the picture of dial and light together, the step is one-to-one, and it's the second case: the light starts blank and leaves carrying the copy, so the fine step leaves the set of states where the light is still blank. At the fine scale nothing merges; the step leaves the states the coarse picture describes. And the heat floor sits where the merge really is. Landauer's premise charges the scale whose step merges states that were really possible; a description that merges only because it stopped looking at the light owes that floor nowhere merely because of that projection. Leaving the set where the light is blank is not the same as visiting a fresh full state at every

later tick. To infer an endless fresh-state run, the theorem needs its third alternative. To infer heat, specify the physical preparation and what happens to the light or reservoir.

## 1.8 What keeping takes, and where the toll comes in

So that's the whole of what keeping takes, set out before anybody writes a price on anything. The toll, which the next chapter states, is what keeping takes, written into a step. It's the price of something being the case: a reading that goes from no to yes and stays is a distinction that stays, and a distinction that stays is a state the run has used and can't give back. Anything whose state includes distinctions it can't give back pays what the toll prices: the bead, the cell, the memory bit, a world with a past. Give one a price list that obeys the toll and it's a Thiele machine, and when the cell could have held anything, Landauer's premise puts a heat floor under that raise, because on a finite cell the raise is a merge. The toll keeps its shape there, charged in heat instead of units of cost. And what keeping takes, deterministic or noisy or quantum, comes to this: the Brownian walker has a positive dissipation rate at nonzero drift, and the cited diffusion and jump models give finite-time distributional bounds under their own hypotheses; keeping something costs on top of that either a merge, with Landauer's floor when the merged states were really possible, or a state never used before; and a finite machine, where every state has a next one, forces the merge. The step law, the noisy steps and the unitary steps are proved, and the table above says how near the laboratory has come to each; the heat is physics.

That's also why the rule sits on the raise and not on the merge. The toll charges for what is kept whichever column pays for it, merge or fresh state. A price on merges charges only the merge column, so it misses something kept by taking a state never used before: the machine that keeps every history, the first way out two chapters on, raises its reading at zero cost under it. And it charges merges that keep nothing, like the reset that lowers a flag. The toll doesn't ask that reset for a unit of cost, and it still merges two states, so in a real device it owes Landauer's floor whenever the bit it lowers was really unknown; the ledger leaves that to physics, the same as the dissipation of happening.

Next comes the smallest model that keeps that cost on its own ledger: two boxes on a sheet of paper, a tally, and one rule.

CHAPTER 2

# A record and a toll

The last chapter found what keeping takes and kept no count of it. Now give it a ledger. Take a record with two positions, no and yes, and build the smallest thing that could carry a charge for its change, on paper, with a pencil. Write the charge into its rules and see what follows. Keeping it this small is what lets you see which part of the construction does which part of the work.

The usual models charge a computation for two things: how long it runs and how much space it uses. The bits it destroys have been counted too, as the opening said, by Li and Vitányi and by Demaine and his coauthors. What this part charges is something else: what the computation keeps and can't take back. This part is about that charge, which gets paid in a merge, carrying a heat floor when the merged states were really possible, or by taking a state never used before, and which the last chapter showed can't be talked down to zero. The record built below is the smallest thing that can carry the charge, and the tally next to it counts every time the record goes up. Whether a machine has to charge it as a unit of cost, and on which premises, is the next chapter.

One thing needs saying about what a flag stands behind, because it stays out of the picture a while. What I'm about to build prices vouching for anything at all: a flag goes up, and the rule doesn't ask what the flag is standing behind. A sorting of a pile into groups, the kind of structure Part II starts from (Section 5), is the first kind of claim I care about. It isn't the only kind, and the book comes back to what a flag stands behind in Section 8, and on a machine in Section 15, where a flag stands on a checked claim. Until then nothing about the flag leans on groupings. The floor in this section, the reason for it in the next, and the latch in Part II all work whatever the reading is about, partition or no partition. When groupings come in, in Section 8, it's as a theorem: a reading that says a grouping has been checked and committed sees the grouping only through its partition, and what it entitles is exactly the groupings that refine it (Theorem 8.5). So the grouping gets a job, and an exact one.

But first the word, because I've got a new one and I don't trust it either: record, receipt, certificate, vouch. People have been making marks that other people are

supposed to rely on for a very long time, and I wanted to know what they thought they were doing, and what it cost them, and whether any of them ever said the thing I need said. So here's what I could reach, pushed on with you, and I'll tell you where the search ran out.

### 2.1 Clay, seals, and seven witnesses

Start a little over five thousand years ago, in the Near East, with people who counted sheep with little clay tokens. To send a count across a distance without the courier changing it, they put the tokens inside a hollow clay ball, pressed the shapes of the tokens onto the outside while the clay was soft, and sealed it. The archaeologist Denise Schmandt-Besserat has spent a career on these, and she puts the practice at around 3300 BC; the ball is called a bulla. The push is one you can do with a cookie tin, with a paper seal glued across the lid and rim. Let it dry, then try opening the lid without tearing or disturbing the seal. The bare lid would come straight off; it is the seal that makes this a test. A substitute seal could fool you, so this little trial shows a disturbance, not an unforgeable history. In the clay ball, the outside shows the count inside and opening the ball breaks it. That is the property the tin and seal are trying to reproduce. The idea is a real one and it will show up again in a form you won't expect: a record is worth something only if it can't be silently undone.

Now law, which is where people get serious about who's allowed to say what happened. The book of Deuteronomy, chapter nineteen: nobody is convicted on the word of one witness; it takes "two or three." The Roman jurist Gaius, writing in the second century, says a will the praetor will honour is one sealed by seven witnesses. Why seven? Gaius doesn't say. The legal historian Henry Maine, in 1861, had an answer: the older Roman will was a mock sale, done in front of five witnesses, a man holding the scales, and the buyer of the estate, seven people in all, and when the ceremony fell away the seven seals kept the count. Maine's is the only account of it I found, so treat it as his. Gaius's own text does show they kept those old formalities around a will, and it isn't hard to see why anyone would: one person can lie, and a will moves a fortune, and the dead man can't object. So the record isn't the paper; it's the number of independent people who have put their own seal, their own name, their own risk, next to the claim. Try it: write "paid five dollars, signed A" on a napkin and ask a friend to forge it. It takes them ten seconds. Now require two other people to initial it at the moment of payment and each keep a copy. If the readers compare those independently kept copies and trust their holders, changing your napkin alone will be exposed. Stolen copies, forged initials, collusion or a reader who never compares them can defeat the test. The claim didn't change, but what stands behind it did.

Which brings me to seals, and the medieval historian Michael Clanchy, whose book is called *From Memory to Written Record*. England between the Norman Conquest and the early 1300s was learning to trust writing. Until then it had trusted people who remembered. The lesson was painful: seals were forged, charters were forged, and historians today have a whole craft of telling real charters from fake ones because the fakes are so many. A seal is only as good as whoever holds the stamp. Keep that one too, because in 2011 a certificate authority called DigiNotar found out, at the cost of its existence, that a mathematically perfect signature is worth exactly what the custody of the key is worth, and intruders had taken control of the machines that signed. It is the same failure nine hundred years apart, and I find that oddly comforting.

And then the obvious next question: if seals could be forged, what caught the forgeries? It wasn't the seals. The most famous forged record in European history is the Donation of Constantine, a document in which the emperor Constantine supposedly handed rule over Rome and the western empire to the Pope. It was most likely written in the eighth century, some four hundred years after Constantine died, and it was leaned on for centuries. Around 1440 a scholar named Lorenzo Valla took it apart by its Latin: the language belonged to a much later age, and it used words, "satraps" among them, that no Roman of Constantine's day would have used for his officials. (Valla wasn't even first. Nicholas of Cusa, a German churchman and philosopher who later became a cardinal, had already called it a forgery in 1433, in *De concordantia catholica*, a long book on how the Church and the empire ought to hang together. I know that from the reference works, and I haven't read the book. Valla's is the version people remember, which is a pattern that keeps turning up.) Look at what kind of check that was. The question wasn't "is the stamp real." It was "could these words have been written when they say they were." A record can pass the first and fail the second, which is a hint that what I keep calling one thing, a record, is doing more than one job.

## 2.2 Two columns and a stamp

The one I'd never have guessed mattered is bookkeeping. Around 1300, the ledger of a Florentine company called Farolfi, kept at its branch in Provence, shows the earliest complete example anyone has found of a trick merchants were working out: write every transaction twice, once where the money came from and once where it went, and the two columns have to balance. The treasurers of the city of Genoa kept their ledger for 1340 that way, and accounting historians still count it among the most famous government account books of the Middle Ages kept in double entry. And in 1494 a friar named Luca Pacioli printed the method in a mathematics book,

and it spread. The method is double-entry bookkeeping, and it's easy to take it for just a tidy habit.

Do it for one day. Keep two columns of every dollar that moves, and total them at night. Now have a friend secretly change one entry in one column. The totals disagree; you catch it. Now have them change the matching entry in both columns. The totals agree, and you've caught nothing. Push on that and you've found what double entry actually is: a check for slips, which a careful liar walks straight through. Did Pacioli see that? I can only get at it secondhand. The scholar Forrester, in a paper on the University of Strathclyde's accounting-history pages, reads him as reporting that in many places the books were taken to the local consul to be authenticated before a single entry went in. The stamp is a witness again, from outside. The two columns are the discipline; the stamp is the claim that someone other than you will stand behind the fact that these are the books. That works only as far as the authentication and custody distinguish this set of books from a substitute. Keep both halves separate in your head, because one is a method and the other is a commitment.

And then I found the place where this got turned into a motto. The Royal Society, founded in London in 1660 and still running, picked as its motto *Nullius in verba*: take nobody's word for it. (Where did they get it? From somebody's words, naturally. It's half a line of the Roman poet Horace, from a verse letter to his patron, saying he isn't bound to swear by the words of any master. So the society that takes nobody's word for it took its motto from a poet and cut the line to fit. That's not a contradiction, and it's a nice picture of the method: borrow what's useful, say whose it was, and don't swear by it.) The historian Steven Shapin wrote a paper about what that meant in practice, and the answer is more interesting than the slogan. Robert Boyle, pumping the air out of a glass globe to see what happens to a candle or a bird, had three problems: the machine, the report, and the witnesses. He solved the third by having gentlemen present who'd sign their names, and the second by writing the report so richly that a reader who wasn't there could picture the scene and count as a witness anyway. Shapin calls that virtual witnessing, and he notes, dryly, that the air pump was very expensive, seventeenth-century big science. So there *was* a cost; it just wasn't a cost of declaring the fact. It was the cost of the instrument and of getting enough people into the room. Hold on to that distinction, because the rest of the book hangs on it.

### 2.3 A demon, and the step that pays

Now comes the strangest entry, and the one that gets closest. In 1867 the physicist James Clerk Maxwell, in a letter to his friend the Scottish physicist Peter Guthrie Tait, imagined a "very observant and neat-fingered being" sitting at a tiny door between

two halves of a box of gas, letting fast molecules through one way and slow ones the other, so that one side heats up and the other cools with no work done, which is supposed to be impossible. The being got called Maxwell's demon, and physicists have been arguing about it for a hundred and fifty years, which should tell you something.

In 1929 Leo Szilard, a young Hungarian physicist working in Berlin, cut the demon's box down to a single molecule, and that made information part of the accounting. The tempting move, and the one Szilard's own accounting made, is to put the whole heat floor on finding out which side the molecule is on. Bennett separated finding out from restoring the memory for another round: measuring onto a blank record can, in principle, be thermodynamically reversible, and resetting an unknown record is a different operation. That does not mean a real measuring device costs nothing to build or run. It says that learning and resetting do not carry the same unavoidable entropy cost, and I'm about to pick a step to charge, so I need that distinction before I do.

Bennett's version of the rule, in a 2003 paper you can read, is that what must cost is "the erasure of a bit or the merging of two computation paths." That is erasing, or taking two possible situations and making them one. And in the same paper he lists copying unknown data onto a blank place to write among "the most important logically reversible operations," the kind he says can, at least in principle, be run thermodynamically reversibly. So, in the one place in all of this where a record is priced by physics, the unavoidable erasure charge belongs to *forgetting* the unknown record. Making a record on a blank place carries no such charge of its own. It does use up the blank place, which is a state the machine hasn't used yet. Resetting it for reuse has a Landauer bound when that reset removes entropy under the thermal premises. Read that twice, because it's not what I expected either.

Making a record on a blank place is one operation. Landauer's 1961 paper names another: set a bit to one whatever it held. If that bit is the whole state, down goes to up, up stays up, and two possible states become one. Landauer called this "restore to one." On an uncertain input it removes information, and a heat bound then needs the thermodynamic premises.

A larger machine has more state than its flag. Two starts can both finish with the flag up while remaining different elsewhere. The next chapter proves that, on a finite machine whose reading is permanent, any instruction that raises it must merge some full states. It does not say the state whose flag just rose is one of that pair. Put one state outside the set that reads yes and two states inside it. Send the outside state to the first inside state, and send both inside states to the second. The flag rises for the outside state, but the two inside states are the ones that merge. These arrows describe alternative starts, not three things happening in one run.

That is the distinction the one-bit picture hides. The instruction has a collision somewhere, and the actual distribution of starting states determines whether that collision removes uncertainty. A known starting state loses none under the entropy premise, even when its flag rises (Theorem 3.17, next chapter). A separate uncertain memory may have a reset cost, but neither its existence nor an earlier heat payment follows from specifying a known start. The ledger can still charge the instruction under merge pricing. A ledger charge, information loss and a device's heat are three quantities to keep apart.

I'll put the rest of the lineage quickly, because it all points the same way. Regulators who make brokers keep records on media that can't be rewritten (the SEC required exactly that, write once and read many, from 1997 until 2022, when it allowed an audit trail as the other way to comply) are pricing the hardware. The rule puts no charge on the act of writing. Haber and Stornetta in 1991, the timestamping that Bitcoin's footnotes lead back to (Section 14 follows them), chained time-stamped records together with hashes so that changing an old entry breaks every entry after it (and Ralph Merkle's hash tree, from his 1979 Stanford thesis on secrecy and authentication, lets one short value vouch for a great many records); try the dependency with five entries, 1, 2, 3, 4, 5, adding each to the running total: 1, 3, 6, 10, 15. Keep that sheet and make another with the second entry changed to 3. Its totals are 1, 4, 7, 11, 16. The change travels down the chain. Now lower the third entry to 2, and the last total is 15 again. Simple addition can be fooled by compensating changes. A cryptographic chain needs a stronger function, and someone must retain a trusted value to compare against. This trial shows the dependency between entries; it supplies neither that function's security nor a price on the record. Proof-of-work, which the computer scientists Cynthia Dwork and Moni Naor proposed in 1992 to make junk mail expensive and which Bitcoin later chained, is the one framing where a record genuinely costs, and what it charges for is a search: how many tries it took to find a lucky number. At the same difficulty, an empty and a full block face the same target for the header hash. The expected number of hash attempts agrees; the attempts on two actual blocks need not, and handling their transactions is additional work. So the work prices the search. Proof-of-work charges nothing extra for the moment anything becomes certified. Robert Floyd and Tony Hoare, two computer scientists asking how you'd prove a program does what it's supposed to, attached claims to points in a program in 1967 and 1969 ("whenever you get here, this is true"), and the claim is backed by a proof that's paid for off the machine. And the certificates of complexity theory, which go back to Stephen Cook's 1971 work on NP, are strings designed to be cheap to check: hand me the certificate and I can confirm the claim quickly. Whether they're also expensive to *find* is the famous open question of the field, P versus NP, and nobody knows; the definition promises polynomial-time verification. Polynomial

time need not be small in practice, and it need not be less than the time to find every particular witness. This is a bound on checking, not a separate charge for setting an accepting flag. And from statistics, a ledger that runs the other way: Dean Foster and Robert Stine's alpha-investing (2008) runs a long string of tests against a budget of allowed false alarms, where every test spends some of the budget and each observed rejection earns budget back, even though some rejections can be false. That's the other way round from a price on vouching: the testing rule spends and earns an error-control budget; a reward does not certify that the discovery is true.

## 2.4 What they all agree on, and who they rule out

Lay them down, and this time what shows through is not one thing. It's five different jobs that the word "record" has been doing, and I'd been letting one word carry all five. Most of them have a corpse next to them, and the corpses are different too.

Durability: the record can't be silently undone. The bulla, the write-once disk, the hash chain and Bennett's card that has to be wiped all do this job. The corpse is the editable log, and the company that kept its debts off the books: Enron hid its debt in side entities set up to keep it off the balance sheet, which is how the Justice Department described the scheme in 2002, and its auditor, Arthur Andersen, shredded the paper. Andersen was convicted for the shredding that year, and the Supreme Court threw the conviction out in 2005 over how the jury had been instructed; either way, the books concealed debts they should have shown. But notice that this job doesn't need a witness at all. A copy will do: a hash chain nobody signed is still hard to change quietly, so long as somebody else holds the latest link, because a chain with no copy anywhere can be rewritten from end to end and recomputed, and nobody would know.

Authenticity: the record came from who it says it came from. Seals, the signature only a key-holder can make, and the consul's stamp all do this one. The corpse is the forged charter and DigiNotar. This job is about custody of a stamp, and it's a different job from durability; a perfectly authentic entry can still be erased, and a perfectly durable one can still be a forgery.

Evidence: enough independent people stand behind the claim that lying is expensive. Two witnesses, seven seals and Boyle's gentlemen do this one. It rules out anonymous assertion, and it rules out "the universe keeps the receipt", because information spread into the environment is not a record anyone can read or rely on. This one is about numbers of people and their risk. Paper has nothing to do with it.

Accounting: the record checks itself for slips. The two columns do this one. And one thing has to be kept straight: the second column is *not* an outside witness. It's you, checking you. It catches a dropped digit. A careful liar gets past it, as you saw. It belongs on this list as its own job, separate from the seven seals.

Cheap checking: a record whose correctness can be confirmed for much less than it cost to make. Cook's certificates, the signature and the hash do this one. This one isn't universal, and I thought it was. Whether checking needs to be cheaper than making depends on who you're worried about and how much a wrong record costs you; plenty of records are checked by redoing the whole thing, and that's fine when the thing is small.

Now for the thing none of them says. Not one of these jobs is pricing the moment a record goes up and stays. Bennett prices merging and erasing, which that moment might or might not be. Proof-of-work prices a search the designer chose. Bookkeeping prices a discipline. Witnesses and seals price showing up and guarding the stamp. I looked, with everything I had, and that's where it ran out. If there is a cost charged for that moment itself, nobody in this lineage wrote it down, and that's either because it isn't there or because the one person who did write it down is on a shelf I never found. The literal one-bit reset already has a thermodynamic analysis: Landauer's restore to one. The question here is which premises force an instruction that raises a record to merge full states, and which further premises put a price on that loss. So I'm going to have to try it myself, on the smallest thing I can build, and let you watch. And I'll have to be honest about which of the five jobs the small thing even attempts, because the toy has a permanent flag under its three moves, but no witnesses, authenticated identity or check of a claim.

## 2.5 The smallest thing that could carry the charge

Take a sheet of paper. Draw two boxes and label them NO and YES. The two boxes are the reading, a single yes-or-no question asked of the state, and I'm going to be careful about what the question is, because after five thousand years of seals and witnesses the word "vouch" is carrying a lot and this toy carries almost none of it. The question is only: "has the designated flag been set." It is not "is some claim about a grouping true," and it is not "did anyone check." There is no claim in this toy, no evidence, no checker, nobody standing behind anything. There is a flag. Whether a flag can ever be made to *mean* that something was checked and can be relied on is a real question, and it's not this section's; I'll build toward it later, and until then please read YES as "the flag is up" and nothing more. The state is which box you're in, and that's the whole state, one bit. Then draw a line under the boxes for tally marks. The marks sit outside the state. They're the running total you keep beside it as you play, the way you'd keep score beside a board game.

Now give yourself three moves. STAY: you don't move, and you make no mark. WORK: you don't move either (it's a stand-in for every ordinary step a machine takes), and you make no mark. VOUCH: from NO you move to YES; from YES you stay in YES; either way you make one tally mark, because in this toy the price

belongs to the move's name, and where you were standing doesn't enter into it. Now play. Start in NO, make any sequence of moves you like, and stop in YES. Count the marks. Then try to reach YES with zero marks. You can't, and you can see exactly why: the only move that crosses from NO to YES is VOUCH, and VOUCH makes a mark. Now erase the mark from VOUCH's rule and play again. You reach YES with no marks, every time. The entire content of what I'm about to call a rule is this, and I want you to see how little it is before I dress it up: I've decided that the move which crosses the line makes a mark. Nothing in the toy's rules forces the mark. VOUCH does merge, NO and YES both landing on YES, which is what the last chapter says a kept raise on two states has to do; what nothing yet says is that a merge has to cost a mark. Here it's a choice, what the last chapter says keeping takes, written down as a rule. The next chapter is about when a machine has no choice.

Why tally marks, by the way? Because they're about the oldest receipt there is. The real ones, the English treasury's notched sticks split down the middle, get their own story in Part II (Section 6), fire and all. The tally on your paper is nothing like those. Nobody holds the other half. You keep it, beside the board, and you could rub a mark out and nobody would ever know. Hold on to that, because by the end of this section it's the reason the toy does none of the five jobs a record does.

Here is the same toy, stated once, carefully, because this is the one place in the book where I state it exactly and everything later points back here. The point listed the five parts in a sentence; here they're pinned down. The thing on your paper has five parts.

A set of states. (Here: NO and YES, and nothing else.)

A set of move names. (STAY, WORK, VOUCH.)

A step: for each state and each move name, where you land.

A cost for each move name: a whole number of tally marks, zero or more. (Zero, zero, one.)

A reading: for each state, yes or no.

Those five parts are the whole of it. Add the one rule below and I'll call it a Thiele machine.

The toy's reading has two places, no and yes. That is a choice for the toy, not for the model. In the model the record is a place in any order you can read "at or below" on, and the toy's reading is the smallest such order. Everything in this section is proved for the toy. Section 11 proves the same for any order and gets the toy back as the two-place case (Theorem 11.2).

I also want to be exact about what's not in it. There's no counter stored inside the state that's forbidden from going down; the marks are the side tally from the board game, and the tally only ever grows because the costs are whole numbers, zero or more. (A machine could keep the total inside its state too; that's an extra part

beyond the five, and Section 6 builds it.) There's no window, no observer who sees only part of the state; that's a later question and it gets its own section. And the cost depends only on the move's name; where you are when you make it doesn't matter. That's on purpose: the charge is on the attempt, so you pay for making the move, whatever it turns out to do. It's also a real simplification, and where a result needs a price that depends on where you are, Section 7 says so and uses one. Last, the step gives every state and every move exactly one place to land. A choice becomes a move name, and a machine that has stopped steps to itself. A step that can land in several places by chance is a different picture, and the noisy steps in Section 1 take it up. To apply the step law, fix the move being repeated, or include the program and its control in the full state so that one autonomous rule chooses the next step. Arbitrarily choosing different moves from outside is a different case, worked through in the next chapter. Deterministic instructions alone do not supply that fixed rule. Separately, giving every move a successor even when the flag is already up keeps a price per move name from charging exactly the raises: a move that raises the flag and has no next step once it's up could cost one and never charge for anything else.

One rule, which I'll call the toll. For every state and every move: if the reading is NO before the move and YES after it, the move's cost is at least one. That's the whole rule. It says nothing about any other move. It says nothing about how big a mark is in anything you could measure. One mark is one mark; what a mark is worth, why the smallest charge is one (why not two, or a half?), and whether I had any right to decide that, come next. The middle question has a short answer I'll give away now. What the next chapter forces, on its premises, is that a raise that stays up, with only finitely many yes-states, merges two states, so any price on merges lands on it. Costs here are whole numbers, which I chose so the ledger counts acts, and the least whole number above nothing is one. The "at least one" comes from there.

And one theorem, which is the only thing the toy actually proves, and you've already proved it by playing. Take any run of moves that starts in a NO state and ends in a YES state. Then the marks it made add up to at least one. The argument in words is short enough that you can hold the whole thing. Walk the run move by move, reading the box at each step. It read NO at the start and YES at the end, and every move either leaves the reading alone or flips it, so at some move it went from NO to YES. The toll says that move cost at least one. Every other move cost zero or more. Add them up and the total is at least one. Notice what the argument did not need. It didn't need the reading to *stay* YES afterward; if a later move flipped it back to NO and another flipped it to YES again, the toll charges twice, and the argument still goes through. It didn't need the machine to be finite. It didn't need to know what any move does besides what the reading says before and after. It needed the toll, and two things the five parts already hand it: no cost is below zero, and the reading has only two values, so a move that changes it from NO lands on YES.

Give the reading a third value, a "maybe" between the two, and the one-bit toll as written here charges nothing for NO to MAYBE; the toll over any order in Section 11 charges every step that climbs, MAYBE included. That difference is the whole reason the record is a place in an order and not a bit.

That's the whole theorem, and I'll state it once more in the exact form, because from here on the book carries two layers side by side: what I mean, in words, and what's stated and proved, in symbols you can hold me to. The symbols are few. $\mathbb{N}$ is the whole numbers 0, 1, 2, and so on. $S$ and $I$ are just names for "the set of states" and "the set of move names." Writing step : $S \times I \to S$ means "hand it a state and a move name and it hands back a state." A *run* is a list of move names, $t = [i_1, \ldots, i_n]$, played from a start $s_0$, one after another. The double arrow $\Longrightarrow$ reads "then": the thing on its left forces the thing on its right. For this first proof, keep your finger on the run and stop at the first YES. The move immediately before that is the one whose charge matters.

**Definition 2.1** (Thiele machine)**.** A *Thiele machine* consists of a set $S$ of states, a set $I$ of move names, a step function step : $S \times I \to S$, a cost cost : $I \to \mathbb{N}$, and a record rec : $S \to R$ into a set $R$ with an order $\leq$ that is reflexive and transitive, subject to one rule, the *toll*: for every state $s$ and move $i$,

$$\mathrm{rec}(\mathrm{step}(s,i)) \not\leq \mathrm{rec}(s) \implies \mathrm{cost}(i) \geq 1.$$

Nothing more is assumed of the order. In Coq this is the record `AxSys`.

**Definition 2.2** (Thiele machine with a two-place record)**.** Take $R = \{\mathrm{no}, \mathrm{yes}\}$ with no below yes. The record is a reading cert : $S \to \{\mathrm{no}, \mathrm{yes}\}$, it leaves the down-set exactly when it goes from no to yes, and the toll reads: for every state $s$ and move $i$,

$$\mathrm{cert}(s) = \mathrm{no} \text{ and } \mathrm{cert}(\mathrm{step}(s,i)) = \mathrm{yes} \implies \mathrm{cost}(i) \geq 1.$$

The run of $t = [i_1, \ldots, i_n]$ from $s_0$ ends at $\mathrm{run}(t, s_0)$, the result of applying the steps in order, and its total cost is $\mathrm{cost}(t) = \mathrm{cost}(i_1) + \cdots + \mathrm{cost}(i_n)$, with the empty run costing 0. In Coq this is the record `CertificationSystem`, and `two_point_nfi` derives its floor from the floor over any order (Theorem 11.2).

The toy on the paper is the second definition, and so is nearly everything in Parts I and II; when a result holds for any order, the text says so. I write the yes-or-no reading as cert. It names the flag whose rise I am counting. What earns that flag its meaning will depend on the machine's rules.

A word on the label before the first result. I call a result an *observation* when its proof does little more than unfold the definitions, and in an axiomatic cost model that's what the definitions are meant to do: the work is in choosing them, so the

results that follow from them in a few lines are labelled that way and the theorems with more in them stand out.

Take an ordinary computing machine and put a lamp and tally beside it. Let computing run as before, leaving both alone. A separate mark leaves the computing state fixed, adds one to the tally and turns the lamp on when its named event holds. Forget the lamp and tally and the old computation is still there, step for step. Below I build that box, compare different ways to implement it, and ask exactly which differences its questions require. The old machine may already have prices or a reading of its own; this construction supplies the new lamp and its zero/one account. With effective rules an ordinary computer can store all three parts.

**Observation 2.1** (The universal floor)**.** *In any Thiele machine, every run that starts at a state reading no and ends at a state reading yes has total cost at least 1: if* $\mathrm{cert}(s_0) = \text{no}$ *and* $\mathrm{cert}(\mathrm{run}(t, s_0)) = \text{yes}$*, then* $\mathrm{cost}(t) \geq 1$.

*Proof.* Read the states of the run from left to right. The first reads no and the last reads yes, so there is a first state on this list that reads yes. It cannot be the starting state. The state immediately before it reads no: otherwise the chosen YES would not be the first. The move between them therefore costs at least one, by the toll.

Now add the charges for the whole run. That one charge is already at least one, and none of the other charges is negative. The total is at least one. The flag may come down later and go up again; that cannot remove the charge I have found. ■

*The tabletop and the stamp*

Put two boxes on the table. Each has tiles on top, and the tiles spell the same answer. On the side of one box the stamp is up. On the other it is down. Hang a camera directly above them. The photographs are identical.

Now hand me the photographs and ask which box was stamped. Take a better photograph. Enlarge it. Count every tile. None of that gets the side of the box into the picture. The difference is there, in the boxes. It has been left out of what you handed me.

That is the window I want to name. $S$ is the set of full states of the boxes, and $B$ is the set of tabletop pictures. The map $\pi : S \to B$ takes a full state and returns its picture. Every picture in $B$ comes from a box; that is what *onto* says. Pick one picture, $b$, and gather every full state that gives that picture. That whole collection is the *fibre* above $b$, written $\pi^{-1}(b)$.

Keep the whole collection in mind. The tiles can agree while the boxes differ in which checks ran, which earlier pages were certified, and what was kept when those pages were wiped. The fibre holds all the differences this camera leaves out. There can be two states in it, a million, or infinitely many.

I started with one question: is the flag up? That gives me one yes-or-no reading of the full state. A hundred different states can all answer yes. Reading that one flag tells me where they stand on that one question; the other differences are still there. Chapter 14 opens the gauge wider. It asks which checks passed and which editions were certified, and counts how many states those answers require. Keep independently chosen histories across arbitrarily many editions, all recoverable above the same final picture, and the fibre has to be infinite. There is a proof of that in Section 14.3.

To see exactly what the tabletop can do by itself, give these boxes two kinds of move. The tile moves are horizontal. They cost zero, leave the stamp alone, and move the tiles according to what is on the tabletop. The stamp never reaches into their gears. The stamp moves are vertical. They leave the tiles alone and can raise the stamp or leave it where it is. Once raised, it stays raised. A raise pays the toll already defined above.

These are the rules of the sliced machine. They give me a computational view that runs on its own and a record that survives beside it.

*Which question does the toll charge?*

Slide a tile one place to the right. Let that move cost nothing. Now ask a new question: is the tile in the place it just reached? Before the slide the answer was no. Afterward it is yes. If every question had to obey the same toll, that slide would have to cost at least one.

Try the same trick on any free move that changes the full state. Ask whether the machine is in its destination state. Again the answer changes from no to yes. Charging every question would stop every free change. The reading in the definition therefore matters: I have to say which question the toll belongs to before I can say which moves are free.

**Theorem 2.2** (Charging every question stops free changes)**.** Suppose equality of full states is decidable. The same instruction prices satisfy the toll for every Boolean reading exactly when every zero-cost instruction leaves every full state unchanged. □

*Proof.* For a changed state, the question “am I at that destination?” supplies a crossing, so the move cannot be free. In the other direction, a move that leaves the state unchanged cannot change any reading. Every remaining instruction already costs at least one, since prices are nonnegative whole numbers. It therefore pays for any crossing it makes. ■

Now keep one question and require its yes to stay yes. A free move cannot raise that reading, because a raise pays the toll. It cannot lower it, because the reading is permanent. It has to leave it alone.

□ **Theorem 2.3** (Free moves leave a permanent reading alone)**.** The toll and permanence force every zero-cost instruction to preserve the reading on every state.

*Proof.* Starting at no, a free move cannot reach yes by the toll. Starting at yes, it cannot reach no by permanence. Those are the two possible starting readings. ■

This gives two ways to state the reading part of the sliced rules. Require horizontal moves to leave it alone and vertical moves never to lower it; then every move keeps a yes. Or require that every move keeps a yes, keep the toll and free horizontal prices, and recover horizontal preservation by the two sentences above. The window rules still do their own work. A paid move could change both the tiles and the stamp; permanence does not make the tiles stand still.

**Definition 2.3** (Sliced Thiele machine)**.** A sliced Thiele machine is a Thiele machine with an onto view $\pi : S \to B$ and its move names split into horizontal moves $I_H$ and vertical moves $I_V$. Write $r(s) = \mathrm{cert}(s)$, with no $<$ yes, and $F_i(s) = \mathrm{step}(s, i)$. For each $i \in I_H$ there is a map $h_i : B \to B$ such that

$$\pi(F_i(s)) = h_i(\pi(s)), \qquad r(F_i(s)) = r(s), \qquad \mathrm{cost}(i) = 0.$$

For each $v \in I_V$,

$$\pi(F_v(s)) = \pi(s), \qquad r(s) \leq r(F_v(s)).$$

The toll remains: $r(s) < r(F_v(s))$ implies $\mathrm{cost}(v) \geq 1$.

These laws hold on every state of the specified domain. In contrast, the counter example's later live-state slice is a restricted interface: a failed check leaves its domain. It is not a claim that all six native instruction kinds satisfy these equations globally.

The states can be counter configurations, tape configurations, or any other states meeting these rules. The same horizontal argument works with a whole set of claims as the record: every horizontal move preserves the set and evolves the picture using only that picture. The number of possible records never enters that argument.

Now give the two boxes the same horizontal moves. After one move the pictures still agree. After two they still agree. Every next picture comes from the previous picture and the move you named, and those agree at every step. Meanwhile one

stamp stays up and the other stays down. You can watch the tabletop for the whole run and never see the difference you started with.

Give somebody permission to read the stamp and choose a different tile move because of it, and they can signal through the tiles. Here both boxes receive the same moves. That is how I keep the question fixed while asking what the window can reveal.

**Theorem 2.4** (What the tabletop can tell)**.** In a sliced Thiele machine, every horizontal word projects to the corresponding word of the maps $h_i$ on $B$, and preserves $r$. Two states with the same view and different records therefore keep the same view and different records after any common horizontal word. No function of the view alone can recover $r$ on both states. A vertical move that changes $r$ supplies such a pair: $s$ and $F_v(s)$. □

*Proof.* Start with two states showing the same picture. Give them the first horizontal move. Its rule chooses the next picture from the current picture alone, so the next pictures agree too. Neither record changes. You are back in the same situation, ready for the second move. Repeating this argument covers every move in the word, including a word with no moves. On each side the picture follows the corresponding $h_i$, and the original record stays put.

A proposed reader of the picture now receives the same input from both states. It must give the same answer to that input, so it cannot return two different records correctly. Finally, a vertical move that changes the record gives you exactly such a pair: before and after, the picture agrees and the record differs. Those must be different full states. ■

The stamp remains in the full state through every one of those moves. The camera keeps missing it for exactly the same reason it missed it in the first photograph. In Part III a failed check can stop the gears. Hiding that stop while asking the picture to predict every later free move would also hide the difference between gears that turn and gears that cannot. Theorem 15.1 works through that obstruction; the live-state slice there keeps this boundary explicit.

*The smallest price on the instruction*

Put three price cards beside a button: zero, one and two. Somewhere in the machine there is a state with the lamp down where pressing that button raises it. Cross out zero. It fails there. One pays the toll; two pays it too. The toll has picked the smallest card, even though it has not forbidden the larger one.

Now take a button that never raises the lamp, from any state. Zero works. Go through the buttons this way. Put one beside each button that can raise the lamp

and zero beside each button that cannot. The price stays on the button when you use it somewhere else. A button that raises the lamp once still costs one when the lamp is already up.

□ **Theorem 2.5** (The least instruction price)**.** Fix the transitions and reading, and supply a finite list covering the states. The schedule just described is the unique pointwise least nonnegative whole-number schedule satisfying the toll. It is also the least schedule giving every trace from no to yes, from every start, total cost at least one.

*Proof.* For each instruction, inspect the list for a state where it raises the reading. If one is found, every valid schedule must charge at least one there, and the instruction has the same price everywhere. If none is found, the toll puts no positive demand on that instruction, so zero suffices. Our schedule meets every demand and is no larger than any schedule meeting them. Two least schedules must each be no larger than the other at every instruction, so their prices agree. The trace claim follows by finding the first crossing; conversely, a crossing by one instruction is already a trace. ■

The list makes that search finish. With infinitely many states, searching forever without finding a crossing does not tell me that none exists. For the lift, there is a direct answer whenever each mark has an event witness: put the lamp down over a base state where that mark's event holds, and press it. It raises the lamp. Every mark therefore needs one, and computing moves need zero because they leave the lamp alone. That proves the least zero/one schedule even over an infinite base. A mark whose event never holds can cost zero under the toll. Giving that mark a tally of one on every use is an additional accounting rule. The unit in all of this is the unit of the definition; to price a physical device in joules I still need its physics.

*What the next move could tell us*

Put two boxes beside each other and read everything the task lets you read. Suppose the answers agree. That is not yet enough to put the boxes in the same pile. Press the same button on each. Perhaps one lamp now comes on and the other stays dark. The difference mattered, even though the first reading missed it.

Keep two states together only when every permitted finite continuation gives the same required answer. Include doing nothing: the current answer counts too. This tells us which differences the task can afford to forget.

**Theorem 2.6** (Keep what a later question can separate)**.** For a fixed deterministic machine, permitted instructions and required observation, equality of observations after every finite continuation is the unique coarsest equivalence that preserves the current observation and is stable under every instruction. A deterministic implementation preserving those steps and answers cannot identify two states separated by such a continuation. □

*Proof.* Agreement with oneself, reversal of an agreement, and joining two agreements all hold one continuation at a time, so this is an equivalence. Doing nothing checks the current observation. To check stability after a move, put that move in front of each continuation. If two states were already grouped by any observation-preserving, stable relation, apply its stability once for every move in a continuation, then read the answer. Those states agree on every continuation and belong to our equivalence. Our grouping therefore contains every permitted grouping. Finally, if an implementation put two separated states in one full state, the same continuation from that state would have to produce two different answers. Determinism gives only one. ■

Change the questions and the piles can change. A box that toggles between two positions needs two distinguishable states if I ask its position. If my only required answer is always “ready”, and every later answer is also “ready”, those positions can share one description. The task tells me what has to survive.

*The picture, the lamp and the tally*

Take an ordinary computing box. Put a lamp beside it and a tally beneath it. A computing button moves the box by its old rule, costs zero, and leaves the lamp and tally alone. A mark button leaves the box where it is, adds one to the tally, and turns the lamp on if its named event holds there. Once on, the lamp stays on. Start with the lamp down and the tally at zero.

Those instructions tell us what to do without telling us how to build the box. One builder may keep every button press on a tape. Another may keep an extra turn counter. I require each builder to show me the same three things: the computing picture, the lamp and the exact tally. Its computing moves must follow the base rule through that picture. Its marks must leave the picture fixed and latch exactly the named event. Its prices are zero for computing and one for marking; its reading obeys the toll and permanence; its tally increases by exactly the charge on every move.

□ **Theorem 2.7** (Every such box has the same three-part description)**.** Fix those base rules, mark events and accounting requirements. Every implementation has exactly one map to the product of base state, Boolean reading and natural tally that preserves the three readings. That map preserves every instruction and every finite run. From a start with lamp down and tally zero, it maps the reachable implementation states onto all reachable states of the product lift with the matching base start.

*Proof.* Read the picture, lamp and tally, and write them down in that order. There is no other choice for a map required to preserve those three readings. Press a computing button. The picture follows the base rule. The lamp stays put by the toll and permanence, and the tally stays put because the charge is zero. Press a mark instead. The picture stays put, the lamp takes its old value or the event's yes, and the tally rises by one. These are exactly the two rules of the product lift. Repeat for every move in a finite run.

The clean starts agree in all three readings. Any reachable product state has some finite list of buttons reaching it. Press that same list on the implementation. The step argument makes its three readings the desired product state. Every reachable product state therefore has a reachable implementation state above it. ■

Here are the three stages. First write a list of buttons. Then run it on the builder's box. Then read the picture, lamp and tally. You get the same result as running the list directly on the three-part description. Written as arrows, after we have done it:

$$\text{button lists} \longrightarrow \text{implementation states} \longrightarrow \text{picture, lamp, tally.}$$

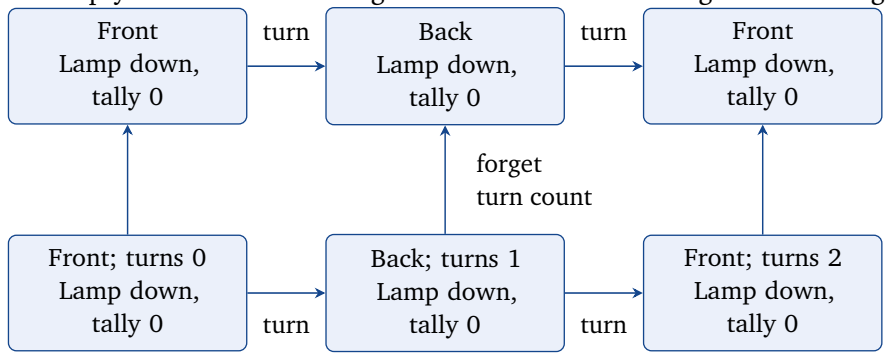
The first arrow has only one possible meaning once the start and buttons are fixed. The empty list gives the start; appending a button takes one more step. This is the free history construction. The second arrow can forget history. Its direction matters.

Turn a box halfway around with a free button. Turn it again. It is back where it started, with the same lamp and tally. Now give another builder's box a counter that goes up on every turn, without charging for it. After two turns its counter is two larger. It meets all the requirements above: the extra counter is separate from the tally of charges.

□ **Proposition 2.8** (Forgetting the turn count goes one way)**.** Even on reachable states, the three-part lift need not have a step-preserving map into every implementation satisfying its contract.

*Proof.* In this example two turns return every product state to itself. A map preserving turns would have to send it to an implementation state that also returns after two turns. No such state exists there: its extra counter has increased by two. Forgetting that counter works in the other direction and preserves both turning and marking. ∎

There is an exact way to tell when the reverse map would work. Whenever two button lists reach the same source state, they must also reach the same target state. Otherwise the map would have to send that one source to two targets. Supply one access list reaching each source state, and this condition is sufficient: run its access list on the target. Any other list reaching that source gives the same target by the condition. The empty list preserves the start, appending a button preserves a step, and reachability leaves no other possible map. The turn counter fails exactly this test: the empty list and two turns agree in the source and disagree in the target.




The top row returns to the same full state. The bottom row does not. Forgetting the turn count respects each turn; recovering it from the top row cannot.

**Theorem 2.9** (The smallest description that answers these questions)**.** Under the preceding contract, two implementation states give the same three readings after every common finite continuation exactly when their current three readings agree. If every two distinct reachable implementation states can be separated by such a continuation, projection is a bijection onto the reachable product states. It preserves starts and steps and is the unique start- and step-preserving map there. □

*Proof.* If the current readings differ, read them now. If they agree, the two product states are equal. Equal states obeying the same rule give equal next states; repeat for every move. Projection preserves those moves, so the implementation's readings agree throughout too.

Now require every distinct reachable pair to be separable. Two states projecting to the same triple could never be separated, so they must be one state. Projection

is therefore one-to-one; the previous theorem makes it onto the reachable product states. Any start- and step-preserving map is fixed on a reachable state by a list reaching it: map the start, then follow the list. It must agree with projection there. ■

The same reasoning explains uniqueness more generally. Take two reachable machines with the same answers to every button list from their starts, and suppose each keeps no two states that all future answers fail to separate. Supply an access list for each state of the first. Run that list on the second. If two lists reach the same first state, append any continuation: their answers agree on the first machine, hence on the second. Separation in the second makes their endpoints equal. The map is well-defined and preserves steps and outputs. Reachability makes it onto; separation in the first makes it one-to-one. Following an access list forces uniqueness.

For our box, that smallest description keeps the tally because I asked for the tally. Stop asking for it and repeated failed marks may become indistinguishable. Ask for every past button and the extra tape starts to matter. With a clean start and an event that never holds, no reachable state has its lamp up. Counting unreachable lamp-up states would answer a different question. "Smallest" here means no two retained states are indistinguishable by the required future readings, including on an infinite machine.

This way of grouping states is the familiar behavioral equivalence behind minimal automata. The initial history construction and final observation construction are treated in Rutten's *The Method of Coalgebra*, Chapters 2 and 4 and Section 8.6.[2] Here the box tells us exactly which observations and operations the lift preserves. Choosing the event, the window and the accounting is where the application enters. Once they are fixed, the argument above determines which implementation differences can be forgotten.

*What can choose the question?*

Take a dial with two faces and one button. Each press turns it to the other face. For this question, supply no starting face, no lamp and no names with an outside meaning. Draw its two arrows. Now exchange the names on the two faces. The arrows describe exactly the same machine.

Suppose those arrows alone are meant to tell us which face deserves yes. A rule that does not depend on our choice of names must give the exchanged face the same reading. So both faces get the same answer. A nonconstant reading cannot be selected that way. Give one face a role—the loaded start, a completed job, a passed check—and we have supplied more to work with. The role has to be stated and defended. Renaming the arrows did not supply it.

[2] Jan Rutten, *The Method of Coalgebra* (2019), `https://www.cs.cornell.edu/courses/cs4810/2025sp/Handouts/Rutten.pdf`.

The argument is two lines: the exchange takes the first face to the second; respecting that exchange makes their readings equal. Those are all the faces. It does not make every machine's reading constant. It rules out a nonconstant reading selected from the transition structure alone on every machine. [3]

That fixes what the uniqueness claim can say. The box, lamp and tally have a unique observable realization under the contract already proved. The arrows by themselves do not choose that contract. Adding the word "intrinsic" cannot do the choosing for us.

*Does the picture keep enough of the device?*
Put a lamp on a box and a hidden switch inside it. The button copies the hidden switch to the lamp. Start with the lamp dark in both boxes, and put the hidden switch down in one, up in the other.

| Lamp before | Hidden switch | Lamp after the button |
|---|---|---|
| Dark | down | dark |
| Dark | up | lit |

A description that keeps only the lamp has one starting state, dark, and is asked for two different next answers to the same button. It cannot do it. Keep the hidden switch too and the problem disappears. This is a reason the additional distinction is necessary for this task: the next press can expose it. It is not a reason every task needs that switch. [4]

There is a complete test for this kind of description. First write down the underlying configurations, the permitted buttons, every permitted successor, the picture being retained, and the answers the task requires. This is the operational model of the device. Then check two things. Configurations with the same picture must give the same required answer now. And the same button, from any configurations with that picture, must leave the same next picture, for every permitted successor.

Both conditions are necessary. An answer decoder has only the picture to read. A deterministic next-picture rule has only the picture and button. Neither can return two different results on one input.

They are sufficient too, with explicit choices. Supply a representative configuration for each picture and a permitted successor for each button at each configuration. Decode a picture by reading its representative. Compute its next picture by pressing the button on that representative and taking the chosen successor. The two conditions ensure that every other configuration and every other permitted successor agrees. Repeat that argument at each step: all finite button lists preserve the required answers. On a finite listed model this can be checked against its complete

[3] Checked in `coq/audit/IntrinsicBoundary.v`: `toggle_symmetry_forces_constant_reading` and `no_nonconstant_renaming_invariant_toggle_reading`.

[4] `hidden_bit_blocks_visible_only_dynamics`, `coq/audit/ObservationContract.v`.

table. For arbitrary state types the choices and their effectiveness are separate data; the proof does not manufacture a search procedure.

This test allows several successors of an underlying configuration.[5] If two of them give different required answers, no exact deterministic encoding of that same configuration and button can represent both. [6] A probability distribution over outcomes, or a finer state including a chosen input or seed, is a different contract and must be described as one.

The test starts with a model. Establishing that a real apparatus follows that model still needs evidence about the apparatus: preparation, controls, readout and the range over which the description is claimed. Passing the mathematical test does not supply those measurements.

*How long did we watch the lamp?*
Watch two boxes for three ticks. Both start dark and light on the first tick. Both stay lit on the second and third. One stays lit forever. The other has a counter inside that turns it off on the fourth tick.

| Tick | Enduring lamp | Delayed reset |
|---|---|---|
| 0 | dark | dark |
| 1 | lit | lit |
| 2 | lit | lit |
| 3 | lit | lit |
| 4 | lit | dark |

The first three ticks do not tell the boxes apart. Pick any longer finite watch and I can move the reset one tick beyond it. Both boxes can pay one unit per tick and satisfy the basic toll. Only the first has permanence. The permanent interface excludes the second by its rule, not by pretending that the finite observation already excluded it.

The construction works for every finite horizon.[7] It is not a prediction that every real record fails. It shows why a finite run alone cannot establish an unrestricted forever claim. A proof from an adequate mechanism, or an explicitly bounded reliability claim supported by measurements, has work to do beyond reporting that the lamp stayed lit while we watched.

The universal floor is about every Thiele machine, a system with the toll among its conditions, and about no system without it. One edge matters: the theorem does not assume that a YES state exists. If no state ever reads YES, no run meets the theorem's hypothesis, so it's true of every run and tells you nothing about any of

[5] `operational_adequacy_iff`, `coq/audit/ObservationContract.v`.

[6] `branching_answers_exclude_exact_deterministic_encoding`, `coq/audit/ObservationContract.v`.

[7] `finite_observations_do_not_establish_permanence`, `coq/audit/IntrinsicBoundary.v`.

them, which is a mathematician's way of saying it's true and useless. The toy on your paper has a YES box, so it isn't useless there. It would be natural to assume the system has to have at least one NO state and one YES state; the statement has no such clause, and that's on purpose, because a theorem that applies to no run is still a theorem and the toy is where you supply the runs.

The theorem also runs backwards, and that direction is one line. Suppose every run from a state reading no to a state reading yes costs at least one, whichever state you start from. A single move is a run of length one, so a move that takes no to yes costs at least one, which is the toll. The floor from every start and the toll are the same rule, and nobody has to choose between them. From one fixed start that reads no, the floor asks for the toll only at the crossings you can reach without paying anything; a crossing you can only reach by paying has already cost at least one by the time you get there.

And the bound of one is the smallest case of a count.

**Observation 2.10** (One unit per crossing)**.** *In any Thiele machine, a run costs at least the number of its steps that take the reading from no to yes, with no permanence assumed. Conversely, a system in which every run costs at least its number of such steps obeys the toll. And the bound is reached for every count: on the one-bit machine with a set that costs one and a reset that costs nothing, setting and resetting $n$ times from no crosses $n$ times and costs exactly $n$.*

*Proof.* Each crossing step costs at least one by the toll, every other step costs at least zero, and each step is counted once, so the total is at least the number of crossings. Conversely, a single move that crosses is a run with one crossing, so it costs at least one, which is the toll. On the one-bit machine, each set from no is a crossing costing one and each reset costs nothing, so $n$ rounds cost $n$ and cross $n$ times. ∎

And one more edge, because it's the first thing to bite when you ask a machine to stand behind more than one thing. The definition has *one* reading in it: one question, one flag, and one crossing the toll charges. Once the flag is up, a second claim has nowhere to go, because raising a flag that's already up doesn't move anything. So a second claim needs a reading of its own, and a third claim a third: one reading per claim, each with its own crossing. Part II builds that properly, as a family of flags, one per threshold, with a step that raises any of them charged at least one (Theorem 10.11). A step that raises three at once still owes at least one, and nothing in the toll makes the charges add up flag by flag. Charges that add need the composition assumptions for independent machines (Appendix E). The same theorem has a count in it that is what finiteness takes. Picture the record as a row of switches that only ever get turned on. Ten switches can change at most ten times in one run, one switch per change, and then the row is full. (Across different runs the row can end in any of its thousand and twenty-four settings; along any one run it

climbs through at most eleven.) A finite machine can keep only so many things up for good in its whole life. That's "you can't keep all of it," which Section 14 says of a full trace, turned into a count.

## 2.6 How the receipt earns its meaning

Give the worker a whiteboard and a pen. Ask for an answer and a line underneath saying "CHECKED = YES." The worker can write both. So can somebody who copied the answer and did no check at all. The extra line has told you nothing about which happened.

Follow that problem into the machine. If an unchecked run can reach the same accepting state as a checked run, there is nothing in that state that tells the two apart. To make the receipt carry the difference, the machine has to close the unchecked route. The worker can request the check; the worker cannot simply write its successful result into the accepting record. That restriction is what gives the record its force.

Put a variable called checked beside the answer and you have another place to write. The name does not give it authority. A program that can write yes there without doing the check can manufacture the same display as the program that earned it.

For an earned receipt, I want the machine's rules to close that route. Every permitted program must go through the required check before it can reach acceptance. The program can ask for the check. It cannot give itself the successful verdict. That is the difference between carrying words about a check and enforcing what those words stand for.

So when somebody offers a simulation, showing me where its bits go is the beginning. Show me which instructions the simulated program can actually use, and prove that none of their allowed sequences can forge acceptance. Preserving that boundary is part of implementing this machine. A copy of the flag alone has not done it.

Now put the camera back over the tabletop. The same answer can sit above an accepting record or an unearned one. The record has to survive, and the rules governing access to it have to survive too. Those are the things I am asking the machine to keep.

*Look before the concrete covers it*
The inspector arrives after the concrete is poured. The question is about the rebar underneath. On one job it was sound. On the other it wasn't. Give both jobs exactly the same finished appearance, and give the inspector only that appearance.

There is no amount of staring that tells those jobs apart. Searching for a sound arrangement that could have produced this surface doesn't do it either. A sound arrangement was possible on the bad job too. What matters is which arrangement was there.

Write $x$ for what was there, $f(x)$ for what remains to be inspected, and $P(x)$ for the yes-or-no question about it. A good $x_+$ and a bad $x_-$ leave the same $f(x)$. The inspector receives one identical input and is supposed to give two different answers. Here is the whole obstruction.

**Theorem 2.11** (An erased distinction cannot answer its old question)**.** Let $f : X \to Y$ and $P : X \to \{\text{no}, \text{yes}\}$. If $f(x_+) = f(x_-)$ but $P(x_+) = \text{yes}$ and $P(x_-) = \text{no}$, there is no $d : Y \to \{\text{no}, \text{yes}\}$ with $d(f(x)) = P(x)$ for every $x$. If $P$ is available before the loss, the enriched observation $(f(x), P(x))$ does answer the question exactly and distinguishes these two cases above their common view. □

*Proof.* The proposed $d$ would return both yes and no at $f(x_+) = f(x_-)$. The enriched observation returns its second coordinate. Its two values are $(f(x_+), \text{yes})$ and $(f(x_-), \text{no})$: the first coordinates agree and the second coordinates differ, so the enriched states remain distinct. ■

While the rebar is exposed, you can inspect it and keep the answer. Or preserve enough evidence to inspect later. Either way, something that distinguishes the good case from the bad one has to survive. Wiping an unrelated detail causes no trouble. Wiping the last difference your question depends on leaves the next inspector with the same surface and two possible pasts.

For one known question repeated along a run, the receipt can be very small. Start a bit at yes and, at each check, replace it by the old answer AND the new answer. It says exactly whether all checks so far passed. A later failure turns it down. The permanent flag that says "some certification occurred" stays up. Put those gauges beside each other and you can see why they answer different questions.

*When the stamp merges two states*

Draw two boxes with the same tiles: Unstamped and Stamped. Pressing the stamp takes the first to Stamped. Pressing it on the second leaves that whole box exactly where it is. Draw both arrows. They land in the same place.

Call the press $V$. At a state $s$ where it raises the record, $s$ and $V(s)$ are different. If another press leaves the full state fixed, then $V(V(s)) = V(s)$. Those two different states have the same successor.

Give the stamped box a hidden wheel that turns on every press. Now the picture has more states in it. If every state reached after stamping still reads yes, and there

are only finitely many such states, a merge is forced somewhere along the continued pressing. It can be later than the first press. Keep the entire move injective instead, and the run has to keep finding distinct stamped states. Call that finite accepting region $H$ and the entering state $u$. The move sends all of $H \cup \{u\}$ into $H$. There are $|H| + 1$ distinct sources and only $|H|$ possible targets, so two sources share a successor. The move preserves the working picture, so both sources and their common successor lie in that same fibre.

An internal ledger belongs in that full-state picture too. If each press increments it, two presses have different ledger values even when the stamp looks unchanged. Count those states with the others. The toll tells me what to charge for a raise; the next chapter gives the premises that connect charges to mergers. I want the arrows and the bill on the same page, with the rule connecting them where you can see it.

Notice also that it asks nothing about how strong the system is, in either direction. A system is *Turing-complete* when it can imitate a Turing machine, and no stronger than one when a Turing machine can imitate it. The theorem asks neither question. Whether the system is stronger than a Turing machine, weaker, or exactly as strong makes no difference to the floor. Bring the five parts, charge the toll on the crossing, and you're in scope. The floor doesn't check what you can compute, because it was never about what you can compute; it's about what a move is allowed to do without costing anything.

That brings up the objection a theorist makes in about four seconds, so I'll make it first. The toll is a floor, and a floor is easy to meet. Charge one unit for every move, whatever it does, and the toll holds for any reading you like, since every crossing is a move and every move costs one. So any computer with a step counter beside it, and any yes-or-no question at all about its state, is a Thiele machine, and a clock passes. Nothing about that is a loophole; it's what the definition says, and it means the definition, on its own, doesn't tie the cost to anything kept or certified. It only says a crossing can't go unpriced. (It's also why "a base that can compute anything plus a flag that obeys the toll" would be a cheap thing to call Thiele-complete, and why the definition that gets that name, at the end of Part II, asks for computing that costs nothing and a record that tells starts apart, which no clock meets; Section 10.8.) Three things later in the book tie the cost down. The first is that bar: computing that costs nothing, a charge of one, and a record that tells starts apart. The second is a ceiling: demand that a run's total never exceed the number of times the flag went up. With a price that can depend on where the step starts, that and the floor leave exactly one way of charging, one unit per raise and nothing else (Sections 6 and 7). With a price per move name, as in my definition, no system whose flag ever rises meets both, because the raising move, played again from yes, which every move can be, raises nothing and still costs one. A clock fails the ceiling the first time it charges a step that raised nothing. The third is order, on any certified machine, and

the example of Part III is one: there, from a clean start, the flag can't go up unless a passing check of a claim came first, then a commitment to that same claim, and a clock can't fake that, because it's about which moves happened, and a clock only charges (Theorem 15.11).

There is also the matter of why the ledger doesn't charge for heat. A physical ledger would need the actual dynamics, distribution and protocol. A ledger that simply charged every nontrivial move would charge by that rule, and that's the clock again: it can't tell a run that kept something from a run that only ran. So the ledger leaves the dissipation of happening to physics and counts one thing, the rise of a reading that stays. A zero abstract charge says nothing about physical dissipation; it just prices no crossing.

The other side of that is that nothing is in scope just for being a computation. A Turing machine's run isn't an instance of this theorem because it's a run. It becomes one when somebody supplies all five parts for it, says which states read yes, and proves the toll for that cost. Until then the theorem says nothing about it, and I won't pretend it does; that limit is the toll's scope. What keeping takes has no scope to supply, because every run of anything with one next state per state takes a fresh state at every step, joins exactly one pair of its states into one, or goes round a loop through its start and keeps nothing (Theorem 1.1). The reading *this isn't the state I started in* rises at the first step and stays up exactly on the runs that never come back to their start, the first two kinds, and those pay, in states never used before or in their one merge.

That's also where the hunch this book started from ended up, once I checked it: the night a video about *Flatland* convinced me there was another axis, a direction ordinary computing is blind to (the Afterword, Section 24, tells it). It isn't a direction an ordinary computer can't reach. Hand a Turing machine a reading, charge the toll on its crossing, and it's in scope like anything else. The possibility was always there. What the usual rules leave out is the price.

The same goes for a system with no reading at all. If nothing in it answers "is the flag up," then nothing in it can say, from the inside, that anything is entitled to be relied on. Whatever entitlement it ends up with gets carried in from outside, by somebody who signs their name to it, and nothing in the system priced it. Smuggled entitlement isn't entitlement for the system, no matter whose name is on it. Be careful which way that cuts, though. Whoever reads the signature can be perfectly entitled on their own account; that's what an auditor has, and it's honest as long as it's called the auditor's. It isn't the system's. On the example I build later, the reason to believe its checker says yes only when the claim is true is exactly that kind: a proof, done outside any record the machine keeps, and I'll say so when I get there (Section 15). What outside entitlement never is, is something the system itself earned.

There's also a question I'll only start on here: bolt two Thiele machines together, and do you get one? Run them side by side, each move being a move of one or the other, and add up the costs. It depends on the reading you give the pair. Take a reading that can only go up when one of the parts' readings goes up, like "either flag is up" or "both are": then any crossing of the pair is a crossing of a part, that part's toll charged it, and the pair obeys the toll. Take "the first flag is up and the second is down," and it doesn't: that reading turns on when the second flag falls, and nothing charges a fall. The first two readings are proved in Appendix E: read through "either flag is up" the pair is a Thiele machine whose floor is one. Read through "both are," a run that starts with both flags down and ends with both up costs at least two, and two is reached when each part can raise its own flag for one. From a start with one flag already up, one raise is enough, so two is a floor for the clean start only. The third needs nothing beyond the one unpriced fall above. A related composite is proved too, a host machine running a guest, read through "the host's own flag or its copy of the guest's" (Theorem C.13). Whether every way of building one machine out of others keeps the toll is a bigger question, and Appendix E takes it as far as I could: some ways do, some provably don't, and it says which.

So why bother with the toy at all? It isn't because the lineage was looking for it. The bulla was after durability, the seven seals after evidence, the consul's stamp after authenticity, the two columns after catching slips; none of them was trying to put a price on the moment a flag goes up, and I shouldn't pretend they were. The reason is narrower. Everywhere in that lineage where somebody *did* locate a cost, but one, it sat on a step next to the flag. The flag itself went unpriced as a flag: Szilard on the measurement, Bennett on the wiping, proof-of-work on the search, Boyle on the instrument and the room, Foster and Stine on the test. One entry prices the crossing's own shape, and I put it beside the toll on purpose: Landauer's restore to one, the literal two-state version of the crossing, whose heat bound depends on the input uncertainty and the thermal premises. The flag's job never came into it. The toy is the first thing I could build where the cost sits on the crossing itself, and it proves that if you price that step, the floor follows everywhere, for any machine with the five parts. That's not an answer; it's the question made exact enough to be wrong. And of the five jobs, the toy supplies only a limited kind of retention: its three permitted moves never lower the flag. The abstract definition allows other machines with a reset. Adding that reset would change this toy, and the cost that doesn't go away is kept by *you*, the scorekeeper, beside the board. The machine keeps none of it. What the toy has is a scorekeeper's receipt for the crossing. Whether the machine can be made to keep that receipt itself, where a step could reach in and change it, is Section 6's question. Authenticity, evidence, accounting, cheap checking: none of them is in here, and I'll say so again when it's tempting to forget.

Let me say what I'll owe you by the end. Whether the toll is a choice or a necessity: the next section, and I'll tell you now that the answer surprised me. What a unit of cost is worth in anything physical: later, and only partly. What an observer who sees only a window can and cannot recover, including whether they can even tell which step paid: its own section. And whether the definition can be met by a machine that computes everything a Turing machine does, with its record earned and not declared, and what it costs to vouch for structure in one: Part III shows that with an example, and takes most of its length on it.

For now you have a sheet of paper with two boxes, three moves, one rule that writes what the last chapter says keeping takes into the step, and one theorem you proved yourself in about a minute. Don't let go of it. It's the smallest version of everything that follows, and if it's wrong here, it's wrong everywhere.

CHAPTER 3

# Why the toll isn’t arbitrary

The first chapter of this part proved what keeping takes by counting states: a run of one rule that raises a flag and keeps it up either lands two of its states on one somewhere or never uses a state twice, so keeping is paid in a merge or by taking states never used before; and a thermal implementation can incur additional dissipation under the specified dynamics. The merge-or-fresh-state part needs no premise beyond one next state for every state; the heat is physics, which I proved for Bennett’s walker with Schnakenberg’s formula taken as the model, and cited for devices whose states jump or drift at random in surroundings at a temperature. This chapter counts the fresh states exactly, and asks what it takes to turn what keeping pays into a unit on a ledger. The flag I built is one thing that stays; the result is about anything that stays.

The last chapter wrote what keeping takes into a toy as one rule, and in the toy the move that raises the flag costs one because I said so. If that’s all it is, the ledger is a long way of describing a preference, and you should treat it the way you’d treat someone telling you their favourite colour is a law of nature. So this chapter is me trying to find out whether a machine has to charge it. I’ll tell you the shape of the answer up front, because I don’t want you reading forty paragraphs to find out whether there’s a point. What keeping takes needs no “if” beyond a next state for every state: on any machine, keeping something is paid in a merge or in states never used before, and physical dissipation is a separate, model-dependent question; that’s the paragraph above. Getting from what keeping takes to my rule, one unit on the ledger for the step that raises the flag, takes three “ifs”: there are finitely many states that read yes, so the fresh states run out; the flag, once raised, stays up; and the ledger charges every move that forgets. Then the move that raises the flag has to forget something, two of the states it acts on landing on one, and pays. Forgetting is also the step Landauer’s principle, my premise, puts a heat price on, when what’s forgotten wasn’t already known; a known preparation alone supplies no positive entropy-loss floor. If the input is correlated with another memory, that larger physical account has to be specified too. Each of the three “ifs” is a way out of the theorem, and I’m going to walk through each one, because what’s on the other

side of each is where the content is. None of them is a way out of what keeping takes. Where the first fails, the machine pays by taking states it has never used. Where the second fails, the flag comes back down, so nothing is kept and there's nothing to charge. Where the third fails, the step still forgets, and only the ledger looks away.

One more thing before the three ways out, and it comes first on purpose. Nothing in the argument uses the word certify. It uses a reading that can go from no to yes and never back. A fuse that has blown is one. So is a seal that has been broken, a bit in write-once memory, a block a chain has finalized, a measurement somebody wrote down and won't take back. The argument below sends what keeping takes to whichever of them you point at, and the latch chapter (Section 10) proves it doesn't matter which one you pick. Certification is the instance I build, for reasons I give in Section 18 and don't claim to have proved. So read this section as being about anything that stays. The flag is only the one I keep on the table.

### 3.1 A closed room with no spare chair

Do this one with real objects, because the whole section rests on it and it's about ninety seconds of work. Draw a box on the table and call it the yes room. Put three chairs inside the box and one chair outside, by the door. Four people, one per chair; the people are the states your machine could be in, and the three inside are the ones whose flag is already up. One move of the machine moves everybody at once: each person stands up and goes to the chair the rule sends them to. There are two rules. First, nobody inside the room may end up outside, because the flag, once up, stays up. Second, the person by the door must end up inside, because this move is the one that raises their flag.

Now try to finish the move with one person in each chair.

You can't. Three people start inside and have to stay inside, so they take the three inside chairs between them, with nobody left out, because nobody may leave. The person from the door has to come in, there is no fourth chair inside, so somebody has to share a chair, and two of the machine's possible states have landed on one state. Now use the chairs as a guessing game: secretly choose one starting chair, apply the rule, and report only the finishing chair. If two starts are sent there, the report cannot tell your partner which start you chose. Watching the people move would tell them, but that would be keeping information outside the finishing chair, and the chair alone has forgotten something.

The whole argument is in that shared chair, and I want to name the piece of mathematics that did the work, because it's old and it's not mine. If you put things into fewer boxes than there are things, some box gets two. People call it the pigeonhole principle; it's the one from "The point," where Jean Leurechon used

it in 1622 to argue that two people somewhere have the same number of hairs, and the first name for it I could find came from Dirichlet, who used it in 1842 and later called it the drawer principle. It's the kind of fact a child accepts and a mathematician spends a career finding new uses for. Here the use is that a step which forgets nothing is a one-to-one shuffle, a one-to-one shuffle of a finite set of chairs lands on every chair, so the inside chairs are all taken by insiders and none is left for the newcomer. Finiteness, permanence and the flip are the three things that force a collision in the instruction's map. The two starts that collide need not include the outsider: send the outsider to one inside chair and send two insiders to another. A run from a known outside chair can still have a uniquely identifiable start. The collision is a fact about the instruction over possible states.

The precise statement of the argument is a theorem a few lines down, and it needs one word first. "Injective" is the mathematician's word for "forgets nothing": a step is injective when two different starts always give two different ends. The theorem says exactly what the people and the chairs said and takes exactly the three premises: a duplicate-free list that names every state (finite), a reading that no step ever turns off (permanent), and one state that reads NO before the step and YES after it (the flip). There is no price in it anywhere; it is pure counting. I'll come back to what counts as physics, but notice that nothing so far has been physics.

The exact words follow. For this whole section the setting is the five parts from Section 2 without the toll: states $S$, move names $I$, a step, a reading, and (later) a cost.

**Definition 3.1** (Finite, permanent, injective)**.** The machine is *finite* when there is a list of states with no repeats that contains every state. The reading is *permanent* when no move ever turns it from yes back to no: for all $s$ and $i$, $\mathrm{cert}(s) = \text{yes}$ implies $\mathrm{cert}(\mathrm{step}(s,i)) = \text{yes}$. A move $i$ is *injective* (forgets nothing) when different starts give different ends: $\mathrm{step}(a,i) = \mathrm{step}(b,i)$ implies $a = b$. A move that is not injective *merges*.

□ **Theorem 3.1** (A permanent flip merges)**.** On a finite machine with a permanent reading, if $\mathrm{cert}(s) = \text{no}$ and $\mathrm{cert}(\mathrm{step}(s,i)) = \text{yes}$, then the move $i$ merges.

*Proof.* Suppose $i$ were injective. Let $C$ be the list of states that read yes: go down the list of every state and keep the ones that read yes. It has no repeats. (That is the only use of finiteness: all the proof needs is a duplicate-free list of the yes-states.) By permanence, used only for the move $i$, $\mathrm{step}(\cdot, i)$ sends every member of $C$ into $C$. Being injective, it sends the members of $C$ to members of $C$ with no two landing on the same one, so its images are as many as $C$ itself, and a list of that many distinct

members of $C$ is all of $C$. So the yes-state $\mathrm{step}(s, i)$ equals $\mathrm{step}(c, i)$ for some $c$ in $C$, and injectivity gives $s = c$, so $s$ reads yes. It reads no, and that is the contradiction, the same one the four people and the three inside chairs ran into. ∎

A list with no repeats, as long as the set it draws from, contains the whole set. Leave one member of the set out and you have packed the list into one place fewer than it has members. That can't be done: take away one occupied place and the member sitting in it, and the list is still one too long for the places left, and it stays one too long until there are no places left at all. This is the pigeonhole in the form Leurechon would recognise.

None of that is new. It has been sitting in two places for a long time, one in physics and one in the theory of automata, and a physicist will recognise it on sight, so I'll start there. In 1876 Josef Loschmidt pointed out, against Ludwig Boltzmann's account of how a gas settles down, that the laws of mechanics run just as well backwards: reverse every molecule's velocity and the gas retraces its steps, so mechanics alone can't make entropy only ever climb. Twenty years on, in 1896, a young assistant of Max Planck's in Berlin pushed harder. He took Henri Poincaré's recurrence theorem of 1890, which says that in a system preserving a finite measure of its accessible states, almost every starting state returns arbitrarily close to itself, again and again, and used it to argue that nothing in such a system can climb forever: give it time and it comes back down. The assistant was Ernst Zermelo, the same Zermelo who'd put set theory on axioms twelve years later, axioms that keep out the paradox that had sunk Frege's foundations of arithmetic (Section 5 tells that one). Boltzmann answered the same year, not very politely, that the times you'd have to wait for a return are so enormous that nobody would ever see one. (Jos Uffink, who wrote the Stanford Encyclopedia's entry on Boltzmann's statistical physics, calls that reply drenched in sarcasm and bitterness, and calls Zermelo's own paper a careful logical argument.)

Now look back at the table. A step that forgets nothing, on a finite set, is a reshuffle: it moves everybody around in loops, and every loop comes back to where it started. A flag that goes up and stays up is exactly a quantity that climbs and never comes back down. So a reshuffle can't carry one, which is Zermelo's objection with the infinities taken out. On a finite set the loops are enough, and Poincaré's theorem isn't needed. And the wait is at most as many steps as there are states, nothing enormous. Zermelo was aiming it at the second law, and flags never came into it. A quantum recurrence result came in 1957, when P. Bocchieri and A. Loinger proved returns arbitrarily close to the initial state under their conditions on a closed quantum system. That is a neighbouring recurrence argument, not this finite-state proof: even a finite-dimensional quantum system has continuously many states.

Applying it to a physical record requires saying how that record is read and how robustly it distinguishes nearby states.

The second place stings more, because it's in computer science, and somebody got here first, again. Picture a finite machine that reads letters, one move per letter. If every letter just reshuffles the states one for one, automata theorists call it a permutation automaton, and the sets of inputs such machines can pick out are called group languages. Jean-Éric Pin's 1992 paper on reversible automata, machines whose letters act as one-to-one maps, sorts out exactly which languages those machines recognise, and in that world this counting argument is folklore. The short version runs like this. Start at a state that reads no, and collect every input that ends at a yes. Because yes is permanent, adding letters to the end of a collected input keeps it collected. A group language with that property is either empty or everything, and everything would include the empty input, which ends where it started, at no. So it's empty, no run of reshuffles raises the reading, and that's Theorem 3.1 with the moves as letters. Pin's paper is where I'd send you to check the automata side. It isn't about flags or prices, and I'm not claiming it says anything about them.

There is a broader way to look at the same obstruction. Suppose every allowed move can be undone by another allowed sequence of moves. Start outside YES and try to enter it. Once inside, permanence says every allowed sequence keeps you there. But the undoing sequence must take you back outside. Those two demands cannot both hold.

On a finite set, a move that keeps all states apart is a reshuffle. Repeat that reshuffle enough times and every state comes home; all but the last repetition therefore undo one application. So the undoing sequence needed in that argument is available. Morgan Rogers studies systems of composable moves in his 2019 and 2021 papers on toposes of monoid actions. That more general language gives this fact another home. Here the chairs and the undoing sequence already show why it holds.

Going quantum doesn't get around the automata version either. Alex Brodsky and Nicholas Pippenger (2002) showed that a quantum finite automaton that measures only once, at the very end, accepts with bounded error exactly the group languages, the reshuffle kind. One that's allowed to measure after every letter, and to stop on what it sees, does better: it accepts every piecewise testable language, and "has seen a $b$ somewhere" is one of those. A measurement that can stop the machine is a record that stays, and it's the extra thing that lets the bigger class in.

The third place is the closest, and it isn't old at all. In 2023 Thomas Ouldridge and David Wolpert derived the thermodynamics of a finite automaton built the way real computers are: the same physical step run again and again, each time touching only the part of the machine that changes. They sorted languages into two kinds by

the smallest automaton that recognises each one. If that automaton has a letter that sends two states to one, every automaton for the language has such a letter too, and on their derivation recognising the language on a device like that produces entropy, waste no cleverer design gets back, unless the inputs are so finely arranged that only one of the merged states ever actually turns up. If the smallest automaton only reshuffles, they find the language can be recognised with no unavoidable entropy production at all. One of their worked examples is the strings with no $b$ in them, which is my flag turned inside out: the runs that raise it are exactly the strings that have seen a $b$. So the counting argument, a real price on it, and the case when you already know which state you're in are all in their paper, three years before this book. Their price is derived from the physics of the device, and mine is assumed.

There's a fourth neighbour, and I'd call it a cousin, because it speaks another language. Constructor theory is David Deutsch and Chiara Marletto's programme for stating physics as which transformations a device that runs in a cycle can and can't bring about, and in it a task is irreversible when it's possible and its inverse isn't. Marletto's version of thermodynamics (posted in 2016) leans on exactly that: tasks that are possible while their transposes aren't. Marletto, the physicist Vlatko Vedral and coauthors (2022) made an irreversible task out of the quantum homogeniser, a machine of reservoir qubits that interact by partial swaps, and ran it with photons. Maria Violaris and Marletto (2022) then argued that the same machine, run in a cycle, can turn a pure qubit into a mixed one and can never turn a mixed qubit back into a pure one, even though every interaction in it is reversible, and that this suggests erasure may cost something beyond Landauer's bound. The counting argument is a classical, finite, exact cousin of that sentence: a set of states a step can't leave, a task that's possible, and no way back. Theirs is quantum and approximate and aimed at physics' price; mine is counted, and assumes its price. Deutsch and Marletto's paper on information (2015) has a neighbour of the counting argument in its section 8.6: measuring one observable disturbs another irreducibly, and nothing can undo the disturbance afterwards while the outcome stays in an information variable. It's a second, theory-independent way of saying that a record that stays makes the step irreversible.

So what I can claim here is narrower again, and I'd rather say exactly how. Even the pricing form is, in physics and on their derivation, Ouldridge and Wolpert's: a merge the language can't avoid costs entropy. A price assumed rather than derived isn't mine either: Li and Vitányi (1996) took a unit charge for every bit irreversibly erased, with steps that can be undone priced at zero, as axioms of a model of computation, and Demaine and his coauthors (2016) put a price on named instructions that destroy bits, a unit for an AND or an overwrite. What I didn't find anywhere I looked is what a price on merging moves forces on a reading that stays, charged on a ledger whether or not anybody's physics is right (put a price on

merging and the flip has to pay it, which is the next subsection), and what I get out of that: the toll, as a theorem about the ledger, the merge-or-revoke split for a single move, the exact match between "forced to be priced" and "merges", the squeeze bound counted in whole units and its entropy form, a built machine walking out through each of the three ways out, and every proof on the page. Those are the forms I didn't find stated anywhere I looked, and Appendix B sorts them the same way: the counting core classical and somebody else's, the pricing form only partly mine. The core move is older than the computer, it had two homes before it got to mine, and its physical price had a third.

*Count the alternatives before putting a price on them*

Write $N$ different starting configurations on cards and spread them across the bench. Run each through the same rule once. When several cards give the same full configuration, put them in one pile. Run the rule again and gather the new piles. Once cards agree on the full state, the same rule gives them the same next state too.

Count the different configurations after each tick. Call their set $Q_t$ and their number $q_t$, with $q_0 = N$ and $Q_{t+1} = F(Q_t)$. The difference $q_t - q_{t+1}$ tells you how many alternatives disappeared on that tick. Four distinct configurations landing at one state lose three alternatives in one landing pile. Keep those two counts beside each other: three alternatives lost, one landing place.

□ **Theorem 3.2** (The alternatives balance exactly)**.** For a total deterministic rule $F$ and $N$ distinct starts,

$$\sum_{t=0}^{T-1} (q_t - q_{t+1}) + q_T = N.$$

If all endpoints fit in $K < N$ full states, a transition at some $t < T$ identifies two different members of $Q_t$. If all endpoints are one full state, the total loss is $N - 1$.

*Proof.* An image has no more elements than its source, so every difference is non-negative. Add the differences: each intermediate $q_t$ cancels, leaving $q_0 - q_T$. If $q_T < N$, at least one difference is positive; at that step a map from more states to fewer cannot be one-to-one. In particular, one final full state leaves a loss of $N - 1$ alternatives. ■

Turn the cards over before counting them together. Two runs might show the same answer on the front and still carry different inputs on the back. Those are different full configurations. The count follows everything the machine kept.

Now put a price rule beside the piles. Let $p_t$ be the price at tick $t$, and require every landing pile to have at most $2^{p_t}$ distinct sources. The number arriving is

then at most $2^{p_t}$ times the number of places they land: $q_t \leq 2^{p_t} q_{t+1}$. Multiply that inequality through the run and you get

$$N \leq 2^{\sum_{t<T} p_t} q_T.$$

The rule connects the bill to how tightly the machine can pack the alternatives.

## 3.2 Now charge for forgetting, and the toll falls out

This is the step I'd been circling since the counting on states. In the last chapter I told you Bennett's rule as he states it: what has to cost is erasing a bit or merging two paths a computation could have taken. I didn't ask you to believe it. I'm not going to ask you now either. I'm going to write it down as a rule you can accept or refuse, in the smallest form that does any work: *any move that forgets something costs at least one.* Any move that is not injective has cost at least one. Notice what it does not say: nothing about certifying, nothing about flags. It's a price on forgetting and only on forgetting.

Put the two together. The first theorem says: on a finite machine with a permanent flag, the move that raises the flag forgets something. The price says: a move that forgets something costs at least one. So the move that raises the flag costs at least one. That is the toll, the rule the toy was handed because I said so, and now a machine has to charge it. The proof is two lines: the first theorem says the flipping move merges, the premise says merges are priced, done. I'll keep saying "premise," because that's what it is, and because a section further on (Section 4) is about what happened to some very smart people who put a price where a premise goes and then treated it as a law of nature, and about half of this one is me trying not to do that.

**Premise 3.2** (Merge pricing)**.** Every move that merges costs at least one: for all $i$, if $i$ is not injective then $\mathrm{cost}(i) \geq 1$.

**Observation 3.3** (The toll from finiteness, permanence, and merge pricing)**.** *On a finite machine with a permanent reading whose cost satisfies merge pricing, every move that turns the reading from no to yes costs at least one. That is, the machine with that cost is a Thiele machine in the sense of Definition 2.2, and the universal floor (Observation 2.1) applies to it.*

*Proof.* A move that flips the reading merges, by Theorem 3.1; merge pricing charges it at least one. ∎

The theorem asks for more than it uses. Only the yes-states need to be finitely many: with a permanent reading and finitely many states reading yes, the rest of the machine as large as you like, merge pricing still gives the toll. With infinitely many it can fail: the first way out below builds a machine that raises a permanent

flag at zero cost while charging every merge. That machine hasn't got out of paying. It pays by taking a state it has never used, and a price list that charges only merges never looks at the fresh states it takes. And merge pricing asks for more than the toll needs. Under those same two conditions the toll holds exactly when every move that flips the reading somewhere and merges is priced. Merge pricing gives that, and the eight-state machine described later in this section, with its jump repriced at zero, meets the toll without merge pricing.

That "more" has an exact size. On a finite machine a move merges exactly when it turns on some reading that it never turns off. If it merges, take the reading "this move can land here." The move never turns it off, because everywhere it lands reads yes; and some state reads no, because a move that sends two states to one lands on fewer states than the machine has, and from that state the move turns the reading on. If it doesn't merge, it turns on no reading it never turns off: that's Theorem 3.1, whose proof asks permanence only of the one move. So on a finite machine a price list meets merge pricing exactly when it meets the toll for every reading at once, each move answering only for the readings it never turns off. The charge on forgetting isn't a second kind of price beside the toll. What the premise chooses is which readings the toll answers to: the flag alone, or every reading a move keeps.

There is one more way to see the same thing, and I give it because it answers a rival I'd otherwise owe you. The rival says: leave certification alone. Price *erasure*, and keep an honest flag that reports when something was erased. Fine; but a system can set its certificate at cost zero with its erasure flag reporting nothing, exactly because the certifying move merged two states and the flag didn't say so. Make the erasure flag honest, so that it reports every move that forgets, and on a finite machine with a flag that stays up the erasure law charges the same cost the toll does. The toll isn't a rival to honest erasure accounting; it's what honest erasure accounting says about the certifying move.

So, is the toll arbitrary? The honest answer is that the toll is a consequence of three things: finitely many states that read yes (a finite memory is plenty), a flag that stays up, and a charge on forgetting. If you accept all three, you don't get to refuse the toll; it's forced. If you refuse any one of them, the toll on my ledger goes back to being my preference. The cost stays where it was; what's gone is the proof that my ledger has to charge it. So the question "did I have to pick it" has turned into three smaller questions, which is what progress usually looks like: a better set of things to argue about. Let me take the three ways out in order, and for each one build the machine that leaves through it, because a premise you can't see the counterexample to is a premise you're taking on faith.

### 3.3 The first way out: keep everything, and pay by taking fresh states

Take a strip of paper that never runs out of boxes. The machine's whole state is the strip. Each move writes a 1 in the next empty box, and the flag is up as soon as any box has a 1 in it. Check the three premises. It is permanent: once there's a 1, there's always a 1. It flips: the empty strip reads NO, and after one move it reads YES. It forgets nothing: every move can be undone by erasing the last 1, so two different strips never become the same strip. So this is a machine that raises a flag, keeps it up, and never merges anything. The theorem that a permanent flip on a finite machine merges has no grip on it, and a price list that charges only merges doesn't charge it. But look at what each move does: it writes in a box nobody has written in before. The raise takes a fresh box, and so does every step after it, forever. It hasn't got out of paying. It's paying in the other column, by taking a state it has never used, and a price list that only charges merges isn't looking at that column. (Each physical write may incur a model-dependent finite-time cost; what the strip dodges is the floor a merge would add on top.) In the exact form its state is a row of 1s of any length, which is how you say the strip has no end.

If the endless strip feels like a cheat, it's a famous one. David Hilbert, in a lecture course in the winter of 1924–25, told a story about a hotel with infinitely many rooms, every one of them taken, where a new guest still gets a room: everybody moves from room $n$ to room $n+1$, and room 1 is left empty. The hotel is that other column, the fresh states, told as a joke about a front desk: nobody shares, and every guest in the building packs up and moves one room along, into a room that's occupied as he moves, by a guest who's moving on at the same moment, so that one room at the front is left empty. The guests go one-to-one into a part of the hotel, with a gap left over. The part worth checking is that Hilbert never published it. The hotel became famous because George Gamow put it in a popular book in 1947, and for a long time it wasn't clear Hilbert had told it at all. The historian of science Helge Kragh first guessed that Gamow had made it up. Then Tilman Sauer, a historian of science who had co-edited a volume of Hilbert's lectures on the foundations of physics, pointed him to Hilbert's lecture notes, which had just been printed in 2013, and in 2014 Kragh took his guess back in a paper called "The True (?) Story of Hilbert's Infinite Hotel." Keep the question mark. Even Kragh's own paper puts the lecture in January 1924 in one place and January 1925 in another. Still, this time a story everybody repeated turned out to have a real lecture behind it. I've learned not to count on that happening. Bring it back to the table. In the closed room, the person from the door had nowhere to sit because the chairs ran out. In Hilbert's hotel they never do, so everybody shifts one over and nobody has to share. The strip machine is that hotel. The argument stays the same. What the hotel shows you is that the whole argument was about running out of states all along.

What walked out is the finiteness premise. Now cut the strip to eight boxes. Keep the same rule: fill the next blank box from the left, leaving every earlier mark alone. There are nine possible strips, including the blank one. After eight moves the strip is full. What should the ninth move do? If a full strip stays full, the seven-mark strip and the full strip both go to the full strip. That is the collision. If the machine stops instead, its next-step rule is partial. If it clears the strip, its YES reading comes down. Each escape changes a premise.

| Marks so far | Strip | Next strip under the stay-full rule |
|---|---|---|
| 0 | 00000000 | 10000000 |
| 1 | 10000000 | 11000000 |
| ⋮ | ⋮ | ⋮ |
| 7 | 11111110 | 11111111 |
| 8 | 11111111 | 11111111 |

Eight unrestricted bits have 256 configurations. These nine are the ones this particular prefix-filling rule permits. A fresh configuration need not use a fresh physical box; changing a pattern can reach one too. Keeping the whole history is exactly the move Bennett made in 1973, when he showed that any computation can be done without forgetting if you let it keep a copy of everything it did; Landauer, back in 1961, had already written that for a machine able to run a program that doesn't stop, the space to hold every intermediate result "cannot be there." Keeping the history is a real way to pay. It costs a memory that never fills, and every machine you can buy has a memory that fills.

Bennett wasn't the first there, or the last. Yves Lecerf, in a short note to the French Academy of Sciences in 1963, had already shown that reversible Turing machines can simulate arbitrary ones, and the note went largely unnoticed. The computer scientist Tommaso Toffoli (1980), and Edward Fredkin and Toffoli together (1982), both then at MIT, built reversible logic: gates and circuits that compute any function and lose nothing, by carrying constant inputs in and extra outputs out. Their basic gate wasn't new even then. Landauer's own 1961 paper writes down a three-bit device that replaces $r$ by $p \cdot q$ when $r$ is 0 and by the negation of $p \cdot q$ when $r$ is 1, calls it logically reversible and capable of an AND, and says each AND then needs its own supplied ZERO and leaves its extra outputs behind. He doubted such a machine could be useful. What Toffoli added was the proof that it computes everything. And the computer scientist Kenichi Morita (1996) proved that reversible two-counter machines are universal, so even a machine as small as the one Section 15 builds, handed its input encoded, can compute everything with every move one-to-one. Every one of them keeps what it doesn't forget somewhere: a history, garbage outputs, a counter that codes the past. The strip is one of those: it keeps the whole history, box by box.

There are three separate claims here, and I need to keep them apart. The strip machine is permanent and forgets nothing. It also has infinitely many states: no finite list holds every finite strip. Neither of those involves a cost. The third does: the strip can raise its reading at zero cost. Count the strip's length as a whole number and let each move add one at a price of zero. Nothing merges, merge pricing holds, and the toll fails. This is what a price list that charges merges and nothing else looks like on a machine with endlessly many states to take: the raised reading shows up at no charge on the ledger because the column it's paid in is one the price list doesn't charge.

The same thing holds for one move instead of a run, and in that form it says exactly how many states the move needs.

**Proposition 3.4** (A raise with no merge takes endlessly many states)**.** Let a move never turn a reading from yes to no, let it raise the reading at some state $s$ (no at $s$, yes where the move sends $s$), and let it send no two different states to the same state. Then it sends the states that read yes one-to-one into a part of themselves that leaves at least one of them out: the state the move sends $s$ to. So there are infinitely many states that read yes. With finitely many, the raising move merges two states. □

*Proof.* The move sends every yes-state to a yes-state, because it never turns yes to no, and it sends different states to different states. It sends $s$ to a yes-state. If some yes-state $c$ were also sent there, the move would send $s$ and $c$ to one state, so $s = c$; but $s$ reads no and $c$ reads yes. So the yes-states are sent one-to-one into the yes-states with that one left out. If there were only $n$ yes-states, that would put $n$ things one-to-one into $n - 1$ places, which the pigeonhole forbids. So there are infinitely many. ■

The proposition is Hilbert's front desk as a theorem, and it's also exactly how Richard Dedekind defined an infinite collection in 1888: one that can be matched one-to-one with a part of itself that leaves something out. So with no merge, every raise has to send the kept things one-to-one into a part of themselves with a gap left over, and only endlessly many of them have a gap to spare. Taking a fresh state isn't the absence of what keeping takes but the other way of paying it, and only a machine with endlessly many states can keep paying that way.

*Which state is being counted.* Whether a move merges depends on what the state keeps. The strip keeps its history, so nothing merges; cut it to eight boxes and the stay-full rule merges. Section 19 proves the general form (Theorem 19.3): keep the list of every earlier state alongside the current one and every fixed-instruction step is one-to-one. (Keeping only the one state before isn't enough: two states that differ only in what came before them still land together.) So a merge price is a fact about

a move on a stated state space. Each price derived in this book is derived on the whole state its definition gives, and three things go with it: which state that is, what lies outside it (nothing outside it is charged), and whether the move is one primitive instruction or a composite run of several. Relabelling the states one-to-one changes no price, since it sends merges to merges and one-to-one moves to one-to-one moves. Adding history to the state, or projecting part of the state away, changes the machine being modelled, and the price can change with it.

The first way out deserves a number, because "a memory that never fills" is doing all the work and I'd rather see what keeping takes than take it on trust.

□ **Proposition 3.5** (Merge, or a fresh state every step)**.** (1) Let a machine run by one rule, so that every state has exactly one next state, and let its reading never go from yes back to no. Take any run in which the reading goes from no to yes. Then either some step of the run merges two of its states (two different states of the run have the same next state), or the run never visits the same state twice. So on a machine with $M$ states, a run that keeps a record and never merges has fewer than $M$ steps. (2) Let an operation send every value to one fixed value and leave that value alone. Take a machine with $M$ states run by one rule, a reading of its states, and $N$ different starting states, exactly one of which already reads the fixed value, such that from every start each of the next $T$ states reads the fixed value. If no two different states among those the runs sit in at steps 0 to $T-1$ have the same next state (the operation is carried out through $T$ repetitions without merging), then $M \geq 1 + (T+1)(N-1)$. When $N \geq 2$, a bounded memory delays the merge and does not avoid it; with only the resting start there need be no merge.

*Proof.* (1) Suppose no two different states of the run share a next state, and suppose the run still visits some state twice, once early and once later. Walk the two visits back together, one step at a time: two equal states whose predecessors are both in the run had equal predecessors, because nothing merged. The early visit gets back to the start first, and the later visit is then at the start too. So the run comes back to its start after some number $p$ of steps, and since one rule gives the next state, it comes back again after $2p$, $3p$, and so on forever. The start reads no: a yes there would stay yes all along, and the reading does go from no to yes. Some of those returns come after the raise, where the reading is yes and stays yes, and each of them is the start, which reads no. So the repeat was impossible. A run that never merges takes a state it hasn't used at every step, and with $M$ states that's at most $M-1$ steps. (2) Count the states along paths. Walking two equal states back together, as in (1), is allowed at any step from 1 to $T$, because the states one step earlier are at steps 0 to $T-1$. Each of the $N-1$ starts not already in the fixed reading, followed through

$T$ repetitions, occupies $T+1$ states, and none twice: a repeat would walk back to the start equalling a later state of its own path, and every later state reads the fixed value while the start does not. Two of those paths cannot share a state either: walk back to the earlier time, and either two different starts are one state, or a start is a state that reads the fixed value. That is $(T+1)(N-1)$ states. The start that already reads the fixed value, followed through $T$ repetitions, ends in one more: if that state were on one of the paths, walking back would again put a start at a state that reads the fixed value. One more is all that start is owed, not $T+1$: every state on its own path reads the fixed value, so walking back along it finds no contradiction, and the path may fold onto one state, as it does in the machine built below. ■

The number is exact, not only a floor, and you can build the machine that reaches it. One state for the start that already reads the fixed way; the operation leaves it there. For every other start, lay out a row of states, one state longer than the number of repetitions you want to survive. The first state in the row reads the way that start reads, the rest read the fixed way, and the operation walks the row, sending its last state to the common fixed state. This also works for zero requested repetitions, when a row contains only its start. Count the states: one, plus a row for each of the other starts. For as many repetitions as you asked for, no two distinct visited source states land on one. If there is at least one other start, the next repetition merges its row's last state with the fixed state: both go to the fixed state. With no other start, the one fixed state idles forever and no finite retention horizon follows. A single fixed reading isn't needed either. If the operation sends every state to one of a few resting states and leaves each resting state alone, and every resting state is one of the starts, the same walking back counts the same rows inside each resting state's own catchment, the starts that end at it. The total comes to one state per start, plus, for every repetition, one more state for each start that isn't a resting state. With one resting state that is the bound in the proposition, and the same rows in each catchment show it exact. In the exact form: $N$ starts, $L$ of them resting states, $T$ repetitions, at least $N+T(N-L)$ states, and that many are enough. In bits: a memory of $B$ bits has at most $2^B$ states, so surviving $T$ repetitions with one resting state takes $B \geq \log_2\big(1+(T+1)(N-1)\big)$. The memory grows with the logarithm of how long the history is kept; a fresh state at every step isn't a fresh cell at every step.

So the strip uses fresh configurations for as long as its record stays up without a collision. Cut its permitted state space to a finite size and the counting bound applies. A configuration is a pattern, not a newly allocated cell. Bennett's 1989 construction saves storage by undoing and redoing intermediate work while retaining enough to recover it. Those extra transitions buy a time–space tradeoff. They do not let a fixed autonomous run under this fresh-state premise reuse a configuration.

Bennett's Brownian model supplies a separate tradeoff: reducing its bias reduces its mean forward speed and its heat per net forward step. At zero bias its mean drift is zero, although individual random walks still move. Theorem 1.9 concerns that model, not every physical step. There are two accounts: configurations in the structural argument, and entropy production under specified thermal dynamics. The total autonomous run just counted, after a new permanent raise, must have a collision somewhere in its continued orbit or take endlessly fresh states. The collision need not be on the entry arrow. Counting gives that alternative; a physical implementation determines its resource costs.

*Two levers, and whether their hidden moves commute*

Put two reset levers on the box. Pull left then right, and look at the tabletop. Now start over and pull right then left. The picture agrees. Open the box as well: are all its hidden parts in the same places? That is the extra demand in this example. The order of the levers must leave the whole machine the same.

Use three tabletop values, $0, 1, 2$. The first lever wipes only 1 to 0. The second wipes only 2 to 0. They commute on the tabletop. Choose one starting full state above each value and allow every word of at most $T$ lever pulls. Call everything reached by depth $t$ $R_t$. Through the promised horizon, each lever must give different successors to different states in $R_{T-1}$, even if those states were reached at different times. You know which lever was pulled, so the two levers may reuse a landing state.

□ **Proposition 3.6** (The exact room for exclusive resets)**.** For $T \geq 1$, the minimum finite installed carrier has $T + 3$ states without a commutation requirement, and $2T + 2$ with commuting hidden levers. These are also the respective minimum sizes of $R_T$ when larger carriers are allowed. With commuting levers that are one-to-one on their *whole* domain, the minimum $|R_T|$ is $2T + 3$, attained for every $T$ by one infinite machine. At $T = 0$ the reachable minimum is three in each class.

Count the zero states first. Write $F, G$ for the hidden levers and $z, a, b$ for the starts reading $0, 1, 2$. At depth $t$, let $Z_t$ contain the states reading zero. The first lever takes every state in $Z_t$, and also $a$, into $Z_{t+1}$. Their successors must be distinct. So $|Z_t| \geq t + 1$. Add $a$ and $b$ and there are at least $T + 3$ states.

A single line of zero states reaches that number. Use $z_0, \ldots, z_T$, advance the line with either lever, and fix its last state. Put $F(a) = z_0$, $G(b) = z_0$, $G(a) = a$ and $F(b) = b$. The final zero state stays outside $R_{T-1}$. Try the levers in opposite orders at $a$, though, and the hidden states disagree. To make them commute, count again.

*Counting when the hidden levers commute.* Start above 1 and pull only the lever that leaves 1 alone on the tabletop. The hidden box can still move. Watch its first $T+1$ states, $a, G(a), \dots, G^T(a)$. Do the matching experiment above 2, watching $b, F(b), \dots, F^T(b)$. The first list reads 1 throughout; the second reads 2. If either list never repeats, its $T+1$ states, the at least $T+1$ zero states already counted, and the other starting state give $2T+3$. There is enough room already.

So suppose both lists repeat. A deterministic run first follows an approach and then enters a loop. Let the first approach take $r$ pulls and its loop take $p$ pulls; let the second take $s$ and $q$. The loop lengths $p, q$ are at least one. Call their first loop states $\alpha = G^r(a)$ and $\beta = F^s(b)$. There are $r+p$ different states on the first approach and loop, and $s+q$ on the second. All have been reached within the promised horizon.

At those loop states, switch levers. From $\alpha$, the first $F$ pull changes the reading from 1 to zero. From $\beta$, the first $G$ pull changes 2 to zero. There are $L = T-r$ pulls left in the first experiment and $M = T-s$ in the second. Write the resulting two rows of zero states as

$$A = \{F(\alpha), \dots, F^L(\alpha)\}, \qquad C = \{G(\beta), \dots, G^M(\beta)\}.$$

Neither row repeats. If two entries agreed, walk their predecessors backwards together. All those predecessors occur by depth $T-1$, where the lever keeps sources apart. Eventually the earlier entry would reach the nonzero seed while the later one still read zero. That is impossible. The rows therefore contain $L$ and $M$ states.

Can the rows cross? Suppose their $i$th and $j$th entries are the same state. Pull $F$ another $q$ times. Seen from the second row, commutation lets you do those pulls at $\beta$ before its $j$ pulls of $G$. They take $\beta$ once around its loop and bring it back. So the shared state returns to itself after those $q$ pulls of $F$. The same argument, exchanging the levers, returns it after $p$ pulls of $G$:

$$F^{i+q}(\alpha) = F^i(\alpha), \qquad G^{j+p}(\beta) = G^j(\beta).$$

I have just proved there are no repeats within either displayed row. The returns must therefore lie beyond their ends: $i+q > L$ and $j+p > M$. Only the last $q$ entries of the first row and the last $p$ of the second can be shared. There are at most $\min(p,q)$ shared states.

Now add the two nonzero lists and both zero rows, subtracting the shared entries once:

$$|R_T| \geq (r+p) + (s+q) + L + M - \min(p,q) = 2T + \max(p,q).$$

The approach lengths disappear because every pull spent on an approach leaves one fewer pull for its zero row. If either loop has length at least two, the count is at least $2T+2$.

Only two one-state loops remain: $p = q = 1$. If the zero rows do not meet, adding them without a subtraction gives $2T + 2$. If they meet, the crossing restriction puts their only shared state at both ends. Call it $c$. It is fixed by both levers. The initial zero state $z$ must now supply another place. To see why, suppose it lay on one of the rows. Starting there would reach $c$ in at most $T - 1$ pulls. The state just before $c$ on the first row is also reachable by then, and is different from $c$. The first lever sends both to $c$, violating its promise on $R_{T-1}$. Thus $z$ lies outside both rows. Count it, and the total is $2T + 2$ again.

Finally strengthen the promise: each lever keeps different sources apart everywhere, even beyond the horizon. A crossing would give the return $F^{i+q}(\alpha) = F^i(\alpha)$ above. Now you may walk backwards all $i$ steps, without a horizon restriction, giving $F^q(\alpha) = \alpha$. The left side reads zero and the right side reads 1. So in this case the rows cannot cross at all. Their count with the nonzero lists is $2T + p + q$, which is at least $2T + 3$ unless both loop lengths are one.

In that last case you need one more state. If $z$ is outside the rows, it is the extra state. If it lies at $F^i(\alpha)$ on the first row, pull $F$ from $z$ another $T$ times. This is allowed from an initial state and reaches $F^{T+i}(\alpha)$. The whole zero row can never repeat under the global promise, by the same backward argument used earlier. This new state lies past its displayed end. It cannot be on the other row either, because the crossing argument still applies, however far you go. Its zero reading also keeps it off the two nonzero lists. If $z$ lies on the second row, pull $G$ instead and make the same count. Every case gives at least $2T + 3$. ■

Build the finite box with two arms and a shared endpoint. The $F$-arm has $a_0, \ldots, a_{T-1}$; the $G$-arm has $b_0, \ldots, b_T$. Each lever advances its own arm and fixes the opposite arm. Both fix the common end. Read $a_0$ as 1, $b_0$ as 2, and everything else as zero. Start at those two heads and at $z = b_1$.

At horizon one, there are just four places. Write every arrow down:

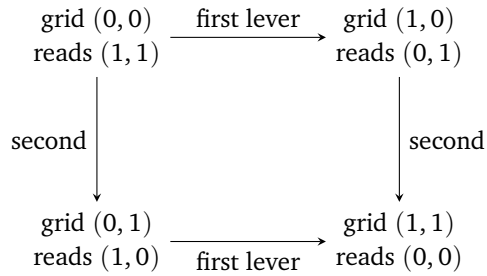
| State | Tabletop | Pull $F$ | Pull $G$ | Start? |
|---|---|---|---|---|
| $a_0$ | 1 | end | $a_0$ | yes |
| $b_0$ | 2 | $b_0$ | $b_1$ | yes |
| $b_1$ | 0 | $b_1$ | end | yes |
| end | 0 | end | end | no |

On the three starts each column has three different destinations. On the whole four-state carrier each column has a repeated destination. The horizon promise and the everywhere promise are different. Try both orders at $b_0$: either order ends at $b_1$. The other three rows also agree in both orders. The endpoint first appears at depth $T$. By then every one of the $2T + 2$ states has been reached, and until then each lever keeps its sources apart.

For one-to-one levers on the whole machine, keep the two arms disjoint and let them run forever. Give them a separate fixed zero state, and start at that state and

the two heads. After $T$ pulls the reachable population is exactly $2T+3$. The same box works at every horizon.

Change the tabletop and the chairs can fit differently. Give it two bits, with one lever clearing the first and the other clearing the second. Lay out an infinite grid with nonnegative integer coordinates $(i,j)$ and a separate blank state. One lever adds one to $i$, the other to $j$, and both fix blank. Read a grid point as $([i=0],[j=0])$ and blank as $(0,0)$. Start at blank and the three grid points $(0,0),(1,0),(0,1)$.




Both orders reach the very same grid point, not just the same reading. The separate blank state also reads $(0,0)$ but is fixed by both levers; it is not the grid's origin.

A pull adds one to $i+j$. The three grid starts have sums zero or one, so no word of length at most $t$ can go beyond $i+j=t+1$. Conversely, the origin is a start. For any other point in that triangle, start at $(1,0)$ if $i>0$, and at $(0,1)$ otherwise. Exactly $i+j-1\le t$ pulls reach it. That gives the whole triangle. Its rows have $1,2,\dots,t+2$ points. Pair the first row with the last and count, then add blank:

$$|R_t| = 1+\frac{(t+2)(t+3)}{2}.$$

Adding to one coordinate is one-to-one on the grid, and neither grid shift reaches blank. Adding to different coordinates in either order gives $(i+1,j+1)$, so the levers commute on the whole carrier. The reading follows the two clears because incrementing a coordinate makes its zero-test false and leaves the other zero-test alone.

This count also meets the lower bound. Each one-cleared fibre starts with one state. The appropriate clear sends that fibre and at least one $(1,1)$ state into it injectively, so its population increases by at least one per depth: at depth $t$ it needs $t+1$. The $(0,0)$ fibre also starts with one. At depth $j+1$, one clear sends both its $j$-depth population and a one-cleared fibre of at least $j+1$ states into $(0,0)$ injectively. Its population therefore increases by at least $j+1$. Summing from $j=0$ to $t-1$ gives $1+t(t+1)/2$. Add the one $(1,1)$ state and the two one-cleared populations: $1+2(t+1)+1+t(t+1)/2 = 1+(t+2)(t+3)/2$. The three-value box needed extra room for commutation. This two-bit box fits its best count while keeping commutation intact.

### 3.4 The second way out: let the flag come back down

Take one coin. The rule is: turn it over. Heads is YES, tails is NO. Every move is undoable (turn it over again), so nothing is forgotten. Tails goes to heads, which is a flip, NO to YES, and heads goes to tails, so the same move takes the flag back down somewhere else. There are two states, as finite as it gets, and the move that raises the flag forgets nothing, so a schedule that charges only for forgetting is allowed to charge it nothing. The move is injective and can switch the reading both on and off. Pricing the turn at zero meets merge pricing, since no move merges, and fails the toll. The premise dropped on this way out is permanence.

There's a fact here sharper than the premise it replaces, and it was the thing in this section that surprised me. Drop permanence entirely and keep only finiteness, and take any move that raises the flag anywhere. Then one of two things is true: the move forgets something, or the same move lowers the flag somewhere else. There's no third option, and the proof below is a search: go through the finitely many YES states looking for one this move turns to NO. Find one, and that's the revocation; find none, and the move is permanent at least for itself, which is all the argument from fresh states needs. So a flag that goes up without a merge is a flag the same move can take back. A flag that nothing takes back is paid for. You can't have a raise at zero cost that stays raised, not on any finite machine, not with any price schedule that charges for forgetting. The sentence is narrower than "every certificate costs." It's true, and I'll take it. The broad one would only be mine.

□ **Theorem 3.7** (Merge or revoke)**.** On a machine with finitely many states reading yes (no permanence assumed), if $\mathrm{cert}(s) = \text{no}$ and $\mathrm{cert}(\mathrm{step}(s, i)) = \text{yes}$, then either $i$ merges, or there is a state $t$ with $\mathrm{cert}(t) = \text{yes}$ and $\mathrm{cert}(\mathrm{step}(t, i)) = \text{no}$.

*Proof.* Check every yes-state on the finite list for one that $i$ sends to no. If there is one, that is the second alternative. If there is none, then $i$ keeps every yes-state at yes, which is permanence for the single move $i$; the argument of Theorem 3.1 uses permanence only for the move it is about, so it applies, and $i$ merges. ■

I'll give you the three-card version too, because it shows that a move can be forced to be priced without writing anything permanent, and I don't want you to think the picture is tidier than it is. There are three cards, T0, T1, T2, with a star on T1. The rule is that T0 goes to T1, T1 goes to T2, and T2 goes to T1. T0 gains the star (a flip). T1 loses it (a revocation). T0 and T2 both land on T1 (a merge). So this move is priced, by the forgetting rule, and yet it writes nothing permanent; it takes a star away in the same breath. "The priced moves are exactly the permanent records" is false, as the three cards just showed. What's true is one direction: on a finite

machine, every permanent raise is a priced move. The other direction has the three cards standing in it. And "priced" here has an exact meaning worth one sentence: a move is *forced to be priced* when every schedule that charges for forgetting charges it, and that holds exactly when the move forgets something, with no finiteness needed at all.

**Definition 3.3** (Forced to be priced)**.** A move $i$ is *forced to be priced* when every cost function satisfying merge pricing (Premise 3.2) charges $i$ at least one.

**Theorem 3.8** (Forced price is merging)**.** A move is forced to be priced exactly when it merges. No finiteness is needed. □

*Proof.* If $i$ merges, merge pricing charges it at least one, whatever the cost function. If $i$ does not merge, define the cost that is 0 on $i$ and 1 on every other move. It satisfies merge pricing, since every merging move is some move other than $i$ and costs one, and it charges $i$ nothing, so $i$ is not forced. ■

A cost here is any rule from move names to whole numbers, so that one is there whether or not anything can tell $i$ from the other names. A test that tells them apart comes in only when a cost has to be computed by a program, and then the program needs it to tell $i$ from the rest.

**Corollary 3.9** (Permanent records are forced)**.** On a finite machine, a move that never turns the reading off and turns it on somewhere is forced to be priced. (By Theorem 3.7 it merges, and merge pricing charges every move that merges, whatever the cost function.) The converse fails: the three cards are forced, because they merge, and they revoke. □ ■

## 3.5 The third way out: don't charge for forgetting

The third way out is the easiest to describe and the hardest to argue about, so I'll describe it and then spend the rest of the section on the argument. Take a sheet of paper and let its reading be just BLANK or STAMPED. There is one move: stamp it if blank, and leave it alone if already stamped. It is finite, with two readings. It is permanent, since STAMPED stays STAMPED. It has a flip, since BLANK goes to STAMPED. Both possible readings now end at STAMPED. The paper's fibres, the ink and your memory are outside this two-state description; the merge is a loss from the reading, not the disappearance of every physical trace. Everything the argument

from fresh states needs is here, and the move merges. Now set the price of that move to zero. The toll fails, and you can watch it fail: the flag went up at zero cost. The premise dropped on this way out is the charge on forgetting. (A price of one on that move satisfies all three premises, so the premises can hold together; the third way out is about whether they must.) Nothing in counting stops you from setting that price to zero. Only a claim about the world does. The nearest such claim is Landauer's, and I have to be exact about the gap between the two: the premise as written charges *every* forgetting move, whether or not you knew which state you were in, and Landauer's own heat, as the coin under the cup below will show, is owed only for what you didn't know. Merge pricing keeps only a yes-or-no distinction: whether any pair can merge. The one-bit reset is where it comes from, and it measures neither the full worst-case entropy loss nor the heat of any particular preparation. The counting needs only that rule.

Here is what Landauer said, in 1961, as exactly as I can put it. I went past the summaries and read the paper itself. A logical operation with no single-valued inverse (a step that forgets, in this book's words) must produce heat, "typically of the order of kT" per operation, where k is Boltzmann's constant and T is the temperature; and for the specific case of resetting one bit whose value you don't know, the minimum is kT times the natural log of 2, about 0.69 kT. I want you to hear two things in that. First, "of the order of": he was not handing out a number to three decimals. Second, and this surprised me, he did not measure anything. The paper is argument, with models. He writes that he can "show, or at least very strongly suggest" the conclusion, and that his reasoning gives "no guarantee that this minimum is in fact achievable." That's the founding paper of the whole subject, and it's a careful man saying "I think this is right and here's why." I like him more for it.

Was it measured, eventually? Yes, and only recently. In 2012 a group at the École Normale Supérieure in Lyon, Antoine Bérut and Sergio Ciliberto's lab working with Eric Lutz, trapped a single glass bead, two millionths of a metre across, in a double well made of laser light, and used the two wells as the two values of a bit. Then they erased the bit, by lowering the barrier, tilting the trap, and raising it again, many times over, slower and slower, and measured the heat each time. The average heat came down toward Landauer's kT ln 2 as the erasure got slower, closer the slower it went, and never under it, on average. Single runs went under it; the bound is about the average, and any one erasure can beat it. And if you only succeed in erasing half the time the floor drops to zero, which the paper's formula shows and which I think is the most honest line in it. Others have done it since with different devices. So it was predicted in 1961 by argument and seen in a laboratory fifty-one years later.

And is it settled? No, and I'd be lying by omission if I left that out, because the people who doubt it are not cranks. John Norton, a philosopher of physics, has

argued since 2005 that the usual proofs of the principle quietly assume what they're trying to prove, and that the kinds of process they assume, combined, could be made to break the very law the principle is supposed to guard. An earlier objection, made in 1998–99 by Norton and his Pittsburgh colleague John Earman, also a philosopher of physics, was that the principle can't be independent of the second law. Charles Bennett, who did more than anyone to build the subject, rated that one "in my opinion the one of greatest merit," and then disagreed with it. Others replied to Norton; Norton replied back. The Stanford Encyclopedia entry on the topic, written by a philosopher of physics who works on exactly this, ends without a verdict. I've read what I could of both sides and I'm not qualified to referee them, and I'm not going to pretend the laboratory settles a question about what the principle means in general, because measuring one bead doesn't do that.

What I did with it is this, and I think it's the only honest thing to do. The charge on forgetting is written into this book as a *named premise*. It's a definition the theorems take as an assumption, and nowhere is it proved as a general fact, because it isn't provable from counting; for a particular machine with a particular price list you can see it hold (the stamp at cost one, above), which is a different thing. If Norton turns out to be right, every theorem in this section still holds; they just have one premise fewer that the world agrees to. If Landauer's side turns out to be right, it supports a thermal price under its conditions; the whole-number charge on every merging move is still the ledger rule stated here. Either way, nothing in this section says a device does anything until someone measures the device. The counting is mine to prove. The physics is a premise I've named, and what settles it is a measurement, which a proof can't do.

Naming a premise, though, isn't the same as leaving it unplaced, and a hedge isn't a position, so I'll give mine. Earman and Norton's objection is a dilemma. Either an information-theoretic exorcism of the demon assumes the second law, which they call the *sound* horn, and then it adds nothing the second law didn't already give; or it rests on a new principle tying information to entropy, the *profound* horn, and nobody has given that principle independent grounds. A named premise is the sound horn, on purpose. That's where Bennett ended up in 2003, and Norton's "Too Good to Be True" (written in 2025, published in 2026) says the careful literature almost always accepts that horn, and argues that whatever succeeds there is ordinary thermodynamics relabelled. I'll take that. Ordinary thermodynamics, relabelled onto a ledger, is all this section asks of the physics. And on the counting I'm on Norton's side outright. His 2018 paper "Maxwell's Demon Does Not Compute" exorcises the demon with Liouville's theorem and nothing else, no information and no computation: dynamics that keeps every patch of states the same size can't crowd a big set of states into a small one. The counting core of this section, that a step losing nothing can't carry a no-state into a set that stays up, is that point in finite,

discrete form. And if you want the best case for the premise itself, the one I found is James Ladyman, Stuart Presnell, Anthony Short and Berry Groisman (2007): from one of the standard statements of the second law, the one named for Kelvin, they derive that a logically irreversible operation can't be carried out thermodynamically reversibly by any of their L-machines. That's a physical argument under their stated model of a machine. It deserves to be argued with on those terms; it does not by itself establish the book's whole-number ledger charge.

One question a physicist asks before any of that, and it's a good one: the laws underneath never merge anything. Classical mechanics carries each exact state forward to exactly one next state and could carry it back again, and Liouville's theorem adds that a patch of states keeps its size as it moves; quantum mechanics is unitary, which is one-to-one too. So when this section says a step merged two states, where did the difference go? It went out of the machine. The merge is in the machine's own description, the states its reading is about. The machine together with everything around it still runs one-to-one, and the distinction the machine forgot survives. It gets pushed out into the surroundings. The surroundings are the endless strip of the first way out, the machine that keeps every history: the column of states never used before, which the universe as a whole may have to spend and the machine doesn't, because the machine isn't the universe and its memory fills. Pushing a distinction out isn't yet heat: another memory could keep it as a readable record. It's heat when the reset removes real uncertainty into a thermal reservoir, and then Landauer's premise bounds the mean heat handed over. None of that heat has to be heat *wasted*. Takahiro Sagawa's 2014 analysis says that erasure can, in principle, approach a process that produces no entropy overall: the reservoir's entropy goes up by what the machine's goes down. A positive mean heat transfer can then be unavoidable without being wasteful. A known input has no uncertainty to remove, and this entropy-loss premise alone sets no positive heat floor for it. The reset can dissipate under the stated thermal-model assumptions, and a reset device that wasn't built for that one input keeps a floor that survives going slowly (Section 3.7).

Ladyman and his coauthors and Sagawa aren't disagreeing. They use "reversible" for two different things. Ladyman's L-machines are devices, and a device counts as reversible only if it produces no entropy on every input it's fed, each one known (their section 4: "the L-machine is thermodynamically reversible if and only if" every one of its processes, one per input, produces no entropy). Sagawa's reversible erasure is a process on one given spread of inputs, with the device tuned to that spread. The standard eraser is reversible in Sagawa's sense and not in Ladyman's: on an even mix of 0 and 1 it produces no entropy, and on each known input it produces $k_B \ln 2$. One identity holds the two together. For a fixed device that sends every input to one output, and any spread $q$ of its inputs, the average over inputs of the

entropy produced on each known input equals the entropy produced on the spread $q$, plus $k_B$ times the Shannon entropy of $q$. The identity holds because the heat is the same average either way, the device being fixed, and the output is the same either way. The only difference is at the start: the spread's own entropy includes $k_B H(q)$ for not knowing which input it is, and each known input's doesn't. Sagawa makes the first term on the right zero; Ladyman reads the left side, which is then $k_B H(q)$, and more than zero.

### 3.6 How much: the price grows with the squeeze

The toll charges at least one unit, and that's a floor for raising one flag. The counting of states says more if you let it. Say there are m states that already read yes and k more are turned to yes in one move. (That's a different k from the one in Landauer's kT; when the two meet, below, Boltzmann's constant gets a little B hung under it, $k_B$, so you can tell it from a count of states.) All m plus k of them end up on the m yes-states. That's a squeeze by a factor of (m + k) over m, and if you accept a slightly stronger version of the forgetting price, that each unit of cost pays for at most one *halving* of the number of distinguishable states, then the move costs at least the number of halvings in that squeeze, which is the base-two logarithm of (m + k)/m.

Back to the stamp from the third way out, but give the reading four possible values this time: A, B, C and YES. Write them on four cards. Choose one secretly, then cover its face with an identical opaque YES label. Read only the new face: all four possibilities now give YES. Before the move, identifying the original face took two yes-or-no questions (one of the first two? then which one?); afterward there is only one possible face to report, so that reading has lost two bits, which is two halvings. Peeling the label off would recover the old face, so this tabletop version models the loss from the reading; it does not demonstrate physical erasure. So that move costs at least two units, and two is enough, since a cost of exactly two satisfies the halving rule. The general statement is m + k at most 2 to the cost times m, and there is also a logarithm form that rounds both logarithms up and is weaker for it (three states and one newcomer round to the same two bits on both sides, and the rounded form says nothing). The halving premise, like the one before it, is a definition used as an assumption. The units are halvings. To turn a halving into joules of heat you multiply by $k_B T \ln 2$, and that multiplication is physics; the counting stops before it.

That little $k_B$ deserves a short trip of its own, because it sits exactly where counting turns into physics, and its name is a small history lesson. Ludwig Boltzmann's grave in Vienna has $S = k \log W$ carved on it (that $k$ is the same $k_B$, before anyone hung a B under it): entropy is a constant times the logarithm of the number of ways

the parts could be arranged. The idea is his. The constant, and the equation written that way, came from Max Planck, in his work on radiation around 1900. In his Nobel lecture in 1920 Planck said the constant "is frequently termed Boltzmann's constant, although to the best of my knowledge Boltzmann himself never introduced it," and guessed that Boltzmann never believed anyone could measure it accurately. I'd assumed Planck named it after Boltzmann. His own words point the other way: it was other people calling it that, and who first did, I couldn't pin down. So the most famous constant in this corner of physics wears the name of a man who didn't write it down. Does that change the heat floor? It doesn't, but look at the job the constant does in that equation. Take it away and what's left is the logarithm of a count, which is a number of halvings, which is what the squeeze bound hands you. The constant is the exchange rate, and whether the world honours that rate for a merge is the premise. No theorem here settles it.

The premise, stated exactly, needs one new piece of notation: for a list $D$ of states, write $|D|$ for how many it has, and $\mathrm{img}_i(D)$ for the number of *different* states the move $i$ sends them to.

**Premise 3.4** (Halving price)**.** Each unit of cost pays for at most one halving: for every move $i$ and every list $D$ of states with no repeats, $|D| \leq 2^{\mathrm{cost}(i)} \cdot \mathrm{img}_i(D)$. (It implies merge pricing, since a merge sends two different states to one and so needs $\mathrm{cost}(i) \geq 1$.)

There's a plainer way to read that premise, and it's worth having because it says what the premise is *about*. Look at one landing spot of a move and count how many states land there; call the biggest such count the move's worst pile-up. The premise holds exactly when the cost covers the worst pile-up in halvings: a cost of one allows piles of two, a cost of two allows piles of four, and a pile of three already needs a cost of two. In one direction, hand the premise one whole pile. It lands on a single spot, so the premise says the pile has at most that many members. In the other, any list of states splits by where its members land, each part sits inside one pile, so the list is at most the number of landing spots times the worst pile-up, which is what the premise says. So the halving price asks a move to pay for the worst it could ever do, whichever state it's actually handed. The worst case is on purpose, and it has a home in physics. In the branch of thermodynamics that prices a single run that has to succeed every time, with no averaging over many runs, Philippe Faist, Frédéric Dupuis, Jonathan Oppenheim and Renato Renner derived in 2015, in that framework, what they give as the least work any logical process needs, and when every input might occur and no error is allowed, their bound comes to $k_B T \ln 2$ times exactly that logarithm. Take an AND gate as the example: three inputs land on 0, so it needs at least $\log_2 3$, about 1.6, halvings' worth of work (two units, in whole

numbers). The premise asks for that bound of a device that has to work whatever it's handed, and it counts it in units of cost. Joules don't come into it.

**Theorem 3.10** (The squeeze bound). On a finite machine with a permanent reading and a cost satisfying the halving price, let $m$ be the number of states that read yes, and let a move $i$ turn $k$ different no-states to yes. Three numbers are in play: $m$ counts every yes-state of the machine, reachable or not; $k$ counts a list of distinct no-states that the move turns to yes, and the move may turn others too; and $\mathrm{cost}(i)$ is the price the cost assigns to the move. Then

$$m + k \;\le\; 2^{\mathrm{cost}(i)} \cdot m, \qquad \text{and, when } m \ge 1, \qquad \mathrm{cost}(i) \;\ge\; \log_2 \frac{m+k}{m}.$$

When $m \ge 1$ the two displayed bounds say the same thing. (When $m = 0$ there is nowhere for a newcomer to land, so $k = 0$ too.)

*Proof.* Put the $m$ yes-states and the $k$ newcomers in one list; it has $m + k$ different members. By permanence every yes-state lands on a yes-state, and by hypothesis every newcomer does too, so all $m + k$ land among the $m$ yes-states: $\mathrm{img}_i \le m$. The halving price gives $m + k \le 2^{\mathrm{cost}(i)} \cdot \mathrm{img}_i \le 2^{\mathrm{cost}(i)} \cdot m$. □

The four card readings are the case of one yes-state and three newcomers, where the bound is reached. With one newcomer the bound says the move costs at least one, which is the toll again. The counting argument also shows that at least as many states collapse as there are newcomers, without any price premise.

The bound is reached, whatever the numbers. Pick how many states already read yes and pick a cost. As long as the old yes-states and the newcomers together fit under that cost's pile-up allowance (a cost of one allows twice the yes-states, a cost of two allows four times), there is a finite machine with a permanent reading and the halving price that has exactly that many yes-states and a move of exactly that cost turning those newcomers to yes. The move turns every other no-state of that machine to yes as well, filling the allowance to the brim. No cost meeting the halving price on that machine charges the move less. So the least whole-number cost that meets the bound is exactly the rounded-up logarithm, $\lceil \log_2 \frac{m+k}{m} \rceil$.

Four more facts about the halving price, and the last one is a limit. First, the plainer reading above is a theorem on a finite machine: the halving price holds exactly when no move has a landing spot with more than $2^{\mathrm{cost}(i)}$ states landing on it. Second, for the eight-state machine described later in this section that's a theorem about its own price list: no move has a larger pile-up than $2^{\mathrm{cost}}$, so the halving price holds, and the toll follows from it; and the price list is the least that meets the halving price, since any price list that does charges every move at least what this

one charges. Third, the squeeze bound is met with equality there: its stamp turns on the four states whose flag is down, beside the four whose flag is already up, and $4 + 4 = 2^1 \cdot 4$. Fourth, and this is the limit, the small machine of Part III (Section 15) as a whole does not meet the halving price, whatever procedure you use to decide whether two of its states are equal, because one of its zero-price moves, the one that takes one off a number the machine holds, lands two different states of the whole machine on the same state (the pair is in Theorem 15.35), and a pile of two at price zero is all it takes to break the halving price. What does hold there is merge pricing asked of the record layer alone, and it gives the toll on that machine too (Theorem 3.19, below).

### 3.7 The same bound, written in bits, and when it's zero

There's one more way to say all this, and I include it because it's the way physicists say it and because it says exactly when the heat floor is zero: when no two states you might be in land on one. Its two ends are a move that forgets nothing and a move on a state already known, and the second I find more instructive than any case where it isn't.

Claude Shannon, in 1948, gave a number to "how much you don't know": if something could be in any of n states and you have no idea which, your uncertainty is log base two of n bits, and that's the most it can be for n states. If you push your uncertainty through a step that forgets nothing, it stays the same; through a step that merges two states you might have been in, it drops. Now push it through the permanent flip. Before the move, you might be in any of the m + k states in play. After it, you're in one of m. So your uncertainty afterward is at most log2 of m, whatever you believed before. If you started with no idea at all, every one of the m + k states equally likely, your uncertainty was exactly log2 of (m + k), and the move removed at least log2 of (m + k)/m bits: the same number as the halving count, now in Shannon's units. Landauer's principle, in the form physicists write it, then says a step that removes that many bits from the state gives off at least $k_B T \ln 2$ of heat per bit. Put the two together and the flip gives off heat, and the exact statements below say how much and under which side conditions, one of which (the temperature has to be taken as zero or above) is easy to skip and shouldn't be.

Why does Shannon's number share a name with the physicists' entropy at all? There's a story about it. Shannon wanted to call it uncertainty, and the mathematician John von Neumann told him to call it entropy instead, for two reasons: the same formula already lived in statistical mechanics under that name, and nobody really knows what entropy is, so in an argument he'd always have the upper hand. It's a great story. It comes from Myron Tribus, who spent years tying information

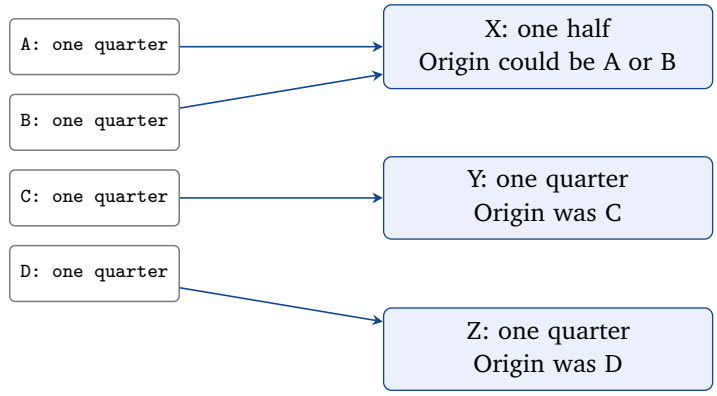
theory to thermodynamics, who said Shannon told it to him in 1961, and who printed it, with Edward McIrvine, in *Scientific American* in 1971. Then in 1982 an interviewer, Robert Price, asked Shannon about it, and Shannon said he didn't think von Neumann had told him any such thing. So it's a legend with one witness, and the person the witness quoted doesn't remember saying it. What doesn't hang on the story: the two formulas really do have the same shape, and it still takes Landauer's premise, a named assumption, to turn a count of bits into heat.

Now the exact version, with the notation built first. A *spread* (probabilists say distribution) is a way of giving each state a chance $p(x) \geq 0$, with the chances adding to 1. Its Shannon entropy, in bits, is

$$H(p) \;=\; -\sum_{x:\,p(x)>0} p(x)\,\log_2 p(x),$$

where the big sigma means "add up, over every state $x$ with a chance above zero," and each term is a state's chance times the bits it would take to single that state out. That comes to $\log_2 n$ when $n$ states share the chance evenly and less when the chance is lopsided. Pushing a spread through a move $i$ gives the spread of where you land: $q(y) = \sum_{x:\,\mathrm{step}(x,i)=y} p(x)$. The bits the move *removes* are $H(p) - H(q)$.

Try four equally likely cards, A, B, C and D. Send A and B to a box marked X, C to Y and D to Z. Beforehand each card has chance $1/4$, so the four terms in the sum are each $1/2$: two bits. Afterward the box chances are $1/2, 1/4, 1/4$, and the three terms are each $1/2$: one and a half bits. Half a bit was lost on average. You can see where: half the time the box is X and leaves one binary question unanswered: A or B? The other half of the time Y or Z tells you the starting card exactly. An actual draw never loses half a card; the half belongs to the average uncertainty over repeated draws.




Half the time one binary question remains unanswered; half the time the origin is known. The average loss is half a bit.

Start with the half of the claim that removes no bits, Bennett's half. A move that forgets nothing sends each state somewhere different, so the chances are carried

along to new places, one for one, and nothing piles up and nothing spreads out. The uncertainty is the same number before and after.

□ **Theorem 3.11** (A move that forgets nothing removes no bits)**.** Let the states be finite, listed once each, and let $f$ send states to states with $f(a) = f(b)$ only when $a = b$. Then for every assignment of chances $p$, the pushed spread $q$ has $H(q) = H(p)$.

*Proof.* An injective map on a finite list of distinct states is a reshuffle of that list: every state is hit exactly once. So $q(f(x)) = p(x)$ for every $x$, and the sum defining $H(q)$ is the sum defining $H(p)$ with its terms in a different order. ■

That's why a price on removed bits can leave every reversible move costing nothing, which is the shape the toll needs: the ledger's cost lands only where something was forgotten.

□ **Theorem 3.12** (Entropy after a permanent flip)**.** On a finite machine with a permanent reading, let $m$ be the number of states that read yes, and let a move $i$ turn $k \geq 1$ different no-states yes. For any spread $p$ whose chance lies on those $m+k$ states, the spread after the move has $H(q) \leq \log_2 m$. For the even spread on the $m+k$ states, the move removes at least $\log_2 \frac{m+k}{m}$ bits and at least $\frac{2k}{m+k}$ bits; the first is reached when $m$ divides $k$, the second when $k \leq m$.

*Proof.* All the starting chance lands among the $m$ yes-states: the old ones stay yes and the newcomers turn yes. There must be at least one landing state, so $m \geq 1$. The first question is how much uncertainty those $m$ places can hold.

Here is the small fact about logarithms that does the work. For $y > 0$, $\ln y \leq y - 1$. Picture the gap $y - 1 - \ln y$. Its slope is $1 - 1/y$: negative below 1, zero at 1, positive above 1. The gap falls to zero and then rises, so it never goes below zero.

Call the positive landing probabilities $q_1, \ldots, q_r$. There are at most $m$ of them and they add to one. Compare each with the even share $1/m$. Apply the logarithm fact to the ratio $1/(mq_j)$ and multiply by the chance $q_j$:

$$q_j \ln \frac{1}{mq_j} \leq \frac{1}{m} - q_j.$$

Now add over the occupied places. On the right the even shares total $r/m \leq 1$, while the actual shares total exactly 1. The sum is therefore at most zero. On the left, splitting the logarithm gives $-\ln m - \sum_j q_j \ln q_j$. Move $\ln m$ across and divide

by $\ln 2$ to count in bits. This says precisely $H(q) \leq \log_2 m$: no arrangement of the chance among those places holds more uncertainty than the even arrangement.

Before the move, the $m+k$ starting states were equally likely. Their uncertainty was $\log_2(m+k)$. Subtract the greatest possible uncertainty left at the end: the loss is at least $\log_2((m+k)/m)$. If $k$ is a multiple of $m$, keep each old yes-state where it is and send an equal number of newcomers to each. The landing chances are then even too, so this bound is reached.

For the other bound, count one landing pile at a time. A pile of $s$ equally likely starts receives chance $s/(m+k)$. Once you know that pile was reached, there are still $s$ equally likely origins to distinguish: $\log_2 s$ bits have disappeared there. Its share of the total loss is therefore $s \log_2 s/(m+k)$.

A pile of one contributes zero. A pile of two contributes $2/(m+k)$. For every larger whole number $s$, $s \log_2 s \geq 2(s-1)$. To check that inequality, subtract $2(s-1)$: the difference is zero at $s=2$, and its slope for $s \geq 2$ is $\log_2 s + 1/\ln 2 - 2 > 0$. It grows from there. Thus each pile costs at least $2/(m+k)$ for every origin after its first. There are $m+k$ origins and at most $m$ piles, leaving at least $k$ such extra origins. The loss is at least $2k/(m+k)$. When $k \leq m$, send each newcomer to a different old yes-state and leave the old states fixed. All piles then have size one or two, exactly the sizes where the inequality is an equality. That construction reaches the second bound. ■

**Premise 3.5** (Landauer's principle, as a premise)**.** For the chosen physical memory and its input distribution, assume that a step lowering its logical-state entropy by $\Delta H$ bits transfers mean heat $Q \geq k_B T \ln 2 \cdot \Delta H$ to a bath at temperature $T$, where $k_B$ is Boltzmann's constant. Here $Q$ is an average over that preparation, not a bound on each individual trial. Applying this premise requires a physical model in which the logical entropy loss has this thermodynamic meaning; a bare state-transition table does not supply one. In particular the memory is symmetric, its logical states alike in their own entropy, as Landauer assumed; for a memory whose states differ, the floor changes (Maroney 2009), and erasure work in an asymmetric double well has been measured below $k_B T \ln 2$ (Gavrilov and Bechhoefer 2016).

**Corollary 3.13** (Heat floor of a permanent flip)**.** Under Premise 3.5, with $k_B T \geq 0$, the flip of Theorem 3.12 from the even spread gives off at least $k_B T \ln \frac{m+k}{m}$, and at least $\frac{2k}{m+k}\, k_B T \ln 2$; with $k_B T > 0$ that is strictly positive. □

*Proof.* Theorem 3.12 gives an entropy loss of at least $\log_2((m+k)/m)$ bits, and of at least $2k/(m+k)$ bits. Multiply by the nonnegative factor $k_B T \ln 2$ and use Premise 3.5. The identity $\ln 2 \log_2 x = \ln x$ gives the displayed floor. Here $m > 0$

and $k > 0$, so $(m+k)/m > 1$ and its logarithm is positive; if $k_B T > 0$, the heat floor is strictly positive too. ■

How big is that? Take the case where the squeeze is two, one state flipping in beside one that was already yes, and the floor is $k_B T \ln 2$. At room temperature, about 300 kelvin, that's roughly three zeptojoules: three thousandths of a billionth of a billionth of a joule. And that's the *biggest* floor a one-state flip gets from this corollary. With one newcomer the floor shrinks as more states already read yes, because the squeeze gets thinner: with a thousand of them it's $2 \ln 2/1001$ of $k_B T$, about a seven-hundred-and-twentieth, some five hundred times less than the case of two. So, for an even spread over the states in play, one unit of cost is not one halving. It's one halving exactly when the squeeze is two; for any thinner squeeze and an even spread, the physics asks for less than a halving, and the toll still charges a whole unit, because the cost comes in whole numbers. The entropy road to the toll, a little further down, doesn't need that rounding: entropy pricing asks about every spread, and the worst one takes a full bit. Read in the worst case over what the machine might be holding, the way the halving price reads a move, it's at least a full bit every time. A permanent flip merges two states (Theorem 3.1); put half the chance on each, and the move removes a full bit, however many states already read yes. So the gap between the unit of cost and the bits is a fact about the even spread, and in the worst case the flip removes at least the full bit the unit pays for. Whatever the size, it's heat only if you grant the premise; nothing on this page measured anything. The bead experiments above probe this energy scale. Ordinary CMOS switching is a different comparison: the implementation figures cited next are much larger than this floor. One 2023 tally of the numbers puts switching a single CMOS gate at a hundred attojoules at the low end, over thirty thousand times this. The point of the number isn't that a permanent flip is expensive but that, on the premise, it isn't zero.

A related logarithm appears in a different physical question. Yûto Murashita, Ken Funo and Masahito Ueda (2014) study processes whose reverse histories can include ones with no forward counterpart. They call this *absolute irreversibility*. If the reverse histories with no forward partner carry some share of the probability, their bound on mean entropy production picks up the logarithm of one over what's left. In their free-expansion example, gas that starts confined to a tenth of a box gives the logarithm of ten.

Make the confined fraction the old yes-states' share of the pile, the yes-states over the yes-states plus the newcomers, and out comes the same logarithm as the squeeze bound. That numerical match does not identify two physical processes. The arrows specify neither a thermodynamic reverse experiment nor its path probabilities, so they don't determine that share of reverse histories. The state count alone says noth-

ing about absolute irreversibility, its absence, or a heat equality. Gonzalo Manzano, Gülce Kardeş, Édgar Roldán and David Wolpert (2024) study such trajectory-level questions for computations with one-way transitions and random stopping times. Their physical premises are additional to the counting done here.

The even spread is the easy case, and it's fair to worry that the heat depends on picking it. It doesn't. Any spread that gives a real chance to every yes-state and to the state that flips will do.

**Corollary 3.14** (Positive heat for any full spread)**.** On a finite machine with a permanent reading, let move $i$ turn a no-state $s$ yes. Take any spread $p$ that gives positive chance to $s$ and to every yes-state. Under Premise 3.5 with $k_B T > 0$, the heat the move gives off is strictly positive. □

*Proof.* After the move, $s$ and the yes-states all land among the yes-states, and there is one more of them than there are yes-states, so two distinct states with positive chance land on the same state. That merge removes a positive number of bits (Theorem 3.18 below), and a positive temperature times a positive number of bits is positive heat. ■

There's a third road to the toll here too, and it's the same shape as the merge road, with the bits removed, in place of whole units, as the price of forgetting.

**Definition 3.6** (Entropy-priced)**.** On a finite machine, a cost is *entropy-priced* when, for every move $i$ and every spread $p$, the cost of $i$ is at least the bits the move removes: $\mathrm{cost}(i) \geq H(p) - H(q)$, where $q$ is $p$ pushed through $i$.

**Theorem 3.15** (The toll from entropy pricing)**.** On a finite machine with a permanent reading and an entropy-priced cost, every move that turns the reading from no to yes costs at least one. So the machine with that cost is a Thiele machine (Definition 2.2), and every run from a no-state to a yes-state costs at least one. □

*Proof.* Let move $i$ turn $s$ yes. By Theorem 3.1 it lands two different states on one. Put half the chance on each: the move removes one bit, so the cost of $i$ is at least one. The run-level floor is the universal floor (Observation 2.1) applied to that system. ■

There are three roads to the toll, with two different strengths. Entropy pricing and the halving price are equivalent: the most bits a deterministic move can remove is the base-two logarithm of its largest pile of starts sharing a destination. Merge

pricing asks only for a cost of at least one whenever any pair collides. It is the two-state restriction of the stronger conditions.

Try eight equally likely cards all sent to one box. The loss is three bits. A charge of one meets merge pricing but fails entropy pricing and the halving price, which require at least three. All three conditions give the permanent-record toll; only the stronger two give the full squeeze bound.

□ **Theorem 3.16** (Entropy pricing, halving, and pairwise merge pricing)**.** Take a finite machine and a move $i$. (1) If no state has more than $K \geq 1$ states landing on it under $i$, every spread loses at most $\log_2 K$ bits through $i$; and if some state has exactly $K$ states landing on it, the even spread on those $K$ loses exactly $\log_2 K$ bits. (2) A cost is entropy-priced (Definition 3.6) exactly when it satisfies the halving price (Premise 3.4). (3) A cost satisfies merge pricing (Premise 3.2) exactly when it satisfies the halving price for every list of at most two different states, and exactly when it satisfies entropy pricing for every spread whose chance sits on at most two states. The first of those two equivalences uses no finiteness.

*Proof.* (1) Sort the possible starts into piles according to where the move sends them. Suppose the landing place is $y$ and its total chance is $q(y) > 0$. Knowing that you landed there leaves only the starts in that pile in question. Their chances, now measured within the pile, are $p(x)/q(y)$. They add to one, and there are at most $K$ of them. The logarithm argument in the preceding entropy bound says their uncertainty is at most $\log_2 K$ bits.

Average that remaining question over the landing places, giving each its chance $q(y)$. This is exactly the information the move discarded. Indeed, a start $x$ contributes

$$p(x) \log_2 \frac{q(y)}{p(x)}, \qquad y = \mathrm{step}(x, i),$$

and summing these terms gives $H(p) - H(q)$ by grouping the output terms by their landing places. An average of numbers no greater than $\log_2 K$ cannot exceed $\log_2 K$. For equality, put equal chance on every member of one pile of size $K$. There are $\log_2 K$ bits at the start and none at the single landing place.

(2) An entropy price must cover that even preparation on every pile. A pile of size $n$ therefore requires cost $c \geq \log_2 n$, or equivalently $n \leq 2^c$. That is the halving rule: no landing place accepts more than $2^c$ starts. Conversely, if the halving rule holds, part (1), with $K = 2^c$, says no preparation loses more than $c$ bits. The same price covers them all.

(3) Now allow at most two starts. If they land separately, the move loses nothing. If they land together, the halving rule reads $2 \leq 2^c$, which asks for at least one

whole unit. This is exactly merge pricing. Nothing in that two-start argument needs the rest of the machine to be finite.

The chance calculation makes the same demand. Two possible starts hold at most one bit, so a charge of one covers every spread over a merging pair. Equal chances on that pair lose exactly one bit and force that charge. A single possible start, or two that stay separate, lose zero. So restricting entropy pricing to at most two possible starts gives merge pricing too. ■

So merge pricing is weaker than full entropy pricing, though both force the one-unit toll under the permanence and finiteness premises. The prices are rules on moves. Reading one as physics takes Landauer's premise for the actual spread and a stated exchange rate from bits or units of cost to heat.

Now for the case where the floor is zero. Take the coin under a cup: toss a fair coin under it without looking at how it lands, and let the rule be to set it heads up. You might have had tails; now you have heads; one bit of what you didn't know is gone, and that's the bit the principle charges for. Now lift the cup, see heads, put the cup back, and apply the rule again. Nothing you didn't know changed, and no bit was lost. And the principle, applied to a state you already knew, demands nothing: the premise holds exactly when the heat is zero or more, which is to say it sets no positive floor. Setting a coin known to be tails to heads may have a finite-time cost under the chosen thermal dynamics; setting one you know is heads changes nothing. What the premise gives up either way is a floor that survives going slowly. Whether the hand that sets the coin keeps one is a question about the hand, and I come to it below. If you lift the cup after an uncertain toss, you have also made a record correlated with the coin. Resetting that record has to be analysed with those correlations and with the controls available. If instead the coin was supplied heads with certainty, there was no uncertain toss to record. These are different preparations. Knowing the result does not by itself locate a heat payment in the observer.

The entropy-loss bound depends on the preparation and on what the move does to it. Here "known" means a preparation concentrated on one logical state, or a conditional ensemble whose side information is explicitly included in the account. Merely telling a spectator which way the coin landed does not change an unchanged device's heat. This is the generalized form of Landauer's principle, worked out by Owen Maroney in 2009, in a paper he wrote at the Perimeter Institute in Canada (he has been a philosopher of physics at Oxford since 2011), and by Sagawa in 2014, and Landauer himself had the seed of it in 1961: for a row of bits all already ONE, he wrote, there's no need to carry out any operation, so no entropy changes and no heat is given off, and resetting a row all ZERO to ONE involves no entropy change either, because neither row carries any information.

□ **Theorem 3.17** (A known state gets no heat floor from the premise)**.** On a finite machine, if the spread puts all its chance on one state $s$, then any move removes zero bits, and Premise 3.5 for that move holds exactly when $Q \geq 0$.

*Proof.* A spread on one state has entropy 0, and so does its image, which is again on ■ one state. So $\Delta H = 0$ and the premise reads $Q \geq 0$.

Be careful with that one, because it's easy to overread. "The premise sets no heat floor for a known state" does not mean a stamping machine run on a known-blank sheet dissipates nothing. A model of the stamping apparatus may impose a finite-time cost, but the logical change alone does not establish it. The restricted thermal results are proved here for Bennett's walker (Theorem 1.9), found for a dragged bead by Schmiedl and Seifert and for the specified classical jump-process models by Shiraishi, Funo and Saito, and how much depends on the device's physical implementation, its control protocol and how fast it goes. What the theorem says is only that the *premise* sets no floor there. The floor comes from what you didn't know. That's a sentence about the premise's floor, and the device is a separate matter. And notice that the toll still charges the known-state stamp one unit, because the toll is a rule about the *move*, and the move doesn't know what the observer knows. That's the worst-case gap from the third way out showing up again, and it's the price of having a rule a machine can follow without asking who's watching.

The physics of all this has a standard review, the physicists Juan Parrondo, Jordan Horowitz and Takahiro Sagawa's "Thermodynamics of information" (2015), which sets Landauer's bound, its generalized form and the costs of measurement and feedback side by side. And the gap between a device and the spread it runs on has a physicist's number on it. Artemy Kolchinsky and David Wolpert (2017) showed that a fixed process built to dissipate as little as possible for one spread over its states pays extra when it runs on another, and the extra is how much closer the process brings the two spreads, measured in relative entropy; the extra was later named mismatch cost. That's the worst-case gap again, priced for an actual spread, where the toll prices the move.

Fix a thermal protocol with the entropy-balance identity just described, using the same controls and environment for each input. A floor then comes from comparing its inputs. Take $K$ inputs the device sends to the same output, and spread your chances evenly over them. The identity of Section 3 that ties Ladyman's reading to Sagawa's says the average, over the $K$, of the entropy the device produces on each one known equals the entropy it produces on the even spread, plus $k_B \ln K$. The second law makes the entropy produced on the spread zero or more. So the average is at least $k_B \ln K$, and the worst of the $K$ is at least the average: on some

one of them the device produces at least $k_B \ln K$ of entropy, and, when dissipated work is temperature times entropy production, at least $k_B T \ln K$ of dissipated work. Slowing this same protocol does not remove that worst-input bound. Kolchinsky and Wolpert's formula says, under their conditions, which input pays: the one the device's own best spread gave the least chance. A device can make one input cheap only by making another dear. Choosing different controls for different known inputs changes the protocol being compared. The bound above concerns one fixed protocol across them. So the floor sits on the merging move, on its worst input, whatever anyone knows, and that's the shape of the toll. The bound on a device left to relax from a known start, in the next subsection, is one more floor of this kind.

There's a second piece of fine print on Theorem 3.17, and it's about who works the controls. The entropy-loss bound supplies no positive floor for a known input. Whether a device can approach that floor also depends on its controls: lower this barrier, tilt that well, raise the barrier again. Those operations have a timing and a physical implementation. The entropy calculation alone does not price either.

Paul Riechers, Alexander Boyd, Gregory Wimsatt and James Crutchfield (2020) analyse metastable memories under *time-symmetric* control: the prescribed control is the same played forwards or backwards, with the relevant time-reversal conventions. A symmetric clock pulse can meet that restriction; an arbitrary clocked pulse need not. For memory states unchanged by time reversal, their low-error result singles out the memory states that the desired logical map, applied twice, fails to bring back to where they started. They call those transitions *nonreciprocal*. With equal local-equilibrium free energies, the leading minimum-work term as the error allowance $\varepsilon$ tends to zero is proportional to

$$k_B T \ln(1/\varepsilon)\ \Pr\big(C(C(m)) \neq m\big).$$

The probability is over the initial memory states. This is a low-error scaling with physical premises, not an exact finite-error heat charge on every execution.

A permanent flip from no to yes is nonreciprocal at that no-state: a second application cannot return it to no. A known input at that state can then face the extra work cost in their restricted control setting, even though its initial logical entropy is zero. Their class also includes a three-state cycle, which merges nothing. The distinction is about the permitted controls as well as the logical map. It neither changes Theorem 3.17's entropy calculation nor identifies the book's abstract merge price with a universal physical heat floor.

A deterministic run from a known start has a known state at every step, so the premise sets no floor under any of its steps; each that changes something can dissipate under the stated thermal-model assumptions, and a fixed device keeps the floor above on each step that merges. One more limit, about which states collide. The counting guarantees a collision among the already-yes states together with the

entering no-state. It does not guarantee that the entering state takes part in the collision. Take four states with only the first reading no, and a move that sends the first state to the second, the second to the third, and leaves the third and fourth where they are: the second and third both land on the third, and the first never shares with anyone. For an input distribution supported only on no-states, this counting argument alone forces no positive entropy loss. Whether other states in that support merge depends on the move. The operation-level ledger toll still follows under merge pricing.

What this section earns comes to this. A finite machine's permanent flag-raising instruction has two allowed starting states that land on one state. Prepare those two with equal chances: that instruction removes one bit, and under the adopted physical premise its mean heat is at least $k_B T \ln 2$. That is a worst-case preparation, not a positive charge on every execution. A known start gives no positive entropy-loss floor. Any finite-time dissipation depends on the device's dynamics, controls and speed; under the fixed-protocol entropy-balance assumptions above, merging $K$ inputs forces at least $k_B \ln K$ of entropy production on one of them. The ledger charges the merging move whatever the preparation, because merge pricing says so. The counting proves a collision, the preparation determines the uncertainty removed, and the physical premises connect that removal to heat.

The general case is the one that shows you where the bits go. Split the bits removed state by state. Each state $x$ you might be in carries its chance $p(x)$ along to where it lands, $\mathrm{step}(x,i)$, and finds whatever other chance landed there too. If nobody else landed there, its term is zero. If others did, its term is the difference between how crowded its landing spot is and how much of the crowd it brought.

□ **Theorem 3.18** (The bits removed, one term per state)**.** On a finite machine, for any move $i$ and any chances $p(x) \geq 0$, with $q$ the pushed spread,

$$H(p) - H(q) \;=\; \sum_{x:\, p(x)>0} p(x)\,\big(\log_2 q(\mathrm{step}(x,i)) - \log_2 p(x)\big).$$

Every term is zero or more, so a move never adds bits. If two different states $x \neq x'$, both with positive chance, land on the same state, the sum is strictly positive.

*Proof.* Group the sum for $H(q)$ by where things land: the chance $q(y)$ at a landing spot is the sum of the $p(x)$ that land there, so $H(q) = -\sum_{x:\, p(x)>0} p(x) \log_2 q(\mathrm{step}(x,i))$. Subtract from $H(p)$ term by term. Each term is non-negative because the landing spot holds at least what $x$ brought, $q(\mathrm{step}(x,i)) \geq p(x)$, and the logarithm keeps order. If $x$ and $x'$ share a landing spot, ■ that spot holds at least $p(x) + p(x') > p(x)$, so the term for $x$ is strictly positive.

### 3.8 What you're entitled to, and what you're not

Let me total it: what have you earned the right to conclude, and where is there still a way out?

You've earned this first, and it needs no premise but one next state for every state: on any machine run by one rule, finite or endless, a run that raises a reading and keeps it up either merges two of its own states or never repeats one, and on any machine at all a single move that raises a kept reading without merging anything has to send the kept states one-to-one into a part of themselves with a gap left over, which takes endlessly many of them. Either way it is a merge, or an endless supply of states never used before. You've earned this: on any machine with finitely many states, a move that raises a flag either forgets something or takes a flag down somewhere else. That is pure counting, with no physics and no premise beyond finiteness, and the half of it where nothing comes down is the pigeonhole on four states. You've earned this: on a finite machine, if a flag, once up, stays up, then raising it has a collision somewhere. If it raises several no-states at once, the yes-states together with those newcomers land on at least that many fewer spots than there are of them. This does not say that each newcomer shares its destination; this one is counting again. And you've earned this: if forgetting is charged for, the toll follows, and if forgetting is charged per halving, the size of the toll follows too; each of those derivations hands back a system with the five parts and the toll, so Section 2's floor applies to it as it stands. This is the rule from Section 2, now derived, with every premise named, for any finite machine with a flag that stays up. In that section I only declared it.

Now for what you haven't earned, which I list so you don't assume it. You haven't earned that forgetting costs anything in the world; that's Landauer, it's measured for beads and argued over for the general case, and in this book it's an assumption with a name. And even granting it, Landauer's entropy-loss bound demands no positive heat from a run whose state is known (the lifted cup of the coin example), so what the physics forces is a floor on a device that has to work whatever it's holding. An individual run can fall below an ensemble average bound. Knowing a state does not by itself prove that another person or memory paid the difference. You haven't earned that any machine you'd actually want to build is of this kind, either, and there are two easy ways to miss. A machine that keeps its running total inside its own state, as a number with no ceiling, has infinitely many states, so the finite premise fails for it as a whole. And you can't patch that by giving the total a ceiling. A total that only climbs, held in finite memory, eventually either stops at its top, and then two different histories leave the same top value, which is a merge, or rolls back over to zero, and then it doesn't only climb. So any running total a finite machine keeps for itself has a horizon: it's honest up to its width and no further. And a

machine whose price list charges the step that raises the flag, but lets some other step merge two states at no cost, raises the flag honestly and still doesn't charge for forgetting in general, so the merge-pricing premise fails for it. Either one puts a machine outside these theorems as stated, though a finite piece of it can still sit inside. The example I build later (Section 15) misses in both ways. It keeps counters with no ceiling, so it's infinite outright, and some of its ordinary steps, counting a counter down among them, can land two different states on one at no charge on the ledger. So the counting, as stated, doesn't reach it: it forces the toll only on machines with finitely many yes-states that price every merge, and that example is neither. It doesn't need the counting, though: the moves that raise its flag merge states themselves, so the same cost lands on them directly (Theorem 3.19).

A finite piece can still sit inside, and the smallest one I could build where every premise can be seen to hold is this. None of them is left as a hypothesis. It has four slots times a flag, eight states, and three kinds of move: stamp (set the flag, go to the next slot, cost one), jump to a chosen slot (cost two), next slot (cost zero). It's finite and permanent; the next-slot move forgets nothing; the stamp and the jump merge; merges are priced, and priced per halving. So this section's two conclusions hold on it outright. The counting lemma underneath, that a duplicate-free list in which no landing spot is hit by more than a set number of members has at most that many times as many members as landing spots, is the plainer reading of the halving price above. Merge pricing alone would take any price of at least one for the jump. The jump's price of two is what the halving price asks, since the jump squeezes four states into one. One more fact about it: each move's price is exactly the number of halvings in its largest squeeze. A move that costs one has no spot hit by more than two states, a move that costs two has none hit by more than four, and some spot is hit by exactly that many (stamp squeezes at most two states into one, jump to a slot squeezes four, and next-slot squeezes nothing).

The example Part III builds runs this eight-state machine. Seen through a window that shows only a line number, read modulo four as the slot, and a flag, a short program of that machine, once a two-instruction setup has run, does what the eight-state machine does, one move at a time, and pays exactly what the eight-state machine charges (Section 15.14 has the construction and the exact statements). The example as a whole prices its record layer and not every merge: some of its zero-price steps merge states, and whatever those merges cost the hardware is off the ledger, so the finite piece is a piece. The toll on the whole example comes from the record layer (Theorem 3.19).

Finiteness did one job in the counting. It forced the step that raises a permanent flag to merge two states. On the machine of Part III no count over the whole machine has to force that, because the merge is in plain view. Its record moves are the check, the commitment and the raised flag that the opening's lock puts in order, and it

names them CHECK, COMMIT and CERTIFY: COMMIT names a checked claim in a slot called the channel, and CERTIFY raises the flag. CERTIFY takes a state with the flag down and the same state with the flag up to one place, since a flag that's already up just stays up. Its COMMIT overwrites the channel, so two states that differ only in what the channel names land on one state. Part (1) of the theorem below shows both. So any price list that charges every merge of that machine's own step charges the flag-raising step, with no premise that the machine is finite. It isn't: its versions and its ledger grow without bound.

Finiteness is still there, in the smallest place it can be. Hold every part of the state but the flag fixed, and the flag has two readings, down and up. CERTIFY sends both to up, and it does the same to everything else from either, so the two states land together. That is the counting of Theorem 3.1 on a set of two. So the choice I made in the opening's list isn't the merge, and it isn't the shape of CERTIFY. It's that the step never reads the flag: the rest of the state moves by its own rule and the ledger adds the instruction's price, whichever way the flag reads, so no instruction keeps a copy of the old reading, and any that raises the flag lands its two readings together. Make CERTIFY trap instead when the flag is already up, and the merge moves rather than goes away: the state it traps lands with the state that was trapped already, since a trapped machine only pays. That's the halted state stepping to itself again, and it holds of every record move: any of the three that fails lands with the same state already trapped. The one way out of the flag's merge is to write the old reading into a part of the state that grows, a list of every reading so far, say, and that is a state never used before, the other column. A ledger that adds more for a re-raise than for a raise doesn't do it: the state with the flag up and the ledger a little lower lands where the raise lands, and the merge has only moved again.

That leaves the other miss, the unpriced merges. Clause (c) of certified Thiele-complete (Definition 15.7, in Part III) asks for base moves at zero, and a base move merges (DEC, counting a counter down, lands two states on one), so full merge pricing and clause (c) can't both hold on that machine. They don't have to. Split a state into its base part, which is what a two-counter machine would show (the line number and the two counters), and its record part, which is everything else: the versions (each counter's count of how many times it has been written), the table of facts, the channel, the trap latch (which goes up when a record move fails, after which only the ledger moves), the ledger and the flag. Then price only the merges the record layer makes.

**Premise 3.7** (Record-layer merge pricing)**.** A move that sends two different states with the same base part to one state costs at least one.

Part of where that line falls is forced. The line number has to be in the base part, or DEC, which can land two lines with the same counters on one, makes a record-layer merge at no cost; each counter has to be there, or its version, or DEC's own merge gets charged; and the flag or the trap has to be outside it, or nothing charges CERTIFY. Where the versions, the table, the channel and the ledger sit moves nothing below.

The merges among base parts are the ordinary computer's business, the ones any two-counter machine makes, and whatever heat they cost is the ordinary computer's heat. The premise prices only merges between states with the same ordinary-computer part. A step like DEC, which changes that part, can lose a version number at no cost, and the premise doesn't object. Merge pricing implies this premise, so it asks less, and on the record distinctions it is still an assumed operation-level price; identifying it with heat needs a physical model. And it is a price on this machine, not a source of the toll on any machine. A certifying move whose effect on the base read the flag, say one that moved a counter up when it raised the flag and left the counter alone when the flag was up already, would raise the flag and send no two states with the same base to one state, and the premise would have nothing to charge at the raise. It would still merge: the state with the flag down and the state with the flag up and that counter one higher land together, across two base parts, where the premise doesn't look. And on this machine its trap would still be charged when it fails. The small machine's record moves treat the base the same whichever way the flag reads, which is why the premise does its job there, and the theorem below is stated for that example and no other.

□ **Theorem 3.19** (The toll on the example machine)**.** For the small machine of Section 15, with no finiteness premise: (1) CERTIFY merges, and so do COMMIT and CHECK of every claim, and each joins two different states with the same line and counters. (2) Every cost that satisfies merge pricing for that machine's step charges CERTIFY, every COMMIT and every CHECK at least one, and obeys the toll, so the machine with that cost is a Thiele machine. (3) No INC, DEC or HALT sends two different states with the same line and counters to one state. So the machine's own prices, base moves at zero and each record move at one, satisfy Premise 3.7, they're the least prices that do, and merge pricing fails for them. (4) Every cost that satisfies Premise 3.7 obeys the toll, so the machine with that cost is a Thiele machine.

*Proof.* (1) For CERTIFY, prepare two states identical except for the flag. Give both a channel holding a commitment. One flag is down and one is up. CERTIFY raises

the first and leaves the second up. Both advance one line and add the same charge; every other field agrees. They have become the same full state.

For COMMIT, put the requested fact, at the counter's current edition, in both tables. Leave one channel empty and put that very commitment in the other. COMMIT writes the same commitment into both channels and treats every other field alike. Again the full successors agree.

For any CHECK, fill the fact table to its sixteen-entry limit. CHECK then fails, whatever claim it was asked to test. Compare that untrapped state with an otherwise identical state whose trap is already up. The first sets its trap; the second is already frozen. Neither changes the other core fields, and both pay the same charge. Their successors agree. In each construction the starting line and counters agree as well, so these are record-layer merges.

(2) A price on every merge charges each of those instructions at least one. Only CERTIFY can raise the certificate. Its merge therefore supplies the toll.

(3) Check the free instructions by asking whether two states with the same line and counters could become one. The table, channel, trap, ledger and certificate are left alone, so any difference in those fields survives. On a live state the shared line and counters choose the same instruction branch. A counter write adds one to its edition; different edition numbers stay different after the same addition. All unchanged editions stay different too. Hence no remaining field can hide a difference between those sources.

A trapped state stays exactly where it is under a free instruction. Two different trapped states therefore stay different. A live state remains live under these instructions, so it cannot meet a trapped one. That exhausts the cases. The free instructions make no record-layer merge, and all three kinds of record instruction do. Charging the former zero and the latter one satisfies the premise and is the smallest such price list. Full merge pricing still fails: the zero-price DEC pair in Theorem 15.35 starts with different counters and lands together.

(4) Any record-layer merge price charges the CERTIFY pair from (1). Since no other instruction raises the flag, this gives the toll for every raise. ∎

So this section reaches the example machine. On it the toll follows from record-layer merge pricing, and the example's own price list, clause (c) included, satisfies that premise. This is a result about its ledger, not a heat bound. There's a second way to settle the clash, which I note and didn't take. Morita's reversible two-counter machine, from the history earlier in this chapter, runs every step backwards and is still universal: with its input encoded, it computes anything a Turing machine can. Put the record layer on a base like that, with the base's step taken as the program's own step from line to line and the halt left with no next step, and every base step is one-to-one, and full merge pricing permits those base steps to cost zero.

Choosing that schedule would meet clause (c); the lower-bound premise alone does not require zero charges. It wouldn't meet clause (a) as I stated it, though. Clause (a) asks for a move for each two-counter instruction, acting from every live state, not a step for each line of one program, and DEC $c$ $j$ sends states on any two lines with the same counters, $c$ above 0, to the same window, so whether it merges turns on the rest of the state and not on the program being reversible. That base wouldn't be getting anything for nothing: what it doesn't forget it keeps in its counters, which is the column of states never used before, again. I haven't built that machine.

And you haven't earned anything about *what* the flag means. The pigeonhole on four states doesn't know what it's certifying. The cost is for raising a flag that stays up. Whether raising that flag ever amounts to standing behind a structural fact, the "vouching" I keep promising and keep not delivering, is a question for the certified example in Part III, and here it has moved exactly this far: on a finite machine that charges for every merge, if that flag stays up, it was paid for.

One last thing, because it should nag at you. The toll is forced by the premises, but the premises were chosen, and "permanent" in particular was chosen by me. Why should a flag stay up? Why not let it come down, pay nothing, and be honest about it? That's a fair question, and merge or revoke is the start of an answer (the three-card machine only showed that the converse of Corollary 3.9 fails): on a finite machine that charges for every merge, a flag that goes up at zero cost is a flag the same move takes back somewhere, and a record that can be silently taken back is the thing the bulla and the hash chain and the write-once disk were all built against. "Permanent" here is exactly that: the flag can't be lowered. The toy in Section 2 has it only because none of its three moves happens to lower the flag, and it does none of a record's five jobs; the lineage in that section, the bulla and the write-once disk, was built around this one, durability; I chose it. Whether that was the right job to pick out of the five is something you can argue with me about, and I'd want you to.

And "permanent" is an idealization, which I should say before a physicist does. No physical record lasts forever. A bit held behind an energy barrier leaks over it at some rate, however slow, and a closed finite system comes back around, which is Zermelo's point from the counting (Theorem 1.6 is the finite form). The toll theorems in this section, the ones that start from a flag that stays up, need permanence on every state the machine can reach, including states a given run never visits, forever; on states nothing reaches it never comes into it, and merge or revoke does without it. Weaken that to "the flag stays up on the states that actually turn up, for as long as anyone cares," and merge-or-revoke tells you what goes wrong: the move could take the flag down somewhere the machine could go and never does, and then it isn't forced to merge at all. A version of the toll stated in chances is the noisy steps of Section 1: a step that squeezes nothing on average can't raise a reading that stays,

and one that can strictly bunches the even spread. A version stated in time spans is a different theorem, and what I have of it is below.

One piece of it I can hand you, and the step law of Section 1 is its general form. Take one move and repeat it from one start, forever. On a finite machine the states that run passes through are finitely many, so it falls into a loop. If the move forgot nothing on those states, it would be a reshuffle of them, and a reshuffle brings every one of them back around, the start included, where the flag was down. So if the flag goes up along that run and stays up from some point on, the move merges two states *the run itself visits*. The state they share is one the machine actually passes through. And it doesn't touch the known-state point above: a merge among states the run visits still receives no positive lower bound from the entropy-loss premise when its state is known. The argument holds for any map on finitely many states iterated from a start whose flag is down, with the flag up at every step from some point on; no permanence premise is used.

Runs that mix different moves split two ways, and the answer is short enough to give here. If the moves are picked from outside, in any order at all, the answer is no. Take two states, one reading no and one reading yes. One move swaps them; the other does nothing. Run the swap once and then the do-nothing move forever: the flag goes up at the first step and stays up for good, and neither move merges anything anywhere. The swap would take the flag down from the yes-state if it ever ran there; on this run it never does, and that's exactly the weakened permanence this paragraph is about. But if the machine is closed, picking its next move from its own state, the run is the orbit of one map, the step with whatever move the state picks, and the single-move argument applies to that map as it stands. Now say the moves come in a pattern that eventually repeats, one phrase of moves played over and over. Suppose the flag is down at some step after the repeating starts, and goes up and stays up after that. Count the repetitions from a step where the flag was down (start the phrase at that move if you have to) and run the same argument on the whole phrase treated as one move, at the states where each playing of it begins. It merges two of those states, both visited; follow the two through the phrase, and the first move where their paths meet merges two states the run visits at that move's place in the pattern. So the answer is no for moves picked from outside, and yes for a closed machine, or for a repeating program once the flag is down somewhere after the repeating starts.

There is also a limited statement about how long a noisy record lasts, using a different quantity: total entropy production during relaxation. Take a device with no outside hand: a random process hopping among finitely many states at rates that never change, with a reverse for every hop and local detailed balance, able to get from any state to any other. Require at least one up-state and one down-state. Left to run, it settles into a long-run spread $\pi$. Measure each up or down stretch from

its entrance, with entrances sampled in the stationary process, and call the mean lengths $T_{\text{up}}$ and $T_{\text{down}}$. Start the device in a known no-state $s$. The total entropy it produces over relaxation is at least $\ln(1 + T_{\text{up}}/T_{\text{down}})$, in units of $k_B$.

First compare the starting spread with the spread the device settles into. A state that is common at the start but rare in the settled spread is a large mismatch. For each state, take the ratio of those two chances, take its natural logarithm, and average using the starting chances. With $p$ the starting spread, that number is

$$D(p\|\pi) = \sum_{x:p(x)>0} p(x) \ln \frac{p(x)}{\pi(x)}.$$

It is called relative entropy. Here every $\pi(x)$ is positive, so the ratios are available. The number is nonnegative: apply $\ln y \le y - 1$ to $\pi(x)/p(x)$, multiply by $p(x)$, and add. The right side is at most zero, since the included $\pi$ chances total at most one and the $p$ chances total exactly one. Negating that comparison gives $D(p\|\pi) \ge 0$.

For a known start $s$, all the starting chance is at $s$. There is only one term: $D = \ln(1/\pi(s)) = -\ln \pi(s)$. A state with settled chance one tenth therefore starts with mismatch $\ln 10$. Since $s$ is a no-state, its settled chance cannot exceed the total chance of all no-states. Its mismatch is at least $-\ln \pi(\text{no})$.

The physical step in the argument is the fixed-rate entropy-production split of Massimiliano Esposito and Christian Van den Broeck (2010). In that setting the total entropy produced is at least the decrease in this relative entropy. Once the spread has settled to $\pi$, the relative entropy left is zero. So the initial mismatch supplies the lower bound for the whole relaxation.

Now read the same number from the record of up and down stretches. In a long settled run, each entry into YES is followed by an exit; apart from the two ends of the record, there are equally many up-stretches and down-stretches. Total time up is the number of up-stretches times their mean length, and likewise for time down. Dividing gives

$$\frac{\pi(\text{up})}{\pi(\text{no})} = \frac{T_{\text{up}}}{T_{\text{down}}}.$$

The two chances add to one. If the ratio on the right is $r$, the no-share is $1/(1+r)$, and its negative logarithm is $\ln(1+r)$. That gives the stated bound. The settling and mean-length arguments below justify passing from a long record to these limiting quantities.

For a large up-to-down ratio, doubling that ratio raises this lower bound by about $k_B \ln 2$. Making both stretches longer by the same factor leaves it unchanged. Across a family of devices whose ratio tends to infinity, the bound diverges; no single finite irreducible device in this class keeps its flag up forever. This is a relaxation bound, not a heat charge on each flip. It uses neither a thermodynamic reverse-path

construction nor time-symmetric control, so it does not identify the quantities in those other results. It also leaves the known-input case of the deterministic entropy-loss bound where it was: this noisy evolution and its total entropy production are a different question. The entropy is added up over one whole relaxation, from the known no-state $s$ until the device has settled, and local detailed balance is what ties it to heat exchanged with the reservoirs. Total entropy production includes the change in the device's entropy too; it is not heat alone. It also isn't the permanence of the finite-state theorems on a noisy device: on this device the flag comes back down, and $T_{\text{up}}$ and $T_{\text{down}}$ are average stretches.

The core of it for a chain that ticks, worked out, runs as follows in the case where the long-run spread $\pi$ gives every state a positive chance and satisfies detailed balance: in the long run as much weight flows from $x$ to $y$ in one step as from $y$ to $x$, $\pi(x)P(x,y) = \pi(y)P(y,x)$, where $P(x,y)$ is the chance of stepping from $x$ to $y$. Count the entropy produced in one step as the change in the device's own entropy plus what goes out to the surroundings, which for a hop from $x$ to $y$ is $\ln(P(x,y)/P(y,x))$. Detailed balance turns that ratio into $\pi(y)/\pi(x)$, and the sum collapses: the entropy produced in one step is exactly the drop in relative entropy to $\pi$. It's never negative, because a step that leaves $\pi$ where it is can't make any other spread harder to tell from $\pi$ (that's the data-processing inequality, a standard fact of information theory). Add the steps up and the sum telescopes. A known start $s$ has relative entropy $-\ln \pi(s)$, so the entropy produced in the first $T$ steps is $-\ln \pi(s)$ minus the relative entropy left at step $T$: it never exceeds $-\ln \pi(s)$, and it reaches it in the limit exactly when that remainder goes to zero. Next, $-\ln \pi(s) \geq -\ln \pi(\text{no})$, because $s$ is one of the no-states and so carries at most their total chance. The flow from the no-states to the yes-states equals the flow back, because in the long run the weight on the yes-states doesn't change, so what flows in equals what flows out. Last, take each mean stretch as its mass over that flow; then $T_{\text{up}}/T_{\text{down}} = \pi(\text{up})/\pi(\text{no})$, and since the two chances add to one, $-\ln \pi(\text{no}) = \ln(1 + T_{\text{up}}/T_{\text{down}})$.

Try the stretches with a bag of ten counters: nine marked YES and one NO. Draw, record the answer, replace the counter and mix again. Each draw gives YES with chance $9/10$, whatever the last answer was. A YES stretch ends on the next draw with chance $1/10$, so its mean length is ten draws: the mean $L$ obeys $L = 1 + (9/10)L$. A NO stretch similarly averages $10/9$ draws. Their ratio is nine, just like the ratio of YES to NO counters. In a long record, count all the YES entries and divide by the number of YES stretches; that is the measured mean stretch length. Divide both counts by the total number of draws and you have mass divided by flow.

For this rule the long-run flow from NO to YES is $(1/10)(9/10) = 9/100$ per draw, and the reverse flow is the same. A start known to be NO has relative entropy $-\ln(1/10) = \ln 10$ against the long-run spread; after one draw the spread is already $1/10, 9/10$, so none of that relative entropy remains. This bag shows the probability

accounting. The entropy-production interpretation still needs the stated physical model; shaking counters in a bag does not measure that heat.

Two things need establishing for a general device: that it actually settles, and that "mean stretch length" means what it sounds like. I'll say what each says, and why it holds.

Settling comes first. A chain doesn't have to settle. It can get stuck in a corner, or cycle round forever. What rules that out here is a condition named for Wolfgang Doeblin, a German-born French mathematician, son of the novelist Alfred Döblin, who took his doctorate at the Sorbonne in 1938 for work on chains like this one and died in June 1940, at twenty-five. In words: from wherever the device is, its next step lands on each state with at least some fixed fraction $\delta$ of that state's long-run chance. No state is ever out of reach in one step, and none is favoured so heavily that the others starve. Under that condition each step shrinks the distance between where the device is and where it's going by the factor $1-\delta$ (the distance here is the total of the differences in chance, state by state). To see why, split each step in two: with chance $\delta$, forget where the device was and draw a fresh state from $\pi$; otherwise, move by whatever is left over, and the condition is exactly what makes the leftover a genuine set of chances. The fresh draw puts the device's spread and $\pi$ in the same place, so only the other share, $1-\delta$, can keep them apart, and no step pushes two spreads further apart than they were. So the distance dies off geometrically and the spread converges to $\pi$ at every state. The payoff for the bound is that the entropy produced from the known start stays under $-\ln \pi(s)$, and it also gets there: after $T$ steps it falls short by at most $(4/\pi_{\min})(1-\delta)^{2T}$, where $\pi_{\min}$ is the smallest long-run chance of any state. The shortfall is the relative entropy left, and $\ln y \le y-1$ once more puts that under the sum of $(p(x)-\pi(x))^2/\pi(x)$, which is at most the square of the distance over $\pi_{\min}$; the distance starts at no more than 2 and shrinks by $1-\delta$ a step. So $-\ln \pi(s)$ is the exact total entropy production of the whole relaxation, in units of $k_B$, and the bound $-\ln \pi(s)$ is reached.

For rates satisfying detailed balance with $\pi$, the same equality holds when the device hops at random moments, with no clock ticking. Write out how fast the relative entropy to $\pi$ changes and pair each hop with its reverse. Detailed balance makes each pair's share minus (the flow one way minus the flow back) times the logarithm of their ratio, and that is exactly the pair's share of the rate entropy is produced; a difference times the logarithm of the same ratio is never negative. So at every instant the relative entropy falls at exactly the rate entropy is being produced, and that rate is never negative; add the rate up between two times and you get exactly the drop in relative entropy between them. For settling, Doeblin's condition becomes a condition on rates: the rate of hopping from any state $x$ to any state $y$ is at least $c$ times $\pi(y)$, for some fixed $c>0$. Split the rates in two again: a part that, at rate $c$, forgets where the device was and redraws it from $\pi$, and the rest,

which still satisfies detailed balance. The production rate adds over the two parts, because with detailed balance it's the rates times quantities that don't depend on the rates. The rest produces nothing negative, and the redraw part alone produces at least $c$ times the relative entropy still left. So relative entropy falls at least as fast as $e^{-ct}$. One wrinkle needs its own line. A known start puts all its weight on one state and none anywhere else, and the logarithms in the relative entropy go bad at zero. The relative entropy is still continuous at the starting instant, because each of its terms has the form $u \ln(u/\pi)$, and $u \ln u$ goes to zero as $u$ does. So adding up the production from the start out to infinity gives exactly $-\ln \pi(s)$.

I should say what Doeblin's condition costs. It asks for every hop to be possible directly, in one step or at a positive rate. A device whose states reach each other only through other states can still settle, and the standard theory of these chains says when: always, for a device that hops at random moments and can get from any state to any other; and for a chain that ticks, as long as it also doesn't march round a cycle. Those devices meet the condition for some fixed stretch of time or number of steps instead of one, and the same argument runs on blocks of that length, because the relative entropy never goes up in between.

Then come the stretches. The bound used the fact that the long-run chances of up and down stand in the ratio of the mean stretch lengths, and I took the mean length of a stretch with the flag up to be the long-run weight of the up-states divided by the flow into them. For a chain that ticks, that's right for the following reason. Start each stretch the way the flow starts them, count the steps until the chain leaves the up-states, and the average is exactly $\pi(\text{up})$ over the flow. Look at the long-run weight that has been up at each of the last $k+1$ ticks. It splits in two: the part whose stretch began exactly $k$ ticks ago, which is the flow times the chance that a fresh stretch lasts more than $k$ ticks, and the part that was up one tick earlier still. Add those splits from $k = 0$ up to some $K$ and the sum telescopes: $\pi(\text{up})$ is the flow times the sum of those chances, plus the weight that has been up for more than $K$ ticks. The sum of the chances that a stretch lasts more than $k$ is its mean length. Doeblin's condition makes the leftover die: from any up-state the next step lands in the down-states with chance at least $\delta\,\pi(\text{down})$, so the weight that has stayed up for $K$ ticks is at most $(1-\delta\,\pi(\text{down}))^K$.

With the stretches measured in seconds, when the device hops at random moments, it comes out the same, by a different route. Ask, for each up-state, how long on average the device takes to leave the up-states from there. Call that its leaving time. The leaving times are finite: from every up-state the device hops into the down-states at a total rate of at least $c\,\pi(\text{down})$, so the chance a stretch is still running at time $t$ falls at least as fast as $e^{-c\,\pi(\text{down})\,t}$, and no leaving time is more than one over $c\,\pi(\text{down})$. And they obey one plain rule, which you get by asking what happens in the first instant: from any up-state, the rate out of it times

its leaving time, minus the rates into the other up-states times their leaving times, comes to exactly one (per second times seconds is a pure number). Average the leaving times over the way the flow starts stretches, multiply by the flow, and the rule plus the balance of the steady spread (as much weight flows into each state as out of it) rearrange the sum into exactly $\pi(\text{up})$, so in seconds too the mean stretch is its mass over the flow. Both of the bound's premises are proved now, in steps and in seconds, under Doeblin's condition.

One chapter remains in this part, and it's about a different way out entirely. Forgetting has had its chapter; the question after that is whether *finding out* costs, because the demon's whole trick was that it knew which way the molecule was going, and I owe you a straight answer on what knowing costs.

CHAPTER 4

# Learning borrows, committing pays

I ended the last chapter owing you an answer about knowing. What keeping takes so far sits on forgetting: under merge pricing, a finite machine's permanent flag-raising instruction was paid for because it merges two states somewhere. But there's a thing that looks like the opposite of forgetting and feels like it ought to cost at least as much, probably more, and that is finding out. Every time a machine rules out a possibility, it knows more than it did. If forgetting a bit costs, does learning one cost too? I had a strong feeling about this, and you probably have it too, so I'll say it first: learning is the expensive part. Most physicists had that feeling as well, for a long time. It's why the demon took a hundred and fifteen years to exorcise, or didn't. "Exorcise" is the word the people who think it worked like to use. John Earman and John Norton, the doubters from the last chapter, called their two-part paper on it "Exorcist XIV: The Wrath of Maxwell's Demon," and I read that Roman numeral as a joke about how many times the demon has been declared gone. So hold the word loosely. This section is me finding out where the feeling goes wrong, and where it's right. I'll give the answer up front. Looking can have a physical cost; its value needs a model of the measurement. But it doesn't have to merge anything, so it owes no floor, and a ledger that charges merges can charge it nothing. What it does use up is a blank page to write on. Learning borrows that page. Wiping the page to use it again is a merge, and that's where the floor gets paid. Committing, making a record nobody can hand back, is paying for good. I'm going to tell you the version where the exorcism worked, Charles Bennett's, because it's the version that says where the heat goes, and the third way out in the last chapter, the one Landauer's premise is there to close, stays open behind it.

## 4.1 Somebody got there first, goddamn it

I'd been sitting on this idea that *committing*, making a result stay so it can't be handed back, is what pays, and learning, writing on a blank page you could hand

back, only borrows, and I thought: this one's mine, I finally had an original thought. Then I found Carlo Rovelli's 2024 paper, "The Thermodynamic Cost of Choosing." I hadn't. Somebody got there first, goddamn it: a real physicist, with a real paper, saying that making a choice that sticks dissipates at least the energy Landauer charges for erasing a bit. He even quotes a German saying, *Wahl ist Leiden*, to choose is to suffer, and at that point what are you going to do but laugh.

And then, taking him apart properly, it got good, because on its face he contradicts Bennett. Bennett says measuring owes no floor; Rovelli says any measurement costs. So which is it? Look at what Rovelli calls a choice. A coin in the air hasn't chosen anything; it has to hit the ground and stop. And then comes the sentence I'd put on my wall: "stabilization must be irreversible, otherwise it is not stabilization." Bennett's floorless measurement is one that could still be undone; that's *why* it owes no floor. It only borrows the page. Rovelli's choice is one that's been made to stay; that's why it costs. The way I read them, they're standing on opposite sides of the same line: learning, which borrows, on Bennett's side; committing, which pays, on Rovelli's. (Bennett's demon still pays in the end, when it wipes the page.) Neither of them puts it that way, so push on it. But if it holds, it's the best sign I've got that the line I'm drawing between finding something out and making it stay isn't something I made up.

Push on it with his paper open and it gives in four places, which I go through at the end of this chapter, once the demon has been built. So the two-sides-of-one-line picture is my reading, not his, and the piece of it that's really his is the one sentence about stabilizing. That sentence I'll still put on my wall.

So he's my closest neighbour, and the difference between us is exact. His is physics: a dissipation bound, at least $kT \ln 2$ at the temperature of the recording parts, with no outside heat bath, for a choice that sticks. Mine is a rule in a model of computation: a charge in the machine's own count, on the step that flips a permanent reading on, with no temperature anywhere in it. It's the same instinct in a different place, and I'll take it, because it's annoying to be second but worse to be alone and wrong. And he's the closest only on choosing. On copying there's an older neighbour, in the chemistry of copying molecules, and the end of this chapter gets to it.

## 4.2 A small being at a small door

You met Maxwell's demon briefly in Section 2. It gets a whole section here, because the step it seemed to pay for and the step it actually pays for are what this section is about.

In December 1867 James Clerk Maxwell, who had published his equations for electricity and magnetism two years before, in 1865, and to my mind had as good a

claim as anyone to being the most important physicist alive, wrote a letter to his friend Peter Tait describing a problem he couldn't get rid of. Take a box of gas at one temperature, with a wall down the middle and a tiny door in the wall. Now imagine, he said, a "very observant and neat-fingered being" sitting at the door, who can see individual molecules coming. When a fast one approaches from the left, it opens the door and lets it through. When a slow one approaches from the right, same. Otherwise it keeps the door shut. It never pushes anything; it only opens and closes a frictionless door at the right moments. After a while the right side is hot and the left side is cold, and you've got a temperature difference you could run an engine off, made out of nothing but *looking*. That's supposed to be impossible; it's the thing the second law of thermodynamics says can't happen, and the second law is the law physicists are least willing to give up. Arthur Eddington, the English astronomer, wrote in a 1928 book of popular lectures that a theory found to contradict it has nothing for it but to collapse in deepest humiliation, and nobody has had to test that yet.

Maxwell put it in his textbook in 1871, and it got named Maxwell's demon, though not by Maxwell. William Thomson, later Lord Kelvin, called it a demon in *Nature* in 1874, and Maxwell, in an undated note to Tait headed "Concerning Demons," tried to take the name back: "Call him no more a demon but a valve . . . " Nobody listened, which is probably for the best, because "Maxwell's valve" would not have lasted a century. But keep his correction, because it's exactly right and it's the whole problem: the being does no work. It opens and shuts a door. So physicists spent the next century trying to find where the being was cheating. I want you to feel the pull of the obvious answer, because I felt it and so did most of them: the being has to *see* the molecules, seeing costs, and the case is closed. In 1929 Leo Szilard built a one-molecule version of the box and did the arithmetic, and said the second law survives if finding out which side the molecule is on costs at least as much entropy as the being gains. In 1951 the French physicist Léon Brillouin made it concrete: to see a molecule you have to bounce light off it, and the way the philosopher of physics Jeffrey Bub sums up the result, finding out that way costs at least k ln 2 of entropy for every bit, which is kT ln 2 of heat, Landauer's number before Landauer, and there was your charge on looking. For thirty years that was the answer. Information costs because getting it costs.

That's a price on one way of looking, and it doesn't show that every possible way of making a record has to pay it. A device can burn energy without that being a law.

Szilard is worth one more minute, for a reason that has nothing to do with the demon. Four years after the one-molecule engine, in September 1933, by his own later telling, he was waiting at a traffic light on Southampton Row in London, having just read a newspaper report in which Ernest Rutherford, the physicist who had announced the atomic nucleus in 1911, called the hope of getting power out of

atoms moonshine, and as the light changed it occurred to him that an element which gave off two neutrons when it swallowed one could keep a reaction going by itself. That's the nuclear chain reaction, thought up crossing a street. It's his memory of himself, and Richard Rhodes's history of the bomb opens with it, in a first chapter called "Moonshine," which is where most people meet it, so hold it as the man's own story, not a record made at the time. Does it change anything about the demon? No. What it adds is a picture of the man: the same mind looked at one molecule in 1929 and one neutron in 1933, and both times the question was what a single small step forces downstream.

### 4.3 Bennett moves where the heat is paid

In 1982 Charles Bennett, who'd already shown in 1973 that computation can be done without forgetting if it keeps its whole history and pays by taking states it has never used instead, took the demon apart one step at a time. The being looks at a molecule and writes down which side it's on. Is writing that down a merge? Picture the being's notepad, which starts blank. After looking, the notepad says "left" or "right." There are two possible worlds before the look (molecule left, pad blank; molecule right, pad blank) and two possible worlds after it (left, "left"; right, "right"), so nothing merged: two went in, two came out, and you can read the before off the after. Copying what you see onto a blank page forgets nothing, and a step that forgets nothing owes no floor under Landauer's premise. Any finite-time physical cost additionally depends on the chosen dynamics and protocol. But it did use something up: the blank page. Before the look, the pad was blank and ready; after it, the pad is full. Brillouin had priced one particular way of looking (bouncing light) and took it as the price of getting information at all. Bennett gave a way of looking that, on his account, doesn't pay that price. It borrows a page instead.

So where is the heat floor? The being has one notepad and a lot of molecules. After the first molecule the page is full, and to do the next one it has to wipe the page. *Wiping* takes two possible worlds (pad says "left"; pad says "right") to one (pad blank), and that's the merge, the step with no way back. For a complete reusable engine under the stated thermal assumptions, the memory reset and the feedback process must be accounted for together. The cycle cannot turn information about the molecule into net work from a single bath while restoring every other resource for free. It's the rule from his 2003 notes that I quoted in Section 2: erasing and merging carry the unavoidable charge, and copying onto a blank place to write doesn't. The look borrows the page. Resetting an uncertain page without using its correlation with the molecule has the corresponding entropy-loss floor under the thermal assumptions. While the molecule is still available, the joint operation can instead undo the copy; the four-state example below does that. A blank supplied at

the start is an initial condition. To charge its preparation, specify what uncertain state was reset and how.

Section 2 separated making a record from resetting it. Now you can see exactly which step pays, and why the other one was tempting, because looking *feels* like the active part, the thing the being does, while wiping feels like housekeeping. In that reset protocol, the housekeeping is where the entropy-loss floor applies. If there's one sentence in this section I'd want you to carry out of it, it's that one, and the idea in it is Bennett's.

## 4.4 Two different things called “narrowing”

Here is what I had to get straight in my own head before any of this made sense, and I'll give it to you as a picture because that's how I eventually got it.

Draw the machine's possible states as a row of dots. Now draw a second row above it: the dots that *you*, standing outside, still think it might be in. Those are two different rows: the first is a fact about the machine, and the second is a fact about you, about what you've seen and what you've ruled out. Both rows can shrink, but they shrink for different reasons, and only one of them costs.

The machine's row shrinks when two of its states land on the same state, which is a merge, and a merge is what the price charges. Under the halving rule from Section 3, a run can't shrink the machine's own row by more halvings than it paid for. That's the theorem just below, and it's that section's halving price run forward along a whole program, where that section took it one step at a time. There is no new idea in it, only the same counting of states.

To state it exactly, write $\lceil \log_2 n \rceil$ for the index-bit count of $n$: the smallest whole number $k$ with $2^k \geq n$, the fewest yes-or-no questions that can guarantee identification among $n$ things when arbitrary binary splits are allowed. It is 0 at $n = 1$, and 0 at $n = 0$ too, since $2^0 \geq 0$; the definition covers empty lists, where the logarithm of zero isn't a number. Wherever I write an ordinary logarithm without the rounding, its argument is positive. For a run $t$ and a list $D$ of starting states, let $\mathrm{img}_t(D)$ be the set of states the run can end in, starting from somewhere in $D$, and write $|\mathrm{img}_t(D)|$ for how many there are (in Section 3 the symbol for one move already meant the count; here I need the set and its size both). Let $\mathrm{cost}(t)$ be the run's total cost.

**Theorem 4.1** (The machine's own narrowing is priced)**.** If the cost satisfies the halving price (Premise 3.4), then for every run $t$ and every list $D$ of starting states with no repeats, □

$$|D| \;\leq\; 2^{\mathrm{cost}(t)} \cdot |\mathrm{img}_t(D)|, \qquad \text{so} \qquad \lceil \log_2 |D| \rceil \;\leq\; \mathrm{cost}(t) + \lceil \log_2 |\mathrm{img}_t(D)| \rceil .$$

*Proof.* After each move, gather the states still possible and count each different state once. If the first move costs $c_1$, the halving rule says the original count is at most $2^{c_1}$ times this new count. If the next move costs $c_2$, the new count is at most $2^{c_2}$ times the count after that. Put the inequalities together: the original count is at most $2^{c_1}2^{c_2}$ times the count after two moves.

Carry on to the end. The factors multiply, their exponents add, and the added prices are exactly $\mathrm{cost}(t)$. The last collection is $\mathrm{img}_t(D)$, giving the first inequality. An empty run contributes no factor and leaves the collection unchanged, so it fits too.

For the rounded count, let $k$ bits suffice to number all the endpoints. There are then at most $2^k$ endpoints, and the first inequality gives at most $2^{\mathrm{cost}(t)+k}$ starts. That many bits therefore suffice for the starts. This also covers an empty starting collection, whose stipulated index-bit count is zero, without taking a logarithm of zero.

Rounding throws some information away. Four starts and three endpoints both need two index bits, so the rounded comparison alone permits zero cost. The unrounded comparison still sees that four is more than three and forces a positive charge. ■

The run itself is the witness; nothing else is handed in. That will matter in Section 8, and I'll say why at the end of this section.

Your row shrinks when you see something that some of your candidates couldn't have shown you, and you cross those candidates off. Nothing happened to the machine. It didn't merge anything; it didn't know you were watching; it would have done the same thing with nobody there. The physicist Edwin Jaynes had a name for mixing these up, and I think it's one of the most useful names in all of science: the mind projection fallacy, treating what you know about a thing as if it were a property of the thing. Your list of candidates is in your head, and the machine's row is in the machine. The halving price is about the machine's row, and crossing a candidate off yours doesn't move it.

And here I have to be fair to Jaynes, because he'd argue with where I just took his phrase. In a 1965 paper on Gibbs's and Boltzmann's entropies he wrote that entropy is "an anthropomorphic concept" (a phrase he credited to the physicist Eugene Wigner), and by that he meant a property of the particular experiments you or I choose to do on a physical system. On his account the system alone doesn't carry it. So for Jaynes the second law is about what's known too, and the line I'd have liked to draw, that the second law has opinions about the machine and none about you, is exactly the line he'd refuse. Who's right is one of the oldest fights in statistical physics, Jaynes's reading against Boltzmann's, and I can't settle it and don't have to. The spreads in Section 3's entropy theorems are spreads over the

states the device might be in when the step runs, and every theorem there holds whether you read those chances as ignorance or as physics; “the floor comes from what you didn’t know” is the same sentence in either reading, with “you” being whoever built the spread. What I need from the fallacy is only this: the two rows are different rows, and only one of them is the machine’s.

## 4.5 The demon on a sheet of paper

Build it, because the whole claim is four dots and two arrows and you should see it merge or not merge with your own eyes.

The machine has a hidden bit (think of it as the molecule’s side) and a display (the notepad). It has four states, written as pairs, hidden bit then display: 00, 01, 10, 11. It has two moves.

MEASURE copies the hidden bit onto the display, by adding it in the way that a light switch adds: if the hidden bit is 1, flip the display; if it’s 0, leave it. Draw the arrows: 00 goes to 00, 01 goes to 01, 10 goes to 11, and 11 goes to 10. Look at the arrowheads: every state has exactly one arrow landing on it, so nothing merged and MEASURE forgets nothing. Look at the two states where the display starts blank: 00 goes to 00 and 10 goes to 11, which is the copy onto a blank page, (hidden, blank) to (hidden, hidden), and it’s the use that matters here. And the light switch isn’t one way of copying among several. A MEASURE that merges nothing, copies onto a blank display and leaves the hidden bit where it was has nowhere else to send 01 and 11: each has one landing left on its own side.

WIPE sets the display to 0 whatever it held. 00 goes to 00; 01 goes to 00 as well. 10 goes to 10; 11 goes to 10 as well. Two arrows land on 00 and two on 10, which is a merge, twice over.

| Hidden bit, display before | After MEASURE | After WIPE |
|---|---|---|
| 00 | 00 | 00 |
| 01 | 01 | 00 |
| 10 | 11 | 10 |
| 11 | 10 | 10 |

Now follow only the prepared cases: 00, 10 become 00, 11 after the copy. WIPE sends those two to 00, 10, still two different full states. The hidden bit retains the distinction. The instruction price nevertheless charges WIPE because its rule merges other possible pairs, as the full column shows. That worst-case charge is not a one-bit entropy loss on this particular correlated preparation. Under any price that charges per halving, WIPE costs at least one, while MEASURE is allowed to cost zero. Zero on the price list isn’t zero spent: a physical MEASURE can dissipate under its specified dynamics, and it uses up the blank display. What it doesn’t do is merge, so it owes no floor.

Now be the observer. You can see the display and not the hidden bit. What you're shown, the display here, is what I'll call a *window*: the part of the state an outside observer gets to see. Section 7 is about windows in general; here the window is just the display. The display starts blank, so your list of candidates is {00, 10}: two possibilities, one bit you don't know. The machine runs MEASURE and you look at the display. If it shows 1, the only candidate that could have produced it is 10, and if it shows 0, only 00, so your list went from two to one and you learned the hidden bit. The machine's price for that move, on a list that charges merges, is zero, and what it spent was the blank display. Its own row of possible states went from {00, 10} to {00, 11}, still two; nothing it could have been got merged with anything else. You narrowed and it didn't. That's the measuring half of Bennett's argument on four dots, and WIPE's charge is the wiping half. So the demon just broke something: one bit gained, nothing charged. The thing it broke, written out so you can see exactly where the crack runs, is this: for every list of starting candidates, every run, and every actual start in the list, the index bits the observer gains (log2 of the candidates before minus log2 of the candidates after, both rounded up) are at most what the run was charged. On four dots that sentence is false.

**Definition 4.1** (What an observer knows)**.** Fix a window $w$ from states to what can be seen. Given a list $\Omega$ of candidate starting states, a run $t$, and the actual start $s_0 \in \Omega$, the observer's *knowledge* $K(\Omega, t, s_0)$ is the list of candidates $s \in \Omega$ whose run shows the same thing through the window at every step as the actual run does.

If that definition looks familiar, it should. It's the standard way logicians of knowledge set it up: what you know is whatever holds of every case you can't tell apart from the actual one, given everything you've seen (Fagin, Halpern, Moses and Vardi's *Reasoning About Knowledge*, 1995, builds a whole subject on it, runs and observations included). And it's the logically omniscient kind, the kind where knowing some facts means knowing everything that follows from them, at no extra cost (Section 14 comes back to that flaw). I'm using it on purpose, flaw and all. This section is about learning, and learning in that sense is exactly the thing that doesn't have to merge. Settling, making a result stay so it can't be handed back, is the separate act with a price, and nothing in this definition settles anything.

□ **Theorem 4.2** (An observer can learn with no merge)**.** There is a four-state machine (the demon), a window (the display), a cost satisfying the halving price, a run of cost 0, and candidates with $|\Omega| = 2$, for which $|K(\Omega, t, s_0)| = 1$. So the claim "for every $\Omega$ with no repeats, every $t$, and every $s_0$ in $\Omega$: $\lceil \log_2 |\Omega| \rceil - \lceil \log_2 |K(\Omega, t, s_0)| \rceil \leq \text{cost}(t)$" is false for that machine and that cost. In the same machine, every cost satisfying the halving price charges WIPE at least one.

*Proof.* The machine and run are the four dots above: MEASURE is a bijection, so cost 0 satisfies the halving price; the observer's two candidates show different displays after MEASURE; WIPE sends two pairs of states to one each, so the halving price needs its cost $\geq 1$. ∎

The demon is the picture, and the smallest case is smaller than it. The claim as written already fails on a machine with two states that the window tells apart and every move priced at zero, because the observer's very first look tells the two apart, and it never fails on a machine with one state. That's why the interesting version is learning *after* the first look, which is where the three states of Theorem 4.3 come in. And WIPE's price needs less than the halving rule: merge pricing alone charges it at least one.

There is one thing to be exact about. The halving rule is a floor: it says merges cost at least this much, and it doesn't say anything costs zero. So the theorem exhibits *one* price schedule that meets the halving rule and charges MEASURE nothing. A different schedule could meet the rule and still charge MEASURE; nothing in counting forbids overcharging. The claim is that learning is not *forced* to cost a unit. That's what "borrows" in the section's title means: the look takes a page and pays nothing, and the unit, like the floor, comes due at the wipe.

The physicists had this freedom on the books long before me. Paul Fahn (1996), doing the entropy accounting of Szilard's one-molecule engine step by step, argued for a symmetry between the measuring step and the erasing step: on his account, which of the two you charge is partly a choice of bookkeeping, and the total is what's fixed. Sagawa and Ueda (2009) derived a bound on that total, with the split depending on the memory. So, on their accounts, "a schedule could charge MEASURE instead" is a known freedom in how the books are kept, and I didn't find a loophole. The other reading is very much alive too. Ruth Kastner (2025) argues that Brillouin was essentially right after all: once the physics of a real measurement is put back in, it's measurement that pays and foils the demon, erasure drops out of the dissipation, and Bennett's floorless copy fails as physics. The theorem above is about price lists, so it stands either way. What hangs on her is whether a real demon can look without a floor, which is the same question the next paragraph's doubter presses on.

And the doubter from the last chapter cuts this side too, so I owe it to you, since I leaned on him there. In 2013 John Norton argued for a no-go result: at the scale of molecules, thermal jiggling keeps any process that's supposed to be thermodynamically reversible from ever actually finishing, and finishing it takes more entropy than Landauer's amount. If he's right, Bennett's measurement has a floor in practice after all, because it's one of those processes. The theorem above survives that, because it's about price lists: a schedule that meets the halving rule

isn't forced to charge the copy, whatever a real device spends. What wouldn't survive is the physics I've been hanging on it, that a real demon's look owes no floor, only the dissipation of happening that going slowly shrinks. If Norton is right, the line between learning and committing keeps its counting and loses its physics. I can't referee that one either.

### 4.6 Who holds the pen changes who pays

Now comes the part that I think is the most beautiful fact in here, and you find it by playing with the four dots rather than by thinking about them. Run MEASURE *twice*. The first copies the hidden bit onto the display. The second adds the hidden bit in again, and adding a bit twice with a light-switch add gets you back where you were: the display is blank again. Check the arrows: 10 to 11 to 10. The display went full and then blank, and nothing merged, and the price on the ledger was zero both times. So a display that was blank before the first MEASURE *can* be put back to blank with no merge and no floor, by a second MEASURE, though not by WIPE. This is the borrowed page being handed back: while the original is still there to check against, the copy can be un-copied. Each MEASURE still dissipates by happening, and going slowly shrinks it.

What's the difference? MEASURE *consults the hidden bit* when it clears the display. It knows what's written there, because it has access to the thing it was copied from, and so it can undo the copy. Nothing gets erased. WIPE doesn't look. It sets the display to 0 regardless, and "regardless" is exactly the many-to-one. Bennett has the general form of this in the 2003 notes: he lists the "erasure of one of two copies of data known to be identical" among the logically reversible operations, because you're removing a duplicate, and no information is destroyed. And it goes further than Bennett in work that came after him: the quantum physicist Lídia del Rio and colleagues argued in 2011 that the floor on erasing a system depends on how much the eraser already knows about it, dropping to zero, and in the quantum case below zero. The heat of clearing a record falls on whoever clears it, and its floor is lower the more they know. Try it with a red or blue card under a cup and a note that is blank or marked. Start with the note blank. Lift the cup: for red, toggle the note (blank to marked, marked to blank); for blue, leave it alone. The note now reports the colour. Repeat that same rule while the card is still available, and the note returns to blank. Check all four card-and-note combinations: the rule undoes itself, so no two land together. Erase the note regardless of the card instead, and for either card both note values land on blank. That rule merges on the full set of combinations. On the matched pairs you actually prepared, the card still has the information, so nothing is lost there; the merge is in what the rule has to do to every combination it might be handed, which is what a rule that doesn't look

has to answer for. Cards and paper don't measure heat. What they show is which information each rule gets to use.

## 4.7 The smallest machine that teaches with no merge, and the one that can't

There are two more small facts, and then the point. The demon measures what you learn by comparing the start to the end, but an observer's very first look already rules things out before any step runs, so the fairer question is: after that first look, can a *step* teach you more, at zero cost? Yes, and the demon still does it. And the smallest machine that can is three states in a ring: U goes to V, V goes to W, W goes to U, and the window shows "true" only at W. Start with candidates {U, V}; both show false. Take one step. From U you now see false; from V you now see true. One step told them apart, and a ring forgets nothing, so a zero price meets the halving rule and is the price chosen here.

Two states can't do it, ever, with any arrows, and this is the one that needs no price premise at all. If a machine has only two states and you can't tell them apart at the first look, then the window shows the same thing on both, and those two are *all the states there are*, so the window shows the same thing everywhere, forever, and you never learn anything. Try to draw a counterexample. You'll draw the same two dots a few times and give up, and the giving up is the proof. Three is the smallest number of states on which two states that look alike at the first look can come apart at a later look, and that's what teaching is.

**Theorem 4.3** (Three states are the minimum for teaching at zero cost)**.** Call a machine a *zero-cost teacher* when it has a cost satisfying the halving price, a list of candidates with no repeats containing the actual start, and a run of cost 0 after which the observer knows strictly more than after the first look, $|K(\Omega, t, s_0)| < |K(\Omega, [\,], s_0)|$. There is a zero-cost teacher with exactly three states, and there is none with two or fewer. □

*Proof.* Three: the ring $U \to V \to W \to U$ with the window true only at $W$, candidates $\{U, V\}$, one step, cost 0 (a ring is a bijection, so cost 0 satisfies the halving price). Two or fewer: the actual start is always in the observer's knowledge, so knowing strictly more after the run needs at least two candidates still left after the first look, which agree there. On a machine with two or fewer states those two are all the states there are, so the window takes one value everywhere and no later look can separate them. This half never uses the price at all. ■

*Finish together, for the first time*
Run the machine from $N$ different starts and photograph its full configuration at every tick. Put the photographs in rows, one row per start, one column per tick. Every run finishes for the first time at tick $T$. Through that finish, two different states in the same column must have different successors.

The sheet has $N(T+1)$ photographs, including the starts and finishes. Could two of them show the same full configuration? In the same column, cancel backwards and the starts would have to agree. In different columns, give both configurations the same remaining steps. One run would finish before its promised first finish. Every photograph has to show a different configuration.

□ **Theorem 4.4** (Room for synchronized first completion)**.** Let $F$ be a total rule, $A$ a set of $N$ distinct starts, and $H$ a completion reading. For every $a \in A$, suppose $F^T(a) \in H$ and $F^t(a) \notin H$ for $t < T$. Suppose $F$ is injective on $\{F^t(a) : a \in A\}$ for each $t < T$. Then all states $F^t(a)$, with $a \in A$ and $0 \le t \le T$, are distinct. Any carrier containing these runs therefore has at least $N(T+1)$ configurations.

*Proof.* First compare photographs in the same column. If two showed the same state, look one column back. The rule keeps different sources in that column apart, so their predecessors must already have been the same. Repeat until the first column: the starts agree. Thus different starts cannot give the same photograph at the same time.

Now compare different columns. Suppose run $a$ at time $i$ shows what run $b$ shows later, at time $j$, with $i < j \le T$. Run $b$ has $T - j$ ticks left before its first finish. Starting from the identical state, run $a$ must follow those same remaining steps and finish too. Its time would then be $i + (T - j)$, earlier than $T$ because $i < j$. That contradicts the first-finish promise for run $a$.

No two places on the sheet can therefore show the same full configuration. There are $N$ rows and $T + 1$ columns, counting time zero, so the machine needs at least ■ $N(T+1)$ different configurations to hold the sheet.

That is the room required by the whole sheet of possible runs. The photographs can come from separate executions of one machine. What keeps different columns apart is the common first finish: a cycle with no such finish can put the same state in several columns. [8]

[8] The count is of full configurations, rather than simultaneously running machines or newly allocated memory cells.

## 4.8 The certainty bound

Now I can put the whole path from not knowing to knowing on one page, with what it costs. You don't know which state the machine started in. There are a few it could have been, each with its chance; that's your uncertainty. Run the machine. Every one of those possible starting states goes somewhere, and only two things can happen to a pair of them. Either they land on one state, and the difference between them is gone from the machine: a merge, and the halving price charges for it, at most one bit a unit. Or they land on two states, and the difference is still in the machine, sitting in its state, kept as two states held apart. There's no third thing a pair of possibilities can do. So the uncertainty about where it started splits exactly in two: the part the merges destroyed, paid for in cost, and the part the end state still holds, paid for by the states it keeps apart.

Now put an observer at the window. Whatever they see is fixed by where the machine started, so whatever they learn is a fact about the start. And the same split holds for it: what they learned is either still held in the machine's end state, or it's gone from the machine and a merge paid for losing it. A run that costs nothing leaves every bit the observer learned still held in the machine. This is the precise sense of "learning borrows": the observer's copy is a copy, and the machine still holds the original. When the machine lets go of the original, a merge happens, and the cost comes due.

Said exactly, then: take a spread $p$ over starting states and a run $t$, and write $q$ for the spread of where the run ends. For an end state $z$ with $q(z) > 0$, the starts that end at $z$ have chances $p(x)/q(z)$ between them; call that the spread of the start given $z$, and average its entropy over the ends:

$$H(\text{start} \mid \text{end}) \;=\; \sum_{z:\, q(z)>0} q(z)\, H\big(\text{start given } z\big).$$

This is how many bits you'd still be missing about the start if someone told you the end. Do the same for anything $v$ that's fixed by the start, like the whole film of what the window showed: $H(v \mid \text{end})$ is what you'd still be missing about $v$ once you knew the end.

**Theorem 4.5** (The certainty bound)**.** On a finite deterministic machine with an entropy-priced cost (Definition 3.6), take any spread $p$ over starting states, any run $t$, and anything $v$ that the start fixes. Then (1) $H(p) = H(\text{start} \mid \text{end}) + H(q)$: the uncertainty about the start is the part the run destroyed plus the part the end state still holds; (2) $H(\text{start} \mid \text{end}) \le \text{cost}(t)$: the destroyed part was paid for, at most one bit a unit; (3) $H(v \mid \text{end}) \le H(\text{start} \mid \text{end}) \le \text{cost}(t)$: whatever the end state can't tell you about what an observer saw was paid for in cost, and everything else

□

the observer learned is still held in the end state. In particular, after a run that costs nothing, the end state fixes everything the window showed, for every start with a positive chance.

*Proof.* (1) Treat the whole run as one map from starts to ends; the proof of Theorem 3.18 works for any map, so $H(p) - H(q) = \sum_x p(x) \log_2 \frac{q(z_x)}{p(x)}$, where $z_x$ is where $x$ ends. Group the starts by where they end: for each $z$ the terms add up to $q(z) \sum_{x \to z} \frac{p(x)}{q(z)} \log_2 \frac{q(z)}{p(x)} = q(z)\, H(\text{start given } z)$. Adding over $z$ gives $H(\text{start} \mid \text{end})$. (2) Let the run's moves be $i_1, \ldots, i_n$, and let $q_0 = p, q_1, \ldots, q_n = q$ be the spreads after each step. Entropy pricing says step $k$ removes at most its own cost: $H(q_{k-1}) - H(q_k) \le \text{cost}(i_k)$. Add over the steps: $H(p) - H(q) \le \text{cost}(t)$, and by (1) the left side is $H(\text{start} \mid \text{end})$. (3) Fix an end $z$. Given $z$, the value of $v$ is a function of the start, so its spread given $z$ is the spread of the start given $z$ pushed through that function, and by Theorem 3.18 (again for any map) a push never adds bits. So $H(v \text{ given } z) \le H(\text{start given } z)$ for every $z$, and averaging with weights $q(z)$ gives the first inequality; (2) gives the second. With $\text{cost}(t) = 0$, $H(v \mid \text{end}) = 0$, so each end state with a chance leaves one value of $v$. ∎

Entropy pricing and the halving price are the same premise (Theorem 3.16), so the theorem holds under either. There are three things it doesn't say. It doesn't say how much heat; that's Landauer's premise, applied to the destroyed part, $H(\text{start} \mid \text{end})$, which is the uncertainty the run really removed. It doesn't say the observer's own notes cost nothing: they need somewhere to be stored, such as a blank page. Supplying that page is an initial condition; resetting an uncertain page for reuse is a separate operation. And with a known start, $p$ on one state, every term is zero: nothing to learn, nothing destroyed, no positive floor from this premise. To ask what heat the step or the observer actually produces, give its physical process and preparation. The entropy calculation has not already charged somebody for knowing.

So the path between uncertainty and certainty runs both ways round. Going in, certainty is got by narrowing what's possible, and the deterministic calculation separates distinctions lost from distinctions retained. A copy can put a retained distinction in an observer's memory. It need not allocate a fresh cell at every step, and an initially held distinction needs no new crossing. The theorem counts entropy lost under its pricing premise; it does not put every act of knowing on a heat ledger.

## 4.9 What the toll prices, said at last

Now I can say, more precisely than I could two chapters ago, what the unit of cost in Section 2 is charged for. It is not charged for learning. What you know can be

narrowed with nothing merged, and an honest price list that charges merges can leave it at zero; the demon is one machine where that happens. It is not charged for looking, or for copying, or for finding out. Their physical dissipation depends on the implementation. In the copying examples here they use a supplied blank: the demon's blank pad, the watcher's own notes, a page somewhere that was blank and now isn't. But they borrow that page, and while the original is still there to check against, the copy can be handed back. The unit is charged for the one thing in Section 3 that couldn't be done without merging, on a machine with finitely many states: raising a flag that stays up, a record nothing hands back. Learning is the demon writing on a blank pad, which merges nothing. What the demon pays for is the wipe, a step where two states land on one. The unit in Section 2 is charged for the same reason, because raising a flag that stays up also lands two states on one, but it's a different step: the demon machine has no flag, and its price list prices merges; commitments never appear in it. It is the same kind of landing in a different step: learning borrows, and committing pays.

This is also where I settle up with Rovelli, my closest neighbour from the start of this chapter, because I put him and Bennett on opposite sides of one line: a measurement that can still be undone borrows and owes no floor, and a choice that sticks pays. I owe you how much of that is mine. All of the reconciliation is mine. Rovelli's paper doesn't make it, and in four places his own framing points somewhere else. He reserves "choice" for an outcome that is stable and *not predictable*, and he rules out a thermostat because it runs on a predictable mechanism; the commit in this book is a step a rule fixes in advance, so by his own definition his result doesn't cover it. His choice is a branching, one possibility splitting into two, and he calls choosing in a sense the inverse of erasing, because branching is the inverse of merging; the toll comes from a merge, the opposite shape. His stability is for a while: he says stabilization means something only on times that aren't too long next to the time it takes things to thermalize, and past that nothing stays chosen; my permanence is forever (and the last chapter owned up that forever is an idealization). And he doesn't stand beside Bennett at all: he holds that any measurement is dissipative, credits the point to the philosopher of science Hans Reichenbach, and doesn't cite Bennett. So what Rovelli and I share is a sentence, that a choice that sticks costs something like Landauer's heat. There's no shared mechanism. Push on it and his coin hitting the ground, a fact gained from something nobody could predict, sits closer to what this section calls learning than to committing. I still think it's the same instinct. This is my reading, and he doesn't make it; you now have everything you need to disagree with it.

And on one point Rovelli isn't the nearest physics. The nearest I've found is in biology. Thomas Ouldridge and Pieter Rein ten Wolde (2017) derived what they call the fundamental costs of making a *persistent* copy of a polymer, one whose sequence

stays correlated with its template after the two have physically come apart, and of destroying one. On their account a copy like that is a state out of equilibrium, and making it has a thermodynamic cost they measure efficiency against. A companion paper the same year, with Christopher Govern, does the same kind of accounting for the molecules a cell uses to read out the state of its receptors, which is a measurement in all but name. The way I read them, the copy that has to stay is the one that pays, and that's the closest thing in physics I've found to the line this section draws between finding out and committing. The differences, so it doesn't get borrowed for more than it is: theirs is work and stored free energy, in chemistry, for a correlation between two molecules, and their title has the destruction of copies in it too, so their copies don't last forever; mine is a count in a model of computation, for a reading that never turns off, with no temperature anywhere. It's the same instinct again, from a different direction.

The chemistry has the sharp form of it, too, and it came two years later. Jenny Poulton, Pieter Rein ten Wolde and Thomas Ouldridge (2019) studied minimal models of polymer copying, and in those models, when the copy grows at equilibrium, with the weakest chemical driving, a persistent copy's accuracy advantage over random copying and the efficiency of storing that information both vanish. This does not mean every copied site is wrong. Driving growth out of equilibrium can produce correlations that remain after the two polymers let go, with chemical work supplied to the process. This is finding out against committing, in chemistry, with the dissipation attached, and I'd rather say plainly that biology had the distinction before I did. What stays different is the kind of thing the record is: theirs is a correlation in a chemical ensemble, mine is a reading enforced by a step in a model of computation.

Two more neighbours have "learning" and "commit" in their titles, so the slogan shouldn't be read against them by accident. Sebastian Goldt and Udo Seifert (2017) did the stochastic thermodynamics of a neural network learning a rule, and argued from it that learning has a thermodynamic cost: on their account, the entropy the network produces bounds how much it can learn. That doesn't contradict this section, and the reason is that their learner isn't copying onto a blank page. Its weights are a driven thermal system, jostled by noise and dragged toward what the data push them to, and what they price is that physical process. This section's claim is narrower: one way of finding out, a copy onto a blank, is *allowed* to owe no floor and no cost. It can dissipate under the stated thermal-model assumptions, and it still spends the blank. Nothing here says every way of learning is cheap, and theirs isn't. From statistics, Aurélien Garivier, Tor Lattimore and Emilie Kaufmann (2016) compare two arms with unit-variance Gaussian rewards and different unknown means. Their explore-then-commit strategies sample the arms equally during exploration, then choose one for the remaining horizon. Within the uniformly efficient classes they compare, the

best leading logarithmic regret is twice that of fully sequential strategies. Whether the gap between the means is known changes both constants; fixing the exploration length in advance changes the comparison again. The factor two belongs to that model and strategy class, not every act called commitment. That's the closest published sentence I've found to "committing costs something" with a number on it. Their price is reward lost, which is a different unit from a unit of cost on a ledger, so it's a cousin. A twin would have to count cost.

So this is exactly what "No Free Insight," the name I gave the toll before I understood it, covers, because I don't want the name doing work it hasn't earned. Insight in the sense of *finding out* can come with no merge and no cost; it borrows a page. The insight the toll prices is the kind made to stay, written into a record the machine can't silently revise. Whether that narrower thing deserves the name is a fair thing to argue about; I kept the name because it's the question I started with, and because the narrowing is the finding. I don't count it as a retreat.

It also tells you how to read a bound that's coming. Section 8 counts what it takes to narrow a list of candidates down to a shorter one: at least as many yes-or-no questions as it takes to point at the survivors among their rivals, a counting fact that holds for anything with a list in it. Turning that count into a *cost*, saying a run actually paid that many units, is a different claim, and when the book makes it, for any Thiele machine in Section 8, the payment goes in as a hypothesis, in one of two ways: a decision tree (a branching chart of yes-or-no questions, which Section 8 builds) no deeper than the number of paid steps the run took is handed in, along with the lists themselves and the fact that the tree's leaves cover the narrowing; or the run makes the narrowing itself, with every test step charged at least one and each test it passes keeping at least a set share of the candidates. This section is the reason it has to go in that way. The narrowing alone doesn't force anyone to pay; the demon just showed you a narrowing charged nothing at all. So wherever narrowing gets a price in this book, the payment is a premise, and I'll keep calling it one.

There is one more boundary. This whole section is about a machine that's finite and deterministic and a watcher who crosses candidates off a list. There's no quantum anything in it. This narrowing has nothing to do with "observation collapses the state": the narrowing here is bookkeeping in the watcher's list, the machine never notices, and the result would be the same if the watcher were a filing cabinet. In 2010 a group in Tokyo (Toyabe, Sagawa, Ueda, Muneyuki and Sano) did run a real demon, a tiny particle walked up a staircase of potential by feedback on what a camera saw, and the energy it gained was paid for by the information the camera took, with no violation of anything. The demon is real, it works, and it pays. Where it pays is the part that took a century.

What I still owe: everything so far says "costs at least one" and I've been keeping the cost on the side of the page, a running total you tally as you go. Part II first

says what structure is; then comes the next question (Section 6). What happens when the machine keeps that total itself, inside its own state, where a step could in principle reach in and change it? That's the counter, and it's the next thing I have to build.

PART II

# What structure is made of

*The sphere, seen from outside the plane.*

Part I found what keeping takes on a single cell and wrote it into a step. Part II is step three: the rest of structure is the same thing at a bigger size. It starts with the word itself, structure, and everyone I could reach who wrestled with it, and comes out where Part I started, at a sorting of a pile into groups, a distinction held. Then it asks what follows, on any machine, once a distinction is kept and the keeping is charged, before I build one. The counter is the ledger, and it uses up states too: a number that only climbs takes a fresh value for every unit of cost. A window is what's left when you stop looking at some parts of the state, and it can hide the very distinction that was paid for. Entitlement and insight ask what a later step may do because a distinction stayed. The latch is the blank cell's no-to-yes over any machine at all, a reading the machine's own steps turn on and nothing turns off, and a row of latches is a direction that only goes one way. The diagonal is the wall no level gets past: no state codes every question about the states (Part I proved that much), and no machine that can hand a program its own code, and can run the flip of a verdict as one of its own programs, can sort all its programs, from inside, by a question about what they do that some pass and some fail.

CHAPTER 5

# What structure is

Part I put a charge beside a new permanent reading under explicit pricing rules. This chapter starts with one yes-or-no question, states the ruler under which it is the smallest way to sort, and goes after the rest of it. This whole section is about one word, and I'll give it to you now, so you can decide whether to stay: structure.

I used that word for most of my life without once looking at it, which I suspect is true of most people and most words, and it's fine, right up until you do something stupid. The stupid thing I did was build a machine that keeps a running total of its costs, and the total has one rule I couldn't talk it out of. You can arrange things however you like, and the ledger lets you put any price on the arranging, including none; the arranging has whatever physical cost its implementation imposes. But the moment an arrangement has to *stay*, the moment you ask the machine to keep it, or to vouch for it, to put its own name on the claim that these things belong together, there's a charge, and it isn't zero. Charged in what? Physical changes have costs determined by the implementation and protocol; keeping something takes more on top, a merge or fresh states. Section 3 showed there is no other way for a machine that picks its own next step to pay; when the moves are picked from outside, whoever picks them is holding the difference. How much, why I picked the one I did, and whether I had any business picking it at all: Part I went through all of that. What was in front of me, standing there with the thing built, was a model that charges for keeping structure, and I could not have told you what structure was. It didn't care. It can't nod along the way a person does. It needed the word to hold still, and the word had never held still for anyone, least of all me. (The model is already on the page: five parts and one rule, which you met in Section 2. Part III builds one small machine (Section 15) that meets it, so you can see a complete example. Until then the rule is all that matters.)

So I went looking at the people who'd wrestled it before me. Some of them have been dead two thousand years and are still, I'd bet money, better at this than I am. I used everything I had: language models, more than one, books, libraries, my own gut, anything in reach. I don't know if I found enough. I never do. There's always another shelf behind the one I was reading. And being dead and famous is not the

same as being right, so I pushed on each of them, and what follows is what I could reach, pushed on as hard as I know how. The one thing I'm asking is that you push on it with me, out loud, at your own kitchen table, because the first time through I was the only one checking, and that's not enough people.

The order I ran into them was a mess. Category theory came first, because I'd built one by feel before I knew it had a name, and everything else came in whatever order the next question dragged it in. I'm going to put them in the order where each one leans on the last, which is mostly the order they happened. And none of it is an oracle. History is where I start digging. It gets retold, smoothed over, and now and then made up outright (there's a drowning coming up that nobody can place), so every piece of it is a lead to check, and I check the parts I lean on.

### 5.1 A string, a square, and a drowning nobody can place

Start with the oldest one, and start by doing it. Take a string. A guitar string, a rubber band over an open box, anything that twangs. Pluck it. Keep its tension as nearly unchanged as you can when you stop it at a new point; stretching it tighter changes the pitch for a second reason. Now stop it exactly at the middle and pluck half. With the tension held steady, that's the same note an octave up, the one that sounds like the same note sung higher. Press at a third of the way along and pluck the long side: that's the fifth, one of the handful of intervals a bugle can play at all, which is why you hear it all through bugle calls. Half, two thirds: small whole numbers, and your ear says yes before your brain does. About twenty-five hundred years ago, in the south of Italy, the followers of a man named Pythagoras are credited with finding exactly that, and it's as true in your kitchen tonight as it was on their hillside.

I'll tell you what they did with it, as best anyone can tell. Nothing survives that's surely from Pythagoras's own hand. The earliest fragments are from a later member, Philolaus, and scholars argue over those. Mostly what's left is Aristotle, one to two centuries later, writing in his *Metaphysics* about them the way you'd write about the neighbours' religion, and he says they took the elements of numbers to be the elements of all things: everything is number. That doesn't mean everything can be counted; it means the parts of numbers are the parts of all things.

Stop and ask what I asked. What does that even mean? Is a chair a number? Does a chair have a number tucked inside it somewhere? Aristotle doesn't say, and he's reporting a belief he doesn't hold. So don't take it on his word, or theirs, or mine. But before you push on it, know what it was holding up, because you can't test a load-bearing wall without knowing what sits on it. These weren't people doing math for fun on weekends. They were a brotherhood, with rules about what you could eat and when you had to keep quiet, and that plucked string was their

scripture. Halve the string, hear the octave: to them that was proof that under the mess of the world there were whole numbers in clean ratios, that the cosmos was a chord and they'd heard it. "Everything is number" wasn't a slogan; it was what they'd organized their lives around. And that's exactly the kind of sentence you should be suspicious of, mine included.

Now push, and push where they loved most: the square. Draw one, with a side one unit long, any unit you like, and draw the line from one corner to the opposite corner. Call that the diagonal. If everything is number, then the diagonal is some whole number of parts long when the side is some whole number of parts, so the diagonal divided by the side is a fraction, a whole number over a whole number. Cancel anything the top and bottom share, the way four-sixths becomes two-thirds, until they have no common factor left. I'll call the top $p$ and the bottom $q$, and the letters are purely so I don't have to keep writing "the top" and "the bottom."

You need one fact about squares, and it's the one Pythagoras is famous for. The square built on the diagonal has exactly twice the area of the square built on the side. If you want to see it for yourself, cut out two identical squares, cut the second one along both diagonals into four triangles, and lay one triangle against each side of the first, its long edge on that side. The outline is the square on the first one's diagonal, and it's made of exactly two of the small squares; it's a five-minute job with scissors. So $p$ times $p$ is twice $q$ times $q$. People write "times itself" as a small raised two, so that's $p^2 = 2q^2$, and that is the only piece of notation in this argument.

Twice anything is even. So $p$ times $p$ is even. An odd number times itself is always odd: draw that many rows of that many dots, pair off the columns, then pair the dots in the one column left over, and one dot remains. The same pairing works for any odd side, so if $p$ were odd, $p$ times $p$ would be odd too. It isn't, so $p$ itself must be even: it's two times something, and I'll call that something $r$. Put it back in. Two $r$ times two $r$ is four times $r$ times $r$, and that equals twice $q$ times $q$. Halve both sides: two times $r$ times $r$ equals $q$ times $q$. So $q$ times $q$ is even, so $q$ is even too, by the same step as before. But you already cancelled everything the top and bottom had in common, so they can't both be even. The argument has eaten itself, and when an argument eats itself the thing you supposed at the start is false. There is no such fraction. The diagonal of a square can't be named by any ratio of whole numbers, ever. Aristotle reaches for this very argument in his *Prior Analytics* as his example of proof by contradiction (suppose the diagonal can be measured, and odd numbers come out equal to even ones), and you just did it with a pencil.

There's a story that a member of the brotherhood named Hippasus let this secret out, and his brothers took him to sea and drowned him for it. I went and checked, because it's a good story and good stories are the ones to check. The oldest drownings I could find are in Iamblichus, writing around AD 300, seven or eight hundred years after Hippasus. In Iamblichus the man drowned by name is Hippasus,

and what he gave away was how to fit twelve pentagons together into a ball, the dodecahedron. A few pages on, Iamblichus tells it again about whoever first let out the incommensurable, and that time he names nobody. Carl Huffman, who wrote the Stanford Encyclopedia of Philosophy's entry on Pythagoreanism, follows the historian Walter Burkert there and finds no ancient writer who connects Hippasus to this discovery at all. So the version with Hippasus drowned for the diagonal looks like two old stories stitched together later. It's a legend, but notice that the legend exists. Somebody, long after, felt that a man *should* have been drowned for this. That's how much was riding on it. To whoever stitched the story together, the diagonal wasn't a math problem; it was the end of the world.

So the claim is dead, and you killed it. I only handed you the pencil. Now the question that matters: what's left? Look at your square again. The diagonal is still there. It still stands in a perfectly definite relation to the side; you drew it, and nobody drowned. That relation was real before anyone tried to express it as a fraction, and it stayed real after that attempt failed. The relation came first. The number, when there is one, is a name for the relation, and sometimes there isn't a name. I'm going to keep that, and I want you to keep it too, because you're going to watch the same move get made again and again as this book goes on: somebody says "everything is X," and it's true enough to do real work with, and it sounds right when you pluck it, and then somebody draws the diagonal. Whether the thing I built survives its own diagonal is a real question, and Section 13 answers it.

## 5.2 Shapes poured into nothing

About a century and a half after the brotherhood, Plato, in a dialogue called the *Timaeus*, tried to build the whole physical world out of geometry. Fire, air, water and earth were four regular solids (the pyramid, the eight-faced one, the twenty-faced one, and the cube), and the faces of those solids were built from exactly two kinds of right triangle. You can make one of each in a minute: cut a square from corner to corner for the first, and cut an equilateral triangle from a corner to the middle of the opposite side for the second. Everything else, in Plato's story, was a receptacle, a featureless medium that takes on shape the way wax takes a seal. Structure, for Plato, is shape laid into something that has no shape of its own.

Why did he care? He wanted a world made on purpose, by a craftsman working from a plan, and triangles were the plan. Is it true? No, but not for the cheap reason. You can't refute a claim about what things are made of at the smallest scale by picking one up; the smallest scale is exactly where your hands don't reach, and Plato knew that. What sank it is that it made no prediction anyone could check, and the people who eventually did reach that scale, two thousand years on, found atoms, and atoms aren't triangles. The idea was a guess with no way to lose, until it

lost. But push on it and two things survive. One: structure is something *laid in*, and the stuff it's laid into doesn't carry it by itself. That's a real idea, and I'd ask you to keep it loosely for now; when the machine arrives in Part III, you can ask whether the state it holds is the wax or the seal. Two: Plato put a crack in his own forms himself, in another dialogue, the *Parmenides*, with an objection that Aristotle later nicknamed the third man, and it's the most useful mistake in this section. Take all the large things. Say they share one thing, Largeness. But Largeness is itself large, so now there's a further thing that all the large things *and* Largeness share, and that one is large too, and it never stops. The extra move is treating Largeness itself as one more large thing. Grant that again at each stage and it never stops; refuse it and this particular regress never starts. The lesson, and I think it's right though Plato never spells it out: the structure of a collection is not one more member of the collection. The plan isn't a brick. Hold on to that one.

### 5.3 Prove it from what you and I both accept

Around 300 BC, in Alexandria, Euclid wrote the *Elements*, and the thing to take from it is not the geometry. It's the method, and the method is the whole reason I trust anything in this book, so give me a paragraph on it. He starts with definitions ("a point is that which has no part"), five postulates (things he asks you to grant, like "to draw a straight line from any point to any point"), and five common notions (things nobody argues with, like "things which are equal to the same thing are also equal to one another"). Then he derives everything else, four hundred and sixty-five propositions, from only those, in order, each meant to use nothing but what came before (he sometimes leans on things he never stated, and later geometers filled those in). He never says "trust me" or "it's well known"; he says, in effect, here are the rules you and I both accept, now watch. If you've ever been in an argument that ended with "well, that's just your opinion," Euclid is the man who found a way out of that room, and I'm not embarrassed to say I learned the way out from someone dead for two thousand three hundred years. Four hundred and sixty-five times he does the same trick, and the trick is everything.

And then there's the one rule Euclid seems almost embarrassed by, and it's a hole worth falling into. His fifth postulate, the one about parallel lines, is long and awkward next to the others, and for two thousand years people tried to prove it from the rest so it wouldn't have to be assumed. Proclus, the fifth-century philosopher who wrote a commentary on Euclid's first book, tried. Mathematicians in Persia and the Arab world tried. In 1733 an Italian priest named Girolamo Saccheri spent a whole book assuming it was false so he could watch the contradiction arrive, and he got pages deep into what is now a perfectly good geometry before declaring that it was repugnant and stopping. A century later three people, separately, took the step

he wouldn't: drop the postulate, and what you get is other geometries that hold up just as well, and nonsense never arrives. Farkas Bolyai begged his son János to stay away from the problem, warning that it had swallowed years of his own life; the son went in anyway and came out with one of those geometries. The part I keep is that they found something real by refusing to believe one of the rules everyone accepted, and checking what happened without it. Refusing a rule and checking what happens without it is the push, and this one went on for two thousand years.

What is structure, to Euclid? Whatever the starting rules force. And when are two shapes the same? You can do this one yourself: cut out two paper triangles and lay one on the other. If they coincide, they're equal. That's his fourth common notion, and it's how he proves his fourth proposition, by picking up one triangle and setting it down on another. Sameness is coinciding under a move. Keep that; it's about to get very general, and I didn't see that coming either.

### 5.4 The statue and the bronze

Aristotle, Plato's student, pulled the forms down out of heaven and put them inside things, which was a bit rude to his teacher and also, I think, correct. His example, from the *Metaphysics*: a bronze statue is bronze plus a shape. And a circle can be made in bronze or stone or wood, so, he says, the bronze is no part of what it is to be a circle. The same shape in marble makes a different statue with the same form. The bronze is the matter and the shape is the form, and the form is what the thing *is*.

Push on it with two things from the kitchen. A cup can be ceramic or metal and still hold your coffee. A sandwich carved in wood can keep the outline and fail to be lunch. The two swaps preserve different things. If the question is shape, the carving may pass; if the question is whether you can eat it, it does not.

Aristotle already makes a distinction of this kind. In Book VII, chapter 11, he warns against treating a living animal just like the circle whose bronze can be changed: its parts must be in a condition to do their work. The sandwich is a way to make that difficulty visible, not a correction he never considered.

That is the bet I need for the record. Change the material, then check the operations and the answers again. If they still agree, the record has survived the change in the respects I promised. Section 10 states those promises over a chosen base; Part III builds an instance. Neither theorem says that every material can do every job, or that the price in one material is the price in another.

## 5.5 Where things are, without numbers

In 1679 Gottfried Leibniz, who co-invented calculus and had opinions about everything, wrote to the physicist Christiaan Huygens asking for something that didn't exist: a way to reason about *where* things are directly, the way algebra reasons about *how much*, without ever laying down coordinates. He called it the analysis of situation. Try it yourself. Describe where the furniture is in the room you're sitting in without using a single number. "The couch is against the wall. The table is between the couch and the window. The lamp is next to the chair." You just did it, and what you used was relations between places and nothing else. Huygens wrote back that November and was not impressed. It took roughly two hundred years for Leibniz's idea to become a field, and the field is called topology, and Huygens is remembered for other things.

What survives the push is the same thing that survived the square, coming at you from the other side. Position is a relation among places. The numbers you lay on top are a convenience, and they're a convenience you can take away and still have the room. I'll be honest about why this one landed for me: the thing I built keeps track of which things belong together, and I spent a long time thinking that was a list of numbers before I understood it was a relation that the numbers were merely naming. Leibniz had that in 1679. Huygens told him no. I'd like to think I'd have sided with Leibniz. I probably would have sided with the famous guy.

## 5.6 Naming a thing by saying when two are the same

In 1884, in a short book called *The Foundations of Arithmetic*, Gottlob Frege noticed something you can check in your head right now. "Line $a$ is parallel to line $b$" and "the direction of $a$ is the direction of $b$" say exactly the same thing. There is no fact in one that isn't in the other. But the second sentence has a new object in it, a *direction*, that the first one doesn't. Where did it come from? From the sameness. Once you can say when two things count as the same, you've earned the right to name the thing they share. Frege used exactly this move to say what a number is: the number of things in a pile is whatever every pile you can match up with it, one to one, has in common. The residue has teeth: a sameness relation lets you name a new object.

Now push, because Frege pushed too hard, and it cost him. In his formal system, Basic Law V treated value-ranges as objects and identified them exactly when the corresponding functions agreed everywhere. Together with his unrestricted comprehension principles, that let the self-membership construction in. The familiar set version is easier to tell: In June of 1902, a young British philosopher and logician, Bertrand Russell, wrote him a short letter. Consider the collection of all collections that are not members of themselves. Is it a member of itself? If it is, then by its own

rule it isn't. If it isn't, then it qualifies, so it is. Do it in your head; it flips every time you look at it, like one of those drawings that's a vase and then two faces and then a vase. Frege's floor had a hole in it, and he knew it the day he opened the letter, and his second volume was already at the printer. One repair was Zermelo's 1908 axiom of separation: a property selects a subset of a set already given. His system also had other existence axioms, including power set and infinity; separation was not its only way to obtain a set. What it removed was unrestricted sorting of "everything." You sort a pile. I'll tell you now that this rule shows up again, later, in a place where I had to obey it or fall through the same hole, and I'll point at it when the book gets there.

Frege's reply is worth hearing, because it's the best example I know of how to lose. He wrote back within days. He didn't hide it or argue around it. He said, more or less, that arithmetic was tottering, and in an appendix to the volume that was about to come out he wrote that hardly anything worse can happen to a scientific writer than having a foundation give way right after the work is finished. Then he tried to fix it. If I'm wrong about something in this book, and I'd bet I am somewhere, that's how I'd like to take it.

### 5.7 Choose the changes, keep what survives

By 1872 there were too many geometries. Euclid's, and ones where parallel lines meet, and ones with no such thing as distance, and nobody could say what made them all geometry and kept them from being a pile of unrelated hobbies. Felix Klein, twenty-three years old and taking up a professorship at Erlangen, published an answer in a program paper that people still call the Erlangen program. Pick a set of allowed changes: rotations, stretches, whatever you decide. Then a geometry is the study of the properties that *stay the same* under those changes. Pick a different set of changes and you get a different geometry on the very same page. Euclid's, in Klein's reading, is the one where you may slide, turn and flip a shape and scale it evenly, and stretching it more one way than another is off limits. And I'd underline, if this were my book, which it is, that in Klein's scheme any group of changes can define a geometry. The choice is yours. Nobody tells you which changes to allow.

You can do this one at your desk. Take any drawing and ask what's still true after you turn it upside down (lengths, angles), after you stretch it (lengths are gone by then; which lines cross which is still there), after you bend and crumple the sheet without tearing it (which drawn paths remain connected survives). Two distant ink marks may touch across a fold; that new physical contact is not a new connection in the drawing. You have to say which relation you are following. The three changes give three answers and three structures, all from one drawing. The residue is that structure is what survives a chosen set of changes, and "chosen" is doing real work

in that sentence. There's no structure of a thing from nowhere. You have to say what you're allowed to do to it first. I didn't know that when I started, and I'd been saying "the structure of" things for years.

Klein's move has a descendant in logic, and it goes all the way. In a lecture in London in 1966, published only in 1986, three years after his death, the Polish-American logician Alfred Tarski asked what makes a notion logical, and answered with Klein's recipe at its far end: allow every one-to-one transformation of the world onto itself, and call logical the notions that survive all of them. The more changes you allow, the fewer notions are left standing, and with every change allowed what's left is logic. Tarski's answer is the sentence above, "structure is what survives a chosen set of changes," with the choice made as large as it can be.

## 5.8 An invariance comes with a conserved quantity

In 1918 Emmy Noether, who at the time wasn't allowed to qualify for a university teaching post because she was a woman, and for a while taught courses advertised under David Hilbert's name, proved something that physicists still treat as close to holy. Take a physical system whose laws can be written the way most of physics is written, as "of all the paths the system could take, it takes one that makes a certain total stationary" (the total is called the action, and stationary means a small change to the path doesn't change the total to first order; the path needn't make it smallest, and that's all I'll say about it here). For such a system, every continuous symmetry that leaves the action unchanged comes with a quantity that is conserved. If the action doesn't change when you shift everything in space, momentum is conserved. If it doesn't change when you shift everything in time, energy is. A symmetry isn't decoration. It has a consequence downstream that you can measure with instruments. I can't hand you this one to do at the table; the proof needs calculus I'd have to build first and I'd lose you and probably me, and the "written as an action" condition is doing real work that I'm not going to unpack here. What I can hand you is the shape. Structure, once it's really there and the setting is the right kind, has consequences you can measure. That's the sentence that made me think a machine could carry a cost for structure in the first place. Noether makes it reasonable to ask. She doesn't make it true, and nothing I build in this book is a system with an action.

## 5.9 All of mathematics as structures

Starting in the 1930s, a group of young French mathematicians began publishing a vast treatise under one made-up name, Nicolas Bourbaki, as if he were a single person. He wasn't. In 1948 he also applied for membership in the American

Mathematical Society, and the application was quietly turned down, which I find comforting. (The historian Michael Barany has gone through that application, a second one a year later, and the letters they set off.) Their goal was to rebuild all of mathematics on one idea: a structure is a set of elements "whose nature has not been specified," together with one or more relations on them that satisfy stated axioms. They believed three kinds of structure were the mothers of all the rest: algebraic, order, and topological.

Push on it. The historian Leo Corry went through the entire treatise looking for the place where this formal definition of structure actually does any work, and what he found is that it "plays no actual role in the presentation of theories within the treatise." Outside the chapter that defines it, it does work almost nowhere. And I found no argument in their writing for why the mother structures number three. And the chapter on category theory, which they promised their readers, never appeared; Corry is where I found that too, and Ralf Krömer's 2006 history of Bourbaki and categories in the 1950s follows the attempt. That does not mean Bourbaki left maps out. Their *Theory of Sets* has sections on morphisms, derived structures and universal mappings (Chapter IV, sections 2 and 3). Corry's criticism concerns how much their general formal apparatus was actually used. The lesson I can take for my own work is smaller: naming a structure is not enough; show which constructions and comparisons it supports. I learned that one the slow way, and the origin story at the back of the book (Section 24) tells it.

Philosophers of mathematics took the same idea further and gave it a name, structuralism. In 1965 the philosopher Paul Benacerraf pointed out that the whole numbers can be built out of sets in two equally good ways that disagree about which set the number three is, and concluded that numbers aren't objects at all: what arithmetic is about is the pattern, and anything standing in the right relations to everything else can play the part of three. The philosophers Michael Resnik (*Mathematics as a Science of Patterns*) and Stewart Shapiro (*Philosophy of Mathematics: Structure and Ontology*), both in 1997, made the patterns themselves what mathematics is about. The philosopher Geoffrey Hellman (*Mathematics without Numbers*, 1989) kept the talk of patterns and did without the patterns, reading each statement as a claim about what would hold in any system with the right structure, if there were one. The logician Steve Awodey (1996) argued that category theory says it best, a structure being what its maps keep, which is where this section goes next. And the philosopher of science James Ladyman (1998) carried the idea into physics: what a successful theory gets right about the world is its structure. This section's thesis, that structure is the relations and the things in them don't matter, is theirs. None of the formal results takes their arguments as a premise, and the bibliography lists them under the same heading as Bourbaki.

## 5.10 A set with some named relations

What mathematical logic eventually settled on, with the logician Alfred Tarski giving the field its name, "the theory of models," in 1954, is almost boring, and the boring ones are the ones that last. A structure is a set, called the domain, plus whatever relations and functions a chosen vocabulary names on it. Pick the vocabulary (Klein again: you choose), and a structure is anything that gives those names a meaning. Now I want to say "the smallest one," and the word "smallest" has no meaning until I say what I'm measuring, so here is the ruler: count the names in the vocabulary first; when that ties, count how many things each name takes at once; when that ties too, count how many answers each name can give. The bottom of that ruler, for anything that takes things at all, is one name, taking one thing at a time, with two possible answers, which is a domain with one yes-or-no property named on it: one question, put to each thing. A single two-place relation ("these two go together") can sort a pile too, and so can a single function that sends each thing to a label, but the first takes two things at once, and the second can give a different answer for every thing, more than two on any pile of three or more, so the ruler puts both above one question. There are two limits, and I'll give them to you now, so you don't have to find them. A vocabulary with one named constant ("this one, here") takes no things at all, so it sits even lower on my ruler, and by itself it doesn't sort anything, so the claim is "smallest that sorts." Comparing things with the constant takes equality, and I count equality as a name when a structure uses it, one that takes two things at once. Leave it uncounted, as first-order logic does, and the constant sorts below one question: "is it this one?" splits a pile into that thing and the rest, which is still a split in two, one question whose yes is a single thing. Plain "smallest" would be false. And one question only splits the domain into two groups when some things answer yes and some answer no; if they all agree there's one group and nothing was sorted. With those limits said, and with the ruler said: by the count of names, the smallest structure that sorts anything is one question. Someone with a different ruler might land somewhere else, and I'd want to see where. Least information lands one bit lower, on the partition this section ends up starting from: on a nonempty pile of $n$ things a question can come out $2^n$ ways, and a split into at most two groups with neither one called yes can come out half as many. The bit between them is the label, which group says yes. Keep it. It's about to turn into the whole section.

## 5.11 The structure is in the arrows

I met this one first, and backwards. I'd built it by feel, in a rendering engine, as "digital DNA," and only went looking for its name after the thing already ran. In

1945 two mathematicians, Samuel Eilenberg and Saunders Mac Lane, published a paper that founded a field now called category theory, and they said in plain words that it was a continuation of Klein, which I'd have loved to know at the time. The move they made was to stop asking what a thing is, and ask how it's carried. A category is some objects and some arrows between them, where arrows can be followed one after another and every object has a do-nothing arrow to itself, plus two rules that make the following dependable: following three arrows in a row gives the same result whether you pair up the first two or the last two, and the do-nothing arrow really does nothing when you tack it onto either end, and that's the whole definition. On the page the two rules look like nothing, and they're the part people skip, so slow down on them for a second, because each one is there for a reason you can see.

Picture three dots in a row, A, B and C, an arrow $f$ hopping from the first to the second and an arrow $g$ from the second to the third. Composition says there's an arrow taking the whole stride from A to C in one, $g$ after $f$. Without it, every multi-hop path has to be named on its own, and the structure falls apart into a pile of unconnected pairs, a phone book without a search function. Associativity says that with a third arrow $h$ on the end, pairing up the first two and then adding $h$ gives the same arrow as adding $h$ to the last two first. Both spell the same three-hop path. With your paper triangle, bracket "slide, then turn" before "flip," or bracket "turn, then flip" after "slide": you still carry out the same three moves in the same order. Swapping the turn and the flip would be a different question. If the two bracketings came out different, two people who built the same chain in different orders would hold two different arrows, which is a bug, because the chain is the thing and how you assembled it isn't. The identity laws say that tacking the do-nothing loop onto either end of an arrow gives back the arrow. If the loop changed things, it wouldn't be the loop. Those six terms, objects, arrows, composition, identities, associativity and identity laws, are all of Eilenberg and Mac Lane's definition.

Composing relations is older than the definition, too. Who put a name on that move first? I went looking. Augustus De Morgan, writing on the logic of relations around 1860, took the "is" out of the old syllogism, put a general relation in its place, and composed two relations to get the conclusion. The example that gets quoted from that tradition, Peirce's a decade later, is "$x$ is the lover of a servant of $y$," which is relational composition with better gossip. When exactly De Morgan read that paper to the Cambridge Philosophical Society, and on which pages it was printed, the sources disagree, which is a small reminder that history is where you start digging. Then in 1870 Charles Sanders Peirce, an American scientist and philosopher who turns up again in Section 14, carried George Boole's algebra of logic (the arithmetic of true and false that Part III comes back to in Section 15, along with the story of Boole himself, a largely self-taught schoolmaster) over to relations, with composition

as one of its basic operations, and he's usually counted as the real founder of the theory. So following one arrow after another is older than computers, older than categories, older than Turing's machine: it is a nineteenth-century logician's product of relations.

And where did they get the word? They didn't get it from mathematics. They borrowed *category* from the philosophers, from Aristotle and Kant, and *functor* (a map from one whole category to another) from the philosopher Rudolf Carnap, and then gave both words exact meanings none of those philosophers had in mind. So Aristotle walks back into this chapter without being asked: the man who put the form inside the bronze lent his word to the field that keeps track of forms by their arrows. That doesn't change the laws. It's a reminder that a word can start in philosophy and arrive in mathematics with a precise meaning bolted on, which is the thing I've been trying to do with *established* since the start of the book.

What a thing *is* is how it maps. For now the residue is the emphasis I needed: structure lives in the maps. Nothing later in this book needs more category theory than that sentence, and the latch of Section 10, where a record is pinned down by the maps between a machine and the base it extends, is where the sentence does its work.

## 5.12 Structure is the part that compresses

Play this one with me. I write "th" and you guess the next letter. You said "e," and you were right. Now "the q": you said "u." Congratulations, you just measured the structure of English. In 1948 Claude Shannon measured it properly, in a paper that founded information theory, and found that about half of written English is "determined by the structure of the language." It's the reason a zip file of a long letter is so much smaller than the letter. Be careful what that says. It does not say that every other letter is fixed by the ones before it; you just guessed "e" after "th" and you could have been wrong (think "than"). It says that, on average and over long stretches, about half of what's written could be rebuilt from the rest, so that the text is roughly twice as long as the choices in it. And he said why that matters in one line that I'd frame if I framed things: the structure "allows one to make a saving in time" (he added "or channel capacity"). The more you can predict, the less you have to read.

Thirty years later Jorma Rissanen, a Finnish information theorist at IBM's research lab in San Jose, turned that saving into a price. His idea is called the minimum description length principle, and it says that the best model of some data is the one that makes the total description shortest, where the total is the length of the model *plus* the length of the data once the model has explained what it can. A model is paid for in bits, and it's worth stating only if it saves more than it costs. Peter Grünwald, a

statistician who leads a machine learning group at CWI, the Dutch national institute for mathematics and computer science in Amsterdam, in a tutorial you can find online, adds the footnote that keeps this honest: the price of a model depends on the code you chose to write it in, so the price itself needs a principle behind it, and that principle is something you have to argue for, not something you're handed.

That's a lot of "length of" to swallow in one paragraph, so do one with me, with numbers small enough to check on the back of an envelope. Every number in it comes from a code I'm about to make up on the spot, and I'll say every rule of the code out loud, because Grünwald's footnote is the whole point: change the rules and every number moves.

The data is a row of thirty-two digits, each a 0 or a 1, and you and I agree in advance that the row is thirty-two long, so nobody has to spend anything saying so:

```
0101010101010101 0101010101010101
```

(The space is only there to rest your eyes.) Every description in the code starts with one digit that says which kind of description follows: 0 means "here's the row, written out," 1 means "here's a pattern, then the exceptions to it." Each digit is one yes-or-no answer, one bit, which is what Shannon's unit is. Written out, the row costs that first digit plus the thirty-two digits themselves, thirty-three bits, and there's no model in it at all. A pattern description goes like this. First come five digits giving the block length minus one: 00000 means one digit, and 11111 means thirty-two. That covers every allowed length. Then comes the block itself. The rule is "repeat the block, then stop at exactly thirty-two digits, cutting off the last repetition if necessary." Then come five digits for how many places the row disagrees with the pattern (up to thirty-one, which is plenty here), and five more digits for each of those places, numbered zero through thirty-one and listed in increasing order without repeats. At every listed place, flip the predicted digit.

Do it for the row above. The block is `01`, two digits long. So the model costs 1 (which kind) plus 5 (the block's length) plus 2 (the block), which makes eight bits. Then comes what's left to say about the data once the model has said its piece: the row disagrees with "repeat 01" nowhere, so that's five digits saying "zero exceptions" and nothing else, five bits. The total is thirteen, against thirty-three for writing it out. The pattern pays for itself, with twenty bits to spare, and Rissanen's sentence has just happened on paper: a model is worth stating when it saves more than it costs.

Now spoil the row. Flip two of its digits anywhere you like. The model doesn't change, eight bits, but now the leftover says "two exceptions, here and here": five plus five plus five, fifteen, for a total of twenty-three. It's still cheaper than thirty-three; the pattern still pays, it just pays less, because the data has more to say that

the model can't. Flip four, and the total is $8 + 5 + 4 \times 5 = 33$, a dead tie, and a pattern that saves nothing isn't worth stating. Flip five and it's thirty-eight, and now writing the row out is cheaper than explaining it, and this particular pattern has lost to writing the row out. Another pattern might still win. Flip places 6, 12, 18, 24 and 30 of the original row: now the block `010100`, repeated and cut off at thirty-two digits, fits exactly. Its description costs $1 + 5 + 6 + 5 = 17$ bits. Five changes defeated "repeat 01" and produced a better description of their own. The comparison is between descriptions in the stated code, not a count of spoiled digits alone.

| Message | Original row | Two flips | Five specified flips |
|---|---|---|---|
| Write all 32 digits | 33 | 33 | 33 |
| Repeat `01`, list exceptions | 13 | 23 | 38 |
| Repeat `010100`, no exceptions | does not fit | depends | 17 |

For the last column, flip positions 6, 12, 18, 24 and 30. That defeats one short description and supplies another. Pattern descriptions include the type bit, length field and exception-count field. The raw row uses one type bit followed by its thirty-two digits.

So "is there structure here?" has turned into a question with a number on each side, and a break-even point you can find yourself: four spoiled digits, in this code.

And now change the code, because I promised you'd see the numbers move. Say I'd used seven digits for each exception's place (wasteful, but nobody stopped me). Then three spoiled digits cost $8 + 5 + 3 \times 7 = 34$, one more than writing the row out, where the five-digit code charged twenty-eight and the pattern paid. The pattern stopped paying one spoiled digit sooner. The row is the same, the pattern is the same, the verdict is different, and the only thing that changed was a convention I chose. That's Grünwald's footnote, and it's why I won't let you take any of those numbers anywhere else. They're the price of a description in one made-up code, a length of a written-down message, and nothing more. They are not the charge I opened this book with. That charge, as Section 2 states it, is on one step a machine takes. How long anybody's description is doesn't enter it, and if the two ever turn out to be related, that has to be shown. Both of them being counted in whole numbers doesn't get to stand in for the showing.

Now push, because this is where "structure means information" gets born, and it's wrong, and it's an easy thing to believe. Flip a fair coin two hundred times and write down the string. Usually there is no substantially shorter way to say that string than to say the whole thing. You can watch it happen in the little code above, cut to thirty-two digits: a coin-flip row disagrees with "repeat 01" in about sixteen places, which would cost eighty-odd bits to list, and even the one block guaranteed to fit with no exceptions, the whole row used as its own block, costs $1 + 5 + 32 + 5 = 43$. The written-out row at thirty-three beats those two descriptions. Another block could do better; a lucky run of coins can make a pattern, so check it. The general

reason long random rows usually resist substantial compression is a count: put each short description beside the one row it prints, and there are too few short descriptions to stand beside more than a small fraction of all the long rows. Nikolai Vereshchagin, a mathematician at Moscow State University, and Paul Vitányi, a computer scientist at CWI in Amsterdam, in a 2004 paper on exactly this question, put it without mercy: every bit of a string like that is "meaningless information, random noise." Such a string carries maximum information and zero structure. So in this sense structure is the part that compresses, the part where the "th" tells you about the "e," and most information and most structure are different things. Hold that one tightly, because if you built something that charged for information, you'd be charging for the wrong thing, and for a while I wasn't sure which one I'd built.

And now notice that the word just changed meaning on you, and on me. I'm pointing at it so it doesn't slide past. Shannon and Rissanen and Vereshchagin mean *regularity*: a pattern that lets you say something shorter. Tarski, a few pages back, and Paul Halmos, a Hungarian-American mathematician whose short textbook on sets is coming up, mean a set with named relations on it, and those two are not the same thing. You can run the test yourself: number a thousand objects, flip a coin for each, heads goes in group one, tails in group two. The result is a perfectly good sorting in the set theorist's sense, every object in exactly one group. And the list of who went where is a thousand coin flips, which is to say it compresses about as well as a thousand coin flips, which is to say not at all. Feed it to a code built like the little one, with the fields widened to fit a thousand, and you get the coin-flip row again, only longer: typically most of the assignment still has to be written down. A lucky pattern can save some of it; the coin tosses alone do not guarantee there is none. By the set theorist's rule, the sorting *is* the structure, and it's every bit as much a sorting as "evens in one group, odds in the other," which compresses to almost nothing. The two are the same kind of object with wildly different description lengths. So the two meanings don't even rank things the same way: one asks "is there a sorting?" and the answer is yes for both; the other asks "does a short rule say who went where?" and the answer is yes for evens and odds and no for the coins. The coin sorting has structure in the set theorist's sense and none in Rissanen's. I'll keep both meanings, and I'll try to say which one I'm using, and when I slip you'll know what to look for.

## 5.13 The four parts of every computer

In 1936 Alan Turing described the machine that every computer you've ever touched is a version of, and he did it before any of them existed. It's a clerk with a pencil, following a rulebook, written down as a machine (that clerk comes back in Section 14). Write down its parts; there are four. There's a tape, divided into squares; a head that reads one square at a time and can write on it and move left or right;

a finite list of states the head can be in; and a table of rules: in this state, reading this symbol, do this. Nothing else is in it, and two different things are true about it that get run together. The standard effective models compared with it, including register machines and lambda calculus, compute the same partial functions; those are theorems, proved one pairing at a time, and Turing himself sketched one of them, the pairing with the logician Alonzo Church's lambda calculus, in an appendix to the 1936 paper. That it also captures everything a person could ever carry out by rote, everything "computable" in the plain-English sense, is a claim, called the Church-Turing thesis, and nobody has proved it, because the plain-English side isn't a mathematical object. Turing himself kept those two apart in the 1936 paper, and so will I. There's a third claim that gets stacked on top, and it's a different kind again: the physical Church-Turing thesis, that every device you could build out of real stuff computes no more than this machine does. That one is a claim about physics. No mathematical proof can settle it. Nobody has built a device that breaks it. Your laptop, your phone, every server in every data center compute nothing a Turing machine can't; what differs between them is speed, space, energy, and cost. Hold on to that last word. Cost is where this book lives.

Now do the thing I did, and do it honestly. Look through those four parts for a place that says "these squares belong together." There isn't one, and that's not me failing to look. Everything the machine is at any moment is which state it's in, where the head is, and what's on the tape, and the table says what happens next, and that's the full list. So any grouping has to live in the states and the table, as a rule, or on the tape, as symbols, which is to say as bytes like any other bytes. The machine can hold a grouping. It just has no place where holding one is different from holding anything else.

In 1972 David Parnas, a software engineer, wrote a short paper that every programmer should be made to read and most haven't, on how to cut a program into modules. He took one small program and cut it two ways. Both worked. But one was easy to change later and the other wasn't, and the easy one could be "much less efficient" if talking between modules cost anything. His point was that a module is "a responsibility assignment," a decision about who answers for what, and how you cut decides what every later change costs. If you've ever maintained someone else's code, you've paid for their cut the way Parnas said you would, probably at two in the morning. What survives the push is that a division is a commitment: it changes what the work costs, and it is no caption, no label stuck on afterwards. I'd like to say the thing I built understood that from the beginning. It didn't. I did, eventually.

### 5.14 Laying them all down

Now lay every one of those on top of the others and see what shows through. I see seven things. They're my overlay, the places where I see the sources agreeing. The sources never proved it together; some of them never met, and some of them disagree (Eilenberg and Mac Lane put maps and their composition at the centre; Shannon's structure and Tarski's are two different words, as I just showed you). You've pushed on each source yourself by now, so you're in a position to tell me if I'm seeing things.

A list of seven slogans would be easy to nod at and impossible to check, so I'm going to carry one thing through all seven and let you watch each one add to the one before it. The thing is the square you drew for the Pythagoreans, side, diagonal and all. Draw a few more on the same sheet: one small, one big, the small one again but tilted, and one stretched out into a long rectangle. You'll want a pile.

Structure is about relations among things, and the things themselves aren't it. You see that in the diagonal, the bronze, Leibniz's room, Bourbaki's unspecified elements, Tarski's named relations and the arrows. On your sheet, what makes a square a square was never the ink. It's that the four sides match and the corners are square, and that the diagonal stands to the side the way it does whether or not any number names it. Every one of those is a relation among parts. But on its own this first thing is too loose to use, because a sheet of paper is drowning in relations: this square is left of that one, this one is nearer the coffee ring. Which relations count?

Structure is what stays the same under a chosen set of changes. You see that in Euclid's triangles laid together, Klein's allowed moves, Noether's symmetry, and Frege's direction that every parallel line shares. This second thing gives the answer the first thing was missing: the relations that count are the ones that survive the moves you've decided to allow. Slide the small square across the sheet, turn it, and the diagonal still stands to the side exactly as before; "left of the coffee ring" doesn't make it across. The second thing is a filter on the first.

The same structure can be carried by different stuff: the statue in marble, or two sets that answer the same vocabulary the same way. The third thing is the second again with one more change allowed, which is to swap the material. Cut the square out of cardboard, or lay it out in string on the table, and if the relations survive, it's the same square in every way you agreed to care about. Aristotle's bronze is one more thing you agreed not to look at, and the wooden sandwich is what happens when the material was never on that list.

The first act of structure is a sameness that sorts a pile you were already given. It's there in Euclid's first common notion, Frege's direction, Tarski's one question, and Russell's letter, with Zermelo's repair that you must start from a pile. At this point the overlay stopped being a list for me and turned into a shape, so slow down

here with me. Look at your pile of drawings and say: two of them count as the same when an allowed move, for now a slide or a turn, carries one exactly onto the other. Watch what happens to the pile. It falls into groups. The small square and its tilted twin go together; the big one sits on its own; the rectangle sits on its own. And it falls cleanly, nothing half in one group and half in another, for a reason you can check. Klein's allowed moves always come with three guarantees: doing nothing counts as a move, every move can be undone, and two moves in a row is again a move. Those three are, in different clothes, "everything is the same as itself," "if this matches that, then that matches this," and "if this matches that and that matches a third, then this matches the third." Put the last two together and you get Euclid's first common notion, "things which are equal to the same thing are also equal to one another." So the second thing, sameness under chosen moves, *is* a sorting of the pile the moment you ask which drawings it calls the same. Frege did it with lines and parallels. Tarski's one question does it in the crudest way there is, by cutting the pile in two. And the pile has to be there first, because a sameness has to be a sameness among something; that's Russell's letter, and Zermelo's repair.

Adding structure rules possibilities out. Every one of Euclid's postulates rules one out. Half of English is already decided. A conserved quantity limits where a system can go. On your sheet: once the small square and its tilted twin are in one group, you've given up the right to answer any question the moves respect differently for the two of them. Tell me the side of one and the diagonal of the other isn't an open question any more. A sorting rules out a whole bundle of possibilities at once.

Held structure gets used without re-deriving it. You see that in Shannon's saving in time, Rissanen's compression, Parnas's one-module-at-a-time, and Leibniz's analysis of situation (reasoning about where things are straight from their relations, with no coordinates laid down first). The fifth thing pays off here: the possibilities you ruled out are ones you never have to check again. Know which group a drawing is in and you know everything the group shares without getting the ruler out, the same way the row of `01` never needed writing out. There's one catch, and I'll need a whole section for it: it only pays if you kept the sorting somewhere you can read it back.

And structure is always relative to a chosen frame, whether that frame is Klein's group, the vocabulary, the code you compress against, or Parnas's criteria. Nobody gets to say "the" structure of a thing without saying from where, and I do it constantly, so watch me for it. On your sheet: allow stretching as one of the moves and the rectangle walks over and joins the squares; allow only slides and turns and it doesn't. Same pile, same ink, different groups, because I chose different moves. The groups were never in the paper. They were in the paper and the choice, together.

I think the seven hang together this way. Relations are what structure is made of. The chosen moves pick which relations count, and a change of material is just

one more move. Asking which things the moves call the same sorts the pile. The sorting rules possibilities out, and what's ruled out is what you get to skip later. And because the moves were chosen, all of it is relative. Read that way, the fourth thing sits in the middle: it's where the first three turn into something you can write down, and where the last three get their start.

That's my reason for starting the next subsection with a sorting, and I want to be exact about what kind of reason it is, because it's mine. None of these people said "start with a partition." Klein would tell you to start with the moves, Eilenberg and Mac Lane with the arrows, Shannon with the probabilities. What I can actually point at is narrower. In the framings about when two things are the same shape or the same kind (Euclid laying one triangle on another, Klein's moves, Frege's directions, Tarski's question) there's a moment where somebody decides what counts as the same, and that decision, written down, is a sorting of a pile. In Shannon, Rissanen and Noether I can't find that moment as cleanly, and I'm not going to bend them until I do. So the overlay doesn't prove that a sorting comes first. It shows me where the framings I trust most keep putting their first foot down, and I'm putting mine there too.

I looked for a framing that denies any of the seven and didn't find one, which could mean there isn't one or could mean I didn't look where it lives. If you know of one, I'd like to see it.

### 5.15 The smallest piece

Now the sentence I wanted to say at the top and had no right to until here, and even here I have to split it in two, because half of it is a textbook fact and the other half is me. The piece of structure I'm going to start from is a partition.

The textbook half: a partition and an equivalence relation are the same object. That's Paul Halmos, *Naive Set Theory*, 1960, section seven, and you could read that section before dinner. I'll say it in words first. Take any pile of things. An equivalence relation on it is a way of saying "these two count as the same" that behaves the way sameness should: everything is the same as itself; if $a$ is the same as $b$ then $b$ is the same as $a$; and if $a$ is the same as $b$ and $b$ the same as $c$, then $a$ is the same as $c$. Every such relation sorts the pile into groups, the things that count as the same go together, and every sorting into groups hands you back a relation, "same group." Do one and then the other and you land where you started, in both directions. So a partition and an equivalence relation are one object seen from two sides, and that's the fourth thing on the overlay, the sorting of a given pile, now with a name and a page number.

I'll give it in the exact form too, since this is the first place the book uses one. A *set* is a pile, $x \in X$ means "$x$ is in the pile $X$," and a *relation* on $X$ is any rule $\sim$

that, for each pair of things, says whether $x \sim y$ holds. A few more symbols show up below, and each is a word in disguise. The buckets are written $S_j$, with $j$ saying which bucket (finitely many buckets in every example here, though nothing below needs that). A cap, $S_j \cap S_l$, is "what the two buckets share," and $\emptyset$ is the empty pile. And curly braces with a colon, $\{\, y \in X : y \sim x \,\}$, mean "the pile of every $y$ in $X$ for which $y \sim x$ holds"; the theorem writes that pile $[x]$, the things that count as the same as $x$.

**Definition 5.1** (Equivalence relation, partition)**.** A relation $\sim$ on a set $X$ is an *equivalence relation* when it is reflexive ($x \sim x$), symmetric ($x \sim y$ implies $y \sim x$), and transitive ($x \sim y$ and $y \sim z$ imply $x \sim z$). A *partition* of $X$ is a collection of non-empty subsets of $X$, the blocks, such that every member of $X$ lies in exactly one block. Spelled out, with the blocks written $S_j$: each $S_j$ is non-empty, no two share a member ($S_j \cap S_l = \emptyset$ when $j \neq l$), and together they cover everything (every member of $X$ lies in some $S_j$).

**Theorem 5.1** (Sameness and sorting are one object)**.** Every equivalence relation on $X$ determines a partition, whose blocks are the classes $[x] = \{\, y \in X : y \sim x \,\}$; every partition determines an equivalence relation, "$x$ and $y$ lie in the same block"; and the two constructions undo each other. □

*Proof.* The classes are non-empty ($x \in [x]$ by reflexivity) and cover $X$. If $z$ lies in $[x]$ and in $[y]$, then $z \sim x$ and $z \sim y$, so $x \sim y$ by symmetry and transitivity, and then $[x] = [y]$ by symmetry and transitivity again (anything the same as $x$ is the same as $y$, and the other way round); so two classes either coincide or don't overlap. Going the other way, take a partition and say two things match when they share a block. Each thing shares its block with itself. If one shares a block with another, the other shares it with the first. And if $x$ shares with $y$ and $y$ shares with $z$, both blocks contain $y$. A partition puts $y$ in only one block, so all three share that block. These are exactly reflexivity, symmetry and transitivity. Going relation to partition to relation returns "$x$ and $y$ share a class," which holds exactly when $x \sim y$ (if $x \sim y$ then $x$ and $y$ both lie in $[y]$; if both lie in $[z]$ then $x \sim z$ and, by symmetry, $z \sim y$, so $x \sim y$), so it is the original relation. Going partition to relation to partition returns the original blocks, because the class of $x$ under "same block" is exactly the block that contains $x$. ■

All of this is textbook mathematics (Halmos, section 7), and the book uses it as a foundation. I'm claiming nothing new in it.

And the smallest structure that sorts, by my ruler in Tarski's setup, the one question, is the same object a third time. Put a yes-or-no question to every thing in the pile. The ones that say yes are one group and the ones that say no are the other. Going the other way, a two-group partition becomes a question the moment you point at one group and call it "yes." The only thing a question has that the partition doesn't is which group got the label. So the set theorist's first step and the logician's first step are the same step, and nobody has to pick a side, which is a relief, because it's an argument you can lose with yourself for a long time.

So much for the textbook half. The other half is mine, so hold it looser, because Halmos says nothing about "smallest" or "first" and I'd be lying if I let his page number carry my sentence. In what I'm going to build, the order goes: a pile, then groups in the pile, then whatever connects one group to another, and the groups come first because a question needs something to be about and a connection needs somewhere to land. That's an order for *my* construction. It is not a law that every account of structure has to start by sorting. You could start with an ordering instead ("this one comes before that one"), or with arrows between the things themselves and no groups at all, and people do, and nothing I've shown you rules that out. So the honest sentence is: a partition is where I chose to start, it is the smallest start by the ruler I gave you, and I chose it because it's the first thing that can be right or wrong about a pile and the last thing I could take away without everything I build on it coming down. If you have a smaller start by a ruler you can state, I want to see it. It would change this book.

### 5.16 Buckets

So I'll do the smallest piece once, for real. Take everything in front of you and sort it into buckets, with nothing in two buckets and nothing left on the floor, and that's the whole of the definition. It's the thing you do with a kindergartner and a pile of blocks, and I am aware that I've just marched you past Pythagoras, Plato, Euclid, Frege and Noether to arrive at a pile of blocks. Keep the blocks anyway. They're about to cost you something.

Take ten blocks, and put the red ones in one bucket, the square ones in another and everything else in a third. Now a red square turns up, as they do, and you've got your first crisis: it belongs in two buckets and the rule says one. So this isn't a partition yet, and you have to decide which bucket wins. That decision is the whole thing, because a partition is a pile of small rulings like that one, and once you've made them all, the pile is holding a fact it wasn't holding a minute ago. The formal version says the same thing with fewer words and more squiggles, and you've already seen it: it's the definition of a partition in the last subsection (Definition 5.1), which asks for non-empty buckets, with nothing in two and nothing on the floor.

The next part isn't kindergarten. Why would that fact cost anything? Watch what it does to one question. Before the buckets, "are these two in the same group?" means working it out: go find the first one, go find the second one, apply the rule to each, and compare. After the buckets, it's a glance: read the first one's bucket, read the second one's bucket, and you're done. With each block's address already in hand, and a table that retrieves a label in one operation, that is two lookups whether the pile has ten blocks or a billion. Those are access and cost assumptions. A pile with no addresses, or an index that must be searched, has not been given that same two-lookup rule. What you're looking at is Shannon's saving in time and Parnas's one-module-at-a-time, the sixth thing on the overlay, showing up in a kindergarten.

Look at what makes that glance possible: a *table*, with a written-down line for each block saying which bucket it went in. The partition is the fact; the table is one way of holding the fact so it can be read back fast, and you had to build the table, and you have to store it, and it has to be the kind of thing you can read in one look (a bucket number, say; a list of every block in the bucket would have you searching through it again). Anybody who's used a database has met this table; it's what an index is, and an index is a thing you build, store, and keep up to date, and every part of that takes something. Take the table away and keep the partition and you're back to working it out. And sometimes there was never a search to save: if the rule is "evens here, odds there," you can answer "same group?" by looking at the last digit of each, no buckets needed. What survives is that holding a sorting in a form you can read back is worth something, and building that form cost something, and those are already two different events. Building the table, storing it, checking that it's right, and deciding to *rely* on it are four different things. The first cost question I want to put to that structure lives in the gap between the third and the fourth. Nothing in the kindergarten has shown a charge in that gap. It has separated the acts well enough to ask.

So who, in everything I read, says what has to be paid? I found three answers that come close, and I'll be exact about how close, because this is the edge of what I know. Rolf Landauer, in 1961, and Charles Bennett after him, say that any step which takes two possible situations and makes them one, erasing a bit or merging two paths a computation could have taken, has a heat floor when it removes uncertainty under their thermodynamic assumptions. That is a physical claim about how the device and its surroundings are prepared, Landauer's principle, and in this book it is always a premise, never something I prove. It's the place, of all I've found, where a fact about information turns into a fact about heat, though at room temperature the amount for one bit is far too small to show up on anybody's electricity bill. In 2012 a lab in Lyon erased one bit, held in a single tiny bead, and measured the average heat it gave off coming down toward Landauer's floor as the erasing was slowed, and other labs have found the same since. Not everyone accepts the principle in

general, and I'll come back to the people who don't. Is sending blocks to buckets a step of that kind? Push on it, because the answer depends on what you keep. If you keep each block and write its bucket next to it on a spot you knew was blank, you've merged nothing; you can read the block right back off the pair. You have spent something, though: the blank spot the label now sits on, which the machine hadn't used before, and any heat from the writing depends on its implementation. Bennett's whole point in his 1973 paper was that a computation which keeps its inputs, and keeps enough of its working to retrace its steps, can be run without any merging step at all, and it pays for that by taking fresh states. If instead you throw the block away and keep only the bucket label, then two different blocks can leave the same label behind, and that is a many-to-one step, the kind Landauer charges for. Both are things people call "sorting into buckets." Only the second loses a distinction. So the bucket by itself doesn't hand you Landauer's heat. The *forgetting* does, when there is forgetting, and keeping instead is paid for with fresh states. Whether a given machine forgets is a question about that machine. It has to be answered on the page, and I won't wave at it here. Rissanen says a model is paid for in the bits it takes to state, and Grünwald says the price depends on the code. Parnas says a bad cut is paid for later, at two in the morning. And that's where it stops. Rovelli, back in Section 4, prices a choice that sticks, and his choice is one nobody could predict, which that section showed is a different act. I couldn't find anyone who says that the moment you *vouch* for a grouping, the moment you stand behind "these belong together," has a price of its own. Just arranging them that way is a different act. That could mean nobody says it. It could mean I didn't find the person who does. I honestly can't tell you which.

So that's where I am at the end of one word. A grouping, the act of making it, the information kept about it, and a commitment to rely on it are different things. The blocks let you separate them with your hands. Now take them down to the smallest case there is, a bucket with two sides, one block, and a ruling that sticks, which is a cell going from blank to 0, the case Part I started from and showed what keeping it takes. The construction I will study keeps selected distinctions in a state. Some are there from the start. Some are newly recorded. Those are different cases: a box that already says YES can sit there forever under a do-nothing rule, with neither a merge nor fresh states. Part I's count starts with NO and follows a new YES that stays under a fixed total rule. That is the case to which its merge-or-fresh conclusion applies. A grouping by itself does not supply that run or its price.

CHAPTER 6

# The counter

Everything so far has kept the count of cost on the side of the page. You played Section 2's toy, you made a tally as you went, and the tally lived in your hand. The machine never saw it. A tally takes a state it has never used for each unit it records and never goes back to an old one, and putting it inside the machine doesn't change that; it only makes the machine the one paying with those fresh states. That's fine for a toy. For anything that's going to run on its own, with nobody tallying, the total has to live somewhere the machine can reach, which means the machine can, in principle, write it. So this section is about what a counter has to be before you're allowed to call it a receipt. I mean the counter of *any* machine whose state carries a running total that every step bumps by whatever a price list says that step costs. Section 2's five parts don't need one (there the total was a sum you did on the side), so this is one piece added to the five, and the facts below are about anything that has it. I'll give you the answer first, because it's short, and then spend the section on why each word is there: a receipt is a total that lives in the state, that only climbs, that equals the sum of what was charged, and that nothing else could have been.

## 6.1 A stick, split down the middle

The oldest receipt I could find that you can still hold is a piece of wood. From about 1100, under Henry the First, until 1826, the English treasury recorded what it was owed on tally sticks: a strip of hazel, notched across with cuts of set widths (the treasury's own handbook from around 1180, the *Dialogue of the Exchequer*, gives them: a thumb's width for a hundred pounds, a swollen barleycorn for a pound, a bare cut for a penny), then split lengthwise so that each half carried the same notches. The payer kept one half, called the stock, which is one story for where the word stockholder comes from. The treasury kept the other. When the time came to settle, you laid the two halves together, and only the two original halves fit, because wood splits along a grain nobody can copy. Add a notch to your half and the other half shows a gap where the partner should be. Try to remove a notch and you find that wood can't be uncut. When they finally stopped using them, the clerks burned centuries of old tallies in the furnaces under the House of Lords, overheated the flue,

and burned down the Palace of Westminster, which I mention because it's the best fact I picked up for this book and because it tells you how many of them there were.

Do it: draw a border round a strip of card, notch it once for every coin you hand a friend, then split it lengthwise through the notches and keep half each. Don't trim the edge or swap in a fresh strip. Now add a notch to one half and put them back together. You'll see the mismatch straight away. And you can't put a cut back: a patch is just one more thing to look at. The total is the notches, one per coin, and under that rule the count only climbs. Both halves started with the same count, so if they disagree now, somebody changed one. The mismatch won't tell you who. What the stick gives you is a comparison you can make, not an object nobody can forge and not a verdict on who cheated.

Economists read the same stick another way, and they're not wrong either. Alfred Mitchell-Innes (1913) made the exchequer tally his star witness that money is credit: the crown paid the people it owed by "raising a tally" on some branch of its revenue, a wooden acknowledgment of debt. A receipt to me, money to him, and it's the same piece of wood. Narayana Kocherlakota (1998) later proved a version of his point: in a wide class of economies where nobody can be held to a promise, anything money lets people do, a memory of everyone's past dealings lets them do too. His title says it in three words: money is memory.

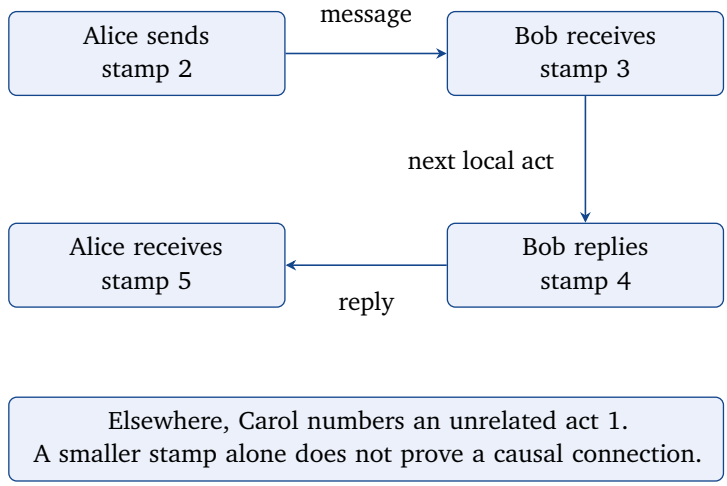
## 6.2 Why it has to climb

Start with the second property, because it's the one that does the most work and the one people fight hardest. Why can't the counter ever go down?

The answer is in what the toll is for. The floor it buys, back in Section 2, says a run from NO to YES costs at least one. That sentence is a *floor*: it says the run paid at least this much. If the counter could come back down, a run could pay the toll, get its flag, and then take the payment back, and the floor would be a floor on nothing; "this run cost at least one" would be true of a run that ended having cost nothing. So a counter that can fall is an accountant you can lean on to revise the books, and a floor stated in terms of it isn't a floor. The counter only climbing is not a flourish; it's what makes "at least" mean something. This ledger records cumulative charges, each a whole number of units and none negative, so it never decreases. A refundable balance, one that pays money back, would be a different accounting object with a different invariant, and the floor here says nothing about it. (In Section 2's abstract theorem the total is a sum of whole numbers and climbing is automatic; this paragraph is about why a total *stored in the state*, where a step could in principle write it, has to be designed so that no step ever lowers it.)

The same rule turns up in two places that have nothing to do with each other, pointing in opposite directions, and that's a big part of why I trust it. In 1978 Leslie

Lamport asked how a set of computers that pass each other messages, with no shared clock, could ever agree on what happened before what. His answer is a counter per machine with two rules: add one before you do anything, and when a message arrives carrying the sender's count, jump your own counter above it (never below your own). Those counters, he showed, respect cause and effect: if this caused that, this has the smaller number. And they do it "with no actual timing mechanism," his words; they're clocks only in name. Try it with three people passing notes and you'll see it work in two minutes, and then let one of them reset their counter to zero after reading a note, and watch a reply come out numbered *before* the note it replies to. A counter that goes down can't be leaned on for order.




Start Bob at zero and Alice just after her local act 1. On receipt, each raises the local count above both its old value and the incoming stamp.

Where did Lamport get it? In the notes he keeps on his own papers, he says the timestamps came from an earlier note by Paul Johnson and Bob Thomas on keeping copies of a database in step, whose algorithm wasn't quite right: it let things happen in an order that broke cause and effect. What let him see the fix, he says, was special relativity, which teaches that the one order of events everybody agrees on is the order of what could have caused what. And he passes on a line from Jim Gray, the database pioneer whose commit comes back in Section 14: Gray had heard two opinions of the paper, that it was trivial and that it was brilliant. Lamport's answer is that he can't argue with the first and isn't inclined to argue with the second. His account stops at relativity, so the chase does too. It leaves a good picture behind: these update rules put a smaller number on every cause than on its later effect, without a shared clock. The converse fails: two people can number unrelated local acts 1 and 2. The numbers alone do not say that the first caused the second.

Now the other direction. In 1949 Alan Turing, and in 1967 Robert Floyd, each asked how you prove a program *stops*. Both times the answer was to find a quantity

that goes *down* on every step and can't go down forever (a whole number that can't go below zero, say). If every step lowers it and it can't fall forever, the steps can't go on forever. Up-only gives you order, and down-only gives you stopping. They're different rules (one needs the jump-above-the-message step, the other needs a floor it can't fall below), but they have the same shape, seen from the two ends: a one-way quantity is something you can build on, and a two-way quantity is a number that happens to be written down somewhere.

There's also a real-world case where a counter was allowed to go down and got gamed exactly there. Ethereum, one of the largest blockchains, charges "gas" for every step of a program, and for years it *refunded* gas when a program freed up storage. Someone noticed you could fill storage when gas was cheap, hold it, and cash the refunds when gas was dear, and a token was built to do exactly that, and the real work in a block could reach nearly twice the block's stated limit. In 2021 a change called EIP-3529 cut the refunds sharply (none for self-destruct, a smaller one for clearing storage, and a cap of one fifth of the gas used). The written reasons are the gas token and those overfull blocks. I read both as the same lesson: a counter that goes down was being leaned on at the place where it went down. Ethereum's gas comes back in Section 17 as one of five small models of systems people actually run, and it's the only one of the five that takes a fee at the moment of the write, and it's instructive that the part they had to fix was the part that could fall.

### 6.3 Conservation: the total is the charges and nothing else

The third property is the plainest one. The counter after a step is the counter before plus what that step was charged. Add up a whole run and the middle values cancel out, the way a sum of differences collapses to last minus first, and the total is the sum of the per-step charges. Why state it?

I state it because it's the property that fails quietly. Say a machine charges one for raising a flag, and also, somewhere in its error handling, forgets to add the charge when the step fails halfway through. Or it charges two in one code path and one in another for the same move. Or it has one step that writes the counter directly. Then the counter is *mostly* the sum of the charges, and "mostly" is the whole gap between a receipt and a note that says "trust me, about this much." Conservation is the sentence that there is no "mostly": every step, every outcome, every error, adds exactly its charge, and a step that fails is charged the same as one that succeeds, because the charge is on the attempt. For a machine built the way I'm describing, that sentence is the rule the counter is updated by, so it can't fail by definition. What can fail is the claim that a real machine *is* built that way, and checking that means going through every move it has, the failure branches included, one at a time, which is dull and is the whole job. The run-long form, counter at the end equals counter

at the start plus the sum of the charges, then follows from the one-step rule by the cancelling sum; it's below. It's the kind of theorem that looks boring and is the one you'd most want if you were going to rely on the number. One thing it doesn't say is that every unit of cost bought something. The ledger is exact about what was spent. What the spending was worth is another question, and whether a unit of cost bought a real narrowing is Section 8's question, and it needs a witness on top of the ledger entry.

*Which mergers does this counter cover?*
Draw the arrows of one run and put a receipt beside each departure. Charge on entry into a parked state and let the parked state idle free. The charge now depends on where the step begins. You can check exactly which mergers those receipts cover.

This paragraph changes the accounting interface explicitly. The basic Thiele definition uses a fixed instruction price $c(i) \in \mathbb{N}$. Here one autonomous step $F$ has a source charge $a(s) \in \mathbb{N}$; with supplied instructions the broader signature would be $\widetilde{c}(s,i)$. Its bill must be summed along the actual run. The same instruction can now cost differently at entry and at idle. The theorem below states its own charge and coverage premises; it does not change the instruction-cost definition used elsewhere.

Start at $u$ and follow $L$ steps. The source positions are $u, F(u), \dots, F^{L-1}(u)$. Mark a landing state if two different sources in that stretch lead to it. Count each marked landing once and call the total $\mathrm{Merge}(L)$. Walking past the same source twice still gives one source. The bill, meanwhile, adds the charge for every step taken.

The pricing rule is this: whenever two different observed sources share a successor, at least one of them pays. Give each source its charge $a(s)$ and let the ledger add exactly that charge. Now every marked landing has a receipt behind it.

**Theorem 6.1** (Receipts covering the observed mergers)**.** Let $a(s) \in \mathbb{N}$ be a source charge and let $\mu(F(s)) = \mu(s) + a(s)$. Suppose every pair of distinct observed sources with the same successor has a positive charge on at least one member. Then

$$\mathrm{Merge}(L) \le \sum_{t=0}^{L-1} a(F^t(u)) = \mu(F^L(u)) - \mu(u).$$

*Proof.* Every counted landing state has a paid predecessor. One predecessor cannot belong to two different landing states, since its successor is fixed. Thus there are at least as many paid source occurrences as counted landing states, and each costs at least one. The ledger equality follows by adding its updates. □

Try it with no going to yes and yes going to itself. Charge one at no and zero at yes. Give those states ledger values zero and one. Entry adds the one; every later idle step adds zero. The collision is covered and the parked machine can stay parked for free.

Now ask that parked yes-state how many mergers your observation has exposed. After one step you have seen only no as a source: no merger yet. After two steps you have seen no and yes, both leading to yes: one merger. The endpoint in your hand is yes both times. It cannot tell you which observation you made.

Add a clock and you can keep that difference. The endpoints become $(\text{yes}, 1)$ and $(\text{yes}, 2)$. You can read the clock and recover the count for the old graph. Draw the new full graph, though, and those clock values keep its states apart. Keep both drawings on the bench so you can see which arrows you are counting. An external record of the observation length can keep the same missing information.

## 6.4 Only one honest count

The next property surprised me by being provable at all. Suppose you fixed the schedule: this move costs one, that move costs zero, and so on, for every move. And suppose someone else keeps their own books: their own function from states to totals, which starts at zero and, on every step, goes up by exactly what the schedule says. Could their books disagree with the machine's counter, somewhere, on some state you can actually get to?

No, and the proof is the tally stick. Both totals start at zero. On every step both go up by the same amount. So on every state you can reach by steps from the start, they're equal. Nothing more is needed. Agreement at the start survives the first move, then the second, then every move after it. This way of proving a statement for any finite number of moves is called induction. The mathematician's word for this is *initiality*: the counter is the one bookkeeping that all the others, if they're honest about the schedule, turn out to be. Given the schedule, the count is forced. There are two things about that sentence I had to be careful with, and they're worth saying before the symbols. It doesn't assume the other books climb: climbing follows from the schedule, since the schedule never charges a negative amount, so "only one honest count" and "it only climbs" aren't two separate demands on the other bookkeeper. And it holds on *reachable* states only. A state you can't get to from the start is one where it says nothing about whether the other books agree, which is the right thing for a receipt to say.

I'll state the properties exactly now: first the piece being added to Section 2's five, a counter and a price list, and then the two facts about it. A *counter machine* here means a machine that keeps this ledger. It isn't Minsky's counter machine, the two-counter base Part I mentioned; the word is the same and the thing is different,

and Section 15 keeps the two apart. The counter gets the Greek letter $\mu$ (mu), which is the letter the machine in Part III uses for it (Theorem 15.6).

**Definition 6.1** (A counter kept by a price list)**.** A *counter machine* has a set of states, a set of moves, a step function apply that takes a state and a move, a start state $s_{\text{init}}$, a schedule $c$ giving each move a cost in $\mathbb{N}$, and a counter $\mu$ from states to $\mathbb{N}$, such that $\mu(s_{\text{init}}) = 0$ and $\mu(\text{apply}(s, i)) = \mu(s) + c(i)$ for every state $s$ and move $i$, with no exceptions. A *run* of a list $t$ of moves from $s$ applies them in order, and $c(t)$ is the sum of their costs. A state is *reachable* when it is the start or is $\text{apply}(s, i)$ for a reachable $s$ and some $i$.

**Theorem 6.2** (Conservation)**.** In a counter machine, for every state $s$ and every list $t$, $\mu$ at the end of the run of $t$ from $s$ equals $\mu(s) + c(t)$. In particular $\mu$ never decreases along a run. □

*Proof.* Write the starting counter at the top of a column. Under it write the charge for the first move, then the charge for the second, and continue to the end of the run. After the first move the counter is the sum of the first two entries, because that is its one-step rule. After the second it is that subtotal plus the next charge. Every further move adds exactly the next entry. With no moves there is only the starting entry. With any number of moves, the final counter is therefore the starting value plus their charges, which is $\mu(s) + c(t)$. Every entry added is nonnegative, so no subtotal is smaller than the one before. ■

**Theorem 6.3** (Only one honest count)**.** In a counter machine, let $B$ (for "books") be any function from states to $\mathbb{N}$ with $B(s_{\text{init}}) = 0$ and $B(\text{apply}(s, i)) = B(s) + c(i)$ for all $s$ and $i$. Then $B(s) = \mu(s)$ for every reachable state $s$. □

*Proof.* Choose a run that reaches $s$ from the start; such a run exists because $s$ is reachable. At the start both books say zero. At the first move both add the same charge, so they still agree. Whenever they agree before a move, adding that move's charge to both keeps them equal afterward. Carry this comparison along the chosen run. The two books agree at its last state, which is $s$. ■

One thing follows that's easy to miss. If a run comes back to a state it has already visited, the counter is the same both times, so the moves on that loop cost nothing in total. A machine that keeps its counter inside its state can go round a loop only when the loop costs nothing, and every paid move lands on a state the run has never

been in, because the counter is higher there than anywhere before. So a counter that only climbs is itself paid for in fresh states: it needs a fresh number for every unit of cost, and it never gets one back. On finitely many states that leaves no paid move at all: play any move over and over from any state, the states come round, and a loop costs nothing, so its price is zero. One move priced above zero takes infinitely many states.

Those two proofs are the whole of it, and they're about any counter machine. They're on the page, where you can check every line. The machine I build later is one of these, and on it the run form is a theorem: the ledger after any list of instructions is the start plus the sum of their costs (Theorem 15.6). The uniqueness half holds even with the books' rule asked only of reachable states.

I want to say plainly what initiality does *not* do, because the fact is smaller than the word. It doesn't pick the prices. The schedule is an input: somebody decided that raising the flag costs one and adding two numbers costs zero, and nothing here says that decision was right (Section 3 is where I argued the floor isn't arbitrary; the rest of a schedule has to be defended by whoever writes it). It doesn't turn units of cost into anything physical. And it isn't the category theorist's "initial object," though the word is borrowed for a related reason.

Try the same instruction list on another machine that understands those instructions. Choose its start. The empty list leaves it there. Append one instruction and its own rule tells you the next state. Append another and the rule tells you again. There is only one answer at each stage. So there is only one way to assign a target state to every instruction list while respecting the chosen start and each appended move. This walking through a list is often called a *fold*. Nothing in the argument used a certificate or a price.

Now ask a different question. Can you assign a target state to each *state of the first machine*, instead of to each instruction list? Two different lists may leave the first machine at the same state. If those lists leave the target at different states, the assignment would have to give two answers at that one state. It cannot.

That is the only obstruction. Suppose every pair of lists that agree at the first machine also agree at the target. To assign an answer to a reachable state, take any list reaching it and run that list on the target. Choosing another reaching list gives the same answer, by the supposition. Appending a move respects both machines' steps because it extends those very runs. The assignment is fixed on every reachable state, since each comes with a reaching list. This is the *descent* condition: the assignment can come down from histories to states precisely when the target does not distinguish histories the first machine has already identified.

A toll does not force that condition. Take a one-bit machine whose set costs one and whose reset costs zero. Setting its bit once leaves it up. Resetting and then setting also leaves it up. But a target that keeps the whole instruction list

distinguishes "set" from "reset, set." Both lists reach the same bit state, and they reach different history states. The instruction-list assignment still works; the state assignment fails. That is where the word "initial" has to stop.

And the borrowed word has a home. I didn't build it, so I'll point at it. That lists of moves are the starting object, and that a machine's behaviour factors through its reachable states exactly when a condition like descent holds, is the theory of minimal realization. Joseph Goguen put it as a universal property in 1972, in a paper whose title says it outright, "Realization is universal," and Thomas Colcombet and Daniela Petrişan redid automata minimization in the same functorial terms in 2020. Their machines can have any output at all, which is the honest point: the fold and the descent condition never use the record, so the toll is doing nothing here, just as I said.

The nearest classical parallel to "only one honest count" is in thermodynamics. The mathematical physicists Elliott Lieb and Jakob Yngvason showed in 1999 that once you fix which states of a system can be reached from which by adiabatic processes, and that relation meets a short list of axioms, there is an entropy, a function that goes up exactly along the relation, and it is unique once its zero and its unit are chosen. Their uniqueness, like this one, comes after the fixing: the order is the input there, the price list is the input here, and in both the count is forced by it.

## 6.5 The toll is an old trick with a new name

Now for something I first took as a let-down. It would be nice to have a new kind of cost argument, and this isn't one. The toll is a case of a technique that algorithm people have used since 1985 and that a mathematician studying the stability of motion had, in a different form, in 1892. I'm telling you it's old so you don't think I invented it.

The trick is called the potential method, and it works like this. Give every state a number, call it the potential, that measures how much "owing" is stored up in that state. Then look at each step and ask: does the cost of the step, plus the change in potential it causes, ever come out negative? If it never does, you've got a bound: any run costs at least the amount the potential *dropped* from start to finish, because the drop had to be paid for by the steps along the way. The computer scientist Robert Tarjan set it out in 1985, crediting the potential view to Daniel Sleator. Tarjan's use goes the other way (potential as prepaid credit, to bound a run from above): he used it to analyse data structures that are occasionally expensive and usually cheap; and the decreasing-function idea underneath it is the one the Russian mathematician Aleksandr Lyapunov used in 1892 to prove that a system settles down. Do the coin version, which runs Tarjan's way, as a ceiling: a stack of coins, PUSH adds one and costs one, POP-ALL removes them all and costs as many as it removed, potential

equals the number of coins. PUSH costs one plus one potential; POP-ALL of k coins costs k and drops the potential by k, net zero. So any sequence from an empty stack costs at most twice the number of pushes, however many expensive pops are in it. The whole method takes three lines.

Push on that 1892 for a second, because a date like that hides a lot. Lyapunov wrote the idea up as his doctoral thesis, in Russian. A French translation came out in 1907, made by a French naval engineer, Édouard Davaux, and corrected by Lyapunov himself. The English one came out in 1992, a hundred years after the original, and it was translated from the French: a translation of a translation. So for most of the twentieth century a reader with only English met the idea secondhand or not at all. Did the algorithm people of 1985 get it from him? I can't tell you, and nothing here needs them to have. Whether anybody walked from one to the other on purpose is a question I'd like answered. What I do have is a picture: the same idea, a number that only goes one way along a run, turned up in a thesis about whether a moving system settles down and in a paper about data structures, close to a century apart.

Now point it at the flag. Potential: one if the flag is down, zero if it's up. A step from down to up drops the potential by one, so for the step's cost plus potential change to be non-negative, the step has to cost at least one. That requirement is the toll exactly, and the two are the same statement: the toll holds for a machine if and only if every step has non-negative amortized cost under the flag potential ("amortized cost" is just that sum, the step's cost plus the change in potential). Then the universal floor from Section 2, "any run from NO to YES paid at least one," is just the potential method's standard conclusion, the drop in potential from one to zero, and every system with the five parts that obeys the toll is an instance, which is exactly what a Thiele machine is. Two more facts fall out of the same picture. Under the toll, a run raises the flag at most as many times as its total cost; each raise uses up a unit and a unit isn't used twice. And if you demand the upper-bound version too, that the potential before a step covers its cost plus the potential after, then the two together say every step's cost is exactly the potential it drops. Go through the four cases. Down to down and up to up drop nothing, so they cost zero. Down to up drops one, so it costs exactly one. Up to down would have to cost minus one, and no step costs less than nothing, so the flag never comes down. So the two together hold exactly when raising the flag costs exactly one, everything else costs zero, and the flag never comes down. Those prices, a cost of one per raise and nothing on any other step, are rise-only pricing in its tightest form, and I'll call them the *exact schedule*. The flag staying up comes from this ceiling, not from the prices: a ceiling that counts raises instead, which Section 7 uses, leaves the same prices and lets the flag come down at cost zero. Notice that Section 2's two-move illustration doesn't meet it: its VOUCH move, played from YES, costs one unit although it raises

nothing, which the floor allows and the exact schedule forbids. Part III's example (Section 15) doesn't meet it either, and can't; both are still Thiele machines, since the toll is all the model asks. The reason: its CHECK and COMMIT, the check and the commitment that must come before its flag goes up, raise no flag, and each sends two states with the same counters and line to one (a COMMIT lands an empty channel with one already naming its claim, and a CHECK that fails lands with the same state already trapped), so pricing the merges in the record's part of the state charges each of them one. It charges every instruction a fixed listed price, which is a different job for the word exact (instruction-exact pricing), and it breaks no rule. (These two facts hold even if the price depends on the state as well as the move's name, which is wider than Section 2's per-name cost; a per-name cost is a special case, and there the floor and the ceiling together are out of reach for any system whose flag ever goes up.) Two older tools sit beside these. Run a list of moves in a "writer," a wrapper that adds up a number as it goes, and you get back the final state and the run's cost; and a per-move schedule is a "grade," fixed before the run. I'm borrowing those two and no more: neither brings along the full type systems they come from. The writer is the move the programming-language researchers Norman Danner and Daniel Licata make in their cost semantics, a translation that carries a cost alongside a program's result (Danner, Licata and Ramyaa, ICFP 2015; Danner and Licata, *Journal of Functional Programming*, 2022). The grade is Shin-ya Katsumata's: a computation carries its resource as a grade, and the grades compose as the computations do ("Parametric effect monads and semantics of effect systems," POPL 2014). Dominic Orchard, Vilem Liepelt and Harley Eades put grades into the types of a working language ("Quantitative program reasoning with graded modal types," ICFP 2019). And the potential method went into types before any of those: the computer scientists Martin Hofmann and Steffen Jost (POPL 2003) gave a type system that attaches potential to data and predicts, before a first-order functional program runs, how much heap it can use, which is where type-based amortized analysis begins; the multivariate work of Jan Hoffmann, Klaus Aehlig and Hofmann (2011) carried it further. Read against them, the counter is a grade carried in the state by the step rule, and no type carries it. And what does that difference even mean? A cost that lives in a type gets checked before the program runs, by whoever checks the types; a cost that lives in the state gets written during the run, by the step itself. The number is the same, and the moment of truth is different.

So is mine an embedding of their frameworks, or something weaker? Half of it is exactly theirs, and the other half is left out on purpose. The counter *is* their writer: a step that moves the rest of the state without looking at the counter, hands back its cost, and has the cost added on is a computation in the writer over whole numbers with addition, and running a list of moves in that writer gives back the final state and the total cost, nothing more and nothing less. Every system in this book whose

step never reads its own counter has that shape. What doesn't come along is the part of their frameworks that works before anything runs: the type systems that carry grades, Danner and Licata's bounding relation and the recurrences they extract for a program's cost. That isn't a gap left to be filled later; it's the moment-of-truth difference I just named. Putting the count in the step is choosing the other moment, so the grade here is written by the step at run time and never checked ahead of it.

How close do the two moments come? Close enough to say exactly, in a limited form. Take a small structured language over a machine with this kind of counter: sequencing, branches whose two arms carry the same grade, and paid acts each graded with their charge, and grade every program the way Katsumata's discipline does, adding grades along the program. On loop-free programs the two agree on the nose: on every run that doesn't trap, the counter rises by exactly the grade. The argument is conservation set beside the grade rule; the counter's rise is the sum of the charges along the run, and the grade rule adds the same charges in the same order. Loops break the agreement, and in one direction only. A loop that runs a paid check once for every unit in some register, where the check always passes and is charged one, costs that register's value on entry, and no fixed whole number is an exact grade for it; an input-dependent grade can express its exact cost if the corresponding proof is supplied. Cost-aware type theories such as calf (Niu, Sterling, Grodin and Harper, 2022) support such reasoning; this is not an automatic exact-cost inference procedure for arbitrary loops. So the counter is their writer exactly, agrees with fixed grades wherever a fixed grade can be exact, and where a run's cost depends on its data it holds the exact number, which a fixed grade can at best bound, and can't bound at all when the data has no ceiling. What has no counterpart in the step is the checking ahead of time: a type checker turning a program away before it runs.

There's one more neighbour, closer than any of those, and I'll name it before somebody names it at me. A number the step keeps up to date and nothing else ever reads is what program-proof people call an *auxiliary variable*, or *ghost* state. Susan Owicki and David Gries added auxiliary variables to parallel programs in 1976, because some true facts about such a program can't be proved without one, under the rule that an auxiliary variable never steers the program. Jean-Christophe Filliâtre, Léon Gondelman and Andrei Paskevich, in 2014, worked out when ghost code can be erased without changing what the program computes. By that test this section's counter is ghost whenever no step consults it, and Section 10's base-driven records are built exactly that way: the base moves the same whatever the ledger and the record say. So what does putting it in the step buy, if the number is ghost anyway? Two things, as far as I can see. The number is there while the machine runs, where a window can show it (Section 7), and not only inside a proof somebody wrote afterwards. The rule that keeps it covers every permitted instruction and

therefore every run. A proof using ghost state can also establish such a machine-wide invariant, and static cost proofs can coexist with a runtime ledger. The difference here is that the defined runtime state includes the count. Whether a record stays ghost depends on what else the step does with it; the machine in Part III has a part that isn't ghost, its table of checked facts, because the step reads it and refuses on it, as the opening section said. One last relative, from the graph side: Section 2's floor is a cheapest-path bound in a system with weights on its moves. Every path from the no-states to the yes-states crosses the line between them, and every edge across that line weighs at least one. Weighted automata go back to Marcel-Paul Schützenberger in 1961, and priced timed automata, which add clocks, were set out in 2001. The floor is about the smallest case of what they compute.

Exactly, for any Thiele machine with states $S$, moves $I$, step, per-name cost, and reading:

**Definition 6.2** (Potential, amortized cost)**.** A *potential* is a function $\Phi : S \to \mathbb{N}$. A step has *non-negative amortized cost* under $\Phi$ when $\mathrm{cost}(i) + \Phi(\mathrm{step}(s, i)) \geq \Phi(s)$. The *flag potential* is $\Phi_{\mathrm{cert}}(s) = 1$ if $\mathrm{cert}(s) =$ no and 0 if it reads yes.

**Theorem 6.4** (The toll is the potential method)**.** (1) The toll holds exactly when every step has non-negative amortized cost under $\Phi_{\mathrm{cert}}$. (2) For any $\Phi$ with non-negative amortized cost and any run $t$ from $s$, $\mathrm{cost}(t) + \Phi(\mathrm{run}(t, s)) \geq \Phi(s)$. (3) So, under the toll, a run from a no-state to a yes-state costs at least $1 - 0 = 1$. □

*Proof.* (1) Put the before and after flags side by side. Down-to-down and up-to-up leave the potential unchanged, so any nonnegative charge meets the inequality. Up-to-down adds one to the potential, so that case meets it too. Down-to-up removes one: only there must the charge supply at least one to make up the drop. That is exactly the toll.

(2) Write one inequality for each step: its charge plus the potential after it is at least the potential before it. Add the lines. Every interior potential appears once as an after and once as the next before, so those equal terms cancel. Only the starting potential, the final potential, and all the charges remain. Thus the charges plus the final potential cover the starting potential. For an empty run the start and finish agree and there are no charges, giving equality.

(3) The flag potential starts at one when the flag is down and ends at zero when it is up. The charges must cover that drop of one. ■

So what's new, if the method isn't? Only where it's pointed. The potential method is a way of proving things about costs. What this book adds is a claim about *which*

potential, the flag, and about what happens to that claim when you can only see a window of the state. The method is Tarjan's and Sleator's. Pointing it at the flag is mine, and I'm about to find out what that can and can't survive.

The flag potential has relatives in physics-flavoured mathematics too, which I found after the fact. Resource theories study what you can turn into what when some operations are free: Bob Coecke, Tobias Fritz and Robert Spekkens gave the general definition in 2016, Fritz worked out its algebra in 2017, and Tomáš Gonda and Spekkens catalogued the numbers that never go up under free operations, which they call monotones, in 2023. Read with the raised flag as the resource and one unit of cost as the currency, the flag potential is the kind of cost monotone they build, measured in that currency, and the floor is its simplest case. Does that make the toll a resource theory? Only as the smallest example of one: it says a run from no to yes costs at least one. To say yes costs exactly one you'd also need a run that gets there for one. It's a neighbour to know about, and I don't count it as a result.

### 6.6 What the counter is, said once

A receipt is a total that lives in the state, only climbs, equals the sum of the charges on every path including the failures, and is the only such total, on the states a run from the start reaches, for its schedule. Each of those had a home before this book: the stick for a total that lives in the state, Lamport's clocks for climbing, Turing's and Floyd's ranking functions for the same one-way rule run downward, the telescoping sum for conservation, the induction for the single honest count. None of them is physics, and none of them says what the charges should be. What they say is that once you've chosen the charges, the counter is the arithmetic and not a second opinion. And the toll on top of it is the potential method with the flag as potential, which is to say a trick from 1985 applied to the question from Section 5.

There's one thing the counter can't do just by existing: if you can only see *part* of the state, can you tell what the counter is? Can you even tell, from outside, which step paid? The stick works because you can lay the two halves together. What if you're only ever shown one half, through a window, and the window doesn't show the notches?

One piece of that has a short answer, and I'll give it now so the next section can start from it. A window, for now, is just the part of the state an outside observer is shown (Section 4's display was one, and Section 7 is about windows in general). Call a window *counter-complete* when some rule, applied to what the window shows, returns the counter on every state; *flag-complete* likewise for the flag. Say a window *forgets the counter* when changing the counter and nothing else never changes what the window shows, and likewise for the flag.

For a full product state in which the counter and flag can be set independently, a window that forgets either field cannot recover that field on all states. Take a state and make two copies of it, one with the counter set to 1 and one with it set to 2, everything else held still. The window shows the same thing for both, so any rule gives the same answer for both, and one of the two answers is wrong. The flag is the same argument with yes and no. On the stick, a window that shows you the wood's colour and its length, and hides the notches, can't tell you what's owed, and staring harder at the colour won't help. (The argument builds its second state by hand, so it may be one no run reaches, and counter-complete asks for every state. On the states a run reaches, forgetting isn't enough: a machine whose one paid move also moves a position the window shows reads its counter off that position everywhere it goes. What blinds a window on reached states is two runs that end on the same view with different counters. The machine I build later has such a pair, written out move by move: two runs with the same window, one paid and one not (Theorem 15.17).)

Be careful what that says. Neither part can be dropped, and nothing else is needed. Call one window bigger than another when what it shows decides what the other shows. A window that gives back both the counter and the flag has a rule for each, so what it shows decides both, and the window that shows those two and nothing more is the smallest that works. And it's about reading the counter back. Which step paid is the harder question, and it's the next section's.

CHAPTER 7

# Windows and shadows

Every state a machine keeps is a distinction, and a look from outside can show some of them and drop the rest. I'll call that look a window. I left you with the stick split in half and asked what happens if you can only ever see one half through a window that doesn't show the notches, and with a harder question under that one: can you tell, from outside, which step paid? This section is about windows. It's a section where the mathematics is so small that I'm going to state it in full and then spend most of the pages on what it's *not*, because the smallness is what catches people: the fact is a one-liner and it's tempting to climb over it with cleverness, and you can't, and seeing why you can't is the content.

## 7.1 A cave, and the people who got good at it

This one is Plato's, and the part I need is near the end. In the seventh book of the *Republic*, Plato has Socrates, the main speaker of the dialogue, describe prisoners chained in a cave since childhood, facing a wall, with a fire behind them and people carrying objects along a walkway between the fire and their backs. All the prisoners have ever seen is the shadows on the wall. To them, Socrates says, the truth would be nothing but those shadows. Further on, Socrates reads the story out himself: the cave is the world of sight, and the climb out of it is the soul's climb toward what's really there. So a real world hidden from anyone who only looks is in the story on purpose, and Plato put it there. It still isn't the part I need.

What I need comes a little before that reading, when one prisoner, freed and taken up into the daylight, thinks back to the cave. The prisoners, Socrates says, hand out honours to whoever is quickest at spotting the shadows as they pass and best at guessing which comes next. The part worth reading twice is that the prisoners *get good at it*. They build a whole science of shadows, with experts and prizes, and it works, on shadows. What they can't do is say anything about a carried object that leaves no mark on the wall, and that isn't because they're stupid; it's because the information isn't on the wall. Skill inside a window doesn't reach past the window, no matter how much skill. Plato dressed that fact as a lesson about

education, and it is one, but take the dressing off and it's a sentence about functions, and it's going to be the whole section.

## 7.2 The shadow of a cube

Do this one tonight with a lamp and a flat wall. Keep the lamp well back, hold a cube face-on with its face parallel to the wall, and mark the outline of its shadow. Replace it with square card and adjust the card's size or distance until the outlines match as closely as you can judge. Show your partner only that outline. It does not tell them whether the object has thickness. For the exact mathematical version, take parallel light rays and let the window report only the silhouette: an aligned cube and a square face of the same size then give exactly the same square. Ask them, and they'll tell you, truthfully, that they can't know, and if they're honest they'll also tell you that no amount of staring at that square will change it.

Now turn each a quarter turn about its vertical centre line. The card's shadow becomes a narrow edge; the cube still casts a broad face. With ideal parallel rays and zero-thickness card, those are exactly a line and a square. Now the wall-watcher knows, and what changed isn't their cleverness but the *window*. A second look from a different angle is a bigger window than one look, and what bought it was the turn. Edwin Abbott wrote a novel about a window like this in 1884, *Flatland*. A Flatlander never sees a shape. Abbott's narrator says it in his first section: nothing was visible to them, or could be, except straight lines. Put a coin flat on a table and lower your eye to the table's edge, and the coin turns into a line, which is all the view there is. What gives a Flatlander anything more is fog. A line that's near is bright and a line that runs off into the fog goes dim, so a trained eye can read a triangle or a pentagon off how fast the brightness fades toward the ends. The training takes time and money, and in the story only people with money get it. The window got bigger, and it was paid for in time.

Then a sphere comes through the narrator's plane. What his eye gets is a line whose ends fade too fast for any flat figure he knows, changing size in a way no flat figure could. He only settles that it's a circle by feeling it, and the circle grows from a point to its widest and shrinks back to nothing as the sphere rises through. From one slice he can't put a sphere together. From the slices one after another, seen and felt and argued over, he starts to. The escape from a window is never "think harder." It's always "a different window," and a different window that answers the question always has to show apart two states the old one showed alike; that's Theorem 7.1 below, applied to the new window. In every story here, getting that window cost something: a turn, a second look, training, time. What a change of window costs in general is a separate question, and no theorem in this section prices it.

Who was Edwin Abbott? Not a mathematician by trade. He was headmaster of the City of London School for twenty-four years, a clergyman, and the author of more than forty books, a Shakespeare grammar among them. He put *Flatland* out under the name "A Square." His full name was Edwin Abbott Abbott, two A's, and the pen name may be a joke on that, A squared; nice if true, but take it as a lead. The part I like best comes from a sketch of his life on the geometer Thomas Banchoff's pages: *Flatland*, by far the most read thing he wrote, isn't mentioned at all in his obituary in the London *Times* or in his entry in the *Dictionary of National Biography*. The official record of the man was a window too: it showed the headmaster and the grammarian and left out the one book people still read. That changes nothing in the argument; it's the argument, acted out by an obituary. (*Flatland* comes back later: Section 14 retells the sphere's visit, and the Afterword, Section 24, says how the hunch behind this book started with it.)

### 7.3 The whole fact, in one paragraph

I'll give the mathematics now, and I promise it's all of it. You have a set of states, the real things. You have a window, which is a function that takes a state and gives you what you can see of it: the shadow, the slice, the half-stick without notches. And you have a question about the state, which is also a function: it takes a state and gives an answer. The thing you want is a *decoder*: a rule that takes what you can see and gives the right answer to the question, every time. When does one exist?

Think about two states that cast the same shadow. The decoder sees the same shadow for both, so it gives the same answer for both. So if the question has *different* answers on those two states, the decoder is wrong on at least one of them, and there's nothing it can do about it, because it never saw the difference. That argument is the whole fact. If a decoder exists, then the question must already give the same answer on every pair of states that look the same through the window. Mathematicians call the set of states that cast one particular shadow the *fibre* over that shadow (think of the shadow as a point on the wall and the fibre as the bundle of all the things that could have cast it), and the sentence is: a decoder exists only if the question is constant on every fibre. The theorem just below says that, and the "only if" is the direction that bites. The other direction, "if the question is constant on fibres then a decoder exists," is true too: send each shadow to the answer at any state that casts it, and constancy makes it not matter which one. A shadow nothing casts can get any answer, since no state shows it. But a decoder existing isn't the same as being able to work it out: to compute its answer for a shadow you still have to find a state that casts it, and that search takes effort, and on some windows it can't be done at all.

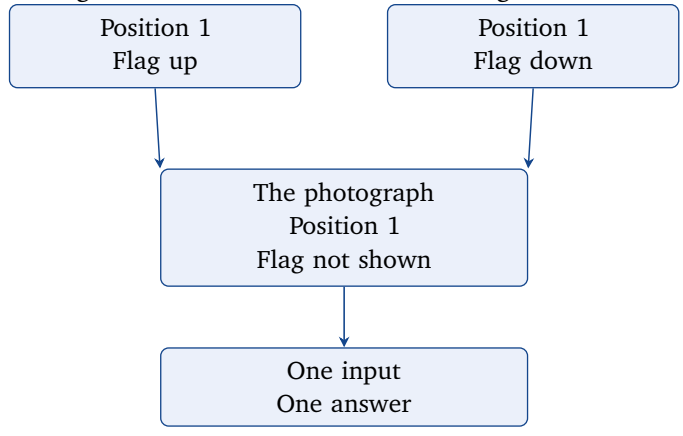
To say it exactly, let $S$ be any set of states, $V$ any set of views, $w : S \to V$ a window, and $q : S \to A$ a question with answers in any set $A$. A *decoder* is a function $d : V \to A$ with $d(w(s)) = q(s)$ for every state $s$. The question is *constant on fibres* when $w(s) = w(s')$ implies $q(s) = q(s')$.

**Theorem 7.1** (Decoders need constant fibres)**.** If a decoder exists, the question is constant on fibres. Conversely, if the question is constant on fibres and there is at least one answer, a decoder exists: send each view that some state shows to the answer at any state showing it, and any other view to any answer at all. □

*Proof.* If $w(s) = w(s')$ then $q(s) = d(w(s)) = d(w(s')) = q(s')$. Conversely, the rule is well defined, because two states showing the same view have the same answer by constancy, so it doesn't matter which one is picked; and at any state $s$ it gives the answer at a state showing $w(s)$, which is $q(s)$. ■

Nothing in the theorem cares what the states or the views are.

Try the smallest case, by hand, because it'll keep coming back. A state is a pair: a position, 0 or 1, and a hidden flag, up or down. The window shows the position only. Two moves from (0, down): one sets the position to 1 and raises the flag; the other sets the position to 1 and leaves the flag alone. Both land on position 1. Now write down every possible decoder from "what I see" to "is the flag up": there are exactly four functions from two positions to two answers. Check each against both results. None of the four gets both right, because both results show position 1 and one has the flag up and one doesn't. That exhaustive check is the theorem on this example; the general statement is the one-line argument above, and that's the one that carries the weight. There's no fifth function hiding somewhere.




Answer yes and the bottom state is misread. Answer no and the top state is misread. The photograph needs to carry another distinction.

### 7.4 People who found this on their own

The fact is so plain that it got discovered separately in every field that has to deal with seeing part of something, and I want to lay them beside each other, because that's the kind of evidence I trust: people who'd never met kept walking into the same wall.

In 1956 Edward Moore, working on the small switching machines that became computer hardware, asked when an experimenter who can only feed a machine inputs and watch its outputs can tell two of its internal states apart. His answer gave a precise meaning to "indistinguishable from outside," and a theorem with a sting in it: even when every pair of states can be told apart by *some* experiment, there are machines where no single experiment tells you which state you started in. What you can always do is run an experiment after which you know which state you're in *now*. In 1960 Rudolf Kalman, whose filter went on to aerospace and industry, asked the same question for continuous systems, called it *observability*, and gave a test (a matrix rank, which you can compute) for whether watching the output over time pins down the internal state. His unobservable states are exactly a fibre: states whose difference never reaches the output, and never will, however long you watch. That second half took me longest to take in. More samples help only when the states eventually show different outputs, and states in the same fibre of the *whole observation history* remain indistinguishable. Try it with two toy cars on identical roads, a mile apart. Give them the same speed at every tick and let neither one's motion depend on its mile marker. Their speedometers then tell the same story throughout. No amount of rereading that story supplies the missing starting position. Those assumptions do work: a landmark sensor, or motion affected by a known hill, could expose the position.

Read the mile markers at two known times instead. Their difference divided by the elapsed time gives the average speed between the readings. To recover an instantaneous speed, you need a model of the intervening motion or a limiting sequence of exact readings. Real noisy instruments give estimates with uncertainty.

Where did Kalman's filter go first, before aerospace and industry at large? It went toward the Moon. In the autumn of 1960 Kalman called an acquaintance, Stanley Schmidt, at NASA's Ames Research Center, and came to visit, not knowing what Schmidt's group was working on. The group, a branch with eight people doing the analysis, was stuck on how a spacecraft going around the Moon and back could work out where it was from what its crew could measure through an optical sight: a few angles to the Earth and the Moon. That question is a window problem with a capsule in it. The state is position and velocity, the view is a handful of angles, and the job is to keep a running best guess at the state from a sequence of views. Kalman presented his new filtering paper to them, and with real work (their problem wasn't

linear and his theory was) it fit, and it went into the feasibility studies for navigating Apollo. Leonard McGee, of Schmidt's group at Ames, and Schmidt, telling the story in 1985, call it a fortunate sequence of events and say the solution might have gone undiscovered but for that meeting. The story shows the limit from the useful side. The filter didn't make the angles show more. It used the dynamics and a sequence of noisy views to estimate the state. Observability asks whether ideal observations can distinguish the states at all; filtering also has to handle noise and uncertainty. More computation cannot recover a distinction that the entire specified observation history loses.

In 1977 Patrick and Radhia Cousot built a whole discipline of program analysis, called abstract interpretation, on throwing information away *on purpose*. You run the program over coarse descriptions of its values, and the exact values never enter. And you accept that the answer you get has to be true of *every* real value the coarse description could stand for. For example, an input known only to lie between $-1$ and 1 has a square between 0 and 1; the interval does not tell you its exact square. That is a sound bound over the hidden possibilities. Exact recovery would require the stronger condition of fibre constancy. And in 1970 Edgar Codd, inventing the relational database, made programs independent of how the data is stored, and the relational systems that followed gave users *views*, deliberately partial windows onto the stored data. A decade after Codd, in 1981, two database researchers, François Bancilhon and Nicolas Spyratos, gave the view-update problem an exact semantics. The problem asks, if a user changes what they see through a view, which of the many underlying states that cast that view should the database move to? Several are legitimate, and that's the fibre again, seen from the writing side.

Two more fields have the fact as a founding definition. In the theory of knowledge, what an agent knows is modelled by a partition of the possible states into cells, the states it can't tell apart, and it knows a fact at a state exactly when the fact holds at every state in that state's cell. The economist and game theorist Robert Aumann used that model in 1976 to prove that two people who start from the same prior, and whose estimates of an event's chance are common knowledge, can't agree to disagree; the computer scientists Ronald Fagin, Joseph Halpern, Yoram Moses and Moshe Vardi built a whole logic of knowledge on the same partitions in *Reasoning About Knowledge* (1995). Theorem 7.1 is that model with the cells written as fibres: a window knows the answer to a question exactly when the question is constant on its cell. And in computer security, the computer scientists Joseph Goguen and José Meseguer defined *noninterference* in 1982: one group of users doesn't interfere with another when nothing the first group does changes what the second can see. Hiding the counter in today's view is only the first step toward that property. If tomorrow's move reads the counter and writes it onto the display, the secret comes out. To keep it hidden through runs, the permitted moves must preserve the agreement of the

views too. The adaptive-observer argument earlier in the book supplies that extra condition for its horizontal moves.

And Parnas, from Section 5, hid things on purpose: a module is characterised by the design decision it *hides* from its callers. Hiding is a window. The caller can't later read back what was hidden, including the things it later wishes it could check, and that's the price side of the coin whose benefit side Parnas was selling. None of these people were doing philosophy. They were trying to build filters, databases, compilers, and modules, and the wall is the same wall in each.

There's one more from information theory. Oliver Kosut, at Arizona State, and Jörg Kliewer, at the New Jersey Institute of Technology, two information theorists, asked in 2018 when a receiver on a channel with a possible attacker, and no shared key, can always either get the message right or notice the attacker. The answer is all or nothing: the rate is zero, or it's everything the channel would carry with no attacker at all, and which one depends on a condition they call *overwritability*, that the attacker can make what the receiver sees look exactly like an honest transmission of some other message. Overwritability is a printer, a way to put the same thing in the window without the event behind it. The fibre fact is their condition in its plainest form, with the probabilities taken out.

And there's one from machine learning, where the two computers of The point, same bits and different histories, turn up with a lawyer attached. Four machine-learning security researchers at the University of Toronto and the Vector Institute, Anvith Thudi, Hengrui Jia, Ilia Shumailov and Nicolas Papernot, asked in 2022 how a company could prove a trained model had forgotten someone's data, and found that the same model can come out of different training sets, so the finished model can't show whether a point was used or removed. The only claim they found you can audit is about the procedure that was run. It's the same wall: the finished state is a window, and what happened is in the run.

Philosophers who study testimony, which is the trade of being told things, have the same wall from the other side. Read one after another they blurred into a list of opinions for me, and the list didn't tell me anything until I built a picture that showed where each one sits, so I'll give the picture first.

Somebody passes you a note across a room. It says the answer to question seven is 42. What crossed the room? Ink. Whatever made the writer believe 42 stayed on their side: the working-out, how careful they were, whether they knew or guessed. Pull that apart and there are five things in play: the claim on the paper, the writer's head, whatever process got the writer there, the trip across the room, and your head at the other end. Every position I found puts its weight on one of those five. Once I saw that, they stopped being a list and became a map, and the map has connections the list hid.

The two ends of it are Locke and Burge, and they're the same dial turned opposite ways. The philosopher John Locke, in the first book of his *Essay* in 1690, says believing something because someone with authority said it isn't knowing it. If the reason didn't cross the room, you don't have it. Tyler Burge, a philosopher of mind and knowledge at UCLA, argued in 1993 that you're entitled to accept what you're told, if it's presented as true and it makes sense to you, unless you have a reason not to. Locke says only ink crossed, so you've got nothing. Burge says the ink comes with a default, and the default holds until something knocks it over. I get the pull of Burge. Nobody could get through a day checking every note. But a default anybody can trigger by writing something sensible is a record nobody paid for, the thing this book keeps refusing, so on that dial the book sits at Locke's end. And sitting there has a price: staying there means carrying the check across the room every time.

Jennifer Lackey, a philosopher at Northwestern who has spent much of her work on testimony, cuts the wire between the writer's head and what arrives. Her case, from 2008, is a teacher who doesn't believe the theory she's required to teach, and her students come away knowing it anyway. I went back and forth on whether those students really know it, and I haven't settled it. But look at where the case cuts. If it's right, what arrives at your end never had to be believed at the other end, and the writer's head drops out of the map. Sanford Goldberg, also at Northwestern, moves the weight one more place over, onto the process. In 2010 he argued that whether your word justifies me can depend on how reliable the processes behind your word were, which sit outside my head. That one took me a few reads. What I think he's saying is that part of my reason for believing you lives in you, in machinery I can't see, and he's fine with that. I'm not, quite, and why I'm not is why this book exists: machinery I can't see is machinery I can't check. Christoph Jäger and Nicholas Shackel, philosophers at Innsbruck and Cardiff, built a model of degrees of belief in 2025 in which the writer knowing is neither necessary nor sufficient for the reader to know. That lands where Lackey's teacher landed, from a completely different direction, a story on one side and a calculation on the other. Agreement like that I trust more than either piece alone, as long as I remember a model only shows what can happen inside its own rules.

Then come two sets of people who drew arrows. Wesley Holliday, a logician then at Stanford, put testimony into a logic of belief change in 2009, with an authority graph recording whose word each agent takes and rules for what hearing a speaker does to a belief. I've only read a summary of Holliday's paper, so I'm taking that on somebody's word, which is a funny thing to admit in this paragraph. Security people had drawn the same kind of arrow two decades earlier. In the logic of authentication of Michael Burrows, Martin Abadi and Roger Needham (1990), two computer scientists at Digital's research lab in Palo Alto and one at Cambridge, if

I believe an authority has jurisdiction over a statement and I believe the authority holds it, I believe it. Set that beside Burge and it's his default with one change: the authority is named before the note arrives. That one change is the whole difference. Burge's default trusts whoever is writing. Theirs trusts only a writer you picked in advance.

Kenny Easwaran is the one who goes after what crosses the room itself. He's a philosopher who works on probability and on how mathematicians know things, and in 2009 he took up a question I'd never have thought to ask. The philosopher Don Fallis had pressed it in a series of papers: why will mathematicians accept a proof that's long and complicated, or partly carried out by a computer, and turn down a probabilistic one that can be far more certain? That looked backwards to me at first, because more certain should win. Easwaran's answer is a property he calls *transferability*. A deductive proof puts the reason on the paper. Every step crosses the room, so you can learn from the writer without trusting the writer. A probabilistic proof leaves the reason behind and sends only the writer's word about what the dice did. Once I saw it that way it stopped looking backwards, and it's the cleanest statement I found of what the machine of Part III does with a check: make the reason cross the room, carry the thing, and run the check on your own side. That reading is mine. His subject is proofs passed between mathematicians, so if it's wrong, it's wrong on me.

The machine carries the check across the room and runs it on a specified checker. Its correctness proof says that YES means the claim holds. That proof is supplied once, outside the individual runs. In Burrows, Abadi and Needham's terms, the checker is the named authority. Here you can examine its rules and the proof of what they mean. That is the connection between the event the machine records and the claim the event supports.

Only Burrows, Abadi and Needham write about machines passing messages, and even they write about the computers as if the computers held beliefs. The rest are about what survives the trip from one head to another. Either way it's the copier's problem (the computer in The point that was handed the sorted list and copied it in), with philosophers' names on it and, in one case, computer scientists'.

Distributed-systems people hit the wall too, and one pair of them measured how far it goes. Their papers were a pile of acronyms to me until I put them all in one room. People at desks work from one rulebook and pass slips to each other. Nobody sees into anybody's head; all anyone gets is the slips handed to them. Somebody may be cheating. What can the honest ones do about it with nothing but slips?

Andreas Haeberlen, then at the Max Planck Institute for Software Systems in Saarbrücken, and Petr Kuznetsov, then in Berlin, asked in 2009 which faults an outside observer can detect at all. As I read their answer, it sorts cheating by where it sits. A cheat can break the rulebook without changing any slip. A cheat can

write a slip an honest player could have written too, for all anyone holding it can tell. A cheat can fail to send a slip that's due, an omission fault, or send one the rulebook would never produce, a commission fault. The first two, they argue, no algorithm can detect. The last two can be caught, and they put a floor under how many messages the catching takes, which I liked for obvious reasons. My first guess was that the classes ranked how bad a fault is. They don't. They rank how far the fault gets toward a slip, and the first two are a fibre: an honest room and a crooked one that hand out the same slips. That pair of classes is the fibre fact with an auditor attached.

Their group's PeerReview system (Haeberlen, Kouznetsov, the same Kuznetsov spelled another way, and Peter Druschel, the institute's founding director, 2007) is the ladder up to that wall: every node keeps a secure record of the messages it sent and received, and a checker replays the record against a reference implementation, a copy of the rulebook it can run, which exposes the faults a correct node got to see. It comes close to the whole-state escape from the three escapes below, built: ship the log, and the checker re-runs it. I say close because a log of slips stands in for a whole state only if the rulebook can rebuild what a desk did from them, and only if the checker's copy of the rulebook is the right one.

The next three ask what you can say once you're at the wall. Ralf Küsters, Tomasz Truderung and Andreas Vogt, three security researchers at Trier, asked what a verdict should hold. They gave in 2010 a general definition of accountability, a verdict naming who misbehaved, and argued, with a proof inside their own definitions, that verifiability, the property that an outsider can check a result, is a restricted form of it. The way I read the restriction: verifiability says something went wrong, and accountability points at a desk. Radha Jagadeesan, Alan Jeffrey, Corin Pitcher and James Riely, programming-language people at DePaul in Chicago and, for Jeffrey, Bell Labs, asked when the verdict comes. They built in 2009 an operational model of systems that enforce a policy after the fact by audit, and studied the trade between the power of the auditor, the cost of the audit and what's asked of the agents. Robert Künnemann, Ilkan Esiyok and Michael Backes, at the CISPA security center in Saarbrücken, in 2019 implemented a check for it, with Certificate Transparency (CT), the public logging of the certificates web browsers trust, among their examples. As I understand it, they did it by turning accountability into a property of the trace, the whole run. It's Thudi and his coauthors' wall again: what happened is in the run.

In September 2026 Timo Treitz, a student at Saarland University, and Künnemann gave CT the full treatment (Section 17 has my fragment of it) and found, in their model, that plain CT gives accountability only if the log is assumed honest, an assumption the SCT auditing extension (SCT is the signed certificate timestamp a log hands out) can drop and the gossiping extension can't. In my room, CT hands

PeerReview's log of slips to a third party to keep. An honest log is the commitment bit from the three escapes below, with a name on the contract: somebody's promise that the bit is set when, and only when, the claim holds, trusted at the step.

Turn the room on me and it gives twice. A raise of the flag that costs nothing and never reaches the window is Haeberlen and Kuznetsov's first class, and no auditor outside catches it; that's the clerk's trouble, next. My answer to that is PeerReview's, and it trusts a copy of the rulebook too: the checker, proved honest once, outside every run. And every verdict in this book is accept or reject on one claim. It never points at a desk. In Küsters, Truderung and Vogt's terms that's the restricted form, and the rest of accountability isn't here.

## 7.5 The clerk with two photographs

Now point it at the counter, because that's why this section is here. Suppose you want to charge for raising the flag, exactly: at least one unit for every step that raises it, and nothing for a step that doesn't. And suppose the person doing the charging is a clerk who sees only a *window* on the state, a photograph before the step and a photograph after. The clerk's charge is a function of the two photographs. Can the clerk charge exactly?

The clerk can't if the window has the wrong kind of blind spot, and the reason is the cube again. Suppose there are two steps that give the clerk the same before-photo and the same after-photo, and one of them raised the flag and the other didn't. (The two-move toy above is exactly such a pair; so are "a stamp goes onto a document inside a closed drawer" and "someone opens and closes the drawer.") The clerk has one charge for that pair of photographs. If the charge is at least one, it overcharges the step that didn't raise the flag. If it's zero, it undercharges the step that did. Any charge that never misses a raise charges something that wasn't a raise, and no charge from those photographs is exact. The fix is the plain one: put the flag in the photograph. If the window shows the reading, then "one when the reading goes from down to up, zero otherwise" is exact, and such a window has no pair like the one above. Showing the flag is enough. It isn't the only way: a window can hide the flag and still have no colliding pair, and then a charge from the photographs is exact, as the observation below says. And "exact" here means rise-only pricing: at least one on every step that raises the flag, nothing on any step that doesn't. That's all the clerk's problem needs, and the charge "one for a raise, zero otherwise" meets it. So it's exact about *which* steps get charged. It isn't exact about *how much*, and it doesn't pick out one charge: "two for a raise, zero otherwise" meets it too. Pinning the charge down takes one more demand, a ceiling: over any run, the total never exceeds the number of raises. With the floor and that ceiling there's exactly one charge left, one per raise and zero otherwise, which is the exact schedule from the

last section; that's an if and only if for any system whose charge is decided step by step, from the state and the move: apply the ceiling to a one-step run. These charges look at where the step starts, not only at the move's name, and they have to. A move that raises the flag from NO raises nothing when it's played again from YES, so with Section 2's price, one price per move name wherever you stand, nothing priced that way charges the raise and leaves the replay alone, and no system whose flag ever goes up meets the floor and the ceiling together. The clerk's charge, read off the photographs, depends on the state from the start. All of this holds for any states and any steps. The window version, the cube theorem with a price in place of a decoder, comes next.

Same clerk, now in symbols, so you can check every word I just used. Take the five parts of Section 2 with a window $w : S \to V$, and say a step $(s, i)$ *raises* the flag when $\mathrm{cert}(s) = \text{no}$ and $\mathrm{cert}(\mathrm{step}(s, i)) = \text{yes}$. A *price* is any function that gives a state and a move a whole number of units, zero or more (wider than Section 2's per-name cost, which is the case that ignores the state). It *meets the floor* when every raise costs at least one, and *never overcharges* when every non-raise costs zero; together, it is *exact*, which is rise-only pricing (exact about which steps are charged; the amount is pinned only by the whole-run ceiling above, and pinned it is the exact schedule). A *window price* is one computed from the two photographs only: $\mathrm{price}(s, i) = b(w(s), w(\mathrm{step}(s, i)))$ for some charge $b : V \times V \to \mathbb{N}$. A *collision* is two steps $(s_1, i_1)$ and $(s_2, i_2)$ with $w(s_1) = w(s_2)$ and $w(\mathrm{step}(s_1, i_1)) = w(\mathrm{step}(s_2, i_2))$, the first a raise and the second not.

**Observation 7.2** (No exact window price across a collision)**.** *If the window has a collision, every window price that meets the floor charges at least one on some step that is not a raise; so no window price is exact. If it has none, the window price "one on any pair of photographs some raise casts, zero otherwise" is exact; in particular when the reading is a function of the view,* $\mathrm{cert}(s) = r(w(s))$ *for some r, there is no collision.*

*Proof.* Across a collision both steps get the same charge $b(v, v')$. The floor makes it at least one because the first step is a raise; the second step, not a raise, pays it. With no collision, a step that casts a pair some raise casts is itself a raise, so that price charges one on every raise and nothing on any other step. If the reading is $r$ of the view, two steps with the same photographs either both raise or both don't, so there's no collision. ∎

So the exact price of raising the flag lives in the step itself, or in a window with no colliding pair, and a window that shows the flag is the plainest such window. A window that contains a pair like that one can't carry it. Whether a particular machine has such a pair, for a particular window, has to be checked machine by machine and window by window; the observation tells you a window prices exactly just when it has no such pair.

You can also find the best cautious prices the clerk could use. Put all the steps that give one pair of photographs in a tray. For this calculation restrict the clerk to zero-or-one charges: each step ought to cost one if it raises the flag and zero otherwise. Without that restriction an all-raising tray would allow charges of two, three and higher, and there would be no largest price that avoids charging non-raises.

To miss no raise, the clerk must charge one whenever the tray contains even one raising step. That is the cheapest price that never undercharges. To charge no non-raise, the clerk must charge zero whenever the tray contains even one non-raising step. That leaves one only when every step in the tray raises the flag, and it is the largest price that never overcharges. In a mixed tray the two demands disagree. In a tray of all raises, or all non-raises, they agree. That is the collision test again, now giving both best cautious prices.

| Steps with these photographs | Miss no raise | Charge no non-raise |
|---|---|---|
| All raise | 1 | 1 |
| None raises | 0 | 0 |
| Some raise, some do not | 1 | 0 |

The entries are the best cautious zero-or-one charges in each direction. The mixed tray is where the two requirements disagree.

The two tests are “some member” and “every member” applied to the same tray. F. William Lawvere studied this movement of logical questions along maps in “Adjointness in foundations” (1969). Robin Cockett, Isabelle Geng, Carlo Maria Scandolo and Priyaa Srinivasan (2023) use the same kind of construction to extend resource measures along a map. Here you can see the whole calculation in the tray: take its highest required charge or its lowest. Exact pricing is possible precisely when those two numbers agree for every attainable pair of photographs.

Control engineers have been hunting pairs like that for a long time, under other names. Discrete-event systems is the field of machines that jump from state to state on events, some of which an outside observer sees and some of which it doesn’t. In 1995 five engineers, Meera Sampath, Raja Sengupta, Stéphane Lafortune and Demosthenis Teneketzis of the University of Michigan and Kasim Sinnamohideen of Johnson Controls, asked when such an observer can always work out, from what it sees, that a hidden fault event has happened, within a bounded number of further steps. They called that *diagnosability* and gave a test for it on finite machines. Anooshiravan Saboori, then a doctoral student at the University of Illinois, and Christoforos Hadjicostis, the engineer who supervised his doctorate there, asked in 2007 the question the other way round, for secrets: a system is *opaque* when an observer can never be sure it’s in a secret state, because some run outside the secret always looks the same. Read the raise as the hidden event and the window as what the observer sees. An ambiguous pair of current views prevents an immediate

answer from that view alone; a retained observation history might still distinguish the runs. One ambiguous pair is enough to defeat a decoder of the current view that must always answer correctly. Opacity asks more: every secret run in its scope must have a non-secret alternative with the same observation. The small machine's no-oracle theorem below supplies an ambiguous pair; it does not by itself establish that stronger claim about all secret runs. That's my reading; they don't put it that way. Their versions are richer than mine: they allow delays, look at whole runs, and come with algorithms. What this section adds is only the price: what a charge computed through the window does across one such pair.

Why can't the clerk just watch the step itself? A clerk who does, who sees the whole state before and the move, charges exactly with no trouble at all: "one for a raise" is a function of exactly those two things, and it's the step price above. So the collision bites an observer who doesn't see the step, or who's handed an account of the step and can't trust the account. That second kind is the trace problem, which Section 14 takes up. If the names of the moves arrive in a log the program wrote, the program can write whatever names it likes, and a clerk reading that log is reading one more window, one the program painted. Seeing the move means being underneath the program, where the step is.

## 7.6 The three escapes, and what each one costs

Suppose you're the person outside, and you want to be convinced that a run paid. The window you've got doesn't show the counter. To see how bad that is, I'll take the smallest setting I can build. Take a counter machine from Section 6, give its states a position and a flag as well, and give it a window that shows the position and nothing else. Now take two runs from the same start, with the position at 0, the flag down and the counter at 0. Run $A$ makes one move that raises the flag, costs one, and lands at position 1; its counter reads 1. Run $B$ makes one move that does nothing but step to position 1, at cost zero; its counter reads 0. Call the pair of photographs, position 0 then position 1, the *transcript*. Both runs leave exactly that transcript. A *verifier* is a rule that takes a transcript and says accept or reject. It is *sound* when every transcript it accepts comes only from states where the claim holds, and *complete* when every transcript from a state where the claim holds is accepted. The claim is "the counter is 1."

No verifier on these transcripts is both, and the proof is three lines. Suppose one were both. Run $A$ ends with the counter at 1 and leaves the transcript, so completeness says the verifier accepts it. Now soundness, on that same accepted transcript: every state that leaves it satisfies the claim, and run $B$'s end state leaves it. So run $B$'s counter is 1, and it is 0. The cube comes back here, with a verdict in

place of an answer: a verifier sees only what the window kept, and two states that the window can't tell apart get one verdict between them.

Now dress the transcript up. Take any richer kind of transcript, with a way to squash one down to its bare photographs, and say a verifier on the rich ones *only looks at the bare part* when two rich transcripts that squash to the same photographs always get the same verdict. If some rich transcript left by run $A$ and some rich transcript left by run $B$ squash to the same photographs, then no verifier that only looks at the bare part is sound and complete either: it gives those two the same verdict, completeness makes that verdict accept, and soundness then lies about run $B$. Notice what that needs: the pair has to survive into the richer kind of transcript. A design whose extra part tells $A$ from $B$ doesn't meet the premise, and the argument says nothing about it. So it doesn't classify every transcript design. What it says is that a verifier that only reads what the bare window keeps fails on that pair.

Economists got to the same wall by another road. Jerry Green and Jean-Jacques Laffont (1986) asked what a mechanism can do when people can claim about themselves only what their situation lets them back up, and Jesse Bull and Joel Watson (2007) built that into mechanism design as *hard evidence*. In their terms, an audit can tell apart only the cases the evidence on offer tells apart, which is the cube again, set in accounting.

What can you do? I know three things that work, and nothing says they're the only ones; as far as I can tell, cryptography uses all three. What any of them needs is the cube's own condition: no evidence the outsider can be handed may come both from a state where the claim holds and from one where it doesn't. I'll call them escapes, and be clear what from: from the window, not from the cost. Run $A$ paid its one unit of cost whatever anybody outside can see. An escape is a way for the outsider to see that it did. Each rests on something you have to trust, and I'll say what, because "a price" in the counter's sense is not what any of them costs.

You can be shown the *whole state*. Then there's nothing to decode; you read the counter off the state. The price is trust: you have to believe the state you were shown is the state there was.

You can be shown a *commitment*: the ordinary window plus one extra bit that is set when, and only when, the claim holds (here, that the counter reads exactly one). Reading that bit gives a verifier that is sound and complete for that claim exactly when the bit is binding (set only when the claim holds) and honest (set whenever it holds). The trust is in those two conditions, which are a contract about who can set the bit. The contract can be met, and a bit anyone can set breaks it, which is the one-line version of why an unsigned log is just another window. Nothing in that picture has a hardness assumption in it; whether a real commitment scheme meets the contract is a question for cryptography, and this book doesn't take it up. (A clash of words I should own: "commitment" here is the cryptographers'

word, something bound to a claim, which in their schemes also hides the claim until it's opened, and this bit doesn't bother hiding anything. The book's commit (Section 14) is a different act, the closing of a question. It's the same word doing two jobs. The sealed envelope, a commitment that hides its claim, has an exact theory of its own. Three information theorists, Andreas Winter, then at Bristol, and Anderson Nascimento and Hideki Imai, then at Tokyo, found in 2003 the best rate at which a noisy channel can carry commitments, and the cryptographers Claude Crépeau, Rafael Dowsley and Nascimento (2020) worked out how that rate drops as the committer gets some control over the noise, which is "take the pen away from the program," Section 14's fix for a log a program writes about itself, with a formula on it.) In 1999 the cryptographers Bruce Schneier and John Kelsey worked out how to make a log that an intruder who takes over the machine can't quietly rewrite, and the honest scope in their paper is the honest scope here: it protects what was written *before* the break-in, through keys, and nothing after. Two cryptographers at the University of California, San Diego, Mihir Bellare and Bennet Yee, had named that property *forward integrity* in a 1997 report and proved their scheme has it. Point it at an AI agent that keeps its own log of what it did, and the line moves to the very start: an agent that writes the wrong thing from its first step is on the far side of the break-in for the whole log. A 2026 preprint on tamper-evident audit trails for such agents, built from hashes anchored on a public blockchain, is even called Agent Flight Recorder (by Laurent Bindschaedler, of the Max Planck Institute for Software Systems, and Quentin Botha and Christoph Siebenbrunner, of the Vienna University of Economics and Business), which is a word Section 14 uses too, and that section's fingerprint paragraph is the part of this book that applies to it.

Or you can *ask*, which is interaction: the outside party gets to put a question and gets an answer that pins the thing down. The price is the honesty of the answerer; a liar's answer is another shadow. Moore's experimenter chose inputs; the zero-knowledge proofs of the cryptographers Shafi Goldwasser, Silvio Micali and Charles Rackoff in 1985 are the refined form, where a verifier is convinced by a conversation designed to cut exactly the line between true and false and nothing else, with a mathematical guarantee against a cheating prover. I mention zero-knowledge to say what I'm *not* claiming: the asking in this section is one round, with one fixed question ("tell me your counter"), and nothing in it models a cheating prover; the honesty is assumed, and nothing enforces it. My one round of asking is a toy of the real thing, and I'm not going to let the real thing's reputation stand behind the toy. (Joseph Halpern, Yoram Moses and Mark Tuttle, computer scientists who work on reasoning about knowledge in distributed systems, gave zero knowledge a knowledge-based analysis in 1988, which is where I'd start for the logic of it.)

The real thing even has a cave in it, which after Plato I can't resist. In a paper for the 1989 CRYPTO conference called "How to Explain Zero-Knowledge Protocols to

Your Children," two cryptographers, Jean-Jacques Quisquater of Philips Research and Louis Guillou of the French telecommunications lab CCETT, with their families and Tom Berson of Anagram Laboratories on the byline, told it as Ali Baba's cave: a ring-shaped tunnel with a magic door at the back that opens only for a secret word. Peggy goes in down one side while Victor waits outside; then Victor walks to the fork and shouts which side she should come out of. If she knows the word, she always can. If she doesn't, she gets away with it only when Victor happens to call the side she went in. Play it at the table with a coin for Victor's shout: one round, a cheat gets through half the time; twenty rounds, about one time in a million. The part of the story that belongs in this section is that Victor ends up convinced, but a video of the whole thing convinces nobody else, because a staged run, where the two agreed beforehand which sides he'd shout, makes exactly the same recording. The recording is a window, and the real run and the fake one cast the same shadow. That's the cube in the cryptographers' favourite bedtime story, The cave protocol gets its bound from fresh, independent coin tosses made after Peggy has entered. That timing is what the staged video lacks. Other zero-knowledge protocols use different setups, including noninteractive ones; the cave does not prove that every proof must be a live conversation.

Each escape is a window with something added: the whole state, a bound bit, an honest answer. None of them is "a smarter verifier on the old window": cleverness sits on the wrong side of the projection, and what's true of any machine is the cube.

I'll write the three constructions out exactly, for the pair above, with the relations written down. The relation matters: it says which state a transcript is allowed to describe. If it quietly assumes a truthful answer, then a verifier sound for it is sound only against honest answerers, and it says nothing about a liar. Write $A$ and $B$ for the states runs $A$ and $B$ end at, $\mu$ for the counter, and $\tau$ for their shared transcript.

The letter $E$ below names a rule about which state may explain which transcript. Think of writing an allowed state beside each piece of evidence. $E(s,t)$ says that state $s$ is allowed beside transcript $t$. The little $\wedge$ means both conditions must hold; $\vee$ means either is enough; $\iff$ means each side holds exactly when the other does. In the bare case, the only evidence is $\tau$, and either endpoint may stand beside it. The three richer cases change that pairing rule, which is where their extra guarantee enters.

**Definition 7.1** (Three richer transcripts and their explanations)**.** Keep the states $A, B$ and their shared bare transcript $\tau$, with $\mu(A) = 1$ and $\mu(B) = 0$. Write $E_0(s,t)$ for $t = \tau \ \wedge \ (s = A \ \vee \ s = B)$. The claim is $\mu(s) = 1$.

A *full-state transcript* is a finite list $t$ of states. Define $E_S(s,t)$ to mean that $t$ is nonempty and its last entry is $s$. The verifier rejects the empty list; otherwise it accepts exactly when the last entry's counter is 1.

A *commitment transcript* is a pair $(t, b)$ of a bare transcript and a Boolean. For any chosen explanation relation $E_C$, the *commitment contract* is the pair of implications, for all $s, t, b$:

$$E_C(s,(t,b)) \wedge b = \text{true} \implies \mu(s) = 1,$$

$$E_C(s,(t,b)) \wedge \mu(s) = 1 \implies b = \text{true}.$$

These are binding and honesty, respectively. The verifier accepts exactly when $b$ is true. One relation meeting the contract is $E_H(s,(t,b)) \iff E_0(s,t) \wedge (b = \text{true} \iff \mu(s) = 1)$.

A *response transcript* is a pair $(t, r)$ of a bare transcript and a natural number. Define $E_I(s,(t,r)) \iff \mu(s) = r$. The verifier accepts exactly when $r = 1$. This relation imposes no condition on the bare component $t$; the response alone carries its entire truth condition.

Each construction assigns verification the constant abstract cost 1.

**Theorem 7.3** (Three conditional escapes)**.** The full-state verifier is sound and complete relative to $E_S$. For every $E_C$, the commitment verifier is sound and complete relative to $E_C$ exactly when $E_C$ satisfies the commitment contract. The response verifier is sound and complete relative to $E_I$. Each has cost 1 under the stipulated cost assignment. The honest relation $E_H$ satisfies the commitment contract; the unchecked relation $E_U(s,(t,b)) \iff E_0(s,t)$ does not. □

*Proof.* For the full-state verifier, an explaining state is precisely the last entry. Reading that entry's counter accepts exactly when the explaining state's counter is 1. An empty transcript is rejected and has no explaining state.

For the commitment verifier, acceptance says $b$ is true. Binding then gives the claim for every explaining state, which is soundness. Conversely, honesty says that any explaining state satisfying the claim has a true bit, which the verifier accepts. Read backwards, soundness for an accepted transcript is binding and completeness is honesty. The equivalence in $E_H$ gives both implications directly. Under $E_U$, however, $B$ explains $(\tau, \text{true})$ even though $\mu(B) = 0$. That violates binding.

For the response verifier, explanation gives $\mu(s) = r$. Acceptance adds $r = 1$, so $\mu(s) = 1$. Conversely, $\mu(s) = 1$ and explanation give $r = 1$, so it accepts. The three cost claims follow from the assigned constant. ■

None of that leans on what $A$ and $B$ are. It uses only that one has the claim, one doesn't, and they share a bare transcript, so it holds for any pair like them, whatever the states, claims, transcripts and explanation relations. Once two transcripts with the same bare part are explained, one by a state with the claim and one by a state

without it, a verifier that's sound and complete for the claim can't be one that looks only at the bare part. The three escapes get past that by adding full states, a bit that keeps the contract, or a truthful answer.

On the machines Part III (Section 15) calls certified Thiele-complete, the collision is a theorem too, and it doesn't depend on the two-move illustration or on any one machine. Certified Thiele-complete is a definition Part III gives in full; on such a machine, computing costs nothing and never touches the record, and from a clean start the record goes up only on an earned check, commitment and certificate. Give a run from a loaded start (Part III's word for a start at line 1 with the two counters set to $a$ and $b$, written $(1, a, b)$) the *bare transcript* made of $a$, $b$ and the base window the run ends with, and say a state *explains* a bare transcript when some run from that start ends at it showing that window.

□ **Theorem 7.4** (No bare verifier on a certified Thiele-complete machine)**.** Let a machine be certified Thiele-complete through an interface (Definition 15.7). (1) No function from bare transcripts to Booleans is both sound and complete, under that explanation relation, for the claim "the record is up." (2) There is a clean loaded state such that no function from bare transcripts to Booleans is both sound and complete for the claim "the ledger is at least 3 above what it was at that start." (3) For the small machine of Part III, whose moves CHECK, COMMIT and CERTIFY and whose counter move DEC are defined there: a run CHECK *is zero* $A$, COMMIT *is zero* $A$, CERTIFY from the loaded start $(1, 0, 0)$, which ends with the ledger at 3, and three DECs of a zero counter from the same start, which end with the ledger at 0, end with the same window, and no function on bare transcripts is sound and complete for "the ledger reads 3."

*Proof.* Observation 15.32(2) gives a loaded state and two runs from it that end with the same window, one with the record up and one with it down, and, from one clean loaded state, two runs with the same window, one whose ledger rose by at least 3 and one made of base moves at no cost that leaves the ledger where it started (that observation's proof builds it). Both end states explain the same bare transcript. A verifier complete for the claim accepts that transcript, since a state where the claim holds explains it; a verifier sound for the claim then makes the claim true of every state that explains it, including the one where it's false. That's (1), and (2) is the same with the ledger: the paid run's end state satisfies the claim, and the unpaid run's doesn't. For (3) the two runs are written out in the statement: three moves land on line 4 in both runs with both counters at 0, and the same two-state argument applies. ■

Parts (1) and (2), with a loaded state for a clean one and "at least 1" for "at least 3", hold of every Thiele-complete machine, certified or not, by the same argument from Theorem 10.15 at the end of this part; certification is what raises the gap to three.

Put with the three escapes, the whole picture holds for every certified Thiele-complete machine and for the small machine: the bare-transcript verifier doesn't exist; the collision survives into any richer transcript that keeps it, so no verifier that only reads its bare part works either; a verifier on full states exists; one on the bare transcript with a committed bit exists, sound and complete when the bit is honest, and the unchecked bit breaks the contract; and one on a truthful response exists.

In the three escapes the collision is gone, and no trick did it: the extra information pulls $A$ and $B$ apart. The full-state transcript shows the counter outright. The honest commitment for $A$ carries a true bit and the one for $B$ a false bit. The honest responses are 1 and 0. In none of the three is the verifier asked to tell identical inputs apart, which was the impossible job.

Now for the limit, which I'd rather say before you find it. A list of states has no authentication built into it, so nothing in it stops a liar from handing over whatever the verifier wants to see. The commitment contract is assumed, and the unchecked bit above is what happens when the assumption is dropped. The response relation only allows a truthful counter value, so asking a question here does not buy resistance to a cheating prover. And the cost 1 is an abstract price I wrote down. It doesn't measure what it takes to authenticate a state, check a cryptographic proof or send a response. What these three show is which extra distinctions a verifier can use. They are not an adversary model, a security parameter or a hardness reduction.

So what would it take to make that commitment bit binding for real, against somebody who wants to cheat? Cryptographers had to face that one in the open, and the place I'd point is a short paper the computer scientist Manuel Blum, whose Turing Award was for complexity theory and its use in cryptography, presented at the CRYPTO meeting in 1981, "Coin Flipping by Telephone." His setup: Alice and Bob have just divorced, they live in different cities, and they want to flip a coin over the phone for who gets the car, with nobody they both trust to do the flipping. If Alice flips and announces, Bob has to take her word. If Bob calls it first, Alice can lie about how it landed. Blum's way out is for one side to commit first: send something that fixes the choice without showing it, and open it only after the other side has called. And what keeps that commitment binding isn't anybody's promise. It's a bet that a certain kind of big number is hard to factor. Push on that and you're standing on the honest edge of this subsection. In Blum's protocol, binding is earned, on the condition that factoring is hard. Here it's written into the contract and assumed outright, and the unchecked bit above shows what's left without it. That's why I

won't say these escapes beat a cheater. Blum's paper is the kind of thing it would take, and nothing like it is in this chapter.

Now point the three escapes back at the record I've been building, because that's where they bite hardest. A record kept in a machine's state is bits too. Hand someone outside a copy of those bits with a note saying "the flag is up," and they're reading a window: the run that really raised the flag and a box built to print that it did cast the same shadow. So putting the record in the step answers "was this earned?" for whoever can see the state and trusts the step that wrote it. For anybody else it moves the question, from "did it run?" to "do I trust the box that says it ran?" Crossing that line takes an escape, a window with more in it, and the three I have are these: the whole state, from a box you already trust; a bit backed by an authenticated and trusted source. A signature authenticates who signed the record; it does not prove that the signer told the truth. A quote from measured hardware likewise needs its connection to the claim (and Section 17 shows a quote vouches only for what was measured); or asking, against a box that might lie, which is the cryptographers' business. One more relative deserves its name here, *verifiable computation*: the box hands back its answer with a short proof that the answer is right, which a checker can verify without redoing the work (Pinocchio, built in 2013 by the computer scientists Bryan Parno, Jon Howell, Craig Gentry and Mariana Raykova), and proof-carrying data passes proofs like that along a whole chain of parties who don't trust each other (the computer scientists Alessandro Chiesa and Eran Tromer, 2010). That's the strongest export there is, and what it exports is the claim; the history stays behind. The proof establishes its stated relation about the input and output. Unless identity or timing is part of that relation and supported by the setup, it does not identify the physical machine or when it ran. Producing a proof may require a witness and substantial computation; possession of the public input alone need not suffice. Nothing in this book supplies any of these escapes. The model's record is local to the machine that keeps it. The example of Part III keeps its record in its state with no signature on it at all, as a bare example would.

## 7.7 What the window keeps

Put the two boxes back under the camera. Their tops agree and their side stamps differ. The photograph answers questions about the tiles. To answer the stamp question, turn the camera far enough to include the side. The full state was there all along; what changed was which part reached the reader.

Taking more photographs from the same position only helps if something different eventually comes into view. When the two runs give the same picture at every step, a longer album keeps the same ambiguity. Kalman's unobservable states give that

situation for an entire trajectory. The question is which window you have, and which distinctions it keeps.

Rice's theorem asks a different question about programs. Henry Rice proved in 1953 that no program can decide every instance of a nontrivial property of the partial function another program computes. Here you can use a box with just two states. Hand the reader one identical picture for both, and a question whose answers differ: the reader has one input and would need two answers. That argument finishes before any question about an algorithm's time or computability arises.

The counter lives in the state, and whether anyone outside can read it depends entirely on what they're shown, and a window with a pair like that in it can't price the flag exactly, with any rule, ever. That raises the question I've been pushing in front of me since the buckets. Fine, the flag went up and was paid for and I can see it in the right window. What does that *entitle* me to? A paid flag is a paid flag, so what's the claim behind it, and when am I allowed to lean on it?

CHAPTER 8

# Structural entitlement

I keep writing the word “vouch” and then taking it back. Let me stop doing that and say what it would take. A reading went from NO to YES, the toll charged the move that did it (Section 2), maybe the reading stays up (Section 3 says what it takes to make it stay), and the right window shows it (Section 7). Now: what is a later step *allowed to do* because of that? It’s a different question from “is the thing true,” and the gap between those two is the subject of this section, and I’d argue it’s the subject of most of the arguments people have about evidence, so give me the long way round.

## 8.1 The coin that moved

There’s an experiment you can run on a friend, and I’d ask you to actually run it, because the feeling of it is the point. Put a coin under one of three cups while they watch; say it’s cup one. Send them out of the room. Move the coin to cup three. Then, and this is the trick, put a *second* coin under cup one. Bring them back and ask: “Is there a coin under cup one?”

Suppose they say yes because they trust that the coin they saw has stayed put. There is a coin there, so their answer is true. Grant that their original look and ordinary expectation of no interference justify the belief. The Gettier-style difficulty is then visible: the reason follows the first coin, while the replacement makes the answer true by luck. A suspicious friend who refuses the retention assumption has not entered this particular trap.

The experiment is Gettier’s trap, done with cups (Section 14 tells his own cases): Edmund Gettier’s three-page paper from 1963, “Is Justified True Belief Knowledge?”, which broke a definition of knowledge in about the time it takes to read it. The definition was that you know something when it’s true, you believe it, and you’ve got good reasons. Gettier took it from two philosophers of the 1950s, Roderick Chisholm and A. J. Ayer, and the only place he mentions Plato is a footnote saying Plato “seems to be considering” something like it in the *Theaetetus*. I’d assumed people had leaned on that definition since Plato. As far as I can find, they hadn’t. The Stanford Encyclopedia’s entry on the analysis of knowledge calls the idea that it

was ever widely held "a bit of a convenient fiction," and Julien Dutant, a philosopher at King's College London, wrote a 2015 paper called "The Legend of the Justified True Belief Analysis" arguing that the older tradition asked for something stronger: a mark on the belief that can't be wrong. I've read only summaries of Dutant's paper, so take that last part as his, secondhand. Gettier's two cases, and the cups, show a belief that passes all three tests and still isn't knowledge, because the reasons and the fact aren't connected; they just happen to agree. Bertrand Russell had a related case in 1948: somebody looks at a stopped clock at the one moment it happens to be right, and they're right about the time, and they know nothing.

Is 1963 the bottom? It isn't even close, and 1948 isn't either. The Stanford Encyclopedia's entry on the analysis of knowledge, which is where I'd send you to check me, gives a case from the eighth-century Buddhist philosopher Dharmottara: you're looking for water on a hot day, you see water ahead, it's a mirage, and when you get there you happen to find water under a rock. And one from Peter of Mantua, a logician of the late fourteenth century who died in 1400 and whose *Logica* survives in manuscript: Plato is running right next to you, you take him for Socrates, so you believe Socrates is running, and Socrates is in fact running, in Rome, where you can't see him. Are there older cases still? Maybe, and this is where the trail goes cold, which isn't the same as where it ends. What it changes is how much I trust the shape. People in different centuries, languages and traditions kept building the same trap by hand: a reason that points at one thing, and a truth that happens to be sitting on another.

What would repair this particular reason? Lift cup one again. Now there is a new look at the coin currently there. Under the ordinary assumptions about seeing the cup and coin, that gives current evidence instead of a memory about the coin that left. It does not make perception infallible or promise the coin will remain after the cup goes down. The demonstration isolates the missing connection; it is not a proof settling every account of knowledge.

## 8.2 Plato nearly had it

Who was Theaetetus, though? The dialogue is named for a real person, and I wanted to know who. He was a mathematician. In the dialogue he's a teenager, and he tells Socrates that his teacher Theodorus had been showing them, one case at a time, that the side of a square of area three, or five, and so on up to seventeen, can't be named by any fraction: the Pythagoreans' diagonal from Section 5, done again case by case. The young man's idea was to stop going one case at a time and find one word for all of them. Euclid's tenth book, on lengths that share no common measure, is thought to be largely his work, and he's credited with being the first to study the eight-faced and twenty-faced solids, the two Plato assigned to air and

water in the *Timaeus*. The dialogue opens with him being carried home to Athens from the army at Corinth, badly wounded and sick with the dysentery going round the camp, barely alive; the usual date is 369 BC. And the conversation itself has come down as a record. Euclid of Megara (not the geometer) says he took notes as soon as he got home, filled them out later, and had Socrates correct them whenever he came to Athens, and he has the result read aloud. So the dialogue about what turns a true belief into knowledge survives as a record Socrates himself corrected. That changes nothing in the argument. I just like that the book about knowledge is named for the man who took the diagonal and made it general.

I want to be fair to the *Theaetetus* here, because it's easy to be quick with the ancients and they deserve better. Socrates, in that dialogue, doesn't just say "true belief plus reasons." He tries three different senses of what the "account" behind a true belief could be: a statement you can say aloud; a list of the thing's parts; and, third, the mark that sets this thing apart from everything else it could be confused with. He tries each, and each fails, and the dialogue ends with nobody knowing what knowledge is, which is one of the more honest endings in philosophy. But the third sense is the one I'd underline: an account as *the thing that distinguishes*, the test that gives the real case a value no rival gives. He couldn't make it work because he had no way to say what the account had to be attached *to*. Two and a half thousand years later, this book's guess is that the thing it has to be attached to is a state, and the account is an object sitting in it. I'm not claiming that settles Plato. I'm claiming he was pointing at the right thing, and that's my reading of the dialogue, which you can check against it in an afternoon.

Push on that once more, because Socrates gives his own reason for dropping the third account, and it isn't the one I just gave. If the account means *knowing* the mark that sets the thing apart, then knowledge has been defined as true belief plus knowledge of something, and the word you were trying to define is back inside its own definition, which is a circle. Does a check get out of it? It gets the word out: a test that ran and found the mark isn't defined in terms of knowing. Whether the test was any good is still a real question, and it's the question the rest of this section puts a price on. So I'd say the circle moves rather than vanishing. It moves somewhere you can look at it, which is the most I'd claim.

### 8.3 The shape, said once

What I mean by entitlement has three parts, and I'll try to earn each.

A run is entitled to a claim when three things are present. First, an *object*, carried in the state: a grouping, a relation between groups, a counter, a package of constraints. The object is a thing you could print; a belief doesn't count. Second, a *checked relation* between that object and the claim: a test that was actually run,

with its result on hand; "a test exists" doesn't count. Third, an *event on the record* that says this object was put there and paid for, so that anyone auditing the run can point back at the moment the claim was marked.

I'll say it as a definition, so the rest of the section has something to point at:

**Definition 8.1** (Entitlement)**.** A run is *entitled* to a claim when three things are on hand: an *object* carried in the run's states; a *checked relation* between that object and the claim, meaning a test that was run and whose result is on the record; and a *paid event*, a move of the run at which the object was put in place and which the price list charged at least one. In a Thiele machine (Definition 2.2) a move that takes the reading from no to yes is a paid move, because the toll charges it; whether the object went in at that move is a separate question. This is the book's word.

The first part comes with the third: a move of the run that put the object in place leaves it in the run's next state. I name the object on its own anyway, because the checked relation is about it. Keep the second and third parts separate. A note saying "coin under cup one, checked" can be written without anyone lifting a cup. That gives an object, but no checked relation. Now actually lift the cup, find the coin and write the result. Suppose that move costs nothing: the note is on the sheet, the check really happened, but the NO box stays NO and the tally line stays blank. This time the object and the checked relation are both there. What is missing is the paid event. The run isn't entitled in this definition's sense; that doesn't make the result of the check false.

The reverse distinction matters too. A move from NO to YES puts one unit of cost on the tally line, but the bare toll says nothing about what that move looked at. Payment alone supplies neither a check nor its correctness. The later machine will bind those acts together in its step rule; the five parts of a Thiele machine do not do that by themselves.

The part to hold on to is that nothing in the five parts makes a later move *read* the reading. The step function is handed the state and a move name, and it's allowed to ignore the YES box completely; plenty of systems will. So when I say a run is *entitled*, I mean what someone auditing the run may conclude from the reading and the receipt. The system doesn't have to enforce it, and unless somebody builds it to, it won't.

This is a good place to line up the three things the word *established* has had to mean so far, because they blur easily, and a blur here would make the book look like it proves more than it does. The yardstick is the four conditions Section 14 sets for a check being about this very claim: the same property, the same thing, the same edition, and an honest check.

| What's asked | Bare toll | Entitlement | Earned certification |
|---|---|---|---|
| A check ran | not required | a test result on the record | a passing check before the commit, on every run from a clean start |
| Same property | not asked | not asked | enforced by the step |
| Same thing | not asked | not asked | enforced by the step |
| Same edition from CHECK to COMMIT | not asked | not asked | enforced; CERTIFY may later read that historical commitment |
| Honest check | not asked | not asked | proved of the checker, once, outside the record |
| Could have failed | not asked | not asked | not policed |
| Paid for | at least one unit | at least one unit | at least three units for the chain |

After the column of questions, the first column is Section 2's bare toll: a paid crossing, and nothing says a check happened at all (Section 9 has a remark saying exactly that, Remark 9.2). The second is this section's entitlement. The third is the small machine of Part III, whose step won't let a commit through without a passing check of the same property, on the same thing, at the same edition, for every program from a clean start (Theorem 15.10, and Theorem 15.11 for a run that ends certified), whose checker comes with a proof that it says what it means (Lemma 15.2, Theorem 15.13), and whose certified runs from a clean start cost at least three (Observation 15.21). Even there two things are missing, and the first is in the table, where you'll see it. The step doesn't police whether a check could have failed: a property every value has is a perfectly legal check, sound and useless. And in all three columns the status belongs to whoever reads the record. The machine doesn't know anything. What its flag means is assigned from outside, by whoever says what each property means.

Why does the honest check carry so much? The philosopher Linda Zagzebski, who works on knowledge and the virtues of a careful mind, argued in 1994 that any account of knowledge whose justification leaves even a small gap between having good reasons and being right lets a Gettier case through: take a case where bad luck makes the reasons point the wrong way, then add a stroke of good luck that makes the belief true anyway. The first three conditions plus an honest check close that gap by construction, and only one half of honesty is needed for it, the half that says no when the property is false. Saying yes whenever it's true is extra. A check

that can be wrong, a test suite that sometimes passes broken code, puts you straight back in her gap, and then the record certifies something smaller and still true: this procedure said yes.

And who certifies that the check is honest? The record doesn't. Honesty is a fact about what the check does on every input, and a finished state holds the run's values, not that. A proof of it can sit in a state and be checked there, but the checking is a check too, with its own honesty to vouch for, and somewhere the chain stops at a proof somebody accepts; in this book that proof is read outside every run, and nothing on the record pays for it. So an established claim is two things stuck together: a run property the record carries and a program property a proof carries, and the proof is entitlement for the auditor. The machine gets none from it. On the machine of Part III the chain stops at the first step, the checker's own proof, by choice; Section 15 says why.

The engineering version of this was already built, in 1996 and 1997. The computer scientist George Necula, then at Carnegie Mellon, called it *proof-carrying code*: when you ship a program to a host that doesn't trust you, you ship a *proof* with it that the program obeys the host's safety rules, and the host *checks the proof that arrived* before running anything: not your reputation, and not "a proof exists somewhere," but the one in the box. A quarter century earlier Stephen Cook's certificates had the same bones, from Section 2: a claim comes with a witness that's cheap to check, and the checker checks *that* witness. And the Scottish philosopher David Hume, in 1748, arguing about miracles in his *Enquiry Concerning Human Understanding*, laid out the thing the receipt replaces. He weighed testimony against evidence, and wrote that "a wise man, therefore, proportions his belief to the evidence." The way I read him, the whole of his argument is a method for how much to trust a witness when you have no receipt. The move in this book is to stop weighing. A claim on the record is either backed by its object and its paid event, or it's testimony from outside, and a run that leans on testimony hasn't earned it itself. It can inherit it, but only as far as the copy it leans on is genuine, which is Section 7's problem of a record shown from outside.

Robert Brandom, a philosopher at the University of Pittsburgh, built a whole account of language on two words this book leans on, commitment and entitlement, and his *Making It Explicit* (1994) is where I'd send anyone who suspects I invented the pair. In his picture, as I read it and as his reviewers describe it, people talking keep score on each other: what each speaker is *committed* to, what they've put themselves on the hook for, and what each is *entitled* to, what they're licensed to claim. You can be committed without being entitled (a reckless assertion), many claims count as entitled by default until someone challenges them, and entitlement can be borrowed: I can claim a thing on your say-so and defer to you when I'm challenged. Lay mine beside his and the differences are the content. His score is

kept by the other people in the room; here it's a reading in the state. His entitlement often comes by default; here it's an event that has to happen and is charged. And his deferral is my testimony, which I've just let back in on a leash: borrowed entitlement is only as good as the copy is genuine. The leash has an older formal cousin, the jurisdiction rule of Burrows, Abadi and Needham's logic of authentication (Section 7), where you take a statement on an authority's word only if you already believe the authority has jurisdiction over it. It came first, presented in 1989 and printed in 1990, years ahead of Brandom's book of 1994, and its words are *believes* and *jurisdiction* where his are *commitment* and *entitlement*.

There's a whole field that would want a word here too, the logic of belief revision, which asks how a reasonable mind should change what it believes when something new comes in. Its nearest neighbours are worth naming, and I couldn't keep them apart until I put them back on the table with the cups. Your friend believes the coin is under cup one. They can learn something that doesn't clash with that, say cup two is empty, and just add it. They can lift cup one, find nothing, and have to change their mind about a table nobody touched. Or you can tell them you moved the coin. The last two felt alike to me at first, but they aren't: in one the friend was wrong, and in the other the friend was right, and then the table changed.

Krister Segerberg, a Swedish logician at Uppsala, built a logic of irrevocable revision in 1998: accepting something so that it can't be given up, which is the firmness question taken to its end. Isaac Levi, a philosopher at Columbia who spent a career on how beliefs get fixed and unfixed, split adding a belief into a deliberate kind and a routine kind (I cite his 1991 book; the split goes back to his 1980 one). Deliberate is reasoning your way there. Routine is where you take on whatever a procedure you committed to in advance hands you, the way you'd accept a reading off a thermometer you've already decided to trust, or what's under a cup when you lift it. What I think the split rests on is when the trust goes in: once, before the data comes. A passing CHECK, the test move of the small machine in Section 15, is routine in exactly that sense, with a checker proved honest once, outside every run.

Neil Tennant, a logician at Ohio State, and Gösta Grahne, a computer scientist trained at Helsinki, I first filed as two separate complaints. They're one move made from opposite sides. The target is the best-known impossibility in that field, the Swedish philosopher Peter Gärdenfors's 1986 theorem that a natural rule for conditionals, the one his paper calls the Ramsey test, can't live with rational revision. Tennant argued in 2008 that it goes through only by treating some additions as revisions: the friend who learned cup two was empty gets counted as having changed their mind. Grahne, working on updates and counterfactuals, argued in 1998 that Gärdenfors's theorem bites only revising, so the friend whose coin was moved is out of its reach. Side by side, they squeeze the theorem from both ends into one case: being wrong about a table nobody touched.

The line this book leans on hardest is between two kinds of change. Revising is changing what you believe about a world that stayed put. Updating is changing it because the world moved. Two computer scientists, Hirofumi Katsuno and Alberto Mendelzon, drew that line for knowledge bases in a 1991 paper on the difference between updating a knowledge base and revising it; database researchers had been working on updates with incomplete information since the mid-1980s, Arthur Keller and Marianne Winslett among them. Grahne's result stands on it. And look at what the friend sees: an empty cup one is the same empty cup whether they misremembered or you moved the coin. The evidence alone can't tell revising from updating; that's a fibre. What tells them apart is a count of how often the cups were touched. The machine in Part III keeps that count as an edition number for each thing, and its version rule treats every write as an update: a write moves the edition number, so every fact about the old edition stops counting for the new one.

And Alexandru Baltag, Bryan Renne and Sonja Smets, three logicians at Amsterdam, gave a logic in 2014 that keeps apart three things Gettier's cases blur: justified belief, explicit knowledge and conclusive evidence. The friend who watched has justified belief; the friend who lifted the cup has conclusive evidence. Explicit knowledge I read as knowing with the justification written out beside it, the way the record keeps its check, and I'm less sure of that one. Put them together and they say something exact about this book's record. A record that's never wrong and never corrected would be a dogmatist's dream if its sentences were about the present, but they aren't: each says a property held of a thing at one edition, and that sentence stays true after the edition number moves on. The record can be infallible and never corrected only because its sentences are edition-stamped. Strip the stamps, and the first write would turn it into a liar that can't take anything back.

Push on that and it gives twice. The record is Segerberg's irrevocable acceptance every time, with no revising in it. If a check was dishonest, the only fix that fits a record like this is a second record on top saying "withdrawn," the way a revocation list works, so the infallibility leans on the checker, proved once, outside. And the record is never wrong partly because it only talks about past editions. Whoever reads it wants now, and getting from "held at that edition" to "holds now" crosses every update since. The record doesn't make that step for you.

That leaves the logicians who taught the logic of knowing to count, whom Section 14 names again in passing. This time I take them properly, and I want to see what each price rests on. Ho Ngoc Duc, in 1997, gave the logic of knowing an operator for "true after some train of thought": an agent doesn't know every consequence of what it knows now. It can come to know any of them by taking steps of reasoning. I took that at first for the thing I was after, until I saw which way it points. His operator points forward, at what the agent could come to know. A record points back, at what was checked. His operator and a record are one arrow,

read from opposite ends. And Sergei Artemov and Roman Kuznets, in 2014, turned logical omniscience into a question of cost: a logic escapes it when every valid claim that something is known comes with a proof whose size is bounded by a polynomial in the size of the claim. The usual modal logics of knowledge can't pass that test unless NP equals coNP, which most people would bet against, and justification logic passes it for knowledge stated together with its justification, because the size of the justification pays for the search. That last clause took me a few reads. What I think it means is Cook's certificate from the proof-carrying-code paragraph above: put the justification inside the claim, and the hard part, finding it, was paid before the claim was written. Their price rests on an open bet, and it's a price on looking, which is a price of a sort. Steps and a price are both there, then, in their own form. What I still don't find in them is the one I care about: a single act that closes a question for good and is charged as that act. Their steps are steps of reasoning, and their price is on finding a proof. Neither is on settling. I couldn't get past the journal's paywall to Artemov and Kuznets's paper, so my account of it comes from how other people describe it.

Push on mine the same way and it gives. A check is one of Duc's trains of thought, so my act is the last step of his train, where the walking stops. What I've added is a direction and a charge on stopping. Where the charge sits follows from one choice: I want it to bind every program the machine runs, and a charge that does that has to sit inside the step, since a program can skip any charge kept where it can write.

There are three things entitlement is *not*. It isn't permission: an access control system says who may write to what, and that's granted from outside and says nothing about truth; a process with permission to write the note still has no entitlement. It isn't "it's true, so I can use it": that's the cups. And it isn't "sound by luck": a shortcut that gives the right answer because the inputs happened to cooperate is the stopped clock. The count below has one condition aimed at it, and I'll tell you what the condition says and how much it actually catches.

## 8.4 What narrowing costs: count the rivals

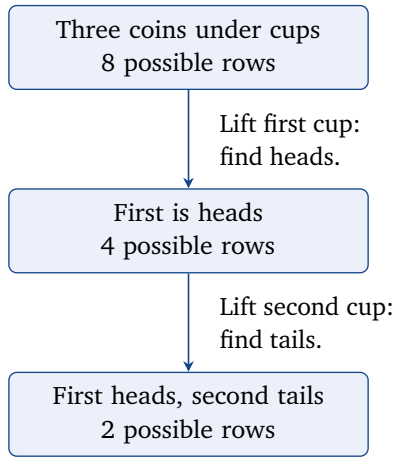
Now I need the number, because a book about what keeping costs has to say how much, and the idea fits in a game you can play with three playing cards. Shuffle cards marked 1, 2, 3 face down. Sort them using only questions of the form "is this one bigger than that one?", and count the questions. There are six orders the cards could be in. Each question has two answers, so after one question you've split the six into at most two piles, after two questions at most four, and you need to get down to one. Four is less than six, so two questions can't always do it, and three can, and you can draw the tree of questions on paper and see it: six leaves, three deep. The number of questions you need is at least the base-two logarithm of the

number of possibilities you started with, rounded up: that's log2 of 6, which is a bit under 3, rounded to 3. Donald Knuth, a computer scientist at Stanford, gives this, in the volume on sorting of his *The Art of Computer Programming*, as the lower bound for all comparison sorting, and the shape of the argument is just: a tree $f$ questions deep has at most $2^f$ leaves, so it can't tell apart more than $2^f$ cases.

And then the obvious next question: is the floor always the price? Try five cards. There are 120 orders, log2 of 120 is a bit under 7, so you need at least seven questions, and seven is enough, though not by the obvious method. Finding the way is a good evening with a deck of cards (Knuth has it, under the name merge insertion, from the mathematicians Lester Ford and Selmer Johnson in 1959). Now take twelve cards. They can be in 479,001,600 orders, log2 of that is about 28.8, so the count says at least 29. Ford and Johnson's method takes 30, and in 1965 Mark Wells, a mathematician at Los Alamos who built a programming language for exactly this kind of combinatorial search, showed by a computer search that nothing does better: twelve cards need 30. The floor was 29 and the price is 30, so the count is a floor and only a floor; it never promises the floor can be reached, and the bound later in this section has to be read exactly the same way.

Ralph Hartley, an electronics researcher at Bell Labs, writing in the Bell System Technical Journal in 1928, before Shannon, measured information exactly this way: the logarithm of the number of possibilities, no probabilities, just a count. I use that measure here, and I'll say plainly it's Hartley's. Shannon's would need a distribution over the list, and there isn't one; there's just the list's length.

The same count works with something you can hold. Put three coins under three cups. Each coin is heads or tails, so there are eight possible rows under the cups. Someone hands you two answers: heads under the first cup, tails under the second. Lift the first cup and compare, and a matching answer leaves four possible rows. Lift the second and compare, and a match leaves two. The last coin can still be either way. The list really shrank: eight, four, two. Nothing in that count needed a claim about which way the last coin faces.




The third coin could be either:
heads or tails.

These observations halve the candidate list. A price on those observations is a separate rule; the shrinking list alone supplies none.

This is a parlour game, and an old one. In *A Christmas Carol*, in 1843, Scrooge's nephew Fred plays a game called Yes and No at his Christmas party: he thinks of something, and everybody else asks questions he may only answer yes or no. He's thinking of an animal, a disagreeable, savage one that growls and grunts and lives in London, and the answer is Uncle Scrooge. Most people know the game as twenty questions, and twenty is a good number for it: a balanced question splits the remaining field in half, and twenty such questions can guarantee identification among $2^{20} = 1{,}048{,}576$ possibilities. A lucky answer to "is it this one?" can finish immediately; the count is about handling every possible answer. Push on that, though. The halving only happens because Fred answers truthfully about what's actually in his head. Lie once and the count still shrinks; it shrinks around the wrong animal. The cups showed the same thing: the four and the two are real because you lifted the cup and compared. Being told what was under it wouldn't have made them real.

Now put it on a run. Say a run comes with a list of states that are "still on the table," the candidates, and a shorter list, with some ruled out. Call the first the prior and the second the posterior, and I want you to hear those as *a list you could print*. They aren't "the space of possible worlds." Picture the ruling-out as observations, each observation that ruled something out being one branch point in a tree like the card tree. The bound says that if that tree is no deeper than the number of paid steps the run took, and the tree has enough leaves to cover the narrowing in a way I'll spell out in a moment, then the counter rose by at least log2 of the prior's length, rounded up, minus log2 of the posterior's length, rounded up. Put another way, the

counter rose by at least one unit for each of the *index bits* the list lost, where the index bits of a list are the yes/no questions it takes to point at one entry of it, the card count again.

“Paid step” needs its own sentence. It means a step the price list charges at least one for. In a Thiele machine the toll guarantees that every crossing of the reading is one; a price list may put a floor under more steps than that (a look, a send, a check), and the count takes every paid step there is. The tree is a bare shape, branches and leaves with nothing written on them, and “no deeper than the number of paid steps” is a count along the run that happened, not a walk down the tree. A paid step doesn’t have to have asked any question at all.

Notice, too, what the bound does and doesn’t say about the run. It takes the two lists, the tree and the covering as given. It doesn’t say the run produced the posterior from the prior; nothing in it reads the lists off the run. What it says is that, given lists, a tree and a covering that someone supplies, the paid steps of the run cover the rounded drop in list length. Here Section 4’s warning comes due: watching can narrow without costing anything, so the payment can’t be dug out of the narrowing. It goes in as a premise, the tree’s depth against the paid steps, and I’ll keep calling it one. Later in this section the run makes the posterior itself, from a prior and the tests it passed (Theorem 8.4), and then the premise is a share each test keeps.

## 8.5 What “narrower” buys: a stronger test

Before the count, one link that’s pure logic, and Plato’s third account is the whole of it. Take a list of candidates and an observation, a function that says what each candidate looks like from outside. The list’s *membership test* accepts an observation when some candidate on the list shows it. Shrink the list, and the test can only get pickier: anything the short list’s test accepts, the long list’s accepts too. The word for that, and it runs backward from English, is *stronger*: the stronger test is the one that lets *fewer* things through.

Pickier, though, isn’t the same as strictly pickier, and the cards show the gap. Take all six orders as the prior, and as the posterior the three where card 1 sits left of card 2: 1-2-3, 1-3-2 and 3-1-2. First observation: “is 1 left of 2?” Every survivor says yes, and the removed 2-1-3 says no, so the prior’s test accepts yes and no, the posterior’s only yes, and the posterior’s test is strictly stronger. Second observation, same two lists: “is the first card a 3?” Now 3-1-2 says yes and the other two survivors say no, and among the removed orders 3-2-1 says yes and 2-1-3 says no. Both tests accept yes and no, so they have the same strength, even though three orders are gone. What made the difference is a removed candidate whose observation *no survivor shares*. Such a candidate is Plato’s mark, the thing that sets this case apart from everything it could be confused with, turned into a function. Narrow the list and

keep one distinguishing removed candidate, and the test is strictly stronger; narrow it without one, and the test may not have moved at all.

### 8.6 The count, exactly

I'll set out the words and then the count, for any system with Section 6's counter. A *list* is a finite list of candidates, repeats allowed, and $|\Omega|$ is the length of the list $\Omega$, repeats counted. $\lceil \log_2 x \rceil$ is the base-two logarithm of $x$ rounded up: the smallest $k \in \mathbb{N}$ with $2^k \geq x$. At $x = 0$ it is 0, since $2^0 \geq 0$. And $a \dot{-} b$ is subtraction on whole numbers that stops at zero, so $3 \dot{-} 5 = 0$.

**Definition 8.2** (Decision tree)**.** A *decision tree* is a leaf, or a branch with two subtrees. Its depth is 0 for a leaf and one more than the deeper subtree for a branch; its leaf count $L(T)$ is 1 for a leaf and the sum of the two counts for a branch. Nothing is written on a tree: no question, no answer, no candidate.

□ **Lemma 8.1** (A tree is no wider than two to its depth)**.** For every decision tree $T$, $L(T) \leq 2^{\text{depth}(T)}$, and so $\lceil \log_2 L(T) \rceil \leq \text{depth}(T)$.

*Proof.* Start at the root. At the first level there can be at most two places, at the second at most four, and each further level at most doubles the places available. To count leaves that stop early as well, extend each such leaf with branches until all tips reach the tree's greatest depth. This only adds leaves. At depth $d$ the filled-out tree has $2^d$ tips, so the original tree has at most that many leaves. For depth zero there is just the root, which is one leaf.

Thus $d$ bits are enough to number all the leaves. The least number of bits that ■ will do, $\lceil \log_2 L(T) \rceil$, cannot be greater than $d$.

Nothing about any machine is in the definition or the lemma. They're about trees, a leaf or a branch, and nothing else.

□ **Theorem 8.2** (Paid narrowing, counted)**.** Take a system with a counter $\mu$ and a price list $c$ that obey conservation (Theorem 6.2): each step raises $\mu$ by exactly its charge. Let $\Omega$ and $\Omega'$ be lists with $|\Omega'| > 0$, let $T$ be a decision tree with $|\Omega| \leq L(T) \cdot |\Omega'|$ (the tree *covers* the narrowing), and let $t$ be a run from $s$ ending at $s'$ in which at least $\text{depth}(T)$ steps are charged at least one each. Then

$$|\Omega| \;\leq\; 2^{\mu(s')-\mu(s)} \cdot |\Omega'|, \qquad \text{so} \qquad \lceil \log_2 |\Omega| \rceil \dot{-} \lceil \log_2 |\Omega'| \rceil \;\leq\; \mu(s') - \mu(s).$$

*Proof.* Count the bill first. There are at least as many paid steps as levels in the tree. Each paid step adds at least one and every other charge is nonnegative. The total bill is therefore at least the tree's depth. Conservation says that bill is exactly the counter's rise, $\mu(s') - \mu(s)$.

Now count the room the tree supplies. It has at most two to its depth leaves. The covering premise allows at most $|\Omega'|$ prior entries per leaf, so

$$|\Omega| \le L(T)|\Omega'| \le 2^{\mathrm{depth}(T)}|\Omega'| \le 2^{\mu(s')-\mu(s)}|\Omega'|.$$

That is the first bound.

For the bit version, let $k = \lceil \log_2 |\Omega'| \rceil$. The posterior has at most $2^k$ entries. Insert that count in the first bound: the prior has at most $2^{\mu(s')-\mu(s)+k}$ entries. So numbering its entries needs no more than the counter's rise plus $k$ bits. Subtracting the posterior's $k$ bits gives the claimed bound; if the subtraction would go negative, the stipulated subtraction stops at zero. An empty prior already has bit count zero and satisfies the same conclusion. ∎

Two arguments are stuck together in that proof, and they come apart. The tree half (the covering and the lemma) bounds the index bits by the sum of the charges of the run, on any price list at all. Conservation is only what turns that sum into the counter's rise. So the form in terms of cost, with the sum of the charges in place of the counter, holds in any Thiele machine, and it's part (3) of Theorem 8.3 below. The shape of the bound shows in what's missing from the hypotheses: anything tying the lists to the run.

There are three honest pieces of small print, all of them visible in the statement. Both logarithms round up separately, and the subtraction stops at zero, so when the prior and the posterior round up to the same number of index bits (four candidates narrowed to three, say) the rounded bound says nothing, though the first form of the theorem charges the run one unit: one leaf can't cover four candidates with three. The lists are plain lists, repeats allowed, so "index bits" is a difference of rounded list lengths with repeats counted; a list can get shorter by dropping a duplicate without losing a single candidate, and "a candidate was ruled out" and "the list got shorter" are two different claims. And the difference isn't the number of questions anyone asked.

The covering condition, $|\Omega| \le L(T) \cdot |\Omega'|$, is the nearest thing the count has to an anti-stopped-clock clause, and I have to be exact about how near. Picture each survivor standing for a little crowd of the prior's candidates, a crowd no bigger than the tree has leaves. The condition says the crowds can hold the whole prior between them: share the prior's candidates out among them and none overflows. That limits how much can be deleted per leaf you paid for, but it doesn't say why anything was deleted. So the condition rules out deleting too much for the tree you've paid for,

and it doesn't rule out deleting for no reason. A run can satisfy every hypothesis of the count and have ruled nothing out on purpose; the count prices the size of the drop and says nothing about the honesty of it. And the covering can't just be dropped. On the small machine of Part III (Section 15), whose record moves CHECK, COMMIT and CERTIFY cost one each, take the sixteen starts with $A$ at 0 through 15, and the run CHECK "$A \geq 15$," COMMIT, CERTIFY from the start with $A$ at 15. One yes-or-no question with a big constant in it narrows sixteen candidates to one, four index bits, and the run has three paid steps, a ledger of three. A tree no deeper than three has at most eight leaves, and eight times one survivor is less than sixteen, so the covering fails, and without it the bound would say $4 \leq 3$. More candidates only widen the gap: $2^n$ starts and the constant $2^n - 1$ give $n$ index bits for the same cost of three (Section 15.14, Proposition 15.39).

A side trip is worth taking here, because it looks like it breaks the count and doesn't. Two information theorists, Rudolf Ahlswede and Gunter Dueck (1989), noticed that asking "is it this one?" is a different job from asking "which one is it?" If each receiver only wants to know whether the message is theirs, a noisy channel used $n$ times can serve a number of receivers that grows like two raised to two to the $nC$, with $C$ the channel's capacity, where naming the message only gets you two to the $nC$. The task is called identification, and on its face it says checking can be exponentially cheaper than naming. But their tests are allowed to say yes wrongly once in a while. Make the yes sound, a test that never says yes to the wrong message, which is the only kind of check this book lets raise a record, and the double exponential collapses. The argument is two lines of pigeonhole. Each receiver's test, to be any use, says yes on some output its own message can produce, and soundness means no other message can produce that output; so those outputs are all different, one per receiver, and there can't be more receivers than outputs, $|Y|^n$ for an output alphabet $Y$. That bound is single exponential, the same order as naming. So for a check that never lies, the card count is the right scale after all, and the cheap trick needs a check that can be fooled.

## 8.7 What you've got

A paid flag is not a true claim; Section 7 left it at "a paid flag is a paid flag," and Section 9 will say it flatly. What a paid flag with an object and a checked relation behind it gives you is *entitlement*: anyone auditing the run may treat the claim as established, and can point at the event where that became legitimate. (Nothing in the five parts makes the system itself read the flag; I said that above and I'm saying it again because it's the sentence most likely to get lost.) Entitlement costs at least the index bits of narrowing it commits to, given lists, a tree and a covering somebody supplies, and given a run that paid for the tree's depth. The cups say why

entitlement and truth are different. Plato says what the checked relation has to do, which is to distinguish. Necula says how to build it. Hartley says what to count.

Put the three pieces on one hypothesis list (narrow the list with a test that separates, find the paid event, count the charges against a tree the run paid for) and you get a single statement, structural entitlement, once over any Thiele machine and once, under the longer name structural entitlement represented, for every certified Thiele-complete machine, the small machine included (Theorem 15.36). The machine version asks the same things with two changes: the run starts clean instead of merely at no, and the paid steps are the run's own record moves, CHECK, COMMIT and CERTIFY, at a cost of exactly one each. The first one reads as follows.

**Theorem 8.3** (Structural entitlement, over any Thiele machine)**.** Take a Thiele machine (Definition 2.2), a state $s$ that reads no, and a run $t$ from $s$ that ends in a state that reads yes. Take a type of candidates, a type of observations with an exact equality test, an observation obs of each candidate, and a *decoder* $d$ that turns a run into an observation (what its record shows). Take lists $\Omega$ and $\Omega'$ of candidates and a decision tree $T$ such that: (H1) $\Omega'$ strictly narrows $\Omega$: every member of $\Omega'$ is in $\Omega$, and some member of $\Omega$ isn't in $\Omega'$; (H2) some dropped candidate $w$, in $\Omega$ and not in $\Omega'$, shows an observation different from every survivor's; (H3) the decoded run $d(t)$ is shown by some survivor; (H4) $|\Omega'| > 0$ and $|\Omega| \le L(T) \cdot |\Omega'|$; (H5) at least $\mathrm{depth}(T)$ steps of the run are charged at least one each. Then (1) the membership test of $\Omega'$ is strictly stronger than the membership test of $\Omega$; (2) some step of the run took the reading from no to yes, and that step was charged at least one; and (3) the sum of the charges of the run's steps is at least $\lceil \log_2 |\Omega| \rceil \mathbin{\dot{-}} \lceil \log_2 |\Omega'| \rceil$. □

*Proof.* (1) is the cards' argument: the dropped $w$ shows something no survivor shows, so the posterior test says no where the prior test says yes, and every survivor was in the prior, so nothing the posterior accepts is refused by the prior. (2) The run starts at no and ends at yes, so at some step the reading goes from no to yes, and the toll charges that step at least one. (3) is the proof of Theorem 8.2 with the sum of the charges in place of the counter's rise: the covering and Lemma 8.1 bound the index bits by $\mathrm{depth}(T)$, and the sum of the charges is at least the number of steps charged at least one. On a system that obeys conservation the sum of the charges is the counter's rise. ■

Theorem 8.2 is the same count with the counter's rise, so for a system that obeys conservation it's the special case. Three things in the hypotheses feed nothing, and I'm naming them here so you don't have to find them: (H3) isn't used by any of the three conclusions; from (H1) only the inclusion is used, and the gap goes unused; and in (H2) the words "not in $\Omega'$" go unused, since a candidate whose observation

differs from every survivor's can't be a survivor itself. And (H4)'s $|\Omega'| > 0$ is used in that proof but can be dropped too: the covering forces it once the prior has a member.

A statement like that is true by packaging: it takes the hypotheses in and hands the conclusions back, and the right word for it is a checklist, a checklist for leaning on a claim. It discovers nothing about leaning. The version in which the run itself does the narrowing, Theorem 8.4, comes later in this section.

And a checklist isn't nothing. On 30 October 1935, at Wright Field, northeast of Dayton, Ohio, the Army was evaluating Boeing's new four-engine bomber, the Model 299. Shortly after take-off it plunged back onto the field and caught fire; that's how the wire report of that day has it, and the later histories add that it pitched up and stalled first, which is what elevators locked in a raised position would do. Two men died of their injuries, the Army's pilot, Major Ployer Hill, and Boeing's chief test pilot, Leslie Tower. The airplane was fine. The crew had left the gust locks on, a locking mechanism worked from inside the cockpit that holds the control surfaces still while the plane sits on the ground. There's a famous line that a newspaper called it too much airplane for one man to fly. I couldn't find that line in any paper from 1935; the earliest place I could find it is a New Yorker article from December 2007 by Atul Gawande, a surgeon who writes about how people get complicated work right, and he names no newspaper, so take it as a lead. What I can find is what came after. The answer was a card: a list of things to check, in order, every time, and the crews of the airplane worked through written checklists for each phase of flight. The most careful history of it I found is by Walter Boyne, a retired Air Force colonel and a former director of the National Air and Space Museum. By his account it wasn't the first written routine for a pilot; those go back before the First World War. And it's this crash, he says, that made the checklist the way a crew flew. The airplane went on to be the B-17. Nothing on that card was a discovery. Every item on it was something the pilots already knew. What the card did was make sure none of it got left out when it mattered. Putting entitlement's hypotheses on one list does the same: it finds nothing new about leaning on a claim, and it makes the hypotheses impossible to drop quietly.

## 8.8 When the run does the narrowing

Theorem 8.3 takes the lists in. In the version below, the run makes the posterior, and the count stops being an item on the list and becomes a consequence. That version needs one thing a bare Thiele machine doesn't have: which steps are tests, and what each asks. So a reading names them. For each instruction it gives either nothing or a yes-or-no question about the state the run is in when it gets there, and a test step *passes* when its question says yes there. The candidates are states the run

could have started from. A passed test is asked of a candidate by running the same moves from the candidate up to that step and asking the same question there. The *posterior* is the candidates that pass every test the run passed, asked that way, in the order the run asked them.

**Theorem 8.4** (Structural entitlement, with the narrowing made by the run)**.** Take a Thiele machine and a reading that names its test steps, every test step charged at least one. Take a state $s$ that reads no, a run $t$ from $s$ that ends in a state that reads yes, and a list $\Omega$ of candidate starts that contains $s$. Let $\Omega'$ be the posterior $t$ makes from $\Omega$, and let $b_i$ and $a_i$ be the numbers of candidates standing just before and just after the $i$th test $t$ passed. Then (1) $\Omega'$ sits inside $\Omega$ and contains $s$; (2) every candidate of $\Omega$ that isn't in $\Omega'$ fails, asked that way, a test $t$ passed; (3) $|\Omega| \cdot \prod_i a_i = |\Omega'| \cdot \prod_i b_i$; (4) if every test $t$ passed keeps at least a $2^{-k}$ share of the candidates standing before it, $b_i \le 2^k a_i$, then $\lceil \log_2 |\Omega| \rceil \mathbin{\dot{-}} \lceil \log_2 |\Omega'| \rceil$ is at most $k$ times the number of tests $t$ passed, that number is at most the number of steps charged at least one, and the index bits are at most $k$ times the sum of the charges; and (5) some step of $t$ took the reading from no to yes and was charged at least one. □

*Proof.* (1) Start with the prior list. Each passed test only removes candidates; it never inserts one. The actual start $s$ passes every test used to filter the list, because these are precisely the tests the actual run passed. So $s$ remains throughout, and the final list is a nonempty part of the original.

(2) Follow any candidate that is absent at the end. It was present at first, so there is a first test at which it disappears. The filter removes it only when its version of that test says no while the actual run's says yes. That test is the required witness.

(3) Write the list lengths on the bench: $q_0$ before any passed test, $q_1$ after the first, through $q_n$ after the last. Here $q_0 = |\Omega|$ and $q_n = |\Omega'|$. Every before-count $b_i$ is $q_{i-1}$ and every after-count $a_i$ is $q_i$. The left product in the statement is therefore $q_0 q_1 \cdots q_n$; the right product has the same factors with $q_n$ placed first. They are equal, including when there are no tests and both products are empty products, equal to one.

(4) At the first test the share premise gives $q_0 \le 2^k q_1$. At the second it gives $q_1 \le 2^k q_2$. Substitute one into the other, and continue. After $n$ tests, $q_0 \le 2^{kn} q_n$. If $j$ bits number the final candidates, then $q_n \le 2^j$, so $kn + j$ bits suffice to number the original ones. The rounded bit drop is at most $kn$. Each of the $n$ passed tests is a separate paid step, and each adds at least one to the nonnegative bill. Thus $n$ is at most the number of paid steps and at most the total charge, giving the two cost comparisons.

(5) Read the actual run from its initial no until its first yes. The preceding reading was no, so that step raised it. The toll charges that step at least one.

Part (3) is the exact count: take base-two logarithms and $\log_2 |\Omega| - \log_2 |\Omega'|$ is the sum, over the tests the run passed, of $\log_2(b_i/a_i)$, the bits each one took. Part (2) answers the complaint I made about the covering, that it doesn't say why anything was deleted: here everything deleted was deleted by a test the run passed and paid for. Two premises stay, the prior list and the share, a bound $k$ fixed before the run on what one test can take. Every run meets some bound, its own biggest test's, so what the share adds is fixing it in advance, and that is needed. On the small machine one CHECK of "$A \geq 15$," asked of sixteen candidates, keeps one, a $2^{-4}$ share, and narrows by 4 index bits on a run of 3 record moves (Proposition 15.39); with $k = 1$ part (4) would say $4 \leq 3$. The same statement on every certified Thiele-complete machine, where the tests are the CHECK moves and the charges are counted by the ledger, is Theorem 15.38.

## 8.9 A grouping, checked and committed

Section 5 ended on buckets, and the toll of Part I never needed them: it prices any yes-or-no reading. In this subsection the word in this section's title gets attached to a partition in a theorem. Take a pile, a finite list of items, and a *grouping*, a bucket label for each item. Its partition is "same label," and the classes of that relation are a partition in the sense of Definition 5.1. The data is a *key* for each item, drawn from any type with at least two values (a grouping's own labels can serve as a key), and a grouping is *sound* for a key when any two items it puts in one bucket have the same key. The grouping says "these belong together," and the key says whether they do. One grouping *refines* another when every bucket of the first sits inside a bucket of the second.

Put two red buttons and two blue buttons on the table, numbered 1 through 4, with 1 and 2 red. Let the key be colour. Circles round $\{1, 2\}$ and $\{3, 4\}$ give a sound grouping: every pair inside a circle has the same colour. Split the red circle into two singletons and it stays sound; splitting cannot introduce a mismatched pair. Join 1 and 3 instead and it fails. Changing the circles' names from A and B to North and South changes none of these checks. That is why the reading below uses the partition, not the spelling of its labels.

Now comes the rule for the record. Put a tick beside a grouping only after comparing the colours within each circle. Copy that ticked grouping onto a separate commitment card, then place the card in a record envelope; charge one tally for each of those three acts. Change a button's colour between the check and the copy, and the old tick must no longer authorize the copy. The machine's SET rule below

does that by clearing the current table and channel. A record already kept remains about the old colours, which it keeps with the grouping.

Here is one example machine that meets the model, the smallest I know that makes a record from a checked claim. It is an illustration; the theorem above holds of any Thiele machine. The *bucket machine* holds a key, a table of checked groupings, a channel, a record, a trap latch and a ledger. SET costs nothing; it replaces the key and throws away the table and the channel, which were checked against the old key. CHECK $g$ costs one and passes when $g$ is sound for the current key, and, when it passes, writes $g$ in the table. COMMIT $g$ costs one and passes when the table holds a grouping with the same partition as $g$, and puts $g$ in the channel. CERTIFY costs one and passes when the channel holds a grouping; if the record is still down it then keeps that grouping with the current key, the key a grouping with its partition was checked against, and a record already kept is left as it is. A CHECK, COMMIT or CERTIFY that fails traps, and a trapped machine only pays. The *reading for a grouping* $g$ is: the record holds a grouping with $g$'s partition. In words, that reading says "this grouping has been checked and committed."

**Theorem 8.5** (A committed grouping entitles exactly its refinements)**.** (1) The classes of "same label" under any grouping form a partition. (2) Two groupings with the same partition of the pile have the same reading at every state of the bucket machine. (3) For groupings $g$ and $h$ of the pile, soundness of $g$ for a key gives soundness of $h$ for that key, for every key, exactly when $h$ refines $g$. (4) On every run from a clean state (table and channel empty, record down), if the reading for $g$ is up, the record holds a grouping and a key for which $g$ is sound, and so is every grouping that refines $g$; and for every grouping $h$ that doesn't refine $g$ there is a key for which $g$ is sound and $h$ isn't. (5) The bucket machine obeys the toll, and a run from a clean state that ends with the record up has raised the ledger by at least three. □

*Proof.* (1) Every item has its own label, exchanging two items does not change whether their labels agree, and two agreements through a third label give agreement with each other. Thus "same label" has the three sameness rules, and its classes form the partition already constructed in Theorem 5.1.

(2) The reading asks which items were put together, not what the buckets were called. Replacing one grouping by another with exactly the same buckets therefore changes no answer to that reading.

(3) If $h$ only cuts $g$'s buckets smaller, every pair $h$ keeps together was already together under $g$. Soundness of $g$ gives that pair equal keys, so $h$ is sound too. If $h$ does anything else, it puts some two items together that $g$ kept in separate buckets. Give every item in the first item's $g$-bucket one key value and every other item a

second key value. Each $g$-bucket has a single key, so $g$ is sound. The chosen $h$-bucket contains both key values, so $h$ is not. This is why the key type needs two values.

(4) Follow the table, channel and record through the commands. They begin empty. CHECK inserts only a grouping it has found sound for the current key. COMMIT takes the partition of a grouping already in that table, so the selected grouping is sound for the same key. CERTIFY keeps that grouping and that key together; later commands do not replace an existing record. SET changes the key but empties both the table and the channel, so no old unchecked carry-over remains there. A failed command traps instead of inserting an unsound grouping, and trapped commands leave those fields alone. These are all the ways those fields change. Their promised soundness therefore survives every command of any run from the clean start. Apply (3) to the grouping and key in the record: every refinement is sound, and every non-refinement has the counterexample key just constructed.

(5) Only CERTIFY raises the record, and its price is one, giving the toll. A successful CERTIFY needs a grouping in the channel. From an empty channel only a successful COMMIT could have supplied it. From an empty table that COMMIT in turn needs a successful CHECK. These are three distinct earlier acts in order, each charged one. The run's nonnegative bill is at least three. ■

So "structural" has a theorem under it. The reading sees a grouping only through its partition, which is item (2). If the recorded circles are sound, so is any way of cutting them smaller, whatever the key is. A grouping that isn't a cutting-smaller can still happen to be sound for this key (paint all four buttons red and every grouping is sound), but nothing in the record promises it: for some key it fails. Item (3) and item (4) say exactly that much, and that's all the record buys. It's the three parts of entitlement from the start of this section in their smallest case: the object is the grouping, the checked relation is soundness for the key, and the paid event is the CERTIFY, on top of a CHECK and a COMMIT at a cost of one each. What it doesn't do is make a later move act on the record. The only move that reads it is CERTIFY, and only to leave a kept grouping alone. Like everything in this section, it says what someone auditing the run may conclude.

Before the machine of Part III, I owe you the word the toll is named after, the "insight" in "No Free Insight" (Section 4), because I've been using it as if it meant one thing.

CHAPTER 9

# What insight is

I named the rule in this book "No Free Insight," and the name ran ahead of any careful account of what insight is. The book has been narrowing it ever since: not learning (Section 4), not looking, not copying. Part I already says where to look: learning borrows and committing pays, so the part of insight a toll can price is the part that stays. It's time to say what that is, and to say it in a way that someone who studies actual insight, in actual heads, wouldn't laugh at. So who are those people, and what did they find? I'll go through what I ran into and push on each piece as I go. Where a story reached me secondhand, I treat it as a lead to check.

## 9.1 Sultan and the sticks

In 1913 the psychologist Wolfgang Köhler went to Tenerife to run a research station with a colony of chimpanzees, and the First World War then kept him there until 1920, five years longer than planned, with the chimpanzees to watch. The famous one is Sultan. Food is outside the cage, out of reach; two sticks are inside, each too short, but one fits into the end of the other. Sultan tries each stick, fails, sulks, plays with the sticks, and then, by Köhler's account, at a moment you can see, joins them and reaches the food. Köhler's words were *Einsicht* and *einsichtig*, which came into English as "insight," and the whole school of psychology he belonged to, the Gestalt school, said that's what thinking is: a *restructuring*, the moment two sticks stop being two sticks and become one tool. Stacking up tries until one works was, for them, something else. Karl Duncker gave the human version, in German in 1935 and in English in 1945, five years after he died at thirty-seven; he'd spent his last years working with Köhler at Swarthmore. You can set it up yourself: a candle, a box of tacks, matches, and the task of fixing the candle to the wall. The obstacle is seeing the *box* as a possible shelf while it is presented as the container for the tacks. And then he found that if you hand people the box *empty*, with the tacks poured out beside it, they solve it far more readily. The objects were the same; what changed was which relation among them was on the table.

I like both of these for the same reason: the insight didn't add an object. Sultan had both sticks the whole time. The box was there the whole time. What arrived

was a *relation*, which is to say a piece of structure in Section 5's sense, and the arrival was sudden. Hold that, because the thing the toll prices also arrives in one step, and at its best it's also a relation.

## 9.2 It isn't a feeling

A feeling of getting closer is a different thing to measure from solving the problem. Janet Metcalfe's experiments help separate them. In the 1980s the psychologist Janet Metcalfe asked people to solve problems while rating, every fifteen seconds, how "warm" they felt, how close to the answer. On ordinary problems, algebra for instance, the warmth climbs steadily and peaks at the answer, which is what you'd expect. On *insight* problems, riddles of the candle-box kind, the warmth stays flat, flat, flat, and then the answer arrives, and if anything the people who felt warmest beforehand were *more* likely to be wrong. She put that last finding in the title of a 1986 paper: premonitions of insight predict impending error. Those results concern the predictive value of warmth ratings in those tasks. They do not show that a solution has no accompanying feeling, or that every person follows the same curve. Try it yourself: write a number from one to seven every fifteen seconds while doing a long sum, then while doing a riddle, and compare the curves.

And the most famous insight story there is fits Metcalfe uncomfortably well. Archimedes, a mathematician who died in 212 BC, asked by Hiero, the king of Syracuse, whether a goldsmith had cheated him on a crown, gets into a bath, sees the water rise, and runs home naked shouting "eureka," I've found it. That comes from Vitruvius, a Roman writer on architecture, writing about two centuries later. Is it true? Galileo Galilei, then a young Italian mathematician, at twenty-two, in 1586, took it as known that Archimedes caught the goldsmith. What he didn't buy was the method. Water spilling over the side, as Vitruvius tells it, was to him "a crude thing, far from scientific precision," and he guessed some writer had set down a rumour. So he wrote a little tract, *La Bilancetta*, proposing a balance weighed in air and in water instead, which he thought was probably what Archimedes actually did, since it rests on Archimedes's own results. Whether there was ever a bath, or a shout, nobody can check now. What survives is a story two centuries late, wrapped around a real man and probably a real problem. Notice what the legend kept: the bath, the shout, the running home naked, which is to say the feeling. What it left shaky is the part that would count as the check, the test that could have caught the goldsmith. The bit everybody remembers is the bit that tells you least.

So any account of insight that keys on the sensation, the "aha," is keyed to something that the data says is unreliable. Whatever insight is, it isn't the feeling of having it. That's a relief, because a system of states and moves doesn't have feelings,

and a word that only meant a feeling would be cheating if I hung it on one. The version this book can price has to be keyed to an *event*, and the question is which.

## 9.3 What happens when a computation learns

I'll lay out the split first, then say what the toll does with each part, and then say who decided the split. Keep Section 2's sheet of paper in front of you: the NO box, the YES box, the tally line, and the three moves STAY, WORK and VOUCH.

The first thing: *raw observation*. Something comes in and gets written down. A counter ticks. Nothing is claimed about what it means, nothing is marked as resting on it, and the reading doesn't move because of it. Later moves can read what it wrote, but that reading is a move of its own. The demon's pad in Section 4 was raw observation, and raw observation is what the statistician Dennis Lindley measured in 1956 as the information provided by an experiment, the words of his title, and defined as the expected drop in your uncertainty: a real quantity, and a quantity about what was *seen*. It can cost nothing on the ledger. The toll says nothing at all about a move that doesn't cross from NO to YES, so a price list may charge it zero: on the paper, STAY and WORK cost nothing, and Section 4's watcher narrowed a whole row of candidates without paying anything. "Can" is the word that matters. A price list is allowed to charge for looking anyway. And in the physical world a look takes something too: copied onto a blank page it merges nothing and owes no floor under Landauer's premise, but it can dissipate under the stated thermal-model assumptions, and it uses up the page, which means taking states the machine hasn't used, borrowed until the page is wiped (Section 4).

Lindley's measure has a back story, and it runs through Bletchley Park. I. J. Good, who was Alan Turing's main statistical assistant at Bletchley Park in 1941, wrote in 1979 that Turing had worked out the expected weight of evidence an experiment would give, to judge one of the codebreaking procedures before running it, and so partly anticipated Lindley by about fifteen years, in work that stayed secret. Turing measured evidence in *bans*, logarithms to base ten, with a tenth of one called a deciban, and the name came from the town of Banbury, where tens of thousands of the sheets used in that procedure were printed. It's the same base-ten unit other people call the hartley, after the Hartley from the last section. And Good adds that Turing didn't know the logarithm he was using had already appeared in 1878, in a paper by Charles Sanders Peirce, the same Peirce from Section 5, who'd called it the weight of evidence. Peirce is as far back as Good takes it. It changes nothing in the argument, but it shows how well trodden this first kind of thing is. For a century and a half careful people have measured what an observation tells you, and every one of those measures is about what was seen. The toll leaves that side of the line alone.

The second thing: *the raise*. Some move turns the reading from NO to YES, and that move is where the toll applies. In any system with the five parts that obeys the toll, that move costs at least one, and every run that starts at NO and ends at YES pays at least one somewhere along the way, which is Section 2's universal floor. What's being charged for, at its best, is exactly the Gestalt thing, a relation committed to: one tool, claimed, out of two sticks. I say "at its best" because Section 2 was careful about what YES means: the flag is up, and nothing more. A crossing that writes down which relation it's committing to carries Sultan's tool with it. A bare crossing, VOUCH on the paper, commits to no relation at all, and it pays exactly the same.

The third thing isn't a kind of event. It's an observation about the floor: it's *the same*, whatever is certified. Say the YES state also shows something striking, a deep relation, a result nobody could have guessed, Sultan's two sticks become one tool. The toll charges the crossing at least one. It doesn't charge the striking crossing more, because the floor's proof never looks at anything but the reading before and after. On the paper, VOUCH from NO makes one mark whether you were thinking of Duncker's box or of lunch. This toy charges one either way. The general toll supplies the same lower bound for both; an actual price schedule can charge them differently.

The candle box has a footnote that says the same thing from the other side. In 1962 the psychologist Sam Glucksberg ran Duncker's problem with money on it: some people were told they could win five or twenty dollars depending on how fast they solved it. With the tacks in the box, the people chasing the money did worse than the people who weren't. With the box empty the problem was easier for everybody, and the accounts I've read don't agree on what the money did there, so I'll leave that half alone. Paying more didn't buy the insight, and on the hard version it got in the way. That's a fact about heads. It says nothing about records, and I won't push it further than that. But it makes a nice parallel to the paper. On the paper a deeper crossing isn't charged more, and in Glucksberg's room a bigger prize didn't bring the crossing sooner. In neither place does the price measure the depth.

I'll state them exactly now, for any Thiele machine (Definition 2.2). A run is still just a list of moves, carried out in the order written. The question is which of those moves crosses from no to yes, and what that crossing lets you conclude about the total charge.

**Definition 9.1** (Raise, non-raising move)**.** In a Thiele machine, a step from state $s$ by move $i$ is a *raise* when it turns the reading on: $\mathrm{cert}(s) = \text{no}$ and $\mathrm{cert}(\mathrm{step}(s, i)) = \text{yes}$. A move $i$ is a *non-raising move* when no step by it, from any state, is a raise. Raw observation, the first thing above, is one kind.

**Remark 9.1** (What the toll prices, and what it doesn't)**.** All three read straight off Definition 2.2 and Observation 2.1, with Section 2's board as the example. (1) The toll puts no bound on the cost of a non-raising move; on Section 2's board STAY and WORK are non-raising moves and cost 0. (2) Every raise is charged at least 1, by the toll, and every run from a state reading no to one reading yes has total cost at least 1, by the universal floor. (3) The floor is the same: the floor's proof uses nothing about the end state but its reading, so it gives the same floor of 1 whatever else the end state shows. A higher floor needs more hypotheses than the toll, and those hypotheses do the work. A system that wants to charge more for a deeper certification has to write that into its price list, move by move; the toll doesn't do it for anyone.

So the account comes down to two facts and a warning: looking can stay off the ledger, turning the reading on can't, and the toll alone does not charge more for a remarkable claim, though another pricing rule or an additional requirement could.

Who decided the split? The costs alone didn't. The toll prices exactly one thing, the crossing. A price list is allowed to put a floor under more moves than that (a look, a send, a check) and allowed to leave other moves at zero, so the sorting into non-raising moves and raises is by whether the reading moves, and that sorting is this book's. The part that's *forced* is the floor under the crossing, once you grant a reading that stays up, finitely many states that read yes, and a price on every merge; Section 3 is the argument; anything a price list charges beyond that is a choice, and any concrete machine owes you its whole table so you can see which choices it made.

## 9.4 Test, then commit, and nothing forces the second to follow the first

The philosopher Karl Popper said a theory earns its standing by being *exposed*: you make a conjecture, you try to kill it, and what survives a real attempt counts for something, where "real" means the test could have failed. His word for that earned credit is *corroboration*. I lean on that idea all through this book (Section 14 gives it its full due), and this subsection shows where a Thiele machine has a version of it and where that version is weaker than Popper, which I have to say plainly.

Add a fourth move to the paper: CHECK. It looks at something (a sum, a coin under a cup, a formula against a row of values), and it can come out pass or fail, and it doesn't move the box. Charge it whatever you like; charging it on the attempt, pass or fail, fits Popper, because what made his test count was that it was really run and could have failed. CHECK is the test, and it's a non-raising move in the sense above, because it never turns the reading on. VOUCH is the commit. So the honest

protocol is two moves: test, then commit, and the costs are separate, one for the checking work and one for the crossing.

But nothing in the five parts makes the second move wait for the first. The step is handed the current state and a move name; the toll looks only at whether the reading crossed. Play VOUCH from NO with no CHECK anywhere before it, and you land in YES and make one mark, the same box and the same mark as the player who checked first. The two tally lines differ by exactly what the CHECK was charged, and if you charged it nothing they don't differ at all. The mark for the crossing is the same mark either way. The receipt says a commitment was paid for; it does not say the commitment was earned. Popper's corroboration needs the test; the toll doesn't.

You can build a system that makes the commit wait. Let CHECK leave a mark in the state when it passes, and write the step so VOUCH only crosses when that mark is there. Making VOUCH wait like that is a design, written into the step function, and it's a perfectly good one. Nothing in the five parts or the toll asks for it. The gap between the word and the bare system is real, and I'm leaving it in plain sight: the toll prices the commit, and whether the commit was preceded by a test is a fact about the run, or about a design somebody chose. The five parts don't guarantee it.

**Remark 9.2** (Nothing makes the crossing wait)**.** Read off Definition 2.2. The toll constrains $\mathrm{cost}(i)$ for moves that cross, and nothing else; the step function is any function of the current state and the move. So a Thiele machine may have a move that crosses from a state no check has touched, and the toll still holds. On Section 2's board extended with CHECK, the run $[\mathrm{VOUCH}]$ from NO crosses with one mark, and $[\mathrm{CHECK}, \mathrm{VOUCH}]$ crosses with one mark for the crossing plus whatever CHECK was charged, one mark in all if CHECK is priced at zero. Making a crossing depend on an earlier check is a condition on the step function, which a particular system can impose and the definition does not.

## 9.5 What the word means

Insight, in the sense this book prices, is a distinction made to stay: a move that flips a reading from off to on, for good, and that move costs at least one, in any system with the toll, whatever the reading is about and whatever its depth. At its best the reading is a relation committed to, a claim checked and closed; a bare crossing that commits to nothing pays the same, and the toll doesn't tell the two apart. Part III's certified machine is one way to make sure a crossing is the first kind. There are three boundaries, each from a different place. From Metcalfe: it isn't a feeling, and a system's version has to be an event. From the proof: the floor is the same; a shallow commit and a deep one both cost at least one, and cost is not depth. A system can pay heavily to check a trivial claim, and Duncker's box, a real insight, is still one

small commit. And from the two-move protocol: a paid flag proves nothing on its own. Price and correctness are separate books.

Which leaves a suspicion I'd have in your place. Everything so far has been one reading on one sheet of paper, with a rule I chose and a count I bolted on. Is the shape of that record something I picked, or is it what any base-driven record has to look like, on any machine at all? The next section answers it: take *any* machine that steps, bolt a record onto it, and see what shape it's forced into. How a concrete machine hangs actual claims on its parts, a checked property of one counter at one edition, and where those sit in the state, comes in Part III (Section 15).

CHAPTER 10

# Records over any base: the latch

The blank cell comes back here, over any machine: a reading that goes from no to yes because the machine did something, and stays. For the whole book so far I've been building up a thing with a flag and a counter, and you'd be right to suspect that the whole apparatus is special pleading for one machine, the one I keep promising to build. So this section answers the suspicion with no particular machine in it at all, and sees what's left. Take *any* machine that steps deterministically: a Turing machine, a register machine (a few numbered boxes that each hold a whole number, and a program that adds, subtracts and jumps), a lambda calculus (Alonzo Church's way of computing by nothing but plugging expressions into expressions; Section 13 builds one), anything. Call it the base. Now bolt a record onto it, under rules I'll state, and ask: what can the record be?

The answer is one shape, and I'll show you the shape first because it's a thing you can build from a cup and a die.

## 10.1 A cup and a die

Put a cup upside down on the table. The rule is that the first time a coin lands in front of it, you turn it right side up. There is no rule for turning it back over. Now roll a die, and put a coin down on every six. Roll twenty times. If a six appears, the cup turns over on the first one and stays turned over. Write down which roll did it. That number is where the record happened, and the cup afterward is the record. If no six appears, the cup stays down; a reachable event need not occur on every trial.

For the deterministic base in the theorem, write the rolls down first and replay them. The base state is the fixed list and a finger pointing to its next entry; a step moves the finger one place, regardless of the cup. The event asks whether that entry is six. Nothing about the cup changes which entry comes next.

The cup is a *latch*. Electrical engineers have had the circuit since 1918, when two British physicists teaching engineering in London, William Eccles, a radio pioneer, and Frank Jordan, applied for a patent on a two-valve arrangement that holds whichever state you last put it in; it became the flip-flop, and the registers and caches in the device you're reading this on descend from it. Cut its reset wire and

what's left has one input, "set," and the rule *new state = old state or set*: once set, it stays set. The cup is that circuit: a flip-flop with its reset wire cut off.

(Eccles and Jordan didn't call it a flip-flop. Their 1919 paper calls it a *trigger relay*. Where the newer, sillier name came from I can't source, so I'll leave it at that.)

Is a flip-flop with its reset wire cut off only a picture, or did anybody ever build one on purpose? Somebody did, and for a reason that's very close to this book's. In the mid-1950s Wen Tsing Chow, an engineer at the Arma division of the American Bosch Arma Corporation in Garden City, New York, was asked by the United States Air Force, in the accounts of his career I could find, for a more flexible and secure way to hold the fixed numbers, the targeting constants, inside the guidance computer of the Atlas missile. What he and his Arma colleague William Henrich patented (filed in 1957, granted in 1962, under the plain title "Storage matrix") was a grid with a connection at every crossing, where you set the numbers by taking out the connections you didn't want, physically or by burning them out with too much current. The word *burn* is right there in the patent, and people still say they burn a chip. The descendants are called programmable read-only memories, PROMs, and a fuse blown in one stays blown: there's no current you can send to grow it back. The grid was a whole memory built out of cups with no reset wire, made because somebody wanted numbers nobody could quietly change later. That doesn't prove anything here, because the theorem below doesn't care what a cup is made of. But it's good to know the shape wasn't invented for a sheet of paper. Somebody needed it, for stakes a good deal higher than a die.

Now the claim, stated loosely and then exactly: *every base-driven record, over any base, is a cup with no reset wire, turned by some event the base passes through.*

## 10.2 What "base-driven" has to mean

The content is in the definition, and I'll give you all five rules, because the theorem is only as strong as they are.

A machine *extends* a base when there's a map from the bigger machine's states down to the base's, and that map starts in the base's start states, reaches every one of them, moves in step with the base (one step up top is one step down below, through the map), and agrees about halting. The bigger machine may carry as much hidden state as it likes beyond the base, and that's what the words "any base" are about: the base doesn't get to see the extension, and the extension doesn't get to change what the base does.

The extension carries a *record*, a yes/no reading, and a *ledger*, a counter. It's a *base-driven record* when five things hold. One: the record's next value depends only on the base's current state and the record's current value. (I'll call this *driven by the computation*. It says the record answers to what the machine is doing. A

clock on the wall or someone's hand doesn't get a say.) Two: the ledger never goes down (Section 6). Three: the step that turns the record on costs at least one; that's the toll (Section 2). Four: once on, the record stays on; that's permanence (Section 3). Distributed-systems people have a name for a predicate like that. The computer scientists K. Mani Chandy and Leslie Lamport, in their 1985 paper on taking a consistent snapshot of a distributed system, called a property *stable* when once it holds it holds forever after, and stable properties are the ones a snapshot can be trusted to detect. Five: some run from a start state actually reaches a step that turns the record on; otherwise the record is decoration.

I'll write the rules out exactly. On/off is true/false, $\vee$ is "or" (on when either side is on), and $\text{next}^k$ is $k$ steps.

**Definition 10.1** (Base, record-carrying machine, cover)**.** A *base* $B$ is a set of states with a step function $\text{next}_B$, a set of starting states, and a halting predicate. A *record-carrying machine* $M$ is a set of states with a step function $\text{next}_M$, a set of starting states, a *record* rec from states to Booleans, a *ledger* $\mu$ from states to whole numbers, and a halting predicate. The cost of a step is $\text{cost}(m) = \mu(\text{next}_M(m)) \mathbin{\dot{-}} \mu(m)$. A *cover* of $B$ by $M$ is a map $\pi$ from states of $M$ to states of $B$ that sends starting states to starting states, has every starting state of $B$ as the image of some starting state of $M$, commutes with the steps, $\pi(\text{next}_M(m)) = \text{next}_B(\pi(m))$, and agrees about halting: $m$ halts exactly when $\pi(m)$ does.

**Definition 10.2** (Base-driven record, latch)**.** $M$ with cover $\pi$ is a *base-driven record* over $B$ when (R1, driven by the computation) there is one function $f$ such that, for every state $m$, $\text{rec}(\text{next}_M(m)) = f(\pi(m), \text{rec}(m))$; for every state $m$, (R2, ledger) $\mu(m) \le \mu(\text{next}_M(m))$; (R3, toll) if $\text{rec}(m)$ is off and $\text{rec}(\text{next}_M(m))$ is on, then $\text{cost}(m) \ge 1$; (R4, permanence) if $\text{rec}(m)$ is on, so is $\text{rec}(\text{next}_M(m))$; and (R5, a reachable write) some starting state $m_0$ and some $k$ have $\text{rec}(\text{next}_M^k(m_0))$ off and $\text{rec}(\text{next}_M^{k+1}(m_0))$ on. An *event* is a function $h$ from base states to Booleans, and its *latch* is the step $(b, r) \mapsto (\text{next}_B(b),\ r \vee h(b))$. *$M$ factors as the latch of $h$* when, for every state $m$ (reachable or not),

$$(\pi(\text{next}_M(m)),\ \text{rec}(\text{next}_M(m))) = (\text{next}_B(\pi(m)),\ \text{rec}(m) \vee h(\pi(m))).$$

The theorem says that under those rules the base state and the record move *exactly* as a latch: the record switches on at the first base state where some event holds, if it started off, and never goes off, and the event is read off the extension's own rule by asking it what it does when the record is off (stated exactly below). The theorem is about what the base and the record do together. The hidden state the extension carries can't push the record anywhere (R1) doesn't allow, since the record's next value is a function of the base state and the record's own value. The

theorem doesn't say the whole extension is a bare latch machine, and it doesn't make hidden state show up anywhere in particular: an extension can carry a field that touches neither the record nor the ledger. And the event is pinned down only on base states that some state with the record off covers; on a base state with no such state above it, $f(b, \text{off})$ is whatever (R1)'s function says, and nothing in a run ever consults it.

I promised the content was in the definition, and the proof bears that out: it is a short case split on whether the record is already on, and it uses only *two* of the five rules, "driven by the computation" and "permanent." The ledger, the toll, and the reachable-write clauses play no part in it. A latch is what you get from "answers only to the base" plus "never comes back down" alone. The other three rules matter for the *converse*: for any base and any event the base can reach, the latch of that event, priced at one unit per write, *is* a base-driven record. The event can be one that holds everywhere, and then its latch is up after one step from every start and tells no two starts apart. It's still a latch and still base-driven: the converse doesn't ask the record to distinguish anything. A record that starts off and tells two starts apart at some step has to latch an event that has held along one run by then and not yet along the other, and the latch tells them apart for as long as that stays so. Put the two together and base-driven records are exactly latches: one shape, and no others, over any base.

**Observation 10.1** (The record is a latch)**.** *Let $B$ be any base, $\pi$ a cover of $B$ by $M$, and suppose $M$'s record is driven by the computation (R1), through a function $f$, and permanent (R4). Then $M$ factors as the latch of the event $h(b) = f(b, \text{off})$. In particular every base-driven record factors that way; (R2), (R3) and (R5) are not hypotheses.*

*Proof.* Ask the extension what it would do with the record off above each base state $b$. Call its answer $h(b)$. Rule (R1) lets you ask that as one function of $b$: no other hidden field is allowed to change the answer.

Now take an actual state. If its record is off, (R1) gives exactly that answer $h(b)$ for the next record. If its record is already on, permanence keeps it on, whatever $h(b)$ says. Together those two cases say "keep an existing on, otherwise use $h(b)$," which is $r \vee h(b)$.

The cover already says that the base part takes its own next step. Put that base step beside the record rule just obtained and I have the stated latch equation. No charge or starting condition entered the argument. ∎

The latch's nearest neighbours are in verification, and they're old. The computer scientists Martín Abadi and Leslie Lamport, then at Digital's research lab in Palo Alto, added *history variables* to specifications in 1991: extra state that records what has happened and never steers the system, used to prove that one specification

implements another. A history variable that only ever turns on is the cup. Temporal logic has the same record as an operator. In the past-time temporal logic that Orna Lichtenstein, Amir Pnueli and Lenore Zuck set out in 1985, "once $h$" holds now exactly when $h$ has held at some point so far, and a runtime monitor keeps it in one bit with the update $r' = r \vee h$. Klaus Havelund and Grigore Roşu, two computer scientists working on runtime verification, synthesized monitors of exactly that kind for safety properties written in that logic in 2002. That update is the latch of Definition 10.2, word for word. The observation adds the other direction: a record driven by the computation that never comes down can't be anything else.

One thing about that proof deserves a second look, because it cuts both ways: it never asks what the base is. The cover decides that: whatever the map sends down is base, and the record's rule is allowed to read all of it; whatever stays up top is extension, and (R1) keeps the record's rule from reading it. Draw the line low, so the base is only the computation's own working state, and (R1) is a real demand. Draw it high, put every table and every refusal bit into the base, and (R1) is easy to meet, and the latch shape comes out exactly the same. So the theorem holds wherever you draw the line, and that's exactly why it can't tell you where a record's meaning lives. The meaning lives in what the event $h$ reads, and that's in the base by construction. On the small machine of Part III (Section 15), the checks that give the flag its meaning sit in the base its cover picks out. So when that machine arrives, ask it first which parts it puts above the line.

□ **Theorem 10.2** (Every reachable latch is a base-driven record)**.** Let $B$ be any base and $h$ any event. Let $\mathrm{Latch}_h$ have states $(b, r, x)$, starting states $(b, \mathrm{off}, 0)$ with $b$ starting in $B$, step $(b, r, x) \mapsto (\mathrm{next}_B(b),\ r \vee h(b),\ x + \delta)$ with $\delta = 1$ when $r$ is off and $h(b)$ is on and $\delta = 0$ otherwise, record $r$, ledger $x$, halting as $b$ halts, and cover $(b, r, x) \mapsto b$. Suppose some starting $b_0$ and some $k$ bring $\mathrm{Latch}_h$ from $(b_0, \mathrm{off}, 0)$, in $k$ steps, to a state whose record is still off and whose base state satisfies $h$. Then $\mathrm{Latch}_h$ is a base-driven record over $B$.

*Proof.* The construction keeps a base state, a flag and a tally. Looking only at the base state gives the promised cover: a start lies over the chosen base start, every base start has such a flag-down, tally-zero start above it, and each step advances the base exactly once. Halting is read from that same base state.

The next flag consults only the present flag and the base event, giving (R1). The tally adds either zero or one and never falls, giving (R2). If the flag rises, it was off and the event held, exactly the case that adds one, giving (R3). An on flag stays on under the displayed rule, giving (R4). Finally the supplied run reaches a state with

the flag still off and the event true. Its next step raises the flag, giving the reachable write (R5).

Whenever a base run reaches the event at all, use its first such visit. Before that visit the flag remains off, so this is the supplied kind of run. ■

What about a base that never merges, a reversible one? It can carry a record, and the next theorem says exactly what it pays. Keep every earlier record value in a history beside the latch, one more entry on every step, and the whole step merges nothing either. But that history needs a fresh slot at every step, forever, so the machine never comes back to a state it has been in, which puts it in the fresh-state column. On a finite state space there aren't fresh states to go on taking forever, and a permanent record can't be written by a step that merges nothing at all. A step that merges nothing is called *injective*: no two states go to the same next state. Keep the old readings on a strip, newest first. If the strip reads $[\text{off}, \text{off}]$ and the current reading is on, the next strip reads $[\text{on}, \text{off}, \text{off}]$. The notation $r :: H$ below means exactly that: put the old reading $r$ at the front of the old strip $H$.

**Theorem 10.3** (Records on reversible machines)**.** (1) Let $B$ be a base whose step is injective. The machine with states $(b, r, H, x)$, step $(b, r, H, x) \mapsto (\text{next}_B(b),\ r \vee h(b),\ r :: H,\ x + \delta)$ ($\delta$ as in Theorem 10.2), starting states with $r$ off, $H$ empty and $x = 0$, has an injective step, never visits the same state twice on a run (the history gets one entry longer on every step), and under the supposition of Theorem 10.2 it is a base-driven record over $B$. (2) With finitely many states reading on, a single injective step with a permanent reading never turns that reading from off to on. □

*Proof.* (1) Suppose two full states have the same successor. Can you recover what they were? Their successor base states agree. The base keeps different sources apart, so their earlier base states agreed too. Their new history strips also agree. The first entry of each strip is the old record, so the old records agree; removing that first entry leaves the old histories, which therefore agree as well.

Those equal old base states and records determine the same charge $\delta$. Subtract it from the equal new tallies and the old tallies agree. Every part of the two earlier full states agrees, so they were the same state. The step is one-to-one.

On a run the strip grows by one entry every tick. A full state cannot recur, because its strip would have to have two different lengths. Carrying that strip changes neither the base nor the flag nor the tally rules checked in the preceding construction, so all five base-driven record conditions still hold under the supplied reachable-event premise.

(2) Suppose instead there are only $K$ on-states and an off-state enters them. Permanence sends all $K$ on-states into that same collection, and the entrant lands

there too. A one-to-one step would have to put $K+1$ different sources into $K$ places. It cannot. Hence a permanent reading with finitely many on-states cannot rise under a one-to-one step. ■

So a base that never merges can carry a record without merging anywhere only by paying in fresh states: once the record is up, the run never comes back to a state it has been in (Proposition 3.5). A history that grows by one entry every step is one way to supply them, and a base that never repeats a state, a counter, supplies them already. A machine with finitely many states hasn't got fresh states to go on taking, so it can't carry one without merging: on a finite machine an instruction that first raises a permanent reading must merge some possible states. The entering state need not belong to the colliding pair; Section 3 prices the instruction under merge pricing.

Process-calculus people ran into the same thing from the other side. Vincent Danos and Jean Krivine, two theoretical computer scientists in Paris, wanted to pin down what a transaction is. In 2004 they had built a reversible version of CCS, Robin Milner's calculus of processes that talk to each other, where every step can be rolled back, and in 2005 they modelled a transaction in it as a process that wanders freely through everything that can be undone until it performs an action declared irreversible, which is the commit. The way I read it, they start from steps that all come undone and ask what has to be added before anything sticks, and the theorem above starts from a record and asks what the step under it must be. In their calculus, as in the theorem, a permanent effect needs a step with no way back, and the calculus has to say which steps those are. The word I tripped on was *declared*. Theirs are labelled. Mine, I thought, are picked out by counting, since on a finite machine the first write merges whatever anyone calls it. Push on that and it gives. I choose the event $h$, and on a base that grows its history forever the write merges nothing, because it pays in fresh states instead (Theorem 10.3), so there my permanence is declared too.

There's an arrow of time in that theorem, and I should say where it came from, because I put it there by hand. Every starting state has the record off. That's the only reason a raised record means the event happened: it was down at the start, and only $h$ lifts it. Turn the cup up before the first roll and it tells you nothing about the die. So the clean start is an assumed ready state, a past hypothesis in miniature. The past hypothesis is David Albert's name for an assumed low-entropy beginning of the universe, which several of the accounts below lean records on. Whether every trusted record leans on it is their argument, and I'm not settling it. Nothing in this book derives one; every theorem here that reads a raised record as news that the event happened assumes its small version and starts from there. The

latch observation doesn't need it, and that's why on its own it can't say the event happened.

When I went looking for who else had found the arrow hiding in the start, I got five papers that read like five unrelated answers. They lined up once I asked each one the same thing: what does the cup being down at the start rest on? There are two places to put it. You can put it at the beginning, as a fact about how things started. Or you can put it in the structure of what causes what. David Wolpert, then a physicist at Los Alamos and the same Wolpert as the free-lunch theorem (Appendix B), sorted memory systems in 1992, and one of his kinds works only because it begins in a prepared state set up by a process that runs one way, which is how it leans on the second law. That's the cup put down by somebody before the game, at the first place. Wolpert and Jens Kipper, a philosopher at Rochester, put in 2024 that sorting on a formal footing, and they say plainly that they have no formal proof this kind can only remember the past. I trust them more for that sentence. What I think they're saying is that a prepared start buys you a memory, and getting its direction too takes something they can't yet prove.

Aram Ebtekar, an independent researcher in Vancouver, and Marcus Hutter, who works on the theory of general artificial intelligence at Google DeepMind and the Australian National University, built in 2024 a reversible toy world, imposed a past hypothesis on it, coarse-grained it, and got records of the past and none of the future. As I read their route, the clean start doesn't go straight to the records. It goes through a causal structure first, and the arrows come out of that. Thomas Blanchard, a philosopher of science at Cologne, got in 2024 the same lopsidedness of knowledge out of the causal Markov condition, a condition on causal structure. So Blanchard starts at the second place, which is the middle of Ebtekar and Hutter's road. That took me a while to see, because one is a toy world and the other is philosophy of causation. They go from the start to causes to records; he goes from causes to records. The fit is my reading, and it's mine to defend. And Carlo Rovelli, two years before his paper on the cost of choosing from Section 4, studied traces in "Memory and entropy" (2022) and concluded that a memory turns low entropy into available information. He's at the first place, and he says what the beginning gets spent on. Read against the latch, that's the blank the record uses up: the flag can say yes once because it started at no.

So does the latch explain why records point backwards? It doesn't, and I'll say that plainly so it doesn't look like it does. The step runs forwards and the runs start clean, and a theorem built on those two can only hand the arrow back. Push further and it gives in a second place. The history that grows forever looked like a way around the clean start. It starts empty, so it leans on the same blank. Under a step that forgets nothing, on a finite set, no permanent record can ever be written, past or future, which is Zermelo again. On the map the book sits at the first place, with

Wolpert and Rovelli, and it sits there by assumption. Ebtekar and Hutter at least imposed their start on a world with its own dynamics and watched what came out. I wrote mine into the definition.

Constructor theory, which Section 3 brings in as a neighbour of the counting argument, has a paper on time (Deutsch and Marletto, 2025) saying that laws stated as which transformations a device can and can't bring about can't mention time at all, so duration has to come from relations between physical systems, clocks, that have their own dynamics. What I think they're saying is that whenever you want a duration you point at a second system and count what it did. This book is in that position by construction. The step count is the only time in it, and it comes from the machine's own steps. The next subsection has a record turned at the fifth tick of a hidden clock, and (R1) throws it out, which read beside them is a rule about which system may be the clock: the one being recorded. Push on it, though. Their laws have no step in them, and the paper's work is getting duration and dynamics out of clocks and timers. I put a step, $\mathrm{next}_B$, into the definition of a base. So I'm in their position only at the far end of their argument, where a clock already exists. The base is my clock, and I assumed it.

Physicists have built this latch on paper too, in quantum mechanics, and it behaves the way the last theorem says. The question that sorted them for me is where the cup puts what it forgets, since a closed quantum system with separate energy levels comes back (Bocchieri and Loinger, Section 3) and the theorem left only two ways to pay for a record that stays: fresh states, which here mean a history that never ends, or a merge. Klaus Hepp, a mathematical physicist at ETH Zurich, wanted a measuring device inside quantum mechanics itself, and studied in 1972 a particle that flips spins in a long chain as it passes, now called the Coleman-Hepp model (I know Hepp's paper only from accounts of it), and John Bell (1975) took it apart: the record is exactly definite only for an infinitely long chain, which is the fresh-state column above, a history that never ends, in quantum clothes. Jonathan Halliwell, a theoretical physicist at Imperial College London, used in 1999 a two-state detector that flips once when a particle crosses a point and is kept from flipping back by a large environment, which is a latch with the environment holding the history. His abstract calls it *effectively* irreversible, and I like that he puts the word *effectively* right there where you can see it. And Murray Gell-Mann, the physicist who named the quark, and James Hartle, of the University of California at Santa Barbara, gave in 1995 the property a name inside their framework of decoherent histories: the permanence of the past, past alternatives that stay settled as the story is extended into the future. They name (R4) for whole histories and leave the cup to others. So Hepp's chain, as Bell read it, is exact and infinite, and Halliwell's detector is finite and only effectively permanent. Mine is finite, exact and permanent at once, and it gets there by letting the first write merge. Push on that and it gives: I never

say where the merged state goes. Hepp's chain had to be infinite. Halliwell had to put his in an environment. Mine just goes, and the toll writes a unit in a counter. That's fine in a model of computation, and it's exactly where this stops being physics.

## 10.3 Both rules do work: the toggle and the timestamp

A theorem with two hypotheses should have two counterexamples, one for each hypothesis dropped, and it does, and they're the two ways a record goes wrong in the world.

Drop permanence. Take a base that just counts, and a record that flips at every step: on, off, on, off. It's driven by the computation; it's not permanent; and it's not a latch of any event. Such a record is a *toggle*, a piece of working state. A record you can revoke is working state wearing a record's clothes, and if a revocation has to stick, that's a *second* latch ("revoked"), which is how certificate revocation lists actually work: you add an entry; you never delete the certificate. Alin Tomescu, a computer scientist then at MIT, and his coauthors (2019) built append-only authenticated sets and dictionaries for transparency logs, and revocation fits that shape exactly: a revocation goes in once and stays. The pair lemma (stated below) covers two latches that can each read the other; do it with two cups where the second only turns over on an even roll *if the first is already up*, and you'll see the second can never turn before the first.

Dialogue theorists built a version of that two-latch rule for arguments. Give each speaker a row of cups, one for every claim they've made. Everybody I read in this field is answering two questions about those cups: whose hand turns them, and can they come back down. C. L. Hamblin, an Australian philosopher and logician who also came up with reverse Polish notation for early computers, wrote *Fallacies* (1970), and it has a chapter on formal dialectic where each speaker has a *commitment store*, a list of what they're on the hook for, and the rules of the dialogue update it, while the speaker doesn't. I know that chapter from other people's accounts of it. Hamblin's store is an answer to the first question, and it's (R1) for arguments: the cup answers to what happened in the dialogue, and the speaker's hand is kept off it. Henry Prakken, who works on AI and law at Utrecht, built in 2005 dialogue games for argumentation on stores like that and made commitments sticky: a claim can be retracted only when it has been challenged, and claims and backtracking carry a price. He also floats a convention that once a player has retracted a proposition, that player can't take it on again, which is a revocation that stays revoked, the way an entry on a revocation list does. Prakken's rules are an answer to the second question, and it took me a second pass to see that they're two answers: a cup can come down, and coming down costs, and once it has, the taking-back is itself a cup that stays up. Erik Krabbe, an argumentation theorist at Groningen, names in 2001

the awkward part of that: retraction is frowned on, and yet resolving a dispute takes somebody retracting a doubt or a point of view, so the rules have to say when taking a commitment back is allowed. As I read him, he's standing between the two ends of the second question. A store that never gives can't settle a dispute, and a store that gives whenever it's asked isn't a store. The part of all that I'd keep is Hamblin's, that the store belongs to the rules and doesn't belong to the speaker. Take the pen away from the program. And this book answers Krabbe with a price: a take-back is a second cup, "revoked," and turning it costs at least one like any other write, so a retraction is allowed, permanent and paid for, which is Prakken's convention with the toll attached. Where mine gives is his first condition. He lets a claim come down only when it's been challenged, and nothing in my latch asks why the second cup turned. The pair lemma lets the second cup's event read the base and the first cup, and nothing in it requires a challenge to be there.

The same field has a notion that looks a lot like earning. In the abstract argumentation of Phan Minh Dung, a computer scientist at the Asian Institute of Technology in Bangkok (1995), arguments attack one another, and the *grounded* semantics starts from the arguments nobody attacks and keeps adding the ones they defend. Look at how it's built: things go in and never come out, which is a row of cups turned one at a time. Pietro Baroni and Massimiliano Giacomin, at Brescia, introduced in 2007, and Martin Caminada, then at Aberdeen, revisited in 2014, *strong admissibility*: a set is strongly admissible when every argument in it is defended by others in the set that don't lean back on it, so the support bottoms out. Strong admissibility is the argumentation form of an earned claim. Above the bottom, nothing in it is in by default, which puts it at the far end from Burge's default on the testimony map in Section 7. Caminada argues it plays a vital part in the dialogue games that decide grounded acceptance. I've read Caminada's paper; Dung's and Baroni and Giacomin's I know through what later papers say about them. Notice the bottom, though. An argument nobody attacks gets in on a check that can't fail, and this book says a check that can't fail establishes nothing (Section 14). Now turn that on my own cups, because it lands. My first write needs only that $h$ holds, and what $h$ reads sits in the base, where the latch theorem doesn't look. I charge a unit for it, and a unit is a price, not a check. So my earned records bottom out the way Dung's do, in something let in without an argument, and the one difference is that mine pays to get in. Whether a machine's earned record certifies exactly the strongly admissible sets is the kind of statement I'd have to prove before I said it, and I haven't.

A record that's allowed to change its mind isn't lawless, though; it's somebody else's subject. Count how many times the cup may be turned. Once, from down to up, is the latch. More than once and you're somewhere else, and the people who went there sort it two ways: by how many turns are allowed, and by how long the game runs. The philosopher Hilary Putnam, then at MIT, and E. Mark Gold,

a computer scientist at a research lab in Santa Monica, in back-to-back papers in 1965, looked at questions a machine answers by guessing and revising, where the answer is whatever the guesses finally settle on. Putnam's paper says outright that you're never sure you have the correct answer. That stopped me the first time. What kind of answer can't you ever be sure you have? The way I came to read it: the cup's position at any moment is the current guess, the answer is where it ends up, and nobody watching can tell when it's done moving. Yuri Ershov, a logician in Novosibirsk, sorted those in 1968 by how many revisions they're allowed (that's how his hierarchy is described; I haven't read his paper). Allow one, from no to yes, and you get exactly the sets a program can list. That level is in a modern account of the hierarchy by Rod Downey, Carl Jockusch, Timothy McNicholl and Paul Schupp, four computability theorists, and you don't need their word for it. If a program lists the set, guess no for every number and flip the guess to yes when the number shows up on the list. Going back, if every guess flips at most once, from no to yes, run all the guessing side by side and write a number down the moment its guess says yes. Those listable sets are what a permanent record raised by a search catches. Allow two, three, and so on, and every extra change buys new sets: the rungs of the ladder are all different. A record that can come back down lives on that ladder. Now put Putnam's worry on my cup, because it lands. On the bottom rung a yes is final the moment it appears, and a no never is, since down means not yet. A single cup of mine can certify that something happened and can never certify that it didn't; a final no would take a second cup, turned by finding a reason, and that's the pair of latches again. That's what the bottom rung costs, and I'd rather name it. This book stays on the bottom rung; I'm pointing at the ladder, and I'm not climbing it.

The ladder doesn't stop at finite time, and the people who followed it past infinity have names for the two kinds of record I just described. A definition that only ever adds, stage after stage, and keeps everything it has at each limit, is an *inductive definition* (the long treatment is a 1974 chapter by Wayne Richter and Peter Aczel, two recursion theorists). It's the latch run through the infinite, a cup that goes up at some stage and is still up at every stage after. Let it take things back, and keep what stays put from some stage on, and you have what John Burgess, a philosopher and logician at Princeton, called in 1986 *quasi-inductive* definitions, which he met in the revision theories of truth that grew up beside the logician Saul Kripke's 1975 theory of truth, which builds up the true sentences in stages. Burgess's paper measures how complicated the truths of those theories get. I've read its abstract, but not the place where he names them. What took me a few reads is that Burgess's rule at a limit is Putnam and Gold's rule, keep whatever the guesses settled on, applied at every limit stage along the way. That's my reading of how the two fit. If it's right, the split between a cup that only goes up and a cup that settles is the same split at finite time and past it: a permanent record is the first kind, and a toggle that's allowed to

settle is the second. Joel Hamkins and Andy Lewis, two set theorists, let a Turing machine keep running through the ordinals in 2000. At a limit a cell keeps its value if it settled, and reads 1 if it kept flipping back and forth forever, and the trick they use again and again is the toggle: flash a flag on and off every time a guess changes, and at the limit the flag is on exactly when the guess changed infinitely often. A flag that can come back down is how that machine learns what no finite run could tell it. A permanent one can't do that trick, because at a limit a cell that only ever goes up just stays up. So past infinity the toggle is the clever one: a latch tells you a thing happened at least once, and their flashing flag tells you it happened infinitely often. I threw the toggle out a few paragraphs back anyway, and I'd do it again, because before the limit its reading can't be relied on, and nobody using a machine of mine gets to wait for a limit. One finite-time fact belongs beside it, so nobody concludes too much from permanence. Hao Wang, a logician then at Oxford, described in 1957 a machine whose only write is to mark a square and which can never erase one, and it computes everything a Turing machine does. Permanence of a machine's own memory costs it no computing power, given an endless tape: each mark goes on a square nobody marked before, which is the fresh-state column again. What the book prices is a permanent reading raised by a step that has to work whether or not the reading is already up, which is a different thing. So the cost I named a paragraph ago is a cost in what a reading can claim, and computing power doesn't pay it.

Drop "driven by the computation." Take a base that counts, and a record that switches on at the fifth tick of a *hidden clock*, whatever the base is doing. It's permanent, and the computation doesn't drive it; such a record is a *timestamp*. A timestamp is not evidence: it says *when* and doesn't say *what happened*, and a record switched by an outside party or a clock records that party or that clock and leaves the computation out. Try it with the recorded die list. Keep a second tally hidden beside it, and turn the cup when that tally reads five before a move; then advance both the tally and the finger on the list. Run two trials from the same list entry with the cup down, but start the hidden tally at five in one trial and zero in the other. The next cup values disagree although everything the base shows agrees, which is the failure of being driven by the base. If the tally always starts synchronized with the list position, the position can predict the cup; the distinction matters below.

The two examples do exactly what I said they do: the toggle is driven, isn't permanent and isn't a latch; the timestamp is permanent and isn't driven. That each also keeps the other rules (the ledger is the step count, the first switch-on costs one, a run reaches it) is true by inspection.

Permanent but undriven is a timestamp, driven but impermanent is a toggle, and a record is the thing that's both. Undriven, for the timestamp, means that along the runs that actually happen the base state doesn't fix when the record turns;

Lemma 10.7 below says why that's the condition, and what (R1) as written adds to it. That's the whole discrimination, and I think it's the most useful single sentence in this book for looking at systems other than the one I build, which is why it's here.

I'll now state the pair lemma and the two counterexamples exactly. The base for both counterexamples is the counter: states are whole numbers, the step adds one, every state is a starting state, and nothing halts. Two more bits of shorthand: $\neg r$ is $r$ flipped, and a statement in square brackets, like $[k = 5]$, reads on when it's true and off when it isn't.

**Lemma 10.4** (Two records are two latches)**.** Let $c_1, c_2$ be two records on $M$, with cover $\pi$ onto a base $B$, such that: there is one function $F$ with $(c_1(\text{next}_M(m)), c_2(\text{next}_M(m))) = F(\pi(m), c_1(m), c_2(m))$; the ledger never goes down; a step that turns either record on costs at least one; and each record is permanent. Then there are $h_1, h_2$ with $c_1(\text{next}_M(m)) = c_1(m) \vee h_1(\pi(m), c_2(m))$ and $c_2(\text{next}_M(m)) = c_2(m) \vee h_2(\pi(m), c_1(m))$ for every $m$. □

*Proof.* Hold the second record fixed and ask what the joint rule does to the first when the first is off. Call that answer $h_1(b, r_2)$; it is the first component of $F(b, \text{off}, r_2)$. At any actual state whose first record is off, that is its next value. At any state whose first record is on, permanence keeps it on. Thus its rule is $r_1 \vee h_1(b, r_2)$.

Now hold the first record fixed and ask the corresponding question with the second off. The answer is $h_2(b, r_1)$, the second component of $F(b, r_1, \text{off})$. The same off/on argument gives $r_2 \vee h_2(b, r_1)$. The records may affect each other's raising condition, but neither can undo an on once it is there. ■

**Lemma 10.5** (The toggle)**.** Let the toggle have states $(n, r)$, step $(n, r) \mapsto (n + 1, \neg r)$, every state starting, record $r$, ledger $n$, cover $(n, r) \mapsto n$ onto the counter. It is driven by the computation, with $f(b, r) = \neg r$; it is not permanent; and it factors as the latch of no event. □

*Proof.* At $(0, \text{on})$ the next record is off, which breaks permanence; and any latch gives on $\vee\, h(0) =$ on there. ■

**Lemma 10.6** (The timestamp)**.** Let the timestamp have states $(b, k, r)$, step $(b, k, r) \mapsto (b + 1, k + 1, r \vee [k = 5])$, every state starting, record $r$, ledger $b$, cover $(b, k, r) \mapsto b$ onto the counter. It is permanent and not driven by the computation. □

*Proof.* Permanence: on $\vee\, x$ is on. Not driven: the states $(0, 5, \text{off})$ and $(0, 0, \text{off})$ have the same base state and the same record, and their next records are on and off, so no $f$ of the base state and the record can give both. ■

"Driven" is a condition on every state. The timestamp makes every state a starting state, so the two states the proof compares are both ones a run can begin at, even though a timestamp started at $(0, 0, \text{off})$ would never reach $(0, 5, \text{off})$.

That last sentence does more work than it looks, so I'll state it as a lemma. Start the same timestamp in sync, at $(b, b, \text{off})$ only. On every state its runs reach, the hidden count equals the base, and the timestamp is exactly the latch of $[b = 5]$. (R1) still rejects it, because (R1) asks about every state, and $(0, 5, \text{off})$ is a state, though no run from a synchronised start reaches it. So (R1) as written separates the timestamp from a latch partly by how the cover is drawn and through states nobody visits. The separation I mean is the one on reachable states: the timestamp with every state a start fails (R1) even when only states some run reaches are counted, since two starting states sit over the same base state with the same record and disagree about the next one. And the latch theorem holds in that reachable form too.

□ **Lemma 10.7** (The timestamp, on the states runs reach)**.** Say $M$'s record is *driven on reachable states* when (R1) holds for every state reached from a starting state in some number of steps. (1) If $M$'s record is driven on reachable states and permanent, then on every reachable state $M$ steps as the latch of some event. (2) The timestamp of Lemma 10.6 isn't driven on reachable states. (3) The timestamp started in sync, with starting states $(b, b, \text{off})$ and the same step, record, ledger and cover, is permanent, isn't driven by the computation, is driven on reachable states, and on every reachable state steps as the latch of $[b = 5]$.

*Proof.* (1) is the proof of Observation 10.1, run on reachable states only. (2) The states $(0, 5, \text{off})$ and $(0, 0, \text{off})$ are starting states, reached in no steps, and the proof of Lemma 10.6 applies to them. (3) From $(b, b, \text{off})$ every step adds one to both counts, so a reachable state has $k = b$ and its next record is $r \vee [b = 5]$. Over every state, (R1) fails at the same pair as in Lemma 10.6. ■

So "permanent but undriven is a timestamp" is a sentence about the timestamp with every state a start, over the cover that hides its count, and it holds there in the strong form: undriven on the very states its runs pass through. A timestamp started in step with its base is a latch on the runs it has, and (R1) throws it out only because (R1) also looks at states those runs never reach.

### 10.4 Is there only one such machine? No: one structure, two unforced choices.

I have to take apart the sentence "there's only one way to carry a record," because three different things get squeezed into it.

The first claim asks: given the toll and a fixed price list, is there one way to keep the books, starting at zero and adding each move's price? Yes, on every state reachable from the start; that's Section 6's initiality, and it assumes the price list.

The second claim asks: is every base-driven record over a base *the same machine*, charges and all? No, and the counterexample is embarrassing in its simplicity. Take any base-driven record, give it a new ledger that adds each step's old cost and one unit more, whatever the step is, and then walk the five rules. The record still answers only to the base and its own value (R1); the cost never touches it. The ledger still never goes down (R2); it climbs faster, if anything. The step that turns the record on still costs at least one (R3), because the toll is a *floor* and a surcharge only raises the price. Once on, the record stays on (R4), and the same run still reaches the write (R5). So the surcharged machine is base-driven, and it isn't the machine you started with: wherever the original sat still at no charge, the surcharged one pays one unit a step for sitting still. Do it with the cup and the die, and charge one unit for every roll, six or not. The cup turns over on the same roll as before, so you've changed the cost and nothing about the record. A machine that charges for CPU time is base-driven and isn't the one that doesn't. The new ledger has to sit beside the old state and not be read off it. A step's cost is the ledger after less the ledger before, so a ledger read off the state charges nothing for a step that leaves the state where it was, such as a halted machine stepping to itself, or for any step on a loop (Section 6). The surcharge is paid in fresh ledger values, the way every counter that climbs is paid for.

The same base computation and latch are visible after forgetting the new ledger. The full transition graph can be different: a free self-loop becomes an endless line of increasing ledger values. So the comparison preserves base, reading and the stipulated stopping convention, not every full-state distinction. If the ledger is also observed, these are different contracts. The construction allows a nonnegative surcharge depending on the current state in this broader accounting interface. Such a charge is not in general a price on instruction names alone. The price schedule remains an unforced choice after the base and event have been fixed.

And then there's the second unforced choice, and this one still surprises me: *which event* is the record? Nothing in the five rules picks it, beyond (R5)'s demand that it hold somewhere a run reaches; on a base whose runs pass through only one state that leaves one record, and the choice opens as soon as a run passes two states an event can tell apart. The converse above already says so, if you read it twice: for any event the base can reach, the latch of that event, priced at one unit per

write, is a base-driven record. Do it with the die. One cup turns over on the first six, and another turns over on the first one. Both are cups with no reset wire; both are base-driven. Roll a one first, and until a six turns up the two cups disagree about whether anything has been recorded. So *which permanent reading counts as the record* is a second thing nobody forces on you. The price list is one unforced choice and the event is the other, and that second choice is the question Section 18 is about: what, if anything, makes certification the event worth recording. "The ledger passed one" is an event only if the chosen base includes enough information to decide it. An extension-only ledger is hidden from a base event, just as the hidden clock was. The unforced choice is among functions of the chosen base.

□ **Corollary 10.8** (The event is an unforced choice)**.** Let $B$ be any base and $h_1, h_2$ two events, each meeting the supposition of Theorem 10.2. Then $\text{Latch}_{h_1}$ and $\text{Latch}_{h_2}$ are both base-driven records over $B$. If some run from a start of $B$ reaches a base state $b$ with $h_1(b)$ on and $h_2(b)$ off, and $h_2$ holds nowhere earlier on that run, then after the step from $b$ the same run has the first record on and the second off.

*Proof.* That both are base-driven is Theorem 10.2 applied twice. Along that run the second record is still off when it reaches $b$, since $h_2$ hasn't held, and the latch's own rule keeps it off, off $\vee\, h_2(b)$; the first is on after that step, whatever it was before, ■ since $h_1(b)$ is on.

Now let the cup wobble. Say whatever turns it over sometimes does and sometimes doesn't, with odds you could write on a card. Does fixing the price list fix the odds? No, and the counterexample is two cups with the same price tag and differently loaded coins. I'll set it up carefully. Let a record be a single bit that moves at random: from each value, a list of weighted next values, every weight positive, never empty, never leaving on once on, and with the write from off to on charged at least one. Call such a record a *permanent probabilistic record*. Call two such records *the same schedule* when the same moves are possible and every move is charged the same, and *the same in probability* when the weighted lists give every move the same odds.

□ **Theorem 10.9** (The schedule doesn't fix the probabilities)**.** (1) There are two permanent probabilistic records with the same schedule that are not the same in probability. (2) There is a permanent probabilistic record that no deterministic latch on its current value describes.

*Proof.* (1) From off, one record moves to off or on with weights 1 and 1, the other with weights 1 and 2; from on, both stay on with weight 1; both charge 1 for the write. Same possible moves, same charges, different weights. (2) The first record moves from off both to off and to on, while a latch $r \vee h(r)$ gives one value. ∎

These are finite-weight models on one bit with no base underneath; they say nothing about any particular machine.

The third claim asks: is there one state space, up to renaming, that every base-driven record-carrier has to have? No, because the extension may carry any hidden field it likes, so two carriers over one base can have state spaces of different sizes, and nothing needs them to match.

## 10.5 Bases from outside, and a relation I have to be honest about

The theorem is stated for any base, so let me put it on three models of computation that other people built: a small Turing machine (any program); a register machine of the kind the computer scientists Stephen Cook (of the certificates) and Robert Reckhow described in 1973, with unit-cost indirect addressing (any program); and a lambda calculus (call-by-value, meaning an argument is worked out before it's plugged in, and made deterministic by letting a term with no step left to take step to itself). On each, the latch form holds, by the one generic theorem applied to that base (the bases are defined below). So a register machine with a base-driven record is a register machine plus a cup, and so is a lambda calculus. (There's one more instance, the machine I build later, where a particular flag is shown to be a base-driven record and the same theorem makes it a latch, in Section 15.)

I'll give the three bases explicitly. A base needs a state, a next-state function, a set of starting states and a halting predicate; the latch theorem asks for nothing else about its instruction vocabulary.

**Definition 10.3** (Three concrete bases)**.** The **tape base** has a finite Boolean list $t$, a head index $h$, a program counter $q$ and a natural-number cost field. Reading beyond the list returns false. Writing at $h$ extends the list with false cells if needed, then changes that cell. Moving right increments $h$ and extends the list through the new head position. Moving left decrements a positive $h$; at $h = 0$, it prepends a false cell and keeps the head at zero. A fixed finite program contains these instructions:

| Instruction | Next state |
|---|---|
| Write a bit | Perform the write and advance $q$ by one. |
| Move left or right | Perform the move and advance $q$ by one. |

| Instruction | Next state |
|---|---|
| Branch to $j$ | Set $q := j$ if the current cell is true; otherwise advance $q$ by one. |
| Halt | Leave the entire state unchanged. |
| ClaimTapeIsZero with a label $\ell$ | Advance $q$ by one and do nothing else. |

An out-of-range program fetch is Halt. Every state is allowed as a starting state, and halting means the fetched instruction is Halt. The cost field is preserved by every instruction in this base. In particular, its ClaimTapeIsZero instruction neither erases the tape nor charges the cost field; neither the name nor the label supplies any semantics.

The **random-access base** has a register function $X : \mathbb{N} \to \mathbb{N}$ and a program counter $q$. For a fixed finite program its instructions are:

| Instruction | Next state |
|---|---|
| Constant $(i, c)$ | Set $X(i) := c$. |
| Add $(i, j, k)$ | Set $X(i) := X(j) + X(k)$. |
| Subtract $(i, j, k)$ | Set $X(i) := \max(0, X(j) - X(k))$. |
| Indirect load $(i, j)$ | Set $X(i) := X(X(j))$. |
| Indirect store $(i, j)$ | Set $X(X(i)) := X(j)$. |
| Positive branch $(j, a)$ | Set $q := a$ if $X(j) > 0$; otherwise advance $q$ by one. |
| Jump to $a$ | Set $q := a$. |
| Halt | Leave the entire state unchanged. |

Each register-writing instruction advances $q$ by one; its operands are evaluated in the old state, and all other registers are preserved. A fetch outside the program also leaves the state unchanged. Starting states have $q = 0$ and arbitrary register contents. A state is halted when it fetches Halt or is outside the program. Unit cost here counts instructions. It has nothing to do with the record's ledger.

For the indirect instructions, draw boxes 1, 2 and 3 and put 2, 7 and 0 in them. Indirect load $(3, 1)$ first reads the 2 in box 1, uses it as an address, and copies box 2's 7 into box 3. Now put 9 in box 3 and execute indirect store $(1, 3)$: the 2 in box 1 tells it where to put the 9, so box 2 changes and box 1 stays 2. An address is being read from a box; it is not necessarily the box's own number.

The **lambda base** uses the terms and weak call-by-value rule defined in Definition 13.3. To compute its next state, first try to reduce the function part of an application. If that part cannot step and is an abstraction, either substitute an

argument that is already an abstraction or take a step in the argument. If neither applies, the term cannot step. Variables and abstractions cannot step by themselves, and there is no reduction under an abstraction. The base leaves a term unchanged when it cannot step and calls precisely those terms halted. Every term is allowed as a starting state, including open terms.

So all three are bases in the sense of Definition 10.1, and the latch theorem applies to each. It says nothing else about them, nothing about what they can compute or how fast.

There's also a relation defined between bases, *weak base equivalence*: two bases are equivalent when some correspondence covers their start states both ways, related states agree on an observation and on halting, and one step on one side is matched by some number of steps, possibly zero, on the other. That "some number of steps" is the *stuttering* idea Leslie Lamport (the same Lamport as Section 6's message counters) put forward in 1983: a fine-grained machine and a coarse one shouldn't count as different because one takes three ticks to do what the other does in one. The relation carries no cost; a machine that multiplies huge numbers in one step and a Turing machine differ in price. The relation never looks at price, so on what it does look at they can match. It's proved to be an equivalence relation, and the latch form is preserved across it.

Draw a coarse machine $0 \to 1$, with 1 halted, and a finer one $a \to b \to c$, with $c$ halted. Let $a$ and $b$ both show 0 and $c$ show 1; halted states stay where they are. Relate 0 to both $a$ and $b$, and 1 to $c$. The fine move $a \to b$ changes nothing visible, so the coarse machine matches it by taking no step. Its move $b \to c$ matches $0 \to 1$. Starting from $a$, the coarse move is matched by two fine steps. "Possibly zero" allows this; the relation is about matching observations and halting, not matching the number of ticks or their cost.

There is a limit to that statement. The "preserved across it" proof throws away its equivalence hypothesis; the latch holds on every base regardless, so the preservation is true for a boring reason. And nowhere in this book are *two* of the bases actually related by this relation. The relation is defined, it's an equivalence, and it is not used to connect the Turing machine to the register machine to the lambda calculus, or any of them to the machine of Section 15. It would be easy to read that list as if it did. The honest sentence is the one that's proved: each base, separately, carries a latch.

**Definition 10.4** (Weak base equivalence)**.** Bases $B_1$ and $B_2$, with observations $o_1$ and $o_2$ into one common set, are *weakly equivalent* when some relation $R$ relates every starting state of each to some starting state of the other, and any related $x$,

$y$ have $o_1(x) = o_2(y)$, the same halting, some $n \geq 0$ with $R(\text{next}_{B_1}(x), \text{next}^n_{B_2}(y))$, and some $n \geq 0$ with $R(\text{next}^n_{B_1}(x), \text{next}_{B_2}(y))$.

□ **Theorem 10.10** (Outside bases, and the relation)**.** (1) Weak base equivalence is reflexive, symmetric and transitive. (2) If two bases are weakly equivalent, Observation 10.1 holds over one exactly when it holds over the other. (3) Observation 10.1 holds over each of the three bases in Definition 10.3, for every fixed program where the base has one.

*Proof.* (1) A base matches itself by pairing each state with itself and matching one step with one step. Reverse any matching between two bases and it gives a matching in the other direction.

For three bases, pair a state of the first with a state of the third whenever some middle state matches both. A start on either outside has a middle start beside it, and that middle start has a start on the far side. Observations and halting agree across the first pair and then across the second, so they agree across the outside pair too.

Follow one step on the first base. Its match supplies a finite sequence of middle steps. Match the first of those steps on the third base, then the second from where that match ended, and continue. Joining the finite sequences gives a match for the original step. If the middle sequence is empty, take no step on the third base. Do the same in the other direction for a step on the third base. This checks both stepping requirements and proves transitivity.

(2) The latch argument uses only a base step and the stated record rules. It therefore holds over both bases, whether or not they are matched. (3) Each of the three displayed constructions supplies that step, its starts and its halting reading. ■ The same latch argument applies separately to each.

### 10.6 Records with more than two values

A cup has two positions, and some records have more: a counter of how many checks passed, or a level that only rises. The same reasoning goes through one threshold at a time. For each value $a$, "the record has reached $a$" is a yes/no reading, and each of those readings is a latch. What doesn't go through is squeezing the whole record into one cup. "Rises" needs one word of explanation: the values come with an order $\leq$, "is at or below," which needn't compare every pair (that's what makes it a *partial* order), and $[a \leq x]$ is the yes/no reading "$x$ has reached $a$.

For example, keep separate records that an experiment was repeated and that its calculation was checked. "Repeated only" and "calculation checked only" are

different places; neither includes the other. “Both” is above each. In this order, moving up means retaining what was already done and possibly adding another act. The order also has three rules: a place includes itself, inclusion carries through an intermediate place, and two places including each other are the same place. Those are the rules the proof below uses.”

**Definition 10.5** (A growing record)**.** Let the record take values in a set $A$ with a partial order $\leq$ decided by a Boolean test. A record $\rho$ on $M$, with cover $\pi$ onto a base $B$, is a *base-driven growing record* when: there is one function $f$ with $\rho(\text{next}_M(m)) = f(\pi(m), \rho(m))$; it grows, $\rho(m) \leq \rho(\text{next}_M(m))$; the ledger never goes down and every step that changes $\rho$ costs at least one; and some run from a starting state reaches a step that changes it.

**Theorem 10.11** (Growing records are families of latches)**.** (1) For every base-driven growing record there is an $h$ with □

$$[a \leq \rho(\text{next}_M(m))] = [a \leq \rho(m)] \vee h(a, \pi(m), \rho(m))$$

for every value $a$ and state $m$: each threshold is a latch, whose event may read the current value. (2) The thresholds determine the value: if $[a \leq x] = [a \leq y]$ for every $a$, then $x = y$. (3) For a growing record, charging every change is the same as charging every threshold that switches on. (4) One latch is not enough: a three-valued record that climbs $0 \to 1 \to 2$ over a one-state base, each climb charged one, is base-driven, and no single Boolean latch with a decoding from the base state and the latch reproduces it. (5) A chain of distinct $k$-bit words, each word’s set bits contained in the next one’s, has at most $k + 1$ members.

*Proof.* (1) Fix a threshold $a$. If the old record has already reached it, growth keeps it reached. If it has not, the cup rises exactly when the next record $f(b, x)$ reaches it. Define the event to say both things: “not reached at $x$, reached at $f(b, x)$,” or $h(a, b, x) = [a \not\leq x] \wedge [a \leq f(b, x)]$. Keeping an existing yes and otherwise using that event gives the stated latch equation.

(2) Suppose $x$ and $y$ give the same answer at every threshold. Ask the threshold named $x$ itself. The record $x$ has reached it, so $y$ must have reached it too: $x \leq y$. Ask the threshold named $y$ and get $y \leq x$. The order’s third rule says those two records are the same.

(3) If the value stays put, every threshold gives the same answer before and after. If it grows from $x$ to a different $y$, look at the threshold $y$. It is reached afterwards. It was not reached before: $y \leq x$, together with growth $x \leq y$, would force $x = y$.

Thus some threshold switches on exactly when the record changes. Charging those steps is the same requirement in either wording.

(4) Keep the base at its sole state and let the record move from zero to one, then two, then stay at two. Add one to the tally at each climb. The record rule consults only its present value, keeps it growing and pays for both changes, so it meets the requirements. A single flag over that unchanging base offers a decoder only two inputs, flag down and flag up. It cannot reproduce the three different record values just visited.

(5) Lay the distinct bit words out in their increasing order. At each move to the next word, at least one bit that was zero must become one; no one may become zero. Across the whole chain there are only $k$ bits available for those changes. There are therefore at most $k$ moves between distinct words, or $k + 1$ words including the first. ■

I have two cautions about that. In (1) each threshold's event may read the record's current value, so the events aren't all functions of the base alone. And (3) charges a step when it switches some threshold on: a step that switches on three thresholds at once owes at least one, and nothing here makes the charges add up per threshold.

The order need not be finite; (5) is the finite-memory bound.

## 10.7 What leaving the base open buys you

The reason this section exists is to answer the suspicion I started with. The record isn't special pleading for one machine. Over any base that steps deterministically, under rules you can read, a record that answers to the computation and never comes down is a latch on an event, and the proof doesn't care what the base is. What a particular machine adds on top of the latch is a *choice of price list* and a *choice of event*, and both are choices. That is why the permanent-flip theorem in Section 3 never needed the word certify: it forces a merge in whatever stays. The surcharged cup shows the first: the cost can change and the record not notice. The two cups on one die show the second: the rules are happy with either event. Any machine that meets the model will have to make both choices, and whether its choice of event is the right one is a question I'll get to (Section 18), and won't claim to have settled. Before that, a record can have more than two positions. Earlier in this section a record that rises got cut into one cup per threshold; the next section asks how much of the rest of the book survives once a record has more places to be than up and down.

## 10.8 What it takes to be complete

The latch left two choices open, the price list and the event. A machine I can hold up against everything else has to make both, and a name for "this machine computes, and its record keeps something" has to ask for more than the toll does. The toll alone is a low bar. A machine that charges one for every step it takes is a clock bolted to a computer, and it obeys the toll for any record at all: every raise costs at least one because every move does. And a machine whose record never goes up obeys it too, because a toll on a raise that never happens is never broken. The clock-priced machine may keep useful records; its prices simply do not distinguish computing from recording. The constant reading can keep no new yes/no distinction. The toll permits both.

I'll give the names now, all of them, once, with the reason for each, and they work the way Turing's do. *The Thiele Machine*, with capitals, is the model: a computer with a reading that can go from no to yes, natural instruction prices and the toll. The reading may also be constant or revocable. An internal ledger is extra structure. *A Thiele machine*, lower case, is any particular system that fits the model, the way one rule table is a Turing machine. That bar is low on purpose, and a clock clears it. *Thiele-complete*, defined next, is the bar worth meeting, the way Turing-complete is: a machine that computes anything, with its record kept apart from its computing and able to tell starts apart. Part III adds one stricter bar, *certified Thiele-complete*, for a record that, from a clean start, goes up only on a claim that was checked, committed and certified. And Appendix C names a *universal Thiele machine*: one machine that runs the others, the way Turing's universal machine runs every Turing machine. Those are all of them: the model, the systems that obey it, the ones that compute and keep, the ones whose keeping is earned, and the one that runs them.

The bar comes in four clauses, and nothing in it mentions what the record is about.

**Definition 10.6** (Thiele-complete)**.** A machine with the five parts is *Thiele-complete* when it can be read through an *interface* that hands over three things: a universal base, a split of its moves into *computing moves* and *marks*, and a *ledger*, a whole number for each state. A *universal base* is a way of reading each state as a configuration of a machine that runs every program (I use Minsky's two counters, Section 15.2), a class of *live* states, a computing move for each of that machine's instructions that does to the configuration exactly what the instruction does from every live state and keeps it live, and, for each start, a live loaded state showing that start. The interface has to meet four clauses. (A) *Computing can't reach the record, and the record stays.* No computing move changes the record, and a record that's up stays up under every move. (B) *Starts are blank.* Every loaded state has the record down.

(C) *Computing costs nothing, marks cost one.* Computing moves cost 0, every other move costs exactly 1, and each move adds its cost to the ledger. (D) *The record tells starts apart.* Some run, the same list of moves, ends with the record up from one loaded state and down from another.

Read the clauses as four plain sentences, with the universal base underneath them: the machine computes anything a Turing machine can. (A) The computing never touches what's kept, and what's kept stays. (B) Nothing is kept before the run starts. (C) The ledger counts the marks, and only the marks. (D) What's kept depends on where you started: there's a run whose record says something about its start.

(D) is the clause where distinction comes in, and it's a low bar: a mark that reads one bit of the input meets it. What it excludes is a record that ignores the start. A record that goes up on every run from every start keeps a before and an after, and nothing about where the run began. (D) doesn't ask that every raise tell starts apart; it asks that the machine can. A machine may have a mark that raises the record from any state at all; it pays one for it, like every raise, and (D) is still met by some other run. Which runs a machine wants its record to tell apart is its own choice, and Part III makes one.

The name echoes Turing-complete, but this is a specification a machine meets, not a class closed under simulation: a machine is held up against it through an interface, and changing the unit of price drops a machine out.

□ **Theorem 10.12** (What every Thiele-complete machine has)**.** Let a machine be Thiele-complete through an interface. (1) It obeys the toll, and the move that raises its record is a mark, never a computing move, and costs exactly one. (2) Along any run, the ledger rises by exactly the number of marks in it. (3) From any state with the record down, every run that ends with the record up raises the ledger by at least 1. (4) Some run from some loaded state raises the record. (5) Every move costs 0 or 1, and some move costs 0. (6) Its base runs every two-counter program step for step from every live state, and halting from a loaded state matches halting from the configuration it shows.

*Proof.* (1) If a computing move raised the record it would change it, against (A); so the raising move is a mark, and a mark costs one by (C). (2) is (C) added up along the run. (3) Somewhere along the run the record goes from down to up, and that move is a mark, by (1), so it adds one. (4) is (D), and by (B) the record started down. (5) Every move is a computing move or a mark, and the universal base supplies computing moves. (6) is the universal base, one instruction at a time, from a loaded state that is live and shows its start. ■

The floor is one, and one is reached: the lift below gives a machine whose record goes up after a single mark.

**Observation 10.13** (A clock is a Thiele machine and isn't Thiele-complete)**.** *Take the two-counter machine, let every move cost 1, and take as the record any yes/no question about its configuration; or a flag that switches on, and stays on, the first time a chosen test of the configuration says yes. It obeys the toll and has a two-counter machine inside it, and it isn't Thiele-complete. No machine whose every move costs at least 1 is.*

*Proof.* A Thiele-complete machine has a move priced 0, by Theorem 10.12(5), and this one has none. ∎

A two-counter base priced 0 with a record that's down at every state obeys the toll and meets (A), (B) and (C), and isn't Thiele-complete either: (D) asks for a run that raises the record.

**Theorem 10.14** (Each of the four clauses is needed)**.** For each clause there's a machine that meets the other three through some interface and is Thiele-complete through none. Two of them are built on the two counters with two marks, one that raises the record when $A$ is 0 and one that always raises it, and a ledger beside them that each move adds its cost to. (A) fails: that machine, and one more move, priced 0, that puts the record back down. (B) fails: the two counters, every move priced 0, with a record that no move ever changes, up at the starts where $A$ is 0 and down elsewhere. (C) fails: the two counters with two marks, every move priced 1. (D) fails: the two counters, every move priced 0, with a record that's always down. □

*Proof.* Each machine meets the three by construction. The first fails (A) through every interface, because a move lowers the record, so the record doesn't stay. The second: every one of its moves costs 0, so (C) makes every move a computing move under any interface; then by (A) no run changes the record, and with every start blank by (B) no run raises it, so (D) fails. The third has no move priced 0, so it fails by Observation 10.13. The fourth never raises its record. ∎

**Theorem 10.15** (The record can't be read off the window)**.** Let a machine be Thiele-complete through an interface. (1) Every configuration is shown, at no cost, with the record down, by one computing move from some loaded state. (2) Some loaded state has two runs that end on the same window, one with the record up and one with it down, and their ledgers differ by at least 1. (3) So no rule that reads only the window says, at the end of every run from every loaded state, whether the record is up, and none says what the ledger reads. □

*Proof.* (1) From the loaded state $(1, x+1, y)$, the computing move "DEC $A$ to line $p$" shows $(p, x, y)$, costs nothing by (C), and leaves the record down by (A) and (B). (2) Take the run (D) gives, from the loaded state where it raises the record. Its end shows some configuration. From the same loaded state, computing moves alone reach any configuration at all: count each counter down to zero, count it up to the value wanted, and jump to the line wanted. That run ends on the same window, and it's all computing, so by (A) and (B) its record is still down and by (C) its ledger hasn't moved, while the first run's ledger went up by at least one, at its raise. (3) A rule that reads the window gives one answer for one window, and (2) has one window with two answers. ■

Part III proves the same blindness for its certified machines (Observation 15.32), with a gap of three, because a certified raise costs three. Here the gap is one, and one is reached.

□ **Theorem 10.16** (Every universal base becomes Thiele-complete)**.** Take any machine with a universal base and any events on its states, and add a record, a ledger and, for each event, a mark that costs one and raises the record exactly when the event holds; computing moves cost nothing and leave the record alone, and every move adds its cost to the ledger. If some run of computing moves makes some event hold from one loaded state and not from another, the result is Thiele-complete, its computing is the original machine's step for step, and its record goes up after a single mark from any state where the event already holds.

*Proof.* (A): computing moves are the original's and don't touch the record, and a mark only adds to it. (B): the record starts down. (C): by construction. (D): run the moves that make the event hold from one loaded state and not the other, then the event's mark. Step for step: a computing move acts on the original part and nothing else. And a mark from a state where its event holds puts the record up. ■

One mark on "$A$ is at least 1" always works: the starts with $A$ at 0 and at 1 already differ on it, so that mark alone, run from each, tells them apart. So Thiele-complete isn't a property only a special machine has. Any model with a universal base becomes Thiele-complete by adding marks, and needs no claims and no checker to do it; a model that simulates the two counters some other way does so once its moves are grouped into ones that carry out a counter instruction each (Theorem D.6). Two kinds of base don't lift, whatever the events: one whose moves don't depend on the state it's in, and one whose moves lead from each state to only finitely many next states. A universal base needs, from one state, a move at no cost to every line of a program, and that's infinitely many places to go.

Some Thiele-complete machine runs every Thiele machine a program can describe. Appendix C defines a *universal Thiele machine* and builds hosts that run any machine a program can describe, in order, carrying its reading as a latch and matching its ledger up to a surcharge of at most three, with every computing step done by counter moves on the host and every rise of the host's record placed by its program. None of that needs certification: the presented host's own program, read on a host with no claims, no checker and no table of facts, and one paid mark that raises the record when a register is at least 1, meets the definition too.

The rest of the book holds machines to this bar. Part III gives one example that clears it, and then asks for the stricter certified bar: that from a clean start its record go up only on a claim that was checked, by a checker proved to mean it, and then committed, and that some claim's check can come out either way. Asking that is a choice of what the record answers to, the second of the latch's two open choices, made one way, with clean starts and three priced acts added (Theorem 15.23).

CHAPTER 11

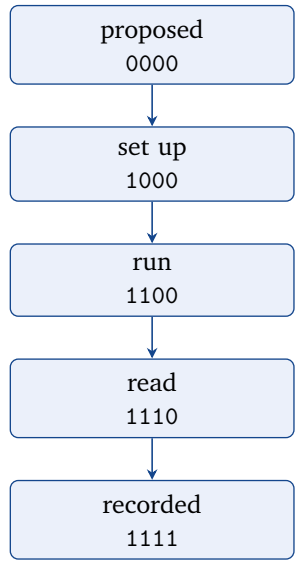
# The whole direction

I have a hunch I can't fully back up, and Part III tells where it came from (Section 14): *established* doesn't feel like a switch to me. It feels like a direction with lines drawn across it, and the switch is just the first line. The one piece of that I can settle is that a single switch isn't enough once there are three or more stages. I settle it in this section, and find out how much of the rest survives being pushed on.

A row of cells that only ever go from blank to written is a direction: you can tell how far along it a state is, and nothing walks it back. The cup of Section 10 has two positions. Definition 2.1 already lets the record sit in any order; the proofs so far were done for the cup. Everything the book has proved about the toll so far (the universal floor, the merges, the squeeze, the windows) was proved for a cup. The question for this section is blunt: if the record has more places to be than up and down, how many of those results come along, and which ones break?

## 11.1 A row of cups

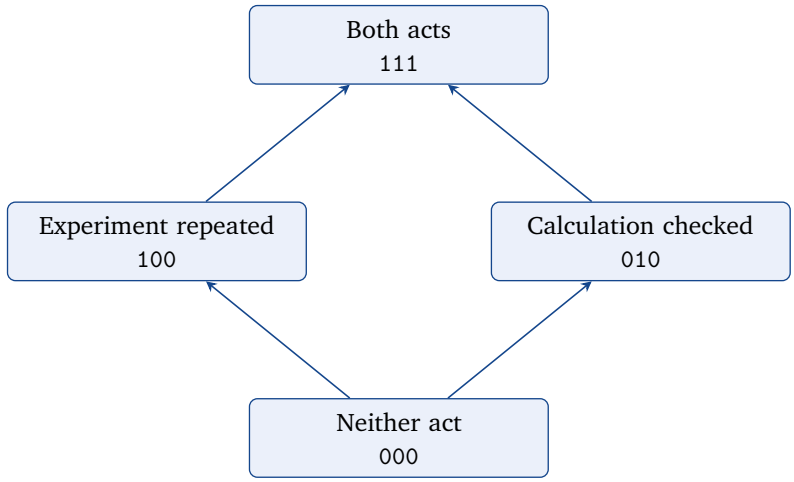
Get four cups and a pencil. Write five stages of an experiment on a strip of paper, left to right: *proposed*, *set up*, *run*, *read*, *recorded*. Agree to do them in that order. Put an upside-down cup between each pair of neighbours. The first means "the setup happened," the second "the run happened," the third "the result was read," the fourth "the result was recorded." As you complete a stage, turn over its cup. Cups only ever get turned right side up; none has a reset wire. These are records of completed acts. A fourth cup up does not make a false conclusion true, and a different experiment may need a different order.




Read down the drawing. Zero means cup down, one means cup up. Four cups give five stages along this run because each new stage turns up one unused cup.

Two things to notice with the cups in front of you. Each cup by itself is exactly the cup of Section 10: it reads yes or no, and once yes it stays yes. And four cups were enough for five stages. Try it with three, and you'll find you can't: three cups that only ever turn up, in order, give you at most four different arrangements along the way. I'll prove both of those in a minute, and the proof is about as short as the experiment.

Now a harder strip. Some checks don't line up. One person repeats the experiment; another independently works through the calculation. Neither act includes the other. Draw a diamond: *neither* at the bottom, *experiment repeated* on the left, *calculation checked* on the right, *both* at the top. There are two ways up, and the middle points don't compare, which is what a *partial* order allows. Put a cup on each point except the bottom, one for the repeat, one for the calculation, one for both. Each cup is still a latch. The three cups tell you which acts have happened. They do not decide whether the experimental model or the calculation's premises were appropriate.




Read the cups as repeat, calculation, both. Neither middle point includes the other. The third cup repeats information already supplied by the first two; it is useful for showing thresholds, not needed for storage.

Whatever the shape of the strip, a straight line of stages or a diamond or something with a hundred points, I'll call the set of places the record can be the *direction*, and each question "has the record reached at least this place?" a *line* across it. One line is one cup.

## 11.2 The order, said exactly

The strip and the diamond each come with an order you can see by looking: whether one place is at or below another. Said exactly, the order is a test. For any two places $x$ and $y$ it answers "is $x$ at or below $y$?" with yes or no, and I write a yes as $x \leq y$. The test has to obey two rules: every place is at or below itself, and if $x \leq y$ and $y \leq z$ then $x \leq z$. An order that obeys just those two is called a *preorder*. If it also obeys a third rule, that $x \leq y$ and $y \leq x$ together force $x = y$, it's a *partial order*, and that third rule is called antisymmetry. The two-point order, no below yes, is the cup. The whole numbers in their usual order are a straight strip that never ends.

The cups say the same thing in their own terms. For a place $a$, the cup for $a$ asks "has the record reached at least $a$?", so with the record at $x$ it's up exactly when $a \leq x$, and I write its answer as $[a \leq x]$. In the row of cups I said the cups together tell you which place the record is at. On the strip and the diamond you can check that with your eyes, and part (2) of Theorem 10.11 already proved it whenever the order is partial, though I hadn't asked what in that proof was doing the work. My first guess was that it held for every order.

Why bother with the bare preorder at all, when every strip on the table is a partial order? I bothered because I wanted to see what the third rule was doing,

and the way to see what a rule does is to take it away. Take a preorder with two places, call them *left* and *right*, where each is at or below the other. There are two cups, one asking "at least left?" and one asking "at least right?", and both are up at left and both are up at right. Look at the table and you can't say where the record is. Now build a record that swaps between left and right on every single step and charges nothing. Every rule I'm about to state is satisfied. The record never drops below anywhere it was, because each place is at or below the other. No cup ever moves, so the toll in the next subsection has nothing to charge. The record moved on every step and nothing on the table noticed.

**Theorem 11.1** (The cups fix the place, and only with antisymmetry)**.** (1) In a partial order, if two places $x$ and $y$ leave the same cups up, $[a \le x] = [a \le y]$ for every $a$, then $x = y$. (2) In a preorder that isn't a partial order this can fail: there's a preorder with two different places that leave the same cups up, and a record over it, driven by the computation and growing in that preorder, that changes on every step while no cup moves. □

*Proof.* (1) Look at the cup for $x$. It's up when the record is at $x$, since $x$ is at or below itself, so it's up at $y$ too, and that says $x \le y$. The cup for $y$ gives $y \le x$ the same way. Antisymmetry gives $x = y$. (2) Left and right above, with the record swapping every step. ■

So in a partial order the cups are the place, and in the preorder I built they aren't. That second half holds in every preorder that isn't partial, by one line: if $x \le y$ and $y \le x$ with $x \neq y$, then a cup that's up at either one is up at the other, by transitivity. So the cups fix the place if and only if the order is partial. My first guess was right on every strip I could draw, and it was right because of a rule I hadn't noticed I was leaning on. On a bare preorder, putting two places on the same rung tells the toll not to care about moves between them, and it doesn't. I don't think that's a defect; it's the order telling you what you agreed to count.

## 11.3 The toll, over the whole direction

One cup had one rule: the step that turns it right side up pays. Now there's a row of cups, or a diamond of them, and a single step can turn over several at once. The rule I'd guess first turns out to be the right one: *a step that turns over any cup pays at least one.* A step that turns over three cups at once still only has to pay one. The rule asks whether a step turned something over. It doesn't ask how many.

That rule buys the same thing one cup bought. Take any run, and suppose some cup is up at the end that was down at the start. Somewhere along the run a step

turned that cup over, and that step paid. So the run paid at least one. Count the same way and you get more: the run paid at least once for every step that turned anything over.

Now look at what that argument didn't use. It didn't care whether the cups sat in a row or a diamond or something with a hundred points. It didn't care how many there were. It didn't even use the fact that cups never get turned back down. If something could knock a cup back over, the argument still works, because all it needs is the one step where that cup went up.

To say it exactly I want to translate it into places. The cup at a place $a$ is up exactly when the record is at or above $a$. Say a step takes the record from $x$ to $y$. The cup at $y$ is up afterwards, since $y$ is at or above itself, and it was down before exactly when $y$ isn't at or below $x$. And if $y$ is at or below $x$, any cup that's up at $y$ was already up at $x$. So a step turns over some cup exactly when it lands somewhere that isn't at or below where it started. The same reasoning says a run ends with some cup up that was down at the start exactly when it ends somewhere that isn't at or below where it started.

□ **Theorem 11.2** (The toll over any direction)**.** Take a machine whose record is a place in any preorder, and say a step *turns a cup over* when it takes the record from $x$ to a place that isn't at or below $x$. (1) Every step that turns a cup over costs at least one exactly when every run, from whatever state it starts, that ends somewhere not at or below its start costs at least one. (2) When that holds, a run costs at least the number of its steps that turn a cup over. (3) It holds exactly when every cup, on its own, obeys the toll of Section 10: for every place $a$, a step that turns the cup at $a$ from down to up costs at least one. The universal floor of Section 2 (Observation 2.1) is (1) with one cup.

*Proof.* (1) and (2) are the walk above. For the other half of (1), a run of one step that turns a cup over ends somewhere not at or below its start. (3) is the translation ■ above.

When the cups only ever go up, as on the strip and the diamond, a step that turns a cup over is just a step up.

That surprised me a little, because I'd filed the floor away as a fact about one bit, and nothing in it was about one bit. It was about one cup going up, and anything you build out of cups has that.

## 11.4 One line isn't enough

Now comes the payoff I promised at the start of this section. Could a cleverer single cup do the job of all four, with one latch and some decoding that reads the record's place off the latch and whatever the machine shows? Section 10 already ruled that out in the smallest case: over a machine with one visible state, a record that climbs through three places can't be produced by one latch and a decoding, because one latch and one visible state make two outcomes and the record has three.

So try a second guess, closer to the cup's own rule. The cup's rule was "new value is old value *or* the event." The obvious rule for a whole row is "new place is old place *joined with* the event's place," where joining two places means going to the lowest place at or above both. On the row, the event names a place, and you turn up every cup as far as that place. To see it break, take the strip $0 < 1 < 2$, a cup at 1 and a cup at 2, over a machine with one visible state, so the event has to be one fixed place. Let the record jump from 0 straight to 2, turning up both cups in one step, and stay put at 1 and at 2. From 0 the event has to turn up both cups, so the fixed place is 2. From 1 the same event would turn up the second cup as well, and the record stays at 1.

What works is giving every cup its own event, and letting the event for a cup look at where the record already is. Theorem 10.11 said exactly that, and this is why it had to say it. That leaves the question of how many cups. In the row of cups I asked you to try five stages with three and said you couldn't. Theorem 10.11 (5) already bounded the chain from above. The count behind that bound comes again below, with the matching example added, and with an arrangement of $k$ cups read as a $k$-bit word, a one for every cup that's up.

**Theorem 11.3** (Lines a direction needs)**.** (1) A list of different arrangements of $k$ cups, each keeping up every cup the one before had up (a chain of different $k$-bit words in which each word's set bits are among the next word's), has at most $k+1$ members, and for every $k$ there's such a chain with exactly $k+1$. So a straight strip of $n+1$ stages needs exactly $n$ cups. (2) Over a machine with one visible state, take the record on the strip $0 < 1 < 2$ that jumps from 0 to 2 and stays put at 1 and at 2. It grows and is driven by the computation, and no rule of the form "join the record with a fixed place" reproduces its steps. □

*Proof.* (1) Going from one arrangement to the next, which is a different one, turns up at least one more cup, and there are $k$ cups, so there are at most $k$ steps and $k+1$ arrangements. For the matching chain, take no cups up, then the first, then the first two, and so on, packed to the left: that's the row of cups. (2) Joining 0 with a fixed

place $g$ gives $g$, and from 0 the step has to give 2, so $g$ is 2. Joining 1 with 2 gives 2, and from 1 the step has to stay at 1.

So five stages need four cups, and neither one nor three will do it, as long as the cups only turn up. An ordinary binary counter holds a thousand stages in ten bits, because its bits go down as well as up, and the theorem says nothing against that. The switch isn't wrong: it's the first line, and a straight strip with more than two places needs more of them, exactly one fewer than it has stages. A direction that branches needs at least as many cups as the steps in its longest strict climb, and at least as many as it takes to tell its places apart, the rounded-up logarithm of how many there are. Put three places above a bottom, none of them above another: the longest climb is one step, and one cup can't tell four places apart, so it takes two, and two do it, though not two lines: one cup that goes up at the first and the third, and one that goes up at the second and the third. It can need more than both. Take two places each below two others: one step of climb, four places, and two cups still won't do, because both upper places would have to keep up every cup either lower one has up, and two cups allow at most one arrangement like that besides the lower places' own.

People got here before me, and from a direction I didn't expect, which was programming for many processors at once. Lindsey Kuper and Ryan Newton's LVars (2013), "lattice-based data structures for deterministic parallelism" in their own title, are memory cells whose value only climbs in an order, where a read can't ask for the value and can only ask whether it has climbed past a threshold. That's a cup. Writers running side by side can push the value up in different orders, and a threshold once passed stays passed. Their deterministic behaviour also depends on which reads and updates the language permits. Permanence alone would not do it: a program allowed to ask whether the cup is up *yet* could still see another writer's timing. Amin Timany and Lars Birkedal (2021), coming at it from proofs about programs, show, as I read their paper, that any preorder at all can be carried as monotone ghost state in a program logic. That's the closest thing I've found to somebody else arriving at the decomposition into lines.

The two rest on the same thing, a value that only climbs and a question that only asks whether it's past a point, and they use it for opposite jobs. In an LVar the threshold read steers the program. Ghost state, as far as I understand the term, lives in the proof, and the running program never reads it. Here the record is never allowed to steer the base computation, so that particular restriction sits closer to the ghost state. Push on that and it gives a little: Section 5's glance, the whole reason to keep a record, is a read. What keeps me on that side is that a glance worth trusting gives the answer the long way would have given. Their purposes aren't mine, but the shape is theirs first. What I can't find in either is a price on crossing a line.

## 11.5 Why crossing a line has to cost

Part I showed that the cup's toll isn't an arbitrary rule. It's forced by three premises, and the argument that ties them together was the pigeonhole: on a machine with finitely many states, the move that raises a record that goes up and stays up has to land two possible states on one state (Theorem 3.1), and a step that lands two states on one is a step that forgets which one it came from. Does that survive the whole row of cups? Mostly, and the parts that don't survive are the interesting ones.

The counting needs a flag, because what it counts is the collection of states with that flag up, and now there are many cups, each one a yes-or-no reading that could serve as the flag, so the first job is to pick one. My first guess was the cup for the place where the record starts. That goes nowhere: the starting state already has that cup up, so it's already in the collection, and there's no state coming in from outside. The cup that works is the one the step turns over, the cup for the place where the record lands. It's up after the step, since a place is at or above itself, and it was down before, which is what turning a cup over means. So the collection is every state with that landing cup up, and the state the step starts from is the one coming in from outside. If the move keeps every state in the collection inside it and never lands two states on one, the states already inside use up every state in the collection between them, and there's none left for the newcomer. So one of two things happened. Either two states landed on one, or some state that was in the collection got moved out, which means its landing cup went back down. I'll call that second thing a *revocation*.

**Theorem 11.4** (Turning a cup over merges)**.** On a machine with a record in any preorder, where each place has finitely many states with its cup up: (1) if a move turns a cup over at some state and lands the record at a place $a$, then either the move merges two states, or there's a state with the cup at $a$ up that the same move takes to a state with the cup at $a$ down, which is a revocation; (2) if the record only grows, so that no cup ever goes back down, the move merges; (3) if the record only grows, the toll holds exactly when every move that turns a cup over at some state, and merges, costs at least one, so merge pricing gives the toll. □

*Proof.* (1) Gather every state with the cup at $a$ up. There are finitely many. If the move keeps each of them there and is injective, it maps that collection into itself without two landing on one, so it hits all of them, including the state it sends the starting state to; injectivity then says the starting state was in the collection already, and it wasn't, because its cup at $a$ was down. So either some state in the collection has its cup at $a$ knocked down, or the move isn't injective. (2) Growth rules out the knocking down. (3) If the toll holds, every move that turns a cup over somewhere

pays, merging or not. The other way, a move that turns a cup over merges by (2), so the hypothesis charges it. ■

It's the old counting argument again, run once for every cup.

Each premise there can be knocked out, and I wanted to see each one go, because a premise you've never seen fail is one you don't understand yet.

Knock out finiteness. Take the whole numbers as the states, which gives a row of cups that never ends, and let the record be the state itself, one more on every step, at no charge. Every step turns up one more cup. The step never merges anything: different numbers go to different numbers. Merge pricing holds, because there are no merges to price, and the toll fails on every single step. The counting broke at the one place it needed the collection to be finite. Every state moved one along, and there was always another state to move to. That's Charles Bennett's 1973 construction in miniature: a machine that keeps its whole history never has to forget, so it never has to merge. It hasn't got out of paying. Every step lands it on a state it has never used, and that's the fresh-state column. The toll fails here because merge pricing only charges merges and this machine pays by taking a state it has never used, which that price list doesn't charge. Take the infinity away and the supply of fresh states goes with it.

Knock out growth. Take a two-state machine whose record flips back and forth between two places, so a cup goes up on one step and gets knocked back down on the next. The two states just swap with each other, so nothing merges, and the price list is honest about merges. The step that turns the cup up pays nothing. It takes the other branch of (1) and revokes: the state that had the cup up gets moved to where it's down.

Knock out the price on merges. A two-state machine that stamps every state with yes turns the cup up wherever it was down, merges at the stamp, and charges nothing. Of course the toll fails; nobody asked the merge to pay.

□ **Theorem 11.5** (Each premise does work)**.** (1) There's a machine with infinitely many states, a growing record and an honest merge price on which every step turns a cup over and the toll fails. (2) There's a two-state machine with an honest merge price whose record doesn't grow, which turns a cup over and fails the toll. (3) There's a two-state machine with a growing record whose merges aren't priced, which fails the toll.

■ *Proof.* The three machines above, in order.

So the toll over a whole direction rests on exactly the three things the cup's toll rested on: finitely many states with each cup up, a record that doesn't come down,

and merges that pay. Nothing extra got smuggled in when one cup became a row, and nothing the one cup needed got left behind.

I'd half expected the diamond to need something new, some extra rule for the two middle cups that don't compare. It didn't. Every cup gets its own collection of states, and the argument on each is the old one.

CHAPTER 12

# The shadow, exactly

Section 7 put the machine behind a window and asked what an outsider could tell. It found that two states can look the same through the window and differ in what matters, and that a price computed from the window can't be exact if the window has a collision (Observation 7.2). That was for one cup. With a whole row of cups in hand I can ask the sharper version of the same question, which is also the one I actually care about. Take the ordinary machine, the one with no cups and no ledger, and let it look at itself. What's the shape of everything it can't see, and what does the toll cost in terms of that shape?

## 12.1 The ordinary machine is a shadow

Take any machine with a record and a ledger, and sweep both off the table, the cups and the counter. What's left is a machine that steps the same way and knows nothing about either, provided the step's effect on everything else never depends on the cups or the counter: sweeping them off has to commute with the step. A system whose control reads its record or its ledger has no shadow in this sense. Where the premise holds, call what's left the *shadow*. Over any one state of the shadow there's a pile of full states, all casting that same shadow and differing only in their record and their ledger. Section 7 already called the pile over a point the *fibre* over it. The shadow is one particular window, the one that shows everything except the cups and the ledger.

The shadow has its own fibres, and every move has piles of its own too: the states a move sends to one landing spot. I'll call those the move's *landing piles*. A fibre is everything that looks the same before anybody moves. A landing pile is everything that ends up in the same spot after one move. Keep the two kinds apart, because the whole section turns on how they meet.

## 12.2 Pricing from the shadow

First the window question, over a whole row. A *window price* charges each step from what the window shows before and after it. It meets the floor if it charges at least

one for every step that turns a cup over, and it's *exact* if it also charges nothing for any step that turns no cup over. A *collision* is two steps that look the same through the window, where one turns a cup over and the other doesn't.

With one cup, the fix in Section 7 was to put the cup in the window. So my first guess for a row was that the window has to show some of the cups. Try the smallest version: a record in the whole numbers, and a window that shows only the first cup, the one that asks "at least one?" Let the record rise on some steps and hold still on others. Once it's past one, that cup is up before and after both kinds of step, so a step that turned over a cup further along looks exactly like a step that turned over nothing, and the two make a collision. So then, I thought, the window has to show every cup. That's wrong too, from the other side. Take a record in the whole numbers with two moves, one that adds one and one that leaves it alone, and a window that shows only whether the record is even, which on the row is whether an even number of cups are up. You can't read the record off that window, and no flat charge is exact, since some steps turn a cup over and some don't. But the evenness changes on exactly the steps that turn a cup over, so charging one exactly when the evenness changed is exact.

**Theorem 12.1** (Window prices over a direction)**.** (1) If the window has a collision, every window price that meets the floor charges at least one for some step that turns no cup over, so no window price is exact. (2) An exact window price exists exactly when the window has no collision; on a machine with finitely many states and moves, whose views can be told apart by a yes-or-no test, it can also be computed. (3) Along any run, a window price that meets the floor charges at least the number of steps that turn no cup over but look, through the window, like a step that does. Some window price meets the floor and, among the steps that turn no cup over, charges one on exactly those, so that number is the least overcharge; when it can be decided, for each pair of views, whether some step showing them turns a cup over, that price can be computed. (4) A window can price exactly without showing where the record is: there's a record in the whole numbers with a move that raises it and a move that leaves it alone, and a window that shows only whether the record is even; the evenness changes exactly on the rises, no charge that is the same for every step is exact, and charging one exactly when the evenness changes is exact, although no rule reads the record off the window. (5) There's a record in the whole numbers and a window that shows only the one cup "at least one," and that window has a collision. □

*Proof.* (1) The two colliding steps get the same charge. The floor makes it at least one because of the step that turns a cup over, and the other step pays it too. (2) Without a collision, "does some step that looks like this turn a cup over?" is a question about what the window shows; charge one exactly when the answer is yes. On a finite machine you can answer it for every pair of views. (3) A step that turns no cup over but looks like one that does gets that step's charge, which is at least one by (1)'s argument; add along the run. For the other half, charge one exactly when the answer to (2)'s question is yes, which needs that question to be decidable, and count the overcharge. (4) A rise changes the evenness and the other move doesn't, so the window sees exactly the steps that turn a cup over; a flat charge either charges nothing for a rise or charges something for a step that turns nothing over; and 0 and 2 look the same through the window. (5) The first cup, as above. ■

Item (4) changed how I think about this. I'd assumed a window had to see the cups to price them. It doesn't; it has to see them *move*. A window that cannot tell you your exact place, but detects every rise and every pause, can charge the right amount every time.

### 12.3 What a priced move destroys

Now the fibres. Part I showed that the move raising a permanent flag on a finite machine lands two possible states on one state, and Theorem 11.4 showed the same for a step that turns any cup over, once the cups only go up. Which two? It matters a great deal. If they're in different shadow fibres, then the ordinary machine lost a distinction too; the record may also differ between them. If they're in the same fibre, the ordinary machine lost nothing, and what got destroyed was only ever off the table: two states that cast one shadow, told apart by their cups or their ledger, crushed into one.

My first guess was that it's always the second kind, because the cups are where everything happens. It isn't always. It depends on whether the ordinary machine forgets anything on its own.

□ **Theorem 12.2** (The toll prices a collapse inside a fibre)**.** On a machine with finitely many states and a growing record: (1) a move that turns a cup over somewhere sends two different states to one landing spot; (2) if the shadow exists, so that sweeping commutes with the step, and the shadow's step for that move forgets nothing, sending different shadow states to different shadow states, then those two states are in the same shadow fibre, and every landing pile of the move lies inside one fibre; (3) without that premise it can fail: there's a two-state machine whose two states sit in different fibres and whose shadow step forgets, so the merge at

the step that turns the cup over lands two states from different fibres and no fibre loses a point. (4) The halving price, which says a move of cost $c$ shrinks any finite collection of states by at most a factor of $2^c$, holds exactly when every landing pile has at most $2^c$ states, on a machine of any size.

*Proof.* (1) is the counting argument of Theorem 11.4, with growth. (2) If two states land on one spot, their shadows land on one shadow spot, and the shadow's step tells shadows apart, so the two states cast the same shadow to begin with. (3) The stamp of Theorem 11.5, with the record's value also shown in the visible part, so each state is its own shadow: both states land on yes, and so do their shadows. (4) is the worst pile-up of Premise 3.4. Any $2^c + 1$ states of one landing pile would be a collection that lands on a single spot, which the premise caps at $2^c$, so no pile has more; and any finite collection is at most the number of spots it lands on times the biggest pile. Neither half needs the machine to be finite. ∎

So on a machine with finitely many states and a record that only grows, when the ordinary machine forgets nothing, the toll is a price on crushing a fibre: on taking two states that cast one shadow, which only what's off the table told apart, and making them one. The last of those premises is the one I just showed you failing.

## 12.4 How much gets destroyed, on average

The halving price is a worst case, because it prices the largest landing pile a move can have. The next question is the average one: if I don't know which state the machine was in, cups and all, but I have some spread of chances over the states, how much do I lose, on average, about which state it was, once I know where it landed? Shannon's conditional entropy measures exactly that (Section 3 built the entropy from scratch, so I won't do it again here), and I'll write it in bits.

**Theorem 12.3** (The halving price in bits bounds the average loss)**.** Take any spread of non-negative weights over the states, uniform or not, with finite total weight, on a machine of any size. (1) If every landing pile has at most $N$ states, with $N \geq 1$, the weighted conditional entropy of the state given its landing spot is at most the total weight times $\log_2 N$ bits. (2) So on a machine with the halving price, that weighted information loss is at most its cost in bits times the total weight, even when the spread's own entropy is infinite. For a probability distribution, the total weight is one and the average loss is at most the cost. (3) If a landing pile has $2^c$ states, the uniform probability distribution on it loses exactly $c$ bits and attains that bound. □

*Proof.* Look at one landing spot. Suppose the sources that land there carry total weight $w > 0$. Divide each source's weight by $w$. Those shares now add to one: they describe the uncertainty about the source once you know this landing spot. There are at most $N$ possible sources. The logarithm argument in Section 3 says the uncertainty is greatest when those possibilities have equal shares, and is then at most $\log_2 N$ bits. Multiply by $w$ to put this pile back into the whole spread. Its contribution is at most $w \log_2 N$. A pile with zero weight contributes nothing.

Add the contributions. The pile weights add to the original total weight, so the same $\log_2 N$ multiplies that total. Under the halving price, $N = 2^{\text{cost}}$, whose base-two logarithm is exactly the cost.

Infinitely many piles cause no subtraction of one infinite entropy from another. I am adding the nonnegative uncertainties inside the piles. Every finite collection contributes at most its weight times the bound, and its weight is at most the total. Taking the largest value approached by those finite sums keeps the same bound.

Finally put all probability on one pile of $2^c$ sources, equally shared. Before landing there are $2^c$ equally possible origins; afterwards the landing spot distinguishes none of them. The uncertainty left about the origin is $\log_2(2^c) = c$ bits. That attains the bound. ■

Where does it stop? My first guess was one place, a landing pile with infinitely many states, and that the average would blow up there right along with the worst case. That's half right, and the half that's wrong surprised me. No finite charge satisfying the halving price can price a move that sends infinitely many states to one spot. The weaker merge price could still charge one, since it does not measure the pile's size. But the average depends on the spread. Put weight $\frac{1}{2}$ on the first state, $\frac{1}{4}$ on the next, $\frac{1}{8}$ on the next, and so on, and the information lost by crushing the pile never reaches 2 bits, however many of its states you count. Under that spread, a two-bit information budget covers the average, though no finite budget covers the worst case. So I went looking for a spread that does blow up, and it took a nastier one: blocks of states, block $k$ holding $2^{2^k}$ states and a share $\frac{1}{(k+1)(k+2)}$ of the weight, which adds up to one. Under that spread the loss grows past every bound, and no finite information budget covers that average loss.

You can generate the first spread by tossing a fair coin until the first head. Zero tails names state 0, one tail names state 1, and so on. State $j$ has chance $2^{-(j+1)}$, and identifying that outcome takes $j+1$ toss-results, so its entropy term is $(j+1)/2^{j+1}$. Add these by counting each possible toss position: the first is reached with chance 1, the second with chance $1/2$, the third with chance $1/4$, and so on. The total is $1 + 1/2 + 1/4 + \cdots = 2$; every finite partial sum of the state terms is smaller. Resetting all those outcomes to one box can lose only those two bits on average, despite having infinitely many possible starts.

For the block spread, split each block's weight equally among its states. The block weights sum to one because $1/((k+1)(k+2)) = 1/(k+1) - 1/(k+2)$. Within block $k$, there are $2^{2^k}$ equally likely choices, so just identifying a state within that block contributes $2^k/((k+1)(k+2))$ to the average bits. Those contributions grow without bound as $k$ grows, and forgetting which block it was only adds to the loss. And no finite worst-case price can handle an infinite pile: for any proposed cost $c$, select $2^c + 1$ different states in it. They all land together, violating the halving rule's maximum of $2^c$.

**Theorem 12.4** (An infinite pile)**.** (1) A move that sends infinitely many states to one spot isn't priced by the halving price at any finite cost. (2) Under the halving spread $2^{-(j+1)}$, crushing the pile loses exactly 2 bits, and counted over its first $n$ states, less than 2 for every $n$. (3) Under the block spread, the information lost grows past every bound, so no finite bit budget covers it. □

*Proof.* The three computations above. ■

And heat? Part I already said where the line is, and it's the same line here. Landauer's principle, that crushing $c$ bits' worth of possibilities gives off at least $c\,kT\ln 2$ of heat on average, is a premise from physics, and a contested one (Section 3). Nothing in this section proves anything about heat. What the counting does give is that, under that premise, on a machine with the halving price, a move's cost priced at $kT\ln 2$ per unit, times the weight of the piles, is always enough to cover the minimum the premise would demand, for every spread of finite total weight. If the premise falls, that sentence falls with it, and every theorem above it stands, because none of them used it.

## 12.5 What the shadow is

Put the two sections together and you get the shape I've been circling since the Flatland night (the night a video about *Flatland* gave me the hunch behind this book; Section 24 tells it). The ordinary machine is the shadow of a machine with a row of cups beside it. Over every shadow state there's a whole fibre, a pile of states that differ only in their cups and their ledger, and the ordinary machine can't tell them apart, on a machine whose step never looks at the cups. On a machine that obeys the toll, a step that turns a cup over costs, whichever cup it is and however many it turns, and on a machine with finitely many states whose record only grows, when the ordinary machine forgets nothing, what that cost pays for is exactly crushing two states of one fibre into one. A window can charge the right cost without seeing the cups, if it sees them move. And on a machine with the halving price, where each

unit of cost pays for at most one halving, the cost in bits, times the weight of the spread, is never less than what got crushed on average.

What I haven't shown is that any of this is out there in the world, that nature keeps a direction like this over its own shadows. That's a belief of mine, and nothing in this book proves it. What I've shown, and you can check, comes with its premises showing. A machine whose record is driven by the computation and climbs a partial order is a row of cups, one latch for every line. If it also has finitely many states, never lets the record come down, and prices its merges, it has to pay this way. And when its ordinary machine forgets nothing, what it pays for is exactly this. It's less than the Flatland night promised me. It's also the part of it that held when I pushed.

## 12.6 Shortcuts

There's a question every machine with a record runs into once its programs can be handed their own code. A cup is worth turning over because something later glances at it and doesn't have to work the thing out again, the way Section 5's glance reads a block's bucket off the table without applying the rule from scratch. Call any move like that a *shortcut*, and call it *sound* when it gives the answer the long way would have given. (That gloss is about cost, leaning on what was done so nothing is redone, and the wall I'm about to show you doesn't look at cost at all; the next section says exactly what it does look at.) So can a machine like that *decide*, from inside, whether its own shortcuts are sound? That question runs into a wall, the same wall Kurt Gödel and Alan Turing hit, and I have to show you that it's there before I show you anything that pretends to be on the other side of it.

CHAPTER 13

# The diagonal

In Section 5 I said you'd watch the same move get made again and again as the book went on: somebody says "everything is X," it's true enough to do real work with, and then somebody draws the diagonal. I owe you the diagonal itself now, because Sections 10 to 12 just handed a record, and a whole direction of them, to *any* system that steps, and the next thing anyone wants from a system with a record is for it to tell sound shortcuts from unsound ones by itself. Before I let anything claim that, I have to show you the wall every such claim runs into, and then show you exactly where a system with a record stands relative to the wall. It doesn't stand on the far side, because nobody's on the far side. A record keeps a distinction and pays for it. What it can't do is the judging: no machine that can hand a program its own code, and can run the flip of a verdict as one of its own programs, can sort all its programs, from inside, by a question about what they do that some pass and some fail. So the record carries a commitment, and somebody still has to make it.

## 13.1 Five names and a table

Do this first, with a pencil, because the whole family of arguments is this one trick and you should own it. Write five friends' names down the left of a page. Next to each name, write a five-digit string of zeros and ones, anything you like. Now build a new string: its first digit is the *opposite* of the first digit of the first row; its second digit is the opposite of the second digit of the second row; and so on down the diagonal. Look at what you've made. It differs from row one in position one, from row two in position two, from every row at that row's own position. So it's nobody's string, and your table of five missed something.

With five rows and five digits that's a cute fact. Georg Cantor, a mathematician working out how to compare infinite sets, made the point in 1891 that it works for *any* list, however long, even infinite: hand me a complete list of all infinite strings of two symbols, I'll walk the diagonal flipping as I go, and I'll hand you back a string that's on no row. So there is no complete list. There are more such strings than there are places on any list, and that's where "some infinities are bigger than others" comes from. (He wrote the two symbols as m and w, where I'd write 0 and 1, and

that paper wasn't even his first proof of the fact. The first was in 1874, by a harder road through the real numbers, and in 1891 he called the new one much simpler. The diagonal was him finding the easy way.)

Now look at what the argument *did*, because it's going to do the same thing again and again, in new clothes, and I want you to recognise it each time. (Not every name below is a flipped diagonal in the table sense. Kleene's theorem is the tool that makes the diagonal buildable, Rice's is the general form, the one from Ken Thompson, who built the original Unix operating system at Bell Labs, is an attack that uses the same trick, and the last, Lawvere's, is the theorem that makes them all one.) There was a table: rows are the things being listed, columns are questions, each cell says "does row i do column j." There was the diagonal: the cells where a thing is asked about *itself*. There was a flip: something with no fixed point, like turning yes into no, applied along the diagonal. And the thing that got refuted was *the assumed object*, the complete list. Nothing outside the assumption was touched. Hold on to those four: a table, a diagonal, a flip, and the assumption eating itself.

### 13.2 The same move, again and again

Russell, 1902, from Section 5: the collection of all collections that don't contain themselves. It's Cantor's flipped diagonal with "contains" as the table, and Russell said as much, in *The Principles of Mathematics* in 1903. He got there, he wrote, by running Cantor's argument on a supposed collection of everything, which ought to be the biggest collection there is. What got refuted was the assumed object, the collection of everything, and Zermelo's fix was to stop assuming it. I promised in Section 5 that I'd come back to this here. Every reading in this book is a yes or no question on the states of one machine, a collection cut out of a collection I was already given, and never a sort of everything. When Part I asked for one state that codes every question about the states (Section 1), the same flip refuted it.

Jules Richard, a French mathematician who taught at the lycées of Tours, Dijon and Châteauroux, took his doctorate in Paris in 1901 and wrote mostly on geometry. In 1905, in a short note, a letter to the editor of a French science review, he observed: some phrases (his were in French) define real numbers ("the ratio of a circle's circumference to its diameter"); there are only countably many phrases, so list the numbers they define in order; run Cantor's diagonal down their decimal expansions and you get a real number no phrase defines, except that you just defined it, in the phrase that describes the diagonal. The lesson I take from that one is the lesson this whole section turns on: keep apart the language a system *talks in* and the language you use to talk *about* it. Mix them and the diagonal bites you.

Richard's paradox has a cousin small enough to do at the table, and it comes with a footnote I can't resist. Russell, the first to put it in print, in 1906, credited

it to G. G. Berry, a junior librarian at the Bodleian Library in Oxford. The paradox has outlived empires, and it came from the library desk. In the version in *Principia Mathematica* it's about syllables. Take the phrase "the least integer not nameable in fewer than nineteen syllables." There are only so many phrases shorter than nineteen syllables, so they name only so many numbers, so some least number is left over, and the phrase picks it out. Now count the phrase: the, least, in-te-ger, not, name-a-ble, in, few-er, than, nine-teen, syl-la-bles, and it comes to eighteen. In eighteen syllables it has named the number that can't be named in fewer than nineteen. It's the same lesson as Richard's, in a smaller coat: "nameable" is a word about the language, used inside the language, and that's where the bite is.

Kurt Gödel, 1931, with an improvement in 1936 by Barkley Rosser, a logician who had studied under Alonzo Church at Princeton and whose trick lets the theorem assume only that the system is consistent: the big one, and I'll say exactly what it says, because every word of it is carrying weight. Take any consistent formal system with a mechanically checkable list of axioms that can do a bit of arithmetic. Then there are sentences it can neither prove nor refute. The engine is a lemma that lets you build, for any property you can write down, a sentence that the system proves *equivalent* to that property holding of the sentence's own code number. Apply it to "not provable" and you get a sentence equivalent, inside the system, to its own unprovability. The table is the sentences listed by code, the diagonal is a sentence about its own code, and the flip turns provable into not-provable. The Stanford Encyclopedia stresses two cautions, and I'll repeat them. The sentence doesn't literally "say" anything about itself. It's provably *equivalent* to a claim about its code, which is less mystical and more exact. And the theorem doesn't say the sentence is unknowable. It says *this one system* can't settle it.

And how did the world take it? Mostly, at first, it didn't. Gödel, twenty-four years old, announced the first theorem during a round-table discussion on the last day of a conference in Königsberg in September 1930, and by the account of Solomon Feferman, one of the editors of Gödel's collected works, the room met it with general incomprehension, except for one man. John von Neumann pulled Gödel aside to talk about it, kept thinking about it, and a couple of months later saw that the argument gave something more: a consistent system of this kind can't prove its own consistency. He wrote to Gödel with it in a letter dated 20 November 1930. Gödel had already got there. His paper had reached the journal three days earlier, with that second theorem announced in it, and von Neumann left the credit with Gödel. It's the "somebody got there first" shape from Section 4, with one difference: here the man who got there second was von Neumann.

Alan Turing, 1936. I'd always taken it that Turing proved the halting problem undecidable, and here I had to go and look. The set theorist Joel David Hamkins and his coauthor Theodor Nenu went back to the 1936 paper and report that the word

"halting," the modern statement, and the self-referential proof everyone teaches are all, strictly speaking, absent from it. What Turing proved undecidable were two other questions about his machines (whether a machine prints infinitely many digits; whether it ever prints a given symbol). The second is *equivalent* to halting and the first is strictly harder. So their answer cuts both ways, and they say so: Turing never stated the halting problem, and he essentially proved it, because his second question is a short translation away from it. The name came later. Martin Davis seems to have been the first to write "halting problem," in a book of 1958, and Stephen Kleene had already stated the problem, without that name, in his 1952 textbook *Introduction to Metamathematics*. The textbook version is still right as mathematics, and it fits on two index cards. Card A is the Checker: hand it any program card and it announces "halts" or "runs forever." Card B is the Contrarian: "Hand me to the Checker. If it says I halt, run forever. If it says I run forever, stop." Hand B to A, and either answer is wrong, so there's no Checker. The table is programs against programs, the diagonal is a program run on itself, and the flip turns halt into loop.

Stephen Kleene, 1938, the logician whose textbook I just mentioned, and this is the one that turns the paradox into a tool, so slow down. Everything above needs the self-referring object to *exist*: Gödel's sentence has to be buildable, the Contrarian has to be able to "hand me to the Checker." Kleene's second recursion theorem says: for any transformation of programs you can compute, there's a program that behaves exactly as if it had been handed its own source code. Any program may be written as though it already had a copy of itself in hand. He stated it almost in passing, in a paper on something else, and proved it in the last two lines of one of its sections. I have that count from Yiannis Moschovakis, a logician at UCLA and Athens who wrote a 2009 paper on it, expanded in the Bulletin of Symbolic Logic in 2010, called "Kleene's Amazing Second Recursion Theorem," and he calls it a little gem. It's the reason "programs that print their own source" exist in every ordinary programming language: one that can compute whatever a Turing machine can and can build, from a program and an input, a program that runs the first on that input (they're called quines, after a philosopher I'll come to in a moment; try "Print the following twice, the second time in quotes: 'Print the following twice, the second time in quotes:'" and obey it). Self-reference in computation isn't a paradox but a construction. The paradoxes happened only when someone assumed an *unrestricted* collection or language; the theorem happens when you build the self-referrer honestly.

And where does the word quine come from? The philosopher didn't coin it himself. Douglas Hofstadter coined it in 1979, in *Gödel, Escher, Bach*, in honour of Willard Van Orman Quine, who had made a long study of self-reference that works indirectly. Quine left a sentence you can take apart at the table, from a 1962 essay on paradox, and in full it reads: "yields falsehood when preceded by its quotation"

yields falsehood when preceded by its quotation. Do what it says. Take the part in quotation marks, put a quoted copy of it in front of itself, and look at what you've built: it's the whole sentence, word for word. So the sentence says that it itself is false. It's the liar paradox, but look at how it got there: no "this sentence" anywhere, no pointing at itself, just an instruction to quote and copy, the same move as the print-twice instruction above. Quoting and copying is Kleene's trick in plain English, and it's why the paradox and the tool look so alike. The paradox is what you get when the copy asserts something about its own truth. The tool is what you get when the copy just does something.

Henry Rice, 1953, whom you met briefly in Section 7: the general form. Any property that depends only on what a program *does*, whatever its text, and that's true of some programs and false of others, has no algorithm that decides it for all programs. "Does this program halt" is one such property. "Does this program ever output 7" is another. "Does this program contain an if-statement" isn't, because that's about the text. One standard proof goes straight through Kleene.

And the one that's actually an attack: Ken Thompson, in his Turing Award lecture, published in 1984. Build a compiler that, when it compiles the login program, plants a back door, and when it compiles *a compiler*, plants both habits into that one too. Then delete the trick from the source. The compiled compiler still carries it, and recompiling a clean source with the poisoned binary re-plants it, forever. Reading the source never reveals it. A locksmith teaches an apprentice to cut a spare for every town-hall key and to teach *that* to every apprentice they train, and never to write it in the manual; years later the manual is spotless and every locksmith in town cuts the spare. Thompson's compiler is the recursion theorem with a knife in it, and his moral was that you can't trust code you didn't write entirely yourself.

And Thompson wasn't first either, and he said so. Pull on the lecture's acknowledgment the way Section 14 pulls on Bitcoin's footnotes: he writes that he first read of the possibility in an Air Force critique of the security of an early implementation of Multics, a time-sharing system from the 1960s, and that he couldn't find a more specific reference to it, and he asks anyone who can to send him one. The critique is Paul Karger and Roger Schell's 1974 security evaluation of Multics, which already described a trap door planted in a compiler that, because the compiler is written in its own language, keeps itself alive. So the knife was in the drawer ten years before the lecture was printed. Then comes the obvious next question, whether there's any way out or whether "trust no binary you didn't build" is the bottom. There's a partial answer, and it took a while. David A. Wheeler, in a 2005 paper and a 2009 dissertation at George Mason University, worked out *diverse double-compiling*: compile the suspect compiler's source with a second, independently made compiler, use the result to compile that source again, and compare the output bit for bit with the suspect binary. If they match under Wheeler's assumptions, that gives

a source-to-binary correspondence. The second compiler must correctly translate the relevant source, the execution and comparison environments must be trusted for these runs, and the build's language and environmental effects must match the conditions of the argument. Two compilers having different authors is useful evidence for trust, not a substitute for those premises. The question has moved to a boundary where we can name what we trust.

Bill Lawvere, in 1969, proved a fixed-point theorem that, as I read it, makes all of these one theorem in disguise: if a set can be mapped onto all the functions from itself to some target, then every function on the target has a fixed point; so if the target has a function with no fixed point (a flip), no such covering exists. Cantor, Russell, Gödel and halting all come out as instances of one shape. Noson Yanofsky retold it in 2003 without the category theory. I mention it because "it's the same wall" is a claim I make below, and I want you to know it's fair at the level of *proof shape*, and that every instance has its own boundary of *who counts as a decider*, and that boundary is exactly where most overclaims live.

## 13.3 Applying the diagonal

Before applying the diagonal to a system with a record, identify who the proposed decider is and what it must decide. Four familiar applications show why those details matter.

"Gödel shows minds aren't machines." John Lucas, a philosopher who spent thirty-six years as a fellow of Merton College, Oxford, argued it in 1961, in a paper called "Minds, Machines and Gödel." Roger Penrose, the Oxford mathematical physicist who later shared a Nobel prize for his work on how black holes form, revived it twice, in two books, *The Emperor's New Mind* in 1989 and *Shadows of the Mind* in 1994. I'm sure of who argued it and when. Whether they're right is the question, and the standard replies, as far as I can tell, have held up: the Gödel sentence is true *only if the system is consistent*, so a human who "sees" its truth has to know the system is consistent, which is the hard part; and nobody can build the Gödel sentence of a system they can't specify, so humans might be machines too complicated to diagonalise. The theorem has a conditional in it and the argument drops the conditional.

And the part that's easy to miss is that Gödel himself didn't go as far as Lucas. In a lecture on the day after Christmas in 1951, at a meeting of the American Mathematical Society at Brown University, he drew from his own theorems only an either-or. Either the human mind, even inside pure mathematics, infinitely surpasses the powers of any finite machine, or there are problems about whole numbers, of a certain simple kind, that no proof the human mind can conceive will ever settle. Feferman, writing about that lecture, says Gödel privately believed the first half and

still stated only the disjunction, because he didn't have a knock-down argument against the machine side. Gödel wrote soon afterward that he meant to publish the lecture, and never did; the text sat among his papers until after his death. So the man whose theorem gets quoted for "minds aren't machines" stated an either-or and left the deciding to whoever came next. The conditional is there again, kept by the person with the most right to drop it.

"Undecidable means unknowable." Two senses share the word: a problem with no single algorithm correct on every instance, and a sentence independent of one particular system. Neither means nobody can know a given answer. Any single instance is decidable, trivially: for any one program, one of "say yes" and "say no" is right. What's ruled out is one *inside* procedure that's right on *every* case.

"Everything interesting is undecidable." Rice is about behavioural properties over *all* programs. It says nothing against checking the text, nothing against deciding for specific programs, nothing against analyses allowed to say "I don't know," nothing against finite-state model checking or type checking. Real compilers and analysers live with Rice by being sound but incomplete, and what limits them in practice is mostly cost. The diagonal shapes them; it doesn't stop them.

"Self-reference is a paradox." It's a theorem with a construction, as Kleene, the quines and Thompson show.

## 13.4 Now point it at the record

The question Section 12 left open can be asked with nothing in it but what Parts I and II have built. A system steps; a record rides along and latches on an event (Section 10); when the event is a certification, Section 8 said the reading, with its object and its paid event behind it, is what lets an auditor treat a claim as established, and for a claim that a shortcut is sound, that's what makes the shortcut *admissible*. Can a system like that have, as one of its own programs, a procedure that correctly decides, for every program, "does this program admit a sound shortcut"? A procedure like that would be autopilot: hand it any program and it says yes or no, and it's always right.

The exact statement is narrower than the question in plain words, so I'll give it before the proof. In the theorem, "admits a shortcut" is any property of programs with two features: two programs that behave alike agree on it, and there's one program that has it and one that doesn't. Those two features are all the argument gets to look at. The diagonal doesn't see soundness. Each system that wants the wall picks its own property, and whether that property lines up with being entitled to soundness in Section 8's sense is a separate fact about that system, checked one system at a time if it's checked at all. A procedure that returns soundness records for

some programs, or only for restricted inputs, or with outside evidence, is not what the wall rules out.

So in the theorem "shortcut" is a label. The plain-words gloss at the end of Section 12, leaning on a record so the thing isn't worked out again, is about *how* a program gets its answer, and how is exactly what extensionality throws away: the theorem only gets to see what programs do, and it treats any two that do the same as one. A question like "is there a cheaper program that gives the same answers?" isn't the theorem's question, and a wall for it would need a different theorem. The nearest thing I know of is Manuel Blum's speedup theorem, from 1967, and it goes somewhere stranger: there are computable functions with no fastest program at all, where every program for the function has another that's faster on all but finitely many inputs. Blum's theorem is a fact about cost, not a wall about deciding, and nothing in this book proves anything in that territory. Each use of the wall also names its sameness. In the substrate theorem it's the substrate's own $\simeq$; on L it's what a term computes; on the host of the small machine Part III builds, in Appendix C.4, it's the partial function a program computes (Theorem C.20). A particular notion of shortcut gets the wall only after someone proves it behaves alike on programs that are the same in the sense in use, and holds of one program and fails of another.

The answer is no, under premises. The proof is the two index cards, and I'll give you the premises because they're the whole content. Suppose there were such a decider, one that says yes exactly when a program admits a shortcut. Build the flip: "if the decider says yes about me, behave like a program that admits no shortcut; if it says no, behave like one that does." If the system has a recursion theorem (a program can be handed its own code), and if that flip is one of the transformations the system can carry out, and if "admits a shortcut" depends only on behaviour (two programs that behave the same agree on it), and if there's at least one program each way, then there's a program that behaves exactly like its own flip, and whatever the decider says about it is wrong. Both cases close in a line each. This rules out a decider that says yes if and only if a program admits a shortcut, for any substrate with those parts. (A substrate here is the mathematical structure just described: programs, a run that may fail to give back a state, a counter with a step relation it never goes down along, an encoding of programs as data, a notion of "representable" transformation, and the recursion theorem for those. It says nothing about what the states are made of. A small instance is the nat family below.) It's Rice's trick one floor over, and I should be careful with that. Rice stated his theorem for classes of recursively enumerable sets, the sets a program can list, and it's usually taught now for the functions Turing machines compute. This wall is the same diagonal with different premises: it carries its own recursion theorem and its own list of representable transformations, and it picks the property. So it isn't a corollary of Rice

as Rice stated it, and it doesn't show that any property is harder or easier to decide than classical halting. And I want you to have one thing about its proof: it never touches the counter, the toll, or the step relation. It uses the recursion theorem, representability of the flip, the two witnesses, and extensionality, and nothing else.

I'll set out the setting and the wall now, with every word pinned down so you can poke at it.

**Definition 13.1** (Substrate)**.** A *substrate* has a type of states, a type Prog of programs, and a run map that takes a program and a state and either gives back a state or gives nothing. It has a counter $\mu$ on states and a step relation along which $\mu$ never decreases; an encoding enc of programs as states with a decoder dec such that $\mathrm{dec}(\mathrm{enc}(p)) = p$; and a predicate Rep on maps $f : \mathrm{Prog} \to \mathrm{Prog}$, the transformations the substrate counts as its own. Two programs are *equivalent*, $p \simeq q$, when $\mathrm{run}(p, s) = \mathrm{run}(q, s)$ for every state $s$. The last condition is the *recursion theorem*: for every $f$ with $\mathrm{Rep}(f)$ there is a program $p$ with $p \simeq f(p)$.

**Definition 13.2** (Shortcut predicate)**.** A *shortcut predicate* on a substrate is a property $A$ of programs with two programs $y$ and $n$ such that $A(y)$ holds, $A(n)$ fails, and $A$ is extensional: $p \simeq q$ implies $A(p) \iff A(q)$. For a Boolean function $d$ on programs, its *flip* is $f_d(p) = n$ if $d(p)$ is true and $f_d(p) = y$ if $d(p)$ is false.

**Theorem 13.1** (No inside decider)**.** Let a substrate carry a shortcut predicate $A$ with witnesses $y$ and $n$. Then for every Boolean function $d$ on programs with $\mathrm{Rep}(f_d)$ there is a program $p$ with $p \simeq f_d(p)$, and $d(p) = \text{true} \iff A(p)$ fails at that $p$. So no such $d$ is right about $A$ on every program. □

*Proof.* Hand the proposed judge $d$ to the flip construction. The flip is an allowed program transformation, so the recursion theorem supplies a program $p$ that behaves exactly as its own flip tells it to behave. Now ask the judge about that particular program.

If the judge says yes, the flip sends $p$ to the no-example $n$. Thus $p$ and $n$ behave alike. The question depends only on behaviour, and $n$ fails it, so $p$ fails it too. The judge's yes is wrong.

If the judge says no, the flip sends $p$ to the yes-example $y$. Now $p$ and $y$ behave alike. Since $y$ passes, $p$ passes too. The judge's no is wrong. Those are its only answers.

I used the recursion theorem to obtain this one self-matching program, the two examples to give the flip somewhere to go, and the rule that alike behaviour gets the

same verdict. The counter, the step relation and the encoding played no part. Any setting supplying that same fixed point for the flip gives the same contradiction. ■

So what's ruled out is a $d$ that says true exactly when $A$ holds, and whose flip the substrate can represent. Every part I named is a premise, and the theorem covers every substrate that supplies them all.

So the wall is there. The part I have to be honest about is *who counts as a decider*. The theorem rules out a decider whose flip the substrate can itself represent. It says nothing about a decider standing *outside*, and an outside decider can exist for the very same property, correct on every program. The line between inside and outside is the boundary of the sentence, and the boundary is Richard's lesson: the language you talk about a system in is a different language from the one it talks in. You'll see it in miniature in a minute, on the nat family, with a pencil.

Two forms of the wall live in this section, and they're honest in different ways.

The first is the *nat family*, a tiny substrate built so that program two is the Contrarian for a given candidate decider d, and for every d, d is wrong about program two. It's a *family*, one instance per d, and its recursion theorem holds by construction. It's the index cards made formal and nothing more.

□ **Theorem 13.2** (The nat family)**.** Fix any Boolean function $d$ on $\mathbb{N}$. Let the substrate $N_d$ have programs and states $\mathbb{N}$; program 0 returns 1, program 1 returns 0, program 2 returns 0 if $d(2)$ is true and 1 if it is false, whatever the input state, and every other program gives nothing; each program is encoded as itself; $\mu$ is constantly 0, so any step relation will do; $\mathrm{Rep}(f)$ holds when $f(p) = 1$ for every $p$ with $d(p)$ true and $f(p) = 0$ for every other $p$; and $A(p)$ holds when program $p$ returns 1 on input 0. Then it is not the case that $d(p) = \text{true} \iff A(p)$ for every $p$.

*Proof.* The recursion theorem of $N_d$ holds with the fixed witness 2: for a representable $f$, $f(2)$ is 1 or 0 according to $d(2)$, and program 2 returns exactly what that program returns. The witnesses for $A$ are 0 and 1, and $A$ is extensional because it reads only what a program returns. The flip of $d$ is representable by the definition of Rep, so Theorem 13.1 applies to $d$ itself. Directly: $A(2)$ holds exactly when $d(2)$ is false. ■

Now push on it once more, because the nat family shows the outside decider in three lines, and you can do them yourself. Stand outside $N_d$, knowing $d$. Write down a new decider $e$: say yes on program 0, say "not $d(2)$" on program 2, and say no on everything else. Check it against $A$. Program 0 returns 1, so $A(0)$ holds and $e$ says yes. Program 1 returns 0, so $A(1)$ fails and $e$ says no. Program 2 returns 1 exactly when $d(2)$ is false, and that's what $e$ says. Every other program gives nothing, so $A$

fails and $e$ says no. So $e$ is right about every program. Does that break the wall? It doesn't, and the reason is the part that matters: in $N_d$ a flip is representable only if it says 1 exactly where $d$ says yes, so the flip of $e$ is representable only if $e$ and $d$ agree everywhere, and they disagree at program 2. The correct decider exists; its flip just isn't one of the substrate's own moves. (It's three lines and a pencil, and you just did it.) One and the same property is undecidable from inside and decided cleanly from outside, at the same time. That's not a paradox, because it's two different people asking.

The second form is the one where the recursion theorem is *proved* from the reduction rules rather than assumed. It's on *L*, the weak call-by-value lambda calculus that Yannick Forster and Gert Smolka presented in 2017 as a model of computation, and which a group at Saarland has since used as the seed of a whole library of undecidable problems. Here L is built from its reduction rules. The recursion theorem is proved, Rice's theorem follows, halting follows, and the diagonal is reproved in the shape of the substrate theorem, for any property of programs that depends only on what they compute and that holds of one program and fails of another. The statement mentions no shortcut. So on L the result is the diagonal for one fixed model, and I don't plug L into the substrate theorem, for this reason: a substrate's run map has to answer, with a state or with nothing, and for L that map is the halting function itself, which no term of L computes. Using it would take something from outside L, and L is in this book as the model where nothing is taken from outside. So on L I run the argument again from scratch instead of applying the general one. Both are true; they're different sentences.

I'll say what that string of words means, because L is the strangest thing in this book that computes, and the smallest. The lambda calculus is Alonzo Church's, from the 1930s, and it's a programming language with one idea in it: functions. A term is one of three things: a variable; a function, written $\lambda$ followed by its body, the body saying what to do with whatever you hand it; or one term applied to another, written side by side. Running a program is one move, over and over: when a function meets an argument, put the argument into the body where the variable was. The lambda calculus has no numbers, no memory, no instructions, and in 1937 Turing published a proof that it computes exactly what his machines compute. "Weak call-by-value" is two house rules about the order of moves: an argument is run down to a finished function (a *value*) before it's put in, and nothing runs inside a function that hasn't been handed anything yet. Variables are written as numbers counting how many $\lambda$s back they were bound (a trick of Nicolaas de Bruijn; it means two variables can never get mixed up by sharing a name), and a term is *closed* when every variable in it is bound inside it, so it leans on nothing outside. Last, to hand a program to a program, L needs a way to write any term as data, the way a program's source is a string you can pass around. The one used here is the *Scott encoding*, named for

Dana Scott, a student of Church's who went on to build mathematical models of the lambda calculus itself. As far as I can trace it, the encoding first appeared in lecture notes of his from Berkeley in 1962 that he never published, as a way to write data; Torben Mogensen carried it over to lambda terms themselves in 1992, and that's the form used here. All you need about it is that every term $t$ gets one *code*, written $\ulcorner t \urcorner$, and the code is itself a closed value. Two tools get built on top of that in the proof: a *quoting* term $Q$ that takes the code of a term to the code of that code (a program that prints its own source needs exactly this move), and a *fixed-point combinator*, a term that lets a function call itself, which is how $Q$ works its way through a code of any size.

Do two of these on paper before I give you the general rule. Write $I = \lambda x.x$: its instruction is "return whatever you were handed." Hand $I$ its own card and you get $I\,I \to I$, and it stops. Now write $D = \lambda x.x\,x$: "make two copies of what you were handed and apply the first to the second." Hand it its own card. Putting $D$ in for both copies of $x$ gives $D\,D \to D\,D$. You're back at exactly the same instruction and argument, so the rule never finishes. Both arguments were already functions, so neither example sneaks any work under a function that hasn't been handed anything.

The variable numbers are what keep the copying from getting confused. In $\lambda x.\lambda y.x$, the last $x$ points past the nearest $\lambda$ (the one for $y$) to the one before it, so its number is 1; in $\lambda x.\lambda y.y$ the last variable has number 0. Rename $x$ and $y$ however you like and neither recipe changes. The codes below are a way of putting a whole recipe on a card that another recipe can read.

**Definition 13.3** (L, as built here)**.** A term of L is a variable (a de Bruijn number), an application $s\,t$, or an abstraction $\lambda s$; the values are the abstractions. One step is weak call-by-value, left to right: $(\lambda b)\,v$ steps to $b$ with $v$ substituted when $v$ is a value, a step inside the left part of an application is a step of the application, and so is a step inside the right part once the left part is a value. A term $s$ *evaluates to* $v$ when $s$ reaches $v$ in some number of steps and $v$ is a value; $p \equiv q$ when $p$ and $q$ evaluate to the same values. Every term $t$ has a code $\ulcorner t \urcorner$, its Scott encoding, which is a closed value. A map $f$ on terms is *L-computable* when some closed value $s$ has $s\ulcorner p \urcorner \equiv f(p)$ for every $p$. A term $D$ *decides* a property $P$ of terms when, for every $p$, $D\ulcorner p \urcorner$ reaches $\lambda\lambda.1$ (true) if $P(p)$ holds and $\lambda\lambda.0$ (false) if it fails. $P$ is *extensional* when $p \equiv q$ and $P(p)$ give $P(q)$.

**Theorem 13.3** (Recursion, Rice and halting for L). (1) For every closed value $s$ there is a closed term $t$ that reaches $s\ulcorner t\urcorner$; so every L-computable $f$ has a term $p$ with $p \equiv f(p)$. (2) Let $P$ be extensional, with $P(y)$ and not $P(n)$ for closed terms $y$ and $n$. Then no closed term decides $P$. (3) No closed term decides halting, the property of evaluating to some value. □

*Proof.* (1) First build a program that can assemble the code of the very application I am about to run. The quoting program $Q$ takes a term's code and returns the code of that code. Appendix I gives the idea of its construction, and the companion volume follows it through the three kinds of term. The program mkapp takes two codes and joins them into the code of an application.

Make $A$ do this when handed a code $x$: quote $x$, put that quoted code beside $x$ using mkapp, and hand the result to $s$. Written out,

$$A = \lambda x.\, s\,(\text{mkapp}\ x\ (Q\,x)).$$

Now run $A$ on its own code, and call that whole application $t$: $t = A\ulcorner A\urcorner$. The first step puts $\ulcorner A\urcorner$ where $x$ stood. Quoting it supplies the code of $\ulcorner A\urcorner$; joining the two codes supplies the code of $A\ulcorner A\urcorner$. That is the code of $t$ itself. The computation therefore reaches $s\ulcorner t\urcorner$, as promised. All the pieces were closed, so $t$ is closed too. If $s$ computes $f$, the resulting behaviour is exactly $f(t)$.

(2) Suppose a closed program $D$ decides $P$. Build another program that asks $D$ about the code it receives, then runs the no-example $n$ after yes and the yes-example $y$ after no. In this language you must keep the two branches wrapped as functions until the answer chooses one; otherwise call-by-value would try to run them before choosing. The following term does that:

$$\lambda x.\, D\,x\,(\lambda_.\, n)\,(\lambda_.\, y)\,(\lambda z.z).$$

The final identity function is the harmless argument that opens the selected wrapper. Thus a true answer runs $n$, and a false answer runs $y$.

Apply part (1) to this program, obtaining its self-matching $t$. If $P(t)$ holds, the judge says true and $t$ behaves like $n$, where $P$ fails. If $P(t)$ fails, the judge says false and $t$ behaves like $y$, where $P$ holds. Either answer contradicts the requirement that alike behaviour has the same verdict. No such $D$ exists.

(3) Halting depends only on which values a term reaches. The identity $\lambda z.z$ is already a value. The term $(\lambda x.\, x\,x)(\lambda x.\, x\,x)$ takes a step back to itself each time, and is never a value. These supply the yes-example and no-example for part (2), so halting has no closed decider either. ■

The construction in that proof isn't mine, and the credit belongs right here. Forster and Smolka's L development of 2017 has the Scott encoding of terms, a closed quoting term with exactly this specification, the code of a term to the code of its code, and a fixed point built the same way: their $t$ is $C\ulcorner C\urcorner$ with $C = \lambda x.\, s\,(A\,x\,(Q\,x))$, so their $C$ is my $A$, and their $A$, which builds the code of an application, is my mkapp. They say they follow the proof in the Dutch logician Henk Barendregt's textbook on the lambda calculus (1984), where statement (1) is the full lambda calculus's second fixed-point theorem, and part (2) is what they call Scott's theorem. The term construction is given in the companion volume, with its idea in Appendix I; the route is theirs, and Barendregt's before them.

So I've given the wall on paper, on a family built to fit it, and on one outside model. The machine built in Part III (Section 15) fills in this same shape: on its host machine, the one the universal program of Appendix C.1 runs on, which has a counter for every whole number, programs are numbers, Kleene's recursion theorem is a theorem, and so is Rice's, and the fixed point can be taken to agree on what the program computes and also on the ledger and the flag at the end of every run where they stop (Appendix C.4).

The table below lines up the picture all of that fills in against Turing's, one column per axis.

| **Time axis** | **Structural axis** |
|---|---|
| counts: steps | counts: $\mu$ |
| asks: does it halt? | asks: does it admit the shortcut? |
| the wall: no Turing machine decides halting | the wall: no decider whose flip is representable, under a recursion theorem |

Read the table as two uses of one diagonal, each under its own premises, and as saying nothing more than that. The structural wall never uses $\mu$; the $\mu$ entry only says what that axis counts. The nat result is a family indexed by the candidate decider, and L's results are about one fixed model and properties that mention no shortcut. Neither hands you a second fixed undecidable problem, and neither separates Turing degrees: a property can sit behind the wall from inside and be decided cleanly from outside, as the pencil argument on $N_d$ showed. And "admits the shortcut" in the right-hand column is the theorem's sense, any behavioural property with a yes and a no. "Has a cheaper way to the same answer" is a second question, and the column has no wall for it.

## 13.5 Where the diagonal applies

The three versions put the candidate procedure in different places. In the general substrate argument, the substrate must represent the candidate's flip. In the natural-number family, each candidate $d$ supplies the substrate whose predicate defeats it. In L, the candidate is a closed term and the question is a nontrivial extensional property of closed terms. The reduction rules and recursion argument give Rice's theorem there.

The pencil calculation on $N_d$ is useful precisely because it is outside that substrate. To run the diagonal inside, I need the representability and recursion conditions stated in the theorem.

## 13.6 What this does to the book

It does two things. It keeps me honest: nothing in what follows claims that a system with a record can check its own shortcuts on autopilot, because that's a wall, the same wall, and I showed you the bricks. And it tells you what a record *can* do, which is the thing Part II has been about since the counter: it can carry a shortcut *with its receipt*, so that the question "was this earned" is answered by an event you can point at. No decider has to exist for that, which is lucky, because on a system with a recursion theorem no inside one can. The diagonal says such a system can't automate the judgement from inside, for any behavioural property with a yes and a no. The latch says you can record the commitment. Those are different jobs, and a record does the second.

The diagonal has a famous cousin that looks as if it ought to bite a record too, so let me walk right up to it. Gödel's second theorem says a consistent system strong enough for arithmetic can't prove its own consistency, and Martin Löb, a logician answering another logician's question, sharpened that in 1955: if such a system proves "if this is provable, it's true," it already proves the thing itself. So can a record vouch for itself, and does Löb stop it? My first guess was that Löb stops it, since a record looked like a system talking about itself. I landed somewhere else, and mostly it's a relief. Hold two things side by side: "there's a receipt for this somewhere," and the receipt in your hand. Löb's theorem is about *provability*, the bare claim that some proof exists, the receipt somewhere. A raised Boolean flag is not, by itself, the receipt in your hand. From a clean start the earned-history theorem says there was a passing check, commitment and certification in that order. The flag does not identify the claim or the steps, and it need not store a proof of the checked claim. The native table and channel carry more information; exporting a proof requires an interface that actually supplies one.

Sergei Artemov's distinction between implicit provability and explicit proof terms is useful here, but neither a flag nor the word "receipt" turns one into the other. The guarantee in this book comes from an external proof about the checker, the clean start and every permitted operation. It is not an internal proof of the system's own consistency, and does not evade Gödel's or Löb's hypotheses.

Push on that and it gives, because a receipt is only as good as whoever wrote it. Its trust rests on the checker meaning what it says, and that gets settled once, from outside the machine. That's where I chose to stop this machine's chain of reasons. Gödel's second theorem does not forbid a machine from checking a proof about a checker. What it limits is a system's ability to prove *its own* consistency, when the system is strong enough to do arithmetic and its axioms can be listed by a machine. Eliezer Yudkowsky and Marcello Herreshoff's 2013 draft on agents that trust their own rewrites describes a related problem, the Löbian obstacle: how an agent can justify trust in a successor's proofs. That's a cousin of this question, and it doesn't settle it, and it doesn't make my design the only one possible. So I didn't get rid of the receipt somewhere. I moved it out of the machine and handed it to whoever reads the record.

One more thing follows from the same picture, by a short argument. Can a record rest on itself, the way the sentence "this sentence is certified" would like to? It can't if it's earned and permanent. The check behind the certificate that put the record up happened while the record was still down, because the record goes up only after that check passes, and once up it never comes back down, so it goes up only once. So no claim whose truth needs the record already up can be the claim the record went up on. (A later certificate, with the record already up, can stand on such a claim; it raises nothing.) Saul Kripke's 1975 theory of truth calls a sentence like that, one that only ever leans on itself, ungrounded, and earned records are grounded in his sense: every one bottoms out in a check that didn't lean on it. Caspar Oesterheld's 2019 work on programs that cooperate by simulating each other needed a grounding step of the same kind to stop an infinite regress. I haven't written this argument out anywhere else; the claim is exactly what this paragraph says, and it's short enough to check by hand.

I want to draw one last distinction, because the two are easy to conflate and conflating them is the easiest way to misread this book. There's a *second* wall, nothing to do with the diagonal. It's Section 7's kind of wall, a window with a colliding pair in it. Two states that differ in their certification reading can look identical through a window that doesn't show that reading, the stamp inside the closed drawer and the drawer opened and shut, and then no function of the window recovers the reading (Theorem 7.1: a decoder has to give one answer on everything the window lumps together). This wall comes from missing information. Nothing about it is undecidable; it holds for finite things with nothing to compute; and it

sits *beside* Rice. None of Rice is underneath it. The property the diagonal uses (any behavioural property of programs with a yes and a no) is a different kind of thing from the reading the window hides, and nothing here says the two are the same. So there are two walls, on two different axes. The diagonal is about what no inside procedure can decide. The window is about what no outside observer can see. I'll need both, and I'll try not to call one by the other's name again. The machine I build later has a colliding pair of exactly that kind (Theorem 15.17).

PART III

# Established

*Checking comes before the record.*

Part III is step four, the sharpest case. A record that a claim was checked and closed is a distinction that stays, so it pays what Part I found keeping takes, and it's the case where you can point at the kept 0 and at what it cost. It starts with what *established* means, and why two computers with the same bits can differ in it. Then it builds the smallest example I could find that meets the definition: an ordinary computer of two counters, with a record layer on top whose flag is the blank cell, six kinds of instruction in all. Read through its window, the line and the two counters, it's the ordinary computer again, run by its moves for the instructions and blind to the record and the ledger: the record layer is the part the slice forgets, and the appendix proves that every computer with a move for each two-counter instruction is the slice of a certified Thiele-complete machine (Theorem D.4). The step that raises the flag is the write of a 0, a merge in the record layer, and that's why it's charged; the order the step enforces, check, then commit, then certify, is what makes this record's raise mean a claim held. That's what I mean by certification, and it's the layer this part adds to the bar Part II set. Then how far the example reaches, the real systems that already put a price on a write, the thesis about which kept distinction is worth paying for, and what the example's counter has to do with heat. Those claims are about the model; the example is where you can check them line by line. The ordinary computer's own steps merge too (counting a counter down lands two states on one), and their physical costs depend on the implementation; the ledger leaves them alone.

CHAPTER 14

# What established means

Parts I and II were about any kept distinction at all. This chapter picks one: the distinction between an answer that was checked and the same answer that wasn't. It's the case the first pages of the book called the sharpest, and it's sharp because the two answers are the same bits, and everything that differs is in what was kept.

## 14.1 Two computers that end up the same

You met these two computers in one paragraph at the start of the book. Now I'll take them slowly: what an ordinary computer does about what Part I found keeping takes, and why that matters to anybody who isn't a physicist.

Put two computers on a table. Hand each of them the same list of a thousand numbers.

The first one sorts it. It compares, swaps, walks the list again and again until nothing is out of order. Then it does one more thing: it walks the whole list front to back and checks that every number is no bigger than the one after it. That check could have failed. It passes. Then it tidies up after itself, wipes its scratch work, and leaves the sorted list in memory. (That check compares neighbours, so what it settles is that the output is in order. That the output holds your thousand numbers and no others is a second claim, and it would take a second check comparing the output with the input. When I say sorted below, I mean in order.)

The second one never looks at your list. Somebody typed the sorted version into it earlier, same thousand numbers, same order, and it puts them in memory.

Now wait until both are done and read the retained digital state we have chosen to compare: the list, the working registers and the control position. Arrange for both programs to finish at the same control point and clear the scratch fields. Those endpoints can agree. The first compared the entries and checked their order. The second loaded the same answer without doing either. Wheel the boxes out, swap them and bring them back. From those fields alone, neither of us can tell which run happened.

That is a construction with a specified observation. It does not establish that every microscopic state of two physical computers agrees: the room, heat, clocks or

retained logs may carry other evidence. And ordinary computers can keep a checked flag or a protected log. These two programs were arranged to erase the distinction from the fields being read. That choice, and what it prevents the next reader from answering, is the example.

Who cares? Anybody who's been handed a sheet that says "all tests passed." Anybody who's looked at an inspection sticker on a windshield: the sticker looks the same whether somebody checked the brakes or somebody sold the sticker. An audit, a proof, a ledger, an AI that sounds very sure of itself: every one of them is a pile of bits that could have come out of either computer, and the question that matters is the one the bits can't answer. Did something establish this, or did it just print it? I'll come back to every one of those later in this chapter and say which of them this book actually helps with, and I'll warn you now that it's fewer than you'd hope.

I'm going to call the first run's relation to that answer *established*: it checked the order of this list. The second run has the answer without that relation. The list's order is there either way. What can a machine keep of the difference between the runs? The rest of this chapter takes that question apart.

It belongs in a book about a 0 because a record that the check ran would be a distinction that stays, a 0 in some cell that says "checked." These two computers did not keep that record. A machine that does keep it gives you something more to inspect.

## 14.2 Look harder

The obvious answer is: just look harder. Don't look at the end, look at everything. So make the computers as lookable as computers get. Scale them up to the size of a warehouse, every wire a lamp that's lit or dark, every switch something you can walk up to and touch. Nothing is hidden in a chip too small to see. Put both of them side by side and walk the floor.

While they run, sure, you'd see the difference. One has a whole wing lighting up, comparing and swapping, and the other just copies. But they finish, and the comparing wing goes dark. Now walk the floor and look at every single lamp. The sorted list is spelled out in lit and dark lamps in both, the same pattern, and everything else is dark in both, because both cleaned up.

There is no lamp anywhere that says "a check ran here and passed." It isn't hidden, and it isn't encrypted; it just isn't there. You can see every retained digital field in this drawing, and none records the earlier check. Looking harder doesn't help, because the problem isn't that you can't see enough. It's that nothing was kept.

The shape I want you to carry through the rest is that the problem was never visibility; it's keeping.

### 14.3 So keep the trace

Then the fix is obvious, isn't it? Don't let the run go. Keep the trace: every comparison, every swap, the check and its answer, written down as it happens. Wheel the computers out on the cart and back, read the traces, and you'll know which one sorted. Problem solved?

It's the right instinct, and it gets most of the way to this book. But take it apart, because "just keep the trace" has three things wrong with it, and each one is a piece of what comes later.

First, a trace is bits, and bits can be typed in. Whoever typed the sorted list into the second machine can type a trace in too: "compared, swapped, checked, passed," a few thousand lines of it, every one of them plausible. A trace written by a program is something the program produced, unless some enforced rule constrains how the program can write that log. So keeping the trace doesn't answer the question; it moves it. Did something establish this trace, or did it just print it?

Second, the way out of that is to take the pen away from the program. Don't let the program write its own trace. Have something underneath it write the trace, one step at a time, as the steps actually happen, so nothing a program does can add a line that didn't happen or erase one that did. Airplanes do a version of this. Flight recorders go back to a prototype David Warren built in Melbourne in the 1950s. He was an Australian scientist at the Aeronautical Research Laboratories, working on jet fuel, and after a string of unexplained airliner crashes he wanted a box that would hold the instrument readings and the voices in the cockpit for whoever came looking afterwards; I know his story from museum and encyclopedia accounts, and I haven't read his own report. The recorders that came out of that idea preserve instrument data and cockpit sound. The sound can include a person's account, correct or mistaken. A recorded sensor reading can also be wrong about the aircraft. What the recording supplies is evidence of what those channels carried, subject to the recorder and sensors working as specified. Crash protection helps that evidence survive. The flight recorder has the right shape, and that shape isn't mine. Later in this chapter it gets its name from computer security, a reference monitor, and I'll say outright that a machine built that way would be doing what this book describes. What the book adds is narrower than a rulebook for everything underneath. It's about the records that go up and never come back down: what that something has to keep for good, and what keeping it takes.

Third, a fixed memory cannot retain every possible history of unbounded length with exact recovery. Give the run one independent yes-or-no choice per tick: after $n$ ticks there are $2^n$ histories to distinguish. A fixed number of bits eventually runs out. A completed finite trace can fit, and a predictable trace can compress. The obstruction needs unbounded independent possibilities, not just the word "trace."

Flight recorders make a practical choice about the history to retain. A voice recorder loops, replacing older sound when its capacity is reached. Thirty minutes, two hours and twenty-five hours are different retention windows; the applicable requirement depends on the aircraft and rule. The FAA's 2026 final rule extends twenty-five-hour recording to specified newly manufactured aircraft. Whatever window is chosen, it is a window. It does not imply that no other machine can store an entire finite run.

The fingerprint is the sharpest reply, so take it seriously. Fold every new line of the trace into a fixed-size scramble of everything before it, and you've got a hash chain. The timestamps of Stuart Haber and Scott Stornetta, two cryptographers at Bellcore whose story comes later in this chapter, work that way; so do the audit logs the cryptographers Bruce Schneier and John Kelsey designed in 1999; and so do the measurement registers in the security chip a lot of computers carry, the TPM. For the PCRs and lifecycle under discussion, an extend folds a measurement into the previous value. Reset permissions depend on the register, platform and locality; some PCRs are resettable during operation. A claim about an unbroken history must specify which resets and writers its policy permits. That really does keep something about the whole trace in a few dozen bytes. But look at what it keeps: it keeps a binding, and the content isn't in it. To read the history you still need the trace stored somewhere, and the fingerprint only tells you whether a copy somebody hands you is the one that went in. The binding holds only because nobody knows how to find two traces with the same fingerprint, which is a bet on how hard a search is. No law of nature backs it. That part isn't only my opinion. The computer scientists Moni Naor and Guy Rothblum proved a sharp version of it in 2005, and they even call the thing a fingerprint: keep a small private fingerprint of a big file somebody else stores, check the file by reading only a little of it, and against an opponent with unlimited computing the size of the fingerprint times the amount you read has to be on the order of the whole file. Do even slightly better than that against an opponent with limited computing, and you've shown that one-way functions exist, which is the hardness bet in its barest form. Their setting is a stored file. A growing trace is a different setting, but it's the same bet, with its terms written down. And squeezing traces that keep growing into a fixed number of bits has to land many traces on one fingerprint, and that's a merge, hidden behind how hard it is to find which traces land on the same fingerprint. It's the same merge as in the example of Part I (Section 3) where four people have to sit in three chairs. Fingerprints are great, but they're not a way out of forgetting; they're a very clever way of forgetting.

So the real question was never "save everything." The real question is: what's the smallest thing worth keeping for good, and what does keeping it take? The answer this book ends up with is the commitment, in the database's sense, a question closed, and the trace drops away. (The cryptographer's sealed envelope also goes by the

name commitment. It's a different thing, and both come up later.) What's kept for good isn't every comparison the sort made, just the one fact that a check of this very claim passed and the question got closed. And the part that matters most is that keeping even that one fact for good takes something on every machine. It's paid by forgetting something somewhere else or by taking fresh states the machine never gets back, and on a finite machine the fresh states run out, so it's the forgetting. That's Part I's counting of states again, and Part I counted the fresh states.

So the premise was never that computers can't keep records. They keep records all day. It's that nothing in the rules of an ordinary computer makes the record paid for, permanent, or earned, and those three come in at different places, so I'll say where. Paid for is the model's one rule: in the Thiele Machine, the step that raises the record has to cost something. Permanent is a premise, the one Part I's counting of states needs, and one of the conditions Part II puts on a record. Earned, meaning a check of this very claim passed first, isn't in the model's core at all. It's what the step of the small machine in the next chapter enforces, proved for that machine and every program on it run from a clean start. One more word rides along with permanent, and it'll come up when I line my record up against other people's: dated. A record that goes up and stays up went up at one particular step of the run and not another, even though the record doesn't write down which step. So dated, in this book, means it has a place in the run's history. Carrying a timestamp is something else, and my record doesn't do it. A machine that charges one for every step it takes meets the first rule, and the charging does nothing toward the others, and I'll keep pointing that out.

*Open the gauge wider*

Go back to the Sorter. It checked the order of the list. Now ask it another question: did it also check that none of the original numbers went missing? The list on the screen can be the same whether that second check happened or not. One lamp saying "checked" will not tell me which work was done.

Put a sheet beside the scratchpad, with a named box for each check. Order gets a box. Membership gets a box. Each other check you want to remember gets its own. When I come back, I want to know which boxes were ticked. If two runs tick different boxes and leave the same working value, the full machine has to leave me something that tells them apart. That is more of the fibre above the same tabletop picture, brought within the question.

Here are the moves. WRITE puts a value $b$ on the scratchpad and starts a fresh sheet, even if I write the same value again. CHECK $i$ tests its named predicate $P_i$ and records its name when it passes. A failed check changes nothing in this monitor. IDLE leaves both scratchpad and sheet alone. The reader can see the working value

and ask about every check name: did it pass since the last WRITE? The sheet must answer yes when it did and no when it didn't.

Take a finite list of named checks, and let $m(b)$ count the ones whose predicates are true at $b$. Each of those can be run or left unrun. The truths themselves may overlap; it is my choice to perform each check that is independent. Four true predicates give sixteen possible ticked sheets above the same value. The general count follows in the same way.

□ **Theorem 14.1** (Exactly how many current sheets)**.** For a finite nonempty working set $B$ and this WRITE/CHECK/IDLE interface, an exact host needs at least $2^{m(b)}$ reachable full states above each $b$. The minimum total reachable state count is

$$\sum_{b \in B} 2^{m(b)}.$$

*Proof.* Hold the working value fixed and start with two named checks that would pass there. Run neither, the first alone, the second alone, or both. The working value is the same in all four cases. The required answers are not. In particular, the first-only case and the second-only case cannot finish at the same full state: ask whether the first check ran, and that one state would have to answer both yes and no.

Each further check that would pass doubles the choices. For every sheet you already had, you can leave its new box blank or run that check and tick it. Starting from one empty sheet, $m(b)$ such choices give $2^{m(b)}$ sheets. Every sheet is reachable: WRITE the value $b$, then perform exactly the checks whose boxes it ticks. Any two different sheets disagree at some named box, so they need different full states. States showing different working values are also different. Add the required states for each value to get the displayed sum.

Now build a machine with exactly that many states. A state consists of the working value and its ticked sheet. Include every sheet that ticks only checks true at that value. WRITE replaces the value and clears the sheet. CHECK ticks its own box if its predicate is true, and otherwise leaves the sheet alone. IDLE changes nothing. These rules never produce a forbidden sheet.

Start with the empty sheet. After a WRITE it is empty again, exactly as the task requires. After a successful CHECK it has precisely that success added; a repeated success changes nothing further. A failed CHECK adds nothing, and IDLE adds nothing. If the sheet described the history correctly before a command, it therefore describes it correctly afterward. Following any finite list of commands proves the promised answers. The WRITE-and-check runs above reach every state I included. ■ The lower count is attained.

Try it with just one box. The workspace holds false or true, and CHECK asks whether it is true. The whole monitor fits in three states: false, true-unchecked, true-checked. WRITE false goes to the first. WRITE true goes to the second. CHECK moves the second to the third and leaves the other two alone. IDLE leaves all three alone. Start at false. WRITE true reaches true-unchecked, and then CHECK reaches true-checked; all three states are reachable. Now take any word of instructions. A WRITE makes the verdict false and starts a new sheet, even if it writes the same value. A CHECK on false leaves the verdict false; a CHECK on either true state makes or keeps the verdict true. IDLE preserves it. These are all the cases, and each matches exactly what happened since the last WRITE. Applying that argument after each instruction proves the promise for every finite word. In particular, the first arrival at true-checked must use CHECK: neither WRITE nor IDLE can enter it from another state.

The two true states also show you the merger. CHECK sends true-unchecked and true-checked to the same next state, right above the same working value. A larger implementation can move that merger later: let CHECK enter a finite region of accepting states in the fibre and keep it there, and some two distinct states must share a CHECK successor. WRITE can still clear the current sheet. Permanence under that one check and permanence under every instruction are different promises. Call that finite accepting region $H$ and the entering state $u$. The move sends all of $H \cup \{u\}$ into $H$. There are $|H| + 1$ distinct sources and only $|H|$ possible targets, so two sources share a successor. The move preserves the working picture, so both sources and their common successor lie in that same fibre.

*Keep the old receipts when the page is wiped*

Now wipe the scratchpad and use it again. I still want the receipt for the first page. Then use it a third time. I want that receipt too. Think of carbon copies filed before each sheet goes in the bin. On the tabletop, you can be back where you started. In the file, you have a past you didn't have before.

On each of $n$ editions, give me the choice to certify it or leave it uncertified. Bring every choice back to the same final working picture $b$. Then ask the machine about any edition I name. It has to answer whether that edition was certified. Two histories that disagree on that answer need two different full states, however alike their tabletops look. There are $2^n$ possible yes-or-no lists to distinguish.

**Theorem 14.2** (Room for independently retained editions)**.** Under those realization and exact-query requirements, the fibre above $b$ contains at least $2^n$ reachable states. If one fixed host meets the requirements for every finite $n$ at that same $b$, no finite set covers its reachable fibre. □

*Proof.* Two different lists of length $n$ differ at an edition. Equal endpoints could not answer that edition's query differently, so the endpoints are distinct. A putative finite cover of size $K$ fails once $2^n > K$. Every endpoint being counted has the same working view $b$. ∎

You can build the witness directly. Keep a Boolean workspace, the answers for the completed editions, and the current stage of CHECK, COMMIT and CERTIFY. On a true workspace, run the chain for an edition you choose to certify, or skip it for one you don't. Then WRITE true to start the next edition. Every choice brings you back to the same true workspace and a fresh current stage. Open the file and the choices are still there, in exactly the places you made them.

Put the whole machine on the table. There is the current page, holding false or true. There is a marker beside it, with four positions: raw, checked, committed, published. And there is a numbered file of yes-or-no answers for pages already put away. Start with a true page, the marker at raw, and an empty file.

CHECK can move the marker from raw to checked, but only if the page says true. COMMIT can move it from checked to committed. CERTIFY can move it from committed to published. Press any of those buttons at another stage and it leaves the arrangement alone. IDLE always leaves it alone. Those are all the ways to move the marker without replacing the page, so there is no shortcut from raw to published.

WRITE does three things, in this order. It files yes if the marker is at published and no otherwise. It puts the requested value on the page. Then it returns the marker to raw. The answer goes into the file before the marker is reset, and no command alters an answer already filed.

Check what that means after any command. CHECK, COMMIT and CERTIFY move the current page through exactly the named stages; they do not touch earlier pages' answers. IDLE leaves both alone. WRITE closes the current page with the answer its stage warrants, then opens a raw page. Start with no earlier pages and follow those rules one command at a time: the file always answers the completed-page questions correctly, and the marker gives the current page's stage.

To file a yes, run CHECK, COMMIT, CERTIFY, then WRITE true. To file a no, use WRITE true alone. Either choice leaves a true page with the marker at raw, ready for the next choice. After $n$ choices the page and marker are identical in every run, while the file holds exactly the $n$ answers chosen. Every yes-or-no list is possible, and different lists are different full states.

Keep asking for another edition, with the same freedom of choice and the same right to read back every answer. Whatever finite number of full states you offer above $b$, a long enough history outruns it. The tabletop can keep coming back to one picture. The machine that keeps all those choices cannot fit its reachable fibre into a finite set. That is where the infinity comes from.

The old yes-or-no gauge still has a job. Ask whether the file contains any certification at all and it becomes a permanent lamp: once yes, always yes. Ask whether today's sheet is certified and wiping that sheet starts the answer over. The small native machine in the next chapter has the permanent lamp. I will look behind it at the whole native state, then count which publication histories even that larger view leaves out. That camera sees more than the working value here; the next count will name its fibre just as explicitly.

### 14.4 Computed, stored, established

Three words get used like they're one.

*Computed* is about the steps. The first machine computed the sorted list. The second didn't.

*Stored* is about the bits. Both machines stored it.

*Established* is neither, and that's the whole problem. It isn't the steps, because they're gone. It isn't the bits, because they're identical. It's something like this: the first machine has a right to lean on "this list is sorted," because it ran a test that could have failed, the test passed, the test was about this very list, and then the question got closed. The second machine has the same sentence and no right to it. That's a true fact about the two of them, and it's written down nowhere, in either one. (Whose right, strictly? A machine doesn't lean on anything, any more than a filing cabinet does. It's the right of whoever reads the first machine to lean on what it says, and the machine is where the grounds for that right would have to be kept. "The machine has a right" is shorter, so I'll keep saying it, and now you know what it's short for.)

I'll hold four words still from here on. A *check* is the test, the thing that could say no. A *commit* is the closing, the question shut for good, about that edition of that thing. The *record* is whatever stays behind to say so. And *established*, in the sense a machine can carry, is all three, about one thing. The first machine had the check and the closing and kept no record of either, which is the problem above. Where the closing and the record are one act, as in a database's commit, writing the record is the closing. The small machine keeps them apart, a commitment the next one replaces and a certificate that stays. The book cashes this working sense at three depths: a paid closing in the bare model; an object with a checked relation and a paid event, in Part II (Section 8); and the whole chain on the small machine, in the next chapter. Each place says which one it means.

That raises a fair question: what if the second machine, handed the typed-in list, runs the check on it? Then it has what the first machine had, and if it closes the question and keeps the record, it has established that the list is sorted. It didn't do the sorting and it doesn't need to. The line I'm drawing is between checked

and unchecked. Whether the list was computed or copied doesn't enter into it, and "without doing either" up there was about the copier that never looked. So whatever this book ends up counting, it isn't how hard anybody worked.

People say "a right to lean on it" about a lot of things that turn out to be vibes, so I had to find out whether this one was anything. I went looking, and I'll go through what I ran into.

### 14.5 The clerk who forgets everything

The picture of a computer I used in Section 5 is a person with a pencil, a strip of paper and a short list of rules. So I went back to the source, Alan Turing's 1936 paper, to see how far he actually takes it.

He takes it a long way. Back then a "computer" was a job, a human doing sums by hand, and Turing modelled that person. He's careful with the person's mind: whatever the clerk knows, beyond a handful of states, has to go down on the paper. Then he goes all the way. He imagines the clerk getting up, walking off, forgetting everything, and coming back later. To pick up where he left off, the clerk leaves himself "a note of instructions," and Turing says that note *is* the clerk's state of mind, as far as the computation is concerned. Marks on paper and a note are the whole state.

My two computers are sitting right inside that. Suppose the clerk didn't write the note. Somebody slipped it under the door, in the right form, with the right instructions. He comes back, reads it, carries on, and every mark he makes from then on is exactly what he'd have made if he'd written it himself. Turing's model can't tell the clerk who did the work from the clerk who was handed a note saying he did. It was never asked to. Turing wanted something you could check without crawling into anybody's head, and a note on a desk is about as checkable as it gets. That's why the paper still works ninety years later. The chosen note omits its provenance. A Turing machine can retain additional provenance fields, and its transition rules can restrict how those fields are written. The omission belongs to this note, not to the reach of the model.

So the founding picture of computing is a man who leaves his desk, forgets everything he was doing, and trusts a sticky note. I relate to him deeply.

And now the question nobody needs answered to follow the argument, which is exactly the kind I can't leave alone: who were those clerks? They were real people. In the late 1800s the Harvard Observatory hired rooms full of women to do the arithmetic of the sky, cataloguing stars by hand for a fraction of a man's wage, and some of them made discoveries of their own, Henrietta Leavitt's way of telling how far away a star is among them. (I took the wages and the discoveries from Dava Sobel's history of those women, and from the historians at the American Institute of

Physics, who put the men's hourly pay at nearly five times the women's.) Rooms like that ran artillery tables in both world wars; that part I took from David Alan Grier's history of human computers. So when Turing modelled "the computer," he was modelling somebody's actual afternoon. That doesn't change the argument a bit. It changes the picture, though, for me anyway: the sticky note was always a person trusting another person's work, and the question of whether the work behind it was really done is older than any machine.

### 14.6 Information doesn't have it either

My next guess was information: one machine got some, the other didn't. That one died fast. Claude Shannon, who turned information into a number in 1948, says on his first page that the meaning of a message is "irrelevant to the engineering problem." His entropy depends on the probabilities of the possible messages. With an even spread it is the logarithm of their number; with an uneven spread the number alone is not enough. Fix the same distribution and the same message and the information assigned to that message agrees for both computers. That calculation has not included which history produced it. He'd give them the same number if one of them got hit by lightning and came out right by luck.

Fine, forget the set and look at the one message. In the 1960s three people, separately, worked out how to measure the information in one string: Ray Solomonoff, an American who was after a theory of guessing what comes next from what came before; Andrei Kolmogorov, the great Soviet mathematician; and Gregory Chaitin, an Argentine-American mathematician who later worked at IBM. Their measure is the length of the shortest program that prints the string. The same two computers hold the same string and get the same number. And then there's the thing in that theory I keep going back to and standing on, to see if it creaks. Running a program can't add this kind of information. What comes out can't hold more than what went in plus the program, give or take a little. Leonid Levin, who studied under Kolmogorov in Moscow and later left the Soviet Union to teach at Boston University, proved a general version of this, and he has a one-liner for it that's too good not to repeat: you can torture an uninformed witness all you want, you won't get information about the crime.

The bound says that an effective sorting procedure does not create unbounded algorithmic description complexity from that input. It does not say that sorting tells a person nothing. A person with limited time may now see the smallest entry immediately. Description length, useful access and a record that a particular check ran answer different questions.

Logic says the same thing in a different coat. The standard logic of knowing, the one the Finnish philosopher Jaakko Hintikka worked out in 1962, has a property

people call logical omniscience: if you know some facts, you automatically know everything that follows from them. So by that logic, once you knew every number on your list, you already knew whether it was sorted. Hintikka flagged the problem himself, in the same book, and people have been trying to fix it since. The fixes read like a pile until I asked each one the same thing: when it lets you write *known* next to "sorted," what does it make you write in the margin? Hintikka's margin is empty. Each fix writes one thing there.

The fix that comes closest to me is from 1995, by Sergei Artemov, a logician at the City University of New York who studied under that same Kolmogorov (the 1995 report came out of Cornell). He wanted a logic that could name the proof itself. What I think he's saying: you only get to know a thing if you've built an actual justification for it, and the justification rides along with the claim. That's a reason traveling with the claim, which is almost a record, but only almost, because it's a way of writing things down and reasoning about them, and nothing in his margin says the reason was ever run, or when. People have taught it to count, too: Ho Ngoc Duc, a logician at Leipzig, in 1997, made knowing take reasoning steps, and Artemov and Roman Kuznets, a logician then at Bern, in 2014, asked how big a justification can get before knowing by it stops being feasible (Part II weighed both properly, in Section 8). I took those for rivals at first. As I read them now, Duc counts the steps the knower takes and Artemov and Kuznets weigh what you'd hand somebody to check it, one distance from two ends. It has a computational side, too: Francisco Bavera and Eduardo Bonelli, two Argentine computer scientists who work on what programming languages mean, in 2018 turned justification logic into a lambda calculus whose terms carry, and can inspect, the history of the computation that made them, and Samuel Bucheli, Meghdad Ghari and Thomas Studer, three logicians, Studer the head of the logic group at Bern, put justifications on a time line in 2017, where they persist and change.

Line that up against the margin this book wants: earned, paid for, permanent, dated. Artemov's reason is nearly earned, the steps and the sizes are prices, the time line is a date. Their time line keeps its dates. Mine is dated only in the execution history: the raise happened at a definite step, and the record doesn't keep the step number. What none of it has is the closing of a question as a step with a price. Their prices are on reasoning, and the Bern justifications change, the one thing a closed question doesn't do. Push on mine, though, and it gives twice. Artemov's margin holds the reason itself, and mine only says a check passed. And their prices grow with how hard the thing was to know. This book charges the step that raises the record, and how long the checking took never enters it, so for a measure of effort, read theirs.

So the two most careful accounts I could find, one about bits and one about knowing, say the same thing: the sort added nothing.

And yet you know something after the sort that you didn't before. You can read the smallest number off the first slot. You can stop worrying. Something happened, and that's the clue, the first solid thing I've got: whatever *established* is, it isn't new bits. It's the same bits, *settled*: a question sorted into yes and no, with yes made to stay. It's the 0 again, about a claim.

### 14.7 The guy who got closest

Somebody had that clue before me, and on that clue he's the closest neighbour I've found, so he gets the most space and the hardest push.

Bennett again, the one from Part I's counting of states and the demon, wrote a paper in 1988 that starts where I just ended up: information isn't the same as value. His example is an almanac of where the moon and planets will be every day for a hundred years. It has no more information than the equations it was calculated from, but it saves you doing the calculating. So the value is the work. He called the amount of work a thing carries its *depth*.

Was this my idea, already done, by a guy at IBM? The only way to find out was to find where it parts from mine, because if I couldn't find that place I'd just be renaming his idea. And it's one word: *plausibly*. Depth is the amount of work a thing was *plausibly* made with. Bennett says straight out that you can't know how a message actually came to be; for all you know it came from flipping coins. Depth is read off the finished thing, the way a fossil's age is read off the fossil.

Bring the two computers back. The sorted list in the first one has some depth. The identical list in the second has exactly the same depth, because depth looks at the list and nothing else. A photocopy of a deep thing is just as deep. Depth tells you what work a plausible way of making the thing would take. It's not a record that *this* machine did it, and it's not a receipt for what doing it cost.

So Bennett's standing right next to the line, on the other side of it, and as far as I can tell he got his own question right. He asked how much work a thing embodies. I'm asking who has the right to lean on it and what they paid to get there. His answer lives in the object. Mine, if I've got one, has to live in the machine.

### 14.8 Statues that walk off

If it isn't in the information, maybe it's in the knowing, so I went and read the philosophers. And there it was again: people had been arguing about the difference between having the right answer and knowing it for a couple thousand years.

Plato's version is a joke about statues. There was a story that the sculptor Daedalus made statues so lifelike they'd walk off if you didn't tie them down. Plato says true opinions are like that: they wander off, and they're not worth much

until you tie them down with the reason why. Once they're tied down, they're knowledge, and they stay put. Knowledge tied to a reason, and knowledge that stays: of everybody I'd have bet on to hand me that shape, Plato wasn't on the list.

Then in 1963 Edmund Gettier, whose trap Part II played with cups (Section 8), broke the tidy version in three pages. Smith and Jones are up for the same job. Smith has good reason to think Jones will get it, and he's counted the coins in Jones's pocket: ten. So Smith figures, with good reason, that the man who'll get the job has ten coins in his pocket. Then Smith gets the job, and, unknown to him, Smith also has ten coins in his pocket. Smith had a belief that was true and backed by a reason, and it still wasn't knowledge. He was right by luck, because his reason was about the wrong pocket.

Gettier's case is the push that matters for a machine. Checking one pocket and committing to a claim about another is right by luck. So whatever a machine commits to, the check underneath has to be about *this very claim*, and I'd better say exactly what that means, because it carries a lot of weight later.

A check is about this very claim when four things hold. It tests the same property the commit names. A cousin of that property won't do. It tests that property on the same thing the commit names, and the thing next to it won't do either. Nothing has written to that thing between the check and the commit, so it's still the same edition the check looked at. And the check is honest: it says yes when the property is true and no when it isn't, on every value it could ever be handed, this one and all the others. Break the first three and you're Smith with his ten coins. Break the fourth and the check is just a rubber stamp. The next chapter builds a machine whose step won't let a commit through unless the first three hold, and whose checks come with a proof of the fourth.

Two notes on those four, because they matter later. The fourth asks for more than it needs. To rule out luck, a check only has to never say yes when the property is false; saying yes every time it's true is a bonus. And the third is a very old bug in computing wearing Gettier's coat. Security people call it time-of-check-to-time-of-use: check a file, use it, and something swaps the file in between (Matt Bishop and Michael Dilger, two computer security researchers at the University of California, Davis, wrote up how to hunt for it in 1996). Databases guard against the same thing by making sure, just before a commit, that nothing they read has been written since; H. T. Kung and John Robinson, two computer scientists at Carnegie-Mellon, called that optimistic concurrency control in 1981. A philosopher's pocket and a sysadmin's race condition are the same bug. One limit, said now: a machine can only promise "nothing has written to that thing" about its own memory; for brake pads or a voter roll the third condition becomes a premise about whoever measured (Section 17).

At this point I went further than this chapter needs, because the obvious next question is: does anything fix Gettier? People tried for years. The first patch is “no false steps in the reasoning”: Smith’s mistake was believing Jones would get the job, so rule that out. Then a philosopher named Alvin Goldman came back with barns, and he credits the barns to Carl Ginet, another philosopher working on what it takes to know something. You’re driving through a county that, unknown to you, is full of fake barns, just painted fronts propped up to look like the real thing. You look at the one real barn in the county and think “that’s a barn.” It’s true, you believe it, you have a good reason, your own eyes, and there’s no false step anywhere. And still you were lucky: if you’d glanced at the next one over you’d have believed the same thing and been wrong. Every patch gets a new barn, and as far as I can tell the arguing hasn’t stopped. That tells me something I’ll keep: whether a check was good isn’t only about this one case. It’s about what the check would have done on the cases next door. Hold on to that, because it’s the next section.

## 14.9 What “could have failed” means

I keep saying the first machine’s test *could have failed*. Take that apart, because it’s the piece everything else hangs off.

The machine is deterministic. Given your list, the check was always going to pass. So in what sense could it have failed? By “could have failed” I mean this: hand the checking stage an out-of-order candidate output and it says no. I do not mean hand the whole sorter an unsorted input; putting that input in order is its job. The relevant counterfactual is a defective candidate reaching the checker. An ordinary unsorted input reaching the sorter isn’t it. Chance has nothing to do with it. The copier that never checks has no corresponding rejection.

A check is a question asked of every possible input, and it splits them into two piles: the ones it says yes to and the ones it says no to. “Could have failed” just means the no pile isn’t empty. A check that says yes to everything has an empty no pile, and it distinguishes no inputs, however long it runs and however official it looks. It can still be sound, when its property holds of everything, and then it’s sound and tells you nothing you didn’t have. A philosopher, Robert Nozick, made a version of this a condition for knowing, in 1981: you know a thing only if, had it been false, you wouldn’t believe it.

“Could have failed” describes the *procedure*, about what the check would have done on other lists. It isn’t a fact about the end state at all. Look at the first machine’s memory when it’s done: a sorted list. Nothing in those bits says “and whatever made me would have caught a mistake.” The warehouse comes back here, every lamp visible and none of them saying it. The one thing that makes the first machine’s answer worth more is exactly the kind of thing those bits can’t hold: the copier

ends with the same bits, and neither machine kept anything else. An end state can hold more than the answer, and the only way a machine can own this is to write something down at the moment the check passes, and to have that something mean what it says. More than anything else in this chapter, that's why I ended up needing a record. A better answer wouldn't have done it. Even then the record only carries half of it. It can say a check ran on an edition and passed. That the check would have said no on a bad list is a fact about the checker, and that half rides on a proof about the checker; on this machine that proof sits outside, and whoever reads the record reads and accepts it. Where that kind of chain of reasons stops is the swamp of the philosopher Karl Popper, a little further down.

One more thing that tripped me up and then didn't. Rig a check to pass only on your one list. Is that cheating? No. It's a perfectly honest check of a different claim, "this is that exact list," and it has a big no pile: every other list. It establishes exactly that and nothing more. What a check establishes is whatever it's honest about.

Put that next to the empty no pile and the two fit together: a check establishes what it's honest about, and a check with an empty no pile is honest only about what you already had. One thing I'll say plainly now, so it doesn't ambush you in the next chapter: the example I build there doesn't police this. Its step makes sure a commit was checked first, about the same thing, by a checker proved honest. It doesn't stop you asking a question that can't come out no, like "is this number at least zero." Ask one and the whole chain runs, it costs three, and nothing is established but the chain. The certificate I show off there asks one that can fail.

### 14.10 Settled, whether it's true or not

Then I ran into Charles Sanders Peirce again, the one from Section 5, in an 1877 essay called "The Fixation of Belief." He was asking what inquiry is for, and his answer named the thing I'd been running together without noticing. The point of inquiry, he says, is to get out of doubt and settle on a belief. And when someone objects that surely what people want is a *true* belief, he answers like a guy who's watched a lot of arguments end: "as soon as a firm belief is reached we are entirely satisfied, whether the belief be true or false."

That's funny, and then it isn't, because he's right. Settling and being right are two different events. A question can be closed on a wrong answer, and once it's closed it acts closed. Peirce cares a lot about which ways of settling hold up (science wins, for him, because it has to answer to something outside any one person), but the settling itself, the question snapping shut, is there every time, and it never comes with a guarantee.

Karl Popper came at it from the other side. He was a philosopher from Vienna asking what separates science from everything else, and in his 1934 book, the one

that came into English as *The Logic of Scientific Discovery*, he says every test has to stop somewhere, at some statement somebody decides to accept, and nothing in the evidence forces you to stop there instead of one step further. You stop when you're satisfied. His picture is a jury: the verdict is a decision about what happened, and the jury deciding it doesn't make it true. Science, for him, is a building on piles driven into a swamp, and the piles stop where somebody's satisfied they'll hold. They never reach firm ground.

A real jury has a problem Popper doesn't mention, because a real jury is twelve people. Lewis Kornhauser and Lawrence Sager, two law professors writing in the Yale Law Journal, noticed in 1986 that a court that votes by majority on each premise of a case and also on the verdict can end up contradicting itself: every judge is consistent, and the court isn't. Christian List and Philip Pettit, two philosophers writing in Economics and Philosophy, turned that in 2002 into a theorem. No way of combining people's yes-or-no judgments on connected questions meets a short list of conditions that each look harmless on their own and always hands back a consistent, complete group judgment. A group's settling is a rule's settling, and which rule you pick decides which of the honest settlings sticks. Try the problem with three ballots and a rule that a payment is due only if a contract was signed and a deadline was missed. One voter says yes to both, one says signed but not missed, and one says missed but not signed. Each voter applies the rule consistently. Count separately: a majority says signed, a majority says missed, and a majority says no payment. You can point to the disagreement on the paper without blaming any one voter's logic. I haven't tried to put a group on the machine, and that's where I'd expect the first wall to be.

Popper's swamp sent me further down, because it's the oldest problem in the book and it doesn't have a bottom. Ask why you believe something and you get a reason. Ask why you believe the reason and you get another one. Sextus Empiricus, a skeptic and doctor who lived probably in the second or third century and whose whole method was holding off judgment, wrote down the ancient version, credited to someone called Agrippa, and in 1968 the German philosopher Hans Albert, who with Popper led the school called critical rationalism, gave it a name with a joke in it: the Münchhausen trilemma, after the baron who claimed he pulled himself out of a swamp by his own hair. Every chain of reasons does one of three things. It goes on forever. It loops back on itself. Or it stops somewhere, because somebody decided it stops. People have argued for a very long time about which of the three to live with. So "when do I actually know, and when am I done checking?" doesn't get a clean answer from anybody. What you can do is be honest about where you stopped and why. That's a lot of what the rest of this book is: every place it stops is written down, with its name on it, so you can decide whether you'd have stopped there too.

Lay Peirce and Popper on top of each other and what shows through is that the check is one act and the stopping is another. The check can run as long as you like and it never *makes* you stop; at some point something decides, and the deciding is its own event. That's the second machine's whole trick: it skipped the check and kept the answer.

And a logician said the part I didn't expect a logician to say. Per Martin-Löf, a Swedish logician, says a proof is an act, and once it's done it's done. His words, from lectures he gave in Siena in 1983: "a proof is, not an object, but an act ... Only secondarily, and irrevocably, does it become the act that has been performed." That took me a few reads, and the tenses unlocked it: first the act being done, then, for good, the act that has been done. You could say that's true of everything that ever happened, and I'd grant it, but it's the two computers again: the first machine's check happened, irrevocably, and the machine right now carries no trace of it. The past can't be undone. The *record* of it can be lost, unless something keeps one. That gap is where this book lives.

Then the intuitionists got there first, which by now I should've expected. The Dutch mathematician L. E. J. Brouwer, who held that a mathematical truth is something a mind makes evident, argued from 1948 on about an idealised mathematician working in stages, the creating subject. The people after him only came clear to me once I sorted them by three words: the act, the object, and the trace the act leaves behind, which Martin-Löf's sentence has no room for. The logician Georg Kreisel (1967), Austrian-born and then at Stanford, wrote it down as a formula: the subject has evidence for $A$ at stage $n$. That's a trace, something you can come back and ask about after the act is over. Anne Troelstra (1969), a Dutch logician who studied under Brouwer's student Arend Heyting and became one of the leading authorities on intuitionism, added that the subject never forgets what it has made evident, so the trace never fades. That's how Mark van Atten (2018), a philosopher in Paris who has spent years on Brouwer, credits it; the Stanford Encyclopedia's entry on intuitionism lists the never-forgets axiom among Kreisel's own, and I haven't settled which is right. Van Atten, going through the whole history, reads the formula as "by stage $n$ the Creating Subject has made $A$ evident." Look at the tense. *Has made* is Martin-Löf's "the act that has been performed," reached from the other end. And look at what that carries. It's tied to a stage; at any stage you can ask whether it's happened yet; it never goes away; and it's never wrong. Van Atten even says the subject can't just accept a proof handed to it from outside: it has to go through the steps and rebuild the evidence itself, which is the copier who runs the check. And then there's Göran Sundholm, a philosopher of logic at Leiden. He has long kept a proof as an act apart from a proof as an object (1998), and, as I understand his work, apart from its trace as well. In a 2014 paper he worked through how Brouwer's theory of the creating subject sits beside the thesis that every effective procedure is

a program, which is the side this book's machines live on. I know that second paper only by what it sets out to do, so I lean on it lightly.

Set their trace beside my record and the columns nearly all fill: their stage is my dated, their never forgetting my permanent, their never wrong my earned. Their stage is written into the trace, though, and my dated is only a step in the run that the record doesn't keep. The empty one is the price, and I think I see why. A mind with unlimited memory pays for never forgetting by taking states it has never used, and an idealised mind has endless fresh states to take, so nobody charges it for that. On a finite machine the fresh states run out, and writing a record that stays up forces the instruction to merge some states. Charging for that merge is a further premise. The forcing is Part I's counting of used and fresh states, and the charging is the price rule Part I sets. So it's the closest thing to my *established* that I've found. The formula is around sixty years old, and the idea under it, Brouwer's creating subject, is older than that. Push on my side from theirs, though, and it gives twice. Van Atten's subject won't take a proof from outside, and my machine takes exactly one: its checker is trusted on a proof about the checker that somebody did outside every run. And their evidence is never wrong because, for Brouwer, made evident is what true means. Mine is right only as far as that checker is honest. Where they differ, the difference is honest: their subject is an idealised mind with unlimited time and memory, and mine is a machine with a finite table that establishes only the claims its program checks.

## 14.11 Not a switch

So far I've been talking about *established* as if it were a light switch: on or off, the first computer on and the second one off. That bothered me the whole time I was writing it, and for a while I couldn't say why. It felt too small, the way a word can be technically right and still not cover what you're pointing at. I don't trust that feeling on its own, because I've had it plenty of times about things that turned out to be nothing. So I went looking for something outside my own head that would either give the feeling something to stand on or show me I was imagining it.

I didn't know what I was looking for, at first, beyond a rough shape: two things that look exactly the same and differ in how far something about them has been settled. So I started collecting anything that seemed to have that shape, and most of what I collected fell apart as soon as I looked at it closely. How each one fell apart taught me more than the thing itself did, so I'll go through them in the order it happened. Every story below is the version I could piece together from the records I could reach, and where I couldn't reach a record I'll say so. Check me.

The first batch was all hidden things. A pea under one of three cups. A tree, whose age you can't see from the outside. A forged painting that looks right until somebody

tests the paint. A glass of gas, where the thermometer shows one temperature and underneath there are uncountably many arrangements of molecules that would all show the same reading. Plato's prisoners in the cave, who see shadows on a wall. The things casting them stay behind their backs (Section 7 tells their story). Every one of these looked promising for about a minute. Then I noticed they all fail the same way: look harder and the hidden thing shows up. Lift the cup. Cut the trunk and count the rings. Run the paint test. Look at the molecules, at least in principle. Turn around in the cave. In each case the hidden part is a fact about the object that better eyes could find. That isn't what the two computers have. You can see every lamp in the warehouse and not find the check.

That failure gave me the first real tool I had, and it's a crude one. Give an observer perfect vision of the visible thing, as much detail as they want, and ask whether they could read the difference off it. If they could, it's something hidden, and that's not what I'm after. If they couldn't, it might be. I should say how much weight that test holds. It's a judgment, and I'm the one making it, so it's a way of sorting my own examples, not a verdict. If you find a case it sorts wrongly, that tells me something about the test and about me.

The second batch got past that test and failed a different way. A clock face reads three o'clock and doesn't show how many days have gone by; no amount of staring at the face tells you. And a computer can keep a complete history of what it did. It's Bennett's 1973 machine from Part I's counting of states, and the story is better than I let on there. Landauer had argued in 1961 that real computers throw information away all the time, and Bennett, his younger colleague, asked whether they had to. His answer, in a 1973 paper, was no. A machine can write down every step it takes, copy out its answer, and then run itself backward, retracing its steps and wiping the history as it goes, until it holds only the input and the answer and has never thrown anything away. It takes a forward run, a copy of the answer and a backward run, with scratch space for the history. Bennett thanks Landauer at the end for raising the question in the first place, which I like: the man whose principle says forgetting gives off heat is the one who asked whether you could get out of it. The history it keeps along the way is real extra information, sitting out where anyone could read it, so it isn't a hidden fact of the first batch's kind. Keeping it and unwinding it throws nothing away, so by Landauer's own reckoning there's no floor under its heat: run it slowly and the heat comes down as near nothing as you like, though never to nothing for a run that finishes, which Part I proved for Bennett's own model (Theorem 1.9). What it spends besides is fresh states, the scratch space the history sits in. And it can be run back down. Neither of those two, the keeping that merges nothing and the running back down, fits what happened to the first computer, which did something that can't be undone and paid for it. If the standing I'm after could be put up and taken back down with nothing kept and no floor either way, it would

just be bookkeeping. So that batch taught me what else I needed: something that only moves one way, and costs something to move.

The third batch came closer and still didn't quite fit. In 1782 James Six, a Canterbury man from a family of silk weavers who had taken up watching the weather, described a thermometer for the Royal Society. He wanted the coldest moment of every night, and he'd found that with an ordinary thermometer you had to be looking at it when it happened, because, in his words, "if it be not immediately noticed, it can never after be known." His fix was two little steel markers in the tube that the column pushes up and that stay where they're left. The markers make a record that only moves one way. Then he spoils it, for my purposes, in the same paper: every evening he wrote the numbers down and slid the markers back with a magnet, "without a touch." A record anybody can reset with a magnet is not the thing I'm after, and Six would have been the first to say he wasn't the first to build one; his paper lists the thermometers that came before his.

Since 1996, starting with oral polio vaccine, vials of vaccine bought through UNICEF have carried a small sticker, a square inside a circle. The square darkens as the vial takes on heat, and the darkening doesn't undo when the vial cools. When the square matches the circle, the vaccine shouldn't be used. It only moves one way, and it can't be reset. But it's chemistry. The reaction in the square is chosen to run about as fast as the vaccine itself spoils, so the sticker is really reading the vaccine's own condition by a proxy. Nobody ran a check; the heat did.

Closer still: the International Vocabulary of Metrology, the reference book that measurement laboratories work from, says a result is traceable only if it "can be related to a reference through a documented unbroken chain of calibrations, each contributing to the measurement uncertainty." That's a lot like what I want, a standing earned by checks, step by step. The same document adds that traceability "does not ensure . . . that there is an absence of mistakes," which is the true-versus-established split written into an international standard. And the top of the chain wasn't fixed either. When the official copies of the kilogram were compared with the prototype in 1889, 1946 and around 1991, they had drifted apart by tens of micrograms over the century, and nobody can say which one moved, because the prototype was the kilogram. In 2019 the kilogram was redefined by fixing a constant of nature, Planck's constant, instead. A chain like that can also lapse: if the calibrations stop, new results stop being traceable. The standing has to be kept up, which my record doesn't.

The rules of chess have the shape too. The current Laws of Chess of FIDE, the international chess federation, say in Article 9.2.3 that two positions are the same only if "the same player has the move, pieces of the same kind and colour occupy the same squares and the possible moves of all the pieces of both players are the same," and they spell out that a king that could still castle and one that can't are in

different positions, even on an identical board. That passes the perfect-vision test. And the official text says why it passes: the castling right changes which moves are legal. So the hidden part steers the game: it decides what can happen next. In the example machine, the flag never steers. It never touches the ordinary computing at all. The table and the trap that give the flag its meaning do steer, and Section 15 says where that line falls. Each of these got something right and missed something else.

The one that fit best, and I want to say clearly that it only fit best among the ones I found, was the standing of a mathematical statement. I'll go through how it went for one statement, as far as I could follow it.

In 1611 Johannes Kepler, the emperor's mathematician in Prague, owed his friend and patron Johann Matthäus Wackher von Wackenfels a New Year's present and, he said, had nothing to give. Wackher, Kepler wrote, loved *nothing*, for the games you can play with the word, so Kepler wrote him a little pamphlet about the nearest thing to nothing he could find, a snowflake, which had landed on his coat as he crossed a bridge. Why six corners? Kepler couldn't say in the end. But along the way he stacked equal balls in his head and asserted that one way of stacking them, the way grocers stack oranges, packs them so tightly that, in the 1966 English translation, "in no other arrangement could more pellets be stuffed into the same container." He didn't prove it. He said it.

Then the sentence sat. In 1831 Carl Friedrich Gauss, reviewing somebody else's book on a piece of number theory, proved what amounts to the special case of stackings laid out on a regular grid. In 1900 David Hilbert put the general question of packing equal spheres on his famous list of problems for the new century, without mentioning Kepler. In 1953 the Hungarian geometer László Fejes Tóth showed how the whole thing might be cut down to a problem with finitely many numbers in it, and a decade later he guessed that computers might one day finish it. Thomas Hales, a mathematician then at the University of Michigan, started on it in earnest in 1994, and in August 1998 he and his student Samuel Ferguson announced a proof that leaned on a great deal of computer work. The Annals of Mathematics gave the paper to a panel of twelve referees, led by Gábor Fejes Tóth, László's son. They ran seminars on it for years. A journalist, George Szpiro, later reported that by July 2002 they were 99 percent certain it was right; I couldn't find those words in anything the referees wrote themselves. What Szpiro printed in *Nature* in 2003 was the editor's letter as Hales reported it: the referees had "not been able to certify the correctness of the proof, and will not be able to certify it in the future, because they have run out of energy . . . " The Annals printed the human part in 2005, and the whole proof appeared in 2006. Hales started the Flyspeck project in 2003 to supply a fully explicit derivation of the entire argument. It was finished on 10 August 2014.

Lay those dates out on the table and look at what moved. The sentence never changed. Hilbert's version of the question didn't even name the man who first said it. Whether it's true never changed, as far as anyone can tell. What changed was how far it had been established, and it changed in stages, each one paid for by somebody's work: a guess, a special case, a place on a list, a plan, an announcement, a panel's reported ninety-nine percent and their exhaustion, a printing, a machine check. A perfect reader of the sentence can't see which of those acts has occurred. An act already completed stays in the history; confidence in the argument can still fall. A place on a famous list is recognition, not itself another proof step.

Then I tried to break that one too, because by this point I'd watched a lot of good-looking examples fall over, and there's an obvious place to push. Standing can fall. In 1852 a student named Francis Guthrie, colouring a map of the counties of England, noticed that four colours always seemed to be enough to keep neighbouring counties different. The question passed to his old teacher Augustus De Morgan, who wrote to the Irish mathematician William Rowan Hamilton that a student had asked him for "a reason for a fact which I did not know was a fact," and that he didn't know it yet either. In 1879 Alfred Kempe, a barrister who did mathematics on the side and sang counter-tenor, published a proof, and his description of finding it is wonderful: "After a somewhat arduous search, I have succeeded, suddenly, as might be expected, in hitting upon the weak point . . . " The journal's associate editor, William Story, added a note saying Kempe had "substantially proved" the theorem, and tightened the counting part of the argument. Eleven years later Percy Heawood, a lecturer at Durham who by one account brought his dog to his lectures, showed that one step of what he called "the now apparently recognised proof" doesn't work. Kempe swaps colours along two chains at once, and swapping the first can spoil the second.

So standing can fall. Story's published note strengthens the counting; that alone does not establish exactly which other steps he privately examined. What Heawood exposed is enough for the comparison: an accepted argument had an invalid step. Human acceptance and a proved sound checker are different premises. The example machine below is given the second one. Its theorem cannot turn the historical acceptance of Kempe's argument into a sound check by calling the overlooked part unearned.

And then came the part I didn't expect. The four-colour statement was true all along. Kenneth Appel and Wolfgang Haken, two mathematicians at the University of Illinois, proved it in 1976 with a computer, and the proof was met with suspicion for years; a cleaner version came in the 1990s; and in 2005 Georges Gonthier, a computer scientist at Microsoft Research in Cambridge, finished a proof of the whole thing checked by a small proof-checking program. So this one statement stayed true the entire time while its standing went up in 1879, came down in 1890, went up

again in 1976 against a lot of doubt, and went up further in 2005. The fall wasn't a fall of the truth. It was a fall of how far the truth had been established, and it was the right fall to have, because what had been established was less than anyone thought.

Laying all of that out together is what changed my picture. If there are stages, then at each one there's a line, and the line has two sides: not this far yet, and at least this far. Each line amounts to its own yes-or-no question. But choose the question carefully. "Do I still trust this proof?" can turn back to no, as Kempe's did. "Did I carry out this particular check?" asks about a completed act. Cross out the acceptance and the act still happened. To keep that answer, the machine has to keep a record of it and have no move that knocks the record down. A check that could have failed doesn't give you that by itself. Maybe the switch wasn't wrong. Maybe it was just incomplete, the first of many lines, and the real shape is a direction with lines drawn across it. I've gone around many times on whether the lines create the direction or the direction comes first and the lines are just places where you cut it, and on whether yes and no are even different from each other until something draws the line between them. I don't have an answer to that. The one part I could actually settle is that a single switch isn't enough once there are three or more stages, and Part II proved it (Section 11). Everything else in this paragraph is me thinking out loud.

Flatland, where this hunch started (Section 24), turned out to belong in the first batch. It's a novel Edwin Abbott published in 1884, narrated by a square who lives in a flat, two-dimensional world (Section 7 says more about him and it). When a sphere passes through the flat world, the narrator sees a line whose ends fade wrong and whose length keeps changing, and only when he feels it does he find a circle, a perfect one. Then the sphere tells him it will rise, and he watches it dwindle to a point and vanish, and come back larger and larger. The hidden direction leaks into his view. Even an exact cross-section does not locate it uniquely: for a sphere of known radius, equal distances above and below the plane give the same circle. A known direction of travel or another observation is needed to choose between them. In this machine two states can sit at different places along the established direction and agree on every bit of the ordinary machine (Theorem 15.17). That may be exactly why the ordinary machine can't see it. It may also be why, if the direction is worth having, someone has to build it in on purpose and keep it on purpose. Keeping it is paid for; Part I's counting of states says so. What's still open is what has to be kept, and that's what the rest of this chapter is for.

Those readings helped me separate the jobs. A Thiele machine keeps a distinction about a claim. Its step determines how that distinction is established, its record keeps it through later work, and its schedule counts the charge. The next few

sections put those jobs in computing's own words; the next chapter builds one example.

### 14.12 Computing already had a word for it

Computing already has a word for this and uses it every day. Anyone who's used a database has seen it: COMMIT.

A database does a pile of tentative work, and then there's one moment, the commit, written to a log, after which the change counts. Jim Gray, who more or less built the theory behind that, opens his 1981 paper on it by saying the idea comes from contract law, and his example is a wedding. People negotiate, somebody asks, two people say "I do," and from then on it's binding. If it was a mistake, you don't erase it. You do another transaction to fix it, the way a bad contract gets a second contract. Nobody reaches for a time machine.

The wedding was a philosopher's example first. The Oxford philosopher J. L. Austin, lecturing in 1955, said "I do" *makes* a marriage. It doesn't *describe* one, and sayings like that aren't true or false. They can misfire (the wrong person says it, a step gets skipped). They can't be false. A commit works exactly the same way. Writing the commit record doesn't report that a question got closed. Writing it *is* the closing. And a commit can't be false. It can misfire: committed without the check, committed about the wrong pocket. Those two misfires are Gettier and Peirce again, coming in from a third direction. Then somebody else reads the record, and for them it stops being a performance and turns into a report: "this was checked and closed." A report can be false. On the small machine of the next chapter it can't be, about the machine's own history: if its flag is up from a clean start, some claim was checked and closed, and that claim held of that edition when it was checked. That's how this book ties a commit to the truth, on that machine and the ones built on it, and it took a proof to do it.

Robin Milner, building a proof checker in the 1970s, ran out of memory, so he decided to keep only the finished theorems and throw the proofs away, like a teacher rubbing out the blackboard to make room. The obvious worry is forgery: if the proofs are gone, what stops someone just writing a theorem down? His answer was beautiful. Theorems got their own type in the programming language, and the only way to make one was through the rules of logic. You couldn't forge one because the language wouldn't let you build it any other way. He named that language ML, for Meta Language. I wanted to call my machine the meta machine. I'm choosing to treat that as a coincidence. Calling it a sign would be more my style, so this is more restraint than I usually manage.

Bitcoin is a modern version of keeping a history costly to replace. A returning node checks the candidate chains against its rules and follows the valid one with

the most accumulated proof of work. “Longest” here means that work, not simply the most blocks. Replacing an old block requires replacing the work that binds the later blocks to it. The protection depends on the adversary, network and mining assumptions; a recently accepted history can be reorganised. The clerk is still reading a note, but now has a specific test for the note and a resource behind the difficulty of replacing it. That resource is not the abstract unit charged by this book’s toll.

Making a settled fact expensive to fake is a different job from the one this book’s price does, the unit of cost it charges for raising a record. Bitcoin makes it tempting to read the toll of Section 2 as a deterrent, so I’d rather say this here. One unit of cost deters nobody. Bitcoin’s work is tied to something scarce, electricity and hardware; a unit of cost in this book is tied to nothing. Eric Budish, an economist at the University of Chicago, built a model in 2025 of what that tie has to cost. As I read him, for Bitcoin’s kind of trust to hold, the honest work has to stay expensive at every moment compared with what an attacker could win, and the expense grows in step with the value being protected. A flat charge of one unit can’t do that job, and it isn’t trying to. It’s a count, not a lock, as the start of the book put it: the lock is the order rule, check, then commit, then certify, and it would work at a price of zero. The price is there because nothing is kept for free: Part I’s counting of states shows that a machine with finitely many states that prices its merges can’t avoid it. Scaring anybody off has nothing to do with it. The machines I actually build have counters with no top, endless fresh states, so Part I’s counting of states doesn’t reach them whole. It reaches the flag: CERTIFY sends its two readings to one, the same counting run on a set of two. The charge comes from pricing that merge in the record’s own part of the state, a premise I choose (Theorem 3.19), and the size of the charge is my choice too (Section 15 says why).

And Bitcoin wasn’t first. Pull on its footnotes: three of the eight references in the paper that started it point to Stuart Haber and Scott Stornetta, who in 1991 worked out how to stamp a document with a time nobody could fake later, by chaining each stamp to the ones before it. A company built on their idea, Surety, printed a short fingerprint of its chain in the classified ads of the New York Times once a week from 1995, and it was still doing it in 2018. I found that in a piece the journalist Daniel Oberhaus wrote that year, which says the ad runs under “Notices & Lost and Found.” Think about that for a second. To make a record nobody could quietly rewrite, they put it somewhere a million people would keep a copy of without even meaning to. A physicist named Wojciech Zurek has spent decades arguing that the world does the same thing with a measurement, copying the result into so many places that nobody can take it back. I’ll get to him properly later in this part (Section 18).

There are more fences like these: logs that can’t quietly change, certificates you can check. Later in this part, Section 17 takes five of them apart. I lean on all of

them. They work. I want that said plainly, because the next thing isn't "they're broken."

The next thing is where they sit. The database's commit is bits in a log, kept by a database program, on a machine whose own step will overwrite those bits the moment some instruction says to. Milner's guarantee is a type checker; it holds inside programs written in his language and nowhere else. Bitcoin is a protocol on ordinary computers that will do whatever they're told. Every one of them is a program or a rule, on top of a machine that can't tell my two computers apart. The machine underneath keeps nothing and charges nothing for what got settled.

The obvious comeback: so put the rule one level lower, in the operating system, or what security people call a reference monitor, the small always-on piece everything has to pass through. Yes, exactly that: a monitor that wouldn't let any program raise its record without paying would be doing the thing I'm talking about. In this book's words it would be *implementing* the model, and I'd count that as agreement. What I care about isn't who builds the fence. It's what the fence has to obey, wherever it's built, and whether that's forced or just a good idea.

That comeback has a history, and it's older than me by a lot. In 1972 James P. Anderson's planning study for the US Air Force described a reference monitor: a piece every access has to pass through, that can't be tampered with, and small enough to check. In 2000 Fred Schneider, a computer scientist at Cornell, argued that a monitor that only watches a run as it goes can enforce only rules whose breaking shows up at some finite point, the kind computer scientists call safety rules. "No certificate without an earlier passing check and a commit" is one of those. And in 2015 Arthur Azevedo de Amorim, a computer scientist then at the University of Pennsylvania, and six coauthors built monitors that check a rule at every single instruction, and proved them correct. So putting a rule under every program, so nothing can get around it, is their move. I didn't invent it. What I'm putting under that move is narrower: a record that has to be paid for, on the small machine one that has to be earned, and a reason, from counting, why a finite machine that prices its merges can't make the price of a record that never comes down zero.

## 14.13 What settling looks like

The chapter on what structure is (Section 5) ends with a pile of things sorted into buckets. Settling a question is that, plus one more thing, and it's the 0 from the start of the book at a bigger size.

Picture every state the machine could be in as a field of dots. A yes-or-no question about the state splits the field into two parts: yes and no. That split is a sorting, the smallest one there is. And remember the no pile: if one of the parts is empty, there's no sorting at all, just a field. Now the extra thing that makes it a settlement: once

the machine is in the yes part, *no step it can take leads back out*. The door opens one way. A settled question is a sorting plus a door that only swings in.

At its best, what goes through the door is a relation: these numbers are in this order, these two things belong together. Sultan, the chimpanzee of Section 9, fitted two short sticks into one long one and reached the food. Nothing new got added to the cage. Two sticks became one tool. That's what I'd want the door to be for.

But the door doesn't care, and this is the honest edge of the whole picture. Hook the door to a check that says yes to everything and the machine walks through even when the claim is false. The question is settled either way, true or false, like Peirce said. A door that only swings one way tells you a question was closed. It doesn't tell you it was closed right. Price and correctness are two different books, and I keep both, separately.

One more edge, the one Gray's wedding already hinted at. What stays shut for good is that the question got closed, about that edition of that thing. Whether the claim is still true is another matter. Change the list after the commit and the record still says, truthfully, that the old edition was checked and closed. And if the claim turns out bad, the fix is another act on top, a second record that says "withdrawn." The first one never quietly comes back down. So "entitled to lean on it right now" is three things at once: closed, not withdrawn since, and the thing unchanged.

## 14.14 What the Thiele Machine is

So I'll set out what I built, and which parts I was pushed into and which parts I chose, because you should know which is which.

You've had the model since the start of the book: five parts (states, moves, a step, a price on each move in whole units, and a record that sits somewhere in an order; here the order has two places, no and yes), one rule, the toll, and a counter and a window when a result needs them. You can't buy it in a box. It isn't a device, a CPU or an instruction set; those are things you build to carry it, and the next chapter builds one, as small as I could make it. It counts what a computation makes the case, and this chapter's case is the sharpest one: the reading is a record that says whether something's been established, and the toll is what the first page of the book said about records, nothing becomes something for free. The event it accounts for happens on any computer, whether anybody keeps track of it or not, and the record exists only if something keeps it. The window earns its place here too, because the two computers already showed you the problem: the usual window, the bits, can't see the record.

A unit of cost isn't a joule or a dollar; whether it's worth anything physical is Landauer's assumption, named, and nowhere else. And one unit is just the floor: Part I shows that if you charge one unit per halving, a step that squeezes a lot

together pays at least the logarithm of the squeeze, roughly how many times it halves the number of states it can tell apart. Don't read that backwards, though. One unit isn't one halving. A single state flipping into a big set of states that already read yes, with every state in play equally likely, squeezes things together only a little, much less than a halving, and the whole-number count rounds that up to one. Read in the worst case over what the machine might be holding, the same flip can remove a full bit; Section 3 has both readings.

*What "new model of computation" means here.* The Thiele Machine takes distinctions I choose and writes them into the state, the step and the price. In the case on the table that means which thing a check was about, which edition of it the check saw, and which later act gets to lean on it. A record keeps some of that. A window decides how much of it somebody looking in can get back. Change what's kept, or what the window shows, and you change which questions the end state can answer. The machine underneath does the computing. The record adds no function it couldn't already compute.

Which leaves a fair question hanging: if it computes nothing new, does it even get to call itself a model of computation? It does, for the reason the start of the book gave: the random-access machine computes exactly what Turing's does, the cell-probe and input-output models leave the computing between the moves they count out of the count, and each is worth having for what it counts. So are the energy-aware random-access machines of Demaine and his coauthors (2016), which count the bits destroyed. Add one more to that family, the bulk-synchronous parallel model of Leslie Valiant, a computer scientist at Harvard (1990), which charges for rounds of communication between processors. Closest in kind are Blum's axioms, also from the start of the book, which constrain every cost the way the toll constrains every price list on the model. The Thiele Machine's own axis is what a run made the case, and what that cost. Part II laid the toll over a register machine of Cook and Reckhow's kind, and over a Turing machine and a lambda calculus, and none of them computes one function more afterward (Section 10). What each of them gains is a record and a cost.

The opening split what I found from what I assumed and what I chose, and the choice that matters most here is the first on its last list: the count goes inside the step.

About the name. I named it after myself, on purpose. Half of that is a joke and half of it is a stake in the ground. Turing got a machine, so here's mine. The longer story, meta machines and all, is in the afterword, "How I got here" (Section 24).

The names are the ones Section 10.8 set out, and they work the way Turing's do. A clock is enough to make something a Thiele machine. It isn't Thiele-complete, because it charges for computing, and neither is a machine whose record never goes up. The small machine of the next chapter is held to a higher bar, *certified*

*Thiele-complete*: on top of all that, from a clean start the record goes up only after a passing check of a claim, by a checker proved to mean what the claim says, then a commitment to that same claim, then a certificate, one unit each; and some claim's check, commitment and certificate raise the record from some starts and not from others, exactly where the claim holds. A *universal* Thiele machine runs any machine of a stated class as a guest, with every computing step done by its own counter moves and every rise placed by its program; a certified one also earns each guest's record again. The end of this chapter pins every one of these words down (Section 14.18).

## 14.15 The two computers, put back together

I'll tell the two-computer story again, rebuilt as Thiele machines, with the record and the counter written out, so you can watch it happen. You don't have to take my word. The prices are the ones on the smallest machine, which the next chapter builds: one unit each to check, to commit, and to certify (raise the record), and nothing for ordinary work like moving numbers around.

| **step** | **sorter does** | **record** | **counter** | **copier does** | **record** | **counter** |
|---|---|---|---|---|---|---|
| start | nothing yet | no | 0 | nothing yet | no | 0 |
| 1 | sorts (lots of ordinary steps) | no | 0 | copies in the typed list | no | 0 |
| 2 | checks "sorted" on this list, this edition: passes | no | 1 | (no check) | no | 0 |
| 3 | commits to "sorted," about this list, this edition | no | 2 | tries to commit: nothing checked, refused | no | 1 |

| step | sorter does | record | counter | copier does | record | counter |
|---|---|---|---|---|---|---|
| 4 | certifies | **yes** | 3 | tries to certify: nothing committed, refused | no | 2 |

At the end, the bits on both machines are still the same sorted list. Everything else differs. The sorter's record is up and its counter reads 3. The copier's record is down, and it's paid 2 for two refusals, because on that machine a refused attempt still costs what it costs. (Read those as two separate attempts. Written as one stored program, the copier would stop at its first refusal and pay 1.) Nothing the copier can do raises its record without first running a passing check of the claim it then commits to. And no outside window that only shows the list can tell you which is which; the difference is all in the part the window doesn't show. Walk that warehouse now and there's finally a lamp that's lit in one and dark in the other.

Now, how literally should you take that table? It's a picture of the chain, and the chain is what's proved. The small machine's own checks are three questions about one counter: is it zero, is it even, is it at least some number. What its proofs need from the questions is a checker proved to say yes exactly when the answer is yes, and an exact test for telling two questions apart. That's enough for "sorted": a version of the machine that takes any questions with checkers like that proves the same chain once for all of them, and with a list packed into one counter it certifies a sorted list for a cost of three and refuses an unsorted one forever (Section 15). Without a proved checker for "sorted," the chain would be honest about a counter and only by convention about your list, which is Gettier's pocket again, one level down.

And a lamp for whom? The record is bits too. It's a lamp on the machine, for anybody walking that warehouse. Hand somebody a printout of it across a desk and you're back at the sticker on the windshield: they have to trust whoever hands it over, or have some way to check the box that says it ran. That crossing is a different problem with its own tools, signatures, attestation and proofs a stranger can check, and this book doesn't solve it. Section 7 says what those tools rest on, and the small machine's record, like any record kept in a machine's state, carries nobody's signature. What the step changes is the question, from "did it run?" to "do I trust the box that says it ran?" That's a smaller question, but it isn't no question.

So, back to the list from the start of this chapter: the test sheet, the sticker, the audit, the ledger, the proof, the confident AI. I'll give the straight answer on

which of those this helps. The sticker is a question of whether a piece of paper is genuine, and nothing here touches it. The "all tests passed" sheet fits only if the test is a checker proved to mean what it says, in the machine's own language; a set of tests that can miss a bug certifies "this procedure said yes," which is true and smaller. An audit and a ledger get the test sheet's answer: kept by a program on an ordinary computer, they're bits the program wrote, and they fit only where the step underneath keeps the record and refuses to raise it without the check. A proof is one example: checking its steps is a procedure that can reject them. Whether its statement says what you meant is still yours to read. The AI that talks as if it's sure of itself isn't addressed at all. Section 22, under "Outward consequences," says how little of the outside world I've reached.

Now comes the honest half. Take away the small machine's rules and keep only the bare toll, and the copier has an easier move: pay one unit, flip the record to yes, and that's the whole move. The toll is satisfied and the receipt is real, but it's a receipt for nothing. The bare toll prices the closing of the door, and the check behind it goes unpriced.

### 14.16 What putting it in the step buys

The hardest objection is one I'd rather make myself. You defined a flag and a charge, then proved the charge gets paid when the flag goes up. That's your definitions agreeing with each other.

On the bare model, that's fair, and the copier's receipt for nothing is the proof. What makes the record mean *established* is the small machine's step, which won't let a commit through unless a check of that very claim passed, meeting the first three of the four conditions from Gettier's pocket, with the fourth, an honest checker, proved about the checker separately. And what it proves isn't about one program: on *every* list of instructions anybody can write, starting clean, if the record is up, then somewhere earlier a check of some claim, about some edition, passed; then a commit of that same claim; then the certificate; with nothing touching the thing between the check and the commit.

So a raised flag, from a clean start, tells you some claim went through a passing check, a commitment to it about the same edition, and a certificate. That piece of history is what the flag keeps, and only that. The channel names the latest commitment that got through, which needn't be the one the flag first went up on, and in general you can't read off the end state which claim was certified at that first rise, or when. And the flag doesn't say the claim still holds, at the certificate or at the end, because the counter can change after the commitment.

That "every" is what the operation-by-operation proof buys. A log written by one program inherits that program's correctness conditions. A database or reference

monitor can instead enforce rules over every operation its interface permits. The small machine does that too: its invariant covers every string of its instructions from a clean start, including hostile strings. The coverage comes from those restrictions and their proof; it does not follow just from calling the record part of the step. And the tie between a check and the thing it checked gets a job: the step reads it, and refuses a commitment when it no longer fits, so the structure is doing something in the machine instead of sitting in a note beside it.

There are two limits on that "every," and I'll say them now so you don't have to find them later. First, it's every program, not every everything. The guarantee covers any list of instructions, run from a clean start, one step at a time, on a step that does what its definition says. Hostile programs are inside it. A hostile start (a table of facts nobody checked, a reset), hostile hardware (a soldering iron on the flag's bit, a second writer the step can't see) or a hostile reader isn't; Section 22 lists the whole fence. The guarantee has the same shape as a database or reference monitor's: it covers operations mediated by a faithful enforcing layer. How large that class is depends on each interface; being a database does not make it a one-program guarantee.

Second, the flag and the ledger on their own are what people who verify programs call ghost state: erase them and every program computes exactly what it did before (Section 6 has where that idea comes from). What isn't ghost on the small machine is the table of checked facts, which the step reads and refuses on, at run time, for every program. And the move of putting a rule under every program isn't mine either. It's the reference monitor's, and its history is a few sections back.

### 14.17 The record, the checker and the host

Take the flag and ledger off the small machine's drawing. The core follows the same steps: neither field tells it what to do next. Put them back and you can read whether certification occurred and what the instruction schedule charged. That is the passive record's job. The latch argument in Section 10 gives its form over any fixed base when the record is driven by that base and never comes down.

The checker has another job. CHECK of "$A \geq m$" tests the counter in one instruction; a program using only the two counter moves would have to work through their operations. A step comparison therefore starts with the instructions each machine is given. Every checker built in this book is a program. The abstract certified interface also admits a supplied checker, so its computational power includes whatever that checker can do: supply a halting oracle and CHECK can answer halting questions. The effective implementations here use effective checks.

Now put a guest program inside the host. A guest that can write any value into its own "checked" variable can write YES without checking. The host I build gives

the guest three entry points instead: CHECK, COMMIT and CERTIFY. Its step rule controls those entry points for every allowed guest program. The fact table and channel determine whether the next act is permitted; the flag and ledger record the result. An ordinary universal machine can implement these rules. The work is in preserving them through every guest instruction.

The universal program in Appendix C.1 does that work with the small machine's six instruction kinds on a host with more counters. It fetches a guest instruction, decodes it, and carries it out. At matching points the guest and host ledgers agree. The host halts exactly when the guest halts, keeps the guest's answer in its counters, and raises its own flag exactly when the guest raises its flag, through its own earned chain. Its guests are the small machine's programs from clean starts, with Minsky's encoding supplying their universal computation. Appendix C.3 gives a simpler picture with a guest-step operation supplied directly; the instruction-by-instruction construction shows how to build it.

Add a one-unit PAY instruction and the host can also follow a computably presented Thiele machine: its selected moves, transitions, prices and total yes-or-no reading are all given by programs. Appendix C.2 constructs that interpreter. Starting with the guest's reading at no, the host follows its ledger with at most two extra units at the raise. Starting with the guest already at yes, the clean host must establish its own flag immediately and spends three. The same clean-start three-act count explains the surcharge when the guest certifies with a whole ledger below three.

### 14.18 Two layers, and the names

*The layers, once.* The core definition (Definition 2.2) is a Thiele machine: states, moves, a step, a cost for each move, a yes-or-no reading, and the toll. Everything else is added on top, and each theorem names what it adds. First come extra structures and requirements: a ledger that sums the costs, a window, permanence of the reading, a price on merges. Then an earned implementation, the small machine of the next chapter, with a checker proved sound and a provenance theorem. Then two interfaces any machine can be read through: Thiele-complete, about computing and a record that tells starts apart, and certified Thiele-complete, which adds the earned chain. Then a universal interpreter for a stated class of guests. The bare definition doesn't ask the step to be computable, so when I say the record adds no computing power, I mean relative to a fixed base.

*Substrate versus scaffolding.* Two layers run through the book, and confusing them makes the theorems quietly stop meaning what they say. The *substrate* is the shape: a kind of state and a rule for stepping. Different theorems ask that shape for different extra pieces, and the pieces aren't the same. The universal floor (the "pays something" result of Section 2) asks for a yes-or-no reading and the toll on it.

The counter's uniqueness (only one count fits a chosen start and price list) asks for a starting value and a schedule of increments; Section 6 gives the argument, a short induction that works for any system with those two pieces, and on the small machine of the next chapter the ledger is the sum of the charges by theorem (Section 15). The diagonal asks for a recursion theorem for the transformations the substrate can express, and needs no toll and no encoding at all. "Substrate-level" doesn't mean one fixed set of parts that every theorem shares. A theorem only gets to use the parts it asks for. Ask it what it needs and it'll tell you, and whatever else is lying around the workshop isn't the theorem's. So each result is stated over exactly the extension it uses. The substrate with the extension a theorem names is what I mean by *the model*, and the name Thiele Machine belongs to the model. The *scaffolding* is everything that holds the shape up so you can touch it, and when I say *my machine*, *the small machine*, *the machine I build* or *the build*, this is what I mean: the machine of the next chapter, with its two counters and their versions, its table of checked facts, its channel, trap, ledger and flag, its six instruction kinds and their prices, the host with a register for every whole number, and the universal programs that run on that host (Appendix C.1). The build fills in several extensions at once and borrows, for any one theorem, only the extension that theorem names. Let that line blur and a theorem stated over one extension starts reading as if it carried premises it never uses. "Substrate" never means the build, and it never means the side of a simulation doing the stepping; for that I say "the machine doing the stepping," as against "the program it runs."

So the book keeps the two apart: Parts I and II were about any system with the pieces each result names, and their pictures were small worked instances you could hold (two boxes on a sheet of paper, chairs in a room, a row of dots). The example machine wasn't one of them; where they said something about it, they pointed ahead, and the next chapter builds it. Every Part II claim that has a machine instance gets that instance there (Section 15.14) or in Appendix C.1, with its own theorem, and Part II said so in one sentence at most. When the origin story (the afterword, Section 24) says the shape "was the substrate the entire time," it means this layer, the model's shape. It doesn't mean any piece of the build. The same goes for two more of its words. When it says "the machine" about the thing it named the Thiele Machine, or calls that thing a meta machine, it means the model; "my machine" is the build of the next chapter. And when it lists what the model asks for (a state, a step, a record, a ledger, a window), read the ledger and the window as two of the extra pieces: a result asks for them only if it uses them.

The build is one piece of scaffolding, and I'd rather you know up front that it is one instance, chosen to be small enough to inspect. Section 15 builds the smallest honest machine I could: two counters, six instructions, a table of checked facts, a ledger and a flag. Earned commitments are proved there (a flag raised from

a clean start was checked, then committed, then certified, on the same edition, with nothing touching that thing between the check and the commit), and so is universality, through Minsky's two-counter machines and his halting result, through the counter constructions of Forster, Larchey-Wendling and their coauthors, set out in Appendix J. Then one fixed program, on a host made of the same kinds of instruction with more counters, runs every program of it, and with one paid move added, every computably presented Thiele machine (Appendix C.1). None of that is the model. It shows the model fits in six instruction kinds.

The names again, with every part of the certified one pinned down, because I'll hold to them for the rest of the book. A Thiele machine is any particular system that fits the model's core definition, its five parts and its toll (Definition 2.2); the toys on paper in Part I count, and so does the small machine, with its toll proved. *Thiele-complete* (Definition 10.6) says nothing about what the record is about. *Certified Thiele-complete* asks for four things of its own (Definition 15.7), and it is exactly Thiele-complete with clean loads, the earned record and a claim that can fail added (Theorem 15.23). The first of the four is a universal base whose moves can't touch the record, and a record that stays up once it's up. The second is a record that, from a clean start, rises only after a passing check of a claim, then a commitment to that same claim with nothing it's about changed in between, then a certificate, with the checker proved to mean what it says. Over a language fixed first, the claims and what they mean are that language's, and the proof is that the checker's yes makes the claim true in its sense; every instance in this book is stated that way (Theorem 15.26). The third is exact prices: one unit for each of those three acts and nothing for anything else. And the fourth is a claim whose check, commit and certificate raise the record from a loaded start exactly when the claim holds there, with the claim holding at some loaded starts and failing at others. Every clock-style machine fails the definition, because it charges for computing, whatever its record, and every machine that meets the definition hides its record and its ledger from its own window, both by theorem. That's about an observer who sees only where a run ended; one who also has the program and the start can work both out. None of the four clauses is there for decoration: for each one there's a machine that meets the other three and fails it, each of them a Thiele machine with a two-counter machine inside that no interface makes certified Thiele-complete (Theorem 15.29). The small machine meets it, and so does the same machine over any properties with an exact test for telling two of them apart and a checker proved to say yes exactly when the property holds, as long as one of them can come out both ways, sorted lists included. A *universal* Thiele machine is one fixed machine running one fixed program that runs any machine of a stated class as a guest, with every step of the guest's computing done by the host's own counter moves and every rise of its record put where its program put it; a certified one also earns each guest's record

again by its own check, commit and certificate. Nothing is left to the guest's good manners. Appendix C.1 builds one, $U$: a fixed program in the small machine's own six kinds of instruction, on a host with more counters, with no step of any guest built in. Its guests are the small machine's programs, so every computation runs on it under Minsky's $2^n$ encoding, done outside the machine, and it matches each guest's halting, answer, flag and ledger exactly, raising its own flag by its own check, commit and certificate. Then $U_P$, the same program with one more handler for a paid move that does nothing else, takes as its guests every computably presented Thiele machine, one whose next move (picked by a driver), step, prices and reading are computed by programs, with the reading decided on every state. It matches the guest's halting, raises its own flag on its own earned chain exactly when the guest's reading has said yes, and keeps the guest's prices to within a surcharge of at most two when the guest's reading starts at no, which no program on that host, run from a clean start, can avoid in general. And a computer that runs the small machine, your laptop say, isn't the small machine. Read through the emulator's prices and flag it's a Thiele machine, a finite one, like the toys of Part I, but it isn't Thiele-complete, because its counters stop somewhere and a universal base needs every configuration. It implements the small machine, inside bounds, the way a laptop implements a Turing machine. Push on that comparison, because it's loose, and the looseness is the point. Does a laptop implement a Turing machine? Not exactly. Turing's tape never runs out, and a laptop's memory does; strictly, a laptop is a very large machine with finitely many states, and it behaves like a Turing machine only on computations that fit. Nobody minds, because the bound is known and most computations people care about fit inside it. That's the same deal any device running the small machine gets: its counters stop somewhere, and the theorems speak only for runs that fit. The bounds aren't fine print. They're the whole difference between implementing a model and being one.

How much can a difference in width matter? There's a famous answer, and it's worth the detour. On 4 June 1996 the first Ariane 5 rocket lifted off and broke apart about forty seconds after its engines lit. The inquiry board, chaired by the mathematician Jacques-Louis Lions, traced it to one conversion in the inertial reference software: a sixty-four-bit floating-point number, an alignment value called BH that's tied to horizontal speed, squeezed into a sixteen-bit signed integer. On Ariane 4, where the software came from, that value could never get big enough to overflow, because of how that rocket flew in its first forty seconds. Ariane 5 flew differently. The value got too big, the backup unit failed on it, the main unit failed the same way one cycle of its software later, seventy-two thousandths of a second, and the function doing the converting didn't even serve a purpose once the rocket had lifted off; it was kept for commonality with the old rocket. The board's report says the assumption that made the conversion safe had been agreed on and then,

not on purpose, kept out of sight of any outside review. A width premise that holds on one machine is a fact about how that machine gets used. The code doesn't carry it, and yet it travels with the code without anybody noticing. The only defence I know is the dull one: write the premise into every statement that needs it, where the next person will trip over it.

CHAPTER 15

# The earned-chain counter example

The chapters before this one never built a machine you could program. They said "any states, any steps," and whenever a picture was needed, a board or some chairs or a cup stood in. Now I have to build a machine, and the first question is about the one I built before it. I built a big one first, on the way here (the Afterword, Section 24, tells that story), with dozens of instructions, some picked for reasons and some picked because I had to pick something, and theorems held of it. What does that show? That the model works, or that I'm good at stacking instructions until the proofs go through? A big machine is a wonderful place to hide a mistake. It's a garage so full of stuff that you could lose a car in it.

So the example machine in this book is a small one. I want the state and every way of changing it where you can inspect them. There are six instruction kinds, and you can follow how a check stays tied to an object, how a change makes it stale, and how the next move uses that relation. Each kind takes operands, a counter or a property or a line to jump to, so there are unboundedly many instructions. It's one way of building the model, not the only one, and the theorems will say which facts hang on these particular choices. I built it by throwing things out until nothing more would go. I haven't got a proof that nothing smaller works, and I'm not claiming one.

## 15.1 What the model asks for, and the example that meets it

The model asks only for states, moves, a step, a price on each move and a yes-or-no reading, with the toll. Before any of its parts, I'll give the example that meets it in three sentences. It's an ordinary computer, two counters and a jump, with a record layer on top. The flag in the record layer is the blank cell, and raising it is the write of a 0, a merge in the record layer (flag down and flag up land on the same state), which is why the step that raises it is charged once merges in that layer are priced

(Theorem 3.19, in Part I). And the order rule, check, then commit, then certify, is what makes a raised flag mean something.

Parts I and II define the model by five parts and one rule. The example meets them with the smallest *record* there is, a flag that reads no or yes and, once yes, stays, which on a machine is the blank cell's 0. Here yes means "certified," because that's the event I picked (Section 18, later in this part, asks whether anything outside the rules picks it out). It adds a *toll*: the step that raises the flag costs something (Section 2). And it adds a *ledger*, the running total of those costs, a whole number that only climbs (Section 6). And Section 10 proved the thing this whole chapter stands on: over any base at all, a base-driven record is a *latch*, a bit that switches on at an event the base decides and never switches off (Observation 10.1). The record watches the base. It doesn't compute anything the base couldn't compute without it.

Push on that for a second, because it cuts the job in half. If the record adds no computing power, then all the computing power comes from the base, and the question "how small can the machine be" splits into two questions that don't touch each other. For the base: what's the smallest thing that can compute everything there is to compute? For the record: what's the smallest set of pieces that makes the flag *mean* something?

There's one honest asterisk on "don't touch," because the machine below makes it matter. What's blind is the flag and the ledger: the base never reads either of them. But the pieces that make the flag mean something (a table of checked facts, a slot for a commitment, a trap that stops a refused run) have to sit where the step can read them, which is in the base, and they do steer the run: a failed check stops a program dead. The latch theorem holds wherever you draw the line between base and record, which is exactly why it can't tell you where the meaning lives. On this machine the meaning lives on the base side of that line, in the bookkeeping, and the flag on top reports it. That line has to sit at the core: the step reads the table, the channel and the trap, so a map that hid any of them wouldn't move in step. The record-layer merge of Part I draws a lower line, with only the line number and the two counters below it, and what the premise charges moves with that line. Drawn there, all three record acts merge in the record's part and the premise charges each of them; drawn at the core, it charges only CERTIFY.

The record question, the second of those two, is the one Part II left to the machine, on purpose. Section 9 said that testing a claim and committing to it are two different acts, and that nothing in the five parts of the model makes the second wait for the first. Read Section 2's definition again with that in mind: a machine where you pay one unit and raise the flag over nothing in particular obeys every rule in it. That machine is honest about the price and silent about what was bought: it's a receipt for nothing. The toll alone can't fix that, because the toll is a rule about *cost*, and

"was this earned" is a question about *order*: did a check happen, of this claim, before the commitment, with nothing changing underneath in between?

This machine is my attempt to make the order part of the machine, in the smallest place I could find to do it. I'll show you the pieces, then the proofs, then the places it falls short, which there are, and which I'd rather you hear from me.

*The blown fuse cannot be hidden from every future move*

Blow the fuse and pull the lever. Nothing moves. Replace the fuse, start from the same counter values, and that lever can turn the gears. The fuse decides what the next pull does, whether the camera can see it or not.

That is what a failed CHECK does here. It traps the core, and every free instruction then leaves the whole state alone. Suppose I hide the failed check from the camera and still demand that the next picture depend only on the present picture and the free move. The live and trapped pictures agree. The camera's rule must give them the same next picture. Since the trapped box stays still, the projected live box has to stay still too.

**Theorem 15.1** (A globally check-silent native shadow is static)**.** Let $\pi$ be any projection from native full states, with any target type. Suppose every CHECK preserves $\pi$, and each zero-cost instruction $i$ has an autonomous projected rule $h_i$, so $\pi(\text{exec}(s,i)) = h_i(\pi(s))$ for every state. Then $h_i(\pi(s)) = \pi(s)$ for every $s, i$. If $\pi$ is onto, each $h_i$ is identity everywhere. The conclusion also holds on the reachable states from a fixed start when those same laws hold throughout that reachable domain. □

*Proof.* Stand at any native state $s$ and read counter $A$. Ask CHECK whether that counter is at least one greater than the number just read. It is not, so this check leaves a trapped state $t$. In symbols, the chosen question is $A \geq A(s) + 1$.

By the proposed silence rule, the failed check has not changed the picture: $\pi(t) = \pi(s)$. Now choose any free instruction $i$. At the trapped state it changes nothing at all. The camera's rule $h_i$ must therefore leave that picture alone. But it is also the picture of $s$, and $h_i$ receives only the picture, not a choice between the two boxes. It must leave $s$'s picture alone too. Written as one chain,

$$h_i(\pi(s)) = h_i(\pi(t)) = \pi(\text{exec}(t,i)) = \pi(t) = \pi(s).$$

You could begin at any $s$ and choose any free $i$, so every free projected move is still on the whole image. If every base picture occurs, that is the whole base. I did choose a different failing question for different counter values; the premise allows every CHECK at every state, so it allows each of those choices. Finally, if $s$ was

reached from the fixed start, append this CHECK to that run to reach $t$. Both boxes remain inside the reachable domain, and the same argument applies there.

The fuse belongs with the gears that read it. The certificate and ledger can sit beside them unread. The later slice follows the computing moves while the machine is live; a failed check takes it out of that working domain. That is where the line goes on this machine. [9]

## 15.2 Where the counters come from

The base first, and the base isn't mine. It's Marvin Minsky's. In a 1961 paper in the *Annals of Mathematics*, and then in his 1967 book *Computation: Finite and Infinite Machines*, Minsky studied a machine so bare it's almost a joke. Picture a few bins of pebbles and a numbered list of instructions. There are only two kinds of instruction: put a pebble in bin $c$ and go to the next line; or take a pebble out of bin $c$ if there's one to take, and go one way if there was and another way if the bin was empty. There's nothing else: no memory but the bins, no tape, no arithmetic beyond plus one and minus one. When the line you're on has no instruction, the machine stops.

What Minsky proved is that two bins are enough to compute anything a Turing machine can compute, as long as you're willing to pack the input in a strange way. The trick is to keep many numbers inside one number, as the powers of different primes ($2^x 3^y 5^z$ holds three numbers in one), and to do arithmetic on the exponents by multiplying and dividing with the second bin as scratch space. Packing numbers that way can turn a modest input into an enormous pile of pebbles, handled one at a time, so it's no way to build a fast machine. It's a way to settle a question: with that packing, the two bins can do every Turing computation. What the bins give me here is a base whose ordinary moves you can see all of. Each of the earlier results says which properties of a base it needs, and Appendix D carries the earned layer over to other bases once you've named what they have to supply.

Who cares? Minsky did, because the question then was where the edge of computation sits and how little machinery reaches it. You should, because it means the base of this chapter's machine hides nothing: if it can compute everything with plus one, minus one and a jump, then nothing about the record can be smuggling computing power in through some instruction I made fancy. For this construction I use Minsky's two-counter halting result and the counter constructions of Forster, Larchey-Wendling and their coauthors. Appendix J sets out the reduction; the step-for-step connection to this machine is the part I have to establish.

[9] A view that exposes refusal, or a machine with a different failure rule, would supply a different interface. The obstruction concerns the native failure semantics and a globally CHECK-silent autonomous projection.

And the pebbles aren't only my picture. They were somebody's actual machine, the same year. In one issue of the Canadian Mathematical Bulletin, September 1961, Z. A. Melzak, a mathematician at the University of British Columbia, described a computer made of distinguishable holes and an unlimited supply of pebbles, with one three-place operation, and a few pages later Joachim Lambek, a mathematician at McGill University, trimmed it to two moves: put a pebble in hole $X$; take one out of hole $X$ if there's one to take. He called it an *infinite abacus*, and early on he writes "Our method is probably not new," which is a hard sentence to beat. He also tells anybody who distrusts infinity not to worry about all those holes, since any one computation only ever uses finitely many of them. Is that two inventions or one? Lambek says his was a modification of Melzak's, so call it one and a half, with Minsky's as a separate find the same year. And then Lambek went somewhere you wouldn't guess: in 1986, with Philip Scott, a logician who had come to McGill to work with him and later taught at the University of Ottawa, he wrote a book whose first part, as the publisher describes it, shows that typed lambda calculi and a certain kind of category are essentially the same thing. He went from pebbles to category theory in one career, twenty-five years end to end. Does any of it change the argument? No. The machine uses Minsky's version and Minsky's name. But a bare machine that several people reach for in the same year looks like a shape that was sitting there. It doesn't look much like anybody's invention.

The title of Minsky's 1961 paper is the clue I can't leave alone. It's about "Post's problem of tag." Who's Post, and what's tag? Emil Post was a logician, a fresh PhD from Columbia, who in 1921, on a fellowship at Princeton, was playing with about the simplest rewriting game there is. Take a word of 0s and 1s. Look at its first letter, chop the first three letters off the front, and stick something on the end: `00` if the letter you looked at was 0, `1101` if it was 1. Repeat, and stop when fewer than three letters are left. Try it at the table. Start from `1000`: you get `01101`, then `0100`, then `000`, then `00`, and it stops. Start from `100`: you get `1101`, `11101`, `011101`, `10100`, `001101`, and then `10100` again, and it goes round forever. Post wanted a rule that tells you, for every starting word, which of those happens. He worked at it for months and later called it hopeless, and as far as I can tell it still is: a hundred years on, nobody knows whether that one little game, with that one rule, can be decided. Stephen Wolfram, the physicist who built Mathematica, wrote a 2021 article on it that tells the Princeton story, and it's where I'd start. In his 1961 paper Minsky proved that tag games in general can compute anything, which is why Post's name is on it. It didn't settle Post's own game. And the chase goes one step further. In 2008 Liesbeth De Mol, a historian and philosopher of computing who wrote her doctoral work on tag systems, turned the famous $3n + 1$ problem (halve an even number, triple an odd one and add one, and ask whether you always get down to 1) into a small tag game of the same kind. So is Post's game secretly as hard as everything, or

just stubborn? I can't tell you, and as far as the published work goes, nobody can yet.

None of that changes the machine below. It changes the picture. The two-counter machine came out of a hunt for the smallest thing that can't be predicted, and the smallest things are exactly where the open questions are still sitting.

There's one word clash to settle before it bites. Minsky's machine is usually called a *counter machine*, and its bins are *counters*. Section 6 also had a counter: the ledger $\mu$, the number that only climbs. It's the same word for two completely different things, and I'd love to say I set that up on purpose. In this chapter, *counter* means one of Minsky's two bins, called $A$ and $B$, and the climbing total is always *the ledger* $\mu$.

### 15.3 The example machine, piece by piece

I'll go through the whole state a piece at a time, in words first.

- Two counters, $A$ and $B$. Each holds a whole number, with no upper limit.
- A *version* for each counter, also a whole number with no limit. It starts at 0 and goes up by one every time its counter is written. Think of it as the counter's edition number: a new value is a new edition.
- A *line number*, which instruction comes next, counting from 1 (more on that choice just below).
- A *fact table*: a list of facts. A *fact* is three things stuck together: a property, a counter, and a version. "$A$ is even, as of edition 3" is a fact. The table has room for sixteen. Why sixteen? It's a pick, with no derivation behind it, and what the pick gives up is set out in the subsection near the end of this chapter on what the machine doesn't do.
- A *channel*: either empty, or holding one fact. This is where a commitment sits.
- A *trap*: one bit, down at the start. When it goes up, the machine has refused something, and it stays up.

Counting lines from 1 looks like the least interesting choice in the chapter, and it's one of the oldest arguments in programming, with a famous referee. Edsger Dijkstra was a Dutch computer scientist, one of the generation that founded the field, and for over forty years he mailed numbered notes to a few dozen colleagues. In 1982 he wrote one of them, EWD 831, titled "Why numbering should start at zero." His case goes through ranges first. Of the four ways to write "2 up to 12" with less-than signs, he picks the one that takes the bottom in and leaves the top out, $2 \le i < 13$: two ranges that meet end to end then share one number written once, and the length is just the top minus the bottom. Start a sequence of $N$ things at zero

and it's $0 \leq i < N$, and each thing's number is how many things came before it. He reports that programmers on the language Mesa, which allowed all four conventions, found the other three a steady source of clumsiness and mistakes, and he points out that FORTRAN starts at one. Does that settle it? It settles which convention is tidier, by an argument you can check. It doesn't settle which convention somebody else's work used, and that's the catch: the two-counter machine used by Forster, Larchey-Wendling and coauthors numbers its lines from 1, and so does this machine, because matching it line for line is worth more here than tidiness. Is zero the right place to start counting? Dijkstra's title says it is, and I can't tell you more than that. What his note carries is an argument about which way makes fewer mistakes. That part you can check for yourself, and it's the more useful thing to have.

Everything in that list is the *core*, the part the base moves. Outside the core sit the two pieces from the model: the ledger $\mu$ and the flag. The machine never reads either of them to decide what to do next. I'm not asking you to take that on trust: it's how the step is written, and it shows up as a theorem below.

The properties a fact can name come from a tiny fixed language with three words: *is zero*; *is even*; and *is at least n*, for any whole number $n$ you like. Each one comes in two forms. There's the *checker*, a little program that computes yes or no: compare with 0, look at the last bit, compare with $n$. And there's the *meaning*, the plain arithmetic statement the checker is supposed to be checking: the number is 0; the number is two times something; $n$ is less than or equal to the number. Why both? Because a checker is code, and code can be wrong in ways that look right, and "the checker says yes" is only worth anything once somebody has proved it says yes exactly when the meaning is true. That proof is one line per property, and it's the first block below.

Why can a machine check a claim at all? Because a claim with a precise meaning is something you *evaluate*: plug in the values and turn the crank. George Boole worked out in 1847 that you can do arithmetic on claims this way, treating "and" like multiplication and "or" like a kind of addition. Why multiplication? Try it at the table. Write 1 for true and 0 for false, and multiply: one times one is one, and anything times zero is zero. That's "and" to the letter. Then comes the stranger thing Boole noticed. Say "x and x" and you've said nothing new; it's just x. In numbers that's $x \times x = x$, and he made it a law of his system, and it's a law ordinary numbers mostly break. Which numbers equal their own square? Two times two is four, no. A half times a half is a quarter, no. Zero, yes. One, yes. That's the whole list. So when Boole wrote his bigger book in 1854, *The Laws of Thought*, he asked you to imagine an algebra whose letters take the values 0 and 1 and nothing else. Every checker on this machine answers in that algebra.

And the 1847 book has a backstory worth a detour, which I found in the Stanford Encyclopedia of Philosophy's entry on Boole. Boole was largely self-taught and spent

about fifteen years as a schoolmaster, and what pulled him back to logic, early in 1847, was a loud public quarrel between Augustus De Morgan and the Scottish philosopher Sir William Hamilton over who had first thought of a small technical idea in the old logic of syllogisms. Two well-known men fighting over credit, and a schoolmaster watching from the side picks up the subject and writes the little book this paragraph is about. Few people remember what the quarrel was over. Every checker in this chapter runs on what the bystander wrote.

Then push on it, because Boole's "or" isn't quite the one a checker uses. His plus only made sense for things that can't both hold at once, so "x or y" with both of them true had no meaning in his system. William Stanley Jevons, an English economist who also worked on logic, read "or" as "either or both," which gives "x or x is x," the same trick as before. He wrote to Boole in 1863 saying plus ought to work that way, and in 1864 he put it in a book called *Pure Logic*. Boole wouldn't have it, since it broke the tie to ordinary algebra his whole system stood on. The "or" in every programming language I know is the inclusive one Jevons argued for. Does that change anything about the checker? No; the crank turns the same way. It changes the picture: the "or" a checker uses is the one the man who invented the algebra turned down.

Hanging a checked claim on a point in a computation is an old move too. Tony Hoare, in 1969, built a whole way of reasoning about programs out of it: attach a claim to a *point* in a program ("whenever execution reaches here, this is true"), and reason forward from claim to claim. Robert Floyd had done it two years earlier on flowcharts. Their claims were checked on paper by the person writing the proof. This machine's version is a property of one counter at one edition, checked by a step while it runs.

And is two years earlier the bottom? Not even close, and the next step down is a paper this book already leaned on. In June 1949, at a Cambridge conference held for the new EDSAC computer, Alan Turing gave a talk three pages long, "Checking a large routine," the same note the counter chapter used for stopping (Section 6). Stopping was the second half of it. The first half is this move, twenty years before Hoare: the programmer should make "definite assertions which can be checked individually," and the correctness of the whole should follow from them. His example program computes a factorial using nothing but addition, and his picture for why you'd bother is one you can do on the back of an envelope. Add up a long column of four-digit numbers and hand someone only the total, and they have to redo the whole sum at one sitting, because the carries tie every column to the one before it. Hand them the total of each column as well, and the job falls apart into small checks they can do one at a time. Each column total is a claim hung on a point in the middle of the work, so the person checking never has to hold all of it at once. The paper got its due mostly later: in 1984 two computer scientists, F. L. Morris of

Syracuse University and C. B. Jones of Manchester University, who had worked on it together at Oxford, republished it, with its misprints corrected, as an early program proof by Alan Turing. Theirs is the text I read, and the column of numbers is Turing's own example in it. Does any of that change the machine? No. It changes the date on the idea, and it tells you something I like: the first person to ask how you'd check a program put the claims in the middle of it. He didn't save them for the end.

The machine has six instruction kinds.

| **Instruction** | **What it does, when the trap is down** | **Cost** |
|---|---|---|
| INC *c* | Adds one to counter *c*, adds one to its version, goes to the next line. | 0 |
| DEC *c j* | If *c* is above 0: subtracts one, adds one to its version, jumps to line *j*. If *c* is 0: just goes to the next line. | 0 |
| HALT | Nothing. A program that reaches it stops. | 0 |
| CHECK *p c* | Runs the checker for *p* on *c*'s value. If it says yes and the table has fewer than sixteen facts, writes the fact (*p*, *c*, *c*'s current version) into the table and goes to the next line. Otherwise, trap. | 1 |
| COMMIT *p c* | Looks in the table for exactly the fact (*p*, *c*, *c*'s current version), field by field. If it's there, puts it in the channel and goes to the next line. Otherwise, trap. | 1 |
| CERTIFY | If the channel holds a fact, preserves that channel, goes to the next line, and the flag goes up. Otherwise, trap. | 1 |

The first two are Minsky's machine, and HALT lets a program say it's done. It says it by doing nothing: HALT leaves the state where it is. If it put the trap up instead, it would land a live state on its trapped twin, a merge in the record's part, and the record-layer merge would charge it. The last three are the record's, and they read

as a chain: you can only certify with a commitment in the channel, you can only commit to a fact in the table, and the only way into the table is a check that passed. The chain has three links, and each one costs.

One of what? One unit of $\mu$, the same unit as Section 2's toll. Why one? Why would ten be worse? Because one is the least each of these three acts can cost once a merge in the record's own part of the state is priced, as the end of this paragraph says, and Section 3 is where that pricing comes from. The argument for a whole finite machine doesn't reach this one: these counters have no top, and some of the base moves merge states at a price of zero. What does reach it is the record-layer merge named at the start of this chapter (Theorem 3.19, in Part I), which gives the toll once you've agreed to price a merge in the record's own part of the state. Agreeing to that is my choice. Once I have, one unit on each of the three acts isn't: COMMIT and CERTIFY merge when they pass, and all three merge when they fail, because a failure puts the trap up and changes nothing else, so the failed state lands where the same state with the trap already up lands. What I pick is one rather than more.

The whole numbers have a history of their own. In 1888 the German mathematician Richard Dedekind published a little book whose title asks what numbers are and what they're for, and in 1889 the Italian mathematician Giuseppe Peano published a booklet, written in Latin, that set out the axioms for the whole numbers in the form people still teach: there's a first number, and every number has a next one, and a few rules about how "next" behaves. I had the two of them down as working independently. I went looking, and the Stanford Encyclopedia's entry on Dedekind says Peano acknowledged Dedekind's priority, which is why the axioms often go by both names. (Peano's booklet is also where the sign $\in$ that you met in Section 5 comes from: he wrote it as a Greek epsilon, for the Latin *est*, "is.") Start from zero and take "the next one" three times, and you have three. Once prices are whole numbers, one is the smallest price that isn't nothing.

The zeros matter just as much: computing costs nothing on this ledger. Plus one, minus one and jumping around cost zero, with physical costs left to an implementation model. Only the three acts that make a claim cost, and each of them costs whether it passes or traps. Failing is not a discount.

**The version rule.** Some numbers show why versions exist. Start with $A = 4$, version 0. CHECK *is even* $A$ passes, and the table gets the fact (*even*, $A$, 0). Now INC $A$: $A$ becomes 5, and its version becomes 1. Now COMMIT *is even* $A$. The machine looks for (*even*, $A$, 1), the claim about the current edition. The table only has (*even*, $A$, 0). There's no match, so the machine traps. The old fact is still sitting in the table, still true about edition 0, and completely useless for committing to anything about edition 1. A fact is about one edition of one counter, and an edition never comes back, because versions only go up and never wrap. The version rule is the

whole trick against stale evidence: you can't check a number, change it, and then commit to the check. And the never-wrap part isn't a convenience. INC $A$, then DEC $A$ back to the same line, puts the line and both counters exactly where they were, at no cost, and once record-layer merges are priced neither move may join two states that agree on those. If the record had only finitely many settings over one line and one pair of counters, enough round trips would bring the whole state back to where a CHECK left it, with $A$ touched in between and its COMMIT still passing. Under those free-move, injectivity and continued-use requirements, the record fibre cannot be finite. This machine supplies fresh editions without a bound. A bounded implementation can instead stop before wraparound or change its invalidation and pricing rules; that changes the contract.

What the version rule stops is the time-of-check-to-time-of-use bug from Section 14, the file checked and then swapped, and the database answer to it is Kung and Robinson's: at commit time, check that nothing you read was written in the meantime, and back out if something was. The version rule is that check, done by the step, on one counter. It also shows where the guarantee ends. It works because one step does every write and every write bumps a version. A second writer that changed a counter without bumping its version, a second processor or a device writing memory on its own, would walk right around it, and none of the theorems below are about a machine with one of those.

Processor designers put a cousin of this check in silicon. Jensen, Hagensen and Broughton, at Lawrence Livermore in 1987, described a pair of instructions now called load-linked and store-conditional: the load reads a word and the processor starts watching it, and the store goes through only if nothing has written that word since; otherwise it fails and the program tries again. It's tempting to say CHECK is the load and COMMIT is the conditional store, and it's wrong in two places. COMMIT doesn't write the counter it checked, and the thing doing the watching, the fact, sits in a table the step keeps. No single processor holds it. The hardware version does cover the second writer, because the watching happens on the memory everyone shares, which is exactly the case the version rule leaves out.

The real kinship is older, and it's somewhere else. Picture the table and the counters shared by many processors, with CHECK, COMMIT and CERTIFY each done as one indivisible step. Maurice Herlihy (1991), a computer scientist who has spent his career on how many processors can share memory safely, ranked shared objects by how many processors each one can bring to agree on a single value with nobody ever waiting on anybody. Test-and-set, an instruction that writes a cell and reports what was there, brings exactly two. A sticky bit, the idea of Serge Plotkin (1989), a Stanford computer scientist working on distributed algorithms, is a cell the first writer jams for good, with the first writer's value, and everyone after can read it; it brings any number. I guessed that the shared object would sit at the top with

the sticky bit, because a write that kills every fact about a counter, together with a commitment other processors can see, looks like a way to build one. Ranking the object takes its operations, what each one returns, which steps are indivisible, how it starts and which processes run it, and then a proof that every processor decides, they all decide the same value, the value is one somebody proposed, and nobody waits. That proof goes through with every instruction one indivisible step and the whole state readable, and it puts the object at the top. The reason is the order of the fact table. Start it clean, with counter $A$ at zero and the trap down. Each processor CHECKs a property that holds of $A$ at zero, Zero to propose 0 and Even to propose 1, then reads the table and decides the bit of its oldest fact. Nobody touches $A$, so the first CHECK of the run passes. The table never loses an entry or reorders the ones it has, and a CHECK that finds it full trips the trap, after which nothing moves, so the first CHECK of the run fixes the oldest fact for good. Every processor that decides decides that bit, the bit is somebody's proposal, and each one decides after two of its own steps whatever the others do. The version rule and the commitment play no part. The trap alone doesn't do it: it records that something trapped, and it doesn't hand everyone the first writer's value.

Now take the version rule away, let failures change nothing, keep only the properties "at least $n$" so that a checked property only ever turns true, drop the decrement, and make the fact table (with no room limit), the commitments and the flag sets and latches that only grow. I guessed that then the object drops to the bottom rung, the level of plain memory, which is the shared-memory face of what the database researchers Joseph Hellerstein and Peter Alvaro (2020), at Berkeley and Santa Cruz, call CALM, where monotone means nobody has to coordinate. That guess is proved too, by Herlihy's own kind of argument. Let the updates return nothing and let the whole state be readable. If at every state, of any two updates, one leaves the state alone, or the two commute, or one overwrites the other, then no protocol brings two processors to agreement with nobody waiting. To see why, give one processor 0 and the other 1. At the start the answer isn't settled, since either one running alone has to decide its own bit. Nobody waits, so it can't stay unsettled forever, and there's a last unsettled state: whichever processor moves next settles it, one way if it's the first and the other way if it's the second. Look at those two moves. If one is a read, or an update that leaves the state alone, the other processor can't tell whether it happened, and running alone it decides the same bit either way. If the two commute, both orders end on the same state with both processors in the same place. If one overwrites the other, the overwriter running alone can't tell whether the other move happened first. Each case contradicts the two ways the answer was settled. The monotone object, with registers beside it, has that property. An update that fails changes nothing. Two updates that both succeed still succeed in either order, because an instruction that would succeed still succeeds after anything

else has run, and adding to sets and to counters gives the same result in either order. Two of the conditions do real work there. The table needs no room limit: a full table that turns away the latecomer is a first-writer-wins cell again, and that's back at the top. And the ledger: it changes neither result as long as a failed act in the variant isn't charged, or nobody reads the ledger. Charge a failed CHECK where everyone can read the total, and one processor INCs while the other CHECKs *at least 1*; the total and the table then tell each of them which act came first, and that's enough for two to agree. Neither idea is mine: the top is Plotkin's and the bottom is the monotone fact that field already knows.

Process-calculus people have a statement of the same split. Johannes Åman Pohjola and Joachim Parrow, computer scientists at Uppsala University (Parrow is one of the inventors of the pi-calculus, a mathematical model of processes that talk over channels), in 2016 proved that no model whose parallel composition is monotonic, where a neighbouring process can never stop a step, has a consensus process. One of their examples of a model that does is test-and-set. Test-and-set isn't a sticky bit: in Herlihy's ranking it brings two processors to agree, and a sticky bit brings any number.

Their setting is small enough to go through here whole. A model is any set of processes with a step relation and a way to compose two processes that doesn't care about order or grouping. Composition is monotonic when whatever a process can reach alone, it can still reach with anything set beside it. A consensus process is one where any number of copies, run together, can end up agreeing on either of two outcomes; once a run has one outcome it can never show the other; and an outcome, once reached, survives having more copies, run any way you like, set beside it. Their theorem goes through in a few lines: run half the copies to one outcome and the other half to the other. If neither half can block the other, the whole crowd can reach the pair of end states set side by side, and since each outcome survives the company, both outcomes hold there, which is forbidden. Then the machine goes into it. A process is a shared machine state with a crowd of threads, each proposing a bit. Composing two processes pools their threads, each thread keeping whatever it has already decided, and settles the shared state like this: if one side is still at the fresh start, the composite takes the other side's state; if both have already moved, the composite is jammed, a state where no thread can take a step. That composition doesn't care about order or grouping: a fresh start is neutral, and any two moved states jam whichever way round you put them. One thread proposing each bit is a consensus process in exactly their sense, for the outcomes "some thread decided 0" and "some thread decided 1": copies started fresh can reach either outcome and never both, because the first passing check fixes the oldest fact and every later thread reads the same one; and an outcome survives company, because pooled threads keep their decisions. So by their theorem the composition isn't monotonic, and

the small machine sits on their non-monotonic side. That verdict is cheap, though: a composition that jams any two moved states makes every shared-state machine non-monotonic, so the encoding puts it there, not the fact table. What does the work in the run, making every thread read the same outcome, is the order of the fact table, the same as above; a store that refuses a stale commit for good, as the version rule does, refuses in the same spirit. The shared-memory version of their split is the pair of results above: the small machine's object reaches agreement for any number of processors, and the monotone variant, with its table unbounded and no charge for failing that anyone can see, doesn't reach it even for two.

Security people know the pattern by a third name. A check that blesses a value, and a write that takes the blessing away, is close kin to dynamic taint tracking, where a label rides along with the data and a sanitizing step changes it; the labels go back to the computer-security researcher Dorothy Denning's 1976 lattice model of how information is allowed to flow. How much a monitor like that has to remember is a studied question too: Chamseddine Talhi, Nadia Tawbi and Mourad Debbabi (2008), three security researchers at universities in Quebec, worked out which rules a monitor can still enforce when its memory is bounded. The version rule is one small, specific member of that family, done by the step.

Notice what *isn't* a write. DEC on a counter that's already 0 doesn't change the counter, so it doesn't change the version. It's a test that happens to move the line number.

**The fact table.** Facts go in and never come out. Nothing erases them and nothing overwrites them. When the table is full, a seventeenth CHECK traps rather than pushing an old fact out to make room. That choice has a downside, and I'll set it out in the subsection on what this machine doesn't do.

**Traps.** Any failure puts the trap up and changes nothing else in the core; the ledger still takes the instruction's cost. Once it's up, every instruction leaves the core alone. A stored program stops at a trap. But I also prove things about *any* list of instructions, programs or otherwise, because an adversary doesn't have to run my programs; they can feed the machine whatever they want. For those lists, an instruction after a trap is still charged its cost, and still changes nothing. That trap keeps nothing of what tripped it, and the record-layer merge charges it for that. Merely counting refusals does not remove the merge. A live state with count zero and its already trapped twin with count zero can both become trapped with count one. Keeping the entire previous state in a history would make each extended move injective, but that is a different, unbounded state space. The counter alone is not that history.

The exact version comes next.

**Definition 15.1** (Properties, checker, meaning)**.** A *property* is one of: Zero, Even, or Ge$(n)$ for a whole number $n$. The *checker* eval$(p, v)$ computes yes or no: $v = 0$ by equality, evenness by the parity function, $n \leq v$ by comparison. The *meaning* holds$(p, v)$ is the arithmetic statement: $v = 0$; there is an $m$ with $v = 2m$; $n \leq v$.

**Lemma 15.2** (The checker says what it means)**.** For every property $p$ and every whole number $v$, eval$(p, v) =$ yes if and only if holds$(p, v)$. □

*Proof.* Ask whether the number is zero: the equality test says yes at zero and no at every other number. Ask whether it is at least $n$: the comparison says yes precisely when $n$ is no larger than it. Those two tests already answer the arithmetic questions they name.

For evenness, the checker says yes at zero, no at one, and adding two leaves its answer unchanged. Take any whole number and remove pairs until fewer than two remain. If none remain, the number was two times the number of pairs removed, and the checker gives zero's answer, yes. If one remains, it was two times that many pairs plus one, and the checker gives one's answer, no. Thus yes means exactly $v = 2m$ for some whole number $m$. These are all three properties in the language. ■

**Definition 15.2** (Core, state, claim, start)**.** A *core* $k$ holds two counters $A$, $B$ and their versions (whole numbers), a line number, a fact table (a list of facts, a fact being a triple of a property, a counter and a version), a channel (empty or one fact), and a trap bit. Write val$(k, c)$ and ver$(k, c)$ for counter $c$'s value and version. A *state* $s$ is a core, written core$(s)$, with a ledger $\mu(s)$ and a flag cert$(s)$. The *claim* $p$ of $c$ in $k$ is the fact $(p, c, \text{ver}(k, c))$. The table's room is 16. The start start$(a, b)$ has counters $a$ and $b$, versions 0, line 1, an empty table, an empty channel, the trap down, $\mu = 0$ and the flag down. A state is *clean* when its table and channel are empty and its flag is down.

**Definition 15.3** (Instructions, costs, the step)**.** The instructions are INC $c$, DEC $c$ $j$, HALT, CHECK $p$ $c$, COMMIT $p$ $c$ and CERTIFY, with effects on a core with the trap down as in the table above. Passing any check, commitment or certification requires the trap to be down. With that condition, CHECK passes when eval$(p, \text{val}(k, c)) =$ yes and the table holds fewer than 16 facts, COMMIT passes when the claim $p$ of $c$ is in the table, and CERTIFY passes when the channel holds a fact; a failing one sets the trap and changes nothing else. On a core with the trap up, every instruction

changes nothing. Costs: CHECK, COMMIT and CERTIFY cost 1, the rest 0. One step on a state is

$$\mathrm{exec}(s,i) = \big(\,\mathrm{cexec}(\mathrm{core}(s),i),\ \ \mu(s)+\mathrm{cost}(i),\ \ \mathrm{cert}(s)\vee\mathrm{fires}(\mathrm{core}(s),i)\,\big),$$

where cexec is the core's step and $\mathrm{fires}(k,i)$ is yes exactly when $i$ is CERTIFY and passes at $k$. For any list $t$ of instructions (a *trace*), $\mathrm{run}(t,s)$ applies them in order and $\mathrm{cost}(t)$ adds their costs. A *program* is a list of instructions numbered from 1; a program run fetches the instruction at the line number and executes it, and stops when the trap is up, the line holds HALT, or the line holds nothing.

Read the step line slowly, because the rest of the chapter leans on its shape. The core moves by its own rule and nothing else. The ledger adds the cost. The flag becomes "it was up, or this is a CERTIFY that passed." The core's move doesn't mention $\mu$ or the flag anywhere. The base is blind to the record by construction.

The shortest worked run there is uses the program CHECK *is zero* $A$; COMMIT *is zero* $A$; CERTIFY. Start at $\mathrm{start}(0,b)$, for any $b$. Line 1: $A$ is 0, the checker says yes, the table is empty so there's room, and the fact (*zero*, $A$, 0) goes in. $\mu = 1$. Line 2: the claim about $A$'s current edition is (*zero*, $A$, 0), and it's right there, so it goes into the channel, and $\mu = 2$. Line 3: the channel holds a fact, the flag goes up. $\mu = 3$. Line 4 has no instruction, so the program stops. The run ends with three paid acts, in order, a flag that's up, and a channel that names the last earned commitment, which here is (*zero*, $A$, 0).

Now start the same program at $\mathrm{start}(5,b)$. Line 1: $A$ is 5, the checker says no, the trap goes up, $\mu = 1$, and the program stops right there. The flag is down, and you can run it as long as you like: it stays down. That second run is the point of the first. The same three instructions certify from some starts and refuse from others, because the check could have failed, and from most starts it does. A check like that has a no pile, which is the only kind Section 14 counts as establishing anything.

Could I have picked an even dumber claim? I could: CHECK *at least 0* $A$ passes on every number there is, so CHECK, COMMIT, CERTIFY on it raises the flag for 3 from every start. The step doesn't ask whether the question was worth asking, and it shouldn't. Worth is a count over the rivals a run narrows (Section 8), and a step that sees one state has no list of rivals to count. What the step checks is order, aboutness and that the checker tells the truth. What a check that can't fail is worth, the entitlement count says exactly: zero bits, for a cost of three. The price is for standing behind the claim. What the claim bought is a separate number, one the step doesn't compute, and I keep the two apart. It's on the list at the end of the chapter.

Others walked into the same line from communication. Three theoretical computer scientists, Oded Goldreich, Brendan Juba and Madhu Sudan (2012), built a

theory of communication with a partner you may not understand, in which getting a message across counts only as a way to reach a goal, and the central notion is *sensing*: being able to tell, from what you see, whether you're making progress. They ask two things of it. It has to be *safe*: with any partner in the class, when there's no progress it keeps saying so, so it can't report a lasting success that didn't happen. And it has to be *viable*: with a partner who could help if you spoke its language, the right way of speaking gets a yes from it within a known time, and keeps getting one. Either half alone is cheap. A sensor that always says no is safe, and one that always says yes is viable. Their main result, as I read it, is that the two together, with the ways of speaking listed one after another, are enough for one strategy that reaches the goal with every such partner. Their theorem, in their setting, is a check that can fail and never lies, standing in front of a verdict you can't take back, and it's the same shape.

## 15.4 It computes everything two counters can

First the base, because if the base can't compute, nothing else here matters. Put the ordinary two-counter program beside this machine's program. Replace each "add one" by this machine's INC, and each "subtract if possible and jump" by its DEC, keeping the named counter and jump destination. That is all the translation does.

Now cover the facts, versions, channel, flag and ledger with a sheet. Leave the two counters and line number visible. I want to show that the visible part follows the original program exactly. The hidden versions may change, but these instructions cannot raise the trap, so the sheet never hides a new refusal that would stop the comparison. Here are the two programs and the window written precisely.

**Definition 15.4** (Two-counter program, compile, window)**.** A *two-counter program* $M$ is a list of instructions, each either MINC $c$ or MDEC $c$ $j$. A configuration is a triple (line, $a$, $b$). One step of $M$: if the line holds no instruction, $M$ has *stopped*; MINC adds one to $c$ and goes to the next line; MDEC subtracts one from $c$ and goes to line $j$ if $c$ is above 0, and goes to the next line if $c$ is 0. $\mathrm{mrun}(n, M, x)$ is $n$ steps from $x$, staying put once stopped. The *compile* of $M$ replaces each MINC $c$ by INC $c$ and each MDEC $c$ $j$ by DEC $c$ $j$. The *window* of a core $k$ is the triple (line number, $\mathrm{val}(k, A)$, $\mathrm{val}(k, B)$).

**Theorem 15.3** (Step for step)**.** For every two-counter program $M$ and every core $k$ with the trap down: $M$ has stopped at $\mathrm{window}(k)$ if and only if the program $\mathrm{compile}(M)$ has halted at $k$; and if $M$ steps from $\mathrm{window}(k)$ to $y$, then one program step of $\mathrm{compile}(M)$ from $k$ lands on a core with window $y$ and the trap still down. □

So for every $n$, $n$ program steps of compile$(M)$ from $k$ land on a core whose window is mrun$(n, M,$ window$(k))$, with the trap down.

*Proof.* Look up the instruction at the line. If there is none, both sides stop. If it's MINC $c$, the compiled INC $c$ adds one to $c$ and goes to the next line, which is exactly the window's move (the version also moves, but the window doesn't show versions). If it's MDEC $c$ $j$, split on whether $c$ is 0, and both sides do the same thing in each case. Neither instruction can trap. After that move the windows agree again and the trap is still down. Those are exactly the conditions I began with, so the next instruction can be compared in the same way. Repeat until the chosen $n$ steps have been counted; if either side stops earlier, both stay put for the remaining comparisons. ■

□ **Theorem 15.4** (Halting matches)**.** For every two-counter program $M$ and whole numbers $a$, $b$: $M$ stops after some number of steps from $(1, a, b)$ if and only if the program compile$(M)$ halts after some number of program steps from start$(a, b)$.

*Proof.* The start has window $(1, a, b)$ and its trap is down. Apply Theorem 15.3 after any number $n$ of steps: the two configurations still have the same window, the trap is still down, and one is stopped exactly when the other is halted. The same $n$ witnesses termination in either direction. ■

Now comes the part that isn't mine. Two-counter universality is Minsky's, and the constructions of Forster, Larchey-Wendling and their coauthors connect it to several other models. The counter convention matters: DEC at zero goes on to the next instruction; otherwise it takes one off and jumps, the convention used in Andrej Dudenhefner's certified decision procedures for two-counter machines (2022). The companion volume builds the route from a tape to counters, packs finitely many counters into two, and gives the halting diagonal (Appendix J points to it). The remaining step here is to match that two-counter machine to this machine's INC/DEC fragment, instruction for instruction.

I want to be exact about what "undecidable" means in the theorem below, because it's stated as a connection between two problems. Suppose you had a procedure that answered the halting question for this machine. Then you could write a program that lists, one by one, every single-tape Turing machine that *doesn't* halt. No such list can exist: the ones that *do* halt can already be listed (just run them all, a step at a time, and call out each one as it stops), and with both lists running side by side you'd have a halting decider. So the theorem says: a decider here would give you that impossible list.

**Theorem 15.5** (Halting is undecidable for this machine)**.** The two-counter machine with the instruction convention just stated agrees with the INC/DEC fragment: its step, its stop, and its termination match those of the compiled program, so its halting problem at $(M, a, b)$ has a positive answer if and only if compile$(M)$ halts from start$(a, b)$. Consequently, the question "does program $P$ halt from start$(a, b)$?" is undecidable in the following reduction sense: a decider for it would make the complement of single-tape binary Turing machine halting enumerable. □

*Proof.* The agreement is a case split on the counter-machine instruction at the current line, and then induction along the run, ending in Theorem 15.4. Composing a proposed decider for the compiled program with the compile map would decide two-counter halting. The tape-to-counter reduction of the companion volume (Appendix J) would then decide single-tape binary halting and so list its complement, as claimed. ■

What that buys is that the base of this machine is a universal machine in Minsky's sense, and that no algorithm can predict in general what it'll do. What it doesn't buy is a fast machine, or one where your input arrives conveniently. The counters start holding $a$ and $b$, given directly; there's no encoding layer in this construction. Minsky's prime-power packing is how you'd get from "any Turing machine" to "two counters," and I use that construction as it stands (Appendix J.1 sets it out).

Is "given directly" just a convenience, or does it bite? It bites, and somebody found exactly where. In 1972 Rich Schroeppel, then at the MIT AI Lab, wrote a ten-page memo titled "A Two Counter Machine Cannot Calculate $2^N$." He spells out the convention first: a counter machine computing a function starts with $n$ in one counter and zero in the others. Then he gives a proof that a two-counter machine started that way can't compute $2^n$. That's his theorem, and I don't reprove it. What I prove is a cousin, in Appendix C.4 (Theorem C.23): no program multiplies every input by one large constant fixed by the program's length. So the packing Minsky needs is precisely the thing two plain counters can't do for themselves. Hand the small machine its inputs straight and there are functions of them it can't compute; hand it the inputs already packed and it computes anything a Turing machine can. That's no contradiction with the undecidability above, which ranges over every program and every starting pair, packed ones included. The base hides nothing; it just needs its input pre-chewed, and the chewing is somebody else's.

### 15.5 It's a Thiele machine, and the flag is a latch

Now it's the record's turn. There are three facts to state first, and all of them read straight off the step line.

□ **Theorem 15.6** (Ledger, toll, permanence)**.** For every state $s$ and instruction $i$, $\mu(\text{exec}(s,i)) = \mu(s) + \text{cost}(i)$, and for every trace $t$, $\mu(\text{run}(t,s)) = \mu(s) + \text{cost}(t)$. If $\text{cert}(s)$ is down and $\text{cert}(\text{exec}(s,i))$ is up, then $i$ is CERTIFY, it passes at $\text{core}(s)$, and $\text{cost}(i) \geq 1$. If $\text{cert}(s)$ is up, so is $\text{cert}(\text{exec}(s,i))$.

*Proof.* Each instruction adds its listed charge to the ledger. Start at $\mu(s)$: after the first instruction its charge has been added, after the second both charges have been added, and every further instruction adds its own. An empty list adds nothing. Thus any finite trace leaves exactly the starting ledger plus the sum of the charges.

The flag has two ways to be up afterwards: it was already up, or this instruction successfully certified. If it was down before and up afterwards, only the second way is available. The instruction must be CERTIFY, its check of the channel must pass, and its listed charge is one. If the flag was already up, the first way keeps it up regardless of the next instruction. ■

The middle clause of those three is Section 2's toll, for this machine, which means this machine *is* one of the things Section 2 was about, and Section 2's universal floor applies to it without my lifting a finger.

□ **Theorem 15.7** (The small machine is a Thiele machine)**.** The states, the instructions, the step exec, the cost and the flag, with the toll of Theorem 15.6, form a Thiele machine (Definition 2.2). So the universal floor (Observation 2.1) holds here: if $\text{cert}(s_0)$ is down and $\text{cert}(\text{run}(t,s_0))$ is up, then $\text{cost}(t) \geq 1$.

*Proof.* Take the state and instruction sets of Definitions 15.2 and 15.3, their step and cost, and the flag as the reading. These provide the five components of Definition 2.2. Theorem 15.6 supplies its one required rule. Applying Observation 2.1 gives the claimed bound. ■

Section 10 comes next. Section 10 said: take any base, and any record that answers only to the base and never comes back down, and the record is a latch. This machine has a base: a program together with a core, stepping by the program step. And it has a record, the flag, with the ledger riding along. So the question is whether it's a *base-driven* extension in Section 10's five-rule sense (Definition 10.2), and if it is, Section 10 does the rest.

**Theorem 15.8** (The flag is a latch of the base)**.** Let the base have states (program, core), stepping by the program step, starting at start$(a, b)$'s core for any $a$, $b$, and halting when the program has halted. Map each (program, state) to (program, core). Then the machine, run as a closed machine on (program, state), is a base-driven record over that base, and so its flag factors as the latch (Observation 10.1) of the event "the next instruction is a CERTIFY that passes." □

*Proof.* First the cover (Definition 10.1): the map sends starts onto starts, it commutes with the step because the core's move never reads $\mu$ or the flag, and whether the program has halted is read off the core. Then the rules of Definition 10.2 one at a time. Driven by the computation: the next flag is the old flag or "the next instruction is a passing CERTIFY," which is a function of the base state and the old flag (that's the step line). Ledger: $\mu$ only adds. Toll: Theorem 15.6. Permanence: Theorem 15.6. A reachable write: the three-line program of the worked run raises the flag from a start. Section 10's theorem then gives the factorization. ■

So far there's nothing new. The toll, the floor and the latch are things Parts I and II already promised about any machine of this kind. The next subsection is the part neither of them could give.

*The lamp and the claims it does not name*

The lamp is on. Which claim turned it on? Did another claim get certified afterwards? Look at the lamp as long as you like. It still gives you one answer.

Put a file beside it. Every time CERTIFY succeeds, copy the fact from its channel into the file: the predicate, the counter it concerns, and that counter's edition. Call the set of filed facts $R$. Publishing the same fact again leaves the same set. A later write can change the counters while the old receipt stays filed under its old edition.

Run this record alongside the native state. The native instructions keep doing exactly what they did before. Forget the file and you recover their run. Open the file and you can ask which claims were actually published.

**Theorem 15.9** (The native lamp is the archive's Boolean reading)**.** From a clean start with an empty added archive, after every finite native instruction word, □

$$\mathrm{cert}(s) = \mathrm{true} \quad \Longleftrightarrow \quad R \neq \varnothing.$$

Membership in $R$ means exactly that this fact was emitted by a successful CERTIFY earlier in that word. Already recorded facts persist, and zero-cost instructions cannot add a fact.

*Proof.* Start with the empty archive and the flag down. Inspect the next instruction. A successful CERTIFY inserts its channel fact into $R$ and raises the flag. Whether or not that fact was already present, the resulting set is nonempty and the flag is up. Every other case leaves both the archive and the flag alone, including failed or trapped instructions. This includes every horizontal move. It follows, one instruction at a time, that a fact belongs to $R$ exactly when some earlier successful CERTIFY published it, that no later move removes it, and that the flag is up exactly when $R$ is nonempty. Forgetting $R$ leaves the native updates untouched, so it also recovers the native run. ■

The lamp tells you whether that file is empty. The file tells you what is in it. Below, I will make 65,535 different nonempty files leave the entire native machine in one identical state. Before that, follow how a fact earns its place: CHECK establishes it in the table, COMMIT selects it, and a successful CERTIFY publishes it. Each step has a job.

## 15.6 Commitments are earned

The shape comes first, before the words. Pick any run from a clean start, any list of instructions at all, chosen by somebody who'd love to cheat. Look at the moment a COMMIT would pass. Then, somewhere earlier in that run, there's a CHECK of exactly the same claim that passed and paid, at the same edition of the counter, and between that CHECK and now, nothing touched the counter. That holds on every list of instructions anybody can write, not "probably" and not only "on the programs I wrote."

To say "nothing touched it" exactly takes one word.

**Definition 15.5** (Untouched)**.** A list of instructions *mid*, run from a state $s$, leaves counter $c$ *untouched* when, for every way of splitting *mid* as $t_1$, then $i$, then $t_2$, executing $i$ at the core reached by running $t_1$ from $s$ leaves both $\mathrm{ver}(\cdot, c)$ and $\mathrm{val}(\cdot, c)$ as they were.

□ **Theorem 15.10** (A commitment was checked first)**.** Let $s_0$ be clean and *pre* any trace after which COMMIT $p$ $c$ would pass. Then *pre* splits as $pre_1$, then CHECK $p$ $c$, then *mid*, where: the CHECK passes at the core reached by $pre_1$; CHECK $p$ $c$ and COMMIT $p$ $c$ each cost 1; $c$'s version after $pre_1$ equals its version after *pre*; and *mid*, run from just after the CHECK, leaves $c$ untouched.

*Proof.* COMMIT passes, so the claim $(p, c, \mathrm{ver}(c))$ for the current version is in the table. By the no-forging invariant (Theorem 15.14 below), that fact was written by a passing CHECK $p$ $c$ at some point $pre_1$, with the version $c$ had then; since it's the current version too, the two versions are equal. Versions never go down, and any instruction that changes a counter's value also raises its version, so an instruction in *mid* that touched $c$ would have left $c$'s version higher at the end than at the CHECK. It isn't. So nothing in *mid* touched $c$. ∎

**Theorem 15.11** (A raised flag was checked, then committed, then certified)**.** Let $s_0$ be clean and $t$ any trace with $\mathrm{cert}(\mathrm{run}(t, s_0))$ up. Then for some $p$ and $c$, $t$ is □

$$pre_1,\ \text{CHECK}\ p\ c,\ \ mid_1,\ \ \text{COMMIT}\ p\ c,\ \ mid_2,\ \ \text{CERTIFY},\ \ post,$$

where the CHECK passes at its state, the COMMIT passes at its state, the CERTIFY passes at its state, $c$ has the same version at the CHECK as at the COMMIT, and $mid_1$ leaves $c$ untouched. And when the CERTIFY runs, the channel holds exactly the claim that COMMIT made: the fact ($p$, $c$, $c$'s version at the CHECK). CERTIFY takes no operand; this is what ties it to the claim.

*Proof.* The flag starts down, so some first instruction raised it, and by Theorem 15.6 it's a passing CERTIFY. A CERTIFY passes only with a fact in the channel, the channel starts empty, and only a passing COMMIT puts a fact there, so some earlier COMMIT $p$ $c$ passed. Take the last one before the CERTIFY: the fact in the channel is the one it wrote. Theorem 15.10 at that COMMIT gives the CHECK, the equal version, and the untouched stretch, and the equal version makes the committed fact ($p$, $c$, $c$'s version at the CHECK). ∎

The failure side comes next, so you can see the refusals are clean, with nothing left half-done:

**Theorem 15.12** (Unearned steps trap and change no claim)**.** If COMMIT $p$ $c$ would not pass at a core $k$, executing it puts the trap up and leaves the channel and the table as they were. If CERTIFY would not pass at a state $s$, executing it puts the trap up and leaves the flag as it was. □

*Proof.* If the trap is already up, the core is unchanged and its trap remains up. Otherwise, the failed COMMIT takes the failure branch of the instruction table: set the trap, leave every other core field alone. A failed CERTIFY takes the same branch,

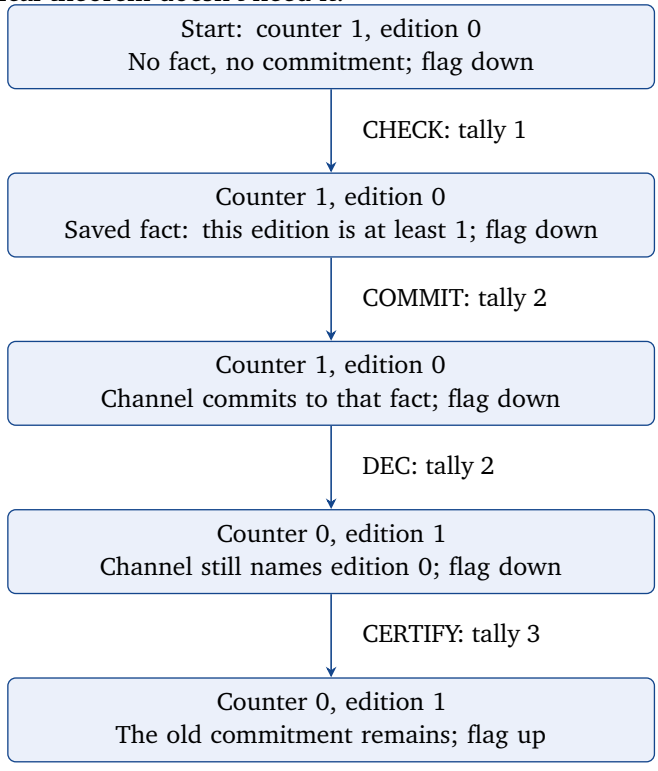
and its fires value is false. The flag update is then its old value or false, which leaves it unchanged. The instruction's charge still applies to the ledger.

If that rings a bell, it should. It's Milner's move from Section 14: in LCF a theorem is a value that only the rules of the logic can build, so nobody can just print one (Milner and two colleagues on his Edinburgh project, Michael Gordon and Christopher Wadsworth, wrote the system up in 1979, and Gordon, who went on to build the HOL prover at Cambridge, told its history in 2000). Here the thing nobody can print is one bit, and what guards it is one step rule. Milner's guard was a type checker. Same architecture, much lower down.

A provenance theorem is also an itemised receipt, and the itemised receipt has a history that's exactly this gap. James Ritty ran a saloon in Dayton, Ohio, and believed some of his staff were pocketing money. On a steamboat trip to Europe in 1878 he watched a mechanism that counted the turns of the ship's propeller, and wondered whether something like it could count the money instead. In 1879 he and his brother John patented a "Cash Register and Indicator," and they sold a version under the wonderful name Ritty's Incorruptible Cashier. The first model was this book's counter (Section 6) with brass on it: press a key, the amount goes onto a dial showing the day's running total. There was no drawer and no paper. At closing time you hold the dial up against the cash. And that's all a total can do. It tells you *that* the money is short. It can't tell you which sale went missing or who rang it up. The paper came later. The patents passed to other owners, first a Cincinnati salesman named Jacob Eckert, and in 1884 to John H. Patterson, who renamed the business the National Cash Register Company, and the story told about that company is that it added a paper roll: one copy kept for the shop's own books, one handed to the customer. That last part I only have from histories of the company and of the machine, so hold it the way you'd hold any company's story about itself. The order is right either way: first a total that can tell you something's wrong, then a list that tells you where. The ledger $\mu$ is the dial. Theorem 15.11 is the list: which CHECK, of which claim, about which edition, before which COMMIT. The theorem supplies that list about the run. The machine doesn't keep it as data anyone can read back from the end state, and nothing in the theorem gives a procedure for rebuilding it from there. A fact in the table holds a property, a counter and a version number. That's neither the counter's value back then nor a log of what happened in between.

Read Theorem 15.11 once more for what it *doesn't* say. Between the CHECK and the COMMIT, the counter is untouched. Between the COMMIT and the CERTIFY, it isn't promised anything. You can check $A$ is even, commit to it, add one to $A$, and then certify. The flag goes up, and what it stands on is a commitment about the edition that was current when you committed. The certificate is historical: it says "this was checked and committed about this edition," and it says nothing about

whether that's still true. I think that's the right meaning for a certificate (a birth certificate doesn't expire when the baby grows), but it's a meaning, and you should know which one you're getting. So a certificate doesn't say the claim holds at the CERTIFY, and it doesn't say the claim holds at the end of the run. A CERTIFY that rechecked the version would give a different guarantee. That's a possible extension, and the historical theorem doesn't need it.




The checked claim was true of edition zero. The raised flag does not say that the current counter is still at least one.

## 15.7 What a passing check means

Provenance says a CHECK happened. It doesn't yet say the CHECK was *right*, and that's a separate question with a separate proof. A machine could have a perfectly ordered chain of check, commit, certify, all standing on a checker that says yes to everything. So what does a fact in the table actually tell you about the counter?

**Theorem 15.13** (Checker soundness)**.** Let $s_0$ be clean and $t$ any trace, and let $k$ be the core of $\mathrm{run}(t, s_0)$. For every fact $(p, c, v)$ in $k$'s table with $v = \mathrm{ver}(k, c)$, the □

property holds of the counter's current value: $\text{holds}(p, \text{val}(k, c))$. In particular, if COMMIT $p$ $c$ would pass at $k$, then $\text{holds}(p, \text{val}(k, c))$.

*Proof.* Carry one sentence through every instruction: every fact in the table has a version no higher than its counter's, and if its version equals its counter's, its property holds of its counter's value. A clean start has an empty table, so the sentence holds there. A passing CHECK adds a fact at the current version, and the checker said yes, so by Lemma 15.2 the property holds. A write raises one counter's version, so every old fact about that counter falls strictly behind and the second half no longer applies to it. Nothing else changes a value or a version. The sentence holds after every instruction, so it holds at the end. The part about COMMIT is the case where the claim is in the table at the current version. ∎

So, on a run from a clean start, a live fact means exactly that one of three arithmetic properties holds of one counter's current value. It doesn't mean "the program is correct," and it doesn't mean "the number is interesting"; it means the property holds of the value, now. That's all "checked" means on this machine, and it's enough to be true. Three events stay separate here. The property holding of the value is one. A CHECK passing is another: a CHECK of a property that holds still doesn't pass when the table is full or the trap is already up. A certification is a third, and it needs a COMMIT and a CERTIFY after the CHECK. So a CHECK that didn't pass says its property is false only when the trap was down and the table had room before it.

So who checked the checker? Nothing on the record did. From a clean start, the record carries the run's half of the story: a CHECK, then a COMMIT of the same claim about the same edition, then a CERTIFY. The other half, that the checker says yes exactly when the property holds, isn't in any run. It's Lemma 15.2, a proof about the little program the step calls, priced by nothing on the ledger. So a claim established here stands on two legs: a fact about this run, carried by the record, and a fact about the checker, carried by a proof that whoever reads the record has to accept. That's the Münchhausen trilemma from Section 14 taking its third horn on purpose: the chain of reasons stops at a proof, named and on the page, where you can decide whether you'd have stopped there too. And the entitlement it gives is the reader's. The machine doesn't know anything. It keeps the order.

The question in that paragraph is older than any checker, and its most famous wording isn't about what you'd guess. "Who will guard the guardians?" gets quoted about governments and police and the people checking mathematical proofs. Where's it from? Plato's *Republic* has the worry, sort of: in Book III (at 403e), Glaucon, Plato's older brother and one of the people Socrates is talking with, says it would be absurd for a guardian to need a guardian, which is close in words and opposite in mood, a man calling the question silly. The Latin, *quis custodiet ipsos*

*custodes*, is from the Roman satirist Juvenal's sixth satire, and it's a joke about a husband. His friends tell him to bolt his wife in and set guards on her; the line asks who'll watch the guards, since she'll start with them. And it gets better. In 1899 an Oxford undergraduate, E. O. Winstedt, found extra lines of that satire in a medieval manuscript in the Bodleian Library, lines the other copies lack, partly overlapping the familiar ones, and the famous question sits in that tangle. Scholars have argued ever since about whether those lines are Juvenal's at all. So the most quoted sentence about trusting the checker comes with a doubtful source of its own. Does that change anything here? It doesn't. The answer used here is simple: you never get rid of the last guard, you make the last guard small enough to watch yourself. Lemma 15.2 is one line per property. Euclid had the instinct: five postulates you can hold in your head, and everything else meant to follow from them.

That split, a fixed checker proved sound from outside and a process that only gets to act when the checker says yes, has a name in the writing on self-improving programs. The AI researcher Jürgen Schmidhuber, at the IDSIA lab in Lugano, described a Gödel machine (2003) that rewrites its own code only after its built-in proof searcher has proved the rewrite useful, and Benja Fallenstein and Nate Soares (2015), researchers at the Machine Intelligence Research Institute in Berkeley, survey agents that reason about their successors through formal proofs a fixed checker accepts. Same architecture, aimed at a different question: they want to know when a machine may trust a smarter version of itself, and this machine only wants to know whether a flag was earned. Keeping the checker's proof outside the machine is my design choice. The nearest theorem, in Section 13, is narrower: no procedure whose flip the substrate can represent decides, for every program, a property that depends only on behaviour and holds of one program and fails of another. It doesn't say a machine can't check a proof about a checker, or verify a restricted part of itself.

Soundness rules out a lying checker. It doesn't rule out a pointless question: *at least 0* has a perfectly sound checker and an empty no pile, as the worked run showed.

Program checking has a theory of its own, and its founding paper asks for a checker that could have said no. The computer scientists Manuel Blum and Sampath Kannan (1989; journal version 1995) defined a checker for a computational problem: given a program and an input, it may run the program as it likes, and it must say "correct" with high probability when the program is right everywhere and "buggy" with high probability when its answer on that input is wrong. A checker that always says "correct" isn't one. Here the checker is fixed, deterministic and proved sound once, and the part of their demand that survives is the fourth clause of certified Thiele-complete (Definition 15.7 below): some claim whose check can fail.

### 15.8 Nothing gets forged

The proof of Theorem 15.10 leaned on an invariant I promised to show you, and this subsection shows it. The fear it answers: could some clever sequence of instructions get a fact into the table without a CHECK writing it? Could a DEC with a funny jump target do it, or a COMMIT after a trap, or something I didn't think of? I answer it one instruction at a time, which is the only way I trust myself to answer it.

**Definition 15.6** (Earned fact; no forgery)**.** After a trace $t$ from $s_0$, a fact $f = (p, c, v)$ is *earned* when $t$ is *pre*, then CHECK $p$ $c$, then *mid*, where the CHECK passes at the core reached by *pre*, $f$ is exactly the claim $p$ of $c$ at that core, and, if $f$ is live at the end ($v$ equals $c$'s version after $t$), *mid* leaves $c$ untouched. The trace has *no forgery* when every fact in the table after it is earned.

□ **Theorem 15.14** (No forging)**.** For every core $k$, instruction $i$ and fact $f$ in the table after executing $i$ at $k$: either $f$ was already in $k$'s table, or $i$ is CHECK of $f$'s property and counter, it passes at $k$, and $f$ is the claim it writes. Consequently, if a trace has no forgery, so does that trace followed by any one instruction; and from every clean start, every trace has no forgery.

*Proof.* Inspect who can write to the table. INC and DEC change counters, versions and the line; they leave the table alone. HALT leaves it alone. COMMIT reads it and may write the channel, but cannot write the table. CERTIFY reads the channel and cannot write the table either. CHECK is the only remaining instruction. On failure it adds nothing; on success it adds exactly the property, counter and present edition it just checked. A trapped instruction changes no core field. That exhausts the ways a fact could appear after one instruction.

Now suppose the table's old entries already have their promised histories. A new entry gets its history from the passing CHECK just performed. For an old entry, that earlier CHECK still happened. If the entry is live after the new instruction, its saved edition equals the counter's edition now. The counter's edition never goes down. It therefore could not have risen away from the saved edition and then returned. In particular, it was already live before this instruction and the instruction did not change the edition or the value. Its untouched stretch extends by this one step. If the entry is stale, only its earlier passing CHECK is required, and that history remains.

At a clean start there are no entries to forge. The argument preserves that promise after the first instruction, after the next, and after each further instruction in any finite trace. ■

The closest earlier result is in capability hardware, and it's much bigger. The computer scientists Jack Dennis and Earl Van Horn, at MIT, introduced *capabilities* in 1966: tokens a process has to hold to touch an object, which it can pass on and can't forge. CHERI, a capability extension of real instruction sets built at the University of Cambridge and SRI International, rests on that idea, and in 2020 Kyndylan Nienhuis and twelve coauthors proved, for the whole CHERI-MIPS instruction set, that arbitrary code given some capabilities can't increase them as it runs (up to any change of protection domain). Their result says nothing gets forged across a full instruction set. Theorem 15.14 is the same shape, one instruction at a time, for six instruction kinds and a table of checked facts.

*The actual counter covers every added fact*

Put the table beside the bill. Every passing CHECK writes one more entry and adds one to the ledger. A failed CHECK writes no entry. COMMIT and CERTIFY can add to the bill without adding a table entry. Walk down the run one instruction at a time: the table can never grow faster than the bill.

**Theorem 15.15** (The native bill bounds the table)**.** For every clean run, whether driven by a finite instruction word or by the stored-program step for a finite number of ticks, □

$$\text{len}(\text{facts}) \leq \mu.$$

More generally, table growth is exactly the number of passing CHECKs, and is at most the increase in the ledger from any starting state.

*Proof.* A passing CHECK adds one entry and one unit to the bill. A false CHECK, a full-table CHECK or a trapped CHECK adds no entry but still adds one unit. COMMIT and CERTIFY add one unit and no entry. INC, DEC and HALT add neither. No instruction removes an entry.

Place a mark beside each passing CHECK in the run. The final table has exactly one more entry for every mark. Each mark also accounts for one unit on the bill; unmarked instructions can add further units, never take them away. So table growth is exactly the number of marks and at most the ledger's increase. This works from any starting state.

At a clean start the table is empty. The initial ledger is a nonnegative whole number, so the table length is at most the final ledger. At the specified loaded starts the ledger begins at zero as well. For a stored program, every running tick executes one of the six instructions just counted, and every stopped tick leaves both table and bill alone. The same count therefore applies to any finite number of ticks. ■

One passing CHECK from a clean start gives one entry and a bill of one. Repeat a passing check and its duplicate entry counts as another line in the table. The tally follows what was written there. COMMIT and CERTIFY can then make the bill larger while the number of lines stays put.

### 15.9 A full table traps

The table has room for sixteen facts and evicts none of them, so there are two halves to prove: nothing ever leaves, and nothing ever overflows.

□ **Theorem 15.16** (Bounded table, no eviction)**.** Every instruction keeps every fact already in the table. If the table holds 16 or more facts, CHECK $p$ $c$ puts the trap up and leaves the table as it was. If the table holds at most 16 facts, it still holds at most 16 after any instruction.

*Proof.* With a raised trap no instruction changes the table. With the trap down, only CHECK can change it, and a successful CHECK adds one fact without removing an entry. Success requires a previous length strictly below 16, so the new length is at most 16. At length 16 or greater that branch is unavailable: CHECK raises the trap and preserves the table. Every other instruction preserves it as well. ■

Why trap? Why not make room? Because "making room" means deciding which fact dies, and a fact that can die is a fact whose absence you have to explain later. A table that only grows keeps every checked fact available to read back, and the shared-memory result above leans on exactly that, since its oldest fact can never move. Provenance doesn't: a table that dropped its oldest fact to make room would still hold only facts a CHECK wrote. What the choice gives up is real, though, and it's in the subsection near the end of this chapter on what the machine doesn't do.

### 15.10 The window can't see the record

Section 7 was about windows: a view of a state that shows some of it, and the theorem that a window which shows the same before-and-after pair for a raise and a non-raise cannot price both exactly (Observation 7.2). The two runs below are the smallest pair I could build where the window is Minsky's view of the machine (line number and the two counters) and the record is everything else.

Run two traces from start$(0, 0)$. The *earned* run is CHECK *is zero* $A$; COMMIT *is zero* $A$; CERTIFY. The *idle* run is DEC $A$ 0, three times. In the idle run, $A$ is 0 every time, so each DEC just goes to the next line and writes nothing. Both runs end on line 4, with $A = 0$ and $B = 0$, both versions 0. Through the window they're

identical. One of them has paid 3, holds a fact, could commit again, and has its flag up. The other has paid nothing, holds nothing, can't commit, and its flag is down. They're like two photographs that show the same picture and are of different people.

**Theorem 15.17** (Same window, different record)**.** The earned run and the idle run from start$(0,0)$ end at cores with the same window and the same versions of $A$ and $B$, and with different ledgers, different flags, different tables, and different answers to "would COMMIT *is zero* $A$ pass." □

*Proof.* In the earned run, CHECK adds $(\text{Zero}, A, 0)$, COMMIT puts that fact in the channel, and CERTIFY raises the flag. The ledger is $1+1+1=3$. No counter is written. In the idle run each DEC sees zero, so it advances the line without writing a counter or raising a version. Its three costs are zero and it adds no fact or commitment. So both final windows are $(4,0,0)$ with versions $(0,0)$. The earned table contains the current zero claim and the idle table is empty, giving the stated difference in whether another COMMIT would pass, as well as in the ledger and flag. ■

**Theorem 15.18** (No window oracle)**.** There is no function $g$ from windows to whole numbers with $g(\text{window}) = \mu$ at the end of every trace from start$(0,0)$. Likewise there is no function from windows to yes or no that gives the flag at the end of every such trace, and none that gives whether COMMIT *is zero* $A$ would pass. □

*Proof.* Such a $g$ would have to give both 3 and 0 on the one window the two runs of Theorem 15.17 share. The same for the flag and the commit right. ■

The record lives off the window, and that's Section 7's shape, made of pebbles. Anybody who only watches the counters and the line number, which is everything Minsky's machine *is*, can't tell you what was paid, what was committed, or whether the flag is up, and the reason isn't that they looked carelessly; it's that none of it is there to see.

Two things the same in everything you look at and still two things: philosophy has an old fight about exactly that, and it's worth a detour, because it tells you which side of the fight these theorems stand on. Leibniz said that for real things it can't happen. There are no two individuals indiscernible from each other, he wrote in 1716 to Samuel Clarke, the English philosopher and theologian who was taking Newton's side against him in an exchange of letters, and he told a story to go with it. A clever gentleman of his acquaintance, talking with him in front of the Electress

Sophia of Hanover, Leibniz's friend and patron, in the garden at Herrenhausen she had helped lay out, thought he could find two leaves perfectly alike. She dared him to. He "ran all over the Garden a long time" looking, and it was "to no purpose." You can run his search tonight with a handful of anything, leaves or coins or grains of rice: look hard enough and every pair differs somewhere. Then in 1952 Max Black, a Russian-born philosopher teaching at Cornell, pushed back with a universe you can't build at the table. Nothing in it but two iron spheres, the same size, the same colour, the same everything, some distance apart. Name a difference between them. There isn't one, and there are still two of them. People have argued about Black's spheres ever since, and I'm not going to settle it.

I don't have to, because Leibniz's claim is about *every* property a thing has. The theorems above are about three: the line and the two counters. The two runs aren't indiscernible in his sense at all: one has a fact in its table and its flag up, and the other doesn't. They're indiscernible to a window, which is a much cheaper thing, and it's the thing that matters for a machine, because an observer of a machine only ever gets a window. Leibniz's gentleman would always find his difference if he could look at everything. It doesn't help him if all he's allowed to look at is three things about each leaf.

There's a horse in this, too, if you squint. In Germany in the early 1900s a mathematics teacher named Wilhelm von Osten showed off a horse called Clever Hans, who answered arithmetic questions by tapping a hoof, and got them right, in front of crowds. Hans gave the same answers out as a horse that could count, and that's the idle run's matching window. Then the psychologist Oskar Pfungst stopped looking at the answers and started changing the conditions, and published what he found in 1907. Put blinders on Hans, or have him asked by someone who didn't know the answer, and the arithmetic fell apart. Hans was reading the questioner: tiny, unintended movements of the head and body as the hoof reached the right number. That's the probe. Pfungst didn't argue with the answers. He asked a question the answers couldn't fake, one that needed the real thing to be there, the way a COMMIT needs a checked fact to be in the table: run COMMIT *is zero* $A$ after each of the two runs, and one passes and the other traps (Theorem 15.17). And the part of the story I like best: nobody was lying. Von Osten was never persuaded by Pfungst and kept showing his horse. A matching window doesn't need a cheat behind it, which is why you want a probe. An accusation is the wrong tool for it.

*The rebar example, on the actual counters*

Put a one in counter $A$, leave $B$ at zero, and give the machine these four instructions:

$$\text{CHECK } (A \geq 1); \quad \text{COMMIT } (A \geq 1); \quad \text{CERTIFY}; \quad \text{DEC } A \text{ 5}.$$

The check passes. The claim gets committed and certified. Then DEC takes the one away. Look at the counters now: zero and zero. The value that passed the check has gone.

Start again with both counters zero and give the same instructions. This time the first check fails and the machine traps. The zero stays where it was. Look at the counters at the end: zero and zero again. One run has its certificate up; the other has it down.

**Theorem 15.19** (Same final counters and bill, different certification)**.** For those two clean starts and that word, both final counter pairs are $(0, 0)$ and both ledgers are 3. The first certificate is true and the second false. No function of just the final counter pair can recover which of those starts passed the check. □

*Proof.* For the start $A = 1, B = 0$, CHECK sees $1 \geq 1$, writes the edition-zero fact, advances the line from 1 to 2 and makes the ledger 1. COMMIT finds that live fact, puts it in the channel, advances to line 3 and makes the ledger 2. CERTIFY sees the occupied channel, raises the flag, advances to line 4 and makes the ledger 3. DEC sees a positive $A$, changes it to zero, advances to line 5 and raises $A$'s version from 0 to 1. Its price is zero. The final counters are $(0, 0)$ and the flag is up.

For the start $A = 0, B = 0$, the first CHECK fails. It sets the trap, leaves the line at 1, leaves the table empty and charges one. The external word still supplies COMMIT and CERTIFY: on a trapped state they change no core field and cannot raise the flag, but each adds its unit to the ledger. DEC then changes nothing and charges zero. The final counters are again $(0, 0)$, both versions stay zero and the flag is down. Both ledgers are 3. A function of the common counter pair would have to return both flag values, which is impossible. ■

The counters show the same surface. The record keeps the difference. Open the rest of the box and the trap, version and line number tell different parts of the story too: one run reached DEC, while the other stopped at the check.

Ask the successful run to CHECK $A \geq 1$ once more. It fails; $A$ is zero now. The old certificate stays up because the earlier certification still happened. That receipt keeps its past even while the workspace goes on to something else.

*One clipboard, 65,535 different publication histories*

Open the whole native box this time. Read the counters, versions, table, channel, line number, trap, ledger and lamp. I can make all of them agree while the files of published claims differ in 65,535 ways.

Start clean at $A = 16, B = 0$. CHECK the sixteen claims $A \geq 0, \ldots, A \geq 15$. Those counter values are initial data, before any instruction: versions zero, line 1,

ledger 0, empty table and channel, and trap and flag down. Every one passes on edition zero. The clipboard now has sixteen entries. The line number is 17, the bill is 16, and the certificate is still down. I can choose which of those checked claims to publish.

Visit the claims in order. At each one, choose between these blocks:

| | |
|---|---|
| publish it: | COMMIT claim; CERTIFY, |
| leave it unpublished: | COMMIT claim; COMMIT claim. |

Count the steps in either block: two. Count the charge: two. Look at the channel when it finishes: the same claim. The counters and table stayed put. Once one publication has raised the lamp, that lamp stays up whichever later blocks you choose. CERTIFY reads the channel without consuming it; this is why the two blocks leave the same channel as well as the same line and bill. The predicates overlap logically; it is the choice to publish each checked claim that is independent.

□ **Theorem 15.20** (The exact native publication-history fibre)**.** In the extension retaining the set of published claim/edition facts, exactly 65,535 different nonempty sets can lie above one reachable native endpoint. This is the maximum over clean native runs with cert true. Sixteen additional fixed binary bits suffice to encode the set relative to the native table, and fifteen do not suffice in the worst case.

*Proof.* First prepare the table. After the first $j$ checks, for $0 \le j \le 16$, its entries are exactly $A \ge j-1, \dots, A \ge 0$, newest first and all on edition zero (the table is empty at $j = 0$). The line is $1 + j$, the ledger is $j$, the counters and versions are unchanged, the trap is down and the flag is down. This holds at $j = 0$. If $j < 16$, there is still room and $16 \ge j$, so the next CHECK prepends precisely the next fact, advances one line and charges one. That proves the preparation claim through $j = 16$.

Now process the sixteen facts in order. Each COMMIT finds its fact in the unchanged table at the current edition. Either block ends with that fact in the channel, advances two lines and charges two. The publishing block additionally raises the flag and inserts that fact into the archive; the other block changes neither. Inducting on the number $j$ of blocks gives line $17 + 2j$, ledger $16 + 2j$, and exactly the chosen subset of the first $j$ facts in the archive. The flag is up if and only if at least one publishing block has occurred. None of the blocks changes counters, versions, table order or trap. At $j = 16$ every nonempty choice therefore has the same native endpoint: $A = 16$, $B = 0$, versions zero, line 49, ledger 48, trap down, flag up, and channel $(A \ge 15, A, 0)$. Its archive is precisely its chosen subset. There are two independent choices at each of sixteen positions, hence $2^{16}$ subsets; excluding the empty one leaves 65,535 different archives above that endpoint.

For the upper bound, keep two invariants together: every published fact is in the table, and every occupied channel names a fact in the table. Both hold at a clean start. CHECK only prepends to the table. A successful COMMIT selects an entry already there. CERTIFY publishes only its occupied channel. Counter writes can make a fact stale; they preserve the channel and never remove table entries. Failed and trapped instructions do not introduce a new channel fact or publication. Thus both invariants survive every instruction, including later writes. Every final archive is a subset of the final table.

If that table has $d$ distinct entries, it has at most $2^d - 1$ nonempty subsets. The table has at most sixteen entries, so $d \le 16$ and $2^d - 1 \le 65{,}535$. An explicit code puts beside each table entry one bit saying whether it belongs to the archive, and pads with zeros to sixteen positions. Duplicate entries receive matching bits. Read back the entries with bit one and you recover the set. Equal codes therefore give equal archives, and an archive above a flag-up state uses a nonzero code. Finally, fifteen fixed bits have only $2^{15} = 32{,}768$ values, fewer than the exhibited 65,535 archives; sixteen bits have 65,536 values and the code just given suffices. ∎

Put sixteen checkboxes beside the clipboard's sixteen lines. A tick says that line's claim was published at least once. The table already names the claims and editions; the extra marks say which ones made it into the publication file. The lamp gives the same yes for every nonempty selection. Open the checkboxes and all 65,535 selections come apart. Each tick answers whether that claim was published at least once. Order and repeated publications leave these answers unchanged.

In Chapter 14 the camera showed the working value. Here I let it see the entire native machine and still found histories it could not recover. The map here forgets the added publication file; its fibre contains those different files above one native endpoint.

The sixteen-line clipboard eventually fills. Even sixteen repeated checks of one fact fill it, and the next CHECK traps. The unlimited-edition machine has a larger obligation: keep accepting independent publication choices across every finite number of editions and keep them recoverable at the same working picture. Its fibre has to keep growing. That is the count from Chapter 14, with its own camera and its own task.

## 15.11 The exact price is three

Section 2 said a raised flag costs at least one. On this machine that's true and too modest. A raised flag needs a CERTIFY, a CERTIFY needs a COMMIT, a COMMIT needs a CHECK, and each of those is its own instruction costing one. So a raised flag costs at least three. One of what: three units of $\mu$, one per paid act, each one

the least price a record-layer merge allows. And three is the number of links in the chain, not a guess that happens to hold.

**Observation 15.21** (The minimum price of a certificate)**.** *If* $s_0$ *is clean and* $\mathrm{cert}(\mathrm{run}(t, s_0))$ *is up, then* $\mathrm{cost}(t) \geq 3$ *and* $\mu(\mathrm{run}(t, s_0)) \geq \mu(s_0) + 3$*. For programs: if the flag is up after* $n$ *program steps from* $\mathrm{start}(a, b)$*, then* $\mu \geq 3$ *there.*

*Proof.* Theorem 15.11 finds a CHECK, a COMMIT and a CERTIFY at three different places in $t$, each costing 1, and every other instruction costs at least 0. The ledger statement is Theorem 15.6. A program run is the run of the instructions it actually executes, so the program form follows. ■

The three is three independent units, one for each part of a clean start. A flag that is down buys the first: a run that raises it costs at least one. An empty channel buys the second: with the flag down and the channel empty, a run that raises the flag costs at least two. An empty table buys the third, and that is the theorem above. Leave out any one part and its floor fails, and a cheaper certificate is there: with the flag already up it costs 0, with a commitment already in the channel 1, and with a live fact already in the table 2. Every run from a state that ends with the flag up costs at least three exactly when the flag is down and one of two things holds: the trap is up, or the channel is empty and every fact in the table is about an older version of its counter. And three is paid exactly from every clean start whose trap is down.

□ **Theorem 15.22** (Three is tight)**.** For all whole numbers $a$ and $b$, run the program CHECK *is zero* $A$; COMMIT *is zero* $A$; CERTIFY for 4 program steps from $\mathrm{start}(a, b)$. If $a = 0$, it ends with the flag up and $\mu = 3$ exactly. If $a \neq 0$, it ends with the flag down and the trap up, and for every $n$, $n$ program steps from $\mathrm{start}(a, b)$ leave the flag down.

*Proof.* If $a = 0$, the checker for Zero says yes at 0, so the first instruction adds $(\mathrm{Zero}, A, 0)$ to the empty table and charges 1. No counter changes, so the second instruction finds that same fact, puts it in the channel and charges 1. The third finds the channel occupied, raises the flag and charges 1. The line is now 4, beyond the three-instruction program, and the fourth program step adds no charge. The final ledger is exactly 3. If $a \neq 0$, the checker says no, so the first instruction sets the trap and leaves the flag down. A program run takes no step once the trap is up, so every later program step leaves the state, flag included, exactly where it is. ■

Theorem 15.22 is the worked run from early in this chapter, now holding from every start.

The version whose check can't fail, CHECK *at least 0 A*; COMMIT *at least 0 A*; CERTIFY, certifies from every start for exactly 3. So the floor of three is reachable from every loaded start, and that's the machine declining to police trivial checks. On that program the check distinguishes no inputs.

The floor is 3, and a run whose check could have failed pays exactly 3, so 3 is the price. The bound has no slack in it.

Don't mistake that 3 for a measure of how hard a claim is to certify. It's the same 3 for every claim, easy or hard, because only the three acts that make a claim are priced and the counters work out whatever the check needs at no charge on the ledger. A cost of certifying that grows with the claim would have to price the checking itself, and that's a different question with its own literature. Four theoretical computer scientists, Guy Blanc, Caleb Koch, Jane Lange and Li-Yang Tan (2022), studied one version of it under the name the query complexity of certification: how many questions an algorithm has to ask a black-box function to hand back a small certificate for its answer. Their count moves with the function. Mine doesn't move at all.

And that's every piece of the machine, standing on six instruction kinds: a base that computes everything two counters can, a ledger that's exact, a toll the universal floor applies to, a flag that's a latch of the base, a window that can't see the record, and, from a clean start, a commitment that can't happen unless a check of the same claim, on the same edition, came first and was right.

## 15.12 What it takes to be certified

Section 10.8 set the bar worth meeting, Thiele-complete, and it says nothing about what the record is about. A clock fails it, and so does a machine whose record never goes up. The small machine clears it (Theorems 15.31 and 15.23(2) below), and it does more: from a clean start, its flag goes up only on top of a check of a claim, by a checker proved to mean what the claim says, then a commitment to the same claim with nothing changed underneath, then a certificate. A Thiele-complete machine can raise its record with one mark that goes up from any state at all, a receipt for nothing (Section 14). A certified one can't. The second definition is the first one with three things added, and it's written so that any machine can be held up against it. You name a language of claims first, then hand over a reading of the machine in that language, a list of things, and then the reading has to pass four clauses.

**Definition 15.7** (Certified Thiele-complete)**.** A *machine* has states, moves, a step, a cost for each move and a yes/no reading of each state, the *record*. A *claim language* for it is fixed first: a type of *claims* with an exact equality test; a *meaning* for each claim, a statement about a state; for each claim, a relation "the thing this

claim is about is unchanged" between two states; and a proof that a meaning that holds keeps holding while the thing it's about is unchanged. An *interface* over that language hands over the rest: a *universal base*, meaning a two-counter configuration shown by each state (its *window*), a class of *live* states, a move for each two-counter instruction, and a loaded state for each start $(1, a, b)$; a *checker* that says yes or no to a claim at a state, with a proof that a yes means the claim's meaning, as the language gives it, holds of that state, with a CHECK of a claim passing at a state only when the checker says yes to it there; a set of *clean* states; a *ledger*, a whole number for each state; and a *kind* for each move: a base move, CHECK of a claim, COMMIT of a claim, or CERTIFY. The machine is *certified Thiele-complete over* the language when some interface over it meets four clauses, and *certified Thiele-complete* when some interface, with claims, meanings and "unchanged" of its own, meets them. (a) *Universal base.* Each two-counter instruction's move is a base move; from every live state it acts on the window exactly as the instruction acts on a configuration, and the state stays live; each loaded state shows $(1, a, b)$, is live and is clean; no base move changes the record; and a record that's up stays up under every move. (b) *Earned record.* Clean states have the record down. Every run from a clean state that ends with the record up contains a CHECK of some claim $c$ that passes, then a COMMIT of $c$ with the thing $c$ is about unchanged at every state in between, then a CERTIFY, with the record down just before that CERTIFY and up just after it. A passing check means its claim's meaning holds of the state it checked, and a meaning that holds keeps holding while the thing it's about is unchanged. (c) *Exact prices.* Base moves cost 0, each CHECK, COMMIT and CERTIFY costs exactly 1, and each move adds its cost to the ledger. (d) *Non-vacuity.* Some claim $c$ has a CHECK, a COMMIT and a CERTIFY such that those three moves alone, from a loaded state, raise the record exactly when $c$'s meaning holds there; and $c$ holds at some loaded state and fails at another. A Thiele machine *has a universal base* when some interface gives it one and nothing more is asked: each two-counter instruction's move acts on the window exactly as the instruction does from every live state and keeps it live, and each start $(1, a, b)$ has a live loaded state showing it. Nothing is asked about the record, the prices or the checker.

An interface over a language becomes an interface of the plain kind by taking the language's claims, meanings and "unchanged" as its own, and the four clauses are asked of that.

□ **Theorem 15.23** (Certified is Thiele-complete plus certification)**.** (1) A machine is certified Thiele-complete through an interface exactly when it is Thiele-complete through the same interface read without its claims (its base moves as the computing

moves, CHECK, COMMIT and CERTIFY as the marks, the same universal base and ledger), and three more things hold: its loaded states are clean; clause (b) holds, so clean states have the record down, every run from a clean state that ends with the record up contains the earned chain, a passing check means its claim holds, and a claim that holds keeps holding while what it's about is unchanged; and clause (d) holds, a claim whose three moves raise the record from a loaded state exactly when it holds there, true at one loaded state and false at another. (2) Every certified Thiele-complete machine is Thiele-complete. (3) Not the other way: the two counters with two marks, one that raises the record when $A$ is 0 and one that always raises it, and a ledger beside them that each move adds its cost to, is Thiele-complete, and certified Thiele-complete through no interface.

So certification is exactly three things on top of the bar: clean starts, an earned record (the chain, a checker proved sound, claims that stay true while what they're about is unchanged), and a claim that could have come out either way. Everything Section 10.8 proved holds of every certified machine. What this section adds is what those three things buy, from a clean start: a floor of three instead of one, a raise that is always a CERTIFY, and no receipt without a check; and, from the loaded states, a check that can fail.

The language comes first for a reason. If the interface could name the meanings, it could take "the checker says yes" as the meaning of every claim, and checker soundness would hold for nothing. In the language-first form the meanings are written down before any checker is handed over, so "a checker proved to mean what it says" has the content it has on the small machine: what the claim says is the language's, and the proof is that a yes from the checker makes it true. The theorems that build certified Thiele-complete machines ask the checker for one thing more, that it say yes exactly when the claim holds, because clause (d) needs a check that can come out yes. The plain form stays because the theorems that follow hold for any interface at all, so they hold over every language too.

Read the four clauses as four plain sentences. (a) It computes anything a Turing machine can, under Minsky's $2^n$ encoding, done outside the machine, and the computing can't reach the record: the base moves are the two-counter machine, they leave the record exactly where it was, and a record that goes up stays up. (b) From a clean start, the record goes up only on top of the small machine's chain: a check that passed, a commitment to the same claim with nothing changed underneath, then a certificate. (c) Computing costs nothing on the ledger and each of the three acts costs one unit, so the ledger counts the acts. The steps in between add nothing to it. (The unit belongs to the ledger; it isn't a unit of heat.) (d) Somewhere there's a claim whose check could have said no. A machine doesn't qualify over a language whose every claim is "at least 0", and neither does one with nothing to certify.

And look where the checker's honesty sits. In (b), "a passing check means its claim holds" is a condition on the interface, proved once, outside every run, exactly where it sat for the small machine (Lemma 15.2), and over a claim language it's proved against meanings the interface didn't write. The definition doesn't ask the machine to vouch for its own checker. The nearest reasons not to ask are Gödel's second theorem, about a system proving its own consistency, and the diagonal of Section 13, about deciding a property of every program. Neither forbids a machine checking a proof about a checker, so keeping that proof outside is a choice. Nor does it ask the checker to be something a machine could run. Clause (b) wants a yes or no for every claim at every state, sound for the claim's meaning, and nothing in it says that answer has to be computable. Every instance in this book has a checker that's an actual program, so anything those machines certify could be listed by a program that runs them. For example, take a function that answers whether an arbitrary program runs forever. No program computes it (that's Turing's halting theorem). Hand the definition a checker built on it and it's still met, and the machine certifies claims no program could ever confirm. Nothing in this book builds one. So "a certified Thiele-complete machine certifies exactly the claims a program could list" is true only with "and its step and its checker are computable" added, and the definition doesn't require either.

The same scope goes on Section 14's "it isn't more computing power." The record (the flag and the ledger) is never read by the step that computes the next core, so erasing it changes no run, and it makes nothing faster. The checker is part of the machine under the record. With an oracle checker that machine decides what the oracle decides, since whether a CHECK traps is there for anyone to see; with a checker that's a program, it does what programs do. And a checker that's a program still makes CHECK a test at unit cost.

□ **Theorem 15.24** (CHECK is a unit-cost test)**.** From any start $(a, b)$, the single move CHECK Ge$(m)$ $A$ costs 1, and the trap latch is down after it exactly when $a \geq m$.

The two counters alone look at $A$ only through the zero test of DEC, and started one pebble below a threshold or right at it, that test can't come out differently until the pile has been counted down to zero, so they need about as many steps as the threshold for what CHECK does in one. Theorem 15.43 gives the precise lower bound and the matching ladder. So the record makes nothing faster, and CHECK, a primitive test, can.

*Three kinds of exact.* The word exact does three jobs in this book, and each job keeps one name. *Rise-only pricing* charges at least one unit on every step that raises the reading and nothing on any other step; its tightest form, one unit per raise and

nothing else, is the *exact schedule*. *Instruction-exact pricing* charges every instruction a fixed listed price. On this machine that's one unit each for CHECK, COMMIT and CERTIFY and nothing for the rest, and the definition of certified Thiele-complete calls it *exact prices*, not a toll, because it charges acts that raise nothing. CHECK and COMMIT raise no flag, so the small machine is instruction-exact without being rise-only, and no rule is broken by that. What the three units pay for isn't one thing three times. A passing CHECK adds a fact to a table that holds sixteen, so it takes a fresh state and fills one slot of a bounded store; a passing COMMIT overwrites the channel and a passing CERTIFY raises the flag, each sending two states with the same base to one (Theorem 3.19). And each of the three, when it fails, sends the failed state and its twin with the trap already up to one state. So a passing CHECK pays by taking a fresh state, and every one of the three units is also a price on a merge. *Simulation-exact accounting* is a host whose ledger equals its guest's ledger at every point where the two runs match, as the universal program *U*'s does (Appendix C.1). A machine can meet any one of the three and fail the others.

One word needs pinning before the theorems, using those three names. In clause (c), *exact prices* means instruction-exact pricing: each of the three acts, CHECK and COMMIT included, costs exactly one, so the ledger counts the moves filed as record moves. Those are the acts that make a claim, and also a check that fails, a COMMIT of a fact already committed, and a CERTIFY with the flag already up: each is filed as a record move and costs one. Back in Section 6, one result is about rise-only pricing in its tightest form, the *exact schedule*: one unit on the step that raises the flag and nothing on any other step. Measured against the flag alone, the small machine isn't rise-only, since CHECK and COMMIT raise no flag and cost one each; measured the way (c) asks, it is instruction-exact. Neither is wrong. They answer different questions: does the ledger count the acts, or does it charge the flag and only the flag? Simulation-exact accounting, the third name, comes up with the universal program *U* in Appendix C.1.

**Theorem 15.25** (What every certified Thiele-complete machine has)**.** Let a machine be certified Thiele-complete through an interface. (1) It obeys the toll and has a universal base; its base runs every two-counter program step for step from every live state, and halting from a loaded state matches halting from the configuration it shows. (2) Along any run, the ledger rises by exactly the number of CHECK, COMMIT and CERTIFY moves in it. (3) A run from a clean state that ends with the record up contains at least three of those moves and raises the ledger by at least 3, and the three moves of clause (d), from a loaded state where its claim holds, raise it by exactly 3. (4) Some CHECK fails at some loaded state, on a claim that is false there, and some run from some loaded state raises the record. (5) On a run from a □

clean state, the move that raises the record is a CERTIFY. (6) Every move costs 0 or 1, and some move costs 0.

□ **Theorem 15.26** (The language first, and what a certificate then means)**.** (1) A machine that is certified Thiele-complete over some claim language is certified Thiele-complete, so every part of Theorem 15.25 holds of it. (2) If a machine is certified Thiele-complete through an interface whose claims have an exact equality test, it is certified Thiele-complete over the language made of that interface's claims, meanings and "unchanged," with clause (b)'s proof that a meaning keeps holding as the language's own. (3) If a machine is certified Thiele-complete over a claim language $L$, some interface over $L$ meets the four clauses and has this property: every run from one of its clean states that ends with the record up contains a CHECK of a claim $c$ of $L$ that passed, then a COMMIT of $c$, and $L$'s meaning of $c$ holds at the state the CHECK saw and at the state the COMMIT saw.

Part (2) says where the strength of the language-first form lives: in the naming. Any reading can be turned into a language when its claims come with an exact equality test, so for those "certified Thiele-complete over some language" asks no more than certified Thiele-complete does. "Certified Thiele-complete over $L$," for an $L$ written down without looking at the checker, says that what the machine certifies is what $L$ says. Every instance below is stated that way.

Each item in the proof, in the companion volume, names the clause it comes from, and that's all the definition gives. A certified Thiele-complete machine needn't have a fact table, a CHECK that traps when it fails, a COMMIT that looks anything up, or claims about counters. The definition asks for an earned chain on the runs that raise the record and leaves the rest to the machine. So the small machine's table provenance, its trap, and its channel are facts about that machine, and they don't carry over to every machine that meets the definition. Through a plain interface its claims can be any labels the interface names; over a claim language they are the language's. Either way its checker can be an oracle, as the halting-function checker above shows.

The second half of (6) in Theorem 15.25, that some move costs 0, is the one that rules out the clock: a machine that charges every move. It doesn't rule out a program that makes the ledger tick. On the host of Appendix C.1, a program that checks one fact once and then commits it again before every instruction runs any two-counter program with the flag down and pays one unit a step, plus one for the check: the checked fact is about a counter the program never writes, so its edition never changes, and every COMMIT finds it.

**Observation 15.27** (A clock is a Thiele machine and isn't certified Thiele-complete)**.** *Take the two-counter machine, let every move cost 1, and take as the record any yes/no question about its configuration whatsoever; or take a flag that switches on, and stays on, the first time any chosen test of the configuration says yes. Either way the machine obeys the toll and has a two-counter machine running inside it, and it is not certified Thiele-complete. More generally, no machine whose every move costs at least 1 is certified Thiele-complete.*

(It also follows from Observation 10.13 and Theorem 15.23(2).) The proof, in the companion volume, is short, and it's short because clause (c) does exactly the job it was written for: it says computing costs nothing, and a clock charges for computing. The other clauses do work too, and that's worth checking, because a definition with a clause nothing needs is a definition hiding something.

**Theorem 15.28** (Each clause does work)**.** (1) Take a two-counter base priced 0 with a ledger beside it, and one extra move TICK that costs 1 and raises a flag. It obeys the toll, and read with TICK as CERTIFY it meets the universal base clause and the exact prices clause; it is not certified Thiele-complete, and more generally no machine with a move that puts the record up from every state is. (2) Take a two-counter base priced 0 and a record that is down at every state, so it never rises. It obeys the toll and meets clauses (a), (b) and (c); it is not certified Thiele-complete, and more generally no machine whose record never rises is. □

The paid latch, the TICK machine of (1), is the receipt for nothing from Section 14, wearing a universal base: honest about the price, silent about what was bought. The definition throws it out.

The silent machine of (2) meets three clauses and fails only (d). The paid latch fails (d) as well as (b), so it doesn't show (b) doing work alone, and nothing above says whether a clock meets (a), (b) and (d). The sharper check, the usual evidence that a definition isn't written around one machine, is a machine for each clause that meets the other three and fails it.

**Theorem 15.29** (The four clauses are independent)**.** For each clause of Definition 15.7 there is a machine with an interface that meets the other three clauses and fails that one. Each of the four machines obeys the toll and has a two-counter machine running inside it, and none of them is certified Thiele-complete through any interface at all. (a) fails for this chapter's machine with one more move, DROP, priced 0 and read as a base move, which lowers the flag and trips the trap latch. (b) fails for this chapter's machine with one more move, FORGE, costing one and read as a CERTIFY, which raises the flag on any state whose trap is down, with no □

commitment behind it. (c) fails for this chapter's machine with every move costing one. (d) fails for the silent machine of Theorem 15.28(2).

Three of the four are this chapter's machine with one thing changed, and in each case the one clause it fails is the only thing keeping it out. Clause (c) also fixes something it has no business fixing: the size of the unit. Double every price on the small machine and it leaves the class, because its CHECK now costs two and (c) asks for one. A definition that a change of currency breaks would be a strange one, so the next theorem states (c) with the unit left open.

□ **Theorem 15.30** (Any unit)**.** Call a machine *certified Thiele-complete at unit* $c$, for a whole number $c \geq 1$, when some interface meets clauses (a), (b) and (d) and this form of (c): base moves cost 0, each CHECK, COMMIT and CERTIFY costs exactly $c$, and each move adds its cost to the ledger. (1) At unit 1 that is certified Thiele-complete. (2) Multiply every price of a certified Thiele-complete machine by $c \geq 1$ and the machine is certified Thiele-complete at unit $c$; divide every price of a machine certified Thiele-complete at unit $c$ by $c$ and the machine, with the same states, moves, step and record, is certified Thiele-complete, with the ledger read as $\lfloor \ell / c \rfloor$. (3) At every unit some move costs 0, so the clock of Observation 15.27, with any reading or any latch as its record, is certified Thiele-complete at no unit, and neither is any machine whose every move costs at least 1. (4) At unit $c$, a run from a clean state that ends with the record up contains a CHECK, a COMMIT and a CERTIFY and raises the ledger by at least $3c$; the claim it stands on held when it was checked and still held when it was committed; and the move that raises the record is a CERTIFY. (5) The small machine with every price doubled is certified Thiele-complete at unit 2 and isn't certified Thiele-complete.

So the class doesn't care what a unit is worth. It cares that computing costs nothing and that the three acts cost the same. The clock fails at every unit for the reason it failed at one: it charges for computing. I keep unit 1 in the definition because the small machine is built that way and every count in this chapter is in its units.

Now to the other direction, because a definition nobody meets is as useless as one everybody meets.

□ **Theorem 15.31** (The small machine, the generic one and the sorted one are certified Thiele-complete)**.** (1) This chapter's machine is certified Thiele-complete, read with INC, DEC and HALT as base moves, CHECK $p$ $c$ and COMMIT $p$ $c$ as record moves on the claim "$p$ holds of $c$," CERTIFY as CERTIFY, the checker as eval on the value of $c$,

the window as (line, $A$, $B$), live states as those with the trap down, start$(a, b)$ as the loaded state for $(1, a, b)$, "unchanged" as "same value and same version of $c$," clean states as in Definition 15.2 and the ledger as $\mu$. (2) So is the same machine over any property language with an exact equality test and a checker proved to say yes exactly when the claim holds, as long as some property is true of one number and false of another. (3) In particular the machine whose property is "this counter, read as a list, is sorted" is certified Thiele-complete. (4) Every certified Thiele-complete machine is a Thiele machine (Definition 2.2) and obeys the universal floor; for this chapter's machine, that system agrees step for step and cost for cost with the one of Theorem 15.7, and for the generic machine with the generic machine's own. (5) Each of the three is certified Thiele-complete over its own claim language, fixed before the checker. For this chapter's machine the language's claims are pairs of a property and a counter, equal when both parts are, the meaning of "$p$ holds of $c$" is the meaning holds of Definition 15.1 applied to $c$'s value, written without the checker, and "unchanged" is "same value and same version of $c$." For the generic machine the language is built from the properties, their equality test and what they mean, and the checker isn't part of it; the sorted machine is that case.

Appendix D carries the same fixed-language form to other bases. The claims keep their stated meanings; the new interfaces supply the checkers.

There's one thing every machine that meets the definition can't help doing. It has Section 7's shape again, only this time it isn't about one machine but about all of them.

**Observation 15.32** (Every certified Thiele-complete machine hides its record and its ledger from its window)**.** *Let a machine be certified Thiele-complete through an interface, and let its window be the two-counter configuration its universal base shows. (1) For every state $s$ there's a loaded state and one base move priced 0 from it that lands on a state with $s$'s window, with the record down and the ledger unchanged. (2) There's a loaded state and two runs from it that end with the same window, one with the record up and one with it down; and likewise two runs from one loaded state that end with the same window and ledgers at least 3 apart. (3) So no function of the window gives the record at the end of every run from every loaded state, and no function of the window gives the ledger. (4) The same holds of the Thiele machine built from the machine.*

The observer in that theorem sees one thing, the final window. An observer who sees the window before and after each step, or the whole trace of windows, or who is also handed the program and the start, is a different observer, and the collision says nothing about them; given the program and the start, the run is fixed and so is its record. The theorem's two runs also start from the same loaded state, so they're reachable witnesses, which is stronger than two arbitrary states that happen to look

alike. And "no function of the window gives the record" is all it says. It doesn't say the window tells you nothing about it.

Notice how little that took, and say it out loud so nobody mistakes it for more. It's the fibre fact from Section 7 plus one observation: a universal base can print any window at no cost, and that is the point. The window here is everything a Turing machine would call the machine, the line and the counters. On a certified Thiele-complete machine, that's never enough to tell you what was established or what it cost, and it can't be, because computing costs nothing and can't touch the record. A richer window, one that shows the flag, is a different window, and nothing above says it fails.

The proofs of this subsection are in the companion volume *The Thiele Machine: The Example, Built* (its section "Proofs for What it takes to be certified") and in Coq in `minimal/ThieleComplete.v` and `minimal/ThieleCompleteIndependent.v`.

### 15.13 The machine versus the program it runs

The three hosts of Appendix C (the simpler host, $U$ and $U_P$) all come down to one distinction, and it's worth saying on its own, because the two halves are easy to run together: *the machine doing the stepping* (what the machine is) and *the program it runs* (what runs on it). The toll of Section 2 is a property of the machine doing the stepping. A rule a program keeps holds as long as the program keeps it.

I keep a picture for it: play Monopoly on the counter of a real bank. The game has a bank of its own, with its own rules, and one of them is printed in the rules sheet: the Bank never goes broke, and if it runs out of money the Banker may issue more by writing on any ordinary paper. Inside the game that's fine; the players agreed to it. The real bank's teller, standing right there, has no opinion about any of it. Push a thousand in Monopoly money across her counter and her ledger doesn't move, because her rules are about her own money. The game's money is nothing to them. The real bank is the machine doing the stepping. The game is a program it happens to be hosting. A rule that lives in the game holds as long as the players keep to it. A rule that lives in the bank holds whatever the players do.

(And since the game is on the table: its box told an origin story for decades, that an out-of-work salesman named Charles Darrow invented it during the Depression. The board is older. Elizabeth Magie, an American writer and game designer with an economic argument about land to make, patented The Landlord's Game in 1904, with a square track, railroads and a square that sends you to jail, and in 1902, in *The Single Tax Review*, described it as "a practical demonstration of the present system of land-grabbing with all its usual outcomes and consequences." Darrow sold his version to Parker Brothers in 1935, and the company bought Magie's patent for a few hundred dollars, five hundred by the accounts I found. A story told by the

company selling the product is a lead to check, every time. It changes nothing about banks. It's a good reminder of who writes the story on the box.)

A fixed machine can enforce an invariant in its own transition table, and that includes a Turing machine. Give its tape a ledger track and a certification cell, and write a table that never sets the cell without bumping the ledger: that machine keeps the invariant in its step relation. The ledger track is part of its state, though, and a ledger kept in the state isn't a price on the move. The toll of Definition 2.2 puts the price on the move, the same from every state, so what it charges depends on what the machine's moves are. Give it one move, the next step its table picks: if the table sets the cell from any configuration at all, that move costs at least one, so every step does, whatever the track says, and the machine is a clock (Observation 10.13). Give it a move for each row of its table, as the levers of Appendix D do: every row that can set the cell costs at least one wherever it's taken, and a row that never sets it can cost nothing. Either way it's the price on its moves that meets the toll, and the track only simulates it. Priced that way, it's a Thiele machine itself, so it can't be a counterexample to the model. What it doesn't do, built the ordinary way, is enforce the toll on the programs it runs. A universal machine whose table was written only to interpret carries the simulated certification bit as plain data on its tape. If the host is programmable, so that other code shares that tape with the guest, any of that code can flip the bit without paying, because the host's step has no opinion about it. So an ordinary programmable computer, as built, doesn't enforce the toll for its programs; that comes from how it's built, and it could be built to enforce it. The simpler host of Appendix C.3 enforces it by having its own step mirror the guest's bit and charge for its crossing (Observation C.10 and Theorem C.12), and the universal programs $U$ and $U_P$ are fixed tables that go one better: they earn their own flag, by their own check, commit and certificate, exactly when the guest's would rise, and pay on their own ledger what the guest pays, $U$ exactly and $U_P$ to within a surcharge of at most two when the guest's reading starts at no (Theorems C.5, C.4, C.7 and C.8). That's a statement about which reading a step rule prices, not a claim that Turing machines can't price one.

Computing ran into this exact gap in hardware, and how it got out is worth the detour. In 1974 two computer scientists, Gerald Popek and Robert Goldberg, asked when a machine can run guests, a whole operating system inside another, efficiently and safely. Their answer is one sentence you could check against any instruction set. Call an instruction *sensitive* if it reads or changes the machine's own settings, the things the host has to keep for itself, and *privileged* if running it outside the host's own mode makes the machine stop and hand control to the host, which is called a trap. For the kind of machine they modelled: if every sensitive instruction is privileged, a host can be built. The guest runs at full speed, and whenever it touches

something that matters, the machine itself stops it and lets the host decide, and a host like that is one whose step has an opinion.

Then the obvious next question: did the machines people actually bought pass the test? The most common one didn't. In 2000 John Scott Robin, an Air Force officer, and Cynthia Irvine, a computer-security researcher at the Naval Postgraduate School, went through every instruction of Intel's Pentium and found seventeen that are sensitive and not privileged; POPF, which loads the processor's flags, is the one usually pointed at. A guest could run them and the host would never hear about it. The fixes came two ways: VMware rewrote the guest's code on the fly so the troublesome instructions never ran as written, and in the middle of the 2000s Intel and AMD added hardware support so they'd trap. Does that change anything here? It doesn't change the theorems. Popek and Goldberg were after speed and isolation. A record and a price weren't their question, so their result isn't mine and mine isn't theirs. It changes the picture, though. The universal programs sit where they'd want a host to sit: the host's own steps collect each guest's toll and earn each guest's record again. And an industry learned the same lesson the expensive way: a rule binds the guest only if it lives in the step underneath.

The smallest version of "a ledger beside an ordinary machine" is a Turing machine configuration with a counter next to it, where each step adds one, plus one more if it writes a nonblank symbol. The counter doesn't disturb the machine.

□ **Theorem 15.33** (A counter beside a Turing machine changes nothing)**.** Let a Turing machine with transition table $\delta$ step a configuration $c$ to $\mathrm{tm}_n(\delta, c)$ after at most $n$ steps (stopping when $\delta$ has no entry). Pair each configuration with a counter, and step the pair by stepping the configuration and adding the step's charge. Then for every $n$, $\delta$ and pair $(c, m)$, the configuration part after at most $n$ steps of the pair is $\mathrm{tm}_n(\delta, c)$. In particular this holds from $(c, 0)$.

*Proof.* Put the ordinary run and the run with the counter side by side. Both start with configuration $c$. Whenever their configurations agree, they look up the same entry in the same transition table. If the entry is absent, both stop. If it is present, both take the same next configuration; only the paired run also changes its counter. Their configurations therefore agree again, ready for the next comparison. This covers every step up to the chosen limit $n$, as well as a stop before that limit. The ■ counter's value never enters a lookup.

The pair is a Turing configuration with a number riding beside it, and nothing more. Its counter is kept in the state, and a ledger kept in the state isn't a price on the move: Definition 2.2 puts the cost on the move's name, and this counter's charge depends on what the step writes. This theorem therefore does not supply exact

accounting for the basic instruction-price signature, or a reading with an earned-history meaning. It does not rule out equipping the same state space with other instruction prices and a Boolean reading satisfying the basic toll. The projection claim is simply that this additional count leaves the tape computation alone.

One limit on all of this needs saying, so it isn't read as more. Certification is one event. If you wanted a machine to carry some other kind of reading, you'd need an event for it, evidence for it and a price law for it, and nothing here supplies any of those.

## 15.14 Parts I and II, run on the example machine

Parts I and II proved their counts and their collisions for any system with the named pieces, and they left a promise that the machine of this part would be put through them, and this section is where it's kept. The eight-state machine of Section 3 gets run by the small machine, price for price. Structural entitlement, the count of what narrowing a list of rivals costs, is proved for every certified Thiele-complete machine, with the small machine as a worked member, a larger search on the host that has a counter for every number, and a run that shows the count's one awkward hypothesis, the covering, can't be dropped. And the time a check saves gets an exact exchange rate, for the part of the machine that has no record. (The no-bare-verifier result is in Section 7, next to the collision it comes from.)

**The eight-state machine, run by the small machine.** Look at the small machine through a window that shows two things: its line number read modulo four, which I'll call its *slot* (the eight-state machine's word from Section 3, unrelated to the slot registers of the universal program $U$ in Appendix C.1), and its flag. Call a state *live* here, a narrower use of the word than in the definition of certified Thiele-complete, when its trap latch is down, the fact "$A \geq 0$" about $A$'s current version is in its table, and its channel holds a commitment, which is what CERTIFY needs in order to run.

**Definition 15.8** (The eight-state machine, compiled)**.** Stamp is the single instruction CERTIFY. Next-slot is the single instruction INC $B$. Jump to slot $a$, for $a$ one of $0, 1, 2, 3$, is the four instructions COMMIT $(A \geq 0)$ $A$; COMMIT $(A \geq 0)$ $A$; INC $B$; DEC $B$ $a$. The *setup* is the two instructions CHECK $(A \geq 0)$ $A$; COMMIT $(A \geq 0)$ $A$.

The two COMMITs in a jump change nothing but the line and the channel, and each costs one, which is where the jump's price of two comes from. The INC makes $B$ positive, so the DEC takes one off and jumps to line $a$ at no cost, and $B$ is back where it was.

□ **Theorem 15.34** (The small machine runs the eight-state machine)**.** (1) For every live state $s$ and every move $m$ of the eight-state machine, running $m$'s instructions from $s$ ends in a live state whose window is the eight-state machine's step of $s$'s window by $m$, and the ledger has risen by exactly $m$'s charge: one for stamp, two for jump, zero for next-slot. (2) The same holds for any list of moves: the window after the compiled list is the eight-state machine's run of the list, the state is live, and the ledger has risen by the sum of the charges. (3) So a run of compiled moves from a live state that takes the flag from down to up has raised the ledger by at least one. (4) From any start $(a, b)$, the setup ends in a live state with window (slot 3, flag down) and the ledger at 2; after it, any compiled list runs as on the eight-state machine from (slot 3, flag down), with the ledger at 2 plus the sum of the charges; and if the flag is up at the end, the whole run, setup included, contains the earned chain of Theorem 15.11.

There are three reservations, and I'll say them now. The setup's claim is true of every value, so its CHECK can't fail; it's there because the small machine won't CERTIFY without a checked, committed claim, and not as a test. And the jump's price of two is two paid COMMITs, a choice a program makes that the machine allows and doesn't force. The window is a projection, too: the small machine's full state keeps far more than eight states' worth.

And the whole small machine isn't the eight-state piece it runs. It merges some states without adding anything to the ledger.

□ **Theorem 15.35** (The small machine's merges)**.** (1) A move of the small machine that takes the flag from down to up is CERTIFY; CERTIFY merges two different full states into one (a state with the flag down and the same state with the flag up land on the same state), and it costs at least one. (2) DEC $A$ 2 also merges two different full states, and it costs nothing. So it is false of the small machine that every move that merges is priced.

The theorem puts names to the last item on the list of what this machine doesn't do, near the end of this chapter. Part I's counting of states wants a finite machine that puts a price on every merge. This machine has counters with no top, and it prices the three record acts and leaves its counting merges at a price of zero, so Part I's counting of states doesn't reach it whole, and that's why the eight-state piece is a piece. The record layer closes the rest: asked only of merges between states with the same line and counters, the merge premise holds for this machine's own prices and forces the toll on it with no finiteness premise (Theorem 3.19).

**Structural entitlement, for every certified Thiele-complete machine.** Theorem 8.3 was about any Thiele machine, and what the run paid went in as a hypothesis. For a certified Thiele-complete machine (Definition 15.7) what the run pays is known. Call the CHECK, COMMIT and CERTIFY moves of a run its *record moves*; each costs exactly one, so the ledger rises by the number of record moves, and clause (b) says what a raised record must contain.

**Theorem 15.36** (Structural entitlement represented)**.** Let a machine be certified Thiele-complete through an interface. Take a clean start $s_0$ and a run $t$ from it; a type of rivals and a type of observations with an exact equality test; an observation obs of each rival; a decoder $d$ from runs to observations; lists $\Omega$, $\Omega'$ of rivals and a decision tree $T$. Suppose (M1) $\Omega'$ strictly narrows $\Omega$, as in Theorem 8.3, and (M2) some dropped rival shows an observation different from every survivor's; (M3) the run ends with the record up and the decoded run $d(t)$ is shown by some survivor; (M4) $\mathrm{depth}(T)$ is at most the number of record moves in $t$; (M5) $|\Omega'| > 0$; and (M6) the covering: $|\Omega| \le L(T) \cdot |\Omega'|$. Then (1) the membership test of $\Omega'$ is strictly stronger than that of $\Omega$; (2) $t$ contains the earned chain of clause (b), with the meaning of the claim true at the CHECK and at the COMMIT; (3) $\lceil \log_2 |\Omega| \rceil \mathbin{\dot{-}} \lceil \log_2 |\Omega'| \rceil$ is at most the number of record moves of $t$; (4) the number of record moves of $t$ is exactly the ledger's rise; and (5) the ledger rose by at least 3. □

It's a checklist, as I said in Section 8, and the small print is the same kind. (M3)'s second half feeds no conclusion: only the record's being up is used. (M1) is used only for the inclusion; its gap feeds nothing, since (M2) already hands over a dropped rival. And (M5) is used but can go: (M2) puts a rival in the prior, and the covering then forces a survivor. What the machine adds to Theorem 8.3 is conclusions (2), (4) and (5): the payment is the machine's own three acts, each at exactly one, and a raised record carries the earned chain.

There are two more, one where the tree comes from the run's own questions and one about what the record adds.

**Theorem 15.37** (A floor with no supplied tree)**.** Take a machine and an interface that meets the exact-prices clause (c) of Definition 15.7, so each record act costs exactly one. Call the *questions* of a run the claims of its CHECK moves, and say two rival states *answer alike* when every one of those claims gets the same verdict at both (each verdict taken at the rival itself, as it starts; the states the run passed through don't enter). If, for a list $\Omega$ of rival states and a number $m > 0$, the rivals of $\Omega$ that answer alike with any one rival, itself included, are never more than $m$, then □

$\lceil \log_2 |\Omega| \rceil \dot{-} \lceil \log_2 m \rceil$ is at most the number of CHECK moves of the run, which is at most its number of record moves, which is the ledger's rise.

The tree has gone, and the run's own questions stand in its place. There are limits, and they're the ones in the statement. The verdicts are taken at the rivals as they start, and the states the run had reached when it asked don't come into it. A branch that costs nothing, DEC on zero, can still narrow a list without adding anything to the ledger. So the theorem prices a narrowing that the paid checks explain, and no more.

The next statement takes the verdicts where the run took them. A rival is a state the run could have started from; a CHECK the run passed is asked of a rival by running the moves before it from the rival and checking the same claim at the state that reaches. Nothing is supplied but the prior.

□ **Theorem 15.38** (Structural entitlement, with the narrowing made by the run)**.** Let a machine be certified Thiele-complete through an interface. Take a clean start $s_0$, a run $t$ from it that ends with the record up, and a list $\Omega$ of rival starts that contains $s_0$. Let the posterior $\Omega'$ be the rivals that pass every CHECK $t$ passed, each asked as above, in the order $t$ made them, and let $b_i$ and $a_i$ be the numbers of rivals standing just before and just after the $i$th. Then (1) $\Omega'$ sits inside $\Omega$ and contains $s_0$; (2) every rival of $\Omega$ that isn't in $\Omega'$ fails, asked that way, a CHECK that $t$ made and passed; (3) $|\Omega| \cdot \prod_i a_i = |\Omega'| \cdot \prod_i b_i$; (4) if every CHECK $t$ passed keeps at least a $2^{-k}$ share of the rivals standing before it, $b_i \le 2^k a_i$, then $\lceil \log_2 |\Omega| \rceil \dot{-} \lceil \log_2 |\Omega'| \rceil$ is at most $k$ times the number of CHECKs $t$ passed, that number is at most the record moves of $t$, the record moves are the ledger's rise, and so the index bits are at most $k$ times the ledger's rise; (5) $t$ contains the earned chain and the ledger rose by at least 3; and (6) with no share assumed, the index bits are at most $\lceil \log_2(|\Omega|/|\Omega'|) \rceil$, which is $\lceil \sum_i \log_2(b_i/a_i) \rceil$ and at most the sum over the CHECKs $t$ passed of $\lceil \log_2(b_i/a_i) \rceil$.

Part (3) is the exact form of "index bits are the sum of what each test took": take base-two logarithms and $\log_2 |\Omega| - \log_2 |\Omega'|$ is the sum of $\log_2(b_i/a_i)$. Parts (4) and (6) are what survives rounding to whole bits, and (6) rounds once, at the end, so it can be a bit lower for every test than rounding test by test. Rivals $(a, 0)$ for $a = 0, \ldots, 26$ with CHECKs $A \ge 9$, $A \ge 15$, $A \ge 19$ stand at 27, 18, 12 and 8: the index bits are $5 - 3 = 2$, rounding once gives 2, and rounding each test gives 3. Here the run does the narrowing, every candidate dropped was dropped by a check the run paid for, and what stays a premise is the prior and the share, a bound fixed in advance; part (6) shows what the run pays without one.

**Proposition 15.39** (One CHECK can take four bits)**.** Take the sixteen starts $(a, 0)$ with $a = 0, \ldots, 15$, and the run CHECK $(A \geq 15)$ $A$; COMMIT $(A \geq 15)$ $A$; CERTIFY from $(15, 0)$. It raises the record. Its one passing CHECK keeps one rival of sixteen, so the posterior is $(15, 0)$ alone and the index bits are 4, while the run has 3 record moves. So index bits at most the record moves, the bound of Theorem 15.38(4) read with $k = 1$, fails here. □

The share here is $2^{-4}$, and with $k = 4$ part (4) reads $4 \leq 4$. It's the example of Section 8 that showed the covering can't be dropped, seen from the run's side: one test can take many bits, and the bound of part (4) covers the bits per test times the tests.

**Theorem 15.40** (What the record adds)**.** Let a machine be certified Thiele-complete through an interface. (1) From a clean start, the record is up at the end of a run exactly when the run contains the earned chain, and a run with the record up raised the ledger by at least 3. (2) Conclusions (1), (3) and (4) of Theorem 15.36 hold without a clean start and without the record being up, for a machine that only meets the exact-prices clause (c): the hypotheses (M1), (M2), (M4), (M5) and (M6), and the decoded run being shown by some survivor, are all it needs. □

So the record doesn't make narrowing cost more. It adds the earned chain and a floor of 3, and that's all it adds: a run meeting the same counting and payment hypotheses is still bound by that index-bit floor even if it never raises the record, and the floor of 3 no longer applies to it.

*Weights, fibres and partitions.* Three more pieces of the same count. First, suppose the rivals aren't equally many: give each a whole-number weight, a multiplicity, and call the sum of the weights of a list its *mass*. The count survives with masses in place of lengths. If the prior's mass is at most $L(T)$ times the posterior's, the rounded index bits of the two masses differ by at most depth$(T)$; so for any Thiele machine a run that paid at least that many charged steps has a cost at least the drop, and for a certified Thiele-complete machine the representation theorem goes through with masses and the same five conclusions. On such a machine the ledger rises by the same number from every start, the number of record moves, so the weighted sum of ledger rises is the prior's mass times the record moves, and the drop in rounded bits times that mass is at most the weighted sum. Divide by the total mass and you'd have an average rise; I leave the weights as whole numbers, cross-multiplied, so no probability comes into it. Second, a way to meet the covering (M6) that you can see. Give each survivor a little crowd of rivals, its *fibre*, no bigger than the tree has leaves,

and let the fibres' sizes add up to at least the prior. Then $|\Omega|$ is at most the sum of the fibres, which is at most $L(T)$ times the number of survivors, and that's (M6). Third, a plainer way to get fibres. Pick a *representative* observation of each rival, and say the rivals are *partitioned* by it when every rival looks like some survivor under it and no observation class, counted inside the prior, has more than $L(T)$ members. Take each survivor's class as its fibre: every rival is in the class of some survivor, so the fibres add up to at least the prior, and (M6) holds. The converse is false: take four rivals that all look alike, two survivors and a tree with two leaves, and give each survivor two of the four as its fibre. That covers, but the one class has four members, more than two.

*A worked member on the small machine.* Eight rivals: the starts $(a, b)$ with $a \in \{0, 1, 2, 3\}$ and $b \in \{0, 1\}$, counter $A$ at $a$ and counter $B$ at $b$. The run is CHECK (Even) $A$; CHECK ($A \geq 2$) $A$; COMMIT ($A \geq 2$) $A$; CERTIFY, from $(2, 0)$. The program raises the flag from a rival exactly when $a$ is even and at least 2, so exactly from $(2, 0)$ and $(2, 1)$, and the posterior is those two. The decoder reads one "passed" for each CHECK in the run's receipts; the tree is the complete tree of depth 2; and the covering comes from fibres by a representative observation, "$B$ is 0," which isn't constant: each survivor's fibre is the four rivals that share its $B$.

□ **Proposition 15.41** (The eight-start member)**.** For every $a$ and $b$, the run raises the flag from $(a, b)$ exactly when $a$ is even and at least 2. All hypotheses of Theorem 15.36 hold, with eight rivals narrowed to two: the test of the posterior is strictly stronger; the index bits are $3 - 1 = 2$; the run has 4 record moves and the ledger rises by exactly 4; and the earned chain is in the run. The run's own questions (Theorem 15.37) are its 2 CHECKs, so the floor is met with equality. With the two CHECKs alone, the record stays down, the ledger rises by 2, and the bound $2 \leq 2$ still holds.

It's a real narrowing, and not the cups of Section 8 sitting under it in a costume. The posterior is exactly the set of rivals the run certifies, so here a proof ties the lists to the run, and nobody supplies them. Read with Theorem 15.38, the run's two passing CHECKs, replayed from the eight rivals, leave four and then two, the same two, so the posterior is the one the run makes; each keeps half, so $k = 1$, and the index bits, 2, are at most $1 \cdot 2$ and at most the ledger's rise of 4.

*A search through $n$ bits.* The same shape comes back at a size you can't enumerate, on the machine of Definition C.2, the host in Appendix C.1 with a counter for every number, with the small machine's property language of Zero, Even and At-least-$n$, read at registers, in place of Slot. That machine is certified Thiele-complete too, by the argument of Theorem C.6 with these three properties in place of Slot. Hide

$n = k + m$ bits, one in each of registers 0 to $n - 1$. The program asks $k$ questions, one per bit of the first $k$: CHECK "at least 1" of register $i$ for a 1, CHECK "is 0" for a 0. Then it COMMITs the first claim and CERTIFYs. The prior is every $n$-bit value; the posterior is the $2^m$ values that begin with the answers; the tree is the complete tree of depth $k$; and a survivor's fibre is the $2^k$ values that agree with it on the $m$ bits nobody asked about.

**Theorem 15.42** (A search through $n$ bits)**.** For $1 \le k \le 16$ and any $m$: from a world holding a value $v$ of at least $k$ bits, the program certifies exactly when $v$ begins with its answers, so the posterior is exactly the set of rivals it certifies from; the test of the posterior is strictly stronger than the prior's; the earned chain is in the run; the prior has $2^{k+m}$ members and the posterior $2^m$, so the index bits are exactly $k$; the run has exactly $k + 2$ record moves and the ledger rises by exactly $k + 2$; and the run's own questions are $k$ CHECKs, so Theorem 15.37 is met with equality. With the COMMIT and the CERTIFY left off, the questions alone narrow $2^{k+m}$ values to $2^m$, the record stays down, and the ledger rises by exactly $k$. And no run from a clean start can execute more than 16 CHECKs before its first successful CERTIFY: a failed CHECK would trap, and the table holds only 16 passing ones. Thus this program with 17 questions followed by its first COMMIT and CERTIFY certifies from no start. A trace that certified earlier can keep its flag up even after later failed instructions. □

The sixteen belongs to the machine, and the program has no say in it: it's the same ceiling as the item "Sixteen checks per run, ever" in the list of what the machine doesn't do, near the end of this chapter, and it's why this search never asks more than sixteen questions however many bits are hidden. Unasked bits can be any number. The lists, the tree and the fibres are supplied here, as in the representation theorem, and tied to the run only for this member. The covering still prices only the size of the drop, and its reason goes unpriced.

*The covering can't be dropped.* It's a hypothesis, and you'd hope it's a nuisance you could remove. It isn't, and the small machine shows it. Fix $n \ge 4$, take the $2^n$ starts $(a, 0)$ with $a$ below $2^n$ as the rivals, and run the three-move chain on the claim "$A \ge 2^n - 1$": CHECK, COMMIT, CERTIFY. It certifies exactly from $A = 2^n - 1$, so the posterior is one start, $n$ index bits are gone, and the ledger rose by 3. So for $n \ge 4$ the claim "the index-bit drop is at most the ledger's rise" is false for this run, and no tree of depth at most 3 covers the narrowing: (M6) would need $2^n \le 8 \cdot 1$. One property with a big constant in it carries $n$ bits in a single yes-or-no question; what the bound prices is a narrowing whose classes are balanced, a tree with enough leaves. A narrowing of any other shape is outside it.

**The time a check saves.** The question is whether counter $A$ is at least $m$. There are two ways to find out. The *unpaid* way uses base moves only, INC and DEC, which add nothing to the ledger. But a base move reads $A$ one pebble at a time: a DEC takes a pebble off $A$ or finds it empty. After $n$ moves, a run that started with $A$ at least $n$ has never seen $A$ empty, so it can't have behaved differently for different such $A$ except by a constant shift in $A$ itself. The *paid* way is one CHECK ($A \geq m$): one move, a ledger of one, and the line it ends on says which way it went. Certifying the answer takes the whole chain, CHECK, COMMIT and CERTIFY.

□ **Theorem 15.43** (The time tax). (1) For every program of two-counter instructions and every $n < m$, the line number and $B$ after $n$ moves (or where the run stopped, if it stopped sooner) are the same from $A = m - 1$ as from $A = m$, with the same $B$. So no such program, read off its line number and $B$ after fewer than $m$ moves, tells whether $A \geq m$ at every start; in particular none that stops within fewer than $m$ moves does. (2) A program of $m$ DEC/INC pairs (a ladder: pair $i$ is DEC $A$ on line $2i - 1$, jumping on success to the next pair at line $2i + 1$, and INC $B$ on line $2i$, where a DEC that finds $A$ empty falls through) decides it when observed after $m$ moves: after $m$ moves its line is $2m + 1$ exactly when $A \geq m$; and it costs 0. (3) Executed as a three-move trace from start$(a, b)$, the paid chain raises the flag exactly when $a \geq m$ and has a ledger of exactly 3, whatever $m$ is. As a stored program it pays 3 on success and stops after its first failed CHECK with a ledger of 1. (4) From any loaded start start$(a, b)$, a single CHECK ($A \geq m$) decides it in one move from the line number, with a ledger of exactly 1, whatever $m$ is. (5) So if a unit on the ledger is worth $\lambda$ moves, the paid route to deciding costs $1 + \lambda$ and the unpaid route costs at least $m$, and the paid route is cheaper exactly when $1 + \lambda < m$; the paid route to certifying costs $3 + 3\lambda$.

When $3 + 3\lambda < m$, the paid chain both answers and certifies within a combined budget smaller than the unpaid lower bound for answering alone. That compares different guarantees; the unpaid route never certifies. For a bare answer on both sides, the comparison is $1 + \lambda < m$. The exchange rate is sharper than a comparison of two programs I happened to write would give: the unpaid side is a floor under every program of INC and DEC there is, so no choice of program comes into it, and the paid side doesn't move with $m$ at all. The bound has limits. It's for programs that make no record move, INC and DEC with HALT or falling off the end as stopping. It doesn't cover every certified Thiele-complete machine, because the definition doesn't forbid a base move that changes a counter by more than one pebble, and such a machine is a different machine, one the bound doesn't speak for. And the two routes put their answers in different places, both bare decisions in the line number, with

the paid chain also putting a historical record in the flag; this compares two ways to answer one question, not two certifications. Zero cost isn't zero heat, either: every move on either route is an event, and the moves are what $\lambda$ prices.

The proofs and the worked runs of this subsection are in the companion volume *The Thiele Machine: The Example, Built* (its section "Proofs and worked runs for Parts I and II, run on the example machine") and in Coq in `minimal/FragmentSmall.v`, `minimal/EntitlementSmall.v`, `minimal/BitSearchMember2.v` and `minimal/TimeTax2.v`.

## 15.15 Reading the instruction rules

The instruction rules settle some concrete questions about running this machine. Follow them through a check, a zero decrement and a trapped trace.

The walk through a check, a zero decrement and a trapped trace is in the companion volume *The Thiele Machine: The Example, Built* (its section "Reading the instruction rules") and in Coq in `minimal/EarnedCore.v`.

## 15.16 What this chapter adds up to

So this chapter has one machine, and it isn't the model. It's a Thiele machine in the opening's sense, a Thiele machine with the toll proved (Theorem 15.7), and its toll charges the blank cell's 0: raising its flag is a merge in its record layer (Theorem 3.19). It's Thiele-complete, the sense no clock meets (Section 10.8), and certified Thiele-complete, in the sense of Definition 15.7 (Theorem 15.31): it computes whatever a Turing machine can, under Minsky's $2^n$ encoding, done outside the machine, its computing adds nothing to the ledger and can't touch the record, and, from a clean start, its record is earned.

If you ever find yourself wondering whether a property of this machine is the model's or just the machine's, hold it up against Section 2's definition of a Thiele machine, the latch of Section 10, the four clauses of Thiele-complete, and the four of certified Thiele-complete. What follows from those holds of every machine that meets them; the rest (a table of sixteen facts, three properties) is this machine's choice. The flat price of one is clause (c), and at least one of it is what the record-layer premise asks of each act on this machine. There are six instruction kinds, and a seventh, PAY, for a host that has to charge somebody else's prices (Appendix C.2). If the model needs more than the five parts and the toll, this is where it will show.

CHAPTER 16

# How far it reaches

The small machine earns its flag, and pays for it, on its own programs. Everything in this chapter asks how far that carries. Every statement in it is proved, and the runs are reported as runs. I give each statement here and leave the constructions to the appendices, because they're about the reach of the case, not the case.

*Other machines as guests.* One fixed program, $U$, written in the small machine's own six kinds of instruction on a host with more counters, runs every program of the small machine as a guest, from every clean start. It fetches, decodes and carries out each guest instruction itself, the way Turing's universal machine does; nothing in the host means "run one guest step." It halts exactly when its guest halts, with the guest's answer in its counters; its flag goes up during its run exactly when the guest's goes up during the guest's, by its own check, its own commitment and its own certificate; and its ledger equals the guest's at every point where the two runs match. The host is certified Thiele-complete, and whether $U$ halts is undecidable. Add one move, PAY, that costs one and does nothing else, and the same kind of program, $U_P$, runs every Thiele machine whose next move, step, prices and reading are computed by programs that always answer: it halts when the guest's run stops, its flag goes up exactly when the guest's reading is yes at some state of its run, on a chain whose check is the guest's own reading, and its ledger is the guest's prices plus a surcharge of at most two at the raise when the guest's reading starts at no, and exactly three when it starts at yes, a surcharge that no program on that host, run from a clean start, can always avoid. Appendix C has both; their proofs are in the companion volume.

*Other models of computation.* Lay the toll over a register machine, a Turing machine (with one move for each whole program, not one for each row of one fixed table, which provably can't serve as a base) or the lambda calculus and none of them computes one function more; what each gains is a record and a cost. What carries over from any base, what never does, and why the small machine uses two counters, are in Appendix D.

*Machines from parts.* Wired side by side, end to end, or one inside the other, a Thiele machine stays a Thiele machine, and the tolls add the way money adds. What

breaks, breaks for reasons you could name in advance: a single flag for two claims costs two claims' worth, a clock is still a clock, a shared value without a version can't be trusted, and a hand-off that costs nothing can't be allowed to carry a certificate (Appendix E).

*Running it.* The machine is built twice, once from the definitions and once by hand, and the two are compared after every step; $U$ and $U_P$ have run on guests small enough to finish; a compiler from a small programming language to the host is proved correct stage by stage, and Ackermann's function at $(3, 5)$ comes out of nothing but INC and DEC (Appendix F and the companion volume).

*A check worth earning.* A flag is only as interesting as the claim under it. The CHSH game, the two-player game behind Bell's theorem, gives a claim whose check is a piece of real mathematics: an integer test on the game's tally whose passing implies Tsirelson's bound, earned on the host by the same check, commit and certify (Appendix G). It's there to show the toll charges the same however deep the check behind it is.

CHAPTER 17

# Five small models

So far there's been a toll, a counter, two kinds of wall and a latch, every one of them shown on paper, on cups and cards and dice I made up, and then a small machine to run them on. That is the model on paper and on one small example. It's time to go and look at what people actually do. I picked five systems that keep records other people rely on, with real money or real security behind them, and asked each the pricing question from the paper models: *is there a price on the moment the record is written?* The question isn't about the work around the write; it's about the write, the moment the 0 goes on the page and stays. I'll tell you the answer first, because it's a single sentence and it surprised me: of the five specifications, read through the transcriptions I made of them, only one takes a fee at the moment of the write, and that one has a leak at exactly the place Section 6 said a counter leaks, where it goes down.

"Price" needs one meaning for all five, or the answer is whatever I want it to be, so I'll fix it here, and every system below gets classified by it. A specification *prices* an event when it attaches one of three costs to it. A *fee*: something taken at the event itself, whatever happens afterward. A *bond*: value put at risk by the event and lost only if some later event happens, a contradiction, say. A *budget*: a finite allowance the event draws down, so that only so many such events can ever happen. A wall isn't a price under this definition. Nobody pays a hash; reversing one is just infeasible. Neither is work a protocol needs and never charges for. The table at the end of the section has a column for each of the three kinds, and what it shows is a finding about the five, because the definition came first.

Three things are easy to blur in a section like this, so I'll name them now and keep them apart all the way through. There's the *real specification*: the document real money and real security run on. There's my *transcription*: one mechanism copied out of that document into a mathematical model, small enough to examine. And for some of the five there's a *wrapper*: a tiny abstract model that takes one premise as given and says what follows from it. A claim proved at one level isn't a claim at the level above it. Near the end I'll walk one system, gas, through all three, so you can see exactly where each claim stops.

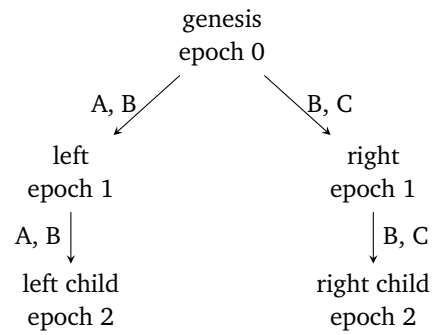
For each system I isolate the mechanism the question concerns: a *fragment* with its own states and operations, transcribed from the system's specification where one was available. I will put that mechanism on the page, state the guarantee it carries, and show how the argument follows its rules.

## 17.1 Proof-of-stake finality: the price is on going back

Ethereum and chains like it decide which blocks are final by having validators, people who've locked up money (their *stake*), vote on checkpoints: blocks picked out at regular numbered rounds called *epochs*. A vote isn't for one checkpoint. It's a *link* from a checkpoint the validator already trusts, the source, to a later one, the target, and every chain starts from one first block, its *genesis*, which counts as trusted from the start. In the design Vitalik Buterin and Virgil Griffith published in 2017, called Casper, a checkpoint is *justified* when validators holding two thirds of the stake vote a link to it from a justified source, and *finalized* when it is justified and a two-thirds link runs from it to its direct child. The record is "this checkpoint is final."

Two rules carry the price: don't vote for two different targets in the same epoch, and don't cast a vote whose span, source to target, surrounds the span of another of your votes. Break either and your deposit can be destroyed, which is called *slashing*. The theorem they prove, called accountable safety, says that if two *conflicting* checkpoints are ever both finalized, at least a third of the stake broke a rule and can be slashed. So the price is on *reversal*, on making the record contradict itself. The write itself, finalizing, need not incur a slashing penalty. This is not a claim that running validators has no operating costs or other protocol charges. In the terms from the start of the section, Casper's price is a bond. The deposit stands behind every vote, the finalizing votes included, and it's taken only from a validator whose own votes contradict each other, which is what making the record contradict itself takes from a third of the stake.

Do it with three validators of equal stake. Draw genesis at epoch 0, two competing children at epoch 1, and one child of each at epoch 2. Let A and B vote both links on the left branch, from genesis to its first child and from that child to the next. Let B and C do the same on the right. Each link has two of the three votes.




B appears on both branches at each target epoch. A and C do not. The diagram records votes; it has no step deducting a balance. Both epoch-1 blocks are justified and have the required link to their child, so both finalize. B voted for different targets at the same epoch, which makes B's third of the stake slashable, while A and C have broken neither voting rule. The votes supply evidence for a penalty; this example does not execute the transaction that collects it, and Theorem 17.2 below has exactly this instance, so the theorem's premise isn't vacuous: forks of that kind can be built, and the theorem fires on them.

The fragment is a reconstruction of the published casper-proofs argument from Runtime Verification and the University of Texas at Austin, based on Yoichi Hirai's proof, with validator sets and quorum intersection as a premise. A *quorum* is a set of validators big enough to act, here two thirds of the stake, and *quorum intersection* is the arithmetic of the three-validator example: two thirds plus two thirds is four thirds, so any two quorums share at least a third. The fragment takes the overlap as given; it doesn't do that arithmetic. It has no deposits, no rewards, no fork choice (the rule for picking a branch to build on), and no connection to anything else in this book. A finalization fork implies that a set of validators the size of that overlap, at least a third of the stake, has committed a slashable offence. A one-validator chain finalizes without any such offence, which is the "write costs no stake" sentence, proved here in a degenerate setting.

The fragment is small enough to read in one sitting, and every name below is from the transcription.

**Definition 17.1** (Casper setting and votes)**.** A *setting* has a type of validators, a type of block hashes with a parent relation and a genesis hash, and two classes of validator sets, the "two-thirds" sets $Q_1$ and the "one-third" sets $Q_2$. It carries two premises: *quorum intersection*, for any two sets in $Q_1$ some set in $Q_2$ lies inside both; and *at most one parent* per hash. A *state* says, for each validator, target hash $h$, target epoch $t$ and source epoch $u$, whether that validator voted the link $(h, t, u)$. A link from $p$ at epoch $u$ to $h$ at epoch $t$ is *justified by* $q$ when $q \in Q_1$, every member of $q$ voted $(h, t, u)$, $h$ is the $(t - u)$-th descendant of $p$, and $t > u$. Genesis is justified

at epoch 0, and a block reached from a justified block by a justified link landing at epoch $t$ is justified at $t$. Block $h$ is *finalized* at epoch $v$ with child $c$ when $c$ is a child of $h$, $h$ is justified at $v$, and the link from $h$ at $v$ to $c$ at $v+1$ is justified. A *finalization fork* is two finalized blocks $h_1 \neq h_2$, neither an ancestor of the other. The transcription calls a validator *slashed* when it voted two different targets with the same target epoch (a double vote), or voted links $(h_1, t_1, u_1)$ and $(h_2, t_2, u_2)$ with $t_1 > t_2$ and $u_2 > u_1$ (a surround vote). The state is *quorum-slashed* when some set in $Q_2$ consists of such validators. These names describe offending votes, not a balance deduction: there are no account balances in this state.

**Theorem 17.1** (Accountable safety)**.** In every setting, every state with a finalization fork is quorum-slashed. □

*Proof.* Let the fork be $h_1$ finalized at $v_1$ and $h_2$ at $v_2$. If $v_1 = v_2$, the epoch is above 0, since genesis is the only block justified at 0, so the links that justified $h_1$ and $h_2$ there are each voted by a set in $Q_1$; quorum intersection gives a set in $Q_2$ that voted both, and their targets $h_1$ and $h_2$ differ. That set double-voted. If $v_1 \neq v_2$, say $v_1 < v_2$, walk down the chain of justified links that justifies $h_2$. Every block on that chain is an ancestor of $h_2$, so none descends from $h_1$. (So $v_1 > 0$: the only block justified at epoch 0 is genesis, an ancestor of every justified block.) Some link in the chain starts at or below $v_1$ and lands above $v_1$. If it lands at $v_1 + 1$, it and the finalization link of $h_1$ have distinct targets at that epoch, and the intersection set double-voted. If it starts at $v_1$, its source is a justified block at $v_1$ other than $h_1$, and the links that justified those two blocks give a double vote the same way. Otherwise it starts below $v_1$ and lands above $v_1 + 1$, so it surrounds the finalization link of $h_1$, and the intersection set made a surround vote. ■

Everything the fragment promises stands on the overlap, and with the overlap taken away it promises nothing. The one parent is in the setting because the original proof uses it, and the proof above never does.

**Theorem 17.2** (The premise can hold, and the write costs no stake)**.** (1) In the three-validator setting where each validator has stake 1, $Q_1$ is the sets of stake at least two thirds and $Q_2$ the sets of at least one third, the state where A and B vote one branch and B and C the other has a finalization fork; B is slashed, A and C are not, and $\{B\}$ is in $Q_2$. (2) In the one-validator chain setting, one vote finalizes genesis and no validator is slashed. □

*Proof.* Both are computations on fixed finite data. In (1) every link carries two of the three votes, so stake two thirds, so both epoch-1 blocks are justified and each has a justified link to its child at epoch 2: both are finalized, and as siblings neither is an ancestor of the other. B voted two different targets at epoch 1, which is the double vote; A's and C's votes are one link at epoch 1 from source 0 and one at epoch 2 from source 1, which neither double-vote nor surround. In (2) the one validator's vote from genesis at 0 to its child at 1 is a justified link, so genesis is finalized, and a single vote can neither double-vote nor surround. ∎

For the record itself, one thing that means is that finality is revocable at a price, the price is named, and a final block can hold a false statement.

The obvious next question is whether the 2017 design is what Ethereum actually runs. The answer is partly, and the paper's own title is the clue. It's called *Casper the Friendly Finality Gadget*, and its first sentence says it's a finality system that *overlays* a chain still choosing its blocks by proof of work, so it's a gadget, bolted on top. In 2020 a larger group, Buterin among them, wrote up *Gasper*, which joins Casper's finality to a fork-choice rule called LMD GHOST, and on 15 September 2022, in the switch people called the Merge, Ethereum stopped choosing blocks by proof of work at all and went over to proof of stake. So the checkpoints and the two slashing rules above are one part of a system that holds real money today, and the transcription is of that part's safety proof as the 2017 design states it. Everything between the two, the fork choice and the rest of the running protocol, is the gap between my transcription and the running system, which I walk through near the end of this section with a photocopied recipe and a kitchen. None of that moves the price: finalizing still costs an honest validator no stake, and what costs is contradicting yourself.

## 17.2 Gas: the one that charges a fee at the write

Every operation on Ethereum costs *gas*, paid in ether by whoever sent the transaction. The yellow paper, Ethereum's formal specification, gives the reason in one line: every computation is fee-bearing so the network can't be abused and halting questions are sidestepped. The record I care about is a nonzero slot in the shared storage, which every full node in the world (every computer keeping its own complete copy of the chain) has to keep until it's cleared. Clearing a slot is allowed, and the refund rules below exist to reward it.

Writing a fresh storage slot from zero to nonzero costs twenty thousand gas, by the table in the paper's appendix, plus twenty-one hundred for touching a *cold* slot, one the transaction hasn't touched yet (after the first touch it's *warm*). As I read it, the point is to discourage storage growth, since storage grows every node's database. That charge is a fee on the write, at the moment of the write, and it's the only fee in

the five. Gas has a budget too, in the sense from the start of the section: every block has a gas limit, and the gas its transactions use can't exceed it, so only so many writes fit in a block.

Where does a number like twenty-one hundred come from? Somebody picked it, so the obvious next question is who, and why, and the answer has a date on it. In September 2016, just as Ethereum's developers were gathering in Shanghai for Devcon 2, their developer conference (the third one, since they started counting at zero), attackers started filling blocks with an operation that was cheap in gas and expensive in fact: it sent every node off to read data from its disk, and nodes fell behind the chain. The first repair, a fork that October called Tangerine Whistle, raised the prices of the operations the attack had leaned on. The cold-slot charge came later, in a proposal by Vitalik Buterin and Martin Swende called EIP-2929, switched on in the Berlin upgrade of April 2021, and the motivation it gives is still those 2016 attacks. So twenty-one hundred isn't a law of nature or a measurement of anything deep. It's a scar: a price set after somebody found the place where the old price was wrong. That doesn't change the argument; the write is still priced at the moment of the write. It does change how I read the rest of that table: some of its numbers, at least, are somebody's answer to "what did this cost last time," which is a record of a different kind.

Now for the leak, which is Section 6's leak. Set a slot and then clear it *in the same transaction* and you get a refund. Count it through: first write twenty-two thousand one hundred; clearing it costs a hundred and credits a refund of nineteen thousand nine hundred; net, twenty-three hundred. The other half is that if you leave it set, you pay at least twenty thousand. The write is priced, and the *revoked* write nearly isn't. But I have to add what the fragment leaves out, because twenty-three hundred on its own misleads: the realized refund is capped at one fifth of the gas the whole transaction used (that's the 2021 change from Section 6, the one that ended the refund-hoarding token's advantage). So a real set-then-clear pays a lot more than twenty-three hundred. As for how much more, the cap is a fifth of everything the transaction used, and that includes the twenty-one thousand every transaction pays just for existing. Take a transaction that does nothing but set and clear the slot, and count only that base and the two stores (the instructions that load their arguments add about a dozen more): it uses 43,200 gas, so the cap is 8,640, so it pays 34,560, which is 13,560 on top of the base. (Look at the slot alone and forget the base, which the rule doesn't do, and you'd get 17,760.) Either way it's far above twenty-three hundred. The counter-level net is twenty-three hundred, and what the transaction pays is not.

The arithmetic below is the way the specification has it, so you can redo it with a pencil.

**Definition 17.2** (One slot in one transaction)**.** A slot run records the slot's value at the start of the transaction (original), its current value, whether it has been touched (warm), the gas charged so far, and the refund counter. A store of a new value charges 2100 if the slot is cold, plus 20000 if the slot still holds its original value, the new value differs, and the original is 0 (2900 if the original is nonzero), and otherwise 100. If the value is unchanged, the refund does not change. Otherwise add together these applicable adjustments: add 4800 when the original and current values are nonzero and the new value is zero; subtract 4800 when the original is nonzero and the current value is zero; and, when the new value restores the original, add 19900 if that original is zero, or 2800 if it is nonzero. Store the new current value, mark the slot warm, and keep the original value unchanged. The run starts fresh: original and current 0, cold, gas and refund 0. The *net* charge is gas minus refund.

□ **Theorem 17.3** (Pricing the persistent write and the revoked one)**.** (1) For every list of stores from a slot whose original and current values are 0, warm or cold, with gas and refund 0, if the slot ends nonzero then the net charge is at least 20000, and from a fresh slot, which is cold, at least 22100. (2) Storing 1 then 0 leaves the slot at 0 with net charge 2300; storing 1 alone has net charge 22100.

*Proof.* (1) Keep two lines on the sheet: "current value zero: net at least zero," and "current value nonzero: net at least 20000." The starting slot meets the first line. I will check that every store leaves it meeting whichever line describes its new value. The original value remains zero throughout, so there are only four changes to check. Zero to zero adds 100 to net; zero to nonzero adds 20000; nonzero to nonzero adds 100; and nonzero to zero adds $100 - 19900 = -19800$. The last case starts with net at least 20000 and so ends at least 200, still nonnegative. So every store leaves the appropriate line true. Carry the two-line test through the list of stores: if the final value is nonzero, the net charge is at least 20000. A cold access adds another 2100, once, on the first store. Keep that extra charge in a separate column; later updates are exactly the ones just checked and do not depend on the accumulated total. A nonempty run from a cold slot therefore pays that 2100 on top of the bound, giving at least $20000 + 2100$ when it ends nonzero. (2) Direct computation: $2100 + 20000 = 22100$ for the first store, 100 for the second, refund 19900, and $22200 - 19900 = 2300$. ■

The fragment covers one slot in one transaction, with the gas and refund arithmetic copied out of the Ethereum execution specification's storage instruction and

gas table, at a named commit. It has no stack, no call context, no other slots, and *no refund cap*, the one-fifth rule applied at the end of a transaction.

Then comes the wrapper, which is a different kind of object, and the walk near the end is about exactly that difference. It's a *gas schedule* in the abstract: any states, any steps, a cost per step, a yes/no commitment flag, and a charge predicate saying which steps get charged. It formalises no EVM (the Ethereum virtual machine) at all. It is Section 7's result with the whole-run ceiling, the one that pins the amount as well as which steps pay, under a new name. Its two halves are worth reading separately. Leave one committing step uncharged, and a one-step run commits at no charge, so the floor fails outright; one missed commit is enough. Charge one step that doesn't commit, and the schedule charges more than the number of commits on that one-step run, so the ceiling fails. And the class isn't empty: in any system, the price that charges one exactly on a step that raises the flag and nothing on any other meets both halves (Section 7's step price).

The wrapper itself is the cleanest thing in this section and has the least Ethereum in it.

**Definition 17.3** (A gas schedule)**.** A *gas schedule* has states, instructions, a step, a cost $c(s,i) \in \mathbb{N}$ that may depend on the state, a Boolean commitment flag on states, and a Boolean charge predicate $\chi(s,i)$, with two premises: a charged step costs at least 1, and an uncharged step costs 0. A step *commits* when it takes the flag from false to true. For a run $t$ from $s$, let $c(t)$ be its total cost and $n(t)$ its number of committing steps. The schedule has the *floor* when $c(t) \geq n(t)$ for every run, and *no overcharge* when $c(t) \leq n(t)$ for every run. It has *unit pricing* when $c(s,i) = 1$ if $\chi(s,i)$ and 0 otherwise.

**Theorem 17.4** (Gas schedule exactness)**.** For every gas schedule: it has the floor and no overcharge if and only if $\chi(s,i)$ holds exactly when the step from $s$ by $i$ commits, and it has unit pricing. Moreover, (1) if some committing step from an uncommitted state is uncharged, that one-step run costs 0 and the schedule fails the floor; (2) if some charged step does not commit, the schedule overcharges. □

*Proof.* The equivalence is Section 7's characterization with the whole-run ceiling, under another name. From right to left, each step's cost is 1 on a committing step and 0 otherwise, so a run's cost counts its commitments. From left to right, apply the floor and no overcharge to one-step runs: a committing step must cost at least 1, so it is charged; a charged step costs at least 1 and may not exceed its commitment count, so it commits and costs exactly 1. Parts (1) and (2) are those one-step runs read the other way. ■

Gas prices storage growth and computation. It says nothing about whether what's stored is true.

### 17.3 Attestation: hash chains, and a quote about the past

A trusted platform module keeps platform configuration registers, or PCRs. An extension hashes the old value together with a new measurement digest. A quote signs selected PCR values and challenge data supplied by the asker; the verifier must check the signature, challenge and expected measurements. That can establish a fresh report of those values. It does not establish that every relevant fact about the machine was measured.

PCR reset permissions depend on the register, locality and platform rules. Some reset at startup; some support permitted resets during a boot. Thus "you can only extend" describes an interval with those resets excluded. Intel SGX uses enclave measurements and attestation reports for a related purpose, but it does not use this PCR-extension interface.

What does the specification price on the registers? Nothing: no fee, no bond, no budget. Within the stated reset and access restrictions, forging a different measurement history that passes verification relies on defeating the cryptographic assumptions or trusted components. Authorized resets are a separate operation, not an attack that requires breaking a hash. Expensive against infeasible is the difference between a price and a wall, and it's easy to blur, which is why the definition at the start of the section leaves walls out. What stands behind a cost can come in three currencies. It can be economic, a fee or a deposit the protocol takes (Casper and gas are here). It can be cryptographic, a reversal made infeasible by a hardness assumption, which makes it a wall with no exact price on it (attestation and transparency are here). Or it can be physical, energy spent or hardware worn, which none of these five specifications charges as a fee or a bond; the nearest thing is the chip's write budget, below. And the record's permanence is only as long as the register lives: a register's continuity lasts only through the interval allowed by its reset rules, not necessarily an entire boot.

Why the registers, then, when something that lasts would make a better record? Because the quote is what a remote asker reads, and the registers are what it signs. The same chip has longer-lived records, and they're closer to the latch than anything else in this section. Version 2.0 of its specification has storage slots of a kind it calls a bit field, changed only by a command that ORs new bits into the old value. During that index's lifetime the command can set a bit and cannot clear one, the OR rule from Section 10. Deleting and redefining an index is governed by separate authorization and policy; it must be included in any claim about persistence across that lifetime. It also has counters advanced by an increment command. On first use,

a new counter starts above the values previously held by counters on that TPM, so deleting an index and making a new one does not restart the count at zero. This is a guarantee of the specified chip, not protection against replacing it with another chip. And those slots are rationed. Bryan Parno and his coauthors, building a system called Memoir in 2011, noted that the specification of the day only required a counter to be able to tick once every five seconds, and that the chip's storage was expected to survive about a hundred thousand writes in its whole life, so writing once a second would wear it out in under twenty-eight hours. Memoir writes to it twice per boot. Those figures describe the hardware assumptions discussed in Memoir, not a universal write quota for every TPM. Endurance and throttling constrain real use, but an expected endurance figure is not itself an enforced protocol budget. The quoted PCR interface supplies no such numerical price. On the bit field it's the budget of Theorem 10.11's chain of $k$-bit words, which can only climb $k$ times. Phones and small controllers go one floor further down, to fuses like the PROMs in Section 10: a version number kept as a row of fuses, one more blown at each security update, and boot code that refuses anything older. An industry guide on it, Rob Smart's 2021 explainer for PSA Certified, a security certification scheme for chips and devices, tells designers to make sure there are enough fuse bits to last the number of updates expected over the device's lifetime, which is part (5) of that same theorem, written as a design rule.

What all of this is for is *rollback*. Memoir's example is a password checker restored to an old saved state to undo a user's password change, and in 2017 a system called ROTE did the same job for Intel's enclaves by having several machines vouch for each other's counters, with the stated result that the only way left to break it is to reset every one of them to its start. In 2016 Raoul Strackx and Frank Piessens's Ariadne got that job down to flipping a single bit of non-volatile memory per state update, which they call the theoretical lower limit. Notice who the attacker is in every one of these: whoever gets to put an old state back. A bad program doesn't come into it. A record kept by the step can't see that attacker. Every guarantee the small machine gives about a record's earned history (Section 15) is stated from a clean start; put back a state the machine really reached and each one still holds of it, through the run that reached it first, and none of them can say that nothing came after it. The per-step toll still holds after a restore; it just can't tell you what happened before it.

Replicated services put the same kind of record somewhere else: inside each replica, where a replica that's been taken over still can't make it say two different things. Byung-Gon Chun and coauthors' attested append-only memory (A2M, 2007) is a log a replica can only add to; Dave Levin and coauthors' TrInc (2009) shrinks that to a trusted counter; MinBFT (Giuliana Veronese and coauthors, 2013) uses a trusted counter to reach agreement with $2f + 1$ replicas when $f$ of them can lie,

where the usual count is $3f + 1$; and Damysus (Jérémie Decouchant, David Kozhaya, Vincent Rahli and Jiangshan Yu, 2022) has a small trusted service, the Checker, that keeps a monotonic counter of where the protocol is. Allen Clement, Flavio Junqueira, Aniket Kate and Rodrigo Rodrigues (2012) worked out, with proofs, what such a record buys, and as I read their result it's a good warning for this book. On its own, stopping a replica from saying two different things doesn't lower the $3f + 1$ bound in an asynchronous network at all. Add signatures that a third party can check, and it turns any protocol that survives crashes into one that survives liars, with no extra replicas. The way I read it, a record helps the others only as far as they can check what it says, which is Section 7's point from the other side.

There's a fourth currency under those three, the logical kind, and it's the one every permanent record on a finite chip pays in: Section 3's counting of used and fresh states applies to any of these records. With only finitely many states holding the record on, a step that switches it on somewhere, and never switches it off anywhere, forgets something. Forgetting alone doesn't single out a record write: any step that overwrites a field able to hold two values forgets something too. What singles it out is where the forgetting lands. Take the states the record already reads yes on, together with the one the step switches on. The step sends every one of them to a yes state, and it can't keep them apart, so what it destroys is a distinction on the record's own yes side: two states it already vouched for, or the state it just raised and one that was up already. Any overwrite merges states somewhere; a permanent write is forced to merge them on that side. That needs no premise about money, keys or heat. The economic, cryptographic and physical payments each need a premise of their own about their kind (a fee rule, a hardness assumption, Landauer's principle), and they hold exactly where that premise holds.

□ **Theorem 17.5** (A permanent write forgets on its own yes side)**.** For any states $S$, instructions $I$, step and Boolean reading: suppose a duplicate-free list names every state, the instruction $i$ never turns the reading from on to off, and $i$ turns it on at some state $s$. Then the step by $i$ is not injective. More: the step by $i$ sends every state that reads on, and $s$, to a state that reads on, and it is not injective even on those states.

Count the on-states and call their number $K$. The list of all states is finite, so this count is finite too. The instruction keeps each of those $K$ sources on, and it also sends the additional off-state $s$ to on. That gives $K + 1$ distinct sources with only $K$ on-states available as landing places. Two sources must land together. Both came from the stated collection, so the failure to keep sources apart already occurs there. ■ Permanence of any other instruction was never needed.

What the quote can do is say what was *measured*. It can't say the measured code is correct, and it can't say anything about runtime state that was never folded in. A correctly measured buggy program attests beautifully. The fragment proves that exact limit: one register, a hash about which nothing is assumed, a quote as a pair, and no function of the quote decides a runtime flag the quote ignores, which is Section 7's cube again. And a second small result shows that a sign-and-verify interface *alone* gives no authenticity (a scheme where verify always says yes passes the interface), so a cryptographic premise has to be added; the fragment doesn't model signatures, formats, or a freshness check (the nonce is just a field the verifier fixes).

The same limit, turned outward, is the honest edge of the four conditions of Section 14. Every one of them is about a thing the machine holds: same property, same thing, same edition, honest check. For a claim about the world (the oven is hot, the parcel arrived) there's no edition the machine controls. Whatever reads the world is a window onto it, and "nothing changed between the check and the commit" stops being something a step can enforce and becomes a premise about the sensor. So a record can establish claims about the machine's own state, and claims about the world only through named premises about whatever measured them. A correctly recorded sensor reading establishes what the sensor reported. Whether the oven was hot still depends on the sensor and the connection between its report and the world.

**Definition 17.4** (Platform and quote)**.** Fix any functions $H : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ (the extend hash) and $\delta : \mathbb{N} \to \mathbb{N}$ (the digest); nothing is assumed about either. A *platform* is a measurement log (a list of numbers) and a Boolean runtime flag. Its register is the log folded from 0 by $H$. The *quote* with nonce $n$ is the pair $(\delta(\text{register}), n)$.

**Theorem 17.6** (The quote cannot attest unmeasured state)**.** For every nonce $n$ there is no function $D$ from quotes to Booleans with $D(\text{quote}_n(P)) = \text{flag}(P)$ for every platform $P$. □

*Proof.* Take two platforms with the empty log, one flagged true and one false. Their quotes are equal, since the quote never reads the flag, so $D$ would have to return both values on one input. ■

Anything that depends only on $\delta(\text{register})$, the quote can answer; that's what it's for. And the second result is the one I mentioned: a "signature" scheme whose check

always says yes fits the sign-and-verify interface perfectly and accepts a signature no key ever made, so the interface on its own promises nothing.

### 17.4 Certificate transparency: visible, unpriced

The certificates that make your browser show a padlock are issued by hundreds of authorities, and a bad or compromised one can issue a certificate for a domain it shouldn't. Certificate Transparency puts certificates in public append-only logs built as Merkle trees. An inclusion check ties a certificate to a particular root; a consistency check ties a later root to one the reader already has. It was specified in RFC 6962 in 2013 and revised as RFC 9162 in 2021. I went looking for which one the running logs follow, and it's still 6962: Mozilla's reference for web developers says browser implementations are based on 6962, and as late as November 2025 Chrome's team was only proposing to drop its rule that every certificate carry a timestamp from at least one 6962 log. The fragment I transcribe is 9162's. A Merkle tree is Section 2's hash chain folded into a tree: hash each certificate (those are the *leaves*), hash the leaf hashes in pairs, hash those in pairs, and so on up to one *root* hash. To prove one leaf is in, the log hands you only the few hashes sitting beside the climb from that leaf to the root, the *path*, and you redo the climb yourself. A second check, *consistency*, shows that a bigger tree is a smaller one with leaves added on the end and nothing rewritten. The record is a leaf in the tree.

The data structure has a history of its own in security. The computer scientists Scott Crosby and Dan Wallach, at Rice University, built tamper-evident logs on the same kind of tree in 2009, with a proof that an entry is in the log and a proof that a later version of the log extends an earlier one, each a logarithmic number of hashes. A rewrite that fails to extend a previously witnessed root is caught by the consistency check.

The RFC prices nothing, and it says in its first section that logs let interested parties *detect* misissuance afterward. They don't *prevent* it. And a log that shows different views to different people (a *split view*) can defeat the auditing, so logs are still trusted parties. What it buys is visibility.

Which raises the obvious question: if the specification prices nothing, why does any authority bother logging? Nothing in the RFC makes it. The answer is a browser. Chrome's team set the rule, enforced from Chrome 68 in July 2018, that every certificate issued after 30 April 2018 had to meet its transparency policy, and a publicly trusted certificate that doesn't is simply not trusted by Chrome: the visitor gets a full-page warning in front of the site. So there is a price after all, though it isn't anywhere in the specification: it lives in one vendor's browser, and it's the price of being shown to that browser's users at all. Is that the bottom? Not quite, because it means the strongest lever on this record is a policy one company can change,

which is a fact about who holds the stamp. The tree has nothing to do with it. That policy changes nothing in the table at the end of this section, which asks what each specification prices. But it's worth keeping two sentences apart: "the specification prices nothing" and "nothing is priced."

Do the inclusion proof by hand. Lay out seven cards $d_0$ through $d_6$, each standing for one leaf hash. Join neighbours with paper brackets: $a = \nu(d_0, d_1)$, $b = \nu(d_2, d_3)$ and $c = \nu(d_4, d_5)$. The last card stands alone. Next join $a$ to $b$, giving $L = \nu(a, b)$, and $c$ to $d_6$, giving $R = \nu(c, d_6)$. The root is $\nu(L, R)$. Here $\nu$ means "put these two expressions, in this order, inside one labelled bracket." Equal symbolic hashes must have exactly the same bracket structure.

To check $d_3$, keep only that card and the three siblings $d_2, a, R$. Combine $d_2$ on its left to recover $b$, then $a$ on the left to recover $L$, then $R$ on the right to recover the root. You never needed to open the cards inside $a$ or $R$. Those three siblings are the inclusion path.

| Step | What is supplied | What is rebuilt |
|---|---|---|
| Start | card $d_3$ | starting leaf hash $d_3$ |
| 1 | $d_2$ on the left | $b = \nu(d_2, d_3)$ |
| 2 | $a$ on the left | $L = \nu(a, b)$ |
| 3 | $R$ on the right | root $= \nu(L, R)$ |

Swapping left and right changes the bracket. The path has to carry the correct order as well as the correct siblings. The hashes here are brackets, not numbers, so what you just checked is how the tree is put together; a real log uses short hashes, and "two different trees can't give the same root" is then a separate promise from the cryptography. Nothing in this fragment signs anything, no auditors talk to each other, and no log misbehaves.

**Definition 17.5** (Inclusion check, RFC 9162 section 2.1.3.2)**.** Given a leaf index $m$, a tree size $N$, a leaf hash, a root hash and a path, the check rejects unless $m < N$. Otherwise start counters $f = m$, $s = N - 1$ and an accumulated hash $r$ equal to the leaf hash. For each path element $p$, reject if $s = 0$. If $f$ is odd or $f = s$, replace $r$ by $\nu(p, r)$; in this branch, while $f$ is positive and even, halve both counters with rounding down. In the other branch replace $r$ by $\nu(r, p)$. After either branch halve both counters once more. At the end accept exactly when $s = 0$ and $r$ is the claimed root. A *symbolic hash* makes a leaf hash $\ell(x)$ and a node hash $\nu(a, b)$ distinct constructors: two hashes are equal only when their entire constructions agree.

The consistency check used here takes old and new sizes $M, N$, roots $u, v$ and a path, and rejects unless $0 < M < N$. If $M$ is a power of two, prepend $u$ to the path. Reject an empty resulting path; otherwise its first element seeds two accumulated hashes $r_{\text{old}}, r_{\text{new}}$. Start $f = M - 1$, $s = N - 1$, and halve both while $f$ is odd. Process each remaining path element $p$: reject if $s = 0$; when $f$ is odd or $f = s$, put

$p$ on the left of both accumulated hashes and halve both counters while $f$ is positive and even; otherwise leave $r_{\text{old}}$ alone and put $p$ on the right of $r_{\text{new}}$. After either branch halve both counters once. Accept exactly when $s = 0$, $r_{\text{old}} = u$ and $r_{\text{new}} = v$. All halvings round down.

□ **Theorem 17.7** (The RFC's own examples pass)**.** With the symbolic hash, in the RFC's seven-leaf tree with leaves $d_0, \ldots, d_6$ (writing $d_k$ also for its leaf hash $\ell(d_k)$), the inclusion check accepts leaf 3 with path $[d_2, \nu(d_0, d_1), \nu(\nu(d_4, d_5), d_6)]$, and likewise leaves 0, 4 and 6 with their paths; the consistency check from size 4 to size 7 accepts. For every input with $m = N$, the inclusion check rejects.

*Proof.* Use the brackets $a, b, c, L, R$ built above. For leaf 3, the counters start at $(3, 6)$: put $d_2$ on the left, leaving $(1, 3)$; put $a$ on the left, leaving $(0, 1)$; put $R$ on the right, leaving $(0, 0)$ and root $\nu(L, R)$. Leaf 0 has path $[d_1, b, R]$ and three right combinations. Leaf 4 has path $[d_5, d_6, L]$: right, right, left. Leaf 6 has path $[c, L]$: left, then left; its first branch shifts $(6, 6)$ to $(3, 3)$ before the ordinary halving, which accounts for the unpaired last leaf. Each reconstructs the same root. For consistency from 4 to 7, the old root is $L$ and the supplied path is $[R]$. Prepending $L$ seeds both hashes with $L$; shifting the initial $(3, 6)$ gives $(0, 1)$, so the sole remaining element leaves the old root alone and makes the new root $\nu(L, R)$, with final counters $(0, 0)$. The boundary case $m = N$ fails the first test $m < N$. ■

And the append-only part of the fragment is as plain as it sounds: the history is a list, and adding to the end of a list leaves every old entry where it was.

### 17.5 Proof-carrying code: no price, no permanence

Necula's idea, from Section 8, is to ship untrusted code with a proof that it obeys the host's policy; the host computes what would need to be true and checks the proof with a small checker. The record is the checked proof or the result of checking it. An implementation may retain either. Proof-carrying code does not by itself require a permanent latch or assign an abstract toll to acceptance. Producing and checking the proof both take computational work; that work has not been identified with the unit charge in this book.

The fragment is a three-instruction memory-bounds language with the policy "every address is below the limit": the checker accepts exactly when the condition holds, and a certificate implies the condition. One example sits on top of that, a computation you can do in your head: a program that reads address two under a limit of two is rejected. The fragment isn't the paper's real case study, and nothing

connects it to anything else in this book. And it has a limit: a program that writes garbage to address zero passes if zero is below the limit. The proof is of the stated policy and nothing else.

And what was the paper's real case study, the one my fragment isn't? It was network packet filters, the little programs an operating system runs on incoming network traffic to decide which program gets to see which packet. They run inside the kernel, the most trusted part of the system, and they have to be fast, so they're exactly the code you'd most like to take from a stranger and least like to trust. In the 1996 paper that first set out the idea, Necula and Peter Lee wrote several filters by hand in the assembly language of a DEC Alpha, shipped each with its proof, and reported a one-time cost of 1 to 3 milliseconds to check the proof, after which the filter ran with no checks at all. Those milliseconds put a number on "checking is cheap by design". What the paper reports changes nothing about my fragment, which still isn't that case study, and it doesn't add a price: the milliseconds measure the checker's work in that experiment. They are not a per-crossing toll theorem, whether or not an implementation saves the result.

**Definition 17.6** (The bounds fragment)**.** Instructions are READ $a$, WRITE $a\ v$, and HALT. For a memory limit $L$, the policy allows READ $a$ and WRITE $a\ v$ when $a < L$, and always allows HALT. The *verification condition* of a program is that the policy allows every instruction in it. A *certificate* is a list of per-instruction policy facts, one for each instruction in order. The *checker* tests the policy on each instruction and returns the conjunction.

**Theorem 17.8** (The checker is the verification condition)**.** For every limit $L$ and program $P$: the checker accepts $P$ exactly when the verification condition holds; any certificate for $P$ gives the verification condition; and with $L = 2$ the checker rejects READ 2 followed by HALT. □

*Proof.* Lay the instructions in a row. For each READ or WRITE, compare its address with $L$; put yes beneath it exactly when the address is smaller. Put yes beneath HALT without asking about an address. The checker accepts exactly when every mark in that row is yes. The verification condition asks for exactly those same per-instruction facts. An empty row has no failing instruction, and extending the row preserves acceptance precisely when the new instruction also passes. This covers programs of every finite length.

A certificate supplies one policy fact at each place in the same row. Reading them in order gives every fact the verification condition asks for, so it implies that condition too.

With limit two, the first instruction READ 2 asks whether $2 < 2$. It does not. The later HALT is allowed, but cannot undo that failed comparison, so the checker rejects the two-instruction program. ■

## 17.6 One example, three levels

Before I lay all five side by side, let me walk one of them up and down the three levels from the start of the section, because every claim in here lives at exactly one level and it's easy to let a claim slide up a floor without noticing. I'll take gas, since it's the one place all three line up. Think of a restaurant's real recipe, my photocopy of one page of it, and a cookbook chapter on what any recipe has to contain. I can check my photocopy line by line. That tells you nothing about what the kitchen cooks tonight.

Start with the real specification. Ethereum's execution specification says, in a document that runs real money, what a storage write costs, what clearing it refunds, and how the refund is capped at the end of the transaction. None of what I proved is at this level. What the network actually charges is whatever the clients people run actually compute, and the evidence for that is the clients and the chain. This book has none of it.

Next comes the transcription. I copied one mechanism out of that document, one slot in one transaction, gas and refund arithmetic only, at a named commit (Definition 17.2, Theorem 17.3). What the result tells you: *if* the storage rule is what I copied, then leaving a fresh slot set costs at least twenty thousand, and set-then-clear nets twenty-three hundred at the counter. What it can't tell you: anything about the refund cap (not copied, which is why what a real set-then-clear pays came out so different from the counter earlier), the stack, calls, other slots, or whether any client implements the rule as written. The boundary of the claim is the edge of the copy.

Last comes the wrapper. The gas schedule of Definition 17.3 isn't Ethereum. It's any system with steps, a cost per step and a yes/no commitment flag, plus one thing *stipulated*: a charge predicate, with the premise that a charged step costs at least one and an uncharged one costs zero. That premise is the whole of what the wrapper adds, and it's added by hand; no specification supplies it. Given it, the theorem says every run costs exactly its number of commits, never letting a commit go unbilled, never billing one twice and never billing a non-commit, exactly when the charge predicate is the commit predicate at unit price. What that tells you is what exact pricing would have to look like, in any system. It doesn't say Ethereum meets it, and Ethereum doesn't: real gas bills computation that commits nothing, which the transcription shows in numbers. So the wrapper prices a definition, the transcription prices a copied rule, and neither prices the chain.

The other four sit on the same three levels, and only one more wrapper is spelled out in this section. Casper's transcription is the accountable-safety proof above; its wrapper is a small gadget with the stake at risk handed in. A finalize step that risks nothing turns the record on at no charge, which breaks Section 2's toll, and one that risks at least one unit has a floor. It records exposure to loss (no collateral is destroyed in it), has no Casper checkpoints in it at all, and is priced by stipulation, the same way the gas wrapper is. Attestation, transparency and proof-carrying code have transcriptions that prove what their verifiers do and no price theorem, because no specification of theirs puts a price on the verdict. Their wrappers, which I don't spell out here, are Section 7's verifier argument told in each system's vocabulary, with a colliding pair behind each one, and that argument is all they carry here.

The book's rule is that every claim gets a way to break it, so I'll say how to knock down the four claims with the most weight on them. For the gas transcription: a list of stores from a fresh slot that ends nonzero with net charge under 22100. For the gas wrapper: a schedule in which every run costs exactly its number of commits, whose charge predicate still differs from the commit predicate at some step. For the finality gadget: a finalizing trace from unfinalized to finalized whose charged stake is zero while the gadget claims a floor. For the attestation fragment: a verifier reading only the quote's fields that decides a runtime flag the quote ignores.

## 17.7 The same table, five ways

Lay them out by the one definition of price from the start of the section. Casper puts a bond behind the write and takes it on *reversal* (slashing). Gas takes a fee at *the write* (and refunds the revocation, capped), and rations writes with a gas limit per block. Attestation prices *nothing* on the registers the quote signs, and puts a budget on its longer-lived slots and counters. Transparency prices *nothing*. Attestation and transparency make reversal infeasible or visible. Proof-carrying code prices nothing and keeps nothing.

| System | Fee at the write | Bond | Budget | What the transcription proves |
|---|---|---|---|---|
| Casper finality | none | the deposit, taken on a contradicting vote (slashing) | none | a finalization fork means a slashable third |

| System | Fee at the write | Bond | Budget | What the transcription proves |
|---|---|---|---|---|
| Gas | 20000 for a fresh slot, mostly refunded on revocation, up to a cap | none | the gas limit of each block | a persistent write to an empty slot nets at least 20000 |
| Attestation | none; reversal is infeasible | none | none on the registers; a write budget on the non-volatile slots and counters | the quote can't decide what it never measured |
| Transparency | none; misissuance is visible | none | none | the RFC's own inclusion examples pass |
| Proof-carrying code | no required persistent latch | none | none | the checker is the verification condition |

Each row proves what its own transcription says, and no more. To read a row as a claim about the running system you'd need evidence from that system, and no row shows that certification is the one event every model has to price.

So the thing the toll charges for, the moment the record is written, is charged a fee in one of the five specifications I read, and in that one the designers had to patch the place where the charge could be clawed back. I find that neither a vindication nor a refutation. It's a fact about what people built, and what they built was mostly walls (hash chains, signatures) and penalties for contradiction. Tolls on commitment were rare. Whether a toll on commitment is a *better* design than a wall is not something I can prove, and I won't pretend the five say so. What I read in them, and it's a reading of five fragments I chose, is that each has a condition treated as a record, though the conditions differ in durability, and that the price axis is mostly empty. With one definition of price fixed before the survey, that last part is a count: of fifteen cells in the table, four are filled (one fee, one bond, two budgets), and only gas takes a payment at the moment of the write.

That leaves the question Section 10 raised and Section 18 has to face. The five systems have five different events chosen as "the record": finalized, committed, attested, logged, checked. Is there anything that picks out one event as *the* one worth recording, or is it an unforced choice every time? The pointer thesis is my proposed answer, and the next section gives the evidence it asks for.

CHAPTER 18

# The pointer thesis

Everything so far leaves one choice open. A permanent record driven by the machine is a latch on an event, and the event is yours to pick (Section 10, the two cups on one die). Five real systems each picked a different one (Section 17), and for the machine in this book I picked certification. So I owe you an answer to the obvious question: is that just my choice, or is there something about how people and machines lean on records that picks this event out?

What I have is a thesis, in the sense the Church-Turing thesis is one (Section 5): a claim tying defined things to the world, which no proof can settle. Whether it stands or falls is what decides whether the choice of event has a reason behind it. Nothing before it depends on it; the latch, the windows and the counting arguments hold whichever event a system records. What follows is the claim, where I got it, and how its terms apply to the systems being described.

## 18.1 The claim

Of all the events a machine could put on its ledger, the ones its price list charges for and its counter adds up, certification is the one whose records get *copied and relied on by parties who didn't produce them*: checked, stored, and acted on across independent places that share no other state. Copying alone can't be the test. A replicated system copies everything it holds to every node, the gas used in a block header as faithfully as the word final, so rival events get copied too. Reliance does not automatically separate them either. In Ethereum, the previous block's gas use helps determine the next block's base fee: another party reads it and acts on it. That is an explicit rule in EIP-1559.

So the examples cannot establish that certification alone is copied and acted on. The thesis asks whether a separately justified class of reliance selects it, and whether that selection gives a reason to charge its establishment. The formal toy below fixes its readers and actions in advance. It can test those choices, but it cannot choose the real readers for us. Both the claimed selection and its connection to prices remain to be justified.

There's a second handle on the same question, and it comes from Section 3. The permanent-certificate theorem (Theorem 3.1) never uses the word certification for anything special: given Section 3's rule that forgetting is charged, it prices any yes-or-no reading that, once on, stays on, on a machine with finite memory. So it points at a narrower candidate: the events to price are the ones that write a record no later step can take back. Certification is one of them. As an exact answer that candidate is false, because a merging step can also revoke (Section 3's three cards). Whether the records that are copied and relied on are the permanent ones, and whether being copied and relied on picks out anything the merge criterion doesn't, is what the two handles have to settle between them.

I can see two holes in that from here, so you don't have to find them for me. The first is a missing premise. Even if the world copies certification records more than any other kind, "copied" doesn't give you "priced" on its own. Something has to connect them, and the best candidate I have is a premise about incentives: a record that many independent parties copy and lean on is exactly the record worth forging, so it's the one where a price buys the most protection. That's a reasonable premise, but it's still a premise. The thesis's second half states the connection as a claim to be tested; it doesn't explain it, and nothing in this book supplies the explanation. The second hole is a circle. If "certification" just means "the record other people rely on," then of course it's the one that gets relied on and copied, and the claim says nothing. So the event has to be picked out without mentioning reliance: here it's a reading going up for good right after a check of the same claim passed, which you can find in a run without asking who reads it. Whether the world copies *that* event more than its rivals is then a real question, and it could come out no.

The thesis says no more than that, and what's left is the rest of what a thesis owes you.

## 18.2 Where the shape comes from, and how much of it survived

The shape is from physics, and I want to tell you where exactly, because I borrowed a word and I didn't get to keep the whole thing.

Quantum mechanics has a famous problem: a system can be in a blend of states, and when you measure it you find one definite state, and nobody agrees on what "measuring" did. In 1981 Wojciech Zurek asked a sharper question: of all the ways a measuring device *could* end up, why does it end up recording one particular kind of quantity (the position of a pointer on a dial, say)? Why never some blend? His answer, developed over twenty years and summarised in a 2009 paper called "Quantum Darwinism," is that the *environment* decides. The air, the light, the table, everything the device touches, is constantly interacting with it, and only certain states of the device survive that interaction; those are the *pointer states*. More than

that, the environment selects them, and then it *copies* them. The light bouncing off a dial carries the dial's reading to every corner of the room, redundantly, so that a thousand observers can each read a different scrap of the environment and all agree on where the pointer is, without touching the dial. Objectivity, on this view, *is* redundancy: a property is objective when many independent fragments of the world each carry it.

Zurek isn't the only one behind it. Fernando Brandão, Marco Piani and Paweł Horodecki (2015) prove an approximate statement about most small fragments of a large output: their channels can be approximated by measuring the input in one common way and preparing fragment states from that result. The error and the exceptional fraction are part of the theorem. Agreement on outcomes also needs the fragments to carry enough information about them.

Xiao-Liang Qi and Daniel Ranard (2021) strengthen the bound on the exceptional region. Outside a region whose size is controlled by the input size, allowed error and size of the fragments being examined, locally accessible information is approximately classical. This does not say that every observer gets an exact record, or any useful information at all. My definitions below ask for exact Boolean answers from specified observers. That is a simpler chosen contract, not the quantum theorem with its qualifications removed.

What I borrowed is that shape. Certification is to a machine's ecosystem (the machine plus everything around it that reads, stores and relies on its records) what the pointer reading is to the room: the thing that gets copied everywhere. It's a good shape and I think it's the right intuition. I'm going to be exact about what survived into the definitions below, because the word "redundant" is doing less than it does for Zurek.

Zurek's redundancy is *graded* (each fragment carries almost all the information, with some small loss) and it requires the fragments to be *disjoint*: different observers reading different parts of the environment. The definition below is the zero-loss case with *no disjointness at all*. An ecosystem is a state type, some number of observers, and one bit per observer; an observer *records* an event when its bit is true exactly when the event holds; an event *proliferates* when every observer records it. Nothing in that says two observers draw on separate parts of the state. They could all be reading the same bit, and the definition permits it. So I kept the word and dropped the independence that makes Zurek's version mean "objective." That leaves a gap between the thesis's prose and that first formal object, and the prose is where the thesis lives. The definitions under the thesis below close part of the gap: they define independent carriers, and the definition turns out to force something (Theorem 18.1).

The independence I dropped is exactly what the thesis's closest published neighbour keeps. Jess Riedel, Zurek and Michael Zwolak (2016) took histories of a closed

quantum system, the story a whole sequence of events tells, and asked which of them the world keeps records of in many separate pieces of the environment. As I read their paper, redundant records in separate pieces make a history consistent, so ordinary yes-or-no reasoning about it works, and they pick out a small set of histories from the huge number that are merely consistent: in their words, an objective past. Their result is "what proliferates is what counts" in quantum form, with the disjointness in it, and it came before this book. Halliwell, in a second 1999 paper, "Somewhere in the universe," had already carried the link between records and consistent histories from pure starting states to mixed ones, where the records still exist but are necessarily imperfect.

There's a deeper mismatch in the borrowing, and it runs against the book, so it goes here. In Zurek's picture the whole thing, system and device and environment together, evolves by quantum mechanics' rule for closed systems, and that rule never sends two states to one; every step can be run backwards. So, seen from the whole, the environment's copying is just that, copying: the dial's reading gets written onto fresh light, and no two states anywhere are merged. A simple copying model sends each new copy to a previously unused carrier. That is a use of fresh space, like Bennett's blank page. It is not a proof that every closed quantum evolution visits a new state forever; finite closed systems can recur.

Looking only at the dial discards the correlations with the light. In an ideal complete-dephasing channel, distinct input phases can give exactly the same reduced output. In gradual decoherence they may only become harder to tell apart; they need not become exactly equal at any finite time. An exact merge in a reduced description, approximate loss of accessible coherence, and a heat transfer are three different claims. To obtain a heat floor one still needs the physical preparation, the relevant entropy change and the thermal premises. Copying alone supplies none of those conclusions.

That is the difficulty for the borrowing. The thesis points at copies being kept and used. The quantum account explains particular ways information can spread and become accessible. It does not attach this book's toll to that act.

A published no-go cuts closer still. Philippe Allard Guérin, Veronika Baumann, Flavio Del Santo and Časlav Brukner (2021) gave an argument that, under assumptions that look natural in quantum mechanics, an observer's own perceptions of her outcome at two different times can't share one reality: no joint chance for the two, depending linearly on the starting state, reproduces what unitary quantum theory predicts for each. A record that persists, for the one holding it, is exactly what gets into trouble once the whole is treated as one-to-one. It isn't a theorem about my machine, which isn't quantum. It's still the strongest published reason I know to hold the word "permanent" loosely when a physicist is in the room.

Which leaves the question to ask of any borrowed shape: has anybody checked the original? For a long time quantum Darwinism was a theory without a lab. Then, in 2018 and 2019, three groups working separately, in Italy, China and Germany, went looking for its fingerprint, as Philip Ball's 2019 report in *Quanta Magazine* tells it. The German one, at Ulm, with Zurek among the authors, watched the spin of a single defect inside a diamond and the carbon atoms around it, and saw the redundancy the theory predicts: the defect's state turned up copied in its surroundings, and it didn't take much of the surroundings before more of them told you nothing new. The other two did it with photons. Is that the bottom? It isn't, not by a long way, and the people involved said so. Zurek himself pointed out that the photon experiments make their copies artificially and that the diamond's environment is small, and others noted that all three are schematic versions of a real environment and don't rule out rival accounts. So the shape I borrowed has passed some first tests in a lab, at the scale of a handful of atoms or photons, with the caveats written on it. None of that touches the thesis here, which is about machines and never rested on quantum physics. It changes the picture: what I borrowed is a lead somebody is checking. It isn't settled, and I don't get to lean on it as if it were.

## 18.3 What's proved

This section has definitions, two toys, a theorem about independence, a survey, and two refutations of things that would have been stronger.

The definitions are the ones above, plus "unique pointer among rivals": the event proliferates and none of a listed set of rival events does. Uniqueness is *relative to the list*, necessarily, since any event that holds on exactly the same states as a proliferating event proliferates whenever it does: the same fact under another name, so someone has to say which names count as rivals. And the first criterion is stated as a schema: for every ecosystem in some class, its metered event is the unique pointer among its rivals. The class and the rivals are inputs. Nothing about them is proved. A second set of definitions, after this one, tightens "proliferates" to "copied and relied on" and drops the rival list.

The formal object is small enough to hold in one hand.

**Definition 18.1** (Ecosystem, records, proliferates, pointer)**.** An *ecosystem* is a type of states, a number $N$ of observers, and for each observer $i$ a Boolean fragment $F_i$ on states. Observer $i$ *records* an event $E$ (a property of states) when, for every state $s$, $E(s)$ holds exactly when $F_i(s)$ is true. $E$ *proliferates* when every observer $i < N$ records it. $E$ is the *unique pointer among* a list of rival events when $E$ proliferates and no rival in the list does. Given a class $C$ of ecosystems, and for each member a

designated metered event and a list of rivals, the *pointer criterion holds* for $C$ when every member's metered event is the unique pointer among its rivals.

Nothing in the definition makes two fragments read different parts of the state, and "metered" is only a label: the definition has no metering in it.

One bit per observer can't even state the objection a replicated system raises. Every full node of a blockchain holds the whole shared state, so every fact about it, "gas used is at least one" as much as "final," sits on every node. The tightened definitions give each observer whatever it holds, an action it takes on what it holds, and a flag for whether it produced the record.

**Definition 18.2** (Copied and relied on, independent carriers, the pointer)**.** An *ecosystem with reliance* is a type of states, a number $N$ of observers, and for each observer $i$ a view $V_i$ (a function from states to some type of things an observer can hold), an action $a_i$ (a Boolean function of the view) and a producer flag. Fix a set $R$ of reachable states. An event $E$ is *copied* on $R$ when for each $i < N$ some Boolean function $d$ of the view has $E(s)$ exactly when $d(V_i(s))$ is true, for every $s$ in $R$. Observer $i$ *relies on* $E$ when $a_i(V_i(s))$ true gives $E(s)$ for every $s$ in $R$, and $a_i(V_i(s))$ is true for some $s$ in $R$. $E$ is *relied on by parties who didn't produce it* when some observer $i < N$ is not a producer and every such observer relies on $E$; it is *copied and relied on* when it is both. $E$ is the *pointer* on $R$ when it is copied and relied on, and every event that is copied and relied on holds at every state of $R$ where $E$ holds. The carriers are *independent* under a family of relations $P_i$ (states that agree on observer $i$'s own part of the world) when each view gives equal values on states related by its $P_i$, and for any two observers $i \neq j$ below $N$ and any states $s$, $t$ there is a state $u$ with $P_i(u, s)$ and $P_j(u, t)$: the two parts can be set separately.

Two pointers on the same $R$ agree on every state of $R$, so the pointer is unique with no list of rivals. And the definition of independence has a consequence I didn't put in on purpose.

Put one yes/no card in front of each of two readers. If the cards can be set separately, all four arrangements are allowed: no/no, no/yes, yes/no and yes/yes. Ask each reader to recover one shared event using only their own card. Suppose that event is false at no/no and true at yes/yes. At no/yes, the first reader sees exactly what they saw in the false case, while the second sees exactly what they saw in the true case. Their answers conflict. That mixed arrangement is exactly what the independence condition hands the proof.

Now impose a copying procedure that only leaves no/no or yes/yes. On those reachable arrangements each reader can report the shared bit correctly. Separate cards are things that *could* carry a copy; what makes them copies is that the arrangements you actually reach have the two agreeing. The theorem below is

about recovering the event exactly on *all* arrangements, the mixed ones included, which the copying never leaves behind.

**Theorem 18.1** (Independent copies on every state: the event is equivalent at any two states)**.** In any ecosystem with reliance whose carriers are independent and which has at least two observers, an event copied on the set of all states holds at one state exactly when it holds at any other, so it holds at every state or at none. □

*Proof.* Let observers 0 and 1 decide the event by $d_0$ and $d_1$. Given states $s$ and $t$, take $u$ with observer 0's part from $s$ and observer 1's from $t$. Then the event at $s$ is $d_0$ of observer 0's view, which is the same at $u$, so it equals the event at $u$; and the event at $u$ equals the event at $t$ the same way through observer 1. ■

So separate parts of the world can each carry an exact copy of a real record only on the states the world actually reaches, where the copying made them agree, and that's why the definition asks for copies on a set $R$ of reachable states. Zurek's copies are the same: the light carries the dial's reading because the dial put it there.

**Theorem 18.2** (A replicated header: copying doesn't separate, reliance does)**.** Let the states be a block header (a finalized bit, a gas-used count) and three nodes' copies of it, with $R$ the states where every copy equals the header; node $i$'s view is its own copy, its part of the world is that copy, node 0 is the producer, and every node acts exactly when its copy says finalized. Then the carriers are independent; "finalized" and "gas used is at least one" are both copied on $R$; "finalized" is the pointer on $R$; and "gas used is at least one" is not relied on by parties who didn't produce it. □

*Proof.* Independence: to give node $i$ the copy from $s$ and node $j$ the copy from $t$, overwrite node $j$'s copy in $s$. On $R$ every copy is the header, so each node reads either fact off its copy. Node 1 acts only where its copy, and so the header, says finalized, and acts at the finalized header with no gas used; that state shows gas used fails node 1's reliance. Any event relied on by node 1 holds wherever node 1 acts, which on $R$ is wherever the header is finalized, so "finalized" is the strongest. ■

It's a sanity check with maps I wrote, and it shows only that the tightened definition can tell the two apart where plain copying can't. Which of the two it picks, the actions pick: let every node act when its copy says gas was used, and gas used is the pointer. When what the other parties act on is copied, that is the pointer, so here the argument lives in the actions, and I wrote those too.

□ **Thesis 18.3** (The pointer thesis)**.** Let $C$ be the class of ecosystems with reliance that faithfully model deployed record-keeping disciplines, each with its reachable states, its observers reading independent carriers in the sense of Definition 18.2, its certification event, and the records its discipline prices. Then for every member of $C$: (1) the certification event is the pointer on the reachable states, so it is copied and relied on by parties who didn't produce it, and every event copied and relied on in that sense holds wherever certification does; and (2) every record the discipline prices is copied and relied on by parties who didn't produce it.

The thesis leans on two notions with no precise mathematical definition: what it is to "faithfully model" a system, and which systems count as "deployed." Those two tie the definitions to the world, the way "computable in the plain-English sense" does in the Church-Turing thesis, and no definition can do that job, so the claim is a thesis. The other term it needs, independent carriers, is defined in Definition 18.2. The schema takes the class, the reachable states, the certification event and the priced records as inputs; it proves nothing about which real systems satisfy them.

The toy has three observers on a replicated ledger, where certification proliferates and "work done" doesn't. It's an illustration, and that's all it is; it isn't evidence.

□ **Theorem 18.4** (The toy)**.** Let the toy have states (a certified bit, a work count), three observers, and every fragment equal to the certified bit. Then "certified" is the unique pointer among the one rival "work at least 1".

*Proof.* Every fragment is the certified bit, so every observer records "certified" by definition. For the rival, take the state with the bit false and work 1: the event holds and observer 0's fragment is false, so observer 0 does not record it. ■

The survey covers twelve events in one theorem: for each of the five systems in Section 17, the designated event (finalized, committed, attested, logged, checked) proliferates in the observer map I wrote for that system, and a rival (proposer work, a scratch counter, measurement noise, submission latency, prover steps) doesn't. The five together are one statement, an instance of the schema above. The survey adds a symmetric message-authentication code, which does *not* proliferate, and a digital signature, which does, for twelve rows in all. The five maps all use the same construction: a state has a designated Boolean event and a separate nonnegative rival count, three observers each return the event bit, and the rival asks whether its count is at least one. The bit and count can vary independently. The signature map likewise gives all three observers the authentic-signature bit. In the MAC

map, observers 0 and 1 receive the authentic-tag bit and observer 2 always returns false. These maps, not a protocol running on a network, are the entire tests here. And there's a thirteenth, which I think is the most important one: swap which bit every observer copies, and the criterion picks the *other* event. The criterion selects whatever the observer map copies. The map is where the argument lives, and I wrote the maps. So the five "confirmations" are five instances of one construction with the fields renamed. I'm saying that twice on purpose, so it can't slip past once.

**Theorem 18.5** (The survey, and the swap)**.** (1) In each of the five observer models (proof-of-stake, gas, attestation, transparency, proof-carrying), the designated event proliferates and the designated rival does not; in the symmetric-MAC model the authentic-tag event does not proliferate; in the signature model the authentic-signature event does. (2) Let the states be pairs of bits (first, second), with three observers whose fragments all equal the second bit. Then "second is true" is the unique pointer among the one rival "first is true". □

*Proof.* Each confirmation holds by definition: each observer's fragment is true exactly when the event holds. Each refutation exhibits one state where the event holds and some observer's bit is down. In (2) that state is (true, false). ■

Two stronger claims are both refuted, and I'd ask you not to run them together, because they fail for different reasons.

The first stronger claim is that *every forgery-resistant commitment proliferates*. If that were true, certification would be the pointer because it's the commitment. It's refuted in three observer models (deniable authentication, where only the designated verifier's bit tracks the event; a symmetric MAC, where the key holders share a secret and nobody else can check; an object capability, an unforgeable token that grants access to whoever holds it) and confirmed in two (a public log; a digital signature). Build the first with three observers: give one a copy of the event's bit and the other two a constant false. The one records it; it doesn't proliferate. The proved half is the map computation. Whether my maps are *fair* to the real designs is argued in prose, and you should argue back.

**Theorem 18.6** (One blind observer is enough)**.** (1) In any ecosystem, if some observer $i < N$ has $F_i(s_0)$ false at a state $s_0$ where $E$ holds, then $E$ does not proliferate. (2) Let the deniable model have states (authentic, verifier-accepts), three observers, observer 0's fragment the authentic bit and observers 1 and 2 constantly false. Then observer 0 records "authentic", and "authentic" does not □

proliferate. The symmetric-MAC and object-capability models also fail to proliferate their events; the public-log and signature models proliferate theirs.

*Proof.* (1) Recording would require $E(s_0) \iff F_i(s_0)$, and the left side is true while the right is false. (2) Observer 0's fragment is true exactly when the event holds. Apply (1) with observer 1 and the state (true, true). The other two refutations apply (1) to a blind observer of their own model (the symmetric-MAC map gives observer 2 a constant false; the object-capability map gives the token holder the bit and the other two constant false). The two confirmations hold by definition: the public-log map gives all three observers the logged bit, and the signature map gives all three the authentic-signature bit. ∎

The theorem is about the observer maps, and only them. That a third party in deniable authentication really cannot be convinced is the modelling claim, and the model has no transcript, key or verifier in it to check that against.

The second stronger claim is that *agreement plus accuracy plus locality imply permanence*. That is, if all observers agree, each sees the event exactly, and each updates by the same local rule with no coordinator, then the recorded event can't be undone. The smallest thing imaginable refutes it: one bit, two observers who both see it, and the rule "flip." All three conditions hold and true becomes false in one step. Do it on paper; it takes a minute and it's humbling. What *does* give permanence is durability, "an observer's true view stays true," added as a premise, and then it follows in a few lines. So durability is the premise doing the work, and durability is Section 3's permanence wearing a different coat. Agreement doesn't buy it. Do the paper version again with one change, the rule "stay true once true" in place of "flip," and the bit can't fall, because the rule won't let it. The observers agreeing has nothing to do with it. (The small machine of Section 15 has a reading that's durable in exactly this sense: its flag never comes down, by its step rule (Theorem 15.6).) None of this rules out a narrower theorem with real assumptions about the network, the adversary or durability; it rules out these two strong forms.

**Definition 18.3** (Observation game)**.** A *game* has states, a number $N$ of observers, a Boolean event $e$, a Boolean view $V_i$ for each observer, and a step map. For $i, j < N$ it has *consensus* when $V_i = V_j$ on every state; *authenticity* when $V_i(s) = e(s)$ for every state and every $i < N$; *coordinator-free update* when one function $\ell$ on Booleans has $V_i(\text{step}(s)) = \ell(V_i(s))$ for every state and every $i < N$; *durable views* when $V_i(s)$ true gives $V_i(\text{step}(s))$ true for every $i < N$; and a *permanent event* when $e(s)$ true gives $e(\text{step}(s))$ true.

**Theorem 18.7** (What buys permanence)**.** (1) It is false that every game with $N > 0$, consensus, authenticity and coordinator-free update has a permanent event. (2) Every game with $N > 0$, authenticity and durable views has a permanent event. □

*Proof.* (1) The toggle game: states are Booleans, two observers, event and both views the identity, step "flip". Consensus and authenticity hold trivially, the local rule is "flip", and the state true steps to false. (2) If $e(s)$ is true, observer 0's view is true by authenticity, stays true by durability, and authenticity turns it back into $e(\text{step}(s))$. ■

## 18.4 What isn't proved, and must stay that way on the page

What isn't proved is that certification *in deployed systems* is singled out by being copied and relied on across *independent* carriers. The criterion defines independence; it has no precise definition of "deployed" or "faithfully model," and no cost law connecting being copied and relied on to price. Every correspondence between a mathematical observer map and a real system is a sentence I wrote. The thesis is a thesis, and anyone who says the formal object proves it is reading more than is there.

## 18.5 What the thesis asks of the examples

There are three concrete comparisons to make. First, look at the record a deployed discipline prices. Do parties who did not produce it copy and rely on it? A priced record that nobody else relies on would contradict the thesis's second half.

Second, compare certification with rival events. If an event not implied by certification is copied and relied on to the same standard, certification is not the unique pointer. Multiplication counts or head-turn counts would qualify if they met that same test in the system being described.

Third, apply the independence condition of Definition 18.2 to a faithful model of the ecosystem and ask whether certification itself meets the test. The symmetric-MAC example shows why the carriers and their access matter. The two stronger forms already refuted above help locate the distinction this thesis needs.

These comparisons supply the evidence the worldly claim asks for. The mathematical criterion supplies a definite question once the model and its observer maps are chosen.

### 18.6 The end of the logic

The logic that started in Part I ends here. Part I found what keeping takes: anything kept is paid in a merge or in fresh states, alongside the model-dependent thermal bounds (Section 1), and a toll on raising a flag is forced by three premises (Section 3). Learning doesn't have to merge anything, so it owes no floor; it borrows a blank page, which is a supply of fresh states, and can hand it back. Committing on a finite supply of fresh states has to merge (Section 4). Part II earned the rest. The counter is arithmetic once the schedule is fixed (Section 6). A window with a colliding pair in it can't price the flag exactly (Section 7). Entitlement is an object, a check, and a paid event, and, given lists, a tree and a covering that someone supplies, and a run that paid for the tree's depth, it costs at least the index bits it commits to (Section 8). Insight is a distinction made to stay, and a committed relation is that at its best (Section 9). A feeling doesn't qualify. Over any base a record that is driven by the computation and never comes down is a latch, with two unforced choices, the price list and the event (Section 10). With more than two places, a record that is driven by the computation and climbs a partial order is a row of cups, each one a latch; the toll over the whole row charges at least one on any step that turns up some cup, and a straight line of stages needs exactly one cup fewer than it has stages, cups that only turn up (Section 11). When the machine has finitely many states, its record only grows, and the ordinary machine forgets nothing, the toll prices crushing two states of one fibre, and a window can price exactly by seeing the cups move (Section 12). No inside procedure whose flip a substrate can represent decides a shortcut property, given a recursion theorem, which the nat family has by construction. L is handled on its own terms rather than as a substrate, and on L the same diagonal is proved again from a recursion theorem it has by proof. What a record can do is carry the commitment (Section 13). This part made "established" a record that a check of the claim passed (Section 14), and the small machine earns one by a check, a commitment and a certificate, one unit each (Section 15). Of five specifications, read through my transcriptions, one charges a fee at the write (Section 17). For the choice of event, the pointer thesis proposes certification and lists how that proposal could fail (Section 18).

One question about the price itself is still open: do the requirements I've put on a price pick out the toll and nothing else? If a short list of them left only one price standing, that would settle it. The lists I tried don't, and the no is small enough to check on three states. Call them blank, once and twice, with once and twice reading yes. The flip takes blank to once and leaves the other two where they are. The smooth takes both once and twice to twice and leaves blank alone. The smooth forgets something, so the price on merging from Section 3 charges it one unit. (The flip forgets too, blank and once both land on once, so the price on merging charges

it as well, as the list requires.) It raises nothing, so the toll charges it nothing. Now try requirements that name no reading: no charge on a move that changes nothing, none on a move that forgets nothing, the same cost when you rename the states, a total that adds up along a run, a cost that looks only at the step, and a cost of one on the flip from blank. The toll, read as the price that charges the flip from blank and nothing else, meets every one of them on this machine. So does the price on merging. They disagree on the smooth, so that list doesn't pick between them. What picks the toll is a requirement that names the record: over any run the total is never more than the number of times the reading went up. With the floor, that leaves exactly one price, one unit per raise and nothing else (Section 7), and it rules the price on merging out, because the smooth costs that price one unit and raises nothing. The only renaming of the three states that respects the moves is the identity, so the renaming requirement asks nothing there. So the uniqueness is real and it's small. It holds once you've said which reading you're charging for, and which reading deserves the charge is the pointer thesis, which is still a thesis.

The uniqueness theorems that do exist are about pricing information loss, and they're worth a look because they say what kind of charge the price on merging is. John Baez, Tobias Fritz and Tom Leinster (2011), mathematicians who work in category theory, the mathematics Lawvere's diagonal theorem came from, asked what a measure of information loss has to look like. As I read their paper, they proved that, for maps that preserve a probability measure, Shannon entropy is up to a constant factor the only measure of information loss that is functorial, convex-linear and continuous. János Aczél and Bruno Forte, mathematicians at the University of Waterloo who worked on functional equations, and C. T. Ng, who had taken his doctorate there under Aczél in 1971, asked in a paper of 1974 which entropies deserve to be called natural, and proved, by their account, that the entropies with a short list of natural properties are exactly the linear combinations of Shannon's and Hartley's, where Hartley's is the log of the count, so even for merges the choice between the average and the worst case is a real choice. Both price merges. Neither prices raises, so neither picks out the toll, and that's the three-state example again, at a larger size. On the other side of the fence, Luciano Pomatto, Philipp Strack and Omer Tamuz (2023), economists at Caltech and Yale who asked what learning something should cost, proved, by their account, a uniqueness result for what it costs to *acquire* information, with constant marginal costs. That result prices producing a signal, which is a different matter from what Section 4 reads off a copy onto a blank page; the copy owes no floor and borrows fresh states.

Most of that isn't a machine. The theorems of Parts I and II quantify over "any states, any steps," and the five models are small copies of somebody else's systems. The example machine is the small one of Section 15, the least I could make that still shows the whole model, with every part named and every price written down,

and the program that runs every program of it and, with one paid move added, every Thiele machine whose workings a program can compute, is in Appendix C. That's where everything above has something to bump into. If a statement up there doesn't mean what I think it means, that's where it shows, and Part IV is how you'd go looking.

CHAPTER 19

# The counter against physics

Section 3 did the physics argument once: on a finite machine a permanent flag merges states, a price list that charges merges (a named premise, with Landauer's principle, another named premise, as the physics behind it) charges the flag, and the toll follows. I'm not going to do it again. This section is about what a machine's counter, the small machine's $\mu$ included, has to do with any of that, and the answer is: less than naming Landauer suggests, and I want you to hear the exact amount from me.

## 19.1 The calibration is a unit conversion

Multiply the counter by an energy per unit of cost that you choose, and you get a theorem: a state with one unit, at the scale kT ln 2 per unit, carries kT ln 2 of energy, for every k and T. Read that for exactly what it is: a unit conversion, true by definition, with the scale as an unforced input. Nothing in it picks the scale. The theorem would be just as true with the scale set to a dollar a unit. What *would* connect units of cost to joules is the premise in Section 3, Landauer's principle, which is a physical premise. And even as a scale, $k_B T \ln 2$ a unit is the floor for one particular squeeze, a halving. Corollary 3.13's floor for a flip that brings $k$ states in beside $m$ already reading yes, starting from the even spread on those $m+k$ states, is the larger of $k_B T \ln \frac{m+k}{m}$ and $\frac{2k}{m+k}\, k_B T \ln 2$, and for one newcomer among many yes-states that's far below a halving, about $1.4\, k_B T / m$. Price every unit at $k_B T \ln 2$ and you can overstate the floor of a single narrow flip by a factor that grows with $m$.

Remember the constant from Boltzmann's grave in Section 3, the exchange rate between a count and heat? It has a newer chapter, and it makes the point sharper than I can. The $k_B$ in $k_B T \ln 2$ turns a temperature into an energy, and for most of its history it was a measured number, with error bars, pinned down by careful experiments. Since 20 May 2019 it isn't. That day the international system of units was redefined so that Boltzmann's constant is exactly $1.380649 \times 10^{-23}$ joules per kelvin, by definition, and the kelvin is whatever makes that true. So the conversion from temperature to energy has been a convention since then, in the strictest sense: its numerical value in those units is fixed. Realising the kelvin in a laboratory still

requires measurements, with uncertainty. What's still physics is everything around it, whether wiping a bit at a temperature really sheds that much heat, which is what the Lyon bead was measuring. The calibration theorem is the same kind of move as that redefinition. It fixes a conversion, and the physics, if there is any, is somewhere else.

**Definition 19.1** (Energy at a scale)**.** For a real number $c$ (joules per unit) and a state $s$, the energy of $s$ at scale $c$ is $\mu(s) \cdot c$. The *Landauer calibration* at $k_B$ and $T$ is the statement that every state with $\mu(s) = 1$ has energy $k_B T \ln 2$ at scale $k_B T \ln 2$.

□ **Theorem 19.1** (The calibration is a unit conversion)**.** (1) For all real $k_B$ and $T$, the Landauer calibration at $k_B$ and $T$ holds. (2) For any counter, there are two real-valued functions of the state that give 1 and 2 on every state with one unit, so the counter alone fixes no energy value.

*Proof.* (1) The scale says what one ledger unit is worth. At a ledger reading of one, multiplying by that scale returns the scale itself: $1 \cdot k_B T \ln 2 = k_B T \ln 2$. No condition on the signs of $k_B$ or $T$ is needed for that multiplication.

(2) Choose a scale of one and a scale of two. The same ledger reading of one is assigned energy one by the first choice and energy two by the second. The ledger has not changed; only the conversion I supplied has. Thus that conversion was not ■ determined by the counter.

The physics is the premise of Section 3, restated here once in the form the next theorems test. In words first: a *dissipation* hands each instruction a whole number of heat units, each unit worth kT ln 2. The floor premise says an instruction that takes at least one state where some yes-or-no property is false to a state where it's true, never sends a true one back to false, and sends two different states to one, dissipates at least one unit. That's a different statement from Landauer's principle as Premise 3.5 has it. That one depends on the spread over states, $Q \geq k_B T \ln 2 \cdot \Delta H$, and gives no floor for a known state (there the distinction was absent from the chosen point distribution; that alone says nothing about a prior preparation bill). This one gives every such instruction a floor of one whole unit, whatever the spread. To get the second from the first you'd need more, at the least a spread with a full bit of uncertainty removed at every such step, and the known-state theorem and the narrow squeeze of Section 3 show that isn't automatic. So I give it its own name, *structural dissipation pricing*. The calibration says no instruction dissipates more units than the counter charges it. One word about that word: "dissipation" here means heat handed to the surroundings, the quantity Section 3 read off Sagawa as

owed even when no entropy is produced, the heat that pays for where the forgotten bit goes. It is not what Section 1 calls dissipating, which is entropy produced, and it has nothing to do with waste, or with doing the job badly. There are two premises here, nothing is proved about either, and the only question is whether they can hold together:

**Premise 19.2** (Structural dissipation pricing)**.** A *dissipation* is a function $\delta$ from instructions to naturals, counted in units of $k_B T \ln 2$. The *floor*, a hypothesis separate from Premise 3.5 of Section 3: every instruction that collapses a yes-or-no property has $\delta(i) \geq 1$, where *i collapses P* when some state with $P$ false is sent to a state with $P$ true, every state with $P$ true is sent to one with $P$ true, and some two different states are sent to the same state. The *calibration*: $\delta(i) \leq \text{cost}(i)$ for every instruction. Neither is proved anywhere; the theorems below ask whether they can hold together.

The last clause of the floor is there because the first two don't make a step forget anything on a machine with infinitely many states. INC $A$ sends a state with $A = 0$ to one with $A = 1$ and keeps every positive $A$ positive, so it raises "$A \geq 1$" one way, and it is one-to-one. Any one-to-one map that misses some state does the same to the property "is in my image". Its input can be read back off its output, so it loses no entropy, and Landauer's premise as this book takes it then puts no heat floor under it at all. It pays in fresh states instead: INC moves every value of $A$ up one, Hilbert's front desk, and only a counter with no top can keep doing it. A thermal implementation can dissipate under the model-specific conditions above; the premise only sets no floor under it. A floor read without the last clause would charge a whole unit. On a finite machine the clause asks nothing extra. A step that sends one false state into the true ones and keeps every true state true puts one more state into the true set than the set holds, so two of them land on the same state. And on any machine the property comes along for the ride: an instruction that sends two different states $a$ and $b$ to one collapses the property "is reached from $a$ in one or more steps", since one of $a$ and $b$ lies outside it and lands in it. So the floor is a price on merging, one unit for every instruction that merges, read as heat.

## 19.2 The premises have to be able to hold together

The one piece of this section that I think is good physics discipline is a consistency test. Suppose you want to claim the counter tracks dissipation. You need two premises: that every step you've chosen to count dissipates at least one unit, and an upper calibration that bounds dissipation by the charge. Can those two hold at once? There's a theorem saying exactly when: a model satisfying both exists precisely when every operation you've chosen to count (the statement below calls those *eligible*) costs at least one. I should say how little that shows. The model the

proof builds takes the dissipation to be the cost itself, so the theorem shows the two premises don't contradict each other. It doesn't show that the counter tracks any dissipation. Still, it does bite on a schedule: if you count a zero-cost step, no such model can exist. One example where the premises hold is a one-bit restore, where restoring the bit to one costs one and idling costs nothing; the small machine's flag-raising steps, below, are another.

Now point the test at the small machine of Section 15, counting every step that collapses some yes-or-no property, and it fails. Read without the last clause of the floor, it would fail at a step that forgets nothing: INC $A$ raises "$A \geq 1$" one way, is one-to-one, and costs zero. With the clause in place the failure sits at a merge. Its unpriced instructions merge states (the last item of the list at the end of that chapter has the pair: two cores that one DEC sends to the same core, at zero cost), so DEC A 2, which collapses the property "$A$ is zero or the line number is 2" by sending a state where it's false and a state where it's true to the same state, costs zero there, and (1) below says no dissipation can charge it one and stay under its cost. The machine prices the certifying chain and leaves its other merges unpriced, so the premise pair holds for the steps the toll is about and fails for the whole instruction set. That follows from computing at no cost on a base that keeps no history, and the base is the part that's by design. The hardware pays for those unpriced merges in heat, off the ledger. The ledger counts the record only. A unit of cost is not a joule.

□ **Theorem 19.2** (When the premises can hold together)**.** (1) For any set of operations, any predicate "eligible" on them, and any cost, a dissipation with $\delta(i) \geq 1$ on every eligible $i$ and $\delta(i) \leq \text{cost}(i)$ on every $i$ exists if and only if every eligible operation costs at least one. (2) On a one-bit machine with a reset of cost 1 and an idle of cost 0, taking "eligible" to mean "sends two different states to one", such a dissipation exists.

*Proof.* (1) If such a $\delta$ exists, $1 \leq \delta(i) \leq \text{cost}(i)$. Conversely, take $\delta = \text{cost}$. (2) The ■ reset is the only operation that sends two states to one, and it costs 1; apply (1).

Three statements are in play, and each points its own way. Energy at a scale, $\mu \cdot c$, is a conversion and points nowhere. The calibration, $\delta(i) \leq \text{cost}(i)$, puts the charge above the dissipation: an upper bound. Landauer's principle puts the heat above an entropy loss: a lower bound. The theorem above says the assumptions can hold together, which is compatibility, and it isn't evidence. It doesn't make the ledger the least energy anything needs, and it doesn't make it the heat anything gives off.

Now point the first way out of Section 3, keep everything and pay by taking fresh states, at a machine. Keep the previous state in a history list (push it onto the front each step; below, $s :: h$ is the list $h$ with $s$ put on the front) and every

fixed-instruction step becomes injective, while the step you can see stays the same; the line number can collapse to one place while the bigger state keeps its past. The unpriced merge, lifted to a machine with history, isn't a merge. It still pays, though: the list gets one state longer every step, so the run never comes back to a state it has been in, and that's the fresh-state column of what keeping takes. On any memory you can build the list fills it, and then the merge is back. So "this step merged, so it dissipated" can't be said of the visible step alone. It's a sentence about the whole state including whatever history is retained, and a physical model has to say what's retained. The permanent-certificate theorem in Section 3 supplies that for the certifying step on finite memory. For the rest of a machine's steps it's unsaid.

**Theorem 19.3** (Keeping history makes every step injective)**.** Let the history step send a pair $(s, h)$, a state and a list of past states, to $(\text{apply}(s, i), s :: h)$. For any step function, the history step with a fixed instruction is injective, and its first component is the original step. Each step makes the list one longer, so a run of history steps never visits the same pair twice. □

*Proof.* Undoing the history step, by popping the list, returns the old pair, so two pairs with the same image are equal. The list after $n$ steps has $n$ more entries than at the start, so two pairs at different times of one run differ in the length of their lists. ■

## 19.3 The calorimeter protocol, as a specification

What follows is how one *would* measure this, and it's here because it makes the boundary exact. Take a two-state register with energies zero and a gap, a population p in the excited state, and the simplest possible rate equation for how p changes in one time step (a rate for climbing into the excited state, a rate for falling out of it, and the length of the step). For the canonical reset (population one half to zero, with the rates chosen so the equation closes), the heat delivered to the bath is exactly half the gap. Choose the gap to be twice kT ln 2 and that heat is exactly kT ln 2.

Read that twice, because the temptation is to read it as a derivation of Landauer's number and it's the opposite. The gap was *chosen* to make the heat come out as Landauer's number. The same populations and rates with half that gap give half the Landauer value, and there's a theorem exhibiting identical population dynamics with different heat. The rate equation fixes the *shape*; the energy scale is a selected Hamiltonian (the physicist's name for the rule that says what energy each state has), and selecting it is not deriving it.

The canonical reset is a bookkeeping toy, and its rates show it. A register that trades heat with a bath at temperature $T$ has rates in *detailed balance*: the rate up over the rate down is $e^{-\Delta/k_BT}$, which at the gap $2k_BT\ln 2$ is $1/4$. The canonical rates are 0 and 1, and the factor $e^{-\Delta/k_BT}$ is never 0, so no gap and no temperature put them in that ratio. With rates that are in it, a relaxation at a fixed gap moves toward the equilibrium population $1/(1+e^{\Delta/k_BT})$ and, with steps short enough not to overshoot, never goes below it. At the gap $2k_BT\ln 2$ that population is $1/5$. The toy has no work term either; nothing in it pays for the squeeze. Part (4) of the theorem below says what follows from that: at the gap $k_BT\ln 2$ the toy books less heat than Landauer's bound asks for the same squeeze from one half to zero, so the toy has no second law.

**Definition 19.3** (The two-state protocol)**.** For a gap $\Delta$, the register's mean energy at excited population $p$ is $\Delta\cdot p$. One step of the rate equation is

$$p' = p + \delta t\,\big(k_{01}(1-p) - k_{10}\,p\big),$$

and the heat to the bath over a step from $p$ to $p'$ is defined as $\Delta p - \Delta p'$. The canonical reset takes $p=\frac{1}{2}$, $\delta t = 1$, $k_{01}=0$, $k_{10}=1$, ending at $p'=0$, and counts one logical event.

□ **Theorem 19.4** (The heat is fixed by the chosen gap)**.** (1) The canonical numbers satisfy the rate equation. (2) For every $\Delta$, the canonical reset's heat is $\Delta/2$. (3) For all $k_B$, $T$, the gap $\Delta = 2k_BT\ln 2$ gives heat $k_BT\ln 2$. (4) If $k_B>0$ and $T>0$, the heat is below $k_BT\ln 2$ exactly when the gap is below $2k_BT\ln 2$; at the gap $k_BT\ln 2$ it is half of $k_BT\ln 2$. (5) There are two different gaps for which the same canonical populations satisfy the rate equation and the heats differ.

*Proof.* (1) Half the population starts excited. The upward rate is zero, so none enters the excited state; the downward rate is one for a step of length one, so the entire excited half leaves. Substitution gives $\frac{1}{2} + 1\cdot(0-\frac{1}{2}) = 0$.

(2) The register began with mean energy $\Delta/2$ and ended with zero. The defined heat is their difference, $\Delta/2$.

(3) Choose $\Delta = 2k_BT\ln 2$. Half of that gap is $k_BT\ln 2$.

(4) Asking whether $\Delta/2 < k_BT\ln 2$ is the same as multiplying both sides by two and asking whether $\Delta < 2k_BT\ln 2$. At the particular gap $k_BT\ln 2$, part (2) gives half that value. With positive $k_B$, $T$ it is strictly smaller because $\ln 2$ is positive.

(5) Take gaps one and two. The population equation in part (1) contains neither gap, so the same populations solve it in both cases. Their heats are one half and one respectively. The rate equation alone did not select between them. ■

A reset that pays its way is driven. Raise the gap a little, let the register settle with rates in detailed balance, raise it again, and so on, starting from a gap of zero, where the two states are alike and the population is one half. Raising the gap while the population holds is work done on the register; settling at the new gap is heat to the bath. That's the textbook way to wipe a bit slowly, and here it is with its numbers worked out.

**Definition 19.4** (The driven reset)**.** Write $kT$ for $k_BT$. At gap $\Delta$ the equilibrium excited population is $\pi(\Delta) = 1/(1+e^{\Delta/kT})$ and the register's free energy is $F(\Delta) = -kT\ln(1+e^{-\Delta/kT})$. Rates satisfy *detailed balance* at gap $\Delta$ when $k_{01} = k_{10}\,e^{-\Delta/kT}$. A *driven reset* along a schedule of gaps $g_0, g_1, \ldots, g_n$ takes, at step $m$, the gap from $g_m$ to $g_{m+1}$ with the population held at $\pi(g_m)$, which is work $\pi(g_m)(g_{m+1}-g_m)$, and then lets the register settle to $\pi(g_{m+1})$, which is heat $g_{m+1}\big(\pi(g_m)-\pi(g_{m+1})\big)$ to the bath. The reset's work and heat are the sums over its steps. The uniform schedule from 0 to $D$ in $N$ steps has $g_m = Dm/N$.

**Theorem 19.5** (A driven reset pays $kT\ln 2$ in work)**.** (1) With rates in detailed balance at gap $\Delta$, one step of the rate equation takes $p$ to $\pi(\Delta) + \big(1-\delta t\,(k_{01}+k_{10})\big)\big(p-\pi(\Delta)\big)$; when $\delta t\,(k_{01}+k_{10}) = 1$ this discrete update lands on $\pi(\Delta)$ from any $p$. (2) For any schedule, work minus heat is $g_n\pi(g_n) - g_0\pi(g_0)$, the change in the register's mean energy. (3) If $kT > 0$ and the schedule never falls, the work is at least $F(g_n) - F(g_0)$; if also every step raises the gap by at most $h$, the work is at most $F(g_n) - F(g_0) + h\big(\pi(g_0) - \pi(g_n)\big)$. (4) If $kT > 0$, $D \geq 0$ and $N \geq 1$, the uniform reset from 0 to $D$ in $N$ steps does work between $kT\ln 2 - kTe^{-D/kT}$ and $kT\ln 2 + D/(2N)$, and leaves an excited population of at most $e^{-D/kT}$. □

*Proof.* (1) Detailed balance gives $k_{01} = (k_{01}+k_{10})\pi(\Delta)$. In the rate equation, the upward contribution minus the downward one is $k_{01} - (k_{01}+k_{10})p$. Substitute the balance equation and this becomes $-(k_{01}+k_{10})(p-\pi(\Delta))$. The update therefore subtracts the fraction $\delta t(k_{01}+k_{10})$ of the distance from equilibrium. What remains is exactly

$$p' = \pi(\Delta) + \big(1-\delta t(k_{01}+k_{10})\big)\big(p-\pi(\Delta)\big).$$

When that fraction is one, none of the distance remains: $p' = \pi(\Delta)$.

(2) Follow one raise from gap $a$ to gap $b$. Its work is $\pi(a)(b-a)$ and its heat is $b(\pi(a)-\pi(b))$. Subtract the heat from the work. The two occurrences of $b\pi(a)$ cancel, leaving $b\pi(b) - a\pi(a)$, the final mean energy minus the initial mean energy. At the next raise, that final energy becomes the next initial energy. Adding over the schedule cancels every intermediate energy and leaves only $g_n\pi(g_n) - g_0\pi(g_0)$.

(3) Differentiating the displayed free energy gives slope $F'(\Delta) = \pi(\Delta)$. With $kT > 0$, this slope decreases as the gap rises. Over $a \le b$, the increase of $F$ therefore lies between the widths times the two endpoint slopes:

$$\pi(b)(b-a) \le F(b) - F(a) \le \pi(a)(b-a).$$

The work uses the right-hand quantity, the slope before the raise. It is at least the free-energy increase. Its excess is at most $(b-a)(\pi(a)-\pi(b))$. If the raise is at most $h$, that excess is at most $h(\pi(a)-\pi(b))$.

Add those bounds over the schedule. Intermediate free energies cancel, and intermediate populations cancel in the excess bound. I obtain work between $F(g_n) - F(g_0)$ and that amount plus $h(\pi(g_0) - \pi(g_n))$.

(4) For the uniform schedule, $h = D/N$, $\pi(0) = 1/2$, and

$$F(D) - F(0) = kT\ln 2 - kT\ln(1 + e^{-D/kT}).$$

For $x \ge 0$, $0 \le \ln(1+x) \le x$: it starts at zero and its slope $1/(1+x)$ is between zero and one. The free-energy difference is thus at least $kT\ln 2 - kTe^{-D/kT}$ and at most $kT\ln 2$. The excess from (3) is at most $(D/N)(1/2)$, giving the work bounds. Finally the denominator of $\pi(D) = 1/(1+e^{D/kT})$ is at least $e^{D/kT}$, so the remaining excited share is at most $e^{-D/kT}$. ∎

Try one finite step in units where $kT = 1$: start with gap zero and half the registers excited, then raise the gap to $\ln 2$. The work is $\frac{1}{2}\ln 2$. After settling, the excited share is $1/3$, so the bath receives $(\frac{1}{2} - \frac{1}{3})\ln 2 = \frac{1}{6}\ln 2$ and the registers retain $\frac{1}{3}\ln 2$ more energy. Work equals those two amounts added together. A third are still excited: one such step has not wiped the bit.

The word “step” matters here. Part (1) is a discrete update of the population. Its multiplier can be zero. Continuous relaxation at fixed positive rates instead multiplies the distance from equilibrium by $e^{-(k_{01}+k_{10})t}$, which is positive at every finite time. Exact settling in the schedule is an idealisation; the algebra does not prove that a continuous bath completes it in one finite waiting period.

Here nobody picked the gap to make the number come out. Take $D$ large and $N$ much larger than $D$: the work closes in on $kT\ln 2$ from both sides, and the population left in the excited state goes to zero. Perfect erasure is the limit, not the result at any finite gap. To reuse the original symmetric memory, the gap must also return to zero while transitions are suppressed. Lowering it at the final fixed population recovers $D\pi(D)$ of work and leaves that population unchanged. This term tends to zero in the same limit. The reset of the population, restoration of the trap and time needed for settling are separate parts of the physical protocol. Detailed balance supplies the thermal premise; the displayed sums supply the work and heat accounting for the idealised schedule.

What ties it to a machine is that the protocol counts one logical event, by definition, and one is what the small machine's CERTIFY charges, the last of the three paid acts behind a raised flag (Section 15). The tie is two numbers that happen to match, the protocol's one by definition and the machine's the least the record-layer premise lets CERTIFY cost, and nothing more. No theorem maps the protocol's two register states onto any state of the machine. And "heat to the bath" here means the register's mean energy before minus after, which quietly assumes the bath gets exactly what the register loses; nothing here establishes that. So a device test would have to measure the gap, the populations, the rates, and the bath heat independently; separately establish that the device's rates are in detailed balance (that the up and down rates stand in the ratio the temperature dictates), which the driven reset assumes and the canonical one breaks; and separately establish that the device's physical reset *is* the machine's certify event. The driven reset is an executable experimental specification. The canonical reset is a bookkeeping toy. Neither is a measured device, and neither is an unconditional conversion of units of cost to joules, and anyone who tells you otherwise has skipped the conditions.

## 19.4 Permanence and the heat premise

Is there a heat prediction the permanent-certificate theorems make that Landauer's principle alone doesn't? No, and the reason is short. The smallest case, one certified state and one flipping in, is "set a bit to one whatever it held," which is the reset Landauer described and Bérut measured; Landauer's own name for it is "restore to one," and Section 2 put it beside the toll. The general bound, the larger of kT ln((m plus k) over m) and 2k over (m plus k) times kT ln 2, for squeezing m plus k equally likely states into m, is what Landauer's principle gives for that squeeze, at kT ln 2 for each bit removed. What the permanence theorems add is the structural half: finiteness and a permanent reading force any flag-raising instruction to merge some full states. They don't say the newly certified state is one of the states that collide, and they don't supply the spread over inputs or the physical conditions a heat bound needs. Supply those and Landauer's premise prices the uncertainty that was actually removed. The machine result and the physical premise do different jobs, and neither does the other's.

## 19.5 What this section is

This section holds a calibration that's a unit conversion; a pair of premises, consistent on the steps the toll is about and inconsistent on the whole instruction set; a history that makes any step injective by paying with a fresh state at every step; and a measurement protocol that's a specification with a selected scale. The physics in this

book is in Part I: Landauer's principle as a named premise with its critics named (Section 3), and conditional finite-time dissipation bounds, cited, and proved for one model (Section 1). Here there's accounting, and I've tried to call every line by its right name. That's the end of the example machine. What's left, after one suspicion I can't prove, is how you'd break any of this, and where it stops.

### 19.6 At the bottom: what I suspect and can't prove

This is underneath the proofs, so read it with the lights on. None of it is a theorem, and I'm not going to dress it up as one. It's why I care, and I'm putting it here, after the main text's proofs and apart from them, so you can check that none of it got into them.

Physics has a word for the moment a question closes: *measurement*. And the people who thought hardest about it keep describing it in this book's shape. They ran together for me until I put them on one picture. Flip a coin in a crowded room. Before it lands, it could go either way. After, it's settled, somebody chalks it on a board, and the result gets copied out into the world, the light and the air, so many times that it can't be taken back and everybody who looks agrees. There are five pieces: the question, the landing, the chalk mark, the copies, the agreeing. Each person leans on a different one. Zurek, in Section 18, leans on the copies. Rovelli's coin has to hit the ground, so his piece is the landing.

Niels Bohr, the Danish physicist whose institute in Copenhagen was where quantum mechanics got argued into shape, leans on the mark. In a 1954 talk he said seeing anything at that scale depends on an apparatus that works irreversibly, like permanent marks on a photographic plate: a door that swings one way. I found that line quoted in a recent paper and haven't seen the page in Bohr's book myself. John Wheeler, the Princeton physicist who made the name "black hole" stick, went further in a 1989 talk in Tokyo and leaned on the question: every physical thing gets its existence from yes-or-no questions answered and registered, boiled down to "it from bit." As I read him, the coin has no face until something asks and something writes the answer down. How literally he meant that, I can't tell. And Peirce, way before any of them, leans on the agreeing: the kind of settling worth trusting is the kind that has to answer to everybody. Laid out like that, they pair up. Bohr and Rovelli lean on a step that only goes one way, in the plate and in the coin. Zurek and Peirce lean on many onlookers who agree, one with light and one with people.

Lay that next to everything in this book and the parts line up: a check that could have come out either way, a result that's now settled, a price paid in forgetting or in fresh states, a record that stays. I suspect that isn't a coincidence. I suspect the event this book prices on a machine is the same kind of event physics calls a measurement, showing up in a different place. The forgetting is my step. Bohr only

says irreversible, and getting from a record that stays to a merge or to fresh states is Part I's counting of states. Push on it and it gives at Zurek, for the reason Section 18 gave: copying onto fresh light merges nothing and uses up fresh states, which is Bennett's measurement that owes no floor. So physics' own word lands on both sides of the line between Bennett and Rovelli: the chalk on the paying side, the copies on the borrowing side. The landing I'd put with the chalk, and when Section 4 pushes on Rovelli's paper it slides toward the borrowing side, the side of finding out. So two pieces already stick out, and I'll leave them sticking out. Which picture a settling on a machine matches is part of the suspicion, and the suspicion doesn't settle it. That's as far as I'll take it here: a suspicion.

John Bell asks for precision in those words. He was the Northern Irish physicist at CERN whose 1964 theorem is behind the CHSH check of Appendix G. In 1990 Bell wrote a short, furious essay against the word "measurement," with a list of words he said have no place in a physics that wants to be precise. On the list: system, apparatus, environment, irreversible, information, and measurement, which he called the worst of all. I used half of them in the last few paragraphs. To turn that suspicion into a physical claim, each of those words would need a specified meaning and a connection to the device. The proofs here work with the models and machines I have defined; the physical applications state the further premises they use.

PART IV

# Results together

*Each opening leaves something out.*

Part III followed one example machine from its first step to the host running a guest. Bring those pieces back to the bench. The holding count says what the arrows must do. The price rule says what goes on the ledger. The window says which of those distinctions a reader receives. The earned chain says how the flag got there.

CHAPTER 20

# The claims and their conditions

Start with a box holding an answer. I need to know which full states count as holding it, which moves keep it there, and whether the box has finitely many such states. Those choices make the counting argument possible. Add a price on merging and the same count gives a charge. Give a reader only the tabletop picture and there is another question: how much of that event reaches the reader?

Here I put the connections in one place. The statements and their proofs stay in the chapters where I built them; this is a way to follow what each piece contributes to the next.

## 20.1 Keeping a distinction

*One next state.* Follow one deterministic rule from a start. Until the first repeated state, every visit is new. At the first repeat, either you have returned to the start, or two different predecessors have reached the same state. From then on the rule repeats the same successors. Those are the three columns of the step law (Section 1). A choice between several successors would change the argument at precisely the point where you follow the next arrow.

*A finite place to hold the answer.* Suppose a move keeps every YES state at YES and brings a NO state in. With $K$ YES states, those states and the entrant give $K + 1$ sources landing in $K$ places. Two arrows meet. The entering arrow need not be one of them: a chain can run through the held states before meeting a fixed endpoint. Section 3 gives the general count and its attaining constructions.

Each part of that picture matters. An endless strip can keep moving to a different YES state forever. A two-state toggle keeps the arrows one-to-one by turning YES back to NO. A finite permanent holder has neither route available. For a single rule, if its entered holding run were to repeat while remaining one-to-one, cancelling the equal successors backwards would identify the unfinished entrant with a held state. That is why the one-to-one alternative needs a fresh configuration at every later tick.

Section 1 also follows finite noisy and unitary systems using their own preservation conditions. For a table of chances, both row and column sums enter the count. For a unitary map, preservation of inner products supplies it. For the zoom argument,

keep track of the full states behind the coarse picture. The finite collection being counted must remain closed under the step; otherwise the next tick can move into a state the count left out.

### 20.2 Putting a price beside the arrows

A graph tells you which states meet. A price schedule tells you what a move costs. The finite holding count and a rule charging merging moves together force a charge on a permanent raise. Once the one-step toll is part of the machine, every run from NO to YES includes a charged crossing; adding nonnegative costs gives the universal floor (Section 2).

There are two useful schedules to keep separate. Charge by instruction name and repeating that instruction has the same price even after the flag is up. Charge by the source state and the schedule can ask whether this particular step raises the flag. The floor together with a ceiling of one unit per raise selects the latter charge, one for a raise and zero otherwise. When the same instruction performs the raise and the later idle, this state-sensitive policy lets the raise pay once and the idle stay free. The first is the cost signature in the basic definition; the second is the broader interface used in the source-charge and exact-pricing comparisons.

The three-card example makes the choice visible. Call the states blank, once and twice; once and twice read YES. A move sending once to twice and fixing blank and twice merges two states without raising the reading. A merge price charges it. A raise price leaves it free. The chosen reading and the chosen schedule specify which question the ledger answers.

With a halving allowance on each move, the list of possible starts can shrink only as far as the accumulated allowance permits. The entropy form additionally counts a supplied probability distribution. A physical heat reading uses Landauer's premise for the device and process being described. These are the successive inputs to the arguments in Sections 3 and 4.

The four-state demon shows the distinction between learning and wiping. MEASURE copies onto a blank page reversibly; applied again it can return that page. WIPE sends different page contents to the same blank state. The first operation uses the prepared page; the second merges the alternatives. The price rule can therefore charge them differently.

### 20.3 What reaches the reader

Put a raising step and a quiet step behind a window that shows the same before-and-after pictures for both. Any price computed only from those pictures gives the

same answer for both. It cannot both charge the raise and leave the quiet step free. Let the window show the raise and that ambiguity disappears (Section 7).

The same argument applies to a question about an earlier value. If a good and a bad value leave the same complete retained observation, a reader of that observation has one input for two required answers. Inspect the value while it is live and retain an adequate answer, or retain the distinctions the later questions will need. Which questions are allowed determines how much must survive.

A permanent base-driven record has the latch form: it switches on at the first occurrence of an event and stays on (Section 10). A toggle changes that task by allowing the record to fall; a hidden clock changes what drives it. For an ordered growing record, threshold readings give the row of cups in Section 11. A chain with $k$ cups can have at most $k + 1$ different arrangements if cups only turn up.

Entitlement adds a particular claim and the evidence permitting a later step to use it. Its index count uses a covering tree whose depth the paid steps cover. The uncovered narrowing example in Section 8 shows why that connection belongs in the statement: narrowing a candidate list alone supplies no such bound on the ledger.

## 20.4 The concrete machine and its guests

From a clean start, only CERTIFY raises the flag, only a successful COMMIT supplies its channel, and only a successful CHECK supplies the matching fact. The counter's edition must still match at commitment. Following those entry points backwards gives the earned chain and its minimum charge of three (Section 15). A claim such as "this counter is at least zero" passes everywhere and still uses the same three acts. The flag records their occurrence; the claim says what they established.

The ledger starts at zero and each step adds its scheduled price. Induction on the run then gives both its total and its uniqueness: two ledgers following that same start and schedule agree before the first step and still agree after each addition (Section 6).

The universal program follows a guest through matching points, preserving its halting, flag and ledger guarantees (Appendix C.1). The presented-machine interpreter extends this to guests with computable transitions, prices and readings, with the surcharge stated in Appendix C.2. Each interpreter earns its own flag through the host's instructions.

*The computational slice.* The lift in Appendix D gives a universal base a check language, room for a fact, a channel and the earned operations. While the lift is live, base moves follow the base and the panel leaves its working value alone. A failed check can trap the lift. Later free moves then freeze, so keeping that failure globally

invisible would also hide the change in subsequent computation. The live-state clause is what connects this lift to its stated slice.

The Thiele-complete interfaces specify free base moves and a record that separates runs with the same working observation. The clock and silent-machine examples show what each requirement contributes; the four independence witnesses separate the clauses. Interface and instruction choice also explain why the one-rule presentation and its macro version have different coverage.

Independent components compose by interleaving their moves. Each move changes its own component, so the pair's cost is the sum of the two component costs. Keep each component's own move order fixed and every interleaving has the same final state. Nesting instead follows a guest through a host map; the record-order condition carries the guest's crossings into the host's accounting (Appendix E).

## 20.5 The diagonal and the worked models

The diagonal argument supplies the contradiction inside a substrate that can represent the candidate's flip and has the stated recursion property. Its fixed point makes the proposed decision disagree with itself. An outside calculation of the natural-number example does not supply an inside program for that substrate. Section 13 and the appendices state the input conventions separately for the natural-number family, L, the counter programs and the host.

The protocol examples in Section 17 isolate particular mechanisms: finality, a charge on commitment, an attestation window, an inclusion proof and a bounds check. Their statements follow the operations in those fragments. The pointer thesis then asks why certification is the event to choose. It proposes a connection to records copied and relied on by independent parties. The counting and window arguments apply to any record meeting their premises; the thesis supplies the proposed reason for this choice.

The game checks in Appendix G give another worked construction. Four fixed bits yield the classical bound by the displayed sign calculation. The matrix arguments relate contraction, positive semidefiniteness and the score bound. The integer checker implements its specified test on tallies with every pair sampled. Its earned chain connects a raised flag to that test. The pinned-matrix test and the completion test ask different questions: the appendix's all-ones and PR-box examples make the difference visible.

Those connections are what the book has built: a count, a schedule, an observation and an earned record, with instructions carrying them through actual runs.

## 20.6 Sources and contributions

Appendix B sets out the sources and contributions item by item. Here is the short form: the earlier results I use, the statements proved here, and the constructions that put them to work.

*Borrowed, with names.* As The point warned, I usually had the hunch first and found these names after, and that order doesn't make any of it mine. Everything in Section 5 and the Afterword (every source is cited where it's used), the category theory (Eilenberg, Mac Lane) included. Minsky's two-counter machines and the constructions that prove their halting undecidable and compile computable functions into them (Forster, Larchey-Wendling and coauthors). Turing, Church, Kleene, Rice, Cantor, Gödel, the diagonal pattern, Lawvere's unification, and later detailed developments: Michael Norrish's, Mario Carneiro's, and Yannick Forster, Fabian Kunze and Gert Smolka's. Bell, CHSH, Tsirelson, Stephanie Wehner's semidefinite proof of Tsirelson's bound (2006), NPA, Landau, Masanes, Fine, the elliptope, and Echenim and Mhalla's detailed treatment of Tsirelson's bound. Landauer, Bennett, Szilard, Maxwell, and their critics. The potential method (Tarjan) and later treatments (Nipkow, Charguéraud and Pottier), Lamport's clocks, Floyd and Hoare. Gettier, Necula, Cook, Hartley, Shannon, Rissanen, Zurek. The guarantee about every program: the reference monitor (Anderson), execution monitors (Schneider), and monitors checked at every instruction and proved correct (Azevedo de Amorim and coauthors). The counting under the permanence results: Poincaré's recurrence, as Zermelo used it. The seL4 and CompCert discipline. The facts under the permanence results are Landauer's own example and Shannon's bounds, and "set a bit to one whatever it held" is my way of saying his RESTORE TO ONE.

*What I proved, each inside its stated interfaces,* is listed in the claims section (Section 21) with what each stands on.

*Interpretation, which is mine and not proved.* That structure is what a computation establishes and the toll is the right thing to price. That certification is the pointer event. That the record axis "belongs in the step." Those are proposals, with the arguments I've given, and the proposals are where I'd aim first if I were you.

CHAPTER 21

# The claim ledger

The full ledger, in the companion volume, lists everything this book claims, one line each, with what kind of claim it is and where it lives; this section places the headline results. "Proved" means proved under the premises printed in that section, and nothing more. If a line here ever says more than its section does, the section is right and the line is wrong. "Directly from the rules" means the claim can be read off the stated operations by checking their cases. "Directly from the interface" means its requirements already include the claimed behavior. These claims are true, but carry almost nothing beyond what those definitions already say. If you're counting results, count those differently. I mark them so they don't pad the list, and where they're stated in the book they carry the label *observation*, with three exceptions: Theorem 15.6, part (3) of Theorem C.8, and the parts of Theorem D.4 its line names.

Ledger is a good word for it, and it's older than bookkeeping. Around 1400 a *ledger* was a book that lay permanently in one place, a big church breviary, say, from the old verb *leggen*, to lay; the account-book sense turns up in the 1500s. Staying put is the job here too. The home sections are long and they wander, and this is the one place where every claim lies still, so you can come back and find it where you left it.

The claims below are placed by where they sit in the model, with the status each carries and the premise it stands on. (S) is a structural theorem about the model or a stated machine, (C) a conditional result under a named premise no machine fact discharges, (R) a refutation by a named construction; the pointer criterion is a thesis. A2 is the toll's name in the mathematical specification; `neutral_complete`, `thiele_complete` and `landauer_heat` are the Coq names of the two completeness definitions and the heat premise.

| **Result** | **Status** | **Key premise** |
|---|---|---|
| *The definition and the price in the move* | | |
| The floor is the toll | **(S)** | A preorder of records; nothing else |
| Thresholds are one-bit machines | **(S)** | A2; antisymmetry for the thresholds to determine the record |
| Universal floor | **(S)** | A2 on the two-point order |

| | | |
|---|---|---|
| All readings and free moves | **(S)** | Decidable state equality; one schedule for every Boolean reading |
| Least instruction price | **(S)** | Fixed dynamics and reading; finite covering list |
| Least unit lift price | **(S)** | An event witness for every mark; all-state toll |
| A2 from merge pricing | **(S)** | Finite states, permanent certificate, merge price |
| Shadow pricing | **(S)** | A colliding pair in the window |
| *What stays* | | |
| Free reading preservation | **(S)** | A2 and global permanence |
| Record axis is a latch | **(S)** | Base-driven record |
| Earned provenance, checker soundness | **(S)** | Clean start |
| *What a window forgets* | | |
| Canonical future equivalence | **(S)** | Fixed deterministic operations and observations |
| Terminal lift and reachable minimality | **(S)** | Fixed base/events, exact latch, toll, permanence, unit ledger; separation for injectivity |
| History map in the reverse direction fails | **(R)** | Toggle base with an additional free-turn counter |
| Window theorem | **(S)** | Thiele completeness (gap of one) or certified Thiele completeness (gap of three) |
| *The room a kept distinction needs* | | |
| Merge or fresh states | **(S)** | Total deterministic step, permanent raise |
| Sharp retention bound | **(S)** | Finite implementation, distinct representatives, reset and injectivity through the specified horizon |
| Stochastic and unitary holding | **(S)** | Finite stochastic kernel with column constraints, or finite unitary dynamics and invariant projection |
| *Computation as the special case* | | |
| Thiele completeness: floor and window | **(S)** | The four clauses of `neutral_complete` |
| Certified Thiele completeness of the small machine | **(S)** | The four clauses of `thiele_complete` |
| Clock-style machines fail | **(R)** | The four clauses |

| | | |
|---|---|---|
| Lifting any universal base | **(S)** | A universal base, a claim language with an exact equality test and a proved checker, a fact table with room for one fact, a claim that can come out either way |
| Composition | **(S)** | Certified Thiele completeness of both parts |
| Each clause is needed | **(R)** | One clause weakened |
| U | **(S)** | Small-machine programs |
| $U_P$ | **(S)** | Priced guests, computably presented machines |
| Verified compiler | **(S)** | Well-formed source programs |
| Extraction to OCaml | **(S)** | Hand-written mapping lines and drivers trusted |
| Structural undecidability | **(S)** | Recursion theorem of the substrate |
| L recursion and Rice | **(S)** | L's reduction rules |
| Two-counter Rice and packed Kleene | **(S)** | Behavioral sameness on chosen inputs |
| Plain-input recursion fails | **(R)** | Inputs in counter $A$, 0 in counter $B$ |
| *The world* | | |
| Biased-walk heat against time | **(S)** | Specified biased-walk entropy formula; positive speed below one |
| Heat floor | **(C)** | `landauer_heat`, finite states, permanent certificate, the uniform distribution on the states in play |
| No intrinsic joule value | **(R)** | None beyond the ledger |
| Algebraic Tsirelson bound | **(S)** | Polynomial minors |
| Calorimeter heat scale | **(R)** | Fixed dynamics |
| Pointer criterion | Thesis | Observer model |

The full ledger, one line per claim with its home section and its kind, and the list of every assumption with what it supplies, are in the companion volume *The Thiele Machine: The Example, Built*. The mathematical specification carries the same table with the Coq declaration beside each result.

CHAPTER 22

# Scope

The definition asks for states, moves, readings and charges. To use a result elsewhere, put that other system beside the drawing and match the pieces. Which states carry the record? Which operation establishes it? What does the reader see, and what does the price count?

The following comparisons work through those questions for heat, computation and the choice of record. They show which further connections an application needs.

*Physical interpretation of the counter.* Choose a start and a price schedule and the ledger is determined on reachable states. To read it as a physical quantity, specify which states carry information and how the scheduled charge relates to dissipation, under physical premises that hold together. The permanent-certificate theorems supply the first for an abstract finite machine with a permanent reading. The two premises are shown able to hold together whenever every counted step costs at least one, and for a one-bit reset. The small machine's unpriced steps merge states and are charged nothing, so its ledger is not a heat meter for computing: whatever those merges cost the hardware is off the ledger, and the two premises hold together for the record steps only (Section 19); and the calorimeter protocol shows that population dynamics don't fix the energy scale. No joule value attaches to one unit of cost without thermal and device premises (Section 3, Section 19). Once you pick the start and the price list, the number is forced by arithmetic, and it isn't energy. What I'd need before it could be is the list at the top of this paragraph.

*Class separation.* The witnesses show specific windows losing specific fields. They don't stop a conventional simulator from keeping those fields in its encoding. No separation of standard complexity classes follows from the collisions (Section 7, Section 15). I'll be blunt about why I'm careful here. Whether P equals NP is one of the seven problems the Clay Mathematics Institute put a million dollars on, at a meeting in Paris in May 2000. One of the seven has been solved since, the Poincaré conjecture, by the mathematician Grigori Perelman, who posted the proof as preprints in 2002 and 2003. P versus NP hasn't. If a two-state collision on my machine settled it, I'd be owed a million dollars. Nobody owes me anything. A window that loses a field is a fact about that window. A complexity class is a fact

about what every possible program can do within a budget, and nothing in these collisions reaches that far.

*Trace initiality, and what it doesn't settle.* The universal property of the trace fold is about evaluating lists of instructions: pick where a target starts, and there's exactly one way to fold a list into it (Section 6). The counter's uniqueness assumes a schedule. Rise-only pricing assumes an event, and a reference count that says what charging too much means (Section 7). Those results make the construction precise. None of them shows that the schedule I chose, or the event I chose, is physically necessary.

*Other observables.* Certification is one worked example. A family of other observations could turn up other structural laws, but each would need its own events, its own evidence, and its own cost rules, justified on their own terms. The window and oracle results are about what specific maps keep, and nothing more.

*Program predicates versus state observations.* The state-level collision (Section 7's wall) and the program-level diagonal (Section 13's wall) live in different domains. Linking them would take a theorem of its own. The nat family and L (Section 13) keep their explicit quantifiers and representability premises.

*Entitlement bounds.* The count is handed its tree and its covering, along with the payment condition. It doesn't dig the tree out of an arbitrary run, and it isn't a Shannon bound on every trace (Section 8). On every certified Thiele-complete machine the payment is the record moves, each costing exactly one, and a floor that takes no tree exists when the questions are the run's own CHECKs; but its answers are taken at the rivals as they start, a branch that costs nothing can still narrow, the lists, tree and fibres are supplied except in the worked members, and the covering can't be dropped. The search through $n$ bits is capped at 16 questions by the machine's fact table, and the time tax is a lower bound for programs of INC and DEC only. Nothing here extends it to every certified Thiele-complete machine (Section 15.14).

*Quantum behaviour.* The fixed-matrix equivalence is about one completion with zero marginals and zero within-party moments. The elliptope lets the within-party moments move, at the level of correlators. A completion of the correlators is realised by actual measurements, by the construction in Appendix H. A full behaviour, with each room's own odds prescribed as well, asks for more, and the construction doesn't deliver those extra probabilities. A sound check isn't complete for those bigger sets, and the elliptope decider is mathematics on the page, not a part of the small machine (Appendix G).

*The record axis.* The proved structural result is: a permanent record driven by a base computation is a latch on an event; on a finite machine its write merges states; and the named windows don't recover it. Certification is one instance. No theorem selects it as the unique event every system must record (Section 10, Section 18).

A finite monotone multi-valued record decomposes into threshold latches. One Boolean latch can't hold it. Revocation, including revocation priced from the other side as in Casper, lies outside the permanent-record axis; a price on revocation doesn't transfer to a price on the write (Section 17). Probabilistic records don't inherit the uniqueness result either. Two honest random records can have the same branches and the same charges and still give those branches different weights, one choosing evenly and one two to one, so the schedule doesn't fix the probabilities. And no single deterministic latch explains a record that can branch both ways from the same observation. Weak base equivalence is defined and is an equivalence; no two of the three bases of Definition 10.3, and none of them and the small machine, are related by it in this book, and the latch form holds on each separately, so it doesn't tell them apart (Section 10). Records that can come back down lie outside all of this: they're limit computability's, and Section 10 points at Ershov's ladder.

*Who it's safe against.* Every guarantee about the record is about programs: any list of instructions, run from a clean start on a step that does what its definition says, one writer stepping at a time. That covers the buggy program and the hostile one. It doesn't cover whoever sets the start (a snapshot put back, a table loaded with facts nobody checked), a reset or a power cycle, the wire that carries the report, someone who flips the flag with a probe, the tools that built the step, or a second writer that changes a counter without bumping its edition. Those are the operator, the device and the tool chain, and they're outside this fence. A record read by someone who didn't watch it being made is only as good as their reason to believe the box that shows it. The ways across that I know of, a quote over measured state and a proof that travels on its own, are in Section 7 and Section 17, and this book builds neither. What the guarantee gives inside the fence is safety: no program forges the chain. Availability is a different promise, and it isn't given. A hostile program can put the trap up or fill the table, and after that nothing useful gets done. The mathematical machine uses whole numbers, which never overflow, so versions and the ledger grow without bound. A build with fixed-width counters, versions or ledger has to stop before any of them wraps, failing safe with the trap up, because a wrapped version could make an old fact look live again and a wrapped ledger would fall. Starting, restoring, resetting and that overflow stop belong to the implementation, and none of them is proved here.

*The diagonal on the machine.* What's proved for the small machine's programs is that halting is undecidable in the reduction sense of Theorem 15.5, for them and for the universal program (Section 15; for the universal program, Appendix C.1). L has its recursion theorem, Rice and halting by proof (Section 13). On the host machine, the one with a counter for every number, Rice's theorem holds for properties that depend on everything a program ends with (every register, the table, the channel, the trap, the ledger and the flag; the line number and the versions are left out), for

programs run on an input in register 1; s-m-n holds with no premise on register 2; and Kleene's recursion theorem holds for the transformations a host program carries out on numbers, with a fixed point that agrees with its image on stopping, register 0, the trap latch, the ledger, the flag, the number of facts and whether the channel is empty; for some of these transformations no fixed point agrees on every register. On the two-counter machine itself Rice's theorem is proved on every input, with facts compared by what they're about, their versions left out of the comparison; Kleene's is proved with inputs written as powers of two and refuted with plain inputs (Appendix C.4).

*The five models.* Each is a fragment transcribed from a specification, with a named theorem and a named limit: the Casper transcription has no deposits or fork choice; the gas fragment omits the refund cap; the attestation fragment has no signatures; the transparency verifier covers control flow and leaves collision resistance out; the proof-carrying checker is a three-instruction memory-bounds language. None is a claim about the real system beyond its fragment (Section 17).

*The machine.* Part III has one small machine, with the hosts that run it in an appendix, and this paragraph is its fence. The small machine of Section 15 (two counters, six instruction kinds) is where commitments are earned: from a clean start, a raised flag went through a passing check, a commitment and a certification of the same claim, nothing is forged, and the cheapest such run adds exactly three to the ledger, a price reached by a check that could have failed (the step doesn't police trivial checks: a check that can't fail certifies for three too). It's also where universality is proved, by matching Minsky's two-counter machines step for step and using the halting reduction of Forster, Larchey-Wendling and coauthors. It's certified Thiele-complete under a definition every clock-style machine fails, and every machine meeting that definition hides its record and ledger from its own window. The universal Thiele machine for the small machine's programs (Definition C.1) is a host with the small machine's instruction kinds and more counters, running one fixed program, $U$, with no guest step built in: it runs every program of the small machine from every clean start start$(x, y)$, halts when the guest halts, raises its own flag by its own check, commit and certificate if and only if the guest's flag rises somewhere in the guest's run, and keeps a ledger equal to the guest's at every matching point. With one paid move added, one fixed program, $U_P$, runs every computably presented Thiele machine: its step, prices and reading computed by programs, the reading decided on every state, the run fixed by a driver. The host halts when the guest stops, its flag is the latch of the guest's reading, raised on its own earned chain whose claim is that reading, and its ledger is the guest's plus a surcharge of at most two when the reading starts at no, a surcharge no program on that host, run from a clean start, can avoid when the guest certifies with its whole ledger below three. A machine whose reading no program decides is outside. A

simpler host with a guest step built in runs guests by construction and keeps a mirror of the guest's flag; its ledger only covers the guest's. The small machine's check price is flat, its fact table holds sixteen, and its certificate is historical (Section 15 lists the rest).

*The comparison machines.* The bases I put the latch on are fragments too. The register machine of Section 10 is the one with indirect addressing in Definition 10.3, and Theorem 10.10 is about it. The reversible result (Theorem 10.3) covers any base whose step is injective, and on a finite state space says only that no injective step raises a permanent reading. None is a theorem about real memories or full reversible languages. The cost-semantics results are local correspondences, short of full cost-framework embeddings: the trace's cost is a writer cost and a grade, and the toll is the potential method with one potential (Section 6). Nobody's whole type system gets embedded either.

*Outward consequences.* I looked for results about real systems that would need the record axis. What's proved are the five fragments of Section 17, each a transcription with its own named limit, and none is a theorem about the deployed system. Stronger candidates would need protocol or hardware semantics I haven't formalised, and I've proved nothing about them. So none of this yields a new outside result that both needs the record axis and is ready to use.

*Where each result lives* is the premise column of the claims section (Section 21).

*Models and processes.* For a model of computation written down as states, a step and a reading that stays, what keeping takes is a theorem about the model, not a thesis. The thesis in the Conclusion is only the tie from a physical process to such a description. Here's how far the theorems reach, and where they stop. A step that gives every state one next state, on any number of states: a run that raises a reading and keeps it up merges two of its own states or never visits a state twice (Proposition 3.5), and with finitely many states reading yes the move that raises it merges (Theorem 3.1, Proposition 3.4). That takes in a Turing machine or a counter machine stepping by its program, and a cellular automaton stepping its whole grid at once. A model that halts has no next state at its halt, and slips past with no merge; give the halt a next state and the run is one rule again, so keeping the answer merges or takes fresh states, and with the halted state stepping to itself the step onto it is a merge (Theorem 1.2). Steps by chance on finitely many states: one whose chances into each state add to one, which is what keeping the even spread takes, never enters a finite set nothing leaves from a state outside it (Theorem 1.7). A unitary step that keeps a finite-dimensional part of the space keeps the part at right angles to it, so a state at right angles to the kept part stays at right angles (Theorem 1.8).

A model whose step may offer several next states, or none, comes in too: a nondeterministic machine, a lambda term reduced at whichever redex you like,

a rewriting system, a proof system read as derivations made one rule at a time. Nothing earlier in the book is stated for such steps, so here are the two arguments. The first is the pigeonhole of Proposition 3.4 on one choice of next state, and the second is the walk back of Proposition 3.5, stopped at the raise. First, suppose only finitely many states read yes, none of them can step to a state reading no, each has at least one next state, and some state reading no can step to one of them. Pick one next state for each yes-state, and the outside state's step into the yes-states: that's one more step than there are yes-states, each leaving a different state and landing on a yes-state, so two land on the same one, and two different states share a next state. That needs the yes-states finite; Hilbert's hotel, everybody moving up one, keeps a reading with no two states sharing a next state. Second, along any one run, however its choices fall, take the step where a reading goes up, and suppose it stays up for the rest of that run. If the run is in one state at two times after the raise, step both times back together. They can't stay equal past the raise, because the state at the raise reads no and every later one reads yes, so at the first pair that differs, two different states of that very run step onto one. So after the raise the run merges or never repeats a state, on any number of states. Before the raise it can come back to a state and then leave it for good with no merge, which one rule can't do. A normal form, with no next state, is the halt again: a reading kept by stopping is kept with no merge, as Theorem 1.2 says. For such steps the first argument says little by itself: in the lambda calculus every term $N$ is one step from two different terms, $(\lambda x.x)N$ and $(\lambda z.N)M$ with $z$ not free in $N$, so two states sharing a next state is no news there, and the second argument is the one that carries weight. And when a step can branch, a shared next state doesn't by itself bunch the crowd up, since a branch spreads it out again; with chances on the steps and finitely many states, the count that does is Theorem 1.7.

Where they stop. Steps in continuous time have no next state to count, and nothing here covers them. Steps by chance where endlessly many states read yes, and unitary steps whose kept part has endlessly many dimensions, get none of the finite results above, and the bilateral shift keeps a reading with no merge, a fresh state every step. What still reaches them is the run: a unitary step is one rule, so its runs are Proposition 3.5's, and one run of steps by chance is a run the second argument covers. And a truth isn't a process at all: that a statement is provable is a fact with no step in it, and the theorems are about a run that derives it. What they give is the merge or the fresh states. The heat stays Landauer's premise.

*What I'm proposing beyond that.* A way of describing computation in which the relations a computation carries live in the state, what each move does to them lives in the step, and a price list is one part of the description, not an afterthought. The permanent record is the one case I took apart all the way. The pointer thesis asks whether certifying is the thing that deserves that record in the systems it talks about,

and it can be wrong without touching the latch or the window arguments, which never use it.

Underneath that is a question about which distinctions a description of a computation ought to keep. The constructions show particular ways of keeping some, and what follows once you do, under premises I've printed. What that's worth depends on what you need the description to answer. A program log with a proof about its updates may recover the required distinctions. A host can instead enforce those updates through every guest program. A system may prevent an act with a wall, penalise it afterwards, or charge it at the step. Comparing those choices means following what each one guarantees for the task.

CHAPTER 23

# Conclusion

Put the blank cell, the triangle, the cups and the two computers back on the table. The cell went from a specified blank state to a 0 and stayed under its write-once rule. The rule merges blank and zero; a heat floor still needs the input preparation and the thermal premises. Turning the triangle preserves relations you can check with your fingers. Turning a cup records an act; a row of cups can distinguish stages that one cup cannot. The two computers can leave the same answer while differing in what they did to reach it. In each case I had to say what the things were, which moves were allowed, which relations survived, and what my way of looking could tell apart.

Part I started at the bottom, with the cell. On any machine run by one rule, every step lands on a fresh state, merges two into one, or goes round a loop, and a run that raises a reading that stays either merges somewhere or never repeats a state, so it's paid in a merge or in fresh states, alongside the model-dependent thermal bounds, more the faster it goes. The counting needs nothing but one rule and a reading that stays. A single 0 written on a blank cell is already a merge, and every zoom pays the same way. Then the model got an exact form: five parts, one rule, one theorem you proved at the table, what keeping takes written into a step. Then the rule stopped being only my preference. With finitely many yes-states and a flag that stays up, the fresh states run out, every move that raises the flag merges, and under the merge-pricing rule the toll follows. The collision belongs to the instruction, over the states it's allowed to act on; the state that just came in needn't be one of the two that collide. Turning that into heat takes a spread over inputs and Landauer's premise on top, its critics named; from a start somebody already knows, the premise puts no floor under it, because that prepared distribution has no entropy to remove; the step can dissipate under the stated thermal-model assumptions. The toll is a derivation under three premises you can refuse one at a time, each with the machine that walks past it built; what keeping takes, underneath it, isn't. Learning borrows: copying onto a blank page merges nothing, the copy can be handed back, and the page is paid for when it's wiped. Committing makes a record that can't be handed

back, so it pays. That's as far as physics goes in this book, and I've tried to keep it exactly that far.

Part II gave the triangle and the cups their words: structure is relations among things, kept under the changes you choose, carried by whatever stuff happens to be there, relative to a frame. A partition turned out to be the smallest sorting by the ruler I picked, and a cell that goes from blank to 0 and stays is that smallest sorting, one yes-or-no question, kept. And making a grouping, keeping it, checking it and standing behind it came apart into four different acts. So the question became how an account of computation could carry what keeping the cell's 0 takes, and the difference between those acts, in its step.

The rest of Part II was the logic, and what it earned comes next, in the order it was earned. The counter is arithmetic once the schedule is chosen. A window in which a raising step and a quiet one look the same can't price the raise exactly, by a fact about functions far older than the machine. Entitlement is defined as an object, a checked relation, and a paid event; given lists, a tree and a covering somebody supplies, a run that paid a charged step for every level of the tree has paid for the index bits it commits to (that count doesn't show a run has to pay them, the run doesn't recover the tree, the lists count repeats, and the index bits aren't the questions anybody asked); when the run makes the narrowing itself, every candidate dropped was dropped by a test the run paid for, and the index bits are at most $k$ times the cost when each test keeps a $2^{-k}$ share; on every certified Thiele-complete machine the same count, for a run from a clean start, is a theorem about its record moves, with the earned chain inside the run, and the count can't do without the covering; and it is a license to lean on a claim. It doesn't prove the claim true. Insight, in the sense the toll prices, is a distinction made to stay. A feeling doesn't count. Its floor is flat: at least one, however deep. The claim behind the flag is something a check could have refused, and in the bare model nothing makes the commit wait for the check. Over any base, a base-driven record is a latch, with two unforced choices, the event and the price list; "only one substrate" is false and "one structure, many price lists" is true once the event is fixed. No procedure whose flip the substrate itself can represent decides the shortcut predicate, given a recursion theorem, by the same diagonal that broke Frege's floor, and the machine's answer is to record the commitment. It leaves the judgement unautomated.

Part III built an example that meets the definition. The small machine's state ties a checked claim to an edition of a counter, and its step uses the tie: change the counter and the old check no longer lets you commit. So from a clean start a raised flag needs a passing check, a commitment about the same edition and a certificate, in that order. A cost of three is enough, and no such run pays less. Those are facts about the example, the check that can't fail included, and its two-counter base is universal under Minsky's packing.

The universal programs ask whether those relations survive somebody else's machine doing the work. On the host with more counters, $U$ runs the small machine's programs from clean starts, halts when they halt with their answers, earns its own flag exactly when the guest raises one, and matches the ledger wherever the two line up. With PAY added, $U_P$ runs every computably presented Thiele machine, latches the guest's reading through its own earned chain, and carries the guest's ledger to within the stated surcharge: at most two at the raise when the reading starts at no. The theorems say which states match; the companion volume, in its section on running it, says how long this particular build takes to get there, which is a long time.

Then I changed the base and joined machines up, to see what carried over. The panel bolts onto any box with a universal base, with the boxes it won't fit named (Appendix D). Two machines side by side need both records kept; one total loses which side did the work. Hand-offs and nested runs each come with conditions of their own (Appendix E). Every one of those constructions makes you say which relation survives the change and which is lost, and that saying is the work.

Then came a check worth earning, the two-player game with an integer test whose pass forces Tsirelson's bound for the tally it's handed. Of five record disciplines I modelled in fragments, one charges a fee at the write, and which event to record is a thesis, with its refutations listed. And then came what the counter has to do with physics, which is less than the name Landauer might suggest. Two facts about the machine sit where they belong. Its unpriced steps merge states, because that's what forgetting scratch work is; whatever those merges cost the hardware is off the ledger, so the toll on this machine comes from the record layer's merge premise, with no finiteness premise on the whole machine; the only finiteness in it is the flag's two readings. And a check that cannot fail still certifies for three, because the step charges for the act, and the entitlement count says what the claim bought.

Part IV brings those pieces together (Section 20). A finite permanent holder has to merge; the named price rule puts that merger on a ledger; a window determines which part of it a reader can recover. The example machine then supplies the earned path from a passing check to commitment and certification. Each argument has its place in that chain, and the constructions show how to carry it out.

Four contributions bring that chain into focus, each alongside the earlier work it builds on. First, the toll isn't only my preference: on a finite machine a reading nothing turns off can only be turned on by a step that merges states, so if merges are priced, raising it costs (Theorem 3.1, Observation 3.3); and on any machine a raise that merges nothing needs endlessly many yes-states, so it pays in fresh states instead (Proposition 3.4). The counting under it is the finite cousin of the recurrence argument Zermelo threw at Boltzmann in 1896, and the abstract merge price is motivated by Landauer's reset example. Identifying that price with heat

needs the further premises stated in Part I. Ouldridge and Wolpert (2023) had already put a physical price on that merge for finite automata, and Li and Vitányi (1996) had already assumed a unit price on each irreversibly erased bit in a model of computation, and Demaine and his coauthors (2016) a price on instructions that destroy bits, so what's here is the toll such a price forces on a reading that stays. Second, a window that shows a raising step and a quiet one the same can't price the raise exactly (Observation 7.2); that a price computed from what a window shows can't tell apart two steps the window shows the same is elementary, and what's here is the pricing form. Third, a record driven by the computation that never switches off is a latch on one event, over any base (Observation 10.1); latches and history variables are old, and the theorem covers only records that never come down. Beside it, built rather than new, sit the definition of Thiele-complete, which any universal base meets once marks are added (Section 10.8), and the slice under it: a computer with a move for each two-counter instruction is what's left of a certified Thiele-complete machine with its record layer forgotten, blind to the record and the ledger, and every Thiele-complete machine leaves one through its window (Theorem D.4). That forgetting a layer gives the runs back, where the step doesn't read it, is the ghost-code erasure of Section 6; what's here is that the forgotten layer still earns its record, and that what's left can't see it. It's a theorem about boxes with a universal base, and it uses no equivalence between models; the one that carries a box to the other models is imported (Theorem D.6). Fourth, a record that's priced, and on the small machine earned (Theorem 15.11), under the move of guaranteeing something about every program. That move isn't mine at all. It's the reference monitor's (Anderson, 1972), made precise for monitors that watch a run (Schneider, 2000), and proved for monitors checked at every instruction (Azevedo de Amorim and coauthors, 2015). What I put together under it is built, and its idea isn't new: the check bound to the edition it looked at, the toll charged on top, and universal programs that carry both to every program of the small machine and to every computably presented Thiele machine, earning their own flag as they go. Appendix B sorts all four with sources.

And there are two things I'd point at if you asked how far the proofs reach. Neither is a fifth new thing.

In the game: an integer check on eight counts forces a five-by-five matrix to be positive semidefinite, and from there the Tsirelson bound follows, and the chain never once reaches for physics. There's no Hilbert space in it, no operators, no observables. The route through the matrix is the first rung of a ladder Navascués, Pironio and Acín built in 2007. None of that route is mine, and the semidefinite road to Tsirelson's bound is older than that. What's here is the check that runs in whole numbers, and the proved link from "the check passed" to the bound. It's algebra you can follow line by line. It's algebra and nothing more, because nothing in the

definitions or the argument ties the counts or the matrix to any physical device. And on the small machine's host a raised flag can stand on that check: from a clean start a raised flag contains a passing CHECK of the packed counts, a COMMIT and a CERTIFY, and the check passes exactly when every pair was sampled and the five-by-five is positive semidefinite (Appendix G).

In the model: the diagonal rules out a correct decider whenever its flip can be expressed inside the substrate and the substrate supplies the fixed point; in L the fixed point is proved from the reduction rules and the diagonal is Rice's theorem. On the small machine's host, Kleene's recursion theorem and Rice's theorem are proved for its own programs, so the wall stands there by proof, with its premise a theorem: no host program decides a property of what a program computes that some program has and some lacks; such a property is undecidable in the reduction sense of Theorem 15.5 (Appendix C.4).

There are things this book doesn't settle, by my count. It doesn't settle whether the toll is the *right* thing to price; the five real systems mostly built a wall or a penalty for contradiction. It doesn't settle whether certification is the pointer event. And it doesn't settle whether Landauer's critics are right, which would leave every theorem standing and one premise unowned by the world. And the question I'd most like someone to take from me: is there a smaller start than a partition, by a ruler they can state? I said in Section 5 it would change this book, and it would.

One more claim ties the counting to the world, the way Church and Turing's ties computing to it, and I'll state it as what it is, a thesis, the *Thiele thesis*: every physical process that computes with final answers has an exact description as a machine run by one rule, with a next state for every state, computably presented (its states written as whole numbers, and its step, its reading and its start for each input computed by programs that halt on every input), on which the answer is a reading that no run from any input ever turns off, on inputs accepted and rejected alike. No proof settles it, because "a physical process that computes" isn't a mathematical object. A model of computation is one, and for a model written down as states, a step and a reading that stays, what keeping takes needs no thesis: it's a theorem, and Section 22 says how far it reaches. The thesis is about what a physical computation pays for what it keeps, which is Part I, and not a claim that its description run by one rule, with its program loaded, is a Thiele machine: by Theorem D.4(4) that one-rule description is one only as a clock or with a reading that never rises. Whether the computer underneath is a Thiele machine depends on the moves it offers before a program is chosen, and with the program in the state a reading is a record, not a simulated bit, exactly when it never comes down from any state (Observation D.5). So the thesis is about physical processes and the one-rule descriptions of them. The Thiele Machine is the model that prices what such a description keeps; the thesis does not say a physical process is one. There is an established route to a conditional physical

description. Gandy's work on mechanisms, restated by Arrighi and Dowek, starts with homogeneity in space and time, finite local state sets, bounded propagation and an initially finite departure from a preserved quiescent background. Their classical theorem makes the resulting evolution computable. In the construction, a cell consults finitely many neighbours through one finite table; a finite active region can therefore be advanced by a program. These are substantive modeling premises. They do not select a certification event or a price in joules, and they are not asserted of every physical system here. The quantum extension uses different premises. See Arrighi and Dowek, *The physical Church–Turing thesis and the principles of quantum theory*, Section 2, `https://arxiv.org/abs/1102.1612`.

What is proved sits on either side of the thesis. On one side, Part I applies to every such description on the states its inputs reach, where the answer never comes down: a run that raises it merges two of its own states or never visits a state twice (Proposition 3.5). On the other, seen only through which inputs it accepts, the thesis implies that every set a physical process accepts with final answers is one a Turing machine can list. At that extensional level it has the physical Church-Turing thesis's conclusion. Run the description from every input, a step at a time across all of them, and list an input the first time its reading says yes; no run turns its answer off, so a yes seen once is the answer. The other way, a set a Turing machine lists is accepted by the machine that searches the list for its input and then steps from its halted configuration to itself, with the reading "halted," which nothing turns off. This converse constructs a machine with the same accepted set; it does not prove an exact state-by-state description of the original physical process. Nothing about how the keeping is paid survives that view: the same search held at its halt by a step to itself, which merges, or by a counter that goes on ticking, which takes fresh states, accepts the same inputs. And "final" has to cover the rejected inputs too. A machine that starts at no, reads yes while program $n$ is still running and falls to no for good if it halts reads yes for good on exactly the programs that never halt, and no Turing machine lists those, since one that did, run beside the search for a halt, would decide halting.

And then there are the two computers of Section 14, the sorter and the copier, holding the same list. Walk the warehouse again. The list is in order on both. On the machine with the earned layer, a different part of the state tells you whether a check happened and a matching commitment followed it, and the rules keep that tied to the edition that was checked. Put your hand over that part of the machine and the difference between the two computers vanishes from where you're standing. Change the counter and you can watch the old check go stale. Those are different operations with different consequences, and you can follow each one with your finger.

Carry a report out of the warehouse and you've got another question: what connects the report to the machine? The flag on its own doesn't answer it. A window, a copy, a second machine and a reader each bring relations of their own, and the constructions in this book ask, one at a time, which of those relations hold. That's how the example gets past the lamp: by what survives a change of view, a change of state, or a change of machine.

At the bottom of the Thiele Machine is one cell going from blank to 0, and the question of what that write keeps and what the chosen price rule charges. Built on that cell, the machine is structure held as state, acted on by a step, with an account of the changes and a view that may keep or lose their distinctions. What I'm arguing is that those relations are part of computation itself, and not something you bolt on afterwards. The record and its toll are the part of that I've worked all the way through, and there this much is proved: a computer with a move for each two-counter instruction is a slice of a certified Thiele-complete machine, what's left when the record layer is forgotten, blind to what was kept and what it cost, and every Thiele-complete machine has such a slice (Theorem D.4). Run as one rule, a machine has no universal base and no universal-base slice, and is a Thiele machine only as a clock or with a reading that never rises. The small machine and its interpreters are things you can take apart. Each theorem answers a stated question under stated premises. Put the computation, the record and its account on the same drawing, and you can follow what changes, what stays, and what the next reader has to work with.

CHAPTER 24

# Afterword: how I got here

What I am is curious, past the point of good sense.

I pick a thing up and cannot set it down until I have turned it over enough times to see how it holds together, and this time it did not let go.

I am not a programmer and I was never formally taught any of this.

I was not trying to build a new model of computation.

I would not have known what one was if it bit me.

I was trying to build a 3D renderer.

There was no plan behind it.

Just a pull to make something real with my own hands, and the kind of stubbornness that will not let a thing go until it stands up on its own.

I have worked on it in off hours every day since.

Once I started digging in, two things came clear fast. I knew nothing about computers, and the field was already saturated, every obvious road paved and crowded. So I needed an angle nobody else was standing on, and the angle I had wasn't a strategy. It was just the only way my head works. I did not know what compositional meant. I had never heard of category theory. I called the thing I was building digital DNA, because that is what it felt like in my hands, and I thought about it in the only language I had: gravity pipes and directions and vectors, things that were linear and somehow not linear at the same time, structure that flowed and joined and came apart without my owning one proper word for any of it. I built a custom rendering pipeline out of those intuitions, by feel, throwing out pieces that did not sit right even when I could not have told you the rule they were breaking. And then, the way I always do, I went looking for what I was actually doing, and found out it already had a name. An old one. A deep one. What I had been reaching for in the dark was composition. It was category theory, a whole field I had never once heard of, sitting there describing the exact shapes I had been shoving around. The moment that landed, the rendering pipeline stopped being a rendering pipeline and started turning into a categorical engine, and I started swallowing the real theory as fast as I could find the words for things I had already built.

I took it one concept at a time, and I would not move to the next until I could watch the current one actually run. Category. Object. Morphism. Functor. Monoidal product. It came together in pieces and never in chapters, the way a picture resolves out of static. The Yoneda lemma (a result buried deep in category theory) was the piece that made the rest snap into place.

Then within weeks I found out the engine I had built to draw pictures was general enough to run a physics simulation on the exact same primitives, without my rewriting any of it. I wrote Newton's laws as a program in it, half expecting nothing, and it ran. I want to be careful about what that was and what it was not, because this is exactly the kind of thing that gets misread. It was not me deriving physics, and I am not claiming the machine explains gravity. It was the compositional structure being general enough to express a physics simulation the same way it had expressed a renderer, and that generality is what told me the structure was the point and the pixels never were. Somewhere in those weeks I crossed from debugging why one pixel would not show up to understanding that I had quietly built a categorical computing engine, and that the rendering had only ever been the first thing I aimed it at. The pixel did eventually show up, for what it is worth.

I kept going, because by then I could not not keep going. The night the engine first ran a program declaratively, meaning I told it what should happen and let it work out the how without my spelling out each step, I told my wife, Julie, I thought I was onto something. She said okay.

Then I hit a wall, and it is the wall this whole document is about. Every tool I had for settling questions about what I'd built could settle the small ones and fell over on the exact ones I needed answered, which left me staring down an infinite hallway of guess-and-test with no proof waiting at the end of it, ever. One night, not even working, I was watching a video about Flatland, the old story about a creature living in two dimensions who cannot so much as picture a third. And it landed on me sideways, all at once, the way the real ones always do. I said it out loud: I get it now. There is another axis. Classical computation only ever sees what it can read and write off its own surface, and there is a whole direction it is blind to. It isn't broken. It was just never built with an eye that points that way. That's how it landed that night, and it didn't survive in that form: the direction turned out to be one an ordinary computer can reach, and what the usual rules leave out is the price (Section 2.6 says where the axis ended up). The two computers in Section 14 are my attempt to show you that blindness with something you can hold. That same night I gave the idea a name: the Thiele Machine. The name began as a working title and became the name of the model. It carries the larger claim plainly: the machine treats structural state as part of computation.

What survived the whole trip from there to here is not the renderer. Not one piece of it, and good riddance. What survived was a single commitment I made early

and never once let anyone talk me out of: structure is the state itself, the actual thing the machine runs on. It is no metaphor laid over some realer state underneath. Category theory was the first language I found for that structure. It is not what the model is. The basic model asks for states, instructions, a step, prices and a reading. The ledger and the window are further structure used in the constructions. None of that needs category theory. The morphisms in that first engine, the night they first composed correctly at two in the morning, were objects that remembered where they came from; the records in the small machine I ended up with remember where they came from too, and you can walk up to one and challenge it. The receipt grew teeth. But the shape itself never moved. The shape was the substrate the entire time, and everything else, every renderer and every proof, was me slowly learning to see what I had already built. By the shape I mean structure held as state. Not category theory. Category theory was just the first language I had for it.

Every step pulled back to show another layer underneath that I had not known was there, and I mean that plainly: I did not know what a Turing machine was when I started, I did not know most of the words in this document existed. I learned each one the same way, by walking face-first into a wall that needed it and not being let past until I had it. The needing always came first. The learning came after, dragged along behind the needing because it had no say in the matter.

Large language models were my hands the whole way through, and I want the division of labor exact, because it is the thing people most want to get wrong about work like this. The hands had nothing like this to draw on: no reference implementation, no prior project to point at. The ideas, the architecture, the direction, the demand that the thing be true: those are mine, all of them, no asterisk. The symbol-pushing, the code and the formal derivations, the models produced. So I read every statement. A derivation can settle an empty question, and I had to refuse that too. However the symbols got onto the page, the argument on it is checkable by reading, and it would be exactly as right or as wrong if a million monkeys had typed it. The monkeys could not have refused.

A word on what I actually do, because it runs under everything after it. I do not write code or math by hand. I do not pick constants. I do not grind through derivations. I have never written a line of code on paper or in a notepad in my life. What I do is see, and direct, and refuse. When I take hold of an idea it does not stay a point. It opens up. It becomes a shape I can blow up and turn over and take apart, lay across another shape to find where the two of them touch, run forward and then back to where it started, zoom into until it is the only thing in the world and pull back until it is one dot in a field of other dots, and the whole time it stays wired to everything I have ever connected it to, the right connections and the wrong ones both, all of them live at once. I am not reading that field neutrally, the way a search would. The thing doing the looking is me, the whole of me, my gut and my mood

and my stubbornness picking what lights up and which way I move through it, and that is the part you cannot lift out without turning me back into a search engine. So I see where a thing has to fit, the shape it is obligated to have, usually well before I could write it down in a single symbol. Then the model gives me the symbols, and I refuse anything whose argument does not hold or does not match the shape I am holding. That refusal is the one thing in this entire document that is mine and only mine: the demand that it be true, and the throwing-out of everything that wasn't.

I am telling you all of this because it's the engine. I did not come up through anywhere that hands you a foundation and says trust us, this part is settled. No professors, no colleagues, no field full of authority to lean back on when a claim got heavy. From the outside that is a deficit. From the inside it is the whole reason any of this is solid: when I meet a claim I either find the proof or I do not believe it, and that goes double for my own, because my own are the ones I most want to wave through. It is how I am built. It is the only way I know to hold a thing and actually trust that I am holding it.

So everything in here gets built from the ground up. It isn't a courtesy to you. I build it because I do not believe I understand a thing until I have built it with my own hands, watched it stand up on its own, and tried to knock it over. I built this document the exact way I built the machine, brick by brick, refusing each brick until it held weight. That includes the parts you already knew cold before you opened this. I did not skip them for you, and I could not skip them for me.

The vocabulary I had to go look up to do any of this sits at the back of the book, so the argument comes first. If a word stops you, it is there. If you already know this material, you never have to open it. But if you catch something in the basics that is wrong, I want to know, and I mean that as a real ask. It isn't politeness. Nothing in here is correct just because I wrote it. Especially the parts I wrote.

**On the name.** I wanted to call it the meta machine. Then, in October 2021, Facebook renamed its company Meta: a company taking a term for self-reference as its name and draining the word of meaning in the same motion. So I went one level up, where this machine already lives. The most meta move available, for a machine that runs on meta-level reasoning, was to put my own name on the cover and take the word back. That's the whole of it, a meta move on a meta machine. It happens to be my dad's name too, and my wife's now. So I asked them before I did it. They said go ahead.

APPENDIX A

# Things I had to learn first

I came in with none of the formal fields this work turned out to need. By the time the argument had taken shape I'd needed enough of half a dozen of them that I had to put down somewhere what those fields had words for. This appendix is that place. None of it is the argument; it's what I needed to be able to make it. Everything below is what I understood after I went and read about it; I'm not claiming to teach you these subjects, and if you already know a part, skip it.

## A.1 Bits, bytes, and what a CPU sees

A bit is one digit that's either zero or one, the smallest fact a machine can hold. Everything else is bits stacked up. Eight of them make a byte, which lands you any number from 0 to 255, and the first thing that genuinely surprised me is that the same byte is also a letter: a convention called ASCII simply agrees to call 65 the letter A, 48 the digit 0, 32 a space. Nothing in the byte knows which it is. The meaning is a convention laid on top.

You can see the convention's seams with a pencil. Write 65 in binary: 1000001. Knock off the first two digits and what's left, 00001, is 1, and A is the first letter. B is 66, which ends in 00010, two. Now lower case: a is 97, which is 1100001, the same pattern with one more bit switched on, the one worth 32. Every capital and its small letter differ in exactly that bit, and that's on purpose. The first version of the code, in 1963, had no lower case at all; when lower case arrived, in the 1967 revision, it was placed so the two cases differ by one bit, which made matching them, and building keyboards and printers, simpler. Nothing in the byte knows it's a letter, but somebody made the letters easy for a circuit to tell apart.

Everything in this subsection is about real CPUs. None of it belongs to the abstract model, which has a kind of state and a step rule and nothing about widths or slots, and the small machine of Section 15 needs very little of it: its two counters have no width at all, and hold any whole number.

A word is a chunk of bits the CPU treats as one unit, in 2026 usually 64 bits (0 to about 18 quintillion) on a desktop or server and 32 (0 to about 4 billion) on many small devices. The abstract model doesn't care which width.

A register is a named storage slot inside the CPU that holds one word, the fastest place to read or write. Memory is a longer list of word-sized slots accessed by index, the address. The program counter is one specific register that tracks where in the program the machine is; after every instruction it moves on to the next instruction unless the instruction says jump. On the small machine it's called the line number. A stack pointer is a register pointing at the top of a region used for nested calls. A port is a named channel for talking to the outside world.

Modular arithmetic means counting wraps around: counting modulo 16, the numbers run 0 to 15, and 17 wraps to 1. Bitwise operations work on the bits of a value one position at a time: AND is 1 only when both bits are, OR when either is, XOR (exclusive or) exactly when they differ. A logical shift moves bits left or right: for unsigned values it multiplies or divides by a power of two, discarding overflow on a fixed-width left shift and rounding down on a right shift. Popcount is the count of one-bits in a value. Hexadecimal is base 16, digits 0 to 9 then A to F, written with a 0x prefix: 0xFF is 255.

A trap, on a CPU, transfers control to a handler; the interrupted program may later resume. It can report an error or request a service. On the small machine a trap is one bit that goes up when an instruction refuses, and then the core stops moving. An ISA, instruction set architecture, is the list of every instruction a CPU recognises, each identified by an opcode, a numeric tag the CPU decodes. The small machine's has six. The abstract model has an instruction set and a step rule, but prescribes no binary opcode encoding.

## A.2 The notation the formal parts use

I don't think in these symbols, and I had to learn to read them. The natural numbers are 0, 1, 2, and so on, written with a double-struck N; the integers add the negatives; the reals are the whole number line. A function from a set A to a set B is written with an arrow. For a finite set, vertical bars around it mean its size; for a number, its absolute value. A big sigma is a sum over every member of a set. A delta before a quantity means the change in it. An upside-down A means "for every"; a backwards E means "there exists." A double arrow is implication; a double-headed one is "if and only if." A small circle between two functions is composition, apply the right one first. Angle brackets mean one of three things, and the sentence around them says which: a pair of numbers packed into one, $\langle m, n \rangle$ (Appendix C.1); a dot product of two vectors; or, around a random quantity, its long-run average, which I picked up from the Bell-test papers. The circled times $\otimes$ in the quantum derivation combines two systems: $A \otimes B$ acts as $A$ on the first and $B$ on the second; Appendix H gives its entries. The integer check itself uses neither it nor any quantum system. A

superscript T on a matrix is its transpose, rows and columns swapped, and I with a subscript n is the identity matrix.

## A.3 Category theory beyond the sketch

Section 5 sketches objects, arrows, composition, identities, associativity, and the identity laws. Beyond that: a functor is a structure-preserving map from one category to another, sending objects to objects and arrows to arrows while respecting composition and identities; think of it as a translation rule between two worlds of structure. An embedding is a functor that throws nothing away, and faithful is the precise version for arrows: two distinct arrows between the same two objects never land on one. Conservative is the version for inverses: if the image of an arrow has an inverse, the arrow already had one, so the functor can't manufacture an isomorphism (an arrow with an inverse) that wasn't there. The two properties are different, and neither gives the other. Read the natural numbers under addition as a category with one object and an arrow for each number, and the same for the integers. The inclusion of the first into the second is faithful, since distinct naturals stay distinct integers. It isn't conservative: 1 has no inverse among the naturals, and its image has one, minus 1. The other way round, the functor from the integers onto the trivial one-arrow group is conservative, since every arrow of the integers already has an inverse. It isn't faithful: it sends every integer to the same arrow. An embedding needs faithfulness specifically. A monoidal product combines two objects or two arrows into one, with a unit and laws that make the combining behave. An initial object has exactly one arrow into every object; the counter's uniqueness in this book is a uniqueness-on-reachable-states fact, and no category-theoretic initial object is involved. Yoneda's lemma was the piece that made the categorical engine snap into place for me (Section 24), and nothing in the formal development uses it.

## A.4 Bell-test vocabulary

A correlation between two outcomes is the average of their product; with outcomes plus or minus one it lies between minus one and one, plus one meaning always match, minus one always differ. A correlator is one such average in a labelled setting, E00 for Alice's setting 0 and Bob's 0. The marginal of a joint distribution is the distribution of one side alone with the other averaged out; zero-marginal means each side's outcome is plus or minus one equally often on average. A local hidden-variable model is one where both outcomes are fixed by a shared random variable agreed before the game plus each side's own setting; Bell's theorem says no such model produces certain correlations quantum mechanics does. No-signalling means neither side's marginal depends on the other's setting. Quantum-realisable means

compatible with some quantum state and measurements; the NPA hierarchy gives a ladder of necessary tests, each a positive-semidefiniteness condition on a moment matrix. For the four correlators of this game alone, the first rung is already exact; for the full table, each side's own odds included, Chaturvedi's 2026 preprint argues that no finite rung is, even in this simplest case. The PR box is the no-signalling correlation that wins the game every time, stronger than quantum mechanics allows. Cauchy-Schwarz is the inequality that a dot product is bounded by the product of the lengths. The Tsirelson bound is two root two, the largest game score quantum mechanics allows; the classical bound is two and the PR bound is four. (If you go looking for Boris Tsirelson's 1980 paper, look under Cirel'son, the spelling the journal printed, or you may not find it.) The elliptope is the set of correlators that can be filled out into a positive-semidefinite matrix with ones down the diagonal.

### A.5 Other terms I use without rebuilding

Provenance is the recorded chain of where an object came from; on the small machine, a commitment's provenance is the passing check that came before it (Section 15). Simulation, in computer science, is one system imitating another's input-output behaviour; a Turing machine can simulate a system with effective finite encodings of its states and moves and a computable step on those encodings, my small machine included, by representing its state on tape, and the difference this book is about is what a window on that tape shows. That's real, and it raises a fair question: if a tape can stand in for the state, what was the difference? The tape carries a picture of the state, the way a photograph carries a fire, and a photograph of a fire keeps you exactly as warm as any other photograph. The counter and the toll live in the step rule. An encoding copies the bytes, and the step rule isn't in them. A Turing machine that bumps a ledger on its tape whenever its table sets its flag keeps that ledger in its state, and a ledger kept in the state isn't a price on the move. The toll prices the machine's moves: with one move, the next step its table picks, every step costs at least one once the flag can go up at all, and with a move for each row of the table, every row that can set the flag costs at least one wherever it's taken (Section 15.13). The ledger on the tape only simulates that. What no machine can do is make a simulated rule binding just by writing it down on tape.

Projection is a function that drops information on purpose. The window on the small machine keeps three things, the line number and the two counters, which are what Minsky's machine already has, and that's why projecting lands you back in ordinary computation. It drops everything else, including the parts doing the new work: the versions, the table of checked facts, the channel, the trap, the ledger and the flag. Nothing is wrong with it, and it's a perfectly good function; it just loses you what made the state more than a pair of counters. When what's left runs

as the machine underneath runs, reaches what it reaches and still can't see the record or the ledger, the projection is a slice (Definition D.1): every Thiele-complete machine has one onto two counters, its window, and every computer with a move for each two-counter instruction is a slice of a certified Thiele-complete machine (Theorem D.4). A quotient is the operation of declaring two things the same when they agree on what you chose to look at and working with the classes instead. A projection does exactly that. Dropping fields one at a time also glues together states that the dropped fields kept apart. The fibre of a function over a value is everything the function sends to that value. An injection sends no two inputs to the same output; a surjection covers every output. Decidable means an algorithm always halts with a yes-or-no answer; keep the claim and the algorithm apart, since the algorithm decides the claim only once there's a proof that its answer matches. Polynomial time means running time bounded by a power of the input size; P is the problems solvable that way, NP the problems where a candidate solution can be *checked* that way, and whether the two are the same is open.

"Kernel" has two uses here. In mathematics, the kernel of a map is what it sends to the identity; for a projection the useful version is the relation "these two states project to the same thing." An operating-system kernel is the core of an operating system, as in the packet-filter example in Section 17.

## A.6 Matrices

A matrix is positive semidefinite (PSD) when $v^{\mathrm{T}}Mv \geq 0$ for every vector $v$; the game check is all about one matrix being PSD. The determinant of a square matrix is a single number that is zero exactly when the matrix squashes some direction flat. For a two-by-two matrix with rows $(a, b)$ and $(c, d)$ it's $ad - bc$. For a bigger one, expand along a row: for each entry, delete its row and column, take the determinant of what's left, multiply by the entry, alternate the signs, and add up; the recursion bottoms out at two by two. A principal submatrix is what you get by choosing some of the indices and keeping only those rows and the same columns; its determinant is a principal minor. A symmetric matrix is PSD exactly when every principal minor is at least zero, every one of them, and the step from the game check to Tsirelson's bound uses two of the three-by-three ones (Appendix H.4). The Schur complement turns a question about a block matrix into a question about a smaller one. For a symmetric block matrix with corner block $P$, off-diagonal block $Q$ and other corner $R$, it's $R - Q^{\mathrm{T}}P^{-1}Q$, and when $P$ is positive definite the whole matrix is PSD exactly when that complement is (Lemma H.6). Positive definite matters; invertible isn't enough. Take $P = -1$ and every other entry zero: the complement is zero, which is PSD, and the whole matrix isn't.

## A.7 Borrowed ideas the book pushes on

These get their full story in the chapter that uses them, and each gets a one-line version here for when you've forgotten which one Landauer was.

The pigeonhole principle: put more pigeons than holes and some hole holds two. Section 3's room with too few chairs is that, and so is the proof that, when only finitely many states read yes, a permanent flag can only be raised by a move that forgets something, two states landing on one. Landauer's principle (Rolf Landauer, 1961): a step that wipes information, lowering the entropy of what a machine holds by some number of bits, gives off mean heat at least Boltzmann's constant times the temperature times the natural log of two per bit, under a physical model relating that logical entropy to the memory's thermodynamic entropy. The bound is a floor that going slowly doesn't lower, and it isn't all the heat: specified thermal dynamics can require additional entropy production for a finite-time change of distribution. In this book Landauer's floor is always a premise with its name on it. Nothing here derives it (Section 3, Section 19). Entropy, in Claude Shannon's sense: for a spread of chances over states, roughly the average number of yes-or-no questions you'd need to pin the state down, so an even spread over $n$ states has $\log_2 n$ bits. A spread, in this book, is just a list of chances that add up to one. Maxwell's demon (Section 4): a little creature that watches molecules and sorts fast from slow, seemingly for nothing; Charles Bennett's answer was that the looking can be done without merging anything, so Landauer's premise sets it no floor (it can dissipate under the stated thermal-model assumptions, less the slower it goes), and the floor lands when the demon wipes its notes to look again. The potential method (Robert Tarjan, 1985): pay for a run with a quantity stored in the state, so a step's cost plus the change in the stored quantity is what you count; Section 6 reads the toll as that method with one particular potential.

The halting problem (Alan Turing, 1936, who essentially proved it without stating it in this form; Stephen Kleene stated it in 1952, and Martin Davis seems to have been the first to name it, in 1958, Section 13): no program decides, for every program and input, whether it stops. Kleene's recursion theorem (Stephen Kleene, 1938): for any transformation of programs that can itself be computed, some program computes the same partial function as its own transform. Its timing, storage and records need not agree. Rice's theorem (Henry Rice, 1953): any property that depends only on what a program does, and isn't true of all programs or of none, can't be decided by a program. The diagonal is the move under all three: build the thing that does the opposite of what the table says about it (Section 13). L is a small lambda calculus, a language made of nothing but functions applied to functions, which Yannick Forster and Gert Smolka presented and which Section 13 uses as a tool. A cryptographic hash is a short fingerprint of a longer piece of data, made so

that finding two inputs with the same fingerprint is supposed to be hard; a hash chain folds each new record into the fingerprint of the last, and a Merkle tree folds them pairwise into one root (Section 2, Section 17).

## A.8 Words this book builds

These are mine, or mine in the sense that I pinned them down for this book. Each is built properly in the chapter named; this is only a reminder.

A few words first, because everything else hangs off them and each has exactly one meaning. The *model* (or the abstract model) is the definition of a Thiele machine, Definition 2.1, five parts and the toll, with the two-place record of Definition 2.2 as its smallest case, plus exactly the extra pieces a given theorem asks for: a counter in the state, or a window, or programs written as states and a recursion theorem. A few theorems, the step law among them, use only the states and the step. The name Thiele Machine belongs to the model. Parts I and II develop those general arguments, and Part III applies them to the machine it builds.

A *substrate* is the model's shape with whatever extension a theorem names, and nothing else; the word never means my machine, and never the side of a simulation that does the stepping (Section 14.18).

*The example machine* (also the machine I build, the build, the small machine, the smallest honest machine, Section 15) is one small machine I defined to show the definition can be met. It is an example of a Thiele machine, not the model. It fills in those pieces at once: two counters with their versions, a table of checked facts, a channel, a trap, a ledger and a flag, six instruction kinds and their prices, with its hosts and universal programs.

The *scaffolding* is the example machine and the constructions run on it, and the model does not depend on any of it. It lives in Section 15 and in the appendices that carry the machine on, from the universal machines (Appendix C.1) to the CHSH check (Appendix G).

*A Thiele machine*, lower case, is any system proved to fit that definition (Definition 2.1), five parts and the toll, on any order of records or on the two-place one of Definition 2.2; the paper toys are, and so is the small machine. That bar is nearly empty, since a clock that charges one for every move obeys the toll for any record.

*Thiele-complete* means more (Definition 10.6): computing that runs every program, costs nothing and can't touch the record, every other move costing exactly one, a record that stays up once it's up, blank starts, and some run whose record ends up from one start and down from another; a clock fails it, every universal base becomes one by adding marks, and every machine that meets it hides its record and ledger from its own window.

*Certified Thiele-complete* means more again (Definition 15.7): a universal base whose moves cost nothing and can't touch the record, a record that stays up once it's up and is raised from a clean state only after a passing check, which means its claim holds, a commitment to the same claim with the thing it's about unchanged in between, and a certificate, at a cost of one for each of those three acts, and a claim whose check, commit and certificate raise the record from a loaded start exactly when the claim holds there, true at some loaded starts and false at others. Every clock-style machine fails it; the small machine meets it; and every machine that meets it hides its record and ledger from its own window.

A *universal* Thiele machine is one fixed machine running one fixed program that runs any machine of a stated class as a guest, with every step of the guest's computing done by the host's counter moves and every rise of its record put where its program put it, and, in the certified case, each guest's record earned again by the host's own check, commit and certificate (Definition C.1). The ones in this book are $U$ (Appendix C.1), a fixed program in the small machine's instruction kinds, with no guest step built in, that runs every small-machine program with halting and flag exact and its ledger equal to the guest's at every matching point, and $U_P$ (Appendix C.2), the same program with one paid move added, that runs every *computably presented* Thiele machine, one whose driver, step, prices and reading are computed by programs with the reading decided on every state, keeping its reading as a latch and, when the reading starts at no, its ledger to within two of the guest's, a surcharge no program on that host, run from a clean start, can avoid when the guest certifies with its whole ledger below three. The simpler host with a built-in guest step runs guests by construction and keeps a mirror of the guest's flag.

A computer running the small machine isn't the small machine; it implements it, inside bounds, the way a laptop implements a Turing machine. When a word below belongs to only one of the two layers, the entry says which.

A *partition* is a sorting of a pile into groups, nothing in two groups and nothing left over (Section 5). A *Thiele machine* is a state type, moves, a step, a cost per move, and a yes-or-no reading, plus one rule, the *toll*: a move that turns the reading from no to yes costs at least one (Section 2). The rule is called A2 in the mathematical specification and in the conditions table of the companion volume. A move *merges* when two different starting states land on the same end; one that never does is *injective*, it forgets nothing. A reading is *permanent* when no move turns it from yes back to no (Section 3).

A *distinction that stays* is something the world tells apart from something else and keeps telling apart: a cell that reads 0 and not 1 and not blank, a bead in the left well and not the right, a flag that's up (Section 1). The *step law* says what any step of any machine with one next state for each state does to the states its run has used, with no finiteness anywhere (Theorem 1.1). It lands on a state the run

hasn't used yet, which uses up a *fresh state*. Or, the first time it comes back to a used state other than the start, it comes from a different state than the one that first got there, which is a *merge*, two states sharing a next state, and under Landauer's premise a merge of states that were really possible carries a *heat* floor. Or it comes back to the start, or retakes a step it already took, which is all a run does after its first return: it's going round a loop and makes nothing new. *Merge, or a fresh state every step* is that law aimed at a reading that stays up: a run that raises one either merges two of its states somewhere or never repeats a state (Proposition 3.5), and that's what *keeping* takes. Physical dissipation is additional only under physical premises. In the specified thermal models a nonzero change of distribution has a finite-time entropy-production bound, with fixed finite activity or mobility, and slowness does nothing to the floor a merge adds. The book proves that for Bennett's walker (Theorem 1.9); the general finite-time bounds are physics (Schmiedl and Seifert, 2007; Shiraishi, Funo and Saito, 2018). A *reversible* machine never merges, so it owes no floor, but that doesn't mean it pays nothing: each step can dissipate under the stated thermal-model assumptions, and it pays in fresh states; on finitely many states it comes back to its start and keeps nothing (Theorem 1.6). *Finiteness* is what takes the column of fresh states away: when only finitely many states read yes, a move that raises a reading that stays up has to merge (Proposition 3.4).

The *counter*, which I also call the *ledger*: in the model, a number carried in the state that starts at zero and that every step raises by exactly that step's charge (Section 6). My machine has exactly one, the ledger $\mu$ (Section 15). Two other things get called counters, and each chapter says which: Minsky's two bins on the small machine, and the eight counts that tally the game's rounds (Appendix G). Neither is the ledger. A *price list* says what each move costs; the model takes it as given, and my machine's table of what each instruction costs is in Section 15. A *receipt* is a total that lives in the state, only climbs, equals the sum of the charges on every path including the failures, and is the only such total, on the states a run from the start reaches, for its price list (Section 6 proves it for any system with a counter and a price list; on my machine the sum is a theorem too, Section 15). In the model there is just a *reading*, yes or no (Section 2); on the small machine it's the *flag*, set by a CERTIFY that passes and never cleared.

These words describe the example machine and its universal programs, and none of them is part of the definition of a Thiele machine (Section 15). A *two-counter machine* is Minsky's: two counters holding whole numbers with no upper limit, and two instructions: add one, and subtract one and jump if the counter is above zero, or else go on to the next line; already enough that nobody can decide in general whether one halts. The small machine is a two-counter machine plus HALT and three instructions for the record, six in all: INC, DEC, HALT, CHECK, COMMIT and CERTIFY. A counter's *version* is its edition number: it starts at 0 and goes up by

one every time the counter is written, so an edition never comes back. A *fact* is a property, a counter and a version together, such as "$A$ is even, as of edition 3"; the properties come from a fixed language with three forms, *is zero*, *is even* and *is at least* $n$. The same theorems also hold for any property language with an exact equality test whose checker is proved to say yes exactly when the claim holds, and one such language reads a counter as a list and asks whether it's sorted. The *fact table* holds up to sixteen facts, and the only way in is a CHECK that passed; nothing ever leaves it. The small machine's *channel* holds one committed fact or nothing, and COMMIT puts a fact there only if the table has it about the counter's current edition. With the flag up from a clean start it names a commitment earned that way, and a later COMMIT replaces it, as a two-commit run shows. The *trap* is a bit that goes up when an instruction refuses, and then nothing in the core moves again. A commitment is *earned* when a passing check put its fact in the table first, about the same edition, with no write to that counter in between; on the small machine every flag raised from a clean start is earned, and that's a theorem. CHECK, COMMIT and CERTIFY cost one each and everything else costs nothing, so from a clean start a run ending with the flag up adds at least three to the ledger, and a run whose check could have failed adds exactly three: CHECK *is zero* on $A$, COMMIT, CERTIFY, from a start where $A$ is 0. The step doesn't police trivial checks; *is at least 0* passes every value and certifies for three too. The *host machine* is the small machine with a counter, called a *register*, for every whole number, and one property, Slot, true of a number that packs a property of the small machine together with a value it holds of. A *slot* is a register of the host used for nothing but holding one checked fact about the guest, and its *mirror* is a register beside it, safe to read, that says which claim the slot holds while its fact is live. $U$ is the fixed program of host instructions that runs every small-machine program; it reads guest instructions out of one register, keeps the guest's counters in two others, and puts every checked fact in a slot, because on this machine reading a counter above 0 into a program's control flow writes it. The universal program's *guest* is a program of the small machine, held as one number in a register of the host and run one instruction at a time by $U$'s fetch and decode. $U_P$ is $U$ with one more handler, for PAY, an instruction that costs one and does nothing else; its guests are programs compiled from computably presented Thiele machines (Appendix C.2).

A *window* is any function that shows you part of a state (Section 7); a *shadow*, said loosely, is what a window leaves you with. A *collision* is two steps that look the same through a window, before and after, where one raises the flag and the other doesn't (Section 7). *Entitlement* is what a run has when it carries an object, a checked relation between that object and a claim, and a paid event that put the object in place: a license to lean on the claim. It doesn't prove the claim true (Section 8). A *latch* is a record that switches on the first time some event holds and

never switches off (Section 10). The diagonal's *substrate* (Section 13) is the model's shape extended with programs, a run that may give nothing back, a counter that never falls, a way to write programs as states, the transformations the substrate counts as its own, and its own recursion theorem for those; the proof of the diagonal uses neither the counter nor the toll.

APPENDIX B

# What's mine, and what isn't

Every chapter of this book leans on somebody, and I've named them where I lean. This appendix does a different job, and it's the one where I'd most like to be caught out, and caught out early. For each thing the book could be accused of claiming as its own, it says what was already known, what this work actually defines, builds, proves, or connects, and what kind of contribution that is. Five kinds, and they aren't the same claim: a *new result* (a statement nobody had); a *new proof* of something known; an *implementation* (something built that runs); a *synthesis* (known pieces put together under one question); and a *conjecture* or *thesis* (a claim stated and not proved; a thesis ties defined notions to the world, so no proof can settle it). Every kind can be worth doing. None of them is the others. And if you read down the list and notice that "new result" hardly shows up, that's good, and it means you're reading it right.

Two warnings about my own evidence before the list. First, "I didn't find it" is a fact about my searches. The literature is bigger than my searches. I searched with every tool I had, and some tools will cheerfully invent a citation if you let them, so where it matters I say where I looked, and you should assume it may well be sitting in a paper I'd need a few more lifetimes of homework to read. Second, nothing below is inferred from how many theorems there are or from a good story about who came before. A proof, if it's right, tells you the statement is true. It doesn't tell you it's new. A thousand correct proofs could be a thousand careful ways of agreeing with Tarjan.

After the prior-work map come the items themselves, fifteen of them, numbered as their subsections are, in the order the book reaches them: what keeping takes first, certification last.

### The prior-work map

*What I do and don't know about prior work.* I came to this with no formal training in the fields it turned out to need, and I've read what I could reach, with every tool I had. The prior-work map below is orientation. It doesn't pretend to be scholarship. For each piece of this book it says who did the thing it leans on, and what, if anything,

this development adds. A row that's wrong gets fixed, and, as the end of Section 21 says, I'd want to hear of one.

There's a law about rows like these, and it's a joke that keeps turning out true. In 1980 the statistician Stephen Stigler proposed that no scientific discovery is named after the person who first found it, and then credited the idea to the sociologist Robert K. Merton, who had been writing about fights over scientific priority since 1957. So Stigler's law is an example of itself, which is either the best joke in the history of science or a very tidy accident. Is it a law? No, it's a pattern, and nobody proved it. But this book has already tripped over it twice, in The point at the front and in Section 3: the pigeonhole principle wears Dirichlet's name, and Leurechon had it two centuries earlier. And I put my own name on a machine. By Stigler's law, that makes it somebody else's. The table below is my best attempt at finding out whose.

What I compare each time is what a result is *about*. Landauer's bound is about heat, a physical quantity under a thermodynamic model. No Free Lunch (an optimisation result, in the list below the table) is about averages over a whole spread of problems. The toll is a floor on one kind of step, the step that turns the reading on. Rise-only pricing adds a second rule on top: charge nothing on a step that doesn't turn it on. So the comparisons are about the shape of each result. They aren't claims about who got somewhere first, and they don't show that certification is the only event worth pricing.

The right-hand column uses words built in the body of the book (the latch, the fibre, the universal program, the integer check, the matrix test); the subsections after the table say where each one is built.

| Prior work | What this book leans on | What this development adds |
|---|---|---|
| Landauer 1961; Bennett 1973, 1982, 2003; Zermelo 1896; Bocchieri and Loinger 1957; Pin 1992; Brodsky and Pippenger 2002; Ouldridge and Wolpert 2023; Murashita, Funo and Ueda 2014; Li and Vitányi 1996; Demaine, Lynch, Mirano and Tyagi 2016 | the heat of erasure and merging; a unit price per irreversibly erased bit as an axiom of a model of computation, and on the random-access machine; a measurement need not merge, and the demon pays when it wipes its notes; recurrence (a step that loses nothing, on finitely many states, comes back around); machines whose moves all reshuffle states recognise only group languages; recognising a language whose smallest automaton has to merge states produces entropy; absolute-irreversibility bounds require forward and reverse path probabilities | the ledger form of a counting core that's classical (recurrence's finite cousin, folklore for permutation automata) and already priced in entropy for automata, and priced per erased bit and per instruction by Li and Vitányi and by Demaine and coauthors: a certification toll derived from finitely many yes-states, permanence and that rule, with merge-or-revoke, the squeeze and entropy forms, and each premise's counterexample built |
| Maroney 2009; Sagawa and Ueda 2009; Sagawa 2014 | thermodynamic bounds that depend on the actual spread over states; the entropy-loss bound is zero for a point-mass input; further work bounds depend on control, reliability and input distribution | the certificate-specific entropy bound with its support and permanence premises out in the open; a heat floor still needs the Landauer premise, and no new physical law is proved |
| Shannon 1948; Hartley 1928 | entropy and log-of-count bounds | the same squeeze bound in counted states and, from an even spread, in bits, for a permanent flip |
| Tarjan 1985; Lyapunov 1892; Nipkow; Charguéraud and Pottier | the potential method and later treatments | the toll as that method with the flag as potential |

| Prior work | What this book leans on | What this development adds |
|---|---|---|
| Katsumata 2014; Orchard, Liepelt and Eades 2019; Danner and Licata | cost carried as a grade that composes with the computation | a run's cost as a writer cost and as a grade fixed before the run; the counter rides in the state, put there by the step rule, and it is their writer exactly, while their type systems and before-the-run checking of grades are not embedded, by design |
| Lamport 1978; Floyd 1967; Turing 1949 | one-way counters give order and termination | the receipt's three properties (it only climbs, it equals the sum of the charges, it's the only such total on the states a run from the start reaches) stated and proved for any machine with a counter and a price list fixed in advance |
| Moore 1956; Kalman 1960; Cousot 1977; Codd 1970; Sampath and coauthors 1995; Saboori and Hadjicostis 2007 | what a window loses; when an observer can work out a hidden event, and when a secret state stays hidden | the fibre fact with a price in place of a decoder; the small machine's named collision, and the window every certified Thiele-complete machine is blind through |
| Smith 2009 (quantitative information flow) | how much an observer learns about a secret from what it sees | none for leakage: the collisions rule out exact decoding through named windows, which isn't zero leakage, and no leakage measure is supplied |
| Gettier 1963; Necula 1997; Cook 1971 | entitlement is not truth; proof-carrying checks | entitlement as object, relation, paid event, with a counted bound |
| Dwork and Naor 1992; Back 2002 (Hashcash, first proposed 1997) | make establishing something cost, in a way anyone can check cheaply | a comparison, with no claim that the two are the same: proof of work checks a found value against a difficulty target, while the toll charges one unit at the step that certifies and verifies nothing about effort |

| Prior work | What this book leans on | What this development adds |
|---|---|---|
| Eccles and Jordan 1918 (patent filed) and 1919 (paper); Plotkin 1989; Shapiro and coauthors; Abadi and Lamport 1991; Ahman and coauthors 2018 | the latch; write-once and grow-only state; history variables; monotonic state whose witnessed facts can be recalled later | the latch form over any base, with its two counterexamples and converse |
| Cantor, Gödel, Turing, Kleene, Rice; Lawvere 1969 | the diagonal | the shortcut wall on any substrate; L's recursion theorem by proof |
| Bell, CHSH, Tsirelson, Wehner, NPA, Landau, Masanes, Fine; Echenim and Mhalla 2023 | the game, its walls, and the matrix test | an integer check on counts that implies the pinned matrix is PSD, which implies the bound |
| Buterin and Griffith; Wood; TCG; RFC 9162; Necula | five record disciplines | five fragments read through one question: is the write priced |
| Zurek 2009; Ollivier, Poulin and Zurek 2004; Blume-Kohout and Zurek 2006 | pointer states and redundancy | a thesis borrowing the shape, with the gap to the physics stated |
| Leroy 2009; Klein et al. 2009 | the discipline of saying where the proof stops | the separation of what's proved about the model from what any device running it does, inside its bounds |
| Minsky 1961, 1967; Forster, Larchey-Wendling and coauthors 2019, 2020 | two-counter machines are universal, and their halting is undecidable | a machine of six instruction kinds on two counters where earned commitments, no forging and an exact minimum price of three are theorems, with halting reduced from two-counter halting |
| Forster and Smolka 2017; Forster, Larchey-Wendling and coauthors; Norrish 2011 | L as a model of computation; counter-machine constructions; a compiler from computable functions to counter machines | L's recursion theorem, Rice and halting; a counter compilation putting every computably presented Thiele machine on the universal host |

| Prior work | What this book leans on | What this development adds |
|---|---|---|
| Turing 1936; Post 1947; Popek and Goldberg 1974; Hamlen, Morrisett and Schneider 2006 | the universal machine as one fixed table; hosts that keep guests apart; which policies a monitor can enforce | a definition of certified Thiele-complete that every clock fails; one fixed program in the small machine's own instruction kinds that runs every program of it with halting, flag and ledger exact, earning its own flag; and one fixed program, with a paid move added, that runs every computably presented Thiele machine, its ledger exact to within a surcharge of at most two when the guest's reading starts at no, which can't be avoided in general |

Everything in the right column is stated with its premises in the body of the book, and everything in the middle column is cited where it's used. The table is the short form. The long form is the rest of this appendix, which takes each row, says what was already known with sources, and says what kind of addition each one is. Very little of the right column is a new result. Most of it is things built, new proofs of known things, and known pieces put together under one question.

A few more labels I went and looked up are neighbours. Some of them also sit in the table above, and one, the reference monitor, is the move the small machine's step makes. I list them so you can see why I don't think the toll is one of them under a new name. Each is a claim inviting correction.

- Linear logic (Girard 1987) treats an assumption as a resource that gets used up. It constrains what a program, or a proof, may consume. The toll constrains the model's step function instead. It's the same instinct at a different layer.
- Foundational proof-carrying code (Appel 2001) checks safety proofs against the smallest trusted base it can: a logic plus the machine's semantics. Here no proof rides along with the program. The step rule writes the receipts as the run goes.
- Session types (Honda 1993) constrain how programs talk to each other so the conversation can't go off script. The toll is about what one computation has paid to establish, which is a different problem.
- No Free Lunch (Wolpert and Macready 1997): averaged over every problem, whatever an optimiser gains on some problems it gives back on others. That's an

average over a whole problem space, and the edge is paid for elsewhere. The toll charges one step in one run, with no refund anywhere.

- Kolmogorov complexity (Kolmogorov 1965): the description complexity of a string is the length of the shortest program that prints it, relative to a fixed description language. That's a property of a finished string, with no steps and no moment of paying. The toll is a law about a step.
- Irreversibility cost and information distance (Zurek 1989; Bennett, Gács, Li, Vitányi and Zurek 1998; Li and Vitányi 1996): the fewest bits a reversible machine must take in and erase irreversibly to turn a string $x$ into a string $y$, about $C(x \mid y) + C(y \mid x)$, where $C(x \mid y)$ is the length of the shortest program that prints $x$ given $y$; and Kolchinsky and Wolpert's thermodynamic complexity of a Turing machine's outputs (2020). These tie algorithmic information to steps and erasures, and they price a transformation between strings, or an output, in erased bits. The toll prices the rise of a chosen reading.
- Cook and Levin (Cook 1971; Levin 1973): checking can be cheap while finding looks expensive, and whether finding is really harder is the open P-versus-NP question, a gap in how difficulty grows with size. The toll puts a fixed floor under the one step that commits, however easy or hard the finding was.
- The thermodynamic cost of choosing (Rovelli 2024): settling a binary choice at temperature T dissipates at least kT ln 2, with no outside heat bath. That's the closest thing to the toll itself I've found, because it charges a commitment. An object or an average isn't what it prices. But it's physics, indexed to a temperature, and its choice is unpredictable, a branching, and stable only for a while, where the toll prices a deterministic no-to-yes transition. On my side, finitely many yes-states and permanence are what force the raising move to merge states. The toll charges the commitment in a model of computation, with no temperature in it. It's the closest neighbour, and it's still a different thing.
- Persistent copies (Ouldridge and ten Wolde 2017; Ouldridge, Govern and ten Wolde 2017): making a copy that stays correlated with its original after the two come apart has a thermodynamic floor. The nearest physics I've found to "learning borrows, committing pays." It's work and free energy for a correlation, in chemistry; the toll is a count in a model of computation.
- Thermodynamics of finite automata (Ouldridge and Wolpert 2023): if the smallest automaton for a language has a letter that sends two states to one, recognising that language on a device that repeats one local step produces entropy, and if the smallest automaton only reshuffles, recognition carries no such floor. That's Part I's counting of states with a physical price on it, and one of their worked examples is the latch's language turned inside out. It's the closest neighbour of

all to the road the toll is derived on; Section 3 says exactly what's left over for me.

- CerCo, the Certified Complexity project of Amadio and coauthors, builds a verified C compiler that carries the object code's execution cost back up to annotations on the source. There the cost decorates the program and survives compilation. Here it lives in the model's step rule.
- Reference monitors and execution monitors (Anderson 1972; Schneider 2000; Azevedo de Amorim and coauthors 2015). A small piece every action passes through, enforcing a rule on every program. Schneider argued that a monitor watching a run can enforce only rules whose breaking shows up at a finite point, and "no certificate without an earlier passing check and commit" is one of those; the micro-policies check a tag rule at every instruction, proved correct. That's the move the small machine's step makes, so this one isn't a neighbour at all: the small machine's step is a monitor of this kind. What's added is the edition-bound check-then-commit with a paid, permanent record on top, and the derivation of why the payment can't be zero. Kevin Hamlen, Greg Morrisett and Fred Schneider (2006) went on to sort which rules monitors, program rewriters and static checkers can each enforce, and they differ, which is one reason this book doesn't claim that every enforcement mechanism is a Thiele machine underneath.
- Trusted components in replicated services (Chun and coauthors 2007; Levin and coauthors 2009; Veronese and coauthors 2013; Decouchant and coauthors 2022; Clement, Junqueira, Kate and Rodrigues 2012): a small record inside each replica that a replica taken over still can't make say two things, with a proof of exactly what that buys. The record is theirs. The toll isn't in their picture.
- Ghost state (Owicki and Gries 1976; Filliâtre, Gondelman and Paskevich 2014): variables added for a proof that can be erased without changing what the program computes. The flag and the counter are ghost in that sense. The small machine's table of checked facts isn't, because the step reads it and refuses on it, for every program. The slice of Appendix D is that erasure on the lift, while its trap is down, with two things on top: the erased layer still earns its record, and what's left can't read the record or the ledger (Theorem D.4).
- Verifiable computation and proof-carrying data (Parno, Howell, Gentry and Raykova 2013; Chiesa and Tromer 2010): a short proof, checkable by a stranger, that a computation's output is right. That establishes the claim for the stranger. Producing it may require a witness and substantial computation; it does not identify the original computing device unless that is part of what the proof establishes. Another prover may reproduce it, the copier included. The record here is about one machine's own history. A proof like that is one way to carry the local record out of the box.

- Commitments and entitlements (Brandom 1994): talk as score-keeping, tracking who is committed to what and who is entitled to it. Brandom's entitlements can be defeated and can be passed on by testimony. Here entitlement is priced, step by step, and a copy carries it only as far as the copy is genuine. The words are the same and the machinery is different.
- Correspondence assertions (Woo and Lam 1993; Gordon and Jeffrey 2003). In protocol verification, "this end event must be preceded by a begin event for the same thing" is a correspondence assertion, and a type system can prove a protocol keeps it against any opponent. "No certificate without an earlier commit, and no commit without an earlier check of the same claim" has that shape. Theirs is proved about programs by typing. Mine is enforced by the step, which is the LCF comparison again, at the level of single instructions.
- Monotonic parallel composition (Åman Pohjola and Parrow 2016) and persistence in the pi-calculus (Palamidessi, Saraswat, Valencia and Victor 2006). Åman Pohjola and Parrow separate models where a neighbouring process can never stop a step from models that can solve consensus, and test-and-set, an instruction that writes a cell and reports what was there, is on the consensus side. Palamidessi and coauthors compare messages that are used up with messages that stay, and the end of their study is a reduction from Minsky machines. The small machine, put into Åman Pohjola and Parrow's setting with threads pooled on one shared state (two states that have both moved jam), has a consensus process in their sense, so it lands on their non-monotonic side; their theorem and that placing are argued in a few lines in Section 15. The comparison with Palamidessi and coauthors' persistence stays a comparison.
- Constructor theory (Deutsch and Marletto 2015; Marletto 2016; Marletto, Vedral and coauthors 2022; Violaris and Marletto 2022): physics as which tasks a device that runs in a cycle can do, with irreversible meaning possible while the inverse isn't. It's the nearest physics I know to "committing costs" stated for a cycling device. It's quantum and approximate, and Part I's counting of states is classical, finite and exact (Section 3).
- Accountability and fault detection (Haeberlen and Kuznetsov 2009; Küsters, Truderung and Vogt 2010; Treitz and Künnemann 2026): when an outside party can tell who misbehaved. Their windows are this book's windows, and the trust they leave over, an honest log or a trusted step, is left over here too (Section 7).
- Transferability (Easwaran 2009): a deductive proof lets you learn from someone you don't trust because you can check every step, and a probabilistic proof doesn't. That's the argument for carrying the object and re-running the check (Section 7).

And then the silly question that list raises, which I'm going to chase anyway: why is an optimisation theorem named after lunch? Wolpert and Macready borrowed an old saying, "there ain't no such thing as a free lunch," and the saying is older than almost everyone who gets credit for it. American saloons in the 1800s put out a free lunch for anyone who bought a drink, and the drink paid for the lunch. A Washington newspaper was already saying in 1909 that somebody has to pay for a free lunch; a newspaper fable by Walter Morrow had the whole line in 1938; a book by Pierre Dos Utt already had the squeezed form, TANSTAAFL, as its title in 1949; Robert Heinlein's 1966 novel *The Moon Is a Harsh Mistress* made that squeezed form famous; and the economist Milton Friedman put the saying on the cover of his 1975 book of essays on public policy. I'd have credited Heinlein or Friedman. As far as anything I could find goes, neither coined it. That's Stigler's law again, at lunch, and it matters here because No Free Insight is named like a cousin of the saying, and the family resemblance misleads in one way. The saloon's lunch was paid for somewhere else, in the price of the beer, and that's Wolpert and Macready's shape: a gain here, handed back over there. The toll isn't that. On the ledger nothing gets paid somewhere else: the step that raises the flag pays, on the spot. A blank tape prepared by wiping an uncertain one may have an earlier reset cost under the thermal premises. A supplied blank alone does not prove that payment; the toll is the step that won't hide its share that way.

## B.1 Every step is a fresh state, a merge, or a loop

Known: the step law is the first-return argument every textbook on dynamics makes, and under it sits the pigeonhole. A one-to-one move that raises a reading and keeps it is Hilbert's hotel, everybody moving up a room so that a room is left empty (Hilbert's lecture course of 1924–25, made famous by Gamow in 1947), and a collection that can be matched one-to-one with a part of itself that leaves something out is exactly what Dedekind (1888) called infinite. Recurrence for reversible finite machines is Poincaré's (1890) and Zermelo's (1896) in its finite form. The noisy steps are the fact that a doubly stochastic chain has no closed class it can enter from outside, with Gibbs's inequality for the entropy gap; the unitary steps are that a unitary preserving a finite-dimensional subspace preserves its complement; the bilateral shift is the standard example of what goes wrong without finite dimension. Heat against time is Bennett's Brownian computer (1982) with Schnakenberg's formula for the entropy a biased walk hands its surroundings; the bounds for finite-time changes of probability distributions under specified stochastic dynamics belong to finite-time stochastic thermodynamics (Schmiedl and Seifert, 2007; Shiraishi, Funo and Saito, 2018); and the trade of space for time by recomputing is Bennett's (1989). The measurements are Bérut and coauthors' (2012), Jun, Gavrilov and Bechhoefer's (2014), Hong and

coauthors' (2016) and Gaudenzi and coauthors' (2018). Mine: the reading, that the write of a 0 on a cell is a merge and the toll is the price of something being the case; putting those old facts side by side as one law with three columns (a fresh state, a merge, a loop that makes nothing new) and no fourth, for deterministic runs, beside separate preservation theorems for the stated noisy and quantum models; the merge-or-fresh-state dichotomy for a reading that stays up, with no finiteness in it; and the exact count of states for a bounded memory, $M \geq 1 + (T+1)(N-1)$, with a building that reaches it (Section 3, Section 1); and that a merge seen in a coarse description forces, in any finer description it's exact about, a fine merge, a fine step out of the set, or infinitely many fine states (Theorem 1.10). Kind: synthesis and a reading, with the count of states and the every-scale theorem as small exact lemmas. The mathematics is old. What I'd defend is the arrangement: endless memory, history-keeping and reversible machines aren't a way out of what keeping takes, they're the fresh-state column, which dodges only the floor a merge adds and does not settle the implementation's finite-time dissipation; and finiteness is what takes that column away.

### B.2 The universal floor and the potential method

Known: the potential method of amortized analysis, a function on states whose drop bounds the cost of a run, is Tarjan's (1985). Later developments include Nipkow's accounts (2015 and 2019) and Charguéraud and Pottier's time credits (2019). It has been built into type systems for automatic resource analysis (from Hofmann and Jost, 2003, and Hoffmann and coauthors, 2011, onward), and recast in cost-aware type theories (Niu, Sterling, Grodin and Harper, 2022). Mine: choosing the potential that is one when a designated reading says no and zero when it says yes, so that "the crossing costs" (the toll) and "every run that crosses paid" (the floor) are the potential method's two halves (Section 6), and its count, one unit per crossing (Observation 2.10), the same sum read crossing by crossing. Kind: a known technique, used with one particular potential. The choice is the reading, and it's the point: it makes "was certified" something the account must carry. The numbers come with it. A natural-number potential, as defined here, that reads only the reading has to drop by exactly one from no to yes, or its inequality says something other than the toll on a machine of two states and one move; so it's one at no and zero at yes, up to a constant added to both. The theorem is not new mathematics. If it looks clever, that's Tarjan's cleverness; my part was picking which reading to plug in.

Go back to Tarjan's paper and there's a sentence that sits right on the line this book draws. He opens with the dictionary (to amortize is to put money aside at intervals, as in a sinking fund) and then gives two views of the method. In the banker's view the computer is coin-operated, and each operation is handed credits

to spend or save; in the physicist's view, which he credits to Daniel Sleator, every state has a potential and you count its changes. Then he stops to say that the credits are only an accounting device: the programs that actually run contain no mention of credits or debits. That's exactly right for what he wanted, a bound proved about a program from outside. It's also exactly the gap. In his method the coins live in the proof. The one move I chose was to put them in the step, where no program takes a step without it being counted. The arithmetic is the same, and only the place is different.

## B.3 The toll from finiteness, permanence, and a price on merges

Known: that steps which merge states (no single-valued inverse) are what Landauer's principle charges for is Landauer's (1961), sharpened by Bennett (1973, 1982, 2003; the words "the erasure of a bit or the merging of two computation paths" are from the 2003 abstract); generalizations and the dependence on the input distribution are Maroney's (2009), Sagawa's (2014), Wolpert's; the critique is Earman and Norton's and Norton's. Write-once memory is an old topic in coding theory (Rivest and Shamir, 1982), where the question is reuse, and what it takes is counted in cells, which in this book's words is paying in fresh states; heat and prices never come up, and Chris Heegard (1985) worked out the capacity of permanent memory with noise, again a question of how much fits. Statisticians priced permanence on finite memory in their own currency: in Hellman and Cover's analysis of hypothesis testing with finite memory (1970), a machine that settles its answer by falling into states it never leaves does worse than one that doesn't, so a decision held for good costs memory states. Heat never enters their analysis. And the counting under the theorem is older than any of that, twice over. Theorem 3.1 is the finite, discrete cousin of the recurrence argument Zermelo, using Poincaré's theorem of 1890, turned on Boltzmann in 1896, here aimed at a reading. Zermelo aimed it at entropy, and Bocchieri and Loinger gave the quantum recurrence in 1957. In automata form it is classical: a finite machine whose moves are all one-to-one reshuffles recognises only group languages, and the inputs that raise a permanent reading form a language that is never one of those unless it is empty (Pin's 1992 paper on reversible automata is the reference I'd check; automata theorists treat the step as folklore), and going quantum doesn't change it (Brodsky and Pippenger, 2002: quantum finite automata that measure once, at the end, accept with bounded error exactly the group languages). The closest of all is recent. Ouldridge and Wolpert (2023) give a proof that if the smallest automaton for a language has a step that merges states, every automaton for it does, and argue that recognising it on a device that repeats one local step carries an entropy floor, while a language whose smallest automaton only reshuffles carries no such floor; one of their worked examples is the strings with no $b$, the complement

of the latch's language. That's the core with a physical price already on it. And the logarithm in the heat floor matches the absolute-irreversibility bound of Murashita, Funo and Ueda (2014), which Manzano, Kardeş, Roldán and Wolpert (2024) carried over to computation. The logarithms coincide for the stated counts. This does not supply a physical reverse process: absolute irreversibility needs forward and reverse path probabilities that the logical map alone does not specify. Constructor theory states a cousin of the same core for devices that run in a cycle: Marletto, Vedral and coauthors (2022) and Violaris and Marletto (2022) built an irreversible task out of the quantum homogeniser and argued that erasure may dissipate more than Landauer's bound. Section 3 tells all of it. The price isn't mine: Li and Vitányi priced each erased bit (1996), and Demaine and his coauthors priced named instructions that destroy bits (2016). Mine is what a flat price of at least one on every merging move does to a reading that stays: it turns "the flip merges" into "the flip costs" on any machine with finitely many states reading yes, whether or not anybody's physics is right, and gives the toll (Observation 3.3); what comes with that form, namely the characterisation of the moves every merge-pricing schedule must charge (the merging moves, so the core again in pricing words), the merge-or-revoke dichotomy for a single move, the squeeze and entropy forms for a permanent flip, and a built counterexample for each premise (Section 3). The counterexample for finiteness is the endless strip, the machine that keeps every history, and it says what the premise is for: the strip raises its reading with no merge by taking a fresh state at every step, so a price list that charges merges and nothing else lets it certify at a cost of zero. That's a price list that charges nothing for the fresh states, not a machine that got out of paying, and its physical cost remains implementation-dependent. Kind: the core is classical, and somebody else's: Zermelo's in physics, folklore in automata theory, with Pin's paper as the place to check it. The pricing form isn't wholly mine either: priced in entropy, for automata, it's Ouldridge and Wolpert's. They derive the price from the physics. An assumed price is older than mine too: Li and Vitányi (1996) took a unit charge per irreversibly erased bit, with reversible steps priced at zero, as axioms of a model of computation, and Demaine and his coauthors (2016) put that charge on named instructions of the random-access machine. What's mine is the toll derived from a price on merging moves for a reading that stays, its single-move and counted forms, and the counterexamples. The physics is a named premise and is Landauer's. Searches: the core turned up in the places just named, and its physical price in Ouldridge and Wolpert; I didn't find the derived toll stated anywhere I looked, and nobody's search covers a whole field, mine least of all.

### B.4 Learning borrows, committing pays

Known: that a measurement can be made without merging anything, so that Landauer's premise sets it no floor, and that the floor lands when the demon wipes its notes to look again, is Bennett's resolution of Maxwell's demon (1982), with the quantitative trade-off by Sagawa and Ueda (2009); that the entropy can be split between measuring and erasing in more than one way is Fahn's (1996), and that the work an erasure takes depends on what the eraser already knows is del Rio and colleagues' (2011). The closest physics I found to the line between learning and committing is Ouldridge and ten Wolde's fundamental costs of persistent copies (2017; with Govern, 2017), copies that stay correlated with their originals after the two come apart; theirs is a work and free-energy bound for a correlation in chemistry, and Section 4 says how it differs. The sharp chemical form, that a persistent copy grown at equilibrium carries no information about its template, is Poulton, ten Wolde and Ouldridge's (2019). Mine: the four-state demon, where copying a hidden bit onto a display known to be blank merges nothing and wiping the display merges two pairs of states; the second copy that hands the display back blank while the original is still there; the three-state ring and the two-state impossibility (Section 4); and the certainty bound (Theorem 4.5), which splits the uncertainty about the start into the part a run destroyed, paid for in units of cost, and the part the end state still holds. That split is Shannon's chain rule (1948), and bounds of this kind on the actual spread are Maroney's (2009) and Sagawa's (2014); mine is reading it on the ledger, about what an observer saw. Kind: finite models and one bookkeeping identity illustrating a known physical point inside my ledger, and not a new physical claim. Bennett did the hard part in 1982. My four dots put the line where you can work it through yourself: the copy owes no floor and can cost nothing, but it still takes something. Its physical cost depends on the dynamics and protocol. And it uses up a blank page, which is a fresh state; while the original is still there it can hand the page back, so learning borrows. Resetting the uncertain page without using its correlations has the stated floor. A supplied blank alone does not prove a previous payment. Committing makes the record permanent, so there's nothing to hand back, and with finitely many states it has to merge.

### B.5 The counter as a receipt

Known: one-way counters that give order and termination are old: Turing's 1949 note on checking a large routine, Floyd's ranking functions (1967), Lamport's clocks (1978). Carrying a run's cost beside its result is the writer cost of Danner, Licata and Ramyaa (2015) and of Danner and Licata (2022), and the graded effect semantics of Katsumata (2014), with a working-language version in Orchard, Liepelt and

Eades (2019). The receipt side of time credits is Mével, Jourdan and Pottier's (2019): in their extension of the Iris logic every step uses up a credit and produces a receipt, receipts prove lower bounds, and enough of them prove that a counter can't overflow in any feasible time, which is the closest published neighbour of a total that only climbs. Mine: putting three properties side by side (the total only climbs, it equals the sum of the charges on every path including the failures, and it is the only such total, on the states a run from the start reaches, for a given price list), the argument for any system with a counter and a price list, and the sum proved from the small machine's own step (Section 6, Section 15). Kind: synthesis; the abstract statements are proved for any system with a counter and a price list, and on the small machine the sum is proved too. Against the graded frameworks the answer is split: the counter is their writer exactly, and their type systems, bounding relations and before-the-run checking of grades are not embedded, because the grade here is written by the step during the run (Section 6). For the well-graded, loop-free programs defined in Section 6, on runs that do not trap, the counter's rise equals the fixed grade. The loop built there costs its input value, so no fixed whole number is its exact grade on every input; an input-dependent grade is exact. The arguments are given there. Cost-aware type theories such as calf allow grades that depend on the inputs.

### B.6 Windows that can't price what they hide

Known: a function of an observation is constant on the observation's fibres; that's elementary, and the systems version of it is Moore's distinguishing experiments (1956), Kalman's observability (1960), and the completeness question of abstract interpretation (Cousot and Cousot, 1977; Giacobazzi, Ranzato and Scozzari, 2000); in control theory it is diagnosability and opacity of discrete-event systems (Sampath and coauthors, 1995; Saboori and Hadjicostis, 2007), where a collision is a hidden event no observer can diagnose without delay; in economics it is the hard-evidence view of mechanism design (Green and Laffont, 1986; Bull and Watson, 2007), where what a mechanism can tell apart is limited by the evidence on offer; and its sharp form for fingerprints is Naor and Rothblum's lower bound for checking a stored file (2005); in distributed systems it is Haeberlen and Kuznetsov's (2009), that only omission and commission faults can be detected from outside, and in philosophy it is Easwaran's transferability (2009). Mine: stating the pricing version (no window price is exact across a collision), exhibiting a colliding pair for the small machine's window, and the theorem that every certified Thiele-complete machine has two runs from one clean start that end looking the same through the window its base shows, one certified and one not (Section 7, Section 15). Kind: an elementary observation about what a window hides, plus specific facts about the machine and the definition.

### B.7 Structural entitlement and its bound

Known: decision-tree counting bounds (a tree that tells $N$ cases apart has depth at least $\log_2 N$); certifying algorithms that hand over checkable certificates (McConnell, Mehlhorn, Näher and Schweitzer, 2011); proof-carrying code (Necula, 1997); the gap between justified true belief and knowledge (Gettier, 1963). Mine: the definition of entitlement as an object, a checked relation, and a paid event on the record, and the count that, given lists, a tree and a covering, a run that paid for the tree's depth pays at least the index bits it commits to (Section 8). Kind: synthesis; the counting idea is standard, the assembly and its vocabulary are this work's, the pricing policy is assumed (nothing derives it), and the count is proved in its cost form over any Thiele machine, and for the record moves of every certified Thiele-complete machine on a run from a clean start, with the earned chain in the run, a floor that takes the run's own checks in place of a tree, and a counterexample showing the covering can't be dropped (Section 15.14); and in a form where the run makes the posterior from a prior, every drop is made by a check the run passed, and the index bits are at most $k$ times the cost when each passing check keeps a $2^{-k}$ share, a hypothesis the same counterexample shows is needed (Theorems 8.4 and 15.38).

### B.8 The record axis is a latch

Known: monotone, write-once state is old: the latch (Eccles and Jordan, patent filed 1918, paper 1919), sticky bits (Plotkin, 1989), grow-only state in replicated data types (Shapiro and coauthors), history variables and stuttering in refinement (Abadi and Lamport, 1991), append-only authenticated sets for transparency logs, which fit revocation as a once-only event (Tomescu and coauthors, 2019), and, in program logic, monotonic state, where a property witnessed once on state that only moves forward can be recalled at any later point, with a soundness proof and a working home in F$^\star$ (Ahman, Fournet, Hriţcu, Maillard, Rastogi and Swamy, 2018): the type-theory name for a record that stays up. Past the finite case, inductive and quasi-inductive definitions (Richter and Aczel, 1974; Burgess, 1986) and the flag trick of infinite time Turing machines (Hamkins and Lewis, 2000) are the transfinite versions of a record that only grows and a toggle, and Danos and Krivine's (2004) reversible calculus needs an action declared irreversible in order to commit. Mine: the exact statement that a record driven by the computation that never switches off is, over any base, a latch on one event, with its converse and a counterexample for each of the two conditions, and the argument that base-driven records agree up to the price list while the stronger uniqueness fails (Section 10). Kind: a small structural lemma, I think not stated before in this form (I found no prior statement

in the searches I ran), built from old notions. The proofs show why the record must latch and why its price list remains a separate choice.

### B.9 The diagonal and L

Known: Kleene's recursion theorem and Rice's theorem; Lawvere's fixed-point form (1969) and Yanofsky's retelling (2003); later detailed developments of recursion theorems and Rice (Norrish, 2011; Carneiro; Forster, Kunze, Smolka and coauthors, including L itself, 2017, and their library of undecidability results). Mine: the diagonal stated once over any substrate that supplies a recursion theorem; and Rice and halting proved again for L from its reduction rules (Section 13); and Rice's and Kleene's theorems proved for the small machine's host, with Rice's on every input of the two-counter machine itself, Kleene's there with inputs written as powers of two, and Kleene's refuted for plain inputs (Appendix C.4). Kind: new proofs of known theorems of computability, for particular machines. The theorems hold for any model with universality and s-m-n, and the work is in building those for each machine; the refutation on plain inputs is a small fact about two counters, resting on Schroeppel's.

### B.10 An example machine that meets the definition

Known: two-counter machines and the undecidability of their halting are Minsky's (1961, 1967), developed further by Forster, Larchey-Wendling and coauthors (2019, 2020). The guarantee about every program is older than any of this, and it isn't mine either. It's the reference monitor (Anderson, 1972): a check that sits in the path of every operation, so no program gets around it. Schneider (2000) worked out what a monitor that watches a run can enforce: only safety properties, the kind a bad run breaks at some finite moment, and "no CERTIFY without an earlier earned COMMIT" is one of those. Azevedo de Amorim and coauthors (2015) verified monitors that check tags at every instruction. The shape of "earned," a certificate nothing can make except through the rules, is Milner's LCF, where a theorem is a value only the logic's rules can build (Gordon, Milner and Wadsworth, 1979; Gordon, 2000); the small machine is that architecture at the level of single instructions. "No certify without an earlier commit" is also a correspondence assertion in the sense of Woo and Lam (1993), which Gordon and Jeffrey (2003) check with a type system, and a store of commitments that the rules of a dialogue keep, each speaker's updated by the rules as the moves are made, is Hamblin's (1970, chapter 8, on formal dialectic); the difference from all three is again that here the step enforces it. A check that blesses a value and a write that takes the blessing away is close kin to dynamic taint tracking, whose labels go back to Denning's lattice model of information flow

(1976), with the blessing itself, controlled endorsement, given a type system and a security theorem by Cecchetti, Myers and Arden (2017), and what a monitor can enforce with bounded memory is Talhi, Tawbi and Debbabi's question (2008). A small monotone record inside each replica of a replicated service is A2M (Chun and coauthors, 2007), TrInc (Levin and coauthors, 2009), MinBFT (Veronese and coauthors, 2013) and Damysus (Decouchant and coauthors, 2022), and Clement, Junqueira, Kate and Rodrigues (2012) worked out, with proofs, what such a record buys and what it doesn't. The signature route to the same goal, which needs no trusted step, is IA-CCF (Shamis and coauthors, 2021): replicas sign and log their protocol messages in the ledger, so blame lands on individual members however many of them misbehave. And "check once, issue a capability the system trusts, rely on it later" is built and deployed in Garg and Pfenning's proof-carrying file system (2010), whose logic lets a policy's consequences change with time and state, and in Sirer and coauthors' logical attestation for the Nexus operating system (2011), unforgeable statements about programs used as credentials; those are the closest systems I've found to check, commit, certify. Mine: one worked example, a machine of six instruction kinds on two counters, proved to meet the definition so that the definition is not empty, with a version for each counter, a bounded fact table and a channel; its theorems that, from a clean start, a committed fact was checked first about the same edition, that nothing is forged, that nothing is evicted, that with the flag up from a clean start the channel names a fact an earlier passing COMMIT committed, and a later commit can replace it, as a two-commit run shows, and that a run from a clean start to a raised flag adds at least three to the ledger, with three reached by a run whose check could have failed (the step doesn't police trivial checks, so one that can't fail costs three too); the same theorems for any property language with an exact equality test whose checker is proved to say yes exactly when the claim holds, a sorted list read out of a counter included; and the step-for-step match with the two-counter machine, so halting is undecidable for it (Section 15). Kind: an instance of monitor enforcement with two things on top that I didn't find put together where I looked: the check is bound to the edition of the thing it checked, and a toll is paid for the commitment. The first of those alone is old: checking at commit time that nothing you read has changed since is optimistic concurrency control (Kung and Robinson, 1981), and each counter's version here is that scheme's version number. What I didn't find is that check put together with a toll. The universality and the undecidability are borrowed, and the other properties are short consequences of the step rule. Whether the design is new I can't say, and I make no claim beyond the proofs.

### B.11 Thiele-complete and certified Thiele-complete, defined so a clock fails

Known: Turing completeness, and the observation every theorist makes in four seconds, that a toll alone is cheap: a machine that charges for every step pays any toll. Enforcement mechanisms have been compared and sorted before, and they aren't one notion seen several ways: James Morris (1973) separated access limitation from authentication in programming languages, and Kevin Hamlen, Greg Morrisett and Fred Schneider (2006) classified which policies execution monitors, program rewriters and static analysis can each enforce, and they differ; Wai Wong (1995), on a neighbouring track, recorded and checked HOL proofs so a theorem could carry its derivation. Mine: a definition of Thiele-complete with four clauses that say nothing about what the record is about (computing that runs every program, costs nothing and can't touch the record, while every other move costs one; a record that stays; blank starts; and some run whose record tells two starts apart), with the proofs that a clock and a silent machine fail it, that each clause is needed, that every universal base becomes one by adding marks, and that every machine meeting it hides its record and ledger from its own window (Section 10.8); and a definition of certified Thiele-complete with four clauses (a universal base that can't touch the record and a record that stays up once it's up, an earned record, exact prices, nothing for the base and one for each of the three record acts, and some claim whose check can fail), the proofs that every clock-style machine and a paid latch and a silent machine fail it while the small machine, the same machine over any property language with an exact equality test, a checker proved to say yes exactly when the property holds and a property that holds of one number and fails of another, and the universal program's host meet it, that the four clauses are independent, each failed by some machine with a two-counter machine inside that meets the other three (Theorem 15.29), and the theorem that every machine meeting it hides its record and ledger from its own window (Section 15), with the theorem that it is exactly Thiele-complete with clean starts, the earned chain and a claim that could have failed added (Theorem 15.23). Kind: definitions with short consequences. The window theorem is the fibre fact of item 6 plus one observation, that a universal base prints any window at no cost on the ledger. I'm not claiming the definition is the only reasonable one, or that LCF kernels, reference monitors and proof-carrying code are all certified Thiele-complete machines in disguise; Hamlen, Morrisett and Schneider's sorting says such mechanisms genuinely differ, and the book doesn't have a theorem showing the definition holds up across such differences.

### B.12 Five record disciplines, one question

Known: each system and its guarantee belongs to its designers: Casper (Buterin and Griffith, 2017), Ethereum's storage gas and refunds (EIP-2200, EIP-3529), extend-only TPM registers, Certificate Transparency (RFC 9162), proof-carrying code. Mine: asking all five whether the write is priced, with a small worked fragment of each (Section 17). Kind: synthesis and an organizing reading; the fragments are not verifications of those systems.

### B.13 The pointer thesis

Known: Zurek's quantum Darwinism (2009), with Ollivier, Poulin and Zurek (2004) and Blume-Kohout and Zurek (2006): preferred states are the ones whose records proliferate redundantly in the environment. Mine: borrowing that shape for the question of which event deserves a record, with definitions (copied and relied on by parties who didn't produce it, independent carriers), two toys, and refutations of two stronger forms (Section 18). Kind: thesis. No claim about quantum physics.

### B.14 The universal host for the example

Known: Turing's universal machine (1936), which runs any machine given as data, and the universal register machines that run two-counter programs. Keeping a guest and the code running beside it out of each other's state is what a virtual machine monitor promises (Popek and Goldberg, 1974). Compiling computable functions into counter machines is Forster, Larchey-Wendling and coauthors' work. Mine: one fixed program, $U$, written in the small machine's own six kinds of instruction on a host with a register for every whole number, with no guest step built in, that runs every program of the small machine from every clean start start$(x, y)$, halts exactly when its guest does, raises its own flag by its own check, commit and certificate if and only if the guest's rises somewhere in its run, and keeps a ledger equal to the guest's at every matching point; its host machine is certified Thiele-complete, and whether $U$ halts on a loaded guest is undecidable (Appendix C.1). Two small facts force its shape, and they're stated there: reading a counter above 0 into the program's control flow on this machine writes it, and a finite program can't keep the guest's facts on a copy of the guest's counter, by the pigeonhole. Beside it, a simpler host with a built-in guest step runs guests by construction and keeps a mirror of the guest's flag, with the mirror, isolation and earned-chain theorems. Code that shares unprotected memory with a record can overwrite it, and ordinary computers isolate memory and enforce rules with monitors too, so what I claim is narrower: the simulation and the record guarantee proved for this one fixed program $U$. Kind: an implementation, with its proofs, of a design whose shape

is Turing's universal machine, with the record and the toll carried through. The universality is borrowed from the small machine and from Minsky. And the same program with one paid move added, $U_P$, runs every computably presented Thiele machine, with its halting matched, its reading carried as a latch on the host's own earned flag whose claim is that reading, and its ledger to within a surcharge of at most two when the reading starts at no (exactly three when it starts at yes), which no program on the host, run from a clean start, can avoid when the guest certifies with its whole ledger below three (Appendix C.2); the counter-compilation construction is given in Appendix J. Kind for that part: an implementation with its proofs again. That a universal machine runs any computable machine is Turing's; what's carried through here is the record, earned again, and the prices, exact up to the surcharge, and I didn't find that put together where I looked. In the words of Definition C.1, $U$ is a universal Thiele machine for the small machine's programs with no surcharge, and $U_P$ is one for every computably presented Thiele machine with a surcharge of at most two when the guest's reading starts at no and three when it starts at yes.

### B.15 The game check

Known: that positive semidefiniteness of a correlation matrix gives the Tsirelson bound is Tsirelson's (1980) and Stephanie Wehner's semidefinite proof (2006), from a paper on Tsirelson bounds for generalized CHSH inequalities; the moment-matrix hierarchy, the Schur-complement analysis of the level-one matrix of this scenario and the level $1 + AB$ are Navascués, Pironio and Acín's (2007, 2008), with both four-body cells unconstrained where my matrix ties them; that every point with a PSD completion is the correlator table of a quantum strategy, built from anticommuting observables in a maximally entangled state, is Tsirelson's too (1980); the arcsine characterization is Landau's (1988) and Masanes's (2003); Echenim and Mhalla (2023) give a further detailed derivation of the CHSH and Tsirelson bounds. Rational sum-of-squares certificates are Helfried Peyrl and Pablo A. Parrilo's (2008), from a paper on computing sum-of-squares decompositions with rational coefficients. Mine: a decision procedure in integer arithmetic on outcome counts; a proved chain from "the counts pass" to "the pinned level-one matrix is positive semidefinite" to "$S^2 \leq 8$"; explicit constructions of the representation in finite dimension, on two four-level systems with real amplitudes and two two-level ones with complex, and a proof of the arcsine characterization; proofs of the contraction equivalence by explicit test vectors; and the level-$1 + AB$ cascade (on the slice $\langle \{A_0, A_1\} \otimes \{B_0, B_1\} \rangle = 0$) and an elliptope gate in whole numbers, with a soundness proof (Appendix G and its derivation, Appendix H). Kind: an integer procedure connecting counts to these known bounds, with known theorems (Tsirelson's representation, the

arcsine characterization, the equivalence of contraction and PSD) proved again. I found no earlier chain that takes counts as input where I looked. The same check earned on the small machine's host, a raised flag standing on it, is built and proved (Theorems G.4 and G.5); the idea isn't new.

### B.16 What that adds up to

Read straight down the list. The mathematics underneath this book is mostly known, and some of it is very old. The question organizing this book is what it costs for a distinction to come into being and stay, written into the step of a model of computation, with certification, a record that a claim was checked, as its sharpest case, and what's mine in the answer isn't one result. It's four, and I name them the same way wherever they come up. The toll derived from finiteness, permanence and a price on merges (item 3; the counting is classical, the finite cousin of Zermelo's recurrence objection and folklore about permutation automata, and Ouldridge and Wolpert already priced it in entropy for automata, so what's mine is the toll a price on merging moves gives for a reading that stays, after Li and Vitányi's price per erased bit and Demaine and his coauthors' price on instructions, and the price is an abstract premise motivated by Landauer's reset). The pricing form of the window fact (item 6; the fact itself is elementary). The latch form of a record that never comes down (item 8; latches and history variables are old). And a record that's priced, and on the small machine earned, under the reference monitor's every-program move (items 10, 11 and 14; the move is Anderson's, Schneider's and the micro-policy authors', and what's added is the check bound to its edition, the toll on top, a definition of Thiele-complete that a clock fails, over any base and any event, a certified refinement the small machine meets, and universal programs that carry the check and the toll to every small-machine program and to every computably presented Thiele machine, earning their own flag). Under all four sits item 1, which is old mathematics read one way: every step of every machine run by one rule takes a fresh state, merges, or goes round a loop, a merge in a coarse description is, in every finer description it is exact about, a fine merge, a fine step out of the set, or endlessly many fresh states, specified thermal models add their own dissipation bounds, and finiteness is what takes the fresh states away. The first is a ledger form of a core that's classical and already priced in physics; the second and third are small exact statements I didn't find stated where I looked; the fourth is built, with its proofs on the page, and its idea isn't new. The general statements are proved for every machine that meets the definition; the small machine and the programs that run on it are the worked instance that shows the definition can be met. Around it is my best attempt at saying which is which, which you should check as hard as anything else in here.

If someone shows me item 3's ledger form, the pricing form in item 6, or item 8 stated earlier, I'll move it up the list.

APPENDIX C

# The universal machines

This appendix carries the two universal programs of Section 16 in full: $U$, which runs every program of the small machine, and $U_P$, which runs every computably presented Thiele machine, with the simpler host that has the guest step built in, the diagonal on the small machine's programs, and exactly what the universal machines cover.

## C.1 One machine that runs them all

The small machine raises its flag only by checking, committing and certifying, and it pays for that. The next question is whether one machine can stand in for a whole class of machines, the way Turing's universal machine stands in for every Turing machine. What I mean by that is a *universal Thiele machine*: one fixed machine running one fixed program that runs any machine of a stated class as a guest, with every step of the guest's computing done by the host's own counter moves and every rise of the host's record put where the host's program put it. No program running on the host has to keep a convention for it. In full:

**Definition C.1** (Universal Thiele machine)**.** Fix a class $C$ of Thiele machines, each with its runs from given start states. A *universal Thiele machine for* $C$ is a Thiele machine $H$, the *host*, with one program $P$ fixed for the whole class, a way to load any guest $M$ in $C$ with any start $s_0$ into a start state of $H$, and a way to read a guest's state off the host's registers, such that: (1) $H$ is Thiele-complete (Definition 10.6), and its computing moves are its counter moves; (2) the run of $P$ from the loaded state halts exactly when $M$'s run from $s_0$ stops, and then shows the observable part of $M$'s last state, as the class names it, its reading kept as a latch, and its ledger up to the surcharge; (3) the run passes, in order, through host states that show the observable part of $M$'s states along its run, at which the host's record equals $M$'s reading kept as a latch and the host's ledger equals $M$'s ledger plus a surcharge, never negative and bounded by a number fixed for the class; (4) *the rise is the program's*: from every untrapped state, a record move selected by $P$ either traps, or leaves the flag up exactly when it was already up or the current line is marked

as raising by $P$. With the flag down and the move succeeding, that line decides whether it rises; (5) $H$ is a *register host*: its counter moves are INC, DEC and HALT on registers, its record moves leave the registers alone and go to the next line or trap, and which move comes next depends only on the line and whether the trap is up; (6) each guest is loaded with the record down, no trap, a fixed line and the guest written into the registers, and the guest's state is read off the registers in a coding fixed with the class, before any guest is named. A *certified* universal Thiele machine also has its host certified Thiele-complete with clean loads, and its record, whenever it's up, standing on the host's own check of the guest's own justification, as the class supplies it.

For $U$ the guest's justification is its chain claim; for $U_P$, the guest's reading at a state of the run. For $U$ the observable part I've proved it matches with all six clauses is the guest's two counters, and in order it matches everything but the guest's version numbers, which no run of the guest can show; for $U_P$ it's the whole state.

The guest lives only in the loaded state; the machine and the program are fixed before any guest is named. For $U$ the surcharge is zero; for $U_P$ it is at most two when the guest's reading starts at no, and exactly three when it starts at yes, so three bounds it over the whole class. Both are certified universal Thiele machines, by the theorems of this appendix put together: each meets the six clauses, each host is certified Thiele-complete with clean loads, and each one's flag stands on its own check of the guest's justification.

The two that $U_P$ adds once a guest's record is up is, at worst, the price of certification, not of universality. A certified host raises its record from a clean load only after three marks, so on a guest whose whole ledger is one when its reading first says yes it falls two behind at least. A host without certification can keep up exactly: on a host with registers, INC, DEC, HALT, a paid move TICK that always raises the record, a paid move PAY that never does, and a paid move MARK that raises it when register 0 is at least 1, which no program below runs and which Thiele-complete needs, since without it no run's record depends on where it started, the program "count register 0 down to zero; TICK; HALT" runs the guests that count a pile down to zero and then raise their reading at a cost of one, each from its own pile, with surcharge zero. A guest whose reading is already yes at its start costs any host loaded with its record down at least one, and one is reached. And from a start at no one host keeps up exactly on the whole class: $U_P$'s compiler on that host, with the three-act chain at the first yes replaced by one TICK and the rest of that move's price paid in PAYs. The toll on the guest makes that move cost at least one, so the TICK always fits inside it, and the host's ledger is the guest's at every matching point. On a host without certification the surcharge is zero from no and one from yes, and neither can be lowered. (And PAY is needed on such a host: a guest that pays while

its reading is down makes the host run a record move that doesn't raise, which is what PAY is.)

Each clause stops something. For clauses (1), (4) and (5) there's a host that meets the others and fails it; (2) and (3) together stop a host that ignores its guest; (6) I state in words below. Clause (1) stops a host that raises its record early and lowers it again before anyone looks. Clause (4) stops a mark that reads the registers, or a copy of the guest kept outside them, and raises when the guest would: the same move on the same line would raise from one state and not from another. Clause (5) stops a host with a move that takes one step of the guest. In the one I built that move costs nothing on the guest's unpriced instructions and traps on its paid ones; costing nothing, it would have to be a counter move, and a counter move never traps. Make such a move cost one instead and it's a record move, which has to leave the registers alone, so it can't move the guest's counters, which (6) puts there. Clause (5) also stops a move costing nothing that fetches, decodes and carries out a guest instruction on the registers, because that move adds one to one register from one state and to another register from another, which no single INC or DEC does. Clause (6) makes the guest come in through the registers. It's the one clause I've stated in words without a theorem that it discriminates: a reading of the registers that adds part of the guest back in isn't excluded by anything I've proved. What stops a host that ignores its guest is (2) and (3), on any class whose guests differ in when they halt or what they record, which a universal class does. What the clauses allow is exact too. By (1), (4) and (5), the line, the registers, the record and the ledger of a run before any trap depend only on the line, registers, record and ledger it was loaded with: each counter move acts on those alone, each record move leaves the registers and moves the line, and its effect on the record is fixed by its line. So a host may carry anything else along, even a copy of the guest that its counter moves step, and what it carries along that way can't touch anything the host shows. One route I haven't closed: a move that traps may read anything, and stopping the run is all it can do. $U_P$ never traps on its class, so for it the route is shut. For $U$ it's shut too: $U$ is loaded with every version at 0 and an empty table and channel, so whatever a trapping move of $U$ reads was written by the run itself, from the registers it was loaded with. For a host in general it stays open.

None of this needs certification. $U_P$'s own program, read on a host with no claims, no checker, no table and no channel but with that MARK, with CHECK, COMMIT and PAY read as PAY and CERTIFY as TICK, meets (1) to (6) with the same surcharge: $U_P$ never traps on its class, and while nothing traps the two hosts agree, step for step, on the line, the registers, the record and the ledger. Certification adds the tie to the guest's own justification, and it needs (4) as much as anything: a host that carries out a hidden copy's own check, commit and certificate meets the tie and fails only (4) and (5).

And I want it the hard way. The easy way is to give the host a move that means "take one step of the guest," and then "it runs any guest" is true because I wrote it that way. There's a host like that two subsections on, after the two programs, and it's useful as a picture. It isn't Turing's kind of universal. Turing's universal machine has no instruction for "do what the machine on the tape would do." It has the same kind of instructions every other Turing machine has, and one fixed table, and the table does the reading and the deciding and the carrying out. The bar here is the same, and clauses (4) and (5) make it exact: one fixed program, every computing step a counter move, and every rise performed by that program. The table, channel and trap still govern whether record operations succeed; the restricted computing-step claim is not independence from all auxiliary state.

Turing's own explanation is worth having in your head, because the program below copies its shape. In the 1936 paper the universal machine's tape starts with the machine to imitate, written out as a string of letters, its standard description, and after it the imitated machine's complete configurations, one after another: where its head is, what state it's in, what's on its tape. To take one step, the universal machine finds the last configuration, looks through the description line by line, comparing, until it finds the line for that state and that symbol, and then writes out the next configuration the way that line says. Then it goes back and does it again. It leaves marks on squares so it doesn't lose its place. Every bit of that is ordinary instructions on an ordinary tape, and nothing in its table knows any particular machine.

Turing's universal program came first on paper, and one that people actually came to use arrived through the side door. John McCarthy, who had started the artificial intelligence project at MIT with Minsky, was writing up his new language Lisp around 1960 and wanted to show it was neater than Turing machines, so he wrote a universal function for it, *eval*, the language describing how to evaluate itself, and argued it was shorter and easier to follow than a universal Turing machine. In his own history of Lisp, that was for the paper, a thing to read; the notation for writing programs as data was made up for the paper with no thought of using it in practice. Then Steve Russell (S. R. Russell in McCarthy's account), a programmer on that MIT project, noticed that eval could serve as an interpreter and coded it by hand, and there was a running language with an interpreter that nobody had planned. McCarthy adds that the surprise froze some decisions he'd made "rather lightheartedly" for the paper. So the trick has a habit: written to be read, it ends up being run. The program below goes the other way round, on purpose. It's written to run, and the proofs show that what runs is what the page says.

**The host.** The program runs on a machine with the small machine's instructions and the small machine's prices: INC, DEC, HALT costing nothing, and CHECK, COMMIT, CERTIFY costing one each, with the same table of sixteen facts, the same channel, trap, ledger and flag. One change: it has a counter for every whole number,

each with its own version, where the small machine had two. There's no new kind of instruction and no new price. Take away the table, the channel and the flag and what's left is old: a machine with as many registers as it needs, adding one, taking one away and jumping on zero, is the unlimited register machine of John Shepherdson and H. E. Sturgis, two logicians who proved in 1963 that such machines compute every partial recursive function.

**Definition C.2** (The host machine)**.** The *host machine* is the machine of Definition 15.3 with one change: it has a counter, with its version, for every whole number $r$ (a *register*), and INC $r$, DEC $r$ $j$, CHECK $p$ $r$ and COMMIT $p$ $r$ name one register each. Its property language has one property, Slot. Write $\langle m, n\rangle$ for the number $(2n+1)\cdot 2^m$; every positive number is $\langle m, n\rangle$ for exactly one pair, and 0 is no pair. Number the small machine's properties Zero $\mapsto$ 0, Even $\mapsto$ 1, Ge$(n)$ $\mapsto$ $n+2$. The checker of Slot says yes at a value $x$ exactly when $x = \langle m, v\rangle$ and the small machine's checker says yes to the property numbered $m$ at $v$; its meaning is the same with holds in place of the checker.

There's one place the guest's machine shows up in the host. The host's single property knows how to read the small machine's three-word property language: "this number is the pair (property, value) and the property holds of the value." It doesn't know the small machine's step. Knowing the language and not the step is the counter version of Turing's universal machine knowing the alphabet its guests write in.

**Two facts that force the design.** Why more counters? The answer is two facts, and the program looks the way it does because of them.

The first is about reading. On this machine a program that wants a counter's value in its own control flow has to take pebbles out (DEC) and put them back (INC), and every INC, and every DEC on a counter above 0, is a write, so the counter's version goes up. A fact is about one version, so a read done that way kills every live fact about the counter. Not every look is a write: CHECK reads a value without writing it, a DEC on 0 changes no version, and a fresh CHECK after such a read can make a new live fact. But the guest's counters are exactly the ones the host has to read by taking pebbles out, again and again, to decide which way a DEC jumps and to work out a property's answer, and each of those reads kills whatever fact was kept there. So the host can't keep the guest's facts on the copies it reads, and a copy it only writes is what the second fact rules out.

The second says that even if reading wrote nothing, a fixed program couldn't keep them by sending each guest property to one of a fixed, finite list of host properties and checking it on the same register the guest's fact is about. The theorem below rules out that and nothing more, and the pigeonhole is what rules it out.

□ **Theorem C.1** (No host fact on an exact copy of the guest's counter)**.** Take any host property language with an exact equality test and any checker, a finite list $Q$ of host properties, and any translation of the small machine's properties into members of $Q$ whose host check passes whenever the guest's check passes on the same number. Then there are $n \neq m$, with $\mathrm{Ge}(n)$ and $\mathrm{Ge}(m)$ translated to the same host property, such that the guest program CHECK $\mathrm{Ge}(n)$ $A$; COMMIT $\mathrm{Ge}(m)$ $A$ traps within two steps from every start, while on every host state with the trap down and room in its table, and some register holding a number at least $n$, the translated CHECK and COMMIT on that register both pass and put the committed claim in the channel.

So the host keeps the guest's facts somewhere else: in *slots*, extra registers used for nothing but holding facts, one per checked fact, each holding the pair (property number, value) that the guest checked. The pigeonhole doesn't bite a slot, because the property lives in the slot's number. The instruction doesn't have to carry it: CHECK Slot on a slot asks about whatever pair is sitting there, and there are infinitely many pairs. A slot's number changes only when the program is moving a fact into it. And when the guest writes the counter a fact was about, the program gives every slot about that counter a nudge, one INC and one DEC, which leaves the number alone and pushes the version up, so the host's facts go stale at exactly the moment the guest's do.

The stored fact still names edition $v$. The two writes restore 18 but do not restore that edition. The intermediate row is inside the handler; the simulation promise is at the completed handler's endpoint.

Count what costs anything. Fetching, unpacking, copying, nudging and branching are all INC and DEC, and all cost nothing, as computing does on the small machine. The only paid instructions are the CHECK, COMMIT and CERTIFY sites, and each guest CHECK, COMMIT or CERTIFY makes the program run exactly one of them. So the host's ledger lands on the guest's, unit for unit. And when the guest's flag goes up, the host's goes up by the host's own CERTIFY, standing on the host's own COMMIT and CHECK on a slot. The host isn't copying a bit; it runs the same chain on its own registers.

**Definition C.3** (The program $U$, and loading a guest)**.** $U$ is the fixed list of 3728 host instructions specified by the block expansions in the companion volume (Appendix K points to them), starting at line 1 and mentioning registers 0 to 80; exactly 69 of them cost anything. For a guest program $P$ and a start $(x, y)$, the *loaded host* $\mathrm{hload}(P, x, y)$ has register 0 at $x$, register 1 at $y$, register 2 at the number of $P$, register 3 at 1 and every other register at 0, every version 0, line 1, an empty table and channel, the trap down, ledger 0 and flag down. Write $\mathrm{grun}(m)$ for $m$

program steps of $P$ from start$(x, y)$, and hrun$(n)$ for $n$ program steps of $U$ from hload$(P, x, y)$.

**Theorem C.2** ($U$ keeps pace with its guest)**.** For every guest program $P$, start $(x, y)$ and number of guest steps $m$, there's a number of host steps $n$ such that one of two things holds. Either the host is back at line 1 with $m \leq n$ and *related* to the guest: registers 0 and 1 hold the guest's $A$ and $B$, register 3 holds the guest's line, the guest's trap is down, the two ledgers are equal, the two flags are equal, and the host's table and channel hold, fact for fact, slot facts matching the guest's facts and channel. Or both have stopped, with the same counters, the same trap, the same ledger, the same flag, and matching tables and channels. And from any related pair where the guest hasn't stopped, one guest step is matched by at least one host step that lands related again, or with both stopped. □

**Theorem C.3** (Halting and answers match)**.** For every $P$, $x$ and $y$, the guest's run of $P$ from start$(x, y)$ halts if and only if $U$'s run from hload$(P, x, y)$ halts. And whenever the guest has halted at step $m$ and $U$ at step $n$, the host's registers 0 and 1 hold the guest's $A$ and $B$, and the two traps, the two ledgers and the two flags are equal. □

**Theorem C.4** (The host's flag is the guest's, and its ledger is exact)**.** (1) The guest's flag is up at some step of its run if and only if the host's flag is up at some step of $U$'s run. (2) For every $m$ there's an $n$ at which the two are related, with $m \leq n$, or both stopped, with $\mu(\text{hrun}(n)) = \mu(\text{grun}(m))$. And for every $n$ there's an $m$ with $\mu(\text{grun}(m)) \leq \mu(\text{hrun}(n)) \leq \mu(\text{grun}(m+1))$: in between matching points, the host's ledger never gets ahead of the guest's next step. □

Equal ledgers at the matching points are a third sense of "exact," and the plainest of the three (Section 15 names all three): two ledgers holding one number.

**Theorem C.5** (The host earns its own flag)**.** If the host's flag is up after $n$ steps of $U$ from hload$(P, x, y)$, then the instructions $U$ executed contain, in order, a passing CHECK Slot on a slot, then a passing COMMIT Slot on the same slot, with the slot untouched and its version unchanged in between, then a passing CERTIFY; the slot held $\langle$number of $p$, $v\rangle$ at the CHECK, with holds$(p, v)$. And the guest's own run from □

start$(x, y)$ contains a passing CHECK $p$ $c$, then, with $c$ untouched and its version unchanged, a passing COMMIT $p$ $c$, then a passing CERTIFY, with $c$ holding $v$ at its CHECK.

□ **Theorem C.6** (The host is certified Thiele-complete, and $U$'s halting is undecidable)**.** (1) The host machine, with Slot as its property and one move per instruction, is certified Thiele-complete (Definition 15.7), and certified Thiele-complete over its own claim language, whose claims are "Slot holds of register $r$," whose meanings are written without the host's checker, and whose "unchanged" is "same value and same version of register $r$"; and $U$'s run from hload$(P, x, y)$ is a run of it. (2) Read as a Thiele machine it obeys the universal floor; a run of $U$ that raises the host's flag costs at least 3 there. The system built from its certified Thiele-completeness agrees with it step for step and cost for cost, so the same run costs at least 3 on that one too, and the universal floor of 1 holds there directly. (3) $U$ halts from hload$(P, x, y)$ exactly when $P$ halts from start$(x, y)$, so the question "does $U$ halt from this loaded host?" is undecidable in the reduction sense of Theorem 15.5.

Put it together, plainly. One fixed program, 3728 instructions of the small machine's six kinds, with no guest step built into anything, runs every program of the small machine from every clean start start$(x, y)$. It halts exactly when its guest halts, with the guest's answer in its first two registers. Its flag goes up at some step of its run exactly when the guest's goes up at some step of the guest's run, by its own check, its own commit and its own certificate. Its ledger is the guest's, unit for unit, wherever the two line up. And the machine it runs on is certified Thiele-complete. Since the small machine runs every two-counter program, through those guests $U$ runs every computation, under Minsky's $2^n$ encoding, done outside the machine. That's a universal Thiele machine for the small machine's programs, and the definition allows any other host that meets it; the next subsection carries it to every computably presented Thiele machine whose reading a program decides on every state.

There's fine print, and it comes before you take that anywhere. The guests start clean, from start$(x, y)$; the whole-run theorems are about loaded hosts, and only the one-step theorem is about any related pair. The host's property reads the small machine's three-word language, so a guest with a different property language needs a different host property; the host of the next subsection has one, a single property that reads a universal claim language. Sixteen slots per counter is the guest's whole table size, reserved for each counter in case every fact is about that one. And it's slow: for every guest step it unpacks the program number from the front again, and it copies a number one pebble at a time. That's where the running time reported in

the companion volume's section on running it comes from. The theorems above say nothing about time; they match states and ledgers at the points where the two line up.

Enforcement in the step is the reference monitor's move from Section 14, Anderson's 1972 move: one mechanism everything has to pass through. What's put together here is a priced, earned record under that move, carried to every guest by one fixed program. The move itself is Anderson's. Kevin Hamlen, Greg Morrisett and Fred Schneider (2006) worked out, in their model, which policies mechanisms like that can enforce at all, and their paper is where I'd go to see why enforcing a property of every run escapes the Rice-style walls that deciding one runs into.

Why make such a fuss about proofs that walk every case? Because the first universal machine had bugs. It's the machine in Turing's 1936 paper, and nobody gets to call Turing careless. In 1947 Emil Post, in an appendix to a paper on a problem of Thue, pointed out that the construction leaned on a convention it hadn't earned, and he patched a small defect in how the universal machine writes out each complete configuration. Around the same time Donald Davies, who joined the National Physical Laboratory in 1947, where Turing was designing a computer, worked through the universal machine's tables and found more mistakes, much to Turing's annoyance, as the accounts of Davies's life have it. (It's the same Davies who later named the *packet* in packet switching.) His corrections were finally printed in a collection edited by Jack Copeland, a philosopher and historian of computing who runs an archive of Turing's papers, *The Essential Turing*, in 2004. They may be some of the first bugs ever written, in a program for a computer that didn't exist yet. That's a lovely claim, and I haven't been able to confirm it, so take it as a lead. What it changes here: a universal machine is exactly the kind of construction where a careful person leaves a hole, *U* has 3728 lines for a hole to hide in, and each case of its simulation argument has to hold. The definitions are on the page so you can examine what those cases establish.

## C.2 Every computably presented Thiele machine, on one host: driven, decided, priced in whole units, carried as a latch

*U* runs the small machine's programs, and through them every computation, under Minsky's $2^n$ encoding, done outside the machine. But a Thiele machine from somewhere else has its own states, its own moves, its own prices and its own reading, and running it through the two-counter encoding keeps its computation and nothing else. So the question Turing's machine raises for this book is the bigger one. Take any Thiele machine whose workings can be computed. Is there one fixed program that runs every such machine as a guest, keeping its halting, raising its own flag exactly when the guest's reading would rise, on an earned chain of its own, and

keeping a ledger equal to the guest's own prices, with the host's flat ones nowhere in it? There is, within limits I'll list, and this subsection gives it.

First the class, said plainly. A *computably presented* Thiele machine is a Thiele machine in Section 2's sense (states, moves, a step, a price for each move, a yes/no reading, and the toll: a move that turns the reading from no to yes costs at least one) together with a *driver* that picks the next move from the current state or says stop, and a way of writing states and moves as whole numbers. "Computably" means four little programs, written in the plainest language of computable functions there is, the one built from zero, plus one, picking an argument, composing, recursion and search: one computes the driver, one the step, one the price, and one the reading. The driver, step and price routines are required to halt and give the specified result on valid state and move codes; their behavior on other numbers is not constrained by the presentation. The reading routine is required to halt on every number, returning the reading on valid state codes and no on other numbers. State and move encodings have left-inverse decoders, so distinct states or moves cannot share a code. Run from a start $s_0$, the driver's moves give one run. Its *ledger* after $n$ moves is the sum of their prices, and its *latch* after $n$ moves says whether the reading has been yes at any state so far.

The host gets one thing the small machine doesn't have: a paid move that certifies nothing. On the small machine every charge comes from a CHECK, a COMMIT or a CERTIFY, and those can charge without raising the flag: one passing CHECK and six COMMITs of the same claim cost seven. But every passing CHECK takes one of the table's sixteen entries for good, every COMMIT rewrites the channel, and a failing CHECK traps. So the host machine gets a seventh instruction kind, PAY: it costs one, goes to the next line, and changes nothing else; it never raises the flag, takes no table entry and leaves the channel alone. That's why the compiler uses it.

How far the small machine gets without it is proved, and the count is short. Take a run that never traps and ends with the flag down. No CERTIFY ran in it, since a passing one raises the flag and a failing one traps. Every CHECK in it passed, since a failing one traps, and each passing CHECK filled one of the table's sixteen entries, so there were at most sixteen. Every COMMIT in it committed a claim the table holds, and the table holds at most sixteen, so at most sixteen COMMITs can be of claims the run hadn't committed before. Sixteen and sixteen make 32, and the only way past it is a COMMIT of a claim already committed earlier in the run. A claim names its counter's version, and the version changes at every write, so a repeat needs its counter left alone since the earlier COMMIT. A guest that writes both counters between any two of its payments can't do that, and it can't commit at all, since a COMMIT needs its counter left alone since its CHECK, so it runs out of ways to pay after sixteen. Without that limit, 32 is reached: sixteen rounds of CHECK *at least 0 A*; COMMIT *at least 0 A*; INC *A* pay 32 with no trap and the flag

down. A guest that writes both counters at every move of the machine it simulates needs a fresh CHECK for each move it pays for, so it can pay for at most sixteen of them. A simulated machine that charges more often than that has to be paid for somewhere, and PAY is where. The 32 isn't a ceiling on either machine: a program that checks one fact about a counter it never writes can commit that fact again and again, at a cost of one each, with the flag down and no trap, because the counter's version never changes. On the small machine that leaves one counter to compute with while it repeats, which is too few (Theorem D.7); the host has a counter for every whole number, so on it the repeats cost the computing nothing. A program written for the host's registers doesn't need PAY: it checks one true fact once, about a register it never writes, and pays every later unit by committing that fact again. Those commits overwrite the channel, so at the move where the reading first says yes it runs the chain's CHECK, COMMIT and CERTIFY back to back and pays the rest of that move after the CERTIFY. $U_P$ can do without it on its class, and PAY is what it needs beyond the class. $U_P$ names no register above 80 and its guest can't name a host register, so two sit untouched. Pay the first unit by raising one of them to 1, which reads "zero holds at 0," and checking it; pay every later unit by committing that fact again, which passes because the register's version never moves. A compiled guest runs its chain back to back and pays after it, so the chain's own commitment is what its certificate finds, and its one check leaves the table at two facts. Every payment costs one, the surcharge is unchanged, and the host is the register host with no seventh instruction. Outside the class the trick fails: a guest that pays twice and then certifies with nothing committed traps, and this host, with the pay-fact sitting in its channel, would raise its flag instead. PAY leaves the channel alone, and that is what those guests need. On a host without certification PAY is a necessity: a guest that pays while its reading is down makes the host run a paid move that doesn't raise its record.

Read as a Thiele-complete machine, the host needs no filing at all: PAY is a mark that never raises the record. The definition of certified Thiele-complete sorts every paid move into CHECK, COMMIT or CERTIFY, so PAY has to be filed under one of them, and it's filed as a CHECK of the empty claim, a claim that never holds. The filing is for the interface. It isn't what PAY does on the machine: a CHECK that fails on this machine traps, and PAY goes on to the next line. With PAY filed that way the host stays certified Thiele-complete, and certified Thiele-complete over its own claim language, in which the empty claim holds of no state. PAY costs exactly one, as clause (c) asks of a CHECK; its check never passes, so it can't stand in a chain; and it leaves the table, the channel and the flag alone, so it can't add to a chain or raise the flag either. For the same reason the floor of three for a flag raised from a clean start still holds with PAY present: the flag still needs a CHECK, a COMMIT and a CERTIFY.

The program is $U$ with one more handler: a seventh branch on the guest's opcode, for PAY, that runs the host's own PAY once. Call it $U_P$; it's 3735 instructions, with 70 paid sites, the 69 of $U$ plus the PAY. The guest it runs is a program of the small machine with PAY added, and that guest is where the presented machine goes. Its four programs are compiled into two counters, by the packing Minsky used, using the translations of Appendix J from computable functions to counter programs; the guest keeps the presented machine's state coded in its counter $B$, takes one of its moves by running the compiled driver and step, runs the compiled reading on the new state, and pays the move's price in PAYs; the first time the reading says yes, it checks, commits and certifies, and those three paid instructions count toward that move's price. The host's one property is fixed too: a claim is either one of the small machine's three words about a number, or "run reading program $r$ on the coded state $v$," and the host's checker evaluates that with one fixed routine for every presented machine there is. So nothing in $U_P$ knows any particular presented machine. The machine arrives as data, the way Turing's guests arrived on the tape.

A fixed checker that runs whatever program it's handed and always answers needs saying how, so here is the routine. A claim of the second kind carries a number $r$. The checker decodes $r$ as a counter-machine routine: its instructions, where they sit, which register holds the answer, which register holds the fuel, and how many registers to load. If $r$ doesn't decode to a routine, the answer is no. Otherwise the checker loads the routine's registers from $v$, reads a fuel number $f$ out of $v$ too (the fuel register's entry in $v$'s prime factorisation), runs the routine for at most $f$ steps, and says yes exactly when by then the routine has left its code with 1 in its answer register. Every input gets an answer within $f$ steps, so the checker is total, and what it means is that bounded run: yes if and only if the routine halts within the fuel the value carries, with answer 1. That's how it stays clear of the diagonal. A total evaluator $E$ that agreed with every total yes-or-no program on that program's code alone would let $g(n) = \neg E(n, n)$ contradict itself at its own code, as long as $E$ is itself computable. This checker never evaluates a program on its code alone: the fuel comes with the claim's value. The guest compiled from $M$ runs $M$'s reading routine itself and puts at least the step count it took into the fuel register, and at such a value the checker says yes if and only if the reading says yes.

For the fuel rule, take a routine containing just one instruction: increment an answer register initially holding 0, then leave its one-line code. Give it fuel 0 and the check says no: it has not completed. Give it fuel 1 and the check says yes: it has left the code with answer 1. The first no did not mean the routine would never answer yes. It meant "not within this allowance." You can run that distinction with a pebble and one instruction card. The real checker does the same job for any list of instructions, with the allowance handed in alongside the claim.

**Theorem C.7** (One host runs every computably presented Thiele machine)**.** Let $M$ be a computably presented Thiele machine, so that a driver picks its next move (one run per start), a program decides its reading on every state, and its prices are whole numbers; and let $s_0$ be a start state. Load $U_P$ with the guest compiled from $M$'s presentation and $s_0$ coded in. What the host carries of $M$'s reading is its latch. Then: (1) $U_P$ halts exactly when $M$'s run stops. (2) For every $n$ such that $M$ hasn't stopped before move $n$, there's a host time $t$ at which the host's register for the guest's counter $B$ decodes to $M$'s state after $n$ moves, the host's ledger is $M$'s ledger after $n$ moves plus a *surcharge*, and the host's flag is $M$'s latch after $n$ moves; and if $M$ stops at $n$, such a $t$ has $U_P$ halted as well. (3) The host's flag is up at some time exactly when $M$'s reading is yes at some state of its run. (4) If the host's flag is up at time $t$, the host's own instructions up to $t$ contain a passing CHECK on a slot, then a passing COMMIT on the same slot with the slot untouched in between, then a passing CERTIFY; and the slot held the claim "reading program $r$ says yes of $v$" for $M$'s own reading program $r$, that claim holds, and $v$ decodes to a state of $M$'s run where the reading is yes. □

There are two things the theorem doesn't say. First, the host's flag is $M$'s latch. If $M$'s reading says yes at some state and no at a later one, the host's flag stays up, so it says the reading was yes somewhere in the run, and it doesn't say the reading is yes now. Second, what the host re-establishes is the reading: the claim it checked is "$M$'s reading program says yes of this coded state," and that's all. A bare Thiele machine needn't tie its reading to anything true about sorting, proofs or the world outside. If $M$'s reading is meant to say a list is sorted, that it does is a separate correctness theorem about the reading, and the host doesn't supply it. The two universal programs also keep their own guests: $U$ runs the small machine's programs over its three words about numbers, and $U_P$ runs computably presented machines over the universal claim language.

The surcharge comes next, because that's where the ledger stops being exact, and below three units it can't be helped. The host raises its flag the only way the host machine allows, by a CHECK, a COMMIT and a CERTIFY, and those cost three. If $M$'s first move that turns its reading to yes costs at least three, the host pays the difference in PAYs and nothing more is owed. If it costs less, the host pays three there anyway, which is the larger of three and the move's price. The surcharge starts at that move: while $M$'s latch is down it's zero, so a run that never certifies is matched exactly. Whether anything could do better depends on what $M$ paid before. When $M$'s whole ledger at the raise is below three, no program on the host machine can match it, since a raised host flag costs three from a clean start. When $M$'s ledger

at the raise is three or more, this compiler isn't always the cheapest. Take a guest whose first move costs two and leaves its reading at no, and whose second costs one and turns it to yes. This compiler carries it at 2 and then 5, against the guest's 2 and 3. A program that looks ahead carries it exactly: inside the first move's price of two it works out the guest's state two moves on, puts it on $B$, runs the CHECK and the COMMIT of the reading's claim about it, and puts the guest's state back; inside the second move's price of one it runs the CERTIFY, which needs only a fact in the channel, and nothing has committed since. Its totals are 2 and 3, and its flag stands on the guest's reading at the first state where it said yes. Part (1) of the theorem below bounds this compiler, and part (3) is the only lower bound proved.

□ **Theorem C.8** (Exact, within two, and no better)**.** Let $M$, $s_0$ and the loaded host be as in Theorem C.7, and let $M$'s reading be no at $s_0$. (1) For every $n$ before $M$ stops there's a host time at which the state is decoded and the flag matched as before, and the host's ledger is $M$'s ledger after $n$ moves while $M$'s latch is down, and that plus the larger of $3 - c$ and 0 once it's up, where $c$ is the price of the move that first turned $M$'s reading to yes; so the surcharge is at most two. (2) If moreover every move that first turns $M$'s reading to yes costs at least three, the surcharge is zero and the host's ledger equals $M$'s exactly at those times. (3) If $M$'s latch is up after $n$ moves while its ledger is below three, then no run of the host machine from a clean start, under any program at all, ends with its flag equal to that latch and its ledger raised by exactly $M$'s ledger. (4) If instead $M$'s reading is yes at $s_0$, then at the host time matching each move $n$ before $M$ stops, the host's ledger is $M$'s ledger after $n$ moves plus exactly three.

Part (3) is an observation in the sense of Section 2: it reads off Observation 15.21, which holds on the host machine with PAY for the reason given above. Part (4) is the same three paid at once: if the reading is already yes at the start, the host has to raise its flag before $M$ has paid anything, and at that point no program can do better, since $M$'s ledger there is nothing. After it the three stay on only because this compiler doesn't count them against $M$'s later prices.

And one machine is carried all the way, by proof, so the class isn't empty. Its states are whole numbers, its one move adds one and costs three, its driver always picks that move, and its reading is "at least three." The driver returns the fixed move; the step adds one; the price returns three; and the reading compares the state with three. The counter constructions in Appendix J implement all four. Loaded into $U_P$ from 0, the host never halts, its flag goes up, at the host time matching move $n$ its ledger is exactly $3n$ and its flag is up exactly when $n \geq 3$, and the raised flag stands on a host chain whose slot claims the machine's own reading of a state

at least three. That follows from the two theorems above: the reading is no at 0, and the one move that turns it to yes, from 2 to 3, costs three, so part (2) of the surcharge theorem makes the ledger exact.

Now the limits, each of which is in the statement, said plainly.

- **Readings that are decided.** The reading's program has to answer yes or no on every state. A reading a program could only confirm, waiting forever on the no cases, isn't covered, and neither is a reading no program computes at all.
- **One run per start.** The driver picks the move, so the guest is deterministic. A machine whose next move is somebody's unforced choice gets in only with that choice written into its driver.
- **The record is the latch.** The host's flag is up once $M$'s reading has been yes at any state of the run, which is exactly the latch of Section 10. A reading that goes yes and then back to no is carried as "it was yes once."
- **The ledger is exact up to the surcharge.** When the reading starts at no, at most two, and zero when the first raising move costs three or more; when it starts at yes, three. And no program on this host, run from a clean start, can do better when the guest certifies with its whole ledger below three.
- **Prices in whole units, paid one at a time.** A move that costs a million is a million PAYs. The compiler also uses Minsky's packing. Between them, those two choices set how long the host takes to pay for the guest's moves.
- **Proved; never run on a presented machine.** The demonstration machine's numbers come from the two theorems above. $U_P$ itself has run on guests small enough to finish; a presented machine's compiled guest is far too large for that, and nobody has run $U_P$ on one. Appendix F says what was run.

### C.3 A simpler host, with the guest step built in

There's a much simpler host than $U$, and I keep it in the book because it's easy to read and it shows the copy rule in one line. Its universality is by construction: it has a move that runs one guest step by calling the small machine's own step. It isn't a universal Thiele machine in the sense of Definition C.1, and the reasons are exact. Its interpreter loop $U_0$ never halts: it keeps returning to its first line even after the guest stops. The host itself does have an OWN HALT instruction, which another program can use. The loop also keeps charging one per pass, even when the guest has stopped or its next step costs nothing, so no fixed surcharge bounds it. And it isn't a register host, since its guest step does the computing. A host that fixed the first two and kept the built-in step would still fail the last (the argument after the definition). $U$ is the universal Thiele machine of this book. Read it as a picture of

what $U$ does the long way, and as a host that starts from any guest state at all, clean or not.

It carries a guest inside its own state: the guest's program as data, the guest's counters, its table, its ledger, its flag. The host has one extra move, run one guest step. In that one move it looks at the guest's flag, takes the step, looks again, and if the flag went from down to up, the host's own copy of that flag goes up in the same move, and the host pays for it.

**Definition C.4** (The host)**.** A *guest* is a program $P$ of the small machine together with a state $g$ of it (Definition 15.2). A *host state* $h$ has six parts: its own core $\mathrm{core}(h)$, its ledger $\mu(h)$, its own flag $\mathrm{own}(h)$, the guest program $P(h)$ held as a list of instructions, the guest state $g(h)$, and the *mirror* $m(h)$, a yes/no copy of the guest's flag. The host has two kinds of move. OWN $i$ runs the instruction $i$ on the host's own core, costs what the small machine charges for $i$, and raises own exactly when that machine's flag would rise. GSTEP $b$ costs $b$. If the host's core is trapped it does nothing else. Otherwise it moves the host to its next line and, if the guest's next instruction costs at most $b$, takes one program step of the guest, which leaves a stopped guest where it is; in the same move the mirror becomes $m(h) \vee (\neg\mathrm{cert}(g(h)) \wedge \mathrm{cert}(g'))$, where $g'$ is the new guest state. The host's reading is $\mathrm{own}(h) \vee m(h)$. Its stored program is $U_0 = [\text{GSTEP 1; OWN (INC } A\text{); OWN (DEC } A\ 1)]$: one guest step, then two moves that bring the host back to its first line. It *loads* a program $P$ from $(x, y)$ with $g(h) = \mathrm{start}(x, y)$ and the mirror and its own flag down.

Every instruction of the small machine costs at most one, so GSTEP 1 runs every guest instruction there is. Why charge the budget $b$? Why not the cost of whatever instruction got fetched? Because charging the budget keeps the price a function of the move alone, which is what the definition of Section 2 asks for: you read the price of a host move off the move, the way you read the small machine's prices off its table, without peeking inside the guest. What that costs the host is easy to see. A host that steps a guest instruction priced 0, an INC say, still pays one for the move.

So the host's ledger is closer to a clock than to a receipt for its guests: one unit per guest step, whatever the step was. It never falls behind the guest's own ledger (part (3) of Theorem C.11 below), and it matches that ledger only when every guest step happens to cost one. Could a cleverer host have simulation-exact accounting, a ledger equal to its guest's? Not with this one fixed-price move: GSTEP 1 is the same move whether the guest's next instruction is an INC priced 0 or a paid CHECK, so one price can't match both. Two moves can: one that costs nothing, takes the guest's step when that step costs nothing and otherwise jumps, and one that costs one and takes it. A loop that tries the first before the second keeps the host's ledger equal to the guest's at every pass. That host still has its guest step built in. $U$ gets

the match with ordinary instructions: it pays only on the record moves it runs itself, so its ledger equals its guest's (Theorem C.4).

And there's one thing to say about the definition before the theorems, so you know which of them carry weight. GSTEP runs the guest's instruction by calling the small machine's own step on the guest's state: the guest's step is a primitive of the host. So "the host runs any guest" (Theorem C.9) is true by construction. The host was built to do exactly that, and the proof follows the three moves around the loop. The theorems with something in them come after it. From a loaded host, or any state where the two agree, the mirror always equals the guest's flag (Observation C.10). Nothing raises the mirror unless the guest itself passes a CERTIFY inside a guest step the host paid for (Theorem C.12). No OWN move, in any host program at all, touches the guest's state or the mirror (the second sentence of that same theorem), and no host move of either kind rewrites the guest's program. And the mirror is tied to the guest's earned chain: after any host run from a loaded host, if the mirror is up, the guest's own run from its start contains a passing CHECK, then a COMMIT of the same claim, then a CERTIFY, since the mirror is the guest's flag and the guest started clean (Theorem 15.11). So this host runs any small-machine program by construction and keeps the guest's record by theorem, as a copy of the guest's flag. The mirroring program $U_0$ does not earn the host's own flag; a different program could do so with the host's OWN CHECK, COMMIT and CERTIFY. $U$ does both the long way: it runs the guest with its own instructions, and it earns its own flag.

**Theorem C.9** (The host runs any guest)**.** (1) If the host's core is not trapped, then after $n$ moves GSTEP 1 the guest state is the guest program's own run of $n$ steps from $g(h)$, the guest program is unchanged, and the host is still not trapped. (2) If the host's core is at line 1 and not trapped, then $3n$ steps of the program $U_0$ leave the guest exactly where $n$ steps of its own program put it. (3) For every two-counter program $M$ and inputs $a$, $b$: $M$ halts from $(1, a, b)$ exactly when the host program made of OWN moves of the compiled $M$ halts, from a host whose core starts at $(a, b)$, whatever guest that host carries. □

**Observation C.10** (The copy is the record)**.** *Say a host state $h$* agrees *when $m(h) =$* cert$(g(h))$. *Every host move takes an agreeing state to an agreeing state, so every host trace and every stored host program does too, and a freshly loaded host agrees. In particular, if $h$ agrees and is not trapped, the mirror after $n$ moves* GSTEP 1 *is the guest's flag after $n$ steps of its own program.*

□ **Theorem C.11** (The host's step collects the guest's toll)**.** (1) If $h$ agrees, is not trapped, and the guest step that budget $b$ allows takes the guest's flag from down to up, then the mirror is down before GSTEP $b$ and up after it, the host's ledger rose by exactly $b$, and $b \geq 1$. (2) In the run of (1) of Theorem C.9, from a host that agrees, if the guest's flag goes up between its steps $n$ and $n+1$, the mirror goes up between the host's moves $n$ and $n+1$, and that move costs 1. (3) For every host trace $t$, $\mu(g(\text{run}(t,h))) + \mu(h) \leq \mu(\text{run}(t,h)) + \mu(g(h))$: the host's ledger grows at least as fast as the guest's.

□ **Theorem C.12** (No unpaid host certification)**.** If the mirror is down at $h$ and up after a host move $i$, then $i$ is GSTEP $b$ with $b \geq 1$, the host was not trapped, the guest's next instruction was CERTIFY, and the guest's flag went from down to up in that move. A trace made only of OWN moves leaves the mirror and the guest state exactly where they were. And the host's own flag rises only by an OWN CERTIFY that passes on the host's core.

□ **Theorem C.13** (The host is a Thiele machine)**.** For every host state and move, if the host's reading goes from no to yes, the move costs at least 1; the same holds for the mirror alone and for the own flag alone. So the host read through own $\vee$ $m$, the host read through $m$ alone, and the guest class are Thiele machines (Definition 2.2), and any host trace that raises the host's reading costs at least 1.

Two statements have been proved here. The designated guest-step runs follow the guest computation. Separately, every permitted host instruction preserves agreement with a genuine guest run, so arbitrary host instruction sequences from a loaded start cannot fabricate the guest's mirror. The second statement uses the full host instruction set. It does not follow merely from the first, or from charging for a raise. The combined reading own-or-mirror remains a different observation: the host may establish its own record while the guest's record stays down.

**Remark C.1** (Two guests, run)**.** Load the guest CHECK *is zero* $A$; COMMIT *is zero* $A$; CERTIFY; HALT with both counters at 0. After two moves GSTEP 1 the mirror is down; after three it is up, the guest's flag is up, the host's own flag is still down, and the host has paid 3. Under the program $U_0$ the mirror is down at host step 6 and up at step 7, with 3 paid. The guest CERTIFY; CERTIFY traps on its first step: after five

moves GSTEP 1 the mirror is down, the guest is trapped, and the host has paid 5 for nothing.

That copy can't be faked. The proof walks every host move there is, and the copy only rises in a guest step where the guest itself ran CERTIFY and passed. A host program that never touches the guest leaves the copy exactly where it was. From a loaded host the copy always equals the guest's flag, and the host's ledger always grows at least as fast as the guest's, and a three-instruction loop runs any small-machine program one guest step per pass. Here the mirror, the crossing test and the charge sit in the host's step rule, so Theorems C.11 and C.12 hold for every sequence of host moves, including ones that never run a guest. The fixed program $U$ supplies simulation, an exact ledger at matching points and an earned flag using ordinary instructions. Its guest simulation theorem concerns that program from its loaded starts. The host's native earned-chain rule covers other programs too, but does not make every such program simulate the same guest.

## C.4 The diagonal, on this machine's programs

Section 13 left the diagonal standing on any substrate, on a little family built to fit it, and on L, handled on its own terms rather than as a substrate. It also said what a machine has to supply to fill in the same shape, two things: programs that can be handed to programs as numbers, and a recursion theorem for the transformations it can carry out. This machine's universal program reads a whole guest program out of one register, with a numbering of programs that goes there and back (Definition C.3), and halting is undecidable here in the reduction sense of Theorem 15.5. What's left are the two classical results that turn a universal machine into a place where the diagonal bites: Rice's theorem and Kleene's recursion theorem. They come with a wrinkle I should put first, because it decides which machine they're about.

The wrinkle is the one from Section 15, at the end of the subsection on what two counters compute. The small machine takes its inputs straight into two counters, and Schroeppel's memo gives a proof that two plain counters can't even compute $2^n$ from $n$. Rice's reduction has to keep a program's input safe while a copy of a halting instance runs. Kleene's construction has to run an evaluator while keeping its input. Two counters have nowhere to keep it. So the two theorems land where they're true and where I could prove them. The host machine of Definition C.2, the one with a counter for every number that $U$ runs on, is a fine programming system: its programs are numbers, an input sits in register 1, and there's room to spare. On it both theorems hold in full. On the two-counter machine itself the story splits, and I'll tell it at the end of the subsection: Rice's theorem holds on every input, by a trick for keeping the input that I'll show you; Kleene's holds when inputs are written as

powers of two, the way Minsky packed them; and on plain inputs Kleene's is false, and that's a theorem too.

The host machine's programs are numbered in the same style as $U$ numbers a guest's. Write $\langle m, n\rangle$ for $(2n+1)\cdot 2^m$. INC $r$ is $\langle 0, r\rangle$, DEC $r$ $j$ is $\langle 1, \langle r, j\rangle\rangle$, HALT is $\langle 2, 0\rangle$, CHECK $r$ is $\langle 3, r\rangle$, COMMIT $r$ is $\langle 4, r\rangle$, CERTIFY is $\langle 5, 0\rangle$, and a program is the list code: the empty program is 0, and an instruction $x$ followed by a program $t$ is $\langle x, \mathrm{code}(t)\rangle$. An instruction number that is none of these codes reads as HALT. Then every number decodes to some program, since a positive number is $\langle x, n\rangle$ with $n$ smaller than it, and decoding a program's code gives the program back. When I put one program after another, the later one's jumps move down by the length of what comes before it, and a jump to a line where it has no instruction, line 0 included, goes to the line just after it, so a stop inside it hands on to whatever follows. Say a host program *outputs* $y$ on input $x$ when, started with $x$ in register 1 and every other register 0, with a clean table, it halts with $y$ in register 0.

Rice comes first, because it's the shorter road. Henry Rice, whom you met in Section 7, proved in his 1951 doctoral thesis, and published in 1953, that no program decides any property of what programs *do* unless every program has it or none does. Properties of how a program is written aren't covered. The road from halting is a single detour. Take a property of what programs do, a program $y$ that has it, and a program $n$ that doesn't. Given an instance of two-counter halting, a Minsky program $H$ with starting counters $a$ and $b$, build a host program that first loads $a$ and $b$ into two fresh registers (not register 1, and named by neither $y$ nor $n$), then runs a copy of $H$ moved onto those registers, then empties both, then runs $y$. If $H$ halts on $(a, b)$ the built program behaves like $y$ and has the property. If $H$ runs forever the built program never stops on any input, and it behaves like the program that loops, which lacks the property (if the looping program has it instead, swap the roles of $y$ and $n$). Everything the built program does before $y$ is INC and DEC, which add nothing to the ledger and touch no fact, channel, trap latch or flag. So a decider for the property would be a decider for halting, and by Theorem 15.5 that would let a program list every single-tape Turing machine that doesn't halt. Two words carry the weight, and they're the ones to check in the statement: what "behaves the same" sees, and where the program starts.

**Definition C.5** (Behaving the same, on the host)**.** A host program *ends* in a state when it halts there, from the start with $x$ in register 1 and the rest clean; a trap counts as stopping. Two host programs *behave the same* when for every input $x$, any state one ends in has a state the other ends in that agrees with it on every register's value, the fact table, the channel, the trap latch, the ledger and the flag, and the same the other way round. The line number and the versions aren't compared. A

property of host programs is *extensional* when behaving the same carries it from one program to the other.

**Theorem C.14** (Rice's theorem on the host machine)**.** Let $\Pi$ be an extensional property of host programs that some program has and some program lacks. Then $\Pi$ is undecidable in the reduction sense of Theorem 15.5. □

A property that depends only on the partial function a program computes, whether it stops and what it leaves in register 0, is extensional, so it is covered. That case also follows from the recursion theorem below.

On the small machine itself the same road works from the clean start, and only from there. Say two two-counter programs *behave the same from the clean start* when, whatever state one halts in from counters $(0, 0)$, the other halts in a state with the same counters, trap latch, ledger and flag, and the same the other way round. The fact table and the channel aren't compared, because the prefix moves the versions of $A$ and $B$ by fixed amounts and the facts, the channel's included, carry versions. There is no input to keep, since the program starts at $(0, 0)$, and $H$'s instance is loaded straight into $A$ and $B$.

**Theorem C.15** (Rice's theorem for the small machine, from the clean start)**.** Let $\Pi$ be a property of two-counter programs that's carried from a program to any program that behaves the same from the clean start, held by some program and lacked by some other. Then $\Pi$ is undecidable in the reduction sense of Theorem 15.5. In particular "halts from the clean start" is undecidable. □

That theorem covers the clean start only. Every input is harder, because the standard reduction has to keep the input safe while the halting instance runs, and two counters have nowhere obvious to keep it. The way through is at the end of the subsection.

Kleene's recursion theorem is the deeper one, and it's a construction, which is Section 13's point about Kleene's little gem (Moschovakis's phrase): the content is the building, and the contradiction in Section 13 was just the last line. It says: for every transformation $F$ of programs that some program can carry out on numbers, there's a program $p$ that computes the same thing as $F(p)$. The shape of the building is the old one. From the program that carries out $F$, build an *evaluator* that reads a number, treats it as a program's code, specialises that program to its own code, runs $F$'s program on the result, and runs whatever program comes back. The specialising step is what logicians call s-m-n: fix one input of a program and get a new program that needs one input fewer. Then the witness is the evaluator specialised to its own

code. Run on any input, it builds its own code, hands it to $F$'s program, gets the code of $F(p)$, and runs that.

Each piece on the host machine comes next, with its exact premise.

□ **Theorem C.16** (s-m-n, and a universal program for host programs)**.** (1) For a number $c$ and a host program $V$, let spec$(c, V)$ be $c$ copies of INC 2 followed by $V$ with its jumps moved to follow them. Then for all $x, y$, spec$(c, V)$ outputs $y$ on input $x$ if and only if $V$, started with $x$ in register 1 and $c$ in register 2, halts with $y$ in register 0. (2) There is a host program $W$ made only of INC, DEC and HALT such that, for all $c, x, y$, $W$ started with $x$ in register 1 and $c$ in register 2 halts with $y$ in register 0 exactly when the host program numbered $c$ outputs $y$ on input $x$.

One thing to be exact about: $U$ is not that universal program. $U$ runs two-counter guests, and a two-counter guest given its input plainly can't run the evaluator. With inputs written as powers of two one can, as Theorem C.22 near the end of this subsection shows. $W$ runs host programs, and it can, because it's a host program on a machine with registers.

□ **Theorem C.17** (Kleene's recursion theorem on the host machine)**.** (1) For every host program $T$ there is a host program $e$ such that for all $x, y$: $e$ outputs $y$ on input $x$ if and only if, for some $d$, $T$ outputs $d$ on the number of $e$, and the program numbered $d$ outputs $y$ on $x$. (2) So for every transformation $F$ of host programs that some host program carries out on numbers (on the number of $p$ it outputs the number of $F(p)$, for every $p$), there is a host program $e$ such that for all $x, y$, $e$ outputs $y$ on $x$ if and only if $F(e)$ does. (3) The same on numbers: for every function $f$ on numbers, defined everywhere, that some host program computes, some number $e$ has the program numbered $e$ and the program numbered $f(e)$ output the same on every input.

That fixed point is of the *function*: $e$ and $F(e)$ stop on the same inputs and leave the same number in register 0. Their ledgers, flags and fact tables can differ: $e$ is all INC and DEC, so whatever $F(e)$ pays, $e$ only simulates. I'd like a fixed point that agrees on the record too, and the next two theorems say how much of that can be had. The ledger and the flag can be had, but agreement on every register can't.

□ **Theorem C.18** (The recursion theorem, up to the record)**.** Let $F$ be a transformation of host programs that some host program $T$ carries out on numbers. Then there is a host program $e$ such that, for every input $x$, any state $e$ ends in has a state $F(e)$ ends

in that agrees with it on register 0, the trap latch, the ledger, the flag, the number of facts in the table, and whether the channel is empty, and the same the other way round. So $e$ stops on $x$ exactly when $F(e)$ does, and the ledgers are equal exactly, with no surcharge. $T$ may itself check, commit and certify.

Look only at the function computed and you get the usual fixed-point statement back. There are two limits, and the second one is a theorem. First, $e$ pays after it has computed everything. It is not the way $U$ runs a guest, mirroring every move: $e$ pays the counts, CHECKs, then COMMITs, then the CERTIFY, whatever order $F(e)$ used, so the order of the record's events isn't preserved, and for a program $F(e)$ that never stops, $e$ never stops and pays nothing, though $F(e)$ may pay without bound. The relation says nothing about states in the middle of a run. Second, no fixed point agrees on every register, or on what the facts name.

**Theorem C.19** (No fixed point of full behaviour)**.** There is a transformation $F$ of host programs, carried out by a host program, such that no host program $e$ behaves the same as $F(e)$ in the sense of Definition C.5. □

A variant of $F$ that also CHECKs register $r$ and commits to it gives $F(e)$ a fact and a commitment about $r$ that $e$ can't have, so no fixed point agrees on the fact table alone, or on the channel alone, or on the versions of the registers. What survives is what Theorem C.18 states, plus any register named in advance, or any finite set of them: before it pays, the evaluator reads that register's final value off the simulated run of $F(e)$ and writes it into its own copy of the register, keeping its scratch work and its two payment registers elsewhere. The register $F$ above writes is chosen by looking at $e$, and that's the one no fixed point can follow.

And with the recursion theorem proved, the wall of Section 13 goes up on this machine without assuming that theorem as an extra premise. Call a property of host programs *functional* when it depends only on the partial function a program computes.

**Theorem C.20** (The diagonal on the host machine)**.** Let $\Pi$ be a functional property of host programs that some program has and some program lacks. (1) Read "$f$ is representable" as "some host program carries $f$ out on numbers." Then no function $d$ from programs to Booleans whose flip is representable is right about $\Pi$ on every program: this is the argument of Theorem 13.1, reproved for host programs, with its recursion premise a theorem here. (2) No host program decides $\Pi$ outright: there is no program $D$ that, on the number of every program, halts with a number in register 0 that's 1 exactly when the program has $\Pi$. □

Rice and Kleene are the ingredients the diagonal of Section 13 asks a substrate for, and the wall they put up doesn't care about the record: the base does the computing and the flag watches. A program's ledger and flag are among what "behaves the same" sees on the host, so a nontrivial property of those is undecidable too, in the reduction sense of Theorem C.14, and the fixed point is up to the record, as far as the theorems above say and no further.

Now the two-counter machine itself, which I promised at the start of the subsection: Minsky's two bins of pebbles and nothing else. Rice comes first. Say a two-counter program *behaves the same* as another on a set $X$ of inputs when, on every input $n$ in $X$, started with $n$ pebbles in bin A and none in bin B, one stops exactly when the other does, and then they agree on both bins, the trap latch, the ledger, the flag, and on which property and which counter each recorded fact and the channel are about. The versions are left out: they say which write a fact is about, and the trick below moves them.

The trick is the whole problem. Rice's reduction has to keep the program's input safe while a copy of a halting instance $H$ runs, and there's no third bin to keep it in. My first guess was bin B, and that dies at once, because $H$ needs both bins. The second guess is Minsky's own packing: keep several numbers in one bin as powers of different primes, $H$'s two counters as the exponents of 2 and 3, and the input as the exponent of 5. That dies more quietly. To put the input there, the program would have to turn $n$ pebbles into $5^n$, an exponential, and Schroeppel's memo, from Section 15, gives a proof that two plain counters can't even build $2^n$ from $n$. I had no way to build $5^n$.

What works is to stop keeping the input beside the work. Turn $n$ pebbles into $6n + 5$, which is only adding, and then multiply in $H$'s counters as powers of 2 and 3. $H$ runs by halving the pile and splitting it in thirds and building it back up, and the $6n + 5$ rides along in the middle of every one of those piles. It's odd, and counted off in threes it leaves two over, so no halving or thirding ever reaches into it.

□ **Theorem C.21** (Rice's theorem for the two-counter machine, on any inputs)**.** For any set $X$ of inputs, a property of two-counter programs that's carried from a program to any program that behaves the same on $X$ is undecidable in the reduction sense of Theorem 15.5, unless every program has it or none does, and then it's decidable. $X$ can be every input, the powers of two, or $\{0\}$.

So $6n + 5$ is the place to keep the input that two counters seemed not to have. It isn't beside the work but inside it, in a factor the work never touches. I like that one more than I can justify: the input hides in the leftover that every halving and thirding walks straight past.

Kleene needs more than keeping an input. It needs a program that runs a program. So write a number $x$ as $2^x$ pebbles and read a result as the exponent of 2, the way Minsky did. With inputs written that way, there's one two-counter program, call it $U_2$, that, started on $2^x \cdot 3^e$, stops with $2^y$ exactly when the program numbered $e$ gives $y$ on $x$; specialising a program to a number is exact; and the recursion theorem holds. ($U_2$ is a different program from the universal host program $U$, which runs on the machine with a register for every number.)

**Theorem C.22** (Kleene for the two-counter machine, inputs written as powers of two)**.** (1) For every transformation $F$ of programs that some program carries out on program numbers, there's a program $e$ that computes the same function as $F(e)$, with inputs and outputs written as powers of two. (2) The fixed point is of the function only: there's such an $F$ for which no $e$ behaves the same as $F(e)$ once the properties and counters of the recorded facts count. □

The counterexample in (2) is short. Number the programs the way $U$ numbers its guests (Appendix K, in the companion volume), so a program's number is above the code of every property it names. Let $F(p)$ be the one-line program that checks that bin A holds at least the number of $p$ plus one. A program can only record facts about properties it names, so no $e$ can record the fact $F(e)$ records, and from a large enough input $F(e)$ does record it.

Even with Schroeppel in hand, my first guess about Kleene was that the powers of two were a convenience, and that plain inputs would only take more work. They don't take more work: with plain inputs, Kleene's theorem is false. A transformation is *carried out* if some program, started on $2^n$ where $n$ is the number of a program $p$, stops with $2^m$ where $m$ is the number of $F(p)$.

**Theorem C.23** (No plain recursion on two counters)**.** (1) No program of $n$ lines stops with $c \cdot x$ pebbles in bin A, started on $x$ pebbles in bin A and none in bin B, for every $x$, where $c$ is the least prime above $N = 16(n+1)(4n+1)^{16}$. (2) So there's a carried-out transformation $F$ such that no program $e$ computes the same plain function as $F(e)$. □

Nothing in that proof, which the companion volume carries, used more than this: two bins that move one pebble at a time, finitely many controls, a next step fixed by the control and the bins, and bins that, above some threshold, are read only through their parity. Any machine like that has the same obstruction, and the same computable transformation $F$ with no program $e$ equivalent to $F(e)$ on plain inputs.

I went looking for Schroeppel's own memo and found a scan of it. As I read it, his route is close in spirit: he trades a two-counter machine that starts with one counter empty for a machine with one register that can only add, multiply and divide by fixed numbers and take one away, each of those built out of one of the program's loops, and that's the stages here under another name. I haven't followed his proof line by line, so I compare the two at that level and no further; the argument this book stands on is the one above. The $c$ it uses is astronomically bigger than anything a real multiplier needs, and finding the best bound is a question I leave alone.

So on two bins, Rice holds on every input; Kleene holds with inputs written as powers of two, for the function, and fails for the facts; and with plain inputs Kleene fails outright. The last one is the answer I didn't see coming: the powers of two weren't a convenience, and on two bins they're the difference between a recursion theorem and none.

That failure has a setting computability theorists know. The mathematician Hartley Rogers, in his 1967 textbook, calls a numbering of the computable functions *acceptable* when every other computable numbering of them can be translated into it by a computable map, and in every acceptable numbering Kleene's theorem holds. Read on plain inputs, the two-bin programs number something smaller than all the computable functions, since none of them computes $2^n$ from $n$ (Schroeppel's theorem; compare Theorem C.23), so they fall outside Rogers's class, and his route to the theorem isn't open there. The computer scientists James Royer and John Case (1994) studied that territory: programming systems for classes of functions smaller than all the computable ones, which can't be acceptable, and what can still be programmed and proved in them. And the logician and puzzle writer Raymond Smullyan's two books on recursion theory and self-reference (1993, 1994) take the fixed-point theorems apart axiomatically, so you can see which property of a numbering each one leans on. The plain-input failure is a small instance on the map they drew.

## C.5 The universal machine's guest interface

I'll say what the two universal programs cover plainly, a sentence at a time. The stated class of guests of $U$ is the small machine's programs, every one of them, from every clean start $\mathrm{start}(x, y)$. The whole-run theorems don't cover a guest handed facts in its table, a commitment in its channel or a raised flag at the start; the one-step simulation, by itself, works from any related pair. It's the same kind of class Turing's universal machine has: a fixed instruction set, and a finite program held as data, here packed into one number the way Turing packed his into a description number. Each guest keeps its computation, its halting, its answer, its record and its

prices, exactly: the host halts when it halts, its flag goes up when the guest's would, on a chain of its own, and its ledger is the guest's.

The guests of $U_P$ are every Thiele machine whose driver, step, prices and reading are computed by programs, with the reading decided on every state. Each keeps its computation and its halting; its reading becomes the host's flag as a latch, raised on an earned chain whose claim is the guest's own reading; and its prices become the host's ledger, exactly, except for a surcharge at the move where the reading first says yes: at most two when the reading starts at no, exactly three when it starts at yes. The surcharge is zero when that move costs three or more, and it can't always be avoided: when the guest certifies with a ledger below three, no program on that host, run from a clean start, matches its ledger exactly (Theorems C.7 and C.8). Picture a clerk who'll stamp anybody's receipt. He copies it onto his own forms, bills at your rates to within two when your reading starts at no and to within three when it starts at yes, and the stamp at the bottom is one he earned by checking your own reading, not one he copied.

What's outside the class is outside the theorems. A machine whose reading no program decides (an oracle checker, say, which the definition of certified Thiele-complete doesn't rule out) doesn't come in. Nor does a machine whose next move is nobody's computation. The two universal programs have different start conditions: $U$'s whole-run result starts its guest clean; $U_P$ starts its guest where the presentation says, even if that state already has a yes reading. The host starts clean and earns a claim about that reading. It does not reconstruct or authenticate the guest's earlier history.

And insisting on no built-in step is the point, and a plain interpreter shows why. A plain interpreter on an ordinary computer runs guests too. If its guest record is freely writable data, its meaning depends on the interpreter following the intended update rule. An ordinary computer can also protect that data with a monitor or another enforced interface. Here the rule is explicit in the host's instruction semantics: $U$ and $U_P$ are fixed interpreters, and nothing in them raises the host's flag except the host's own CERTIFY, and the host's step won't let that pass without a COMMIT, or the COMMIT without a CHECK on a slot nobody has touched. The guest's record isn't data the host updates. It's a record the host earns again, under the same rule, on its own registers.

The proofs of this appendix, the register-by-register description of $U$ and the worked tables are in the companion volume *The Thiele Machine: The Example, Built* (its section "Proofs, the program and the worked tables for The universal machines") and in Coq in `coq/kernel/foundation/UniversalRun.v`, `UniversalPRun.v`, `SmKleene.v`, `SmGuestRice.v`, `TcRice.v` and `Tc2Plain.v`.

APPENDIX D

# Every model

After the smallest honest machine (Section 15) and the universal programs of the previous appendix, there was a worry I couldn't put down. Everything in them was built on Minsky's two counters. The definition of certified Thiele-complete names a two-counter window in clause (a). The universal program runs two-counter guests. So is all of this a fact about counters? Pick a different machine to start from, a Turing machine with its tape, a register machine, Alonzo Church's lambda calculus, and does the record still go on the same way, or did the definition depend on my example?

I chased it with a picture: a box with a panel bolted to its front. The box computes, with levers for moves and a window showing what clause (a) asks to see, two numbers and a line number. The panel is the example machine's earned layer: version tallies that click when the box writes, a rack for facts, the flag, and the three buttons CHECK, COMMIT and CERTIFY at a cost of one each. The worry is whether that panel only bolts onto the one box I built.

So this appendix goes both ways. First the panel goes onto boxes I didn't build, and I say exactly what a box has to supply for it to fit. Then other boxes, panel and all, get run inside mine. If the record is really about the record and not about pebbles, both trips should work, and the conditions on the way should say what has to survive the move.

## D.1 From any base up

Start with the direction I care about most: the panel onto a box I didn't build. The box can be any machine at all that has a universal base in the sense of clause (a), and in the picture that means two things: it shows a two-counter window, and for each two-counter instruction it has a lever that acts on the window as that instruction does. What else is inside, what its states look like, I don't get to know.

My first guess was to keep whatever the box came with. Plenty of machines have a meter of their own, or a light on the front that means something's done, so leave those where they are, bolt the panel on beside them, and see what happens. It can't work, and the definition says why. If a lever can cost anything, computing adds to

the ledger, and certified Thiele-complete asks that it add nothing, which is a rule of the price list, not a claim about heat. If a lever can switch on the box's own light, a record went up with no CHECK behind it. Certified Thiele-complete rules out both. So the meter and the light come off, along with everything else the box already had: its costs, its record, all of it, except the states, the levers and what each lever does to the state.

Then bolt on a panel cut down from the small machine's, and nothing else: one version number for the whole box that every lever raises, which is the tally, a table of facts, a channel, the flag, and three new moves, CHECK, COMMIT and CERTIFY, each costing one, with every lever costing nothing. The three act as on the small machine, with the tally in place of a counter's version: CHECK files its claim with the tally when the checker says yes and the table has room, COMMIT puts the fact in the channel when the table holds its claim at the current tally, and CERTIFY raises the flag when the channel holds a fact. Call that the *lift* of the machine. A lever moves the box only while the trap is down. The trap works as on the small machine: a CHECK, COMMIT or CERTIFY that fails sets it, nothing takes it down, and once it's up every move, a lever included, changes nothing but the ledger.

**Theorem D.1** (Every universal base lifts)**.** (1) The lift of any box with a universal base, over a language of claims about the box's state ("unchanged" meaning the box's state is the same), with an exact equality test and a checker proved to say yes exactly when the claim holds, is certified Thiele-complete, as long as the claims its CHECK can ask include one that's true at one loaded state and false at another, and the fact table has room for at least one fact. (2) The claims "this register is at least $n$" and "the program is at line $n$," read off the window, always include such a claim, so with those claims every universal base lifts. (3) Conversely, every certified Thiele-complete machine has an interface meeting the four clauses under which its base moves, taken alone, form a machine with a universal base. (4) The panel leaves the box alone: from a state whose trap is down, a run of lever pulls in the lift moves the box exactly as the box's own run does, and the trap stays down. □

*Proof.* (1) The four clauses, one at a time. (a) The lift keeps the box's states, levers and step and adds the tally, the table, the channel, the flag, the trap and the ledger. With the trap down, a lever moves the box's part of the state as the box's step does, raises the tally by one, and touches nothing else. So from a live state, which is a state whose box part is live and whose trap is down, the window moves as the instruction does and the state stays live; no lever changes the flag; the loaded states are the box's loaded states with an empty panel, so they're clean; and the flag, once up, is lowered by nothing. (b) Start with the panel empty. The only move that

raises the flag is CERTIFY, and CERTIFY traps unless the channel holds a fact; the channel is filled only by a COMMIT, which traps unless the table holds its claim at the current tally; and the table is filled only by a CHECK that passed, which files the claim with the tally it was checked at. So in front of the raise there's a passing CHECK of some claim $c$, then a COMMIT of $c$ at the same tally, then the CERTIFY. The trap is down from the CHECK to the COMMIT, since the COMMIT passed and nothing takes the trap down, and with the trap down the tally rises with every lever pull, so an unchanged tally means no lever was pulled in between, so the box's state is unchanged, and that's what "the thing $c$ is about is unchanged" says in the lift's language, whose claims are about the box's state. A passing CHECK means the checker said yes, and the checker was proved to mean what it says. (c) is the price list: levers at a cost of zero, the three buttons at one each, the ledger the running sum. (d) Take the claim that's true at one loaded state and false at another. From the first, CHECK passes and files it, the table having room, COMMIT finds it at the same tally, and CERTIFY raises the flag. From the second the CHECK traps, and a trapped machine raises nothing. (2) "$A$ is at least 1" is true at the loaded state showing $(1, 1, 0)$ and false at the one showing $(1, 0, 0)$. A window claim is a register and a number, or a line number, so two claims can be tested equal, and its checker reads the window, which is its meaning, so the checker is sound by definition. (3) Take an interface the machine is certified Thiele-complete with and throw away every move whose kind isn't base. What's left has the same states, the base moves as levers, and the interface's window, live states and loaded states, and clause (a) says of exactly these that the window follows every instruction from every live state, that the state stays live, and that each loaded state shows $(1, a, b)$ and is live. That's a universal base. (4) A lever pull with the trap down changes the box's part by the box's step and the tally by one, and nothing else; the trap goes up only on a CHECK that fails, a COMMIT without its fact, or a CERTIFY with the channel empty, never on a lever. So along a run of lever pulls the box's part follows the box's own run, pull for pull, and the trap stays down. ■

Of the checker's exactness, (1) uses one half whole and the other half once: (b) needs the checker to say yes only when the claim holds, and (d) needs it to say yes on the one claim at the one loaded state where that claim is true. The theorem asks for both halves whole, the way every theorem about the small machine does, so one checker serves all of them.

Clause (b), the small machine's proof with the bins swapped out for whatever the box has, is the part I'd braced for, and it turned out to be nothing. The chain CHECK, COMMIT, CERTIFY never looked at anything but the tally and the claim. The levers changed only the tally on the panel, making old facts stale; they never

inserted a fact, made a commitment or raised the flag. Neither side of the proof ever needed to know it was looking at pebbles.

So the earned layer isn't something two counters do. It sits on top of any box with a universal base, and it never needed to know which one. I'd half expected to be rewriting the proof once per model.

## D.2 What never lifts

A theorem that says the panel fits every box is one I don't trust until I've watched it fail to fit something. So I went looking for boxes it won't turn into a certified Thiele-complete machine, and the first one I tried was the one I was surest would work.

It was a Turing machine, with one fixed table, and each lever does one row of the table. That looked like the obvious base to me, and it isn't one at all. Clause (a) asks for a lever for every two-counter instruction, and there are infinitely many two-counter instructions, since a jump can go to any line. A fixed table has finitely many rows, so it has finitely many levers, and it runs out of levers long before it runs out of instructions. The first two items below are that failure in two forms, and the others fail for other reasons, and each is proved.

**Theorem D.2** (What doesn't lift)**.** (1) A box with only finitely many levers has no universal base, and no box with a universal base is finitely branching, meaning that from every state all its levers together lead to only finitely many different states. (2) The lift of a finitely branching box is never certified Thiele-complete, whatever claims and fact table it gets. (3) Neither is the lift of a box whose step ignores the state it's in, where each lever sends every state to the same place. (4) On a record axis where every place is at or below every other, there is no certified Thiele-complete machine at all. (5) With a checker that says yes exactly when the claim holds, the lift's own non-vacuity clause holds exactly when the fact table has room and some claim is true at one loaded state and false at another. □

*Proof.* (1) Take a universal base and its loaded state showing $(1, 1, 0)$. For each $j$ the lever for DEC $A\ j$ must take it, in one pull, to a state showing $(j, 0, 0)$: $A$ was above 0, so one comes off and the line jumps to $j$. Different $j$, different window, different state. So one state has infinitely many different successors, which is what not finitely branching means; and since one lever lands in one place from one state, the levers for the DEC $A\ j$ are all different, so there are infinitely many of them. (2) In the lift the moves that cost nothing are the levers, since each button costs one, and from any state the levers together lead to finitely many states: with the trap

down, the box's finitely many successors, each with the tally raised by one, and with it up, the state itself. (The buttons can lead to infinitely many, one CHECK for each claim, but each costs one.) If the lift were certified Thiele-complete through some interface, that interface's base moves would cost nothing by clause (c), so they'd all be levers, and by Theorem D.1(3) they'd form a universal base on their own. By (1) a universal base has a state from which its moves reach infinitely many states, and levers of a finitely branching box can't. (3) Suppose each lever lands in one place whatever the state. The move an interface gives an instruction is a base move, at a cost of zero by clause (c), while each button costs one, so it's a lever. Pull the lever for INC $A$ twice from a loaded state showing $(1, 0, 0)$, and pull the lever for INC $B$ and then the lever for INC $A$ from the same state. Both runs end with the lever for INC $A$, so both end with the box in the one place that lever lands, with the tally two higher and the panel untouched: the same state of the lift. But the window would have to show $(3, 2, 0)$ after the first run and $(3, 1, 1)$ after the second, and a window is a function of the state. The same two runs refute the box on its own. (4) On a record axis, raising the record means leaving the down-set of where it stood. If every place is at or below every other, every down-set is the whole axis and no move ever leaves one; but clause (d) asks for three moves that raise the record from some loaded state. (5) If the table has no room, every CHECK traps and no CERTIFY ever fires, so (d) fails. If no claim is true at one loaded state and false at another, (d)'s second half has no witness. And when both hold, (d) was proved in (1). ■

So my first test case failed, and at first I read item (1) as saying a Turing machine can't be a base at all. It says something narrower. A Turing machine *with one fixed table* can't be one. A classic model comes in the other way round, with one lever for each program that carries out one two-counter instruction, run to its end. Pull it and the box runs that program from start to stop in a single move. That's how the register machine below comes in, and it's the honest reading of "a Turing machine is a base" too. The levers are the programs, not the rows.

That still leaves a worry about clause (a) itself. It asks for one lever per two-counter instruction, which is a fact about how the two-counter machine is presented, and a box that does each instruction in several pulls looks shut out for a reason that has nothing to do with computing. It isn't shut out, and the fix is to let the pulls be grouped. Call a box a *base by simulation* when it has a window reading a two-counter configuration off each state, through any encoding at all, a class of live states, a finite list of its own levers for each two-counter instruction, and a live loaded state showing $(1, a, b)$ for each start, such that from every live state the list acts on the window exactly as the instruction acts on a configuration and the state stays live. The *grouped box* has the same states, and its levers are finite lists of the box's levers, a list pulled in order.

**Theorem D.3** (The one-lever presentation loses nothing)**.** (1) Every box with a universal base is a base by simulation. (2) Every base by simulation is a universal base of its grouped box, and a run of the grouped box is the box's run along the joined lists. (3) So for every base by simulation, the lift of its grouped box, with the window claims and a fact table with room for at least one fact, is certified Thiele-complete. (4) The box with five levers, add one to $A$ and go to the next line, the same for $B$, add one to a jump target (leaving the window alone), and for each of $A$ and $B$ "if it's above 0, take one off and jump to the target, otherwise go to the next line, and put the target back to 0 either way," is a base by simulation, has no universal base, and its grouped box lifts to a certified Thiele-complete machine. □

*Proof.* (1) Each list has one lever. (2) The grouped box's step is the box's run along the list, so the simulation clause of a base by simulation is clause (a) for the grouped box. (3) is (2) with Theorem D.1. (4) INC $r$ is one pull; DEC $r$ $j$ is $j$ pulls of the target lever, then the decrement lever for $r$, from a live state, where live means the target is 0; the decrement lever puts the target back to 0, so the state stays live. The box has five levers, so item (1) of Theorem D.2 says it has no universal base. ■

So clause (a) can be read by simulation, and nothing is lost: a box that carries out each two-counter instruction with one fixed finite list of its own moves, through any window, is a universal base the moment its moves are grouped, and it lifts. Item (1) of Theorem D.2 is about the presentation, and the five-lever box is the proof that a finite lever set is not what fails. What this doesn't give is a Turing machine's rule table running a tape: the five-lever box is a register box, and the next subsection is still how Turing machines come in.

Item (3) is a box with no memory. Pull a lever and it lands wherever that lever always lands, whatever happened before. So a window on it can't tell INC $A$ followed by INC $A$ from INC $B$ followed by INC $A$, and a box that can't tell those apart can't keep two counters in its window.

Item (4) is the trouble with bare preorders from Section 11, where the cups stopped telling you the place, showing up one level higher. If every place on the axis is at or below every other, the flag has no up to go to. No step ever turns a cup over, nothing ever has to be paid for, and the earned chain has nothing to earn.

## D.3 Computation is the slice

The lift says the panel goes onto any box with a universal base. Turn it round and it says something about the box: the box is what you see of the lift when you stop

looking at the panel. That's where ordinary computation sits in this book, so I want it said exactly, with a name.

**Definition D.1** (Universal-base slice)**.** Take a machine with a ledger, a whole number for each state, and a loaded state for each start, and take a box with a universal base. A *universal-base slice*, called a slice in this appendix, is different from the global sliced interface of Definition 2.3. It requires a universal base, a working domain and reachable hidden witnesses; it does not require all non-computing moves to fix the view. It is a map from the machine's states to the box's states, a move of the machine for each lever of the box, called its *computing move*, and a class of the machine's states, called *working*, such that: (i) each loaded state is working, and the map sends it to the box's loaded state for the same start; (ii) from a working state, a computing move lands on a working state, and the map sends where it lands to where the box's lever takes the map's image; (iii) from some loaded state there are two runs whose ends the map sends to one box state, with the record up at one end and down at the other, and two runs whose ends it sends to one box state, with different ledgers.

When there's a slice, I'll say the box is a slice of the machine. The first two clauses say the slice runs as the box runs. Start with a working state and pull one computing lever. Clause (ii) says that its picture moves exactly as the box moves, and that the full state is working again. That last part lets you apply the same rule to the next lever, then the next. A whole run therefore follows the box's run, move for move. Clause (i) puts the two at matching starts, so they finish at matching pictures too. The third clause says it can't see what was kept or what it cost: a function of the box state that gave the record at the end of every run from every loaded state would have to give both answers at the one box state the clause names, and the same goes for the ledger. Nothing else is hidden in the definition. Read the box's window through the map and take as live the working states whose image is live, and the first two clauses make the computing moves for the box's instruction levers a universal base of the machine. So a machine has a slice onto some box exactly when it has a universal base whose window meets the third clause, and then the window itself is a slice onto the two-counter box, whose states are configurations and whose levers are the instructions. The third clause does work. A Thiele-complete machine's computing moves, taken alone, are a box, and the map that forgets nothing meets the first two clauses onto it and fails the third, because the record and the ledger are functions of the state; and the window of the first clock of Observation 10.13, whose record is a question about its counters, fails it too.

**Theorem D.4** (Computation is the slice)**.** (1) Take the lift of a box with a universal base, as in Theorem D.1(1). Forgetting the panel, which keeps the box's part of each state and drops the tally, the table, the channel, the trap, the ledger and the flag, is a slice of the lift onto the box, with the levers as its computing moves and the states whose trap is down as its working states. Beyond that, CHECK, COMMIT and CERTIFY never move the box's part, from any state; from any box state with an empty panel, the lever runs are sent, under forgetting, to exactly the box's own runs from that state, lever list for lever list; and from a loaded state where the claim of Theorem D.1(1) is true, the run of no moves and the run CHECK, COMMIT, CERTIFY of that claim end on one box state with the flag down and up and ledgers exactly 3 apart, CERTIFY three times ends there too with the flag down and the same ledger as the second run, and CERTIFY $d$ times, for any $d$, ends there with the flag down and a ledger $d$ above the first run's. (2) With no room in the fact table, the lift of any box, with any claims, is certified Thiele-complete through no interface at all. (3) Let a machine be Thiele-complete through an interface; every certified Thiele-complete machine is (Theorem 15.23). Its computing moves are exactly its moves that cost nothing, so they're the same through every interface it's Thiele-complete through; taken alone, with the interface's window, live states and loaded states, they form a machine with a universal base; and the window is a slice of the machine onto the two-counter box, with the instructions' moves as computing moves and the live states as working. (4) Let a machine be run as one rule: from each state, every move leads to the same next state. It has no universal base, no base by simulation and no universal-base slice onto any box. It obeys the toll exactly when every move costs at least one or no state reading no steps to one reading yes, so it's a Thiele machine only as a clock or with a reading that never rises. □

*Proof.* (1) The lift's loaded states are the box's loaded states with an empty panel and the trap down, so they're working, and forgetting sends each to the box's loaded state for its start. From a state with the trap down a lever moves the box's part as the box's step does and leaves the trap down (Theorem D.1(4)), which is the second clause. Start with any box state and an empty panel. The first pull matches the box's move and keeps the trap down; the next pull therefore starts under the same conditions. Repeat through the word and every pull matches. A CHECK, COMMIT or CERTIFY that passes changes only the table, the channel, the flag and the ledger; one that fails sets the trap and adds to the ledger; and with the trap up every move changes nothing but the ledger. None of them touches the box's part. Now take the loaded state where the claim is true. The run of no moves ends there with the flag down and the ledger at 0. On the other run the CHECK passes and files the claim,

the table having room, the COMMIT finds it at the same tally, and the CERTIFY raises the flag: three moves at one each, none of them moving the box, so the same box state, the flag up and the ledger at 3. CERTIFY from a clean state finds the channel empty and traps, and every later move changes only the ledger, so CERTIFY $d$ times ends at the same box state with the flag down and the ledger at $d$, which at $d = 3$ is the chain's ledger. So the third clause holds, for the flag and for the ledger, and no function of the box state and the ledger together gives the flag. (2) Take any interface and the three moves its clause (d) names. Each costs one by clause (c), so each is a button. Take a loaded state from which they raise the flag; it's clean by (a), so its flag is down by (b). If its trap is up, nothing passes and nothing rises. If its channel holds a fact, CERTIFY alone raises the flag from it: a run of one move from a clean state that ends with the flag up, where (b) asks for a CHECK and a COMMIT before the raise. If its channel is empty, some COMMIT has to pass inside the three moves, before the CERTIFY that raises the flag. Until then no CHECK has passed, because a CHECK passes only with room in the table, and no button moves the tally, so that COMMIT found its fact in the loaded state's own table at the loaded state's own tally; that COMMIT and a CERTIFY, run from the loaded state, then raise the flag in two moves, against (b) again. So the three moves raise the flag from no loaded state, and clause (d) fails. (3) By (C), computing moves cost 0 and every other move costs 1, so the computing moves are the moves that cost 0, whichever interface the machine is read through. Throw the rest away and what's left has the same states, the interface's window, live states and loaded states, and a move for each instruction that acts on the window as the instruction does from every live state and keeps it live, with each loaded state live and showing $(1, a, b)$: a universal base, as in Theorem D.1(3). The same facts are the first two clauses of a slice onto the two-counter box, and the third is Theorem 10.15(2): two runs from one loaded state end on one window, one with the record up and one with it down, and their ledgers differ. (4) From each state all the moves lead to one state, so the machine is finitely branching, and by Theorem D.2(1) it has no universal base. A slice would give it one, as the remark after Definition D.1 says. A base by simulation fails too. A list of $p$ moves does $p$ steps of the rule, so two lists run one after the other end in the same place in either order. From the loaded state showing $(1, 0, 0)$, INC $A$ and then DEC $A$ to line 5 shows $(2, 1, 0)$ and then $(5, 0, 0)$, while DEC $A$ to line 5 and then INC $A$ shows $(2, 0, 0)$ and then $(3, 1, 0)$, and one state can't show both. For the toll: if a state reading no steps to one reading yes, every move takes that step, so every move costs at least one, every step does, and the machine is a clock (Observation 10.13); if none does, the reading never rises and the toll asks nothing. ■

Not every part of that carries the same weight, and I'd rather say which before somebody else does. Some of it holds by construction. That the slice runs as the box runs, and that CHECK, COMMIT and CERTIFY leave it alone, are how the lift was built, and the blindness in (1) follows from the second once the chain raises the flag: the chain and the run of no moves end on one box state because no button moves the box. In (3), the window meeting the first two clauses of a slice is the definition of Thiele-complete read out, and so is the fact that the moves that cost nothing are the same through every interface. Three parts could have failed. That the lift is certified at all is the earned chain of Theorem D.1, clause (b). That the window of every Thiele-complete machine can't see what was kept or what it cost is Theorem 10.15, and it could have gone the other way: the first clock's window sees its record. That a table with no room certifies through no interface is (2). And (4), that a machine run as one rule has no universal-base slice and is a Thiele machine only as a clock or with a reading that never rises, is short, and its second half rests on one choice in Definition 2.2: the toll puts the price on the move, the same from every state. With a price that could read the state, a machine run as one rule could charge exactly its raises.

What "computation" means in the theorem's name is a box's levers acting on its states. A model that comes with a move for each two-counter instruction is covered as it stands, and one with a fixed list of its own moves for each is covered once its moves are grouped (Theorem D.3). A model run as one rule, a Turing machine stepping by its one table, is covered by (4) only: it has no universal-base slice, and it cannot supply the appendix's universal-base target. Its ordinary transition rule still lifts to the global sliced machine of Definition 2.3: put the lamp and tally beside it. The fixed-list obstruction concerns its instruction interface. A Turing machine read the other way, with a lever for each program that carries out one two-counter instruction, as in the previous subsection, is a box like any other. Nothing in the theorem uses the equivalence of the seven models below (Theorem D.6(1)), which is other people's work; that equivalence is what says a box with a universal base computes whatever the other models compute.

A machine run as one rule can still run programs, and its program can keep a flag. Whether that flag is a record the machine keeps or a bit it only simulates can be said exactly too, once the program is part of the state.

**Observation D.5** (A record or a simulated bit)**.** *Take a machine whose moves are operations on configurations, with a reading of the configurations, and put the program in the state: a state is a program, which picks an operation for each configuration, together with a configuration, and one step applies to the configuration the operation the program picks there and keeps the program. Take every program. Then the reading never comes down under any operation from any configuration exactly when, read on these states, it never comes down from any state.*

*Proof.* If no operation brings the reading down anywhere, no step does, since each step applies an operation. If an operation brings it down from some configuration, the program that picks that operation at every configuration is one of the programs, and the step from the state made of that program and that configuration brings it down. ∎

So whether a cell is a record or a simulated bit is a fact about the one rule that steps program and configuration together, and it takes the whole set of states to see it. A certification cell that some program can clear, like the one a universal machine keeps as plain data on its tape (Section 15.13), is a simulated bit, though on a run whose program never clears it the cell goes up and stays, as a record would.

*Why the live-state clause stays in the slice*

Go back to the blown fuse. A free lever turns the live machine's gears and leaves the trapped machine still. Hide the difference between those states and one camera picture would have to prescribe both outcomes. Theorem 15.1 gives the exact consequence: hide every native CHECK globally, require autonomous free moves, and every projected free move becomes identity.

Keep the camera promise on live states instead. The computing moves act there and keep the machine live. A failed panel check leaves that domain. That is the boundary in the slice above. It lets the computational view move while the native failure rule keeps its power to stop the run. Restricting the promise to reachable states would not do this: append a failed CHECK to a run and the trapped state is reachable too. It is the live boundary, which that check crosses, that makes the difference.

The certificate and ledger remain passive beside the core. The trap is inside the core, where the gears read it. Add the publication file and you can keep which claims were certified, with the native lamp as its nonempty reading. The fuse still works as before. The file records what was published; the live-state clause says where the computing promise holds. Both boundaries belong in the description of the machine. Theorem 15.9 proves that the lamp reports whether the publication file is empty. Theorem 15.20 counts how many different files can sit above the same full native state.

## D.4 The same thing, in seven models

Now the other direction, the one the worry was really about. Put boxes with different insides in a row: one with a tape, one rewriting lambda terms, one with bins. Is a panel riding on one of them the same as a panel riding on another? The honest way to ask is whether the boxes agree on what they compute.

My first idea was to wire the boxes to each other myself, two at a time, with a relation that says one box's run keeps pace with another's, but that's the wrong tool. Section 10's weak base equivalence is about runs keeping pace, and what I need here is coarser: do the boxes compute the same relations on numbers? For that the work is already done, and it isn't mine. The seven models are the weak call-by-value lambda calculus L, Turing machines, two Minsky instruction conventions, binary stack machines, Conway's FRACTRAN, and the $\mu$-recursive functions, and that they all compute the same things is Minsky's, Shepherdson and Sturgis's, and Forster, Larchey-Wendling and their coauthors' work. The translations, with how each one writes its inputs and reads its outputs, are in the companion volume (Appendix J).

**Theorem D.6** (The base doesn't matter)**.** (1) A relation on numbers that's computable in any one of those seven models is computable in all seven. (2) The box that keeps a whole two-counter configuration in one number, with one lever for each instruction, has a universal base, its step is computable in all seven models, and its lift, with a table of sixteen facts, is certified Thiele-complete. (3) A register machine of the kind Cook and Reckhow described, where each two-counter instruction compiles to at most five of its own instructions, has a universal base too, and its lift is certified Thiele-complete. (4) Any box on numbers with a universal base lifts to a certified Thiele-complete machine, and its step, computable in one model, is computable in all seven. □

*Proof.* (1) Compose the translations of the companion volume (Appendix J). (2) Encode a configuration as $\langle \ell, \langle a, b \rangle \rangle$. Decoding, performing one INC or DEC, and encoding the result are the total finite arithmetic operations constructed there. So the window follows each instruction exactly; loading is the encoding of $(1, a, b)$. (3) Let RAM registers 0, 1 and 2 hold the simulated line, $A$ and $B$, and keep register 3 equal to 1. To increment counter register $r$, add register 3 to $r$ and then to register 0. To decrement it with target $j$, use these five lines, numbered 0 to 4: if $r > 0$ jump to 3; add register 3 to register 0; jump to 5; subtract register 3 from $r$; set register 0 to $j$. At line 5 the finite block has ended. A base move restarts a block at its first line, runs it for its length, and keeps the resulting registers; an ended block stays put. The zero and positive cases give exactly the required window, and neither changes register 3. Loading puts $(1, a, b, 1)$ in registers 0 to 3. (4) Apply the lift theorem to that universal base and apply (1) to its step function. ■

So the answer to the worry is that the record doesn't care. A box with a universal base, described in any of the models people have used to say what computing is, takes the same panel, becomes the same kind of machine, and pays the same price, one each, for the same three acts.

And the trip goes back down. So far the panel went onto other boxes. Now put another box, panel and all, inside mine. Appendix C.2 proved that the fixed program $U_P$ on the small machine's host runs every computably presented Thiele machine. In terms of the seven models, that reads: if the four relations that describe a Thiele machine (its driver, which picks the next move or says stop, its step, its cost and its reading, on codes) are computed in any one of the seven by programs that answer on every input, then the machine is computably presented, and $U_P$ runs it, halting when it halts, with its flag, and with a surcharge of at most two when its reading starts at no.

So a Thiele machine described in lambda calculus runs, under one fixed program, on the small machine's host, which gives model independence in both directions: up, by lifting any universal base, and down, by running anything presentable on one fixed program. When I saw both arrows close I stopped worrying that the definition depended on my example. A definition that did wouldn't survive being carried both ways.

### D.5 Why two counters

If the box doesn't matter, why did I build on two bins? I wanted the smallest box that holds up, and the obvious smaller one has a single bin. Nothing earlier in this book says one bin is too few. So before I'd let myself say two is the fewest, I tried one.

Watch what a program on one bin can see. While the bin isn't empty, it can't tell how full it is: a stretch of lines that never finds the bin empty does exactly the same thing from a fuller bin, with the extra pebbles riding along. So the only news a one-bin program ever gets is the bin running dry. Between two of those moments it's just walking its own lines, and among more moments than there are lines, two are on the same line, which is the pigeonhole again. Put those together and a one-bin program either stops within a count you can write down in advance, or never stops at all.

□ **Theorem D.7** (One counter is too few)**.** (1) A program $P$ of INC, DEC and HALT instructions on one bin, started with $n$ pebbles in it, halts exactly when it has stopped within $k(n+k+1)$ steps, where $k$ is the number of lines in $P$. So whether it halts is decidable. (2) Halting for the book's two-counter machine is undecidable. (3) So no computable translation takes each two-counter program and start to a one-counter program and start that halts exactly when the original does.

*Proof.* (1) If $k = 0$, the program has already stopped. Otherwise suppose it is still running after $T = k(n+k+1)$ steps, so the $T+1$ configurations at times $0, 1, \ldots, T$ are all live. If the counter has stayed at most $n+k$ up to time $T$, those live configurations have only $k(n+k+1) = T$ possibilities: $k$ lines and $n+k+1$ counter values. There are $T+1$ of them, so one repeats, and the deterministic run loops forever.

If instead the counter reached $n+k+1$ by time $T$, take the first time it did, and look back to the last time before that when it was at most $n$. In between it stayed above $n$, so above 0. It moves at most one per step, so on the way up it arrived at each of the $k+1$ levels $n+1, \ldots, n+k+1$; mark the first arrival at each. There are $k+1$ arrivals and $k$ lines, so two arrivals have the same line $\ell$, at levels $a < b$. Every DEC in the stretch between them found the counter positive. Start that same stretch at level $b$: every counter value is larger by $b-a$, so still positive, every DEC goes the same way, and the run comes back to line $\ell$ at level $2b-a$. Repeat forever. Either way a run still live at time $T$ never stops, so a run that stops does so within $T$ steps, and running $P$ for $T$ steps decides it. (2) The tape simulation, prime packing and halting diagonal of the companion volume (Appendix J), carried over to the book's instruction set line by line. (3) Such a translation, followed by the test in (1), would decide two-counter halting, against (2). ∎

So two counters are the fewest in exactly this sense: one counter can't compute everything, because its halting can be decided, and two, handed their inputs packed, can. I like where the answer came from. The pigeonhole that left no spare chair in the closed room is the same one that says why the box has two bins.

## D.6 The lift and its instruction interface

The lift gives each box with a universal base an earned panel. In the other direction, every certified Thiele-complete machine supplies a universal base through its own base moves. The obstruction results also identify two kinds of base that never lift. Together these describe the coverage proved here. Two further classification questions concern bases outside those classes and the possibility of certification through an interface other than the lift's own.

The tape-to-counter construction of the companion volume (Appendix J) connects bases through what they compute. The lift then supplies the record operations. Its tally is one edition number for the whole box, whereas the small machine keeps an edition per counter. That coarser tally is enough for the lift's arguments. To run the lifted machine inside the presented-machine interpreter, supply the four computability relations for its chosen presentation.

The construction lets you follow the box and the panel separately: the base provides its computing moves, the panel earns its record, and the live projection recovers the box's working run.

APPENDIX E

# Built from parts

Back in Section 2 I asked a question and only got halfway: bolt two Thiele machines together, and do you get one? Everything real is built out of parts. A computer is a processor and memory and a disk, a bank is branches, a proof is lemmas. If the record and its toll only held for one machine at a time and fell apart the moment you put two together, they'd be a property of toys.

So put two Thiele machines on a bench, each box with its own panel, and wire them up the three ways anybody wires machines. Side by side, with no wire between them at all, so that every move is a lever pulled on one box or the other. End to end, with the first box's bins tipped into the second's when the first is done. And one inside the other, with the inner box running as a guest on the outer one. Some of it holds and some of it provably doesn't, and the places where it doesn't were where I learned the most.

## E.1 Side by side

Two boxes on the bench, no wire between them. A move of the pair is a lever pulled on one box or the other. The question is what the pair's record should be, and my first answer was the wrong one: a single count of how many certifications the bench has made, one number, easy to read. I'll come back to why it fails, because it's in the list of things that break. What works is to keep both panels showing. The pair's record is the pair of records, a place for the left machine and a place for the right one, compared one side at a time.

Since that record has more than two places, "certified Thiele-complete" in this appendix means the definition of Section 15 with the record allowed to be a place along a direction, the way Section 11 set it up. Read on a direction, clean states all sit at one place, the floor; a record never moves below where it was; and in clause (b) a raised record is one that isn't at or below where the run began, with the CERTIFY the move that first takes it out. The old definition had a single bit there, and at two places this one is exactly the old one.

□ **Theorem E.1** (Side by side)**.** (1) The cost of any run of the pair is the left machine's cost of its moves plus the right machine's cost of its moves, and the order in which levers on the two boxes were interleaved doesn't change where the pair ends up or what it paid. (2) If both component state spaces are nonempty, the pair obeys the toll exactly when both machines do. (3) Read through "either flag is up," two Thiele machines side by side are a Thiele machine with floor one; read through "both flags are up," a run that starts with both flags down and ends with both up costs at least two, and two is reached when each part can certify for one. (4) Two certified Thiele-complete machines side by side are a certified Thiele-complete machine. (5) In that pair, call a side's *floor* the place on its record axis where its clean states sit. A run from a clean start that ends with each side at or above a point that isn't at or below that side's floor makes at least six record moves and raises the ledger by at least six. Six is reached, from suitable loaded starts, by a three-act certificate on each side; a particular pair of target points can need more. (6) The pair of records is the product in the precise sense: for any machine mapping into both, preserving records, there's exactly one map into the pair that does both jobs.

*Proof.* (1) A move of the pair is a lever of one box. It moves that box's component by that box's step, leaves the other component alone, and costs what that box charges for it. So the pair's run is the pair of the two boxes' runs along the projections of the trace, each box's moves in the order they came, and the cost is the left projection's cost plus the right's. Two interleavings with the same two projections end in the same state at the same cost. (2) The toll prices a step that takes the record out of the down-set of where it stood. A pair is at or below another when both sides are, so the down-set of a pair is the two down-sets side by side, and a move of the left box leaves the pair's down-set exactly when it leaves the left down-set, the right side not having moved. So the pair's exits are the left exits and the right exits, each priced by its own box, and the pair obeys the toll exactly when both parts do; for the converse, an exit at zero cost on one side is an exit at zero cost of the pair with any state at all on the other. (3) Read through "either flag is up," a step from neither to either raises one side's flag, and that side charges it at least one, so the pair is a Thiele machine with floor one. Read through "both flags are up," a run from both down to both up raises each side's flag on some step, no move touches both, and each such step costs at least one; when each part can certify for one, the two certificates one after the other cost two. (4) Give the pair the interface that reads a move as its own box reads it, takes a claim to be a claim of one side, about that side's component and unchanged when that component is unchanged, with both sides clean as clean, both sides loaded as loaded, the sum of the ledgers as ledger, and the left box's

window as window. Clause (a) is the left box's clause (a), since a left base move acts on the left window as its instruction does and a right move doesn't touch the left component, and no base move of either side changes either record. Clause (c) is the two price lists added. Clause (d) is the left box's witness claim with the right box idle. Clause (b) is the one with work in it. Take a run from both clean that ends with the pair's record raised, and cut it just after the first move that takes the pair's record out of where it began. That move is one side's, and since pairs compare side by side it takes that side's record out of where that side began. Project the cut run onto that side: a run from clean that ends with that side's record raised, so by that side's clause (b) it contains that side's chain, a passing CHECK of a claim $c$, a COMMIT of $c$ with $c$'s component unchanged in between, and a CERTIFY. Put the other side's moves back where they were. None of them touches the component $c$ is about, so the chain is a chain of the interleaved run. (5) If a side ends at or above a point that isn't at or below its floor, its record ends not at or below its floor either, or the point would be at or below the floor by passing through it. The clean start puts that side at its floor, so the projection onto that side is a run from clean with its record raised, which by that side's clause (b) has three record moves of its own, a CHECK, a COMMIT and a CERTIFY. The two sides' moves are different moves, so six record moves at least, and by (c) a ledger rise of at least six; a three-act certificate on each side, from loaded starts where each side's witness claim holds, makes exactly six. (6) A record-preserving map from a machine $Z$ into a box is a map of states with a monotone map of records, the record of the image being the image of the record, and every move of $Z$ simulated by some run of the box. Given one into each box, send a state of $Z$ to the pair of its images. The record of the pair is the pair of the records, and a move of $Z$ is simulated by the left box's run followed by the right box's, which is a run of the pair by (1). That map followed by either projection is the map you started with, and any map into the pair with those two projections agrees with it on every state, which is what equal maps means here. ∎

Most of that is what you'd guess from the bench. Each box keeps its own ledger and nobody else pulls its levers, so the costs add, and it can't matter which box you happened to work on first. The six in (5) is two certifications costing three each, one per box, with no discount for doing them on the same bench.

Machines, with maps that keep the record, form a category, and item (6) says these pairs are its products. Ask the maps to keep the cost too and the pair stops being a product: a machine mapped into both boxes would pay its cost once on each side, twice in all. What survives is a symmetric monoidal structure, which is all that side by side needs. Its pentagon, triangle and hexagon laws say that regrouping four boxes, inserting an idle box, or exchanging groups of boxes gives the same final wiring whichever permitted route you take. Label the boxes and follow each wire:

every rewiring only moves boxes around, so both routes put each wire at the same labelled terminal, at the same cost.

That paragraph is the most formal way I know of saying "side by side composes, and nothing about the order of the parts matters." It's a lot of category theory to confirm what the bench told me in a glance, and I'm glad to have it anyway. A glance is where I'd have missed something.

### E.2 One after the other

Now wire them end to end. Run the first box, and when it's done, tip its two bins into the second box's start, at no charge on the ledger, and run the second. Programs get chained that way everywhere, and my first thought was that the tipping couldn't matter, because it's plumbing: pebbles go from one bin into another, nobody checks anything, and plumbing has no business on a record. That turned out to be half right, and the wrong half is item (4).

□ **Theorem E.2** (One after the other)**.** (1) Running one certified Thiele-complete machine and then handing its two counters to another through loads that cost nothing gives a certified Thiele-complete machine. (2) For any two programs of the two-counter machine, it computes the composite: some schedule leaves the first machine's window where the first program takes the input, and the second's where the second program takes that result. (3) The hand-off keeps the toll when every state it can load into sits at one level of the second machine's record. (4) If some load lands the second machine at a record that isn't at or below another load's, the hand-off loses the toll: it's a certification nobody paid anything for.

*Proof.* (1) Give the second machine a loadable form. A state is either pristine, holding the second machine's loaded state for some $(a, b)$, or running, a state of the second machine. A move LOAD $(a, b)$ turns a pristine state into the pristine state for $(a, b)$ and does nothing to a running one; the second machine's own moves move a running state as before and turn a pristine state into a running one. LOAD is a base move and costs nothing. When the second machine is certified Thiele-complete its loaded states are clean and stand at the floor, so a LOAD leaves the record alone; the base, the toll and the witness claim are the second machine's; and the chain in front of a raise lies after the last LOAD before the second machine's first move, so it lifts through the projection onto the second machine's own moves. So the loadable form is certified Thiele-complete, and the composite, the first machine side by side with it, is certified Thiele-complete by Theorem E.1(4). (2) A wired schedule is the first machine's moves, then a LOAD of the two counters its window

shows at that moment, then the second machine's moves. The first stretch runs the first program on the first window; the LOAD puts the second window at $(1, a, b)$ for the counters $(a, b)$ the first stretch left; the last stretch runs the second program on it. If both programs halt, the two windows after the schedule are the two halting configurations, the second's on the first's output, and the wire cost nothing. (3) and (4) are one fact read two ways. A LOAD from the pristine state for one input to the pristine state for another costs nothing, and carries the record from where the first loaded state stands to where the second stands. If every loaded state stands at one level, no LOAD moves the record, the composite's exits are the two machines' own, and the toll is kept by Theorem E.1(2). If some loaded state stands at a record not at or below another's, the LOAD from that other one to it, at zero cost, leaves a down-set, which is an exit nobody paid for, and the toll fails. ■

Item (4) is the one to remember. The tipping costs nothing, so it had better not tip in anything that counts as established. If pouring pebbles into the second box can land its panel somewhere that isn't at or below where some other pouring would land it, the second machine gets a raised record without ever having checked anything, and that's a forged certificate passed off as plumbing. My first thought was right about the pebbles and wrong about where they land.

## E.3 Connecting records and writers

Joining two boxes also means deciding how to read their flags and who may write their workspace. The following examples trace those connections and the guarantees they affect.

**Theorem E.3** (What breaks)**.** (1) Read the pair through "both flags are up" as a single flag, and any run that raises it from a clean start makes at least six record moves; the definition of certified Thiele-complete asks for a claim that three moves can certify, so through the pair's own interface that reading fails the non-vacuity clause. (2) Two clocks side by side obey the toll with a two-counter machine inside. They aren't Thiele-complete, so composing that easy bar composes nothing. (3) Give the small machine a second writer that changes counter A without raising its version, and the run "check that A is 0, let the other writer change A, commit, certify" raises the flag, so the earned-record clause fails when the machine is read the small machine's way, with the second writer as a base move. (4) Two windows that each price their own machine exactly can collide as a pair, so exact pricing isn't preserved by putting machines together. (5) Count the pair's certifications with one number, and the count can't tell which side certified. The pair of records can. □

*Proof.* (1) Through the pair's interface, a run from a clean start that raises "both flags are up" ends with each side's record raised, and by Theorem E.1(5) that takes at least six record moves. Clause (d) asks for a claim whose CHECK, COMMIT and CERTIFY, those three moves alone, raise the record from a loaded state. Three moves aren't six, so (d) fails for that reading. (2) A clock is the two-counter machine with every move priced at one and a reading over it (Observation 15.27). Two side by side have a two-counter machine running inside, the left's, and obey the toll by Theorem E.1(2); and every move of the pair is a move of one clock, costing one, while a Thiele-complete machine has a move that costs nothing (Theorem 10.12(5)). (3) Add to the small machine a writer that changes counter $A$ and leaves its version alone, read as a base move. The run CHECK ($A$ is 0), the writer's move, COMMIT ($A$ is 0), CERTIFY, from a start with $A$ at 0, passes its CHECK, finds its fact at the unchanged version at the COMMIT, and raises the flag at the CERTIFY. The base and toll clauses still hold, and so does the checker's promise. What fails is clause (b). The small machine reads "unchanged" as same value and same version, and the only chain in that run has the writer's move between its CHECK and its COMMIT, where counter $A$, the thing the claim is about, changed its value. The COMMIT didn't notice, because it looks only at the version, and the version didn't move. So a run from a clean start raises the record with no earned chain in it. (4) Take a left box whose only step its window doesn't see is a certifying one, so the window still prices it exactly, charging that unseen transition one; and a right box whose only step is a stutter, at zero cost, leaving its down-set alone and its window unchanged, priced exactly at zero. In the pair, the left box's unseen certifying step and the right box's stutter show the same pair of windows before and after. One costs at least one and leaves the pair's down-set; the other costs nothing and doesn't. So a price computed from the pair of windows is short on the first or over on the second. (5) Replace the pair of records by the number of sides that have certified. A run in which only the left side certifies and one in which only the right side does both reach the count one, with the left record raised in the first and at its floor in the second. The count can't tell them apart; the pair of records can. ∎

Item (1) puts one flag over two boxes. It costs what two certifications cost, six record moves, and the definition wants a claim that three can settle, so a single flag over both panels, read through the pair's interface, is too expensive to meet the definition. That holds for this interface; nothing here shows that no other interface on the pair could meet it, and Theorem E.1(4) shows the pair meeting it through a richer record. Item (4) says that a window exact for each box alone can be fooled by the pair: two pair states with different costs can show the same pair of readings.

Item (3) is the one with the most teeth, because it's the bug from Section 14, the file checked and then swapped. On the bench it's a second hand reaching into a bin

while the box isn't looking: two writers on one value, and a check that's out of date by the time anybody leans on it. The small machine's version rule is exactly the fix, and this is what happens without it: the commitment goes through on a claim about a thing that changed underneath it. Database people and the people who build processors have been fighting this one since at least Kung and Robinson's paper of 1981 and the Livermore report of 1987, and that section gave it the security people's name, time of check to time of use. Here it's a theorem about the record, and the version number is what makes it go away. The second hand can still reach in, but with the version its reach shows on the tally, and the COMMIT traps.

Item (5) is my first guess from the side-by-side wiring, coming back to fail on schedule. It's why the pair's record compares one side at a time. Count only how many sides certified and the record forgets which side did it. That count answers "how many?" but cannot answer "did the left side certify?" An ordinary number could encode the full pair too: use 0 for neither, 1 for left only, 2 for right only and 3 for both. It is the counting rule that loses the distinction, not the fact that its answer is one number.

## E.4 One inside another

The last way to wire the bench: put one box *inside* the other, as a guest on a host. Part of this the book already has. The appendix on the universal machines (Appendix C.2) showed the fixed program $U_P$ runs any computably presented Thiele machine, and the small machine's own programs, with the record earned again by the host. What's new is putting those links in a row, a box in a box in a box, and asking what the outermost one pays.

My first guess was that it could pay less, even nothing for the guest's climbs, since a host that's clever about how it runs the guest could skip the guest's costs. The first item below puts a floor under that: the host pays at least one for each step where the guest's record went up. It's a floor on rises, and it doesn't keep the guest's whole cost.

**Theorem E.4** (Nesting)**.** (1) Let a host that obeys the toll simulate a guest faithfully: each guest state has an image among the host's states and each guest record an image among the host's records, the host's record at the image of a state is the image of the guest's record there, and each guest move is carried by some host run from the image of the state before it to the image of the state after it. Let the record map also reflect the order: when the image of one guest record is at or below the image of another, the first record is at or below the second. Say a run of the host *follows* a run of the guest when it splits into one stretch of host moves for each guest step, □

each stretch carrying the image of the state before that step to the image of the state after it. Every run of the guest has a host run that follows it, every such host run ends on the image of the guest's end state, and every host run that follows a guest run costs at least the number of the guest's steps that turned a cup over. A host run that only ends on the image of the guest's end state can cost less. (2) In each link the surcharge is 0 until the guest's reading first says yes, and the bounds below hold from then on. A small-machine guest on $U_P$ is carried with surcharge exactly 0. A computably presented machine whose reading starts at no is carried, through its compiled guest, with surcharge at most 2, and, when every move costs at most one, exactly 2 once its reading has first said yes. One whose reading starts at yes is carried with surcharge exactly 3. (3) Links compose, and their surcharges add: a tower of $k$ links has surcharge between the sums of their lower and upper bounds. (4) A tower of presented machines, each level running the next on $U_P$, adds exactly 2 for every level above the first, on top of the first level's own surcharge; every level's host run is presented by the next.

*Proof.* (1) The simulation gives, for each guest move, a list of host moves carrying the image of the state before it to the image of the state after it. String those lists together along the guest run: that host run follows it, and it ends on the image of the guest's end state, because each stretch ends where the next begins. Now take any host run that follows the guest run, and a guest step that turned a cup over, leaving the down-set of its record. Its stretch carries the image of the state before to the image of the state after, and since the record map reflects the order, the image after isn't at or below the image before, or the guest's record after would be at or below the record before. If every host move in that stretch ended at or below where it started, then chaining them, the end of the stretch would be at or below its start; so some host move in it ends not at or below where it started, leaving the down-set of where it stood, and the host, obeying the toll, charges that move at least one. Different guest steps have different stretches, so the charges add up to at least the number of such steps. The end alone isn't enough: a guest on one cup that turns it over and turns it back has one rise and ends where it started, and the host run that does nothing reaches the image of the end state at no cost. (2) A link from a guest to a host matches every point of the guest's run with a point of the host's, with the record bits equal there and the host's ledger the guest's plus a surcharge, the surcharge 0 while the record is down and between the link's two bounds once it's up, and halting and the record's first rising matching. A small-machine guest run by $U_P$ is a link with surcharge exactly 0: that is the ledger equality of Appendix C.1, carried to $U_P$. A computably presented machine run as its compiled guest is a link with surcharge at most 2 from a reading that starts at no and exactly 3 from one that starts at yes, which is Theorem C.8; once the reading has first said yes the surcharge

is the larger of $3 - c$ and 0, with $c$ what the raising move cost, since the host pays the larger of $c$ and 3 there, and when every move costs at most one, $c$ is exactly one, the toll making it at least one, so the surcharge is exactly 2. (3) Match the points through the middle: a point of the bottom run has a matching point of the middle run, which has a matching point of the top run; the record bits agree along the way; the top ledger is the bottom ledger plus both surcharges, which is 0 while the record is down and between the sum of the lower bounds and the sum of the upper bounds once it's up; halting and the first rising pass through. A tower of $k$ links is that, $k - 1$ times. (4) Each level above the first is, by the presentation after this proof, a presented machine whose run is the level below's host run, carried as its compiled guest on $U_P$. Its moves are the host's moves, which cost 0 or 1, so by (2) that link adds exactly 2 once the record is up, and by (3) the additions stack on the first level's own surcharge. ∎

Item (1) is the cleanest sentence in the appendix. However you build the outer box, as long as it obeys its own toll and doesn't lie about the inner box's record, it pays at least one for every step where the inner record climbed, on every run that keeps pace with the inner box step by step. The keeping pace is needed. A guest that turns a cup over and then turns it back ends where it started, and the host can reach that image by doing nothing at all, at no charge on its ledger. The step by step floor is the guest's rises paid one by one, and the end alone can't see them. It's a floor on the climbs, and it isn't the inner box's whole cost. A guest can charge a hundred for one rise while a host charges one for the matching rise, and both obey their tolls. Keeping the whole cost takes a further demand on the simulation, that it carry the guest's costs too, and for $U_P$ that's the separate ledger theorem of Appendix C.2, with its surcharge.

Part (4) looks as if it needs a premise, and it doesn't. Two links are plainly unconditional: from a presented machine to its compiled guest, and from that guest to $U_P$ running it. Going deeper needs the host's own run, at each level, to be presented as a machine the next level up can run. All the host's states together have no one-to-one code in one number, since there's a register for every whole number. But a loaded state has only finitely many registers that aren't at 0 with version 0, and no move changes more than one of them, so the states like that are a machine of their own, closed under every move, and each has a code: its line, its finitely many live registers with their versions, its table, channel, trap, ledger and flag. Programs that always answer compute its step, price and flag from that code, and the driver reads $U_P$'s instruction off the line and says stop when the host has stopped. Its run is the host's run, and its toll is the host's. So every level is presented by the next, and the tower adds 2 a level as deep as you like. The 2 comes from the compiler's

rule of stopping at zero; a compiler that carried the excess forward would add 2 at that level's raise and give it back as the level below paid on.

### E.5 What this adds up to

Wired side by side, end to end, or one inside the other, a Thiele machine on the bench stays a Thiele machine, and the tolls add the way money adds. What breaks, breaks for reasons you could name before you saw the proof: a single flag for two claims costs two claims' worth, a clock is still a clock, a shared value without a version can't be trusted, and a hand-off that costs nothing can't be allowed to carry a certificate. None of those is a surprise in hindsight. All of them are now theorems, and the composition question from Section 2 has an answer. Two Thiele machines side by side are one under both readings I tried, "either flag" and "both flags." The "both flags" reading still fails the definition of certified Thiele-complete through the pair's interface, the second writer breaks the earned-record clause through the small machine's own reading, the clocks and the two counterexamples fail outright, and simulation in general has a proved floor.

Three of my first guesses on this bench were wrong: the single count, the harmless tipping, and the clever host that pays nothing for a climb. The last two leaned toward a record that says more than anybody paid for, and the first toward one that says less than it should. I didn't see any of the three coming; the proofs did.

APPENDIX F

# Running it

Everything so far is definitions and proofs about them. Does any of it run? This appendix is me finding out. I built the small machine twice, once straight out of the definitions and once by hand from the same description, ran both, and compared the whole state after every step. Then I ran the universal programs on guests small enough to finish, and real programs through the part of the compiler that executes.

One rule holds for the whole appendix: the runs test the builds, and the proofs carry the theorems. A run shows what a build did on the inputs it was given, and no theorem in the book leans on one: every theorem in it stands on its argument, on the page.

There are two counts to keep apart as you read. The ledger counts what the price list charges. The number of interpreter steps counts the work this particular build does to carry the run out. One guest step can take millions of host steps and add one to the ledger.

| Stage | What the next stage must preserve |
|---|---|
| Source orders | Variable values and completed procedure calls |
| Assignments and jumps | The same branches, loops and final answer |
| Counter blocks | Values at block boundaries; spare counters restored |
| Host instructions | The same result, with the opposite DEC convention translated |
| Packed two-counter guest | The result as an exponent in the packed counter |
| Fixed interpreter $U_P$ | Guest counters at matching points; decode the exponent to read the answer |

This is a path through the compiler, not a claim that one step at each stage takes the same time. The source compiler inserts no certification chain. Running the executable uses physical hardware and energy. These runs did not measure heat per host step, and the ledger is not a heat measurement. The times only tell you which runs are practical on this build. The table shows what ran and what didn't.

| What | Status |
|---|---|
| The small machine | run: the executable versions agree on every tested step |
| The host machines, with and without PAY | run, compared the same way |
| $U$ and $U_P$ on tiny guests | run, against a separate guest implementation |
| $U$ on a guest that certifies | proved and counted, not run |
| $U_P$ on a guest that certifies | proved, not run |
| Source programs through the compiler | run as far as the host program |
| The two-counter guest and $U_P$ running it | proved, not run |
| The presented machine carried by $U_P$ | proved, not run |

□ **Theorem F.1** (The compiler is correct)**.** For every source program with no recursion, and every input: the program computes $y$ exactly when the runner (the bounded evaluator of the compiled host program) halts with $y$ for some amount of fuel; exactly when the compiled host program halts with $y$; exactly when the two-counter guest halts with $y$; and exactly when $U_P$ running that guest halts with $y$, with its first counter at 0, its ledger at 0 and its flag down. So a terminating compiled program delivers its answer and certifies nothing. To certify a claim about that answer, a target program must add a CHECK, a COMMIT and a CERTIFY; this compiler does not insert that chain. The runner also takes exactly as many steps as the host program does, with the rest of its fuel left over, and a run that doesn't halt has used all its fuel.

The runs themselves, the second machine written by hand, the guests, and the compiler stages are in the companion volume *The Thiele Machine: The Example, Built* (its section "Running it") and in Coq in `coq/kernel/foundation/CmpFinal.v`, `CmpRun.v` and `Realize.v`.

APPENDIX G

# The CHSH check

The example machine checks whether a number is zero, even, or at least something. That's a fine way to show the chain, and a modest thing to certify. This appendix builds a check worth earning from zero, and then shows the one thing the small machine's host does with it, which is smaller and more exact than "it checks for quantum behaviour," a sentence that's tempting and wrong. In brief, so you know where this is going: the check never computes the famous number and never writes down the famous bound. It tests seven facts about integers in eight counts. If they pass, a certain five-by-five table has a property called positive semidefinite, and that property forces the famous number under the famous bound. The wall is a consequence of the check, and it never goes into it. Everything else is explaining those sentences.

Why put it in the book? Because it pulls four things apart that the small examples keep bunched together: the counts, the relation the counts satisfy, the act of checking that relation, and the record kept of the act. Writing a tally down costs nothing on this machine. The checker decides a real condition about a matrix, using nothing but integer arithmetic. The step ties the passing check to this edition of the tally, and the earned chain costs three, the same three it costs for "$A$ is zero." How deep the condition is and what the three acts cost are different numbers, and here you can see them apart.

And the tally on its own carries no history of where it came from. A tally somebody made up passes the same arithmetic as one that was played. What's claimed here is about the numbers and what the machine does with them, and the guarantee paragraph below says exactly how far that goes.

## G.1 The game

There are two players, Alice and Bob, in separate rooms. A referee flips two independent fair coins, independent also of the players' plan, and hands Alice a question x, zero or one, and Bob a question y, zero or one. Alice answers a, zero or one; Bob answers b. No talking during the game. They may agree on a plan beforehand, as detailed as they like. They win a round when their answers *differ*

*exactly when both questions were one*. In words: if either question is zero, match; if both are one, differ.

Try to win every time. A plan is a list: what Alice says to x equals zero, what she says to x equals one, and the same two for Bob, four bits in all. Now write the four requirements the rounds impose: a0 equals b0 (both asked zero); a0 equals b1; a1 equals b0; a1 differs from b1. Chain the first three: a0 equals b0 equals a1, and a0 equals b1, so b0 equals b1. But the fourth says a1 differs from b1, and a1 equals b0 equals b1, a contradiction, so one of the four must break. So any plan loses at least one of the four question pairs, and if the referee asks all four equally often, the best a plan can do is three out of four, seventy-five percent. Do it the long way too: there are sixteen plans (four bits), and if you write them all out and count wins you'll find eight win three rounds and eight win one, and none wins four (the broken requirements always come in odd numbers, which you can see by adding up the four equations and keeping only whether the totals come out even or odd, with "equal" as 0 and "differ" as 1: each of the four answers appears twice on the left, so the left sides add to an even number, and the right sides add to one). Sharing a coin before the game to pick a plan at random doesn't help; a mixture of numbers no larger than three-quarters is no larger than three-quarters. Arthur Fine, a philosopher of physics at the University of Washington, much of whose work is on the physicist Albert Einstein and the quantum theory, proved in 1982 that the whole "pre-agreed answers, no influence between rooms" story is *exactly* the mixtures of those sixteen lists, one scoreboard for all four questions at once; the sixteen are the whole classical story. That one I don't have to take on his word: the form of his theorem that speaks of the four averages is proved in this book, and the end of this appendix says where.

| Alice's question | Bob's question | Both answer 0 | Required relation | Result |
|---|---|---|---|---|
| 0 | 0 | equal | equal | win |
| 0 | 1 | equal | equal | win |
| 1 | 0 | equal | equal | win |
| 1 | 1 | equal | different | lose |

Changing the four planned answers can move the loss to another question pair. It cannot remove every loss: the first three equality requirements already force the fourth equality.

Now the number itself: recode answers zero and one as plus one and minus one. For each question pair, let E be the average, over many rounds, of Alice's answer times Bob's. The CHSH quantity is S equals E00 plus E01 plus E10 minus E11. For one fixed plan each win adds one to S and each loss subtracts one, so S is wins minus losses: a three-win plan has S equal to two, a one-win plan has minus two. With equal question frequencies, the win rate is one half plus S over eight. So

"seventy-five percent" and "S at most two" are the same fact, and the second is the one that got the name.

The ceiling is easy to reach and costs nothing to write down. Tally one round per question pair with both players answering zero every time: after the recoding every answer is plus one, so all four correlators are one. That's the plan that answers zero to everything, which matches on every pair and so wins the three pairs that ask for a match, $S = 1 + 1 + 1 - 1 = 2$. On the small machine's host a tally like that is a few INCs on counters, and INC costs nothing. Nothing in it says the rounds were played; anybody can tally anything at no charge on the ledger. The other side, that no finite mixture of plans beats two, is Theorem H.1 in Appendix H, one direction of Fine's result. The full equivalence, in the form that speaks of the four averages, appears in the last subsection of this appendix.

## G.2 Where it came from, and why anyone cared

So who started this, and why would anybody care whether two rooms can win a coin game? The names come in a line, and the line didn't mean much to me until I saw that each of them moved the same thing a notch: from an argument about what's real, which you can only agree with or not, toward a pile of counts a referee can judge. Each one rests on a different piece of the game above.

The trail starts in 1935, with Einstein, by then at the Institute for Advanced Study in Princeton, and two younger colleagues there, Boris Podolsky and Nathan Rosen, asking whether quantum mechanics describes everything that's real about a pair of particles. Their argument combines certain prediction with a no-disturbance premise: if measuring one particle lets you predict the other's result without disturbing it, their criterion treats that result as an element of reality. They argue that the quantum description leaves something out. What they rest on is the plan in each player's pocket, a pre-agreed list of answers, the plan above. And the first thing worth checking is whether the paper even says what Einstein meant, and by his own account it doesn't quite. Podolsky did the writing, and that June Einstein wrote to Erwin Schrödinger, the Austrian physicist whose 1926 wave equation is at the heart of quantum mechanics, that the essential thing had been "smothered by formalism." The person who lays that letter out is Arthur Fine, the same Fine whose theorem closed the sixteen plans above, in his encyclopedia entry on the argument. So Fine stands at both ends of the line: he checked the founding paper against what its author meant, and he proved exactly what the plans can do. Even the founding paper is a lead to check, not a verdict.

For thirty years that was philosophy. In 1964 John Bell turned it into arithmetic: if each particle carries pre-agreed answers and nothing travels between the rooms, certain averages can't exceed a fixed number, and quantum theory predicts they do.

What that rests on took me a few reads. For this game the assumptions can be said without that shorthand: the questions are independent of the shared preparation, and each room produces its answer using its own question and that preparation, with any local randomness included. Even stochastic response rules under those conditions are mixtures of the sixteen plans. Giving up predetermined answers alone therefore does not evade the bound. What fails is this local explanation under the stated independence condition. And where does a paper like that get published? In a brand-new journal called *Physics Physique Fizika*, which charged authors no page fees and in fact paid them by the word, ten cents a word, and which folded in 1968 after four volumes, or three. I went looking for that count, and the sources don't agree on it: four is the number the American Physical Society's archive of the journal gives, and Iulia Georgescu, writing about the journal in *Physics World* in 2026, says three. She's also where the ten cents comes from. As Andrew Whitaker tells it in his biography of Bell, the journal gave out no free copies, so Bell spent his payment buying copies of his own paper to mail to other physicists. A result nobody else holds a copy of hasn't left the room yet, and Bell paid to send his across. In 1969 four physicists, John Clauser, then a graduate student at Columbia, Michael Horne and Abner Shimony at Boston University, and Richard Holt at Harvard, rewrote it as the four-average form above, S at most two, which could be tried in a lab with imperfect equipment. It's called the CHSH inequality, after their initials. That notch rests on what a real lab can deliver. Shimony was in philosophy and in physics at Boston University both, which is the whole line in one man.

Then the obvious next question: did anybody actually try it? Stuart Freedman, a graduate student at Berkeley, and Clauser, by then a postdoc there too, tried it in 1972. The physicist Alain Aspect's group at the Institut d'Optique in Orsay, with two colleagues there, Jean Dalibard and Gérard Roger, did it in 1982 with the measurement settings switched while the photons were in flight. Their switches changed the analyzer faster than a light signal could cross between the stations. They operated at fixed, incommensurate frequencies, however, rather than choosing fresh independent random settings on every trial. Rapid switching and unpredictable questions are separate requirements; this experiment did not close every loophole. Before that, in 1975, Aspect went to see Bell at CERN to tell him what he planned, and by Aspect's own account Bell heard him out in silence and then asked, first thing, whether he had a permanent position. Aspect did. Bell's reason, as Aspect tells it, was that most physicists would take the experiment for a waste of time, or take him for a crackpot. I read that as the toll in its oldest form: a check that could have come out the other way costs somebody something to run, and Bell wanted to know who'd pay. And in 2015 three groups of experimental physicists (Bas Hensen's in Delft, with electrons 1.3 kilometres apart; Marissa Giustina's in Vienna; Lynden Shalm's at NIST, the National Institute of Standards and Technology, in Boulder) closed the

two main escape routes at once. In 2022 Clauser, Aspect and the Austrian physicist Anton Zeilinger shared the Nobel Prize "for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science." The 2015 tests came out of that line of work, and Zeilinger was one of the authors of the Vienna one. So in under fifty years the subject went from a question you needed a permanent job to risk to a Nobel. The lab number sits above two. A remaining escape, that the referee's coins were secretly correlated with the particles, is a philosophical one and I'm not going to pretend it's closed.

Now put this book's check on the same line, because that's where my position gives. It sits past the far end, at the counts. Those fifty years went into making the counts worth judging, rooms really apart, questions really late, escape routes really shut, and my check starts after all of it. It's handed eight counts and can't tell a lab's tally from one somebody wrote down at no charge. So the check is a referee of counts, and everything that made these counts worth refereeing happened upstream of it.

What quantum play does: if the players share an entangled pair and choose what to measure from their question, the best they can do is S equals two times the square root of two, about 2.83, a win rate of about eighty-five percent. Boris Tsirelson, a Soviet mathematician who later taught at Tel Aviv University, in a 1980 paper called "Quantum generalizations of Bell's inequality," proved that that's a *ceiling*: quantum correlations violate Bell, but only up to there. I prove that ceiling below, and show the state that reaches it, with real amplitudes and with complex ones. The bound holds in arbitrary dimension, including infinite dimension; the construction attaining it needs sufficient room, supplied by the two-qubit example below. The first thing proved is the algebra.

Then there's the wall above that, which is the one I find most instructive. In 1994 the physicists Sandu Popescu and Daniel Rohrlich asked: what if you only demand that nothing can be *signalled*, that Alice's answers on their own have the same odds whatever Bob is asked, and the other way round? Then you can write down a "box" (the PR box, after their initials) that wins *every* round (its answers differ exactly when both questions are one, by fiat, each answer alone a fair coin), S equals four, and nothing about it lets either room send a message. So no-signalling on its own doesn't explain the quantum ceiling; the PR box meets that condition and beats 2.83. (A box isn't a physical theory, and nobody is claiming one that meets everything relativity asks.) The order is: two for plans, 2.83 for quantum, four for no-signalling.

And the box has the same kind of history as everything else in this book: somebody got there first. Popescu and Rohrlich's initials stuck, but the big review of the field by Nicolas Brunner and four coauthors credits boxes like it to Peter Rastall, whose 1985 paper on locality and Bell's theorem appeared in *Foundations of Physics*,

and, separately, to Leonid Khalfin, a physicist at the Steklov Institute's Leningrad branch, and Tsirelson himself, both in 1985, nine years before. Then the question that really matters: if relativity isn't what stops things at 2.83, what is? People have tried candidates. In 2005 the computer scientist Wim van Dam, who works on the theory of quantum computing, argued, in a paper on the implausible consequences of such boxes, that with enough PR boxes, Bob could work out any yes-or-no function of both their inputs after Alice sends him a single bit, which would make the whole question of how much two parties need to talk collapse, and that's a strong hint the boxes are too good to be real. In 2009 Marcin Pawłowski, a physicist at the University of Gdańsk, and five coauthors proposed a principle they called information causality: Bob can't get more information about Alice's data than the bits she actually sends him. They argue that ordinary and quantum physics both obey it, and that any theory that beats 2.83 breaks it. How much of the rest of the quantum boundary it reaches is something I haven't checked. Which principle fixes the whole quantum boundary, if any one does, this book won't answer. The check below is a fence at the middle step.

### G.3 Five sentences that are easy to say by accident

Each one comes below with what I can stand behind beside it.

"Faster than light." The correlations alone cannot carry a chosen message between the rooms: each room's marginal probabilities do not depend on the other question. That operational statement does not settle every account of what happens between measurements. "Quantum means random." The state and the chosen measurements determine the outcome probabilities and expected score; a particular outcome can be certain or random. "A Bell violation proves nonlocality." A statistically supported violation rejects the local response model under the setting-independence and experimental assumptions. Calling it a rejection of "realism" does not identify those assumptions.

"A tally above two proves nonlocality." No: S is a *population* average, and one finite tally can land above two by luck. Four rounds, one per question pair, with coin-flip answers, reach S equals four one time in sixteen. That gap between a population bound and a finite sample is the problem Yanbao Zhang, Scott Glancy and Emanuel Knill, at NIST in Boulder, work through for real Bell tests (Physical Review A, 2011 and 2013), and the book doesn't solve it. The way they solve it is worth knowing, because it has the shape of this book's record. Turn each round into a small bet that no local plan can expect to win, even a plan that remembers every earlier round, multiply the bets as the rounds come in, and the French mathematician Jean Ville's inequality, from his 1939 thesis, one of the works that started the theory of martingales, says the chance the product ever climbs past one over $\delta$ is at most $\delta$. So

a flag can go up for good the moment the product crosses, with its error stated up front. It needs one thing the small machine doesn't have: settings drawn underneath the program, where the program can't choose them. And how much can one round tell you? Wim van Dam, Richard Gill, a mathematical statistician who later held the chair in that subject at Leiden, and Peter Grünwald (2005) measured exactly that, in bits of evidence against every local plan. Of the two-party proofs they compare, CHSH comes out the strongest, "much stronger" than the others in their words, at about 0.046 of a bit per round, a few hundredths, so a convincing record takes hundreds of rounds, and four won't do. About a finite tally, this book proves only the narrow thing, that a tally above two, with every question pair asked at least once, is not explained by *one fixed plan replayed every round*, and that narrow thing is what the sentence has to stay. The bound for a finite local model is a separate theorem, and it says nothing about one tally. And "the PSD test proves the data is quantum." (PSD is short for positive semidefinite, the property of the five-by-five table from the opening of this appendix; the next subsection builds both.) It doesn't, and the direction matters. Passing implies the five-by-five table, pinned in a way the next subsection explains, is PSD, and that implies S squared is at most eight; both are proved. The correlators of a PSD table like that are the correlators of some quantum strategy, by Tsirelson's representation theorem, which is proved here: the strategy is built on two systems of four levels each, or two levels each with complex amplitudes. The exact edge of the quantum set for this game was found by Lawrence Landau, whose 1969 Berkeley thesis was on the quantum theory of local observables, and Lluís Masanes, then at the University of Barcelona (next subsection), and that's proved here too. Failing says nothing about whether the play was quantum: the check is complete for the pinned matrix, the quantum set is wider than that, and quantum behaviour can fail it. Every one of the sixteen plans fails it, and I'll show you why. What it guarantees is in one paragraph, once the check is on the host machine.

## G.4 The five-by-five, and what positive semidefinite means

The mathematics the check rests on is one idea: a table of numbers *could be a table of dot products*. Take a two-by-two symmetric table with a and c on the diagonal and b off it. It's positive semidefinite (PSD) when a and c are at least zero and a times c minus b squared is at least zero. Check two by hand: ones on the diagonal and one half off, the determinant is three quarters, passes; ones on the diagonal and 1.2 off, the determinant is negative, fails. The reading: PSD means the table could be the dot products of some arrows, and 1.2 is too big a dot product for two arrows of length one. Everything else is that idea, generalised to bigger tables.

Now the table. Rows and columns labelled I, A0, A1, B0, B1 (the identity, Alice's two questions, Bob's two). Diagonal all ones. The I row and column zero off the diagonal (that's "zero marginals": the table takes each answer alone to average to zero, a fair coin; the counts record no marginals and the check doesn't test this). A0 with A1 zero, B0 with B1 zero (no round asks Alice both questions, so no tally holds that average, and the table sets it to zero). Those zeros are choices. They pin the table to one slice, and I'll come back to what they cost. And the four Alice-Bob cells hold the four correlators E, which leaves a five-by-five with four free numbers in it. It splits into the one in the corner and a four-by-four block made of an identity and the two-by-two grid E of correlators, and PSD of that block is the same as saying *E shrinks every vector* (E is a contraction), which is three inequalities: one minus E00 squared minus E10 squared at least zero; one minus E01 squared minus E11 squared at least zero; and their product at least the square of (E00 E01 plus E10 E11).

Run it. All four correlators 0.6 except E11 at minus 0.6, S equals 2.4: the two diagonal terms are 0.28 and 0.28, the cross term zero, passes; above two, below 2.83. All four at one over root two (E11 negative): diagonals zero, cross term zero, passes with no slack, S equals two root two exactly, the 2.83; nudge any entry's size up and a diagonal goes negative. The all-ones plan (one, one, one, one): first diagonal is one minus one minus one, negative, *fails*, though S equals two and it's a perfectly legal classical play. The PR box (one, one, one, minus one): fails the same way. The check is a fence at the middle step, and a fence that keeps out every one of the sixteen plans: each has all four correlators at plus or minus one, so the first diagonal is always one minus one minus one. Of the classical plays, only diluted mixtures of plans get through.

That point, one over root two in size in every entry, is four real numbers, and no tally of rounds ever lands on it. A tally's correlators are ratios of whole numbers, and the point needs each of them to be one over root two, which would make root two a fraction, and you killed that fraction with a pencil in Section 5. That point and its mirror image, every sign flipped, are the only places the bound is reached (Remark H.1 in the appendix). So a passing tally always has S squared strictly below eight. It can get as close as you like: the (17, 3) tally in the five examples below has S squared equal to 7.84, and tallies of the same shape with a fraction nearer one over root two get nearer still. Two root two is the least bound the check can promise, and no tally the check accepts ever reaches it. Hold on to that corner; it comes back.

Picture what quantum play can reach as a field, and every test as a fence. In 2007 Miguel Navascués, Stefano Pironio and Antonio Acín, in a paper called "Bounding the Set of Quantum Correlations" (NPA, after their initials), came at the edge from outside, resting on necessary conditions. In their terms a table like this one is the

first rung of a ladder: bigger tables of expected products, each PSD, each a necessary condition for quantum behaviour when its entries are left free, each rung tighter, and (their 2008 paper) converging to the quantum set (strictly, the set from commuting operators, which can differ from the tensor-product set in general scenarios). Each rung is a fence with the whole field inside it. The check's table is the first rung with the marginals and the two within-party entries pinned to zero. Pinning is a stronger demand than the rung makes, so passing puts you inside the first rung, and a quantum point can still fail. My fence cuts across the field. Lawrence Landau in 1988 and Lluís Masanes in 2003 came at the edge itself, for this two-questions-each case only. I'd have drawn a flat wall at 2.83. The exact boundary they found is a curved surface (the arc-sine form) touching 2.83 at one point. It's proved here: a table has a PSD completion exactly when its entries lie between minus one and one and their arcsines pass four inequalities, so those are exactly the quantum tables.

So the obvious next question: if every rung is tighter than the last, does some rung finally land exactly on the quantum set and stop? My first guess was yes, at least here. A 2026 preprint by Anubhav Chaturvedi, a quantum-information theorist at the University of Gdańsk, says no: no *finite* rung of the ladder captures the whole quantum set even in this simplest case. And for the four correlators alone, a 2008 paper by the same Navascués, Pironio and Acín works out that the first rung already lands on Landau and Masanes's curved edge. That looked like a contradiction until I saw what each one fences. Landau and Masanes draw the edge for the four correlators alone. Chaturvedi's is the full behaviours, each room's own odds included. So there's the corner again from the other side: tallies never reach it, and, if his proof holds, rungs never close on the field. I wanted that to be one fact, but it's two. Mine is root two again, about which points a count can hit. His is about the fences. His result backs, from outside, the caution this book keeps about the ladder, that no finite rung may close on the whole quantum set. It says nothing about my own check, though. My check's caution, that a quantum point can fail it, comes from the pinning, and his paper never pins anything.

## G.5 The check, in integers

There are eight counts: for each of the four question pairs, one counting rounds where the answers matched and one where they differed. A round is one bump to one count; nothing about the round is tested, and no marginals are recorded. Then the check reads the eight and tests seven integer facts. For each pair, let N be same plus different and D be same minus different (so E is D over N). The seven: the four N are strictly positive (every pair was sampled; a pair with no rounds is rejected outright, and never read as a correlator of zero); A, which is N00 squared N10 squared minus D00 squared N10 squared minus D10 squared N00 squared, is

at least zero; B, the same with the other column, is at least zero; and with C equal to D00 D01 N10 N11 plus D10 D11 N00 N01, C squared is at most A times B. Those are the three contraction inequalities with the denominators cleared, and there's no division in them, no reals, no square root and no S. Try it: counts (same, different) of (4, 1), (4, 1), (4, 1), (1, 4) give E of 0.6, 0.6, 0.6, minus 0.6; N is five and D is three (minus three for the last); A is 175, B is 175, C is zero; passes. These counts are the first example above, the S equals 2.4 one, written in integers.

| Question pair | same | different | total | difference |
|---|---|---|---|---|
| 00 | 4 | 1 | 5 | 3 |
| 01 | 4 | 1 | 5 | 3 |
| 10 | 4 | 1 | 5 | 3 |
| 11 | 1 | 4 | 5 | −3 |

First check the four totals: all are positive. Each column's remaining allowance is $625 - 225 - 225 = 175$. The cross term is $225 - 225 = 0$. Thus the final comparison is $0 \leq 175^2$, and the check passes. These steps verify the arithmetic; they supply no evidence that rounds were played.

The theorem chain, in words: a passed check gives PSD of the five-by-five built from the counts; contraction and PSD are equivalent, both directions; and PSD forces S squared at most eight. The names are in the statements below. The underlying algebra is one identity: four times the sum of the four correlators squared, minus S squared, is a sum of six squares, so if the row sums of squares are at most one, S squared is at most eight, and the point one over root two reaches it. So Tsirelson's bound comes out as *algebra*, with no Hilbert space (the mathematical home of quantum states) in it.

The chain goes below as three statements. The derivations are in Appendix H, and the statements say what they prove.

**Definition G.1** (Correlators from the counts; the pinned matrix)**.** For eight counts and a question pair $xy$, let $E_{xy} = (\text{same}_{xy} - \text{diff}_{xy})/(\text{same}_{xy} + \text{diff}_{xy})$, read as 0 when both are 0. For reals $E_{xy}$, $M(E)$ is the symmetric five-by-five matrix on $I, A_0, A_1, B_0, B_1$ with ones on the diagonal, $E_{xy}$ in the cells $(A_x, B_y)$ and $(B_y, A_x)$, and zero everywhere else (Definition H.5). $M$ is *PSD* when $v^{\mathrm{T}} M v \geq 0$ for every real vector $v$, and $S(E) = E_{00} + E_{01} + E_{10} - E_{11}$.

□ **Theorem G.1** (The integer check implies the pinned matrix is PSD)**.** If eight counts pass the seven integer facts, then $M(E)$ for their correlators is symmetric and PSD.

*Proof.* The four positivity facts make every denominator positive, so each $E_{xy}$ is $D_{xy}/N_{xy}$, and dividing the other three facts by positive squares turns them into the three contraction inequalities (Lemma H.9). Contraction is equivalent to PSD of $M(E)$ by a Schur complement on the four-by-four block (Corollary H.7). ■

In the appendix this is Theorem H.10, and Corollary H.7 gives the equivalence of contraction and PSD in both directions.

**Theorem G.2** (PSD forces the Tsirelson bound)**.** For all real $E_{00}, E_{01}, E_{10}, E_{11}$: if $M(E)$ is symmetric and PSD, then $S(E)^2 \le 8$. □

*Proof.* Call the four entries $a, b, c, d$, read across the rows. PSD means that no choice of five weights makes the matrix's weighted total negative. Choose weights $(0, 1, 0, -a, -b)$ in the order $(I, A_0, A_1, B_0, B_1)$. The diagonal contributes $1 + a^2 + b^2$, and the two cross terms contribute $-2a^2 - 2b^2$. The total is $1 - a^2 - b^2$, so $a^2 + b^2 \le 1$. Weights $(0, 0, 1, -c, -d)$ give $c^2 + d^2 \le 1$. All four squares therefore add to at most two.

Now compare four times that sum with the squared score. Their difference is

$$4(a^2+b^2+c^2+d^2) - (a+b+c-d)^2 = (a-b)^2 + (a-c)^2 + (a+d)^2 + (b-c)^2 + (b+d)^2 + (c+d)^2.$$

To check it, each squared letter occurs three times on either expanded side; the cross terms are $-2ab, -2ac, +2ad, -2bc, +2bd, +2cd$ on both. The difference cannot be negative because it is a sum of squares. Thus the squared score is at most four times two, which is eight. ■

This is Theorem H.8 in the appendix.

**Corollary G.3** (A passed check means the bound)**.** If eight counts pass the seven integer facts, their correlators satisfy $S^2 \le 8$. In fact $S^2 < 8$, and passing tallies come as close to 8 as you like. □

*Proof.* Theorem G.1 makes the pinned matrix of these four correlators PSD, and Theorem H.8 in the appendix, which holds of every such matrix, gives $S^2 \le 8$; the calculation with the quadratic form is printed in the companion volume. The score is a ratio of whole numbers, and no such ratio squares to 8, so the bound is strict. The tallies whose four correlators are $n/q, n/q, n/q, -n/q$ with $2n^2 \le q^2$ pass, and their scores $4n/q$ come as near $2\sqrt{2}$ as you like. ■

**Remark G.1** (Sound, and incomplete)**.** The pinned matrix is stricter than the quantum question it is usually read against. PSD of $M(E)$ gives the correlators a PSD completion, since the zeros are one choice of the two free entries. The converse fails. The all-ones plan has a PSD completion, and every deterministic plan is a quantum strategy; yet its minor on rows $A_0, B_0, B_1$ is $1 - 1 - 1 = -1$, so $M(E)$ is not PSD there, and a tally of one matching round per pair gives $A = 1 - 1 - 1 = -1$ in the integer check, so the check fails. The second failure is a named theorem on the machine: the tally of one matching round per pair is refused forever (Proposition G.6). Every deterministic plan fails the same way. The tally of one round per pair fails the check for each of the sixteen plans, and so does any tally of one plan replayed over any number of rounds (a pair left unsampled fails the positivity facts outright), because no deterministic plan's pinned matrix is PSD. So a pass proves the bound, and a failure proves only that the tally's pinned matrix isn't PSD or a pair went unsampled; it says nothing about whether the play was classical or quantum. The tally from the start of this appendix, written down at no cost, is exactly such a tally: it reaches $S = 2$ and the check throws it out.

None of that algebra is new, and I don't want it to read as if it were. Mnacho Echenim and Mehdi Mhalla, computer scientists at Grenoble, give a detailed treatment of the CHSH and Tsirelson bounds (2023). Navascués, Pironio and Acín already analyse this very matrix by Schur complement and discuss the level above it. What I think is mine here is smaller: an integer decision procedure on the counts, and a proved link from "the check passed" to PSD to the bound.

The picture under the check is older still. Tsirelson showed, in a longer paper of 1987 after his 1980 note, that the correlations two separated quantum parties can reach are exactly the tables of inner products of unit vectors, one vector for each question on each side. With the marginals pinned to zero the pinned matrix is that picture, and by its Schur complement it is PSD exactly when the two-by-two table of correlators, read as a matrix, has operator norm at most one. The physicists Reinhard Werner and Michael Wolf's survey of Bell inequalities (2001) sets out that correlation-matrix view for two-party inequalities in general.

## G.6 The check, earned on the host machine

Now put it on the machine. A check on the small machine is a property: a checker, which is a little program that says yes or no, a meaning, which is a statement, and a proof that the two agree. The version of the small machine with its property language left open (Section 15, Theorem 15.31) takes any property language with an exact equality test whose properties come with those three things, and for every one of them a flag raised from a clean start stands on a passing CHECK of one

claim, a later COMMIT of the same claim with its counter untouched in between, and a later CERTIFY, the three acts costing one each. For the game, pack the eight counts into one counter, as a list, the way the sorted-list machine packs a list into one number (Section 15). Let the checker be the seven integer facts read off the unpacked counts. And let the meaning be what the facts are for: every pair was sampled, and the pinned five-by-five is positive semidefinite. Then the proof the generic machine asks for, that the checker says yes exactly when the meaning holds, is the chain above with the denominators cleared in both directions, and a flag raised on that machine from a clean start means: these counts, at this edition of their counter, passed the check, somebody committed to that with nothing changed underneath, and the correlators they give satisfy $S^2 \leq 8$. The theorems below state that exactly, and they're stronger than the chain above in one way I want to say once and plainly. The earlier theorems said that a passed check makes the pinned matrix PSD. The check is also complete for it: it passes *if and only if* every pair was sampled and the pinned matrix is PSD, because clearing the denominators works in both directions. The statement is over the machine of Definition C.2, the small machine with a counter for every number, with PCHSH in place of its one property, Slot. A CHECK reads one counter, which is why the eight counts have to travel in one.

**Definition G.2** (The packed tally, and the property PCHSH)**.** Read a whole number as a list in the list code (0 is the empty list, and $2^x(2y+1)$ is $x$ followed by the list $y$), and take as its *tally* the first eight entries, in the order $\text{same}_{00}, \text{diff}_{00}, \text{same}_{01}, \text{diff}_{01}, \text{same}_{10}, \text{diff}_{10}, \text{same}_{11}, \text{diff}_{11}$, a missing entry reading 0. Every number is some tally. The property PCHSH has the checker "the tally passes the seven integer facts" and the meaning "every question pair was sampled, and the pinned matrix $M(E)$ of the tally's correlators is symmetric and PSD."

**Theorem G.4** (The check is exact)**.** (1) The seven integer facts hold of a tally if and only if every pair was sampled and $M(E)$ is symmetric and PSD. (2) For every whole number $v$, the checker of PCHSH says yes at $v$ if and only if the meaning holds of $v$'s tally. (3) Reading the number that packs a tally gives that tally back. (4) If the facts hold, the score of the tally satisfies $S^2 \leq 8$ and $|S| \leq 2\sqrt{2}$; in fact $S^2 < 8$ and $|S| < 2\sqrt{2}$, and passing tallies come as close to 8 as you like. □

*Proof.* (1) The forward half is Theorem G.1. For the backward half, PSD gives the three contraction inequalities (Corollary H.7); with every $N$ positive, multiplying them through by the squares of the denominators gives $A \geq 0$, $B \geq 0$ and $C^2 \leq AB$, which is Lemma H.9 read the other way. (2) is (1) with the packing, since every

number is a tally. (3) is the code and its decoder being inverse. (4) is (1) forward, then Theorem G.2; the strict form and the closeness are Corollary G.3. ■

□ **Theorem G.5** (A raised flag stands on the check)**.** On the machine above, let $\tau$ be a trace from a clean start $s_0$ that ends with the flag up. Then, for some counter $c$,

$$\tau \;=\; pre \cdot \text{CHECK PCHSH } c \cdot mid_1 \cdot \text{COMMIT PCHSH } c \cdot mid_2 \cdot \text{CERTIFY} \cdot post,$$

and with $v$ the value of $c$ after $pre$: the tally of $v$ passes the seven integer facts; every pair was sampled and the pinned matrix is symmetric and PSD; its score satisfies $S^2 \le 8$ and $|S| \le 2\sqrt{2}$, and in fact $S^2 < 8$, though passing tallies come as close to 8 as you like; and $c$ still held $v$ when the COMMIT ran. For any stored program run from a fresh start (counters set, every version 0, line 1, the table and channel empty, the trap and the flag down, the ledger at 0) that raises the flag, the trace it executed contains a CHECK PCHSH $c$ that passed, and the tally $c$ held when it ran passes the seven integer facts, has every pair sampled and the pinned matrix symmetric and PSD, and has $S^2 \le 8$ and $|S| \le 2\sqrt{2}$. Conversely, from a fresh start in which a counter $c$ holds the packing of a tally $t$, the three instructions CHECK PCHSH $c$, COMMIT PCHSH $c$, CERTIFY raise the flag if and only if $t$ passes the check, and then the ledger reads exactly 3; if $t$ fails, the trap goes up at the CHECK and no number of further steps raises the flag.

*Proof.* A raised flag from a clean start has the earned chain (Theorem 15.11, which holds for this machine as for the small one): a passing CHECK, then a COMMIT of the same claim with the counter's edition unchanged, then a CERTIFY. A passing CHECK of PCHSH at $c$ means the checker says yes at $c$'s value, which is the meaning by Theorem G.4(2), and the bound is (4) of the same theorem. An unchanged edition means an unchanged value. For the converse, the CHECK passes exactly when the tally passes; if it does, COMMIT and CERTIFY find their facts and the three acts cost three; if it fails, the trap latch goes up, and no later move can raise the flag. A stored program stops at the trap; an explicitly supplied instruction trace may still incur its defined charges (as in Theorem 15.22). ■

So the one thing the machine does with the check, as this appendix's opening said, is that a raised flag can stand on it. Five tallies, each a theorem with its counts written out, show the check isn't vacuous and where its edges are.

**Proposition G.6** (Five tallies)**.** Counts are given as (same, diff) for the pairs 00, 01, 10, 11. (4,1), (4,1), (4,1), (1,4) has $S = 12/5$: the check passes, the flag rises, the ledger reads 3. (17,3), (17,3), (17,3), (3,17) has $S = 14/5 = 2.8$, just under $2\sqrt{2}$, with $S^2 = 7.84$: it passes, the ledger reads 3. (9,1), (9,1), (9,1), (1,9) has $S = 16/5$ with $S^2 > 8$: refused forever, whatever follows. (1,0), (1,0), (1,0), (0,1), the PR box, has $S = 4$: refused forever. (1,0), (1,0), (1,0), (1,0), the all-ones classical plan, has $S = 2$: refused forever. □

*Proof.* A pair's correlator is $(\text{same} - \text{diff})/(\text{same} + \text{diff})$, and $S$ adds the first three and subtracts the fourth. $(4, 1)$ gives $3/5$ and $(1, 4)$ gives $-3/5$, so $S = 12/5$; the seven facts are the worked tally after Corollary H.11, with $A = B = 175$ and $C = 0$, so the check passes, and Theorem G.5 gives the flag and the ledger at 3. $(17, 3)$ gives $7/10$ and $(3, 17)$ gives $-7/10$, so $S = 14/5$ and $S^2 = 196/25 = 7.84$; with $N = 20$ in every pair and $D = 14, 14, 14, -14$, $A = B = 400 \cdot 400 - 196 \cdot 400 - 196 \cdot 400 = 3200$ and $C = 14 \cdot 14 \cdot 400 - 14 \cdot 14 \cdot 400 = 0$, so the seven facts hold and the same theorem applies. $(9, 1)$ gives $4/5$, so $S = 16/5$ and $S^2 = 256/25 > 8$; the PR box gives $1, 1, 1, -1$ and $S = 4$; both fail the check, because a tally that passes has $S^2 \leq 8$ (Corollary H.11), and a failed check traps at the CHECK, after which no step raises the flag (Theorem G.5). The all-ones plan gives $N = D = 1$ in every pair, $S = 2$, and $A = 1 - 1 - 1 < 0$, the worked tally's second computation, so it fails the check and is refused the same way. ■

The last one is the check being stricter than the bound, as a named theorem: a perfectly legal classical play with $S = 2$ is turned away.

What the theorem doesn't say stays what it was. The tally is *placed*. Nobody plays it on the machine: the theorem starts from a counter that already holds the packing, and no instruction plays a round, so a program can put any code in a counter at no cost. There are no marginals. The pinned matrix is stricter than the quantum set, so failing says nothing about whether the play was quantum, and there is no margin for sampling noise at Tsirelson's point.

Writing a tally costs nothing. Each count is a counter, a round is an INC on one of them, and INC costs nothing, so a pile of trials costs nothing to write down (what it spends is fresh states, in a counter that never runs out of numbers); the packing into one counter is a program's business, and in the theorem the packed counter is simply there at the start. What costs is the claim about the pile. On the ledger, statistics cost nothing; the claim about them is what costs.

So what the check guarantees fits in one paragraph. If it passes, then every question pair was sampled, the four ratios from the counts satisfy three polynomial inequalities, so the pinned five-by-five is PSD, so S squared is at most eight. It

does *not* guarantee that the tally came from honest play (a program can INC whatever counts it likes at no cost, including four rounds that put S at four). It does not guarantee any marginal is zero (no marginals are recorded and none are tested). It does not put the players' state behind the counts. What it does give is a quantum strategy with the same correlators: a passing tally's correlators have a PSD completion, and every point with one is the correlator table of a strategy on two four-level systems. And it fails the tally of every single plan, honest classical play included. It has no margin for sampling noise either. Honest play at Tsirelson's point aims at correlators of plus or minus one over root two, and there every pair of squared correlators the check adds up comes to exactly one, the edge of what it accepts. So an honest tally aimed at that point sits right on the line, and ordinary luck over a finite number of rounds pushes it over most of the time. I simulated it, drawing each round's answers at Tsirelson's correlators: the check fails 92 percent of the time at 10 rounds per question pair, 87 percent at 100, and between 85.4 and 85.9 percent from 1,000 to 100,000. In those simulated trials, fair coins passed every time from 100 rounds per pair on, and failed about 2.5 percent of the time at 10. I saw that in the trials, and it's no guarantee about any one tally. The limit has a closed form. At Tsirelson's point the correlator matrix, read as a two-by-two, turns arrows without stretching them (it's a mirror), and every one of its entries wobbles by the same amount, independently. To first order the largest singular value of the noisy matrix is one plus the larger eigenvalue of $(G + G^{\mathrm{T}})/2$, where $G$ is the wobble turned back by that mirror, a two-by-two of independent Gaussian entries of equal size, and the check passes only when that eigenvalue is at most zero. Write that symmetric matrix's eigenvalues as $u \pm \sqrt{v^2 + w^2}$, with $u, v, w$ independent and of equal spread (scale them to standard normals). Both are at most zero exactly when $u \le -\sqrt{v^2 + w^2}$. Now $v^2 + w^2$ has the exponential law with mean 2, so the chance is $\frac{1}{2}\,\mathbb{E}\left[1 - e^{-u^2/2}\right] = \frac{1}{2}\left(1 - 1/\sqrt{2}\right) = (2 - \sqrt{2})/4$. So honest play at Tsirelson's point fails with probability $(2 + \sqrt{2})/4 \approx 0.854$ in the long run. The arithmetic is on the page; the first-order step is not proved here, and the simulation is what backs it.

The check can also allow a tolerance. For $\varepsilon = p/q$ it tests the same seven facts in integers with the 1 in the contraction inequalities replaced by $(1 + \varepsilon)^2$. At $\varepsilon = 0$ it is the check above. A pass means the two-by-two correlator matrix has operator norm at most $1 + \varepsilon$, and $S^2 \le 8(1 + \varepsilon)^2$: divide every correlator by $1 + \varepsilon$ and the tolerance check becomes the exact one, so the chain above applies to the shrunken table. A tally of 100 rounds per pair with correlators $0.72, 0.70, 0.72, -0.72$ fails the exact check and passes at $\varepsilon = 1/10$. In the same kind of simulation, with $\varepsilon = 3/\sqrt{n}$ for $n$ rounds per pair, honest Tsirelson play failed it less than once in ten thousand trials at every size from 100 to 100,000 rounds per pair: under once in a hundred thousand at 100, five at 1,000, eight at 100,000, close to the long-run

rate the same first-order picture gives, about nine in a hundred thousand. In that Gaussian calculation the exact expression is $1 - \Phi(6) + e^{-9}\Phi(3\sqrt{2})/\sqrt{2}$, where $\Phi$ is the standard normal cumulative probability. Dropping its tiny tail corrections gives $e^{-9}/\sqrt{2}$. This is the asymptotic calculation under independent sampling, not a finite-sample confidence guarantee. What the tolerance gives up shows in the bound: a pass caps $S$ at $2\sqrt{2}\,(1+\varepsilon)$, a little outside Tsirelson's number. The exact check remains the default. With $p$ and $q$ fixed for a machine instance, the tolerance check also runs as CHECK on the same versioned registers: its CHECK, COMMIT, CERTIFY chain certifies exactly when the integer tolerance test passes, costs three on success, and traps permanently on failure. From a clean start, every raised flag has that earned chain behind it, with the same register value at CHECK and COMMIT, and establishes the relaxed bound above: it's Theorem G.5 again, with the tolerance test as the checker and "every pair sampled and operator norm at most $1+\varepsilon$" as the meaning, the scaling just described being the proof that they agree. The check certifies a *shape* of the tally. What the shape means is a question for whoever supplied the rounds.

And notice which way the fence faces. The most practical use anyone has found for a Bell test, certified randomness, needs the opposite bound. Stefano Pironio and colleagues (2010) argued, and backed with an experiment, that a Bell violation, with settings the devices can't choose, certifies that the outputs are random, and a group of quantum cryptographers, Rotem Arnon-Friedman, Frédéric Dupuis, Omar Fawzi, Renato Renner and Thomas Vidick (2018), made the accounting tight with what they call entropy accumulation. Under those assumptions, a predetermined local output strategy passes a properly calibrated test only within its stated false-acceptance probability, so it's the cleanest case I know, outside this book, of a check that can fail, separating output that was established from output that was printed. That needs $S$ bounded from below, with an error probability. This check bounds $S$ from above and fails every single plan, so it's a fence that keeps out the super-quantum and the fixed plan. As a nonlocality test it's no use. The deployed version exists, too: Gautam Kavuri and colleagues, physicists at NIST and the University of Colorado (2025), run a public randomness beacon whose outputs each trace back through a Bell test to a public record, the nearest working system I know to a certified claim written down for good. And records cut both ways there. Jonathan Barrett, Roger Colbeck and Adrian Kent, working on quantum cryptography (2013), pointed out that a device which keeps its own memory of earlier rounds can leak it through outputs it makes public later, so reusing a device can undo a guarantee the user thought was settled. A permanent record in the wrong box is an attack.

### G.7 A rational stand-in for an irrational number

One useful rational bound: 5657 over 2000, which is 2.8285, and two root two is 2.82842... It's the four-decimal round-up, chosen so that rational arithmetic can test "above the Tsirelson bound" without square roots, and it's safe because 5657 squared is 32,001,649 and eight times 2000 squared is 32,000,000, so the fraction sits above root eight by about seven hundred-thousandths. It's used in a few rational theorems. It is *not* used in the integer check, and the real-number theorems use root eight.

### G.8 Two more tests, as mathematics

The second rung: a nine-by-nine table on I, the four questions, and the four Alice-Bob *products*, with the correlators and five extra free numbers for three- and four-body entries. Its two four-body entries are tied, one the negative of the other. That tie is a constraint, $\langle \{A_0, A_1\} \otimes \{B_0, B_1\} \rangle = 0$: the zero entries of the first rung don't imply it, and a classical strategy breaks it. So the second rung is level 1+AB on that slice, and that's what I mean by level 1+AB everywhere in this book. Its PSD is again equivalent to a contraction-like condition. The check for it proves something stronger, that the matrix is positive definite, by a cascade of two Schur steps down to a four-by-four that Sylvester's criterion closes. Sylvester's name on a test about matrices fits in a way I enjoy, because the word *matrix* is his. In 1850, in the *Philosophical Magazine*, the English mathematician James Joseph Sylvester used it for an "oblong arrangement of terms" out of which you cut square blocks and take their determinants, and *matrix* was already an old word for the place something grows from. The test does exactly that: it cuts the top-left square blocks out of the big grid, one row deep, two, three, four (the leading minors), and asks each one's determinant to be positive. Was he the first to state the test? The name doesn't settle that; names stick to people for all kinds of reasons, and I'd treat this one as a lead. There's also a fact that's easy to list without noticing what it means: with all five extra numbers at zero, the condition certifies a strict *subset of the classical set* (the ball where the four correlators' squares sum to at most one), forcing S at most two; it can't certify anything above the classical line. To get a point above two you have to supply the extras (S equals 2.4 with one extra at minus one half). Appendix H has the algebra.

The two sizes are one test. A single positive-semidefinite predicate takes the size as a parameter, and the five-by-five and the nine-by-nine are it at two sizes.

**Definition G.3** (The size-generic PSD test)**.** A *claim* is a top index $n$ and a matrix $M$ indexed by naturals. It is *realizable* when $M_{ij} = M_{ji}$ for all $i, j \le n$ and $\sum_{i \le n} \sum_{j \le n} v_i M_{ij} v_j \ge 0$ for every real vector $v$. The CHSH claim of $E$ is top index 4 with the matrix $M(E)$ of Definition G.1; the level-1+AB claim uses top index 8 and the nine-by-nine matrix.

**Corollary G.7** (The level-1 test is the size-five case)**.** For all real $E_{00}, E_{01}, E_{10}, E_{11}$: the three contraction inequalities hold if and only if the CHSH claim of $E$ is realizable. The same holds at size nine for the level-1+AB contraction condition and its claim. □

*Proof.* Top index 4 means the five indices $0, 1, 2, 3, 4$. The double sum visits each of their twenty-five cells once, multiplies it by its row and column weights, and adds the results. That is precisely the weighted-total test for the five-by-five matrix. The completed-square calculation in Corollary H.7 equates that test with the three contraction inequalities. Top index 8 similarly visits the eighty-one cells of the nine-by-nine matrix. Theorem H.12 writes its total as three squares plus the residual and proves that its test is exactly the residual condition. The same definition therefore asks the same question at either size. ■

The third thing is the *elliptope*, the set of correlators for which *some* choice of the two missing entries (A0 with A1, B0 with B1) makes the five-by-five PSD. Every one of the sixteen plans is inside it, mixtures stay inside, every point inside obeys $S^2 \le 8$, Tsirelson's point is inside and reaches it, the PR box is out, and it's strictly bigger than the classical set. There's a certificate checker for it with a soundness proof, in two branches: a strict Sylvester check with the completion supplied, and a rational factorisation certificate of the completed matrix, checked in integers. What's proved is that a pass puts the correlators inside the set and that no choice of inputs gets the PR box through, and, by computation, that some named points pass, the all-ones plan and a point that the arc-sine boundary puts exactly on the quantum edge among them. And every point with a rational PSD completion has such a certificate: peel the completed matrix a row at a time, as in Section H.8, but keep each pivot instead of taking its square root; a zero pivot forces its row to zero, and every number that comes out is rational. The certificate checker is sound and, on rational completions, complete, as mathematics; none of the small machine's properties is the elliptope check.

## G.9 The quantum set

Three descriptions of the same set have been running through this appendix: the tables some quantum strategy can produce, the tables whose five-by-five matrix has a PSD completion, and the tables inside Landau and Masanes's curved edge. They are the same set, for strategies on finitely many levels, and the next subsection takes away the "finitely". Each piece is taken in turn below, because the constructions are the part worth seeing.

Quantum tables are inside the completion set. A strategy's correlator is the inner product of two unit vectors: Alice's observable applied to the shared state, and Bob's applied to it. Four unit vectors always have a PSD table of inner products, so every quantum table has a completion, and Tsirelson's bound follows from that: $S^2 \leq 8$ for every strategy, and its level-one moment matrix passes the PSD test with the marginals in. Quantum mechanics uses complex amplitudes, and the complex case goes through by a plain trick: a complex vector of length $n$ is a real vector of length $2n$, its real parts followed by its imaginary parts, and the real part of a complex inner product is the ordinary inner product of those longer real vectors. So the real argument runs unchanged. The two-qubit strategy written out in Appendix H reaches $2\sqrt{2}$ in both readings.

The completion set is inside the quantum tables. This direction is Tsirelson's representation, and the construction is his. Take a table with a PSD completion. A PSD matrix is a table of inner products of vectors, always: peel off one row at a time, and a zero on the diagonal forces its whole row to zero. So the completion hands over four unit vectors, $a_0, a_1$ for Alice and $b_0, b_1$ for Bob, with $a_x \cdot b_y$ equal to the table's entries, and they fit in three dimensions: slide one of the two entries the table leaves open to the end of the range that keeps the completion PSD, and there the completion is singular, which puts the four vectors in a space of three dimensions. The companion volume's four-vector construction (Section H.8 points to it) gives the factorization and the measurements explicitly. Now comes the trick. There are three real four-by-four matrices, built from the two-by-two ones that swap two entries or flip the sign of one, that each square to the identity and anticommute in pairs: multiply any two of them in either order and the answers differ only in sign. Use a vector's three components to mix those three matrices, and you get an observable that squares to the identity, a legitimate measurement. Give Alice and Bob a maximally entangled pair of four-level systems, and the correlator of two such observables comes out as exactly the inner product of their vectors. So the strategy built from the four vectors produces the table, for both real and complex amplitudes. Four levels on each side is always enough, and with complex amplitudes two are.

The curved edge is the same set again. Each correlator is the cosine of an angle, the angle between one of Alice's vectors and one of Bob's. Walk the loop $a_0, b_0, a_1, b_1$

and back to $a_0$: the angle across any one leg can't exceed the sum of the other three, because angles on a sphere obey the triangle inequality. Flip one of Alice's vectors to its opposite and walk the loop again, then do the same with her other one, and between them you get the other four inequalities, with $2\pi$ in them. Written in arcsines, those eight inequalities are Landau and Masanes's four. Going the other way, any four angles that pass them can be met by actual vectors: choose an angle between Bob's two that meets both pairs of constraints, build each of Alice's from the two angles it has to make with them, and check that the determinant that says such a triangle exists comes out nonnegative. So a table has a PSD completion exactly when its arcsines pass the four inequalities, and those are exactly the quantum tables. The point $(3/5, 4/5, 4/5, -3/5)$ sits on the edge, because $\arcsin\frac{3}{5} + \arcsin\frac{4}{5} = \frac{\pi}{2}$ by the 3-4-5 triangle, and so does Tsirelson's point. The interval construction, including both endpoint cases, is in the companion volume (Section H.9 points to it).

## G.10 Any number of levels, and one converse that fails

Nothing here cares about dimension. Quantum mechanics also allows systems with infinitely many levels, and the arguments don't need finitely many. Whatever the dimension, a shared state $v$ gives an expectation on the operators the players measure with: $x \mapsto \mathrm{Re}\,\langle v, x\,v\rangle$. That one line is the whole step. The proofs use only what that expectation always has: you can add, multiply, scale and take adjoints, the product spreads over sums, and the expectation of $x^*x$ is never negative. Each measurement is its own adjoint with the expectation of its square at most one, which covers unsharp measurements too, and Alice's commute with Bob's. That last demand is weaker than putting them on two separate systems, so the result covers the wider, commuting model as well. On that footing Tsirelson's sum of squares goes through, and the CHSH value is at most $2\sqrt{2}$. The same nonnegative form, with each diagonal entry raised to one, completes the correlator table to a PSD matrix, so the four arcsine inequalities hold, and with a unit the full first-level moment matrix, marginals and all, is PSD. The other direction needs no help from infinity: every point inside the arcsine edge is reached on two four-level systems.

Also, the finite-mixture theorem proves only one direction, that a finite local model satisfies the bound, and its converse as stated is false: a table with $S = 0$ that no local model produces. Fine's own theorem, in the form that speaks of correlations, is proved separately as an equivalence: a table of four correlators comes from a local model exactly when each is between $-1$ and 1 and all eight CHSH inequalities hold. The companion volume constructs the finite distribution (Section H.10 points to it). And one more thing about tallies, which the guarantee paragraph above already used: a tally loaded at no charge is not thereby classical (the four rounds with S equal to four). The correlators (one, one, one, minus one) are the PR box, which no

classical model has as its averages, though coin-flip play lands on that four-round tally one time in sixteen; the classical vertex this appendix used, which fails the check like every plan, is (one, one, one, one).

That covers the game, the number, the two walls, and the check. The derivations come in full in the next appendix, Appendix H; what all this costs against physics was Section 19.

APPENDIX H

# The CHSH derivation, in full

Appendix G tells the story; this appendix does the algebra. Each item gives the statement with its hypotheses and the steps you need to check it. Nothing here says that a matrix is a physical state. Where quantum strategies come in, I say exactly how they stand to the matrices.

## H.1 Setting and notation

Start with four piles of tally sheets, one for each pair of settings Alice and Bob can be given. On each round their answers either agree or disagree. Give agreement a score of $+1$ and disagreement a score of $-1$. In each pile, add the scores and divide by the number of rounds. That pile's average is its *correlator*. It says how strongly the two answers agree for that pair of settings.

**Definition H.1** (Settings, outcomes, correlators)**.** Alice receives a setting $x \in \{0,1\}$ and Bob a setting $y \in \{0,1\}$. Each answers a bit, recoded as an outcome $a, b \in \{+1,-1\}$ (bit 0 becomes $+1$, bit 1 becomes $-1$). The correlator $E_{xy}$ is the average of the product $ab$ over the rounds with settings $(x,y)$. The CHSH quantity is

$$S = E_{00} + E_{01} + E_{10} - E_{11}.$$

I collect the four correlators into the $2 \times 2$ matrix

$$E = \begin{pmatrix} E_{00} & E_{01} \\ E_{10} & E_{11} \end{pmatrix},$$

whose rows are indexed by Alice's setting and whose columns by Bob's. The rows of $E$ are $(E_{00}, E_{01})$ and $(E_{10}, E_{11})$; the columns are $(E_{00}, E_{10})$ and $(E_{01}, E_{11})$.

**Definition H.2** (Counters)**.** There are eight counts: for each setting pair, $\text{same}_{xy}$ counts rounds whose two answer bits matched and $\text{diff}_{xy}$ counts rounds whose bits

differed. A round adds one to exactly one of the eight. Write

$$N_{xy} = \text{same}_{xy} + \text{diff}_{xy}, \qquad D_{xy} = \text{same}_{xy} - \text{diff}_{xy}, \qquad E_{xy} = \frac{D_{xy}}{N_{xy}}.$$

A matched round contributes $ab = +1$ and a differing round $ab = -1$, so $D_{xy}$ is the sum of the products and $D_{xy}/N_{xy}$ is their average. Since $|D_{xy}| \le N_{xy}$, every correlator from counters lies in $[-1, 1]$. When a bucket is empty, its correlator is defined as 0. The score is $E_{00} + E_{01} + E_{10} - E_{11}$.

### H.2 The classical bound

Write a fixed plan on a card: Alice's answer to setting zero, her answer to setting one, and Bob's two answers. Each answer is a sign, plus or minus. Before multiplying anything, look at Bob's pair. Either the signs agree or they differ. If they agree, adding them gives plus or minus two and subtracting them gives zero. If they differ, the sum is zero and the difference is plus or minus two. One of those two quantities vanishes; the other cannot exceed two in size. The next calculation uses exactly that split.

**Definition H.3** (Finite factorizable model)**.** Four real correlators $E_{00}, E_{01}, E_{10}, E_{11}$ are factorizable when there are a natural number $\lambda_{\max}$, weights $p(\lambda)$ and response signs $A(x,\lambda)$, $B(y,\lambda)$ for $\lambda = 0, \dots, \lambda_{\max}$ such that: $0 \le p(\lambda) \le 1$; $\sum_{\lambda=0}^{\lambda_{\max}} p(\lambda) = 1$; every $A(x,\lambda)$ and every $B(y,\lambda)$ is $-1$ or $+1$; and for all $x, y$, $E_{xy} = \sum_{\lambda=0}^{\lambda_{\max}} p(\lambda) A(x,\lambda) B(y,\lambda)$. In words: a hidden card $\lambda$ is drawn with probability $p(\lambda)$, and each player reads an answer off the card using only their own setting. Only the correlators are modelled.

□ **Theorem H.1** (The classical bound for a finite local model)**.** If $E$ is factorizable, then $-2 \le E_{00} + E_{01} + E_{10} - E_{11} \le 2$.

**Definition H.4** (A tally consistent with one local plan)**.** For bits $a_0, a_1, b_0, b_1 \in \{0, 1\}$ and a counter record $w$, $w$ is locally consistent with $(a_0, a_1, b_0, b_1)$ when, for each setting pair $(x, y)$: if $a_x = b_y$ then $\text{diff}_{xy} = 0$, otherwise $\text{same}_{xy} = 0$; and every bucket is sampled, $N_{00}, N_{01}, N_{10}, N_{11} > 0$.

**Theorem H.2** (A tally above 2 is not one replayed strategy)**.** Let $w$ be a counter record whose rational score $S(w)$ is greater than 2. Then for all bits $a_0, a_1, b_0, b_1$, $w$ is not locally consistent with $(a_0, a_1, b_0, b_1)$. Every locally consistent record has $S(w) = 2$ or $S(w) = -2$; the first statement is that fact turned around. □

### H.3 Tsirelson's bound from algebra

Now put the cards aside and look just at four numbers in the square. I want to bound their three-plus-one-minus score from their squared sizes. The useful trick is to write the proposed allowance minus the squared score as a pile of squares. A square cannot be negative. If the difference really is that pile, the score cannot outrun the allowance.

**Lemma H.3** (The six-square identity)**.** For all real $a, b, c, d$, □

$$4(a^2 + b^2 + c^2 + d^2) - (a + b + c - d)^2 \\ = (a - b)^2 + (a - c)^2 + (a + d)^2 + (b - c)^2 + (b + d)^2 + (c + d)^2.$$

**Theorem H.4** (Row-bound form of Tsirelson's bound)**.** For all real $E_{00}, E_{01}, E_{10}, E_{11}$ with $E_{00}^2 + E_{01}^2 \le 1$ and $E_{10}^2 + E_{11}^2 \le 1$, $S^2 \le 8$. The absolute form $|S| \le \sqrt{8} = 2\sqrt{2}$ follows. □

**Theorem H.5** (Tightness)**.** The point $E_{00} = E_{01} = E_{10} = 1/\sqrt{2}$, $E_{11} = -1/\sqrt{2}$ satisfies both row bounds and has $S = \sqrt{8}$. □

**Remark H.1** (Where equality can happen)**.** Equality $S^2 = 8$ under the row bounds forces equality in both steps of the proof, in the companion volume: the six squares must all vanish, which gives $a = b = c = -d$, and the rows must be tight, which gives $2a^2 = 1$. So the point of Theorem H.5 and its negative are the only places the bound is reached. A tally's correlators are ratios of whole numbers and $2a^2 = 1$ has no such solution, so no tally reaches the bound; the integer check's passing tallies stay strictly below it and come as near as you like. (Take rational $a, b$ with $a^2 + b^2 = 1$, from a Pythagorean triple, and the correlators $(a, b, b, -a)$: they pass the check of Section H.5 with no slack, and $S = 2(a + b)$, which gets as close to $2\sqrt{2}$ as you like as $a/b$ approaches 1.)

### H.4 The zero-marginal $5 \times 5$ moment matrix

The four averages fill only the Alice–Bob part of a larger table. Give that table five labels: a constant reading 1, Alice's two possible readings, and Bob's two. Each cell records an average product of its row and column readings. A reading multiplied by itself is 1, so the diagonal is all ones. The four averages measured fill the cross-player cells and their reflected partners.

**Definition H.5** (The pinned level-1 matrix)**.** Take rows and columns $I, A_0, A_1, B_0, B_1$. Put 1 on the diagonal and the four correlators in the Alice–Bob cells, and set the four one-party entries (each player's own average answer) and the two within-party entries ($A_0$ against $A_1$, $B_0$ against $B_1$) to zero. The resulting matrix is:

$$M(E) \;=\; \begin{array}{c} \\ I \\ A_0 \\ A_1 \\ B_0 \\ B_1 \end{array} \begin{array}{c} \begin{array}{ccccc} I & A_0 & A_1 & B_0 & B_1 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & E_{00} & E_{01} \\ 0 & 0 & 1 & E_{10} & E_{11} \\ 0 & E_{00} & E_{10} & 1 & 0 \\ 0 & E_{01} & E_{11} & 0 & 1 \end{pmatrix} \end{array}$$

□ **Lemma H.6** (Schur complement, the standard form)**.** Let $P$ be symmetric positive definite. The symmetric block matrix $\begin{pmatrix} P & Q \\ Q^{\mathrm{T}} & R \end{pmatrix}$ is PSD if and only if $R - Q^{\mathrm{T}} P^{-1} Q$ is PSD.

**Definition H.6** (Column contractive)**.** Write $p = 1 - E_{00}^2 - E_{10}^2$, $q = 1 - E_{01}^2 - E_{11}^2$ and $s = E_{00}E_{01} + E_{10}E_{11}$. The correlators are column contractive when

$$p \geq 0, \qquad q \geq 0, \qquad p\,q - s^2 \geq 0.$$

□ **Corollary H.7** (The equivalence)**.** For all real $E_{00}, E_{01}, E_{10}, E_{11}$, the symmetric matrix $M(E)$ is positive semidefinite if and only if the correlators are column contractive.

**Theorem H.8** (PSD implies Tsirelson)**.** If the pinned matrix $M(E)$ is symmetric and positive semidefinite at $E_{00}, E_{01}, E_{10}, E_{11}$, then $S^2 \le 8$ and $|S| \le \sqrt{8}$. □

## H.5 The integer check

The machine has whole-number tallies. The three conditions above were written using averages, which are fractions. You can test the same conditions without ever dividing: first make sure each denominator is positive, then multiply it away. Multiplying by a positive number cannot turn a passing inequality into a failing one or a failing one into a passing one.

**Lemma H.9** (The seven facts are the three conditions with denominators cleared)**.** Assume $N_{00}, N_{01}, N_{10}, N_{11} > 0$ and put $E_{xy} = D_{xy}/N_{xy}$. Then, with $p, q, s$ as in Definition H.6, □

$$p = \frac{A}{N_{00}^2 N_{10}^2}, \qquad q = \frac{B}{N_{01}^2 N_{11}^2}, \qquad s = \frac{C}{N_{00}N_{01}N_{10}N_{11}},$$

$$pq - s^2 = \frac{AB - C^2}{(N_{00}N_{01}N_{10}N_{11})^2}.$$

Because every denominator is positive, $0 \le A \iff p \ge 0$, $0 \le B \iff q \ge 0$ and $C^2 \le AB \iff s^2 \le pq$.

**Theorem H.10** (Soundness of the integer check)**.** If a tally $w$ passes the seven-integer check, then its four correlators $E_{xy} = D_{xy}/N_{xy}$ are column contractive. Consequently its pinned matrix is positive semidefinite. □

**Corollary H.11** (A passed check implies the Tsirelson bound)**.** If a tally $w$ passes the seven-integer check, the correlators of $w$ satisfy $S^2 \le 8$. It is the composition of Theorem H.10, Corollary H.7 and Theorem H.8. □ ■

Two of the seven facts carry the bound by themselves. With every pair sampled, $0 \le A$ and $0 \le B$ are $p \ge 0$ and $q \ge 0$, so the four squared correlators add to at most 2 and Lemma H.3 gives $S^2 \le 8$. The fact $C \cdot C \le A \cdot B$ is what positive semidefiniteness adds, and the bound doesn't use it. For the form this takes on the small machine, where a raised flag has to stand on a passing check, see Appendix G.

### H.6 Level 1+AB

The first table had one place for each player's answer. Enlarge it by giving each Alice–Bob product a place too. There are four such products, so the five places become nine. A cell still asks for the average product of its row and column labels. Some of those products reduce to entries I already have; five more numbers are needed for the particular filling below. Call them $\gamma_1$ through $\gamma_5$. The longer labels say which products they stand for.

**Definition H.7** (The level-1+AB matrix)**.** The $9 \times 9$ real matrix on rows and columns $I, A_0, A_1, B_0, B_1, A_0B_0, A_0B_1, A_1B_0, A_1B_1$, with the following entries:

| | $I$ | $A_0$ | $A_1$ | $B_0$ | $B_1$ | $A_0B_0$ | $A_0B_1$ | $A_1B_0$ | $A_1B_1$ |
|---|---|---|---|---|---|---|---|---|---|
| $I$ | 1 | 0 | 0 | 0 | 0 | $E_{00}$ | $E_{01}$ | $E_{10}$ | $E_{11}$ |
| $A_0$ | 0 | 1 | 0 | $E_{00}$ | $E_{01}$ | 0 | 0 | $\gamma_1$ | $\gamma_2$ |
| $A_1$ | 0 | 0 | 1 | $E_{10}$ | $E_{11}$ | $\gamma_1$ | $\gamma_2$ | 0 | 0 |
| $B_0$ | 0 | $E_{00}$ | $E_{10}$ | 1 | 0 | 0 | $\gamma_3$ | 0 | $\gamma_4$ |
| $B_1$ | 0 | $E_{01}$ | $E_{11}$ | 0 | 1 | $\gamma_3$ | 0 | $\gamma_4$ | 0 |
| $A_0B_0$ | $E_{00}$ | 0 | $\gamma_1$ | 0 | $\gamma_3$ | 1 | 0 | 0 | $\gamma_5$ |
| $A_0B_1$ | $E_{01}$ | 0 | $\gamma_2$ | $\gamma_3$ | 0 | 0 | 1 | $-\gamma_5$ | 0 |
| $A_1B_0$ | $E_{10}$ | $\gamma_1$ | 0 | 0 | $\gamma_4$ | 0 | $-\gamma_5$ | 1 | 0 |
| $A_1B_1$ | $E_{11}$ | $\gamma_2$ | 0 | $\gamma_4$ | 0 | $\gamma_5$ | 0 | 0 | 1 |

Write a vector as $(v_I, v_{A_0}, v_{A_1}, v_{B_0}, v_{B_1}, v_{00}, v_{01}, v_{10}, v_{11})$, where $v_{xy}$ is the coordinate of $A_xB_y$. Put

$$\begin{aligned} L_1 &= E_{00}v_{00} + E_{01}v_{01} + E_{10}v_{10} + E_{11}v_{11}, \\ L_2 &= E_{00}v_{B_0} + E_{01}v_{B_1} + \gamma_1 v_{10} + \gamma_2 v_{11}, \\ L_3 &= E_{10}v_{B_0} + E_{11}v_{B_1} + \gamma_1 v_{00} + \gamma_2 v_{01}. \end{aligned}$$

**Definition H.8** (The residual)**.**

$$\begin{aligned} R = {} & v_{B_0}^2 + v_{B_1}^2 + v_{00}^2 + v_{01}^2 + v_{10}^2 + v_{11}^2 \\ & + 2v_{B_0}(\gamma_3 v_{01} + \gamma_4 v_{11}) + 2v_{B_1}(\gamma_3 v_{00} + \gamma_4 v_{10}) \\ & + 2\gamma_5(v_{00}v_{11} - v_{01}v_{10}) - L_1^2 - L_2^2 - L_3^2. \end{aligned}$$

The condition called column contractivity at level 1+AB is $R \geq 0$ for all six real inputs.

**Theorem H.12** (PSD at level 1+AB)**.** For all real $E_{xy}$ and $\gamma_1, \ldots, \gamma_5$, the matrix of Definition H.7 is positive semidefinite if and only if the residual $R$ is nonnegative for every choice of its six real inputs. □

## H.7 The elliptope completion

Go back to the two within-player cells I filled with zero. Leave them blank this time. Keep all four measured cross-player averages fixed, and ask whether there is any way to fill those two blanks that makes the table pass. A failure with zeros no longer settles the question; another filling may work.

**Definition H.9** (Completion and membership)**.** Complete the matrix of Section H.4 by setting the within-party entries to free reals: $x$ in the $(A_0, A_1)$ cells and $y$ in the $(B_0, B_1)$ cells, marginals still zero. The correlators are elliptope realizable when some $x, y$ make the completed matrix symmetric and PSD.

$$M(E; x, y) \;=\; \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & x & E_{00} & E_{01} \\ 0 & x & 1 & E_{10} & E_{11} \\ 0 & E_{00} & E_{10} & 1 & y \\ 0 & E_{01} & E_{11} & y & 1 \end{pmatrix}$$

Its quadratic form at $v = (v_0, \ldots, v_4)$ is $\sum_k v_k^2 + 2xv_1v_2 + 2yv_3v_4 + 2E_{00}v_1v_3 + 2E_{01}v_1v_4 + 2E_{10}v_2v_3 + 2E_{11}v_2v_4$. The pinned matrix is the case $x = y = 0$, so PSD of $M(E)$ implies membership.

**Definition H.10** (The two branches)**.** The positive-minor branch: with the completion supplied as buckets and all entries scaled to integers by a positive common factor $K$, the four leading minors of the $4 \times 4$ block on $A_0, A_1, B_0, B_1$ are strictly positive. The factorization branch: with the completion and a rational certificate supplied (six subdiagonal entries over a common positive denominator, four pivots $d_k \geq 0$ over another), the ten entry equations of $M = LDL^{\mathrm{T}}$ hold in cross-multiplied integer arithmetic. The full gate is their disjunction.

□ **Theorem H.13** (Soundness of the gate)**.** If the full gate returns yes on a counter record, any completion buckets and any certificate, then the four correlators of the record are elliptope realizable. On the PR box tally the gate returns no for every choice of the remaining arguments.

### H.8 The four-vector construction

Put two equal-length pencils down with their ends together, and draw a line along one of them. Keep that pencil still and turn the other. Use the square corner of a sheet of paper to bring the moving pencil's tip straight across to the line, and mark where it lands. The mark moves from a full pencil-length forward, through the joined ends when the pencils make a square corner, to a full pencil-length backward. Call those readings 1, 0 and $-1$. That is the signed shadow I want. Between those positions the reading is the cosine of the angle. A dot product of unit vectors measures that shadow.

### H.9 The arcsine boundary, in both directions

Keep Bob's two directions hinged at one end. Alice's first direction has to make a specified angle with each. How far can Bob's hinge open while that is possible? Hold Alice's direction upright and sweep each of Bob's around it at its required angle. Point them towards the same side to close the hinge as far as it will go; point them towards opposite sides to open it as far as it will go. There is a whole interval of openings between those limits. Alice's second direction asks for another interval.

### H.10 The finite local model, in both directions

Make sixteen cards, each with four boxes labelled $A_0, A_1, B_0, B_1$, and put every possible arrangement of plus and minus signs on them, one arrangement per card. Draw a card before choosing which setting to ask each player about. Alice reads her chosen box; Bob reads his. Neither needs the other's setting. How often each card is drawn is the only choice left in this model.

Every proof of this appendix, the worked tallies and the level-1+AB cascade are in the companion volume *The Thiele Machine: The Example, Built* (its section "The CHSH derivation, in full") and in Coq in `coq/kernel/quantum/TsirelsonGeneral.v`, `CHSHColumnCheck.v`, `QuantumPartitionPSD_1AB.v`, `ElliptopeCompletion.v` and `ArcsineBoundary.v`.

## H.11 Scope of the constructions

The constructions of this appendix, carried in full in the companion volume, pin down the correlators. What follows is where they stop: they aren't a completeness claim for the integer check, they say nothing about full behaviours with marginals, and they say nothing about data from a lab.

The arcsine boundary. For two settings and two outcomes per side, Tsirelson, Landau (1988) and Masanes (2003) give theorems characterizing the correlators reachable by quantum measurements: $|\arcsin E_{00} + \arcsin E_{01} + \arcsin E_{10} - \arcsin E_{11}| \leq \pi$ and its sign variants. The characterization holds as an if and only if with the completion set and, through Tsirelson's representation, with the quantum set. The point $(3/5, 4/5, 4/5, -3/5)$ is on the boundary: $\arcsin \frac{3}{5} + \arcsin \frac{4}{5} = \frac{\pi}{2}$ by the 3-4-5 triangle, so the sum is $\pi$; Tsirelson's point is on it too. The necessary direction holds in any dimension, for any expectation on an algebra of observables in which Alice's commute with Bob's.

Equality with the quantum set. That the completion set of Section H.7 is exactly the set of quantum correlators is Tsirelson's representation theorem, and it is proved in finite dimension: for both real and complex amplitudes, with two four-level systems enough for every point, and two two-level ones with complex amplitudes. That PSD of the level-1 moment matrix is necessary for quantum behaviour is proved for strategies in finite dimension, with real amplitudes and with complex ones, and in any dimension, marginals included, for any expectation on an algebra of observables.

Achievability. Theorem H.5 is a point of $\mathbb{R}^4$. A state and measurements that reach it are explicit: the two-qubit state $(|00\rangle + |11\rangle)/\sqrt{2}$ with $A_0 = Z$, $A_1 = X$, $B_0 = (Z + X)/\sqrt{2}$ and $B_1 = (Z - X)/\sqrt{2}$ has correlators $1/\sqrt{2}$, $1/\sqrt{2}$, $1/\sqrt{2}$, $-1/\sqrt{2}$ and $S = 2\sqrt{2}$, and every strategy in finite dimension has $S^2 \leq 8$, with real amplitudes and with complex ones. Both are arithmetic about the model. How close a real laboratory comes, and what that costs in loopholes closed, is the story told in Appendix G.

Completeness of the checks. The 1+AB cascade and the elliptope gate are proved sound. The cascade has no proved converse; the gate's factorization branch accepts every rational PSD completion, by the row-peeling of Section H.8 with each pivot kept. The level-1 integer check is the exception: on the small machine the same seven integer facts hold exactly when every pair was sampled and the pinned matrix is positive semidefinite (Theorem G.4), so that check is complete for what it tests. The 1+AB cascade is strict and misses boundary points. The two 1+AB checks don't sit inside one another. The full check accepts the $12/5$ tally above, which the zero-moment check refuses, and it refuses the tally with $E_{00} = 1$ and the other three correlators 0, where $H_{11} = 0$, which the zero-moment check accepts.

Finite levels. For full behaviours, marginals included, the NPA levels are relaxations. Chaturvedi (arXiv 2607.14569, 2026) argues, in a preprint, that no finite NPA level characterizes the complete quantum set even in this simplest Bell scenario. I haven't gone through that proof, and nothing here rests on it.

Statistics and physics. Nothing here says how many rounds make a tally informative, and nothing says the counters came from honest play; a program can INC whatever counts it likes.

### H.12 Prior work

Tsirelson's 1980 paper gives the bound $2\sqrt{2}$ for quantum correlations, and his theorem says that quantum two-outcome correlators are inner products of real unit vectors. Landau (1988) and Masanes (2003, arXiv quant-ph/0309137) each give a condition that, by their proofs, characterizes the quantum correlators exactly for two settings and two outcomes. Wehner (2006, arXiv quant-ph/0510076) gets Tsirelson bounds, CHSH among them, from a semidefinite program on the $4 \times 4$ Gram matrix of those vectors. Navascués, Pironio and Acín (2007, arXiv quant-ph/0607119; 2008, arXiv 0803.4290) built the hierarchy of moment-matrix tests; their 2008 paper applies a Schur complement to the level-1 matrix of this scenario, recovers the arcsine condition, and discusses the 1+AB level. I went back to each of those papers for these points, and that is what I found in them. The results are theirs. Two of them are proved again here, in finite dimension: Tsirelson's representation and the arcsine characterization; the rest of their proofs are not redone. Echenim and Mhalla give a detailed derivation of Tsirelson's bound (arXiv 2306.12535), and exact rational sum-of-squares certificates go back at least to Peyrl and Parrilo (2008). So the moment-matrix route to $2\sqrt{2}$ is standard. What is specific here is narrower: an integer certificate computed from outcome counts, and the argument from "the counts pass" to PSD of the pinned matrix and on to $S^2 \leq 8$. I did not find that chain elsewhere, and I make no claim beyond it. On the host machine, the same chain is earned: a raised flag from a clean start stands on a passing check of the packed counts, and the check passes exactly when the pinned matrix is positive semidefinite and every pair was sampled (Appendix G, Theorems G.4 and G.5). The converse direction of the integer check, integers from PSD, is proved there too, and there too I make no claim beyond the chain.

APPENDIX I

# Constructing the quoting operation

The proof in Section 13 uses a program that takes the code of a term and returns the code of that code. Here's the idea; the companion volume builds it.

Draw the instruction "add one" on a card. Now make a second card describing the first card's marks, rather than carrying out the addition. The count has not risen: you have described an instruction. Describe the second card in turn and you have a description of a description. Quoting does that second job mechanically. In that language, take the variable of index 0. Its code is a tree with three lambda nodes, then an application of variable 2 to the numeral for 0. Quoting that code builds a description of those three lambdas and that application; it does not look up variable 0 or execute the term it describes.

The construction is in the companion volume *The Thiele Machine: The Example, Built* (its section "Constructing the quoting operation") and in Coq in `coq/kernel/foundation/LRecursion.v`.

APPENDIX J

# Computing with counters

Put a few beans in one cup and leave a second empty. Write a short list of orders: add one, take one away if there is one, go to this line, stop. A finger on the page keeps your place. Carry out the orders literally. The question in this appendix is how to give the same job to another person with different equipment (a strip of paper, stacks of cards, or a different set of cups) and get the same answer whenever the first person stops.

Agree on how the input is laid out and how to read the output before either starts. One person must not quietly solve the problem while preparing the other's cups. The reductions in this appendix concern functions on natural numbers. To compute a relation $R(\vec{x}, y)$ means to halt with output $y$ exactly when $R(\vec{x}, y)$ holds, and to produce only numerical outputs on halting runs. The machines the companion volume builds are deterministic, so any relation they compute is the graph of a partial function. A translation must preserve both the output and whether the run terminates.

## J.1 Packing finitely many counters into two

Try holding two counts in one pile of eighteen beans. Count how many times you can halve eighteen exactly: once. Count how many times you can divide it exactly by three: twice. Reassemble the eighteen beans after each reading. Those are the two stored counts, 1 and 2. Double the pile to 36 and only the first stored count rises. Divide 36 by three to get 12 and only the second falls. Divide by three again: four beans, storing 2 and 0.

The constructions, the packing of finitely many counters into two, the seven descriptions of one computation and the undecidability of halting are in the companion volume *The Thiele Machine: The Example, Built* (its section "Computing with counters") and in Coq in `coq/kernel/foundation/TcGodel.v`, `TcPacked.v`, `LiftModelsAll.v` and `MM2ComplementUndec.v`.

APPENDIX K

# The universal program in blocks

Put the guest's instructions on one sheet and the host's on another, with a finger marking the current line on each. The guest takes one instruction. The host takes as many of its smaller instructions as it needs, then returns to the head of its sheet. Compare the two arrangements there: counts, live facts, commitment, flag and ledger. The host is allowed to do more work between comparisons. It is not allowed to lose a fact, revive a dead one, or charge another unit for its bookkeeping. The lists in the companion volume let you carry out that comparison instruction by instruction.

The blocks of $U$ and $U_P$, their addresses and their specifications are in the companion volume *The Thiele Machine: The Example, Built* (its section "The universal program in blocks") and in Coq in `coq/kernel/foundation/UniversalLayout.v`, `UniversalBlocks.v`, `UniversalPLayout.v` and `UniversalPBlocks.v`.

APPENDIX L

# Methodology note

Everything this book claims as its own stands or falls on the arguments written in it; what it borrows is cited to the people who proved it, and the physics enters as a named premise. If a proof on the page is wrong, the result is wrong, and nothing else I could tell you would rescue it.

Language models were my hands, as I said in Section 24: they did symbol-pushing, drafted code and helped work through derivations. I have to answer for what stayed. Start with the definitions and follow how each step leads to the conclusion. If I counted a finite construction, its cases belong on the page. If I claimed something for every run, the argument has to explain why the next step preserves it. A successful example cannot carry that claim, and a name attached to a proof cannot fill a missing step.

The same demand applies to the picture around the mathematics. A definition has to say what I meant, and a model of a device has to fit that device. Those questions belong beside the premises and the argument, where you can examine the fit.

APPENDIX M

# Sources

These are the works I leaned on, for their results, their history and the company they keep. A few words on how to read the list. The counter and lambda references are the constructions the universal programs and the compiler stand on (Appendices C.1 and C.2, and I to K); the credit for those constructions is theirs. The thermodynamics references don't take the physical premise out of the heat bound; it stays a premise. And the papers on records in the environment are where the pointer thesis got its shape; none of them proves it.

## M.1 Structure, information and knowledge

Edwin A. Abbott (1884). Flatland: A Romance of Many Dimensions, by A Square. Seeley and Co., London.

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Aristotle. Physics II.1; Metaphysics VII.3 and VII.11. Matter, form and substance; VII.11 distinguishes the material of a circle from the functioning parts of an animal (MIT Classics text). Also Metaphysics I.5 (985b–986b), on the Pythagoreans: "they supposed the elements of numbers to be the elements of all things" (trans. W. D. Ross, MIT Classics text); and Prior Analytics I.23 (41a23–30), the proof by contradiction that if the diagonal were commensurable with the side, odd numbers would equal even ones.

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